



Programming Language TMscript

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1. Overview

TMscript is the programming language of Techman Robot applicable to Flow projects and Script projects. Refer to the table below for the main scope of applications.

● Scope of Applications

Application	Flow Projects			Script Projects
	Set Node (& other nodes)	Listen Node (external scripts)	Script Node	
Expression	✓	✓	✓	✓
define/main/closestop/errorstop				✓
Customized Function				✓
Comment		✓	✓	✓
Global variables in the Project	✓	✓	✓	✓
Local variables in the Function		✓	✓	✓
Multiline Input		✓	✓	✓
Conditional Statement		✓	✓	✓
Loop Statement		✓	✓	✓
Branching Statement		✓	✓	✓
Thread Function				✓
General Function	✓	✓	✓	✓
General Function (Script)		✓	✓	✓
Math Function	✓	✓	✓	✓
File Function	✓	✓	✓	✓
Serial Port Class		✓	✓	✓
Serial Port Function	✓	✓	✓	✓
Socket Class				
Socket Function	✓	✓	✓	✓
MDecision Class		✓	✓	✓
Parameterized Object	✓	✓	✓	✓
Robot Teach Class		✓	✓	✓
Robot Motion & Vision Job Function		✓	✓	✓
Vision Function	✓	✓	✓	✓
ScriptListen Function		✓	✓	✓
External Script/Priority Command		✓		
Modbus TCP/RTU Class		✓	✓	✓
Modbus Function	✓	✓	✓	✓
TM Ethernet Slave Function	✓	✓	✓	✓
Profinet Function	✓	✓	✓	✓
EtherNet/IP Function	✓	✓	✓	✓
EtherCAT Function	✓	✓	✓	✓
CC-Link Function	✓	✓	✓	✓
Real Time Remote Server Function				
Compliance Class		✓	✓	✓
TouchStop Class		✓	✓	✓
FTSensor Class		✓	✓	✓
Force Class		✓	✓	✓

2. Expression

2.1 Types

Different data types of variables can be declared in Variable Manager.

byte	8bit integer	unsigned	0 to 255	significant digit 3
int	32bit integer	signed	-2147483648 to 2147483647	significant digit 10
float	32bit floating-point	signed	-3.4028235E+38 to 3.4028235E+38	significant digit 7
double	64bit floating-point	signed	-1.7976931348623157E+308 to 1.7976931348623157E+308	significant digit 15
bool	Boolean		true or false	
string	string			

In function terms, the integer type further goes by int16 and int32. The default is int32.

int16	16bit integer	signed	-32768 to 32767	significant digit 5
int32	32bit integer	signed	-2147483648 to 2147483647	significant digit 10

2.2 Variables and Constants

1. Variables

Users can only use numeric or English character combinations without special characters or other characters, and the name after the combination cannot be a numeric value, a string, or a Boolean value.

Numbers 0123456789

Characters a-z, A-Z, _

Example

```
Int i = 0
```

```
string s = "ABC"
```

```
string s1 = "DEF"
```

```
string s2 = "123"
```

Without double quotation marks, strings will be taken as variables.

```
s = s1 + " and " + s2    // s = "DEF and 123"
```

```
                          // s, s1, s2 are variable, and " and " is a string.
```

In addition to variables, the naming rule also applies to constants, numbers, strings, and Booleans **except that string constants need to be enclosed in double quotes**.

When a variable is generated in TMflow, a prefix is added based on the source. To use the variable for writing or reading, users must enter the full name including the prefix word such as var_s1 or g_s2. For the rules of adding prefixes, refer to the respective description in variable setting pages.

2. Numbers

- Decimal integer, decimal floating-point, binary, hexadecimal integer and scientific notation are supported.

Decimal integer 123
 -123
 +456

Decimal float 34.567
 -8.9

Binary 0b0000111
 0B1110000

Hexadecimal integer 0x123abc
 0X00456DEF

Scientific notation 3.4e5
 2.3E-4

- For binary and hexadecimal notation, there is no floating-point.
- The notation of number is not case sensitive.

For example:

0b0011 equals 0B0011

0xabcD equals 0XABCD, 0xABCd, 0Xabcd etc.

3.4e5 equals 3.4E5

- The system determines the data types of numbers automatically when using numbers as constants. The rule is to conclude bit types from the smaller to the greater such as

```
100          // data type: byte          // 100 is in the value range of data type: byte.
1000         // data type: int
1.11         // data type: float        //1.11 is in the value range of data type: float.
```

To assign the data type, use variable declaration or conversion to do so such as

```
byte b = 100 // variable b = 100 as data type: byte
int i = 100  // variable i = 100 as data type: int
(int)100     // constant 100 as data type: int
(float)100   // constant 100 as data type: float
```

For function calls, the system determines the data types of arguments and selects the respective syntax. If there is no respective syntax, the system determines the compatible syntax with the rule concluding bit types from the smaller to the greater such as

```
GetBytes(100, 0, 0) // {0x64,0x00,0x00,0x00} // 100 and the 0s come with the data type:
byte, but the syntax goes with GetBytes(int, int, int). The system, therefore,
converts 100 and the 0s into the data type: int to proceed with the call syntax.

GetBytes(100) // {0x64} // 100 comes with the data type: byte. in the syntax GetBytes(?) to go
with any data types. Therefore, the system takes 100 as a constant in data type:
byte to proceed the call syntax.
```

- Byte can only present unsigned numbers in 8 bits ranging from 0 to 255. As a result, if a negative sign is assigned to the byte type or through calculation, it will still save the 8-bit unsigned value only.

For example:

```
byte b = -100 // Error // -100 mismatches with the value range of byte.
byte b = 0 - 100 // b = 156
// 0-100=-100 (0xFFFFF9C). Saved 8 bits as 0x9C. The value equals 156.

b = 0 - 1 // b = 255 // 0-1=-1 (0xFFFFFFFF) Saved 8 bits as 0xFF The value equals 255.
b = 255 + 1 // b = 0 // 255+1=256 (0x100) Saved 8 bits as 0x00 The value equals 0.
```

- Int can present signed numbers in 32 bits ranging from -2147483648 to 2147483647. If the calculation exceeds the value range, it will still save the 32-bit signed integer value only.

For example:

```
int i = -2147483648 - 1 // i = 2147483647 // -2147483648 - 1 = -2147483649 (0xFFFFFFFF
7FFFFFFF) Saved 32 bits as 0x7FFFFFFF. The value equals 2147483647.
i = 2147483647 + 1 // i = -2147483648 // 2147483647 + 1 = 2147483648 (0x80000000)
Save 32 bits in the method of signed integer value. The value equals -2147483648
```

3. String

When inputting string constant, double quotation marks shall be placed in pairs around the string to avoid the recognition error of variable and string. ◦

For example

```
"Hello World! "
```

"Hello TM""5" (If " is one of the character in the string, use two (") instead of one (").

- Control character in double quotation mark are not supported.

For example:

```
"Hello World!\r\n" (the output would be Hello World!\r\n string)
```

- Without double quotation marks, the compiling will follow the rules below

1. Numbers will be viewed as numbers
2. The combination of numbers and characters will be viewed as variable as long as the variable does exist.
3. If the variable does not exist, it will be compiled as string with a warning message.
- The combination of string and variable
 1. Inside double quotation marks, variables will not be combined as variables
For example:


```
s = "TM5"           // s = "TM5"
s1 = "Hi, s Robot"  // s1 = "Hi, s Robot"
```
 2. Standard syntax. Double quotation marks need to be placed around the string, and plus sign (+) shall be used to link variables and numbers
Example:


```
s1 = "Hi, " + s + " Robot" // s1 = "Hi, TM5 Robot"
```
 3. Compatible syntax (not recommended). The single quotation marks can be placed around the variables, but a warning message will be sent out
For example:


```
single quotation marks  "Hi, 's' Robot"           // s1 = "Hi, TM5 Robot"
                        "Hi, 'x' Robot"           // s1 = "Hi, 'x' Robot" // Because variable x does not exist, 'x' is
                        viewed as string
```
 4. Single quotation marks do not support element value retrieval with array indexes. The standard format with double quotation marks should be used.
For example


```
string[] ss = {"Techman", "Robot"}
"Hi, 's' 'ss[0]' Robot" // s1 = "Hi, TM5 'ss[0]' Robot"
                        // 'ss[0]' is invalid
"Hi, " + s + " " + ss[0] + " Robot" // s1 = "Hi, TM5 Techman Robot"
```
 5. Single quotation marks cannot be presented by "'". If users would like to input '(variable name)', The standard format with double quotation marks should be used.
For example


```
"Hi, 's' Robot" // s1 = "Hi, TM5 Robot"
// If s1 = "Hi, 's' Robot" is what you want, please use the following syntax.
"Hi, "'" + "s" + "'" Robot" // s1 = "Hi, 's' Robot"
```
- For control character, e.g. new line, please use Ctrl() command.
For example


```
s1 = "Hi, " + Ctrl("\r\n") + s + " Robot" or "Hi, " + NewLine + s + " Robot"
Hi,
TM5 Robot
```
- Reserved characters is similar to variables, no double quotation marks is needed. (But single quotation mark is not supported)
 1. empty empty string, equals ""
 2. newline or NewLine new line, equals Ctrl("\r\n") or Ctrl(0x0D0A)

4. Boolean

True or false value of logic.

Denote true value true
 True

Denote false value false

False

The Boolean value is case sensitive. Misuses of capital letters such as TRue will be taken as a variable or a string.

2.3 Array

- Array is a set of data with the same data type. The initial value is assigned with {}, and every element remains the characteristic of its data type.

For example

```
int[] i = {0,1,2,3}           // elements in number data type
string[] s = {"ABC", "DEF", "GHI"} // elements in string data type
bool[] bb = {true, false, true} // elements in boolean data type
```

- By utilizing index, the value of specified element can be get, the index is start from 0

For example

index	0	1	2	3	4	5	6	7	
array									eight elements in total
	A[0]	A[1]	A[2]	A[3]	A[4]	A[5]	A[6]	A[7]	

Valid index values [0] .. [7].

An error will occur if accessing beyond the range, such as A[8].

- Only one degree array is supported. The maximum index number is 2048.
- The array size may alter according to the return value of functions or assigned values. The maximum element number is 2048. This feature makes array meet the needs of different functions and applications in Network Node.

For Example:

```
string[] ss = {empty, empty, empty} // The initial size of string array is 3 elements
ss = String_Split("A_B_C_D_F_G_H", "_") // After splitting string, the string array has 7
elements
len = Length(ss) // len = 7
ss = String_Split("A,B", ",") // After splitting string, the string array has 2 elements
len = Length(ss) // len = 2
```

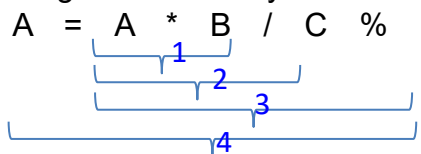
2.4 Operator Symbols

- The operator table is listed below.
- The calculation follows the precedence of operator first then the associativity.

For example

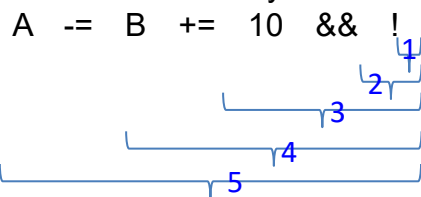
left-to-right associativity

$A = A * B / C \% D \rightarrow (A = (((A * B) / C) \% D))$



right-to-left associativity

$A -= B += 10 \&\& !ID \rightarrow (A -= (B += (10 \&\& (!ID))))$



- The calculation will proceed by the type of the operand.
 1. When both values come as the integer type, the calculation will proceed by the integer type such as


```
int var_a = 10
int var_b = 3
float var_c = var_a / var_b
```

By the operator priority, the calculation goes / first and then =.

$var_a / var_b = 10 / 3 = 3$ (both var_a and var_b are integers)

$var_c = 3$ (The integer 3 assigns to the floating point number var_c)
 2. When one of the two values comes as the floating-point type, the calculation will proceed by the floating-point type such as


```
int var_a = 10
float var_b = 3
float var_c = var_a / var_b
```

$var_a / var_b = 10 / 3 = 3.333333$ (for var_b is a floating-point number)

$var_c = 3.333333$ (the floating-point 3.333333 assigns to var_c)

```
var_c = var_a / 3
```

$var_a / 3 = 10 / 3 = 3$ (both var_a and 3 are integers)

$var_c = 3$

```
var_c = var_a / 3.0
```

$var_a / 3.0 = 10 / 3.0 = 3.333333$ (for 3.0 is a floating-point)

$var_c = 3.333333$

Precedence High to low	Operator	Name	Example	Requirement	associativity
17	++	Postfix increment	i++	Integer variable	left-to-right
	--	Postfix decrement	i--		
	()	Function call	int x = f()		
	[]	Allocate storage	array[4] = 2	Array variable	
16	++	Prefix increment	++i		right-to-

Precedence High to low	Operator	Name	Example	Requirement	associativity
	--	Prefix decrement	--i	Integer	left
	+	Unary plus	int i = +1	Numeric variable, Constant	
	-	Unary minus	int i = -1	Numeric variable, Constant	
	!	Logical negation	if (!done) ...	Boolean	
	~	Bitwise NOT	flag1 = ~flag2	Integer	
14	*	Multiplication	int i = 2 * 4	Numeric variable, Constant	left-to-right
	/	Division	float f = 10.0 / 3.0		
	%	Modulo (integer	int rem = 4 % 3		
13	+	Addition	int i = 2 + 3	Numeric variable, Constant	
	-	Subtraction	int i = 5 - 1		
12	<<	Bitwise left shift	int flags = 33 << 1	Integer variable, Constant	
	>>	Bitwise right shift	int flags = 33 >> 1		
11	<	Less than	if (i < 42) ...	Numeric variable, Constant	
	<=	Less than or equal to	if (i <= 42) ...		
	>	Greater than	if (i > 42) ...		
	>=	Greater than or equal	if (i >= 42) ...		
10	==	Equal to	if (i == 42) ...		
	!=	Not equal to	if (i != 42) ...		
9	&	Bitwise AND	flag1 = flag2 & 42	Integer variable, Constant	
8	^	Bitwise XOR	flag1 = flag2 ^ 42		
7		Bitwise OR	flag1 = flag2 42		
6	&&	Logical AND	if (condition A && condition B)		
5		Logical OR	if (condition A condition B)		
4	c ? t : f	Ternary conditional	int i = a > b ? a : b		right-to-left
3	=	Basic assignment	int a = b	Left side: Numeric variable Right side: Numeric variable, Constant	
	+=	Addition assignment	a += 3		
	-=	Subtraction	b -= 4		
	*=	Multiplication	a *= 5		
	/=	Division assignment	a /= 2		
	%=	Modulo assignment	a %= 3		

Precedence High to low	Operator	Name	Example	Requirement	associativity
	<<=	Bitwise left shift assignment	flags <<= 2	Left side: Integer variable Right side: Integer variable, Constant	
	>>=	Bitwise right shift assignment	flags >>= 2		
	&=	Bitwise AND	flags &=		
	^=	Bitwise XOR	flags ^=		
	=	Bitwise OR	flags = new_flags		

2.5 Data Type Conversion

- Data types can be converted to each other and used in variables/constants or arrays.
- Conversions must be in the same format of the containers such as variable/constant conversions or array conversions. **It is not permitted to convert a variable to an array or an array to a variable.**

Native type	Conversion type	Example	Result
byte	int	int i = (int)100	i = 100
	float	float f = (float)100	f = 100
	double	double d = (double)100	d = 100
	bool	bool flag = (bool)0	flag = false (0 equals false)
	string	string s = (string)100	s = "100"
int	byte	byte b = (byte)1000	b = 232
	float	float f = (float)1000	f = 1000
	double	double d = (double)1000	d = 1000
	bool	bool flag = (bool)1000	flag = true (non 0 equals true)
	string	string s = (string)1000	s = "1000"
float	byte	byte b = (byte)1.23	b = 1
	int	int i = (int)1.23	i = 1
	double	double d = (double)1.23	d = 1.23
	bool	bool flag = (bool)1.23	flag = true (non 0 equals true)
	string	string s = (string)1.23	s = "1.23"
double	byte	byte b = (byte)1.23	b = 1
	int	int i = (int)1.23	i = 1
	float	float f = (float)1.23	f = 1.23
	bool	bool flag = (bool)1.23	flag = true
	string	string s = (string)1.23	s = "1.23"
bool	byte	byte b = (byte)True	Error
	int	int i = (int)False	Error
	float	float f = (float>true	Error
	double	double d = (double>false	Error
	string	string s = (string)True	s = "true" (shown in lower-case)
string	byte	byte b1 = (byte)"1.23" byte b2 = (byte)"XYZ"	1 Error (Unable to convert "XYZ" to value)
	int	int i = (int)"1.23"	1
	float	float f1 = (float)"1.23" float f2 = (float)"A1"	1.23 Error (Unable to convert "A1" to value)
	double	double d = (double)"1.23"	1.23
	bool	bool flag1 = (bool)"true" bool flag2 = (bool)"false" bool flag3 = (bool)"1.23" bool flag4 = (bool)""	flag1 = true (string "true" as true) flag2= false (string "false" as false) flag3 = true (non-empty string as true) flag4 = false (empty string for false)

- The conversion method of arrays is in accordance with the table above. The conversion is performed for each element in the array.

```
string[] ss = {"1.23", "4.56", "0.789"}
```

```
int[] i_array = (int[])ss      // i_array = {1, 4, 0}
float[] f_array = (float[])ss  // f_array = {1.23, 4.56, 0.789}
```

- **Error messages will be returned should the conversions below occur.**
 - Fail to convert to numeric correctly such as Booleans (true/false) or non-numeric strings ("XYZ").

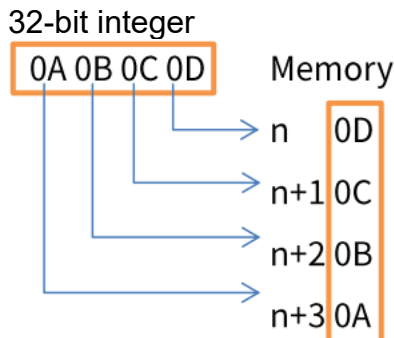

```
int value = (int>true          // Error
int value = (int)"XYZ"        // Error
```
 - Invalid floating-point numbers to convert to floats or doubles such as NaN or Infinity.


```
string dvalue = "1.79769e+308"
float f = (float) dvalue // Error 1.79769e+308 is a valid double type and unable to convert to the
                           float type.
```

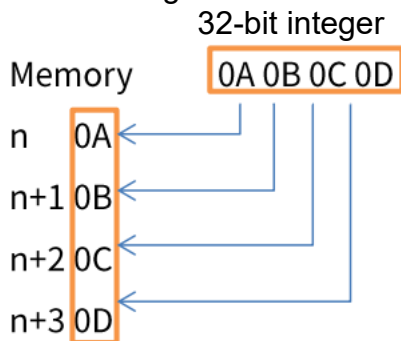
2.6 Endianness and Conversion

Endianness refers to the applications in the memory or the communication networks in which data comes in multiple bytes for the expressions of the order to sort among multiple bytes because the minimal unit comes in bytes.

- Little Endian. Place the low bits of the multibyte at smaller addresses and the high bits at larger ones.



Big Endian. Place the high bits of the multibyte at smaller addresses and the low bits at larger ones.



It is a factor to consider because different orders to sort may cause different conversion results. For example, an integer type is 32 bits. Namely, it occupies 4 bytes. If the int value is 300, the expression in 16-bit is 0x0000012C. If using Little Endian for memory storing or communication networks, the order is

[0] [1] [2] [3]
2C 01 00 00

It fetches 0x0000012C if adopting Little Endian while 0x2C010000 if Big Endian. These are two completely different values, which may cause different results.

TMscript provides a variety of robot programming functions. Among the functions, the conversions of the numeric types, such as int, float, and double, also follow the general endianness rules, which usually apply to file functions, communication-related functions (such as SerialPort, Socket, Modbus), or value-to-byte conversion-related functions. Unless specified otherwise, the function defaults to go by Little Endian. For the string types, it goes by the UTF-8 encoding format that features ASCII character compatibility and fixes endianness without the byte order distinction. For the bool types, it turns the Boolean value true to 1 and false to 0. Refer to the table organized below.

Type	Bit/Byte	Conversion Method	Example	Result
byte	8 bits/1 byte	Little Endian	byte b = 100	0x64
int	32 bits /4 bytes	Little Endian	int i = 300	0x2C 0x01 0x00 0x00

Type	Bit/Byte	Conversion Method	Example	Result
float	32 bits /4 bytes	IEEE-754, Little Endian	float f = 300	0x00 0x00 0x96 0x43
double	64 bits /8 bytes	IEEE-754, Little Endian	double d = 300	0x00 0x00 0x00 0x00 0x00 0xC0 0x72 0x40
bool	bool value	True:1, False:0	bool bf = true	0x01
string	string value	UTF-8	"TM Robot"	0x54 0x4D 0x20 0x52 0x6F 0x62 0x6F 0x74

The value conversion method is an essential basis for communication network applications. Both sides of the applications should recognize the same conversion method to parse the data content correctly. For example,

- socket_send("ntd_a", 9000)**
 By the function definition, it sends **0x28,0x23,0x00,0x00** (**int, Little Endian**) to the device ntd_a.
 It gets various values if the receiving device uses different methods to parse. Such as

int, Big Endian,	0x28 0x23 0x00 0x00 = 673382400
float, Little Endian,	0x00 0x00 0x23 0x28 = 1.2612E-41
string, UTF-8,	0x28 0x23 0x00 0x00 = "(#"
int, Little Endian,	0x00 0x00 0x23 0x28 = 9000
- socket_send("ntd_a", (float)9000)**
 By the function definition, it sends **0x00,0xA0,0x0C,0x46** (**int, Little Endian**) to the device ntd_a.
 It gets various values if the receiving device uses different methods to parse. Such as

int, Big Endian,	0x00 0xA0 0x0C 0x46 = 10488902
int, Little Endia,	0x46 0x0C 0xA0 0x00 = 1175232512
float, Big Endian,	0x00 0xA0 0x0C 0x46 = 1.4698082E-38
string, UTF-8,	0x00 0xA0 0x0C 0x46 = "" // The string ends if encountered 0x00.
float, Little Endian,	0x46 0x0C 0xA0 0x00 = 9000

2.7 Warning

A warning message will prompt, under the condition listed below.

- Double quotation marks does not placed around the string constant.

```
string s = Hello           // warning Hello
```
- There is single quotation mark inside the string constant.

```
string s0 = "World"
string s1 = "Hello 's0'" // warning 's0'
```
- When assigning float value to integer constant, some digits may get lost such as

```
int i = 0
float f = 1.234
i = f           // warning i = 1
```
- When assigning value to variables with fewer digits, some digits may get lost such as

```
byte b = 100
int i = 1000
float f = 1.234
double d = 2.345
b = i           // warning b = 232 // byte can contain values from 0 to 255
f = d           // warning f = 2.345
```
- When assigning a string value to the numeric variable or a numeric value to the string variable, the attempts to convert the string to a number or the number to a string take place automatically. If the conversion is executable, a warning message will prompt, or the project will be stopped by error such as

```
int i = "1234"           // warning i = 1234
int j = "0x89AB"         // warning j = 35243
int k = "0b1010"         // warning k = 10
string s1 = 123           // warning s1 = "123" // Number to string
string s2 = "123"
int x = s2               // warning x = 123 // string to number
// The following code can be compiled with warning, but will be stopped by error when executing.
S2 = "XYZ"
x = s2                   // warning // Stop executing by error // s = "XYZ" cannot be converted to
number
s2 = ""
x = s2                   // warning // Stop executing by error // s = "" cannot be converted to number
```
- String parameters are used as numeric parameters in functions such as

```
Ctrl(0x0A0B0C0D0E) // warning // 0x0A0B0C0D0E is not int type (over 32bit)
// Because there is another syntax, Ctrl(string), the parameter would be applied
to Ctrl(string)
```

Although the project can still be executed with a warning message, correcting all the errors in a warning message is highly recommended to eliminate unpredictable problems and prevent the project being stopped by errors.

- How to fix the error messages
 1. Use double quotations with the string constants

```
string s = "Hello"
```
 2. Use + to link the string constant and the string variable


```
string s0 = "World"  
string s1 = "Hello " + s0
```

3. Specify the type clearly for numerical conversions

```
float f = 1.234
```

```
int i = (int) f
```

// Use (int) for type conversion, i = 1 while processing // It turns the number
in floating-point to an integer.

3. Script Project Programming

3.1 define

This function defines the global variables in the same project. When the project is running, it prioritizes all the variables in the definition section of the project.

Syntax

`define`

For example:

```
define
{
    string text = "Hi TM Robot"
}
```

3.2 main

This function is the first function to call when the project is running, and it is the initial function of the project.

Syntax

`main`

For example:

```
define
{
    string text = "Hi TM Robot"
}
main
{
    Display("Hello Techman Robot")
    Display(text)
}
```

The project begins by running the main function as the result of
Hello Techman Robot

Hi TM Robot *//Since the dashboard shows the last Display content only, the dashboard content is Hi TM Robot.*

3.3 closestop

When the project is in the halt state without any error, it goes to this function. Not until this function ends will the project stop for sure.

Syntax

`closestop`

For example:

```
closestop
{
    IO["ControlBox"].DO = {0,0,0,0,0,0,0,0,0,0,0,0,0,0,0}
    // Set the control box DO to Low.
}
```

After the project is running, pressing the STOP button results in the project running the closestop function.



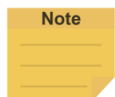
Important

This function does not go with motion commands and will close mandatory after 12 seconds of execution at most.

When the closestop function is running, if errors occur (including other errors occur in the system), it will not run the errorstop function but end the project running directly.

For example:

```
main
{
    Exit()
    // Stop the project. Suppose errors did not occur; it runs the closestop function consecutively.
}
closestop
{
    int zero = 0
    int k = 100 / zero // A dividing by zero error occurred in the closestop function. It ends the project
                    // running directly.
    Display("closestop") // This line does not run due to an error occurring.
```



Note

When the project enters the halt state, the TMflow operation interface will depart from the running state, but it will still continue to run the function. Not until this function ends does the project running stop exactly. Before the project stops running, pressing the Play button to run the project returns errors.

3.4 errorstop

When the project is in the halt state with any error, it goes to this function. Not until this function ends will the project stop for sure.

Syntax

errorstop

For example:

```
main
{
    int zero = 0
    int k = 100 / zero    // A divide by zero error occurs.
}
errorstop
{
    IO["ControlBox"].DO = {0,0,0,0,0,0,0,0,0,0,0,0,0,0,0}
    // Set the control box DO to Low.
}
```

After the project is running, the project will generate errors by dividing by zero and go to the errorstop function.



Important

This function does not go with motion commands and will close mandatory after 12 seconds of execution at most.

When the errorstop function is running, if errors occur (including other errors occur in the system), it will not run the errorstop function but end the project running directly.

For example:

```
main
{
    int k = 100 / (byte)"0.1"    // A dividing by zero error occurred.
}
errorstop
{
    int k = 100 / (byte)"0.1"    // A dividing by zero error occurred in the errorstop function. It ends
                                // the project running directly.
    Display("errorstop ")    // This line does not run due to an error occurring.
```



Note

When the project enters the halt state, the TMflow operation interface will depart from the running state, but it will still continue to run the function. Not until this function ends does the project running stop exactly. Before the project stops running, pressing the Play button to run the project returns errors.

3.5 Customized Function

In addition to the built-in define, main, closetop, and errorstop functions, Script projects provide customized functions for users. The customized Show() function below does not take any parameters nor return any values. (It defines the return type as void.)

For example:

```
main
{
    Show() // Call the customized function Show().
}
void Show()
{
    Display("Hello TM AI Cobot")
}
```

Customized functions support parameter input and output. There are two functions customized in the example below. The Add() can add up the two values users input and return the added up result; the Sub() subtracts the two values users input and return the subtraction result. Note that as long as the return value is not a void type, it requires to return the value of the same type as the definition. Such as the Add() adopts the float type, and the returned variable must be the float type as well.

For example:

```
main
{
    string Add_Result = "Add:" + Add(5.9, 3.6)
    string Sub_Result = "Sub:" + Sub(5.9, 3.6)
    Display("Green", "Yellow", "Result", Add_Result + newline + Sub_Result)
}
float Add(float augend, float addend)
{
    float result = augend + addend
    return result
}
float Sub(float minuend, float subtrahend)
{
    float result = minuend - subtrahend
    return result
}
```



Important

If the function defines the return type, it must return with the same type. If the definition is void, it is not required to use return.

3.6 Comment

Use part of the code as a comment. The compiler ignores the commented code. Users can comment a single line or multi-lines on the code not to execute.

- **Single Line Comment**

Users can comment at the very beginning of the code with `//` to complete a single-line comment, as shown below. The compiler will ignore `a = a + 1`.

```
main
{
    int a = 0
    // a = a + 1
}
```

- **Multi Line Comment**

Users can comment with `/*` as the start of the comment and `*/` as the end of the comment. As below, between `/*` and `*/`, the compiler will ignore `int a = 0` and `a = a + 1`.

```
main
{
    /*
    int a = 0
    a = a + 1
    */
}
```

3.7 Variable

By variable declaration, users can proceed with applications such as calculation, reading and writing, and parameter conveyance. Variables come with the global variables and the local variables.



Important

Variables are case sensitive such as TMrobot and TMROBOT are two different variables.

● Project Variable

Once declared a variable in the definition section, the variable is a global variable in the project. This variable can be read and written in functions of the same project. As shown below, if the define declares the variable b, and the main function runs, it calls Test1() first and sets the variable as 20. When it displays the variable value next, it gets b as 20 for the last set value.

```
define
{
    int b = 0
}
main
{
    Test1()
    Display("B Value: " + b)
}
void Test1()
{
    b = 20
}
// B Value: 20
```

Important

The project variables in the project are different from the global variables in the global variable manager of the robot. The global variables in the project indicate they can operate across functions in the project. Other than across functions in the project, the global variables in the global variable manager of the robot operate across functions of different projects.



Important

Users can create global variables in the global variable manager in the TMflow operation interface only but not with TMscript syntax.



● Local Variable

If variables are declared outside of the definition section such as main, closestop, errorstop, and customized functions, the variables are local variables and good in the functions only.

```
int sum(int s)
{
    int a = 0
}
// Defined two variables int s and int a. When sum() ends, the two local variable are gone.
```

Neither can variable names in the definition section of the same function be duplicated, nor can they replicate with the global variables in the define section and in the global variable manager of the robot.

```
define
{
    int b = 0
    string g_s1 = ""    // Suppose the global variable manager defines g_s1; there will be
                        // variable duplication.
}
void sum(int s)
{
    int a = 0
    int b = 1    // Duplicated with int b in define.
    float a = 0  // Duplicated with int a.
}
```

Local variables are independent of different function sections. Such declared a variable a in both Test1() and Test2(). Because of being in disparate functions, the two variables are independent and valid only in each function section.

```
void Test1()
{
    int a = 0
}
void Test2()
{
    int a = 2
}
```

The scope of the local variables can be valid in a top-down coverage. Users can also declare variables in the conditional statement or the loop statement, and the variables are present in the conditional statement or the loop statement. The variables are absent from the outside of the scope. Therefore, once the variables exit from the scope, the variables will release the variable data. For variables declared by the class, such as SerialPort, Socket, and Modbus, the variables will close the devices.


```

void Test1()
{
    int a = 10
    if (a > 5)
    {
        int b = 20
        Display("A Value: " + a)
    }
    Display("B Value: " + b) // Warning. b will be taken as a string instead of a
    variable.
}

void Test2()
{
    if (true)
    {
        Socket bbb = "127.0.0.1",12345
        socket_open("bbb")
        Sleep(5000)
    } // After a 5-second waiting, variable bbb will be released for exiting the f conditional
    statement scope and close the connection.
    Sleep(10000)
}

```

variable a effective scope

variable b effective scope

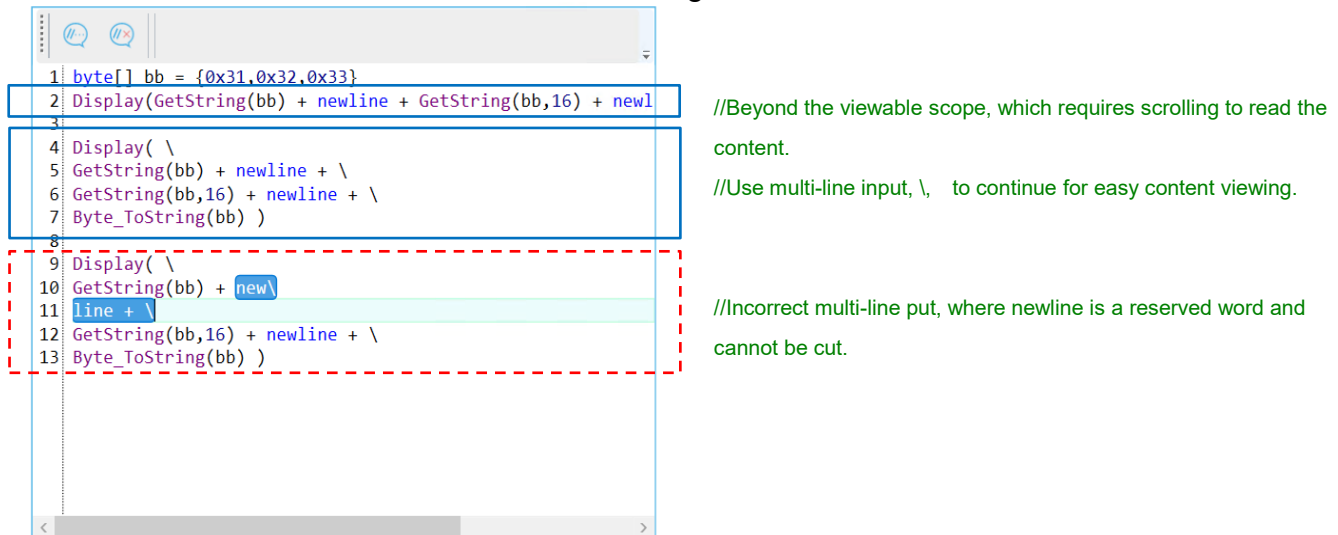
variable bbb effective scope

In the Flow project, if users want to create a global variable in the project, they need to do it in the variable manager or device manager. Users cannot create a global variable in the flow project with the Script. Moreover, because each process node is an independent function, including the Script node, if declaring variables in the node, they are all local variables. When exiting the node, the variables in the node will no longer exist, and the variable data will be released.

3.8 Multi-Line Input

Once an expression to input goes with more contents, it is a disadvantage to maintain or debug. Users can add \ at the end of the line to fulfill the multi-line input of consecutive contents. The multi-line input contents will be taken as an expression in the same line until there is no \, and it goes with a new line character at the end.

While using multi-line input, it is still required to maintain the correctness of wording but not to use \ at random for multi-line continuations among words.



3.9 Conditional Statements

During the project running, it might be necessary to consider different approaches, such as task fails, function, and communication errors, to various situations. Users can use conditional statements to adopt different paths to the result values at this time. Two conditional statements, if and switch, are available so far.

3.9.1 if

The if statement can judge the conditional expression in the brackets. It enforces statement 1 if the condition meets.

```
if (conditional expression)
{
    statement 1      // the condition satisfies.
}
```

Or vice versa. If the condition satisfies, it goes to statement 1. If not, it goes statement 2.

```
if (conditional expression)
{
    statement 1      // the condition satisfies.
}
else
{
    statement 2      // the condition not satisfies.
}
```

Users can also check more possibilities with the if .. else if .. else statement to judge the condition 1, 2, 3, and the last otherwise condition in sequence.

```
if (conditional expression 1)
{
    statement 1      // the condition 1 satisfies.
}
else if (conditional expression 2)
{
    statement 2      // the condition 2 satisfies.
}
else if (conditional expression 3)
{
    statement 3      // the condition 3 satisfies.
}
else
{
    statement 4      // the condition 1, 2, and 3 not satisfy at all.
}
```

Take the example below. If the Score is 100, it displays "Full Score." Between 60 and 99,

"Excellent"; otherwise, "Failed." Users can write the condition with the if statement.

```
void Test1()
{
    int Score = 65
    if (Score == 100)
    {
        Display("Green", "Yellow", "Full Score", "")
    }
    else if (Score >= 60)
    {
        Display("Green", "Yellow", "Excellent", "")
    }
    else
    {
        Display("Red", "Yellow", "Failed", "")
    }
}
// Excellent
```

Note

NOTE:

== of if (Score==100) stands for comparison, and = of int Score = 65, assignment.

The criteria to evaluate conditional expressions are as follows, but boolean values have much to recommend for evaluation.

- Boolean values true or True for the condition satisfies, false or False, not satisfies.
- Numeric values non-zeros for the condition satisfies, zeros, not satisfies.
- String values true or True for the condition satisfies, other string values, not satisfies.

3.9.2 switch

The switch statement is similar in condition judgment to the if statement. The switch statement goes with the other way of writing for similar conditions.

```
switch (variable or expression)
{
    case condition 1 satisfies
        statement 1 // the condition 1 satisfies.
        break
    case condition 2 satisfies
        statement 2 // the condition 2 satisfies.
        break
    default
        statement 3 // the condition 1 and 2 both not satisfy.
        break
}
```

As the example below, it can match various results by the content value of the variable.

```

void Test1()
{
    string di_st = (string)IO["ControlBox"].DI[1] + (string)IO["ControlBox"].DI[0]
    switch (di_st)
    {
        case "00"
            Display("DI[0]=0, DI[1]=0")
            break
        case "01"
            Display("DI[0]=1, DI[1]=0")
            break
        case "10"
            Display("DI[0]=0, DI[1]=1")
            break
        default
            Display("DI[0]=1, DI[1]=1")
            break
    }
}
// DI[0]=0, DI[1]=0

```

Besides, the switch statement supports expressions. Like the example below, if the Score is 100, it displays "Full Score." Between 60 and 99, "Excellent"; otherwise, "Failed."

```

void Test1()
{
    int Score = 65
    switch (Score)
    {
        case >= 100
            Display("Green", "Yellow", "Full Score", "")
            break
        case >= 100-40
            Display("Green", "Yellow", "Excellent", "")
            break
        default
            Display("Red", "Yellow", "Failed", "")
            break
    }
}
// Excellent

```

The differences from the if statement is

- The switch statement retrieves the value once and compares the result value multiple times, and the if statement retrieves every time it compares. This makes the switch's judgment on the DI status more accurate.

- As the reason above, the switch statement cannot apply to the conditions of different natures, and the if statement can. Such as

```
void Test1()
{
    int Payload = 4
    int Length = 130
    if (Payload > 4)
    {
        Display("Green", "Yellow", "Payload", "")
    }
    else if (Length > 70)
    {
        Display("Green", "Yellow", "Length", "")
    }
    // This statement of if .. else if .is unable to be written with the switch statement.
}
```

3.10 Loop Statements

During the project running, it may be necessary to repeatedly calculate certain values or check whether the conditions are comprehended. In these cases, it requires the loop statements that go on iterative processing the code in the statement until the condition is comprehended. Three loop statements, for, while, and do-while, are available so far.

3.10.1 for

The for loop syntax comprises four sections, the initialization section, the loop condition section, the statement, and the iterative operation section.

```
for (the initialization section; the loop condition section; the iterative operation section)
{
    statement
}
```

The execution sequence is as below.

1. The initialization section: When it goes to the for syntax, it executes the initialization section one time. It is usually used to declare variables. (The variables are the local variables in the for syntax.)
2. The loop condition section: It decides whether to go on the for loop condition execution. Once comprehended (true) or not existed, it goes on the for loop execution. It exits the for loop until the condition is not comprehended (false).
3. The execution statement: The statement to execute.
4. The iterative operation section: After the statement execution, it executes the iterative operation section once and then goes back to the loop condition section for the condition judgment.

The example is a basic application of the for loop. It adds the number from 0 to the input K value.

```
int sum(int k)
{
    int result = 0
    for (int i = 0; i < k; i++)
    {
        result += i
    }
    return result
}
```

Users can use multiple for loops together. The example presents the multiplication table by the for loop.

```
void Test1()
{
    string result = ""
    for (int i = 1; i <= 9; i++)
```

```

    {
        for (int j = 1; j <= 9; j++)
        {
            result += j + "X" + i + "=" + i * j + " "
        }
        result += newline
    }
    Display("Green", "Yellow", "multiplication table", result)
}

```

Users can adopt the four sections in for loop optionally.

```

void Test1()
{
    int i = 0
    for (i = 3; i < 4; )
        i++

    // i = 3
    // i < 4    // i++    // i = 4
    // i < 4    // false, exit for
    for ( ; i < 5; i++)
    {
    }

    // Since the value of i is not reset, it will continue using i =4.
    // i < 5    //          // i++, i = 5
    // i < 5    // false, exit for

    for ( ; ; )          // Execution continues because the loop condition section does not exist.
    {
        // ...
    }
}

```

3.10.2 while

The while loop comes with one boolean conditional expression only. Once the condition satisfies (true), it executes the statement until the condition does not satisfy (false) and exit the while loop. Therefore, the number of the while loop execution is zero or more, and it may exit for the condition does not satisfy at the first time.

```

while (conditional expression )
{
    statement
}

```

The example below adopts an arithmetic progression to count the number from 0, 1, 2, 3, ..., and N until the total is 500500 and gets the value of N.

```

void Test1()

```



```

{
    int sum = 0
    int N = 0
    while (sum != 500500)
    {
        N += 1
        sum += N
    }
    Display(N)
}
// N = 1000

```

3.10.3 do while

The syntax of the do while loop is similar to that of the while loop. The while loop checks whether if the condition satisfies first before executing statements in the code block; however, the do while loop executes the command first and then check whether if the condition satisfies. Accordingly, the do while loop executes the statement one time at least.

```

do
{
    statement
} while (conditional expression )

```

For example as below, if using the do while loop, it displays Hello TM Robot.

```

void Test1()
{
    int result = 0
    do
    {
        Display("Hello TM Robot")
    } while(result > 5)
}

```

On the contrary, if using the while loop, it does not display any result.

```

void Test1()
{
    int result = 0
    while(result > 5)
    {
        Display("Hello TM Robot")
    }
}

```

3.11 Branching Statements

3.11.1 break

It applies to exit the last loop statement of for, while, or do while without satisfying the loop condition and exit. As the example below, once i is larger than 10, it exits and ends the for loop.

```
int sum(int k)
{
    int result = 0
    for (int i = 0; i < k; i++)
    {
        if (i > 10)
        {
            break
        }
        result += i
    }
    return result
}

void Test1()
{
    string result = ""
    for (int i = 1; i <= 9; i++)
    {
        for (int j = 1; j <= 9; j++)
        {
            result += j + "X" + i + "=" + i * j + " "
            if (j > 4) break
        }
        result += newline
    }
    Display("Green", "Yellow", "multiplication table", result)
}
```

Exit the closest for i loop directly.

Exit the closest for j loop directly.

3.11.2 continue

It applies to the loop statements of for, while, and do while, but it is different from the break statement in that it ends the closest and current loop and begins the next loop without exiting the loop. As the example below, if it is an even number, it ends the current loop directly and begins the next loop. So, it registers odd numbers only.

```
void Test1()
{
    string result = ""
    for (int i = 0; i < 10; i++)
    {
        if (i % 2 == 0)
        {
            continue
        }
        result += i + ", "
    }
    Display("Odd value: " + result)
}
// Odd value: 1, 3, 5, 7, 9
```

Ends the current loop and goes back to the next loop.



3.11.3 return

The return statement ends the statement execution in the function and returns the value to the function caller. For example,

```
main
{
    int num = sum(10)
    Display(num)
}
int sum(int k)
{
    int result = 0
    for (int i = 0; i < k; i++)
    {
        if (i == 3) return result // Once i == 3, it returns the result directly without further statement
                                // executions.
        result += i
    }
    return result
}
```

3.12 Thread

During the process of project programming, when it comes to asynchronous parallel operations, users need to use threads. By creating new threads for operations, users can make the project go with the multi-tasking concept. However, users also have to pay attention to no priority among threads in execution, and each operates independently when with multiple ones.

The main function to be called first during the project running acts as a thread also. What's different about this thread from others is that the main function called by this thread and other customized functions called in the main function come with the privilege to call robot motion functions. Calling robot motion functions is not granted to other newly created threads. This is to ensure the consistency of the robot motion process because each thread processes independently. It is easy to confuse or interrupt the robot motion processes if granting other threads to use the robot motion functions.

Other than the other threads, the main thread comes with the privilege of the robot motion function call. The other newly created threads are not allowed for the robot motion function call. It is to ensure the consistency of the robot motion process. Since each thread operates independently, if others are for the robot motion function call, it gets the robot motion to be interrupted or confused easily.

In addition to the privilege of the robot motion function call, users can set the thread to whether continuously execute or not, which means if the project pauses the thread will be influenced by the or not. Once setting the thread to continuous execution, even if the project pauses, this thread continues to operate. Such a setting is suitable for communication applications.

When the thread-to-use works in a loop (with the loop statement), adding the additional sleep function to free up the thread-occupied CPU usage is requisite and recommended while working around threads, which is an attention must because excessive CPU usage may result in poor execution efficiency.

3.12.1 ThreadRun()

Create a new thread and use the new one to execute the statement. The previous one will continue proceeding.

Syntax 1

```
int ThreadRun(
    ?,
    bool
)
```

Parameters

?	statement or customized function
bool	Whether the thread will continue the execution without being paused
true	continue the execution without being paused
false	to be paused (default)

Return

int	return the ID of the newly created thread.
-----	--

> 0 created successfully

<= 0 created unsuccessfully

*The ID of the newly created thread goes by the system, and it is not a fixed value. The thread number in the process will not be repetitive, but the number might be recurring once the thread stops.

Syntax 2

```
int ThreadRun(
    ?
)
```

Note

Same as Syntax 1. The default is to set whether the thread will continue the execution to false.

3.12.2 ThreadID()

Retrieve the ID of the current thread.

Syntax 1

```
int ThreadID(
)
```

Parameter

void No parameter

Return

int return the ID of the thread

Note

```
main
{
    int tid = ThreadRun(ThreadTest1(), false)
    Sleep(1000)
    ThreadRun(ThreadTest2(tid), true)
}
void ThreadTest1()
{
    Display("Hello ThreadTest1() " + ThreadID())
}
void ThreadTest2(int k)
{
    Display("Hello ThreadTest2() " + ThreadID() + " " + k)
}
// Hello ThreadTest1() 18
// Hello ThreadTest2() 61 18
```

Create a new thread and and continue executiing ThreadTest1().

↓ The previous thread continues proceeding.

↓ Create a new thread and and continue executiing ThreadTest2().

The following description denotes the execution order of creating new threads for running statements.

```
main
{
    int tid = ThreadID()
```

```

    ThreadRun(ThreadTest3(tid, ThreadID()), false)
}
void ThreadTest3(int k1, int k2)
{
    Display("Hello ThreadTest3() " + ThreadID() + " " + k1 + " " + k2)
}
// Hello ThreadTest3() 65 47 65

```

- The execution order goes by
 1. Retrieve the thread ID in the main thread and assign the value the local variable tid.
 2. Create a new thread and execute the statement of ThreadTest3(tid, ThreadID()).
 3. As tid is a local variable, the value of tid is available to lead in.
 4. Since ThreadID() is a function and it uses a new thread for the function call, the obtained thread ID will be different from the tid.
 5. Again, call the ThreadTest3() function, and lead it and the obtained thread ID into k1 and k2 in the function, respectively
 6. Call the ThreadID() function in the ThreadTest3() function to get the thread ID. Because it is the same thread as the ThreadTest3() function is in, k2 will go with the same ID and k1 with a different ID.

3.12.3 ThreadState()

Retrieve the status of the assigned thread ID.

Syntax 1

```

int ThreadState(
    int
)

```

Parameter

int thread ID

Return

int return the status of the dedicate thread

-1	The thread does not exist.
0	The thread is in execution.
1	The thread is requesting to stop.
2	The thread stopped.

Syntax 2

```

int ThreadState(
)

```

Note

Same as Syntax 1. The default is to set the thread ID with the current thread ID of the function call.

3.12.4 ThreadExit()

Request the assigned thread ID to stop execution.

Syntax 1

```
int ThreadExit(
    int
)
```

Parameter

int thread ID

Return

int return the result of requesting the assigned thread to stop

- 1 The thread does not exist.
- 0 The requested thread stopped execution.

Syntax 2

```
int ThreadExit(
)
```

Note

Same as Syntax 1. The default is to set the thread ID with the current thread ID of the function call.

```
define
{
    int tid = 0
}
main
{
    int st = 0
    int t4 = ThreadRun(ThreadTest4())
    do
    {
        Sleep(100)
        st = ThreadState(t4)
        Display("t4 " + st)    // t4 0
    } while (st != 2)        // Use the loop statement to wait for the thread t4 to stop.
    // Since there is no loop in ThreadTest4, the thread stops after the execution ends.
    Display("t4 " + st + " " + tid)    // t4 2 51
}
void ThreadTest4()
{
    Sleep(1000)
    tid = ThreadID()
}

define
{
    string title = ""
}
main
{
    int st = 0
    title = "t5 " + st        // t5 0
```

```

int t5 = ThreadRun(ThreadTest5())
bool flag = WaitFor((st = ThreadState(t5)) == 2, 1000)
// Use WaitFor to wait for the thread t5 to stop.
title = "t5 " + st + " " + flag      // t5 0 false
// Since there is a loop in ThreadTest5, the thread will not stop.
// Therefore, WaitFor quits after a 1000 ms timeout. So, flag = false.
Sleep(1000)
if (flag == false)
{
    ThreadExit(t5)          // Request the thread t5 to quit.
}
flag = WaitFor((st = ThreadState(t5)) == 2, 1000)
// Use WaitFor to wait for the thread t5 to stop.
// Since the system requested the thread t5 to stop, the WaitFor condition sustains.
Display("t5 " + st + " " + flag, "Hello main() " + ThreadID())// t5 2 true
}
void ThreadTest5()
{
    while (true) // the loop statement
    {
        Display(title , "Hello ThreadTest5() " + ThreadID())
        Sleep(100)
        // While using the loop statement, applying the additional sleep function is
        // recommended to free up the thread-occupied CPU usage.
    }
}

```


4. General Functions

4.1 Byte_ToInt16()

Transform the first two bytes of the assigned byte array to integer, and returns in int type.

Syntax 1

```
int Byte_ToInt16(
    byte[],
    int,
    int
)
```

Parameters

<code>byte[]</code>	Byte array
<code>int</code>	Byte array follows the Little Endian or Big Endian
<code>0</code>	Little Endian (Default)
<code>1</code>	Big Endian
<code>int</code>	Transfer to signed int16 (Signed Number) or unsigned int16 (Unsigned Number)
<code>0</code>	signed int16 (Default)
<code>1</code>	unsigned int16

Return

`int` A signed or unsigned int16 formed by 2 bytes beginning at index [0]. Because only 2 bytes is needed, the index of byte array will be [0][1]. If the data is not long enough, it would be filled to 2 bytes before transforming.

Note

`byte[] bb1 = {0x90, 0x01, 0x05}`

`byte[] bb2 = {0x01}`

// Cause bb2[] does not fill 2 bytes. It would be filled to 2 bytes before transforming.

```
value = Byte_ToInt16(bb1, 0, 0) // 0x0190 value = 400
value = Byte_ToInt16(bb1, 0, 1) // 0x0190 value = 400
value = Byte_ToInt16(bb1, 1, 0) // 0x9001 value = -28671
value = Byte_ToInt16(bb1, 1, 1) // 0x9001 value = 36865
value = Byte_ToInt16(bb2, 0, 0) // 0x0001 value = 1
value = Byte_ToInt16(bb2, 0, 1) // 0x0001 value = 1
value = Byte_ToInt16(bb2, 1, 0) // 0x0100 value = 256
value = Byte_ToInt16(bb2, 1, 1) // 0x0100 value = 256
```

Syntax 2

```
int Byte_ToInt16(
    byte[],
    int
)
```

Note

Similar to Syntax 1 with return value as signed int16

`Byte_ToInt16(bb1, 0) => Byte_ToInt16(bb1, 0, 0)`

Syntax 3

```
int Byte_ToInt16(
```

byte[]

)

Note

Similar to Syntax 1 with little endian input and return value as signed int16

Byte_ToInt16(bb1) => **Byte_ToInt16**(bb1, 0)

4.2 Byte_ToInt32()

Transform the first four bytes of byte array to integer, and return in int type.

Syntax 1

```
int Byte_ToInt32(
    byte[],
    int
)
```

Parameters

byte[] The input byte array
int The input byte array follows Little Endian or Big Endian
0 Little Endian (Default)
1 Big Endian

Return

int An unsigned int32 formed by 4 bytes beginning at index [0].
 Because only 4 bytes is needed, the index of byte array will be [0][1][2][3]. If the data is not long enough, it would be filled to 4 bytes before transforming.

Note

```
byte[] bb1 = {0x01, 0x02, 0x03, 0x4F, 1}
byte[] bb2 = {0x01, 0x02, 0x03}
// Cause bb2[] does not fill 4 bytes. It would be filled to 4 bytes before transforming.

value = Byte_ToInt32(bb1, 0) // 0x4F030201 value = 1325597185
value = Byte_ToInt32(bb1, 1) // 0x0102034F value = 16909135
value = Byte_ToInt32(bb2, 0) // 0x00030201 value = 197121
value = Byte_ToInt32(bb2, 1) // 0x01020300 value = 16909056
```

Syntax 2

```
int Byte_ToInt32(
    byte[]
)
```

Note

Similar to Syntax 1 with little endian input
Byte_ToInt32(bb1) => Byte_ToInt32(bb1, 0)

4.3 Byte_ToFloat()

Transform the first four bytes of byte array to floating-point number, and return in floating-point type.

Syntax 1

```
float Byte_ToFloat(
    byte[],
    int
)
```

Parameters

byte[]	The input byte array
int	The input byte array follows Little Endian or Big Endian
0	Little Endian (Default)
1	Big Endian

Return

float A floating-point number formed by 4 bytes beginning at index [0].

Because only 4 bytes is needed, the index of byte array will be [0][1][2][3]. If the data is not long enough, it would be filled to 4 bytes before transforming.

Note

```
byte[] bb1 = {0x01, 0x02, 0x03, 0x4F, 1}
byte[] bb2 = {0x01, 0x02, 0x03} // Cause bb2[] does not fill 4 bytes. It would be filled to 4 bytes before
                                transforming.

value = Byte_ToFloat(bb1, 0) // 0x4F030201 value = 2.1979466E+09
value = Byte_ToFloat(bb1, 1) // 0x0102034F value = 2.3879603E-38
value = Byte_ToFloat(bb2, 0) // 0x00030201 value = 2.76225E-40
value = Byte_ToFloat(bb2, 1) // 0x01020300 value = 2.387938E-38
```

Syntax 2

```
float Byte_ToFloat(
    byte[]
)
```

Note

Similar to Syntax 1 with little endian input

Byte_ToFloat(bb1) => Byte_ToFloat(bb1, 0)

4.4 Byte_ToDouble()

Transform the first eight bytes of byte array to floating-point number, and return in double type.

Syntax 1

```
double Byte_ToDouble(
    byte[],
    int
)
```

Parameters

byte[] The input byte array
int The input byte array follows Little Endian or Big Endian
0 Little Endian (Default)
1 Big Endian

Return

double A floating-point number formed by 8 bytes beginning at index [0]. Because only 8 bytes is needed, the index of byte array will be [0][1][2][3][4][5][6][7]. If the data is not long enough, it would be filled to 8 bytes before transforming.

Note

```
byte[] bb1 = {0x01, 0x02, 0x03, 0x4F, 1} // Cause bb1[] does not fill 8 bytes. It would be filled to 8
                                         bytes before transforming.
byte[] bb2 = {0x01, 0x02, 0x03} // Cause bb1[] does not fill 8 bytes. It would be filled to 8 bytes before
                                transforming.
value = Byte_ToDouble(bb1, 0) // 0x000000014F030201 value = 2.7769278203E-314
value = Byte_ToDouble(bb1, 1) // 0x0102034F01000000 value = 8.20840179153173E-304
value = Byte_ToDouble(bb2, 0) // 0x00000000000030201 value = 9.73907E-319
value = Byte_ToDouble(bb2, 1) // 0x0102030000000000 value = 8.207852449261364E-304
```

Syntax 2

```
double Byte_ToDouble(
    byte[]
)
```

Note

Similar to Syntax 1 with little endian input

Byte_ToDouble(bb1) => Byte_ToDouble(bb1, 0)

4.5 Byte_ToInt16Array()

Transform byte array to integer every 2 bytes, and return in int[] type.

Syntax 1

```
int[] Byte_ToInt16Array(
    byte[],
    int,
    int
)
```

Parameters

byte[]	The input byte array
int	The input byte array follows Little Endian or Big Endian
0	Little Endian (Default)
1	Big Endian
int	Transfer to signed int16 (Signed Number) or unsigned int16 (Unsigned Number)
0	signed int16 (Default)
1	unsigned int16

Return

int[] A integer array formed by every 2 bytes of byte array beginning at index [0]

Note

```
byte[] bb1 = {0x90, 0x01, 0x02, 0x03, 0x04}
```

// When the remaining part does not fill 2 byte, it would be filled to 2 bytes before transforming.

```
byte[] bb2 = {1, 2, 3, 4}
```

```
value = Byte_ToInt16Array(bb1, 0, 0) // {0x0190, 0x0302, 0x0004} value = {400, 770, 4}
```

```
value = Byte_ToInt16Array(bb1, 0, 1) // {0x0190, 0x0302, 0x0004} value = {400, 770, 4}
```

```
value = Byte_ToInt16Array(bb1, 1, 0) // {0x9001, 0x0203, 0x0400} value = {-28671, 515, 1024}
```

```
value = Byte_ToInt16Array(bb1, 1, 1) // {0x9001, 0x0203, 0x0400} value = {36865, 515, 1024}
```

```
value = Byte_ToInt16Array(bb2, 0, 0) // {0x0201, 0x0403} value = {513, 1027}
```

```
value = Byte_ToInt16Array(bb2, 0, 1) // {0x0201, 0x0403} value = {513, 1027}
```

```
value = Byte_ToInt16Array(bb2, 1, 0) // {0x0102, 0x0304} value = {258, 772}
```

```
value = Byte_ToInt16Array(bb2, 1, 1) // {0x0102, 0x0304} value = {258, 772}
```

Syntax 2

```
int[] Byte_ToInt16Array(
    byte[],
    int
)
```

Note

Similar to Syntax 1 with return value as signed int16

```
Byte_ToInt16Array(bb1, 0) => Byte_ToInt16Array(bb1, 0, 0)
```

Syntax 3

```
int[] Byte_ToInt16Array(
    byte[]
```

)

Note

Similar to Syntax 1 with little endian input and return value as signed int16

Byte_ToInt16Array(bb1) => **Byte_ToInt16Array**(bb1, 0)

4.6 Byte_ToInt32Array()

Transform byte array to integer every 4 bytes, and return in int[] type

Syntax 1

```
int[] Byte_ToInt32Array(
    byte[],
    int
)
```

Parameters

byte[]	The input byte array
int	The input byte array follows Little Endian or Big Endian
0	Little Endian (Default)
1	Big Endian

Return

int[] A integer array formed by every 4 bytes of byte array beginning at index [0]

Note

```
byte[] bb1 = {0x01, 0x02, 0x03, 0x04, 0x05}
```

// When the remaining part does not fill 4 byte, it would be filled to 4 bytes before transforming.

```
byte[] bb2 = {1, 2, 3, 4}
```

```
value = Byte_ToInt32Array(bb1, 0) // {0x04030201, 0x00000005} value = {67305985, 5}
```

```
value = Byte_ToInt32Array(bb1, 1) // {0x01020304, 0x05000000} value = {16909060, 83886080}
```

```
value = Byte_ToInt32Array(bb2, 0) // {0x04030201} value = {67305985}
```

```
value = Byte_ToInt32Array(bb2, 1) // {0x01020304} value = {16909060}
```

Syntax 2

```
int[] Byte_ToInt32Array(
    byte[]
)
```

Note

Similar to Syntax 1 with little endian input.

```
Byte_ToInt32Array(bb1) => Byte_ToInt32Array(bb1, 0)
```


4.7 Byte_ToFloatArray()

Transform byte array to integer every 4 bytes, and return in float[] type.

Syntax 1

```
float[] Byte_ToFloatArray(
    byte[],
    int
)
```

Parameters

byte[]	The input byte array
int	The input byte array follows Little Endian or Big Endian
0	Little Endian (Default)
1	Big Endian

Return

float[] A floating-point number array formed by every 4 bytes of byte array beginning at index [0]

Note

```
byte[] bb1 = {0x01, 0x02, 0x03, 0x04, 0x05}
// When the remaining part does not fill 4 byte, it would be filled to 4 bytes before
// transforming.
byte[] bb2 = {1, 2, 3, 4}

value = Byte_ToFloatArray(bb1, 0)
// {0x04030201, 0x00000005} value = {1.5399896E-36, 7E-45}
value = Byte_ToFloatArray(bb1, 1)
// {0x01020304, 0x05000000} value = {2.3879393E-38, 6.018531E-36}
value = Byte_ToFloatArray(bb2, 0) // {0x04030201} value = {1.5399896E-36}
value = Byte_ToFloatArray(bb2, 1) // {0x01020304} value = {2.3879393E-38}
```

Syntax 2

```
float[] Byte_ToFloatArray(
    byte[]
)
```

Note

Similar to Syntax 1 with little endian input

Byte_ToFloatArray(bb1) => Byte_ToFloatArray(bb1, 0)

4.8 Byte_ToDoubleArray()

Transform byte array to double every 8 bytes, and return in double[] type.

Syntax 1

```
double[] Byte_ToDoubleArray(
    byte[],
    int
)
```

Parameters

byte[] The input byte array
int The input byte array follows Little Endian or Big Endian
0 Little Endian (Default)
1 Big Endian

Return

double[] A floating-point number array formed by every 8 bytes of byte array beginning at index [0]

Note

```
byte[] bb1 = {0x01, 0x02, 0x03, 0x04, 0x05} // When the remaining part does not fill 8 byte, it would
                                              be filled to 8 bytes before transforming.
byte[] bb2 = {1, 2, 3, 4} // When the remaining part does not fill 8 byte, it would be filled to 8 bytes
                           before transforming.
```

```
value = Byte_ToDoubleArray(bb1, 0) // {0x0000000504030201} value = {1.064323253E-313}
value = Byte_ToDoubleArray(bb1, 1)
// {0x0102030405000000} value = {8.207880398492326E-304}
```

```
value = Byte_ToDoubleArray(bb2, 0) // {0x0000000004030201} value = {3.3253575E-316}
value = Byte_ToDoubleArray(bb2, 1)
// {0x0102030400000000} value = {8.207880262684596E-304}
```

Syntax 2

```
double[] Byte_ToDoubleArray(
    byte[]
)
```

Note

Similar to Syntax 1 with little endian input
Byte_ToDoubleArray(bb1) => Byte_ToDoubleArray(bb1, 0)

4.9 Byte_ToString()

Transform byte array to string

Syntax 1

```
string Byte_ToString(
    byte[],
    int
)
```

Parameters

byte[]	The input byte array
int	The character encoding rules applied to input byte array
0	UTF8 (Default) (0x00 END)
1	HEX BINARY
2	ASCII (0x00 END)

Return

string String formed by byte array. The transformation begins from index [0].

Note

byte[] bb1 = {0x31, 0x32, 0x33, 0x00, 0x4F, 1}

byte[] bb2 = {0x01, 0x54, 0x4D, 0x35, 0xE6, 0xA9, 0x9F, 0xE5, 0x99, 0xA8, 0xE4, 0xBA, 0xBA}

value = **Byte_ToString**(bb1, 0) // value = "123" (UTF8 stop at 0x00)

value = **Byte_ToString**(bb1, 1) // value = "313233004F01"

value = **Byte_ToString**(bb1, 2) // value = "123" (ASCII stop at 0x00)

value = **Byte_ToString**(bb2, 0) // value = "\u01TM5\u5730\u4EBA" (UTF8)

value = **Byte_ToString**(bb2, 1) // value = "01544D35E6A99FE599A8E4BABA"

value = **Byte_ToString**(bb2, 2) // value = "\u01TM5?????????" (ASCII)

* \u01 represents the SOH control character, not the string value.

Syntax 2

```
string Byte_ToString(
    byte[]
)
```

Note

Similar to Syntax 1 with UTF8 character encoding rules

Byte_ToString(bb1) => **Byte_ToString**(bb1, 0)

4.10 Byte_Concat()

Concatenate two byte arrays, or concatenate one array with a byte value.

Syntax 1

```
byte[] Byte_Concat(
    byte[],
    byte
)
```

Parameters

byte[] The input byte array
byte The byte value concatenated after the byte array

Return

byte[] The byte array formed by the input byte array and byte value

Note

```
byte[] bb1 = {0x31, 0x32, 0x33, 0x00, 0x4F, 1}
```

```
value = Byte_Concat(bb1, 12) // value = {0x31, 0x32, 0x33, 0x00, 0x4F, 0x01, 0x0C}
```

Syntax 2

```
byte[] Byte_Concat(
    byte[],
    byte[]
)
```

Parameters

byte[] The input byte array1
byte[] The input byte array2, would be concatenated to the end of array1

Return

byte[] Byte array formed from concatenating input arrays.

Note

```
byte[] bb1 = {0x31, 0x32, 0x33, 0x00, 0x4F, 1}
```

```
byte[] bb2 = {0x01, 0x02, 0x03}
```

```
value = Byte_Concat(bb1, bb2)
// value = {0x31, 0x32, 0x33, 0x00, 0x4F, 0x01, 0x01, 0x02, 0x03}
```

Syntax 3

```
byte[] Byte_Concat(
    byte[],
    byte[],
    int
)
```

Parameters

byte[] The input byte array1
byte[] The input byte array2, would be concatenated after the end of array1
int The number of element in array2 to be concatenated
0..the length of array2 Valid number
<0 Invalid. Length of array2 will be applied instead.

> the length of array2 Invalid. Length of array2 will be applied instead.

Return

byte[] Byte array formed from concatenating input arrays.

Note

```
byte[] bb1 = {0x31, 0x32, 0x33, 0x00, 0x4F, 1}
```

```
byte[] bb2 = {0x01, 0x02, 0x03}
```

```
value = Byte_Concat(bb1, bb2, 2) // value = {0x31, 0x32, 0x33, 0x00, 0x4F, 0x01, 0x01, 0x02} //
Concatenate only 2 elements from array2
```

```
value = Byte_Concat(bb1, bb2, -1) // value = {0x31, 0x32, 0x33, 0x00, 0x4F, 0x01, 0x01, 0x02,
0x03} // -1 is invalid value
```

```
value = Byte_Concat(bb1, bb2, 10) // value = {0x31, 0x32, 0x33, 0x00, 0x4F, 0x01, 0x01, 0x02,
0x03} // 10 exceeds the array size
```

```
value = Byte_Concat(bb1, bb2, Length(bb2)) // Length() can be utilized to acquire the array size
// value = {0x31, 0x32, 0x33, 0x00, 0x4F, 0x01, 0x01, 0x02,
0x03}
```

Syntax 4

```
byte[] Byte_Concat(
    byte[],
    int,
    int,
    byte[],
    int,
    int
)
```

Parameters

byte[] The input byte array1

int The starting index of array1

0..(length of array1)-1 Valid

<0 The starting index would be 0

>=(length of array1) The starting index would be the length of array2 (For index over the length of array2, an empty value would be captured)

int The number of element in array1 to be concatenated

0.. (length of array1) Valid

<0 Invalid , length of array1 will be applied instead

>(length of array1) Invalid , length of array1 will be applied instead

If the total number of starting index and assigning elements exceeds the length of array1, the surplus index will be suspended.

byte[] The input byte array2 , would be concatenated after the end of array1

int The starting index of array2

0.. (length of array2)-1 Valid

<0 The starting index would be 0

>=(length of array2) The starting index would be the length of array2 (For index over the length of array2, an empty value would be captured)

int The number of element in array2 to be concatenated
0.. (length of array2) Valid
<0 Invalid. Length of array2 will be applied instead.
>(length of array2) Invalid. Length of array2 will be applied instead.
 If the total number of starting index and assigning elements exceeds the length of array2, the surplus index will be suspended.

Return

byte[] Byte array formed from concatenating input arrays.

Note

byte[] bb1 = {0x31, 0x32, 0x33, 0x00, 0x4F, 1}

byte[] bb2 = {0x01, 0x02, 0x03}

value = **Byte_Concat**(bb1, 1, 3, bb2, 1, 2) // value = {0x32, 0x33, 0x00, 0x02, 0x03}

value = **Byte_Concat**(bb1, -1, 3, bb2, 3, -1) // value = {0x31, 0x32, 0x33}

Syntax 5

```
byte[] Byte_Concat(
  byte or byte[],
  byte or byte[],
  ...
)
```

Parameters (active parameter amount)

byte The input byte value

byte[] The input byte array

It concatenates the content of each parameter in order. It ignores the parameter if it is not a byte or a byte array and continues to concatenate the next parameter.

Return

byte[] Concatenate the parameters in byte and return a new byte array.

Note

byte[] bb1 = {0x31, 0x32, 0x00, 0x4F, 1}

byte[] bb2 = {0x01, 0x02, 0x03}

byte bb3 = 0x5A

value = **Byte_Concat**(bb1, bb2, bb3)

// value = {0x31,0x32,0x00,0x4F,0x01,0x01,0x02,0x03} // Syntax 3

value = **Byte_Concat**(bb1, bb2, "", bb3)

// value = {0x31,0x32,0x00,0x4F,0x01,0x01,0x02,0x03,0x5A}

value = **Byte_Concat**(bb2, 0x10, bb1)

// value = {0x01,0x02,0x03,0x10,0x31,0x32,0x00,0x4F,0x01}

value = **Byte_Concat**(bb1, "AB", bb2, 10)

// value = {0x31,0x32,0x00,0x4F,0x01,0x01,0x02,0x03,0x0A}

// Parameter "AB" is the string type. Ignored.

value = **Byte_Concat**(bb1, "AB", bb2, 1000) // value = {0x31,0x32,0x00,0x4F,0x01,0x01,0x02,0x03}

// Parameter "AB" is the string type. Ignored.

// Parameter 1000 is the integer type. Ignored.

4.11 String_ToInteger()

Transform string to integer (int type)

Syntax 1

```
int String_ToInteger(
    string,
    int
)
```

Parameters

string	The input string.
int	The input string's notation is decimal, hexadecimal or binary
10	decimal or auto format detecting (Default)
16	hexadecimal
2	binary
	String's notation
123	decimal
0x7F	hexadecimal
0b101	binary

Return

int The integer value formed from input string. If notation is invalid, returns 0.

Note

```
value = String_ToInteger("1234", 10)      // value = 1234
value = String_ToInteger("1234", 16)      // value = 4660
value = String_ToInteger("1234", 2)       // value = 0      // Invalid binary format
value = String_ToInteger("1100", 2)       // value = 12
value = String_ToInteger("0x1234", 10)    // value = 4660
                                           // Hexadecimal format by auto detecting
value = String_ToInteger("0x1234", 16)    // value = 4660
value = String_ToInteger("0x1234", 2)     // value = 0      // Invalid binary format
value = String_ToInteger("0b1100", 10)    // value = 12     // Binary format by auto detecting
value = String_ToInteger("0b1100", 16)    // value = 725248   // Valid Hexadecimal number
value = String_ToInteger("0b1100", 2)     // value = 12
value = String_ToInteger("+1234", 10)     // value = 1234
value = String_ToInteger("-1234", 10)     // value = -1234
value = String_ToInteger("-0x1234", 16)   // value = 0        // Invalid hex format
value = String_ToInteger("-0b1100", 2)    // value = 0        // Invalid binary format
```

Syntax 2

```
int String_ToInteger(
    string
)
```

Note

Similar to syntax1 with decimal format or auto format detection

String_ToInteger(str) => String_ToInteger(str, 10)

Syntax 3

```
int[] String_ToInteger(
```

```
string[],
int
```

```
)
```

Parameters

```
string[]  Input string array
int       The notation of element in input string array is decimal, hexadecimal or
          binary
          10      decimal or auto format detecting (Default)
          16      hexadecimal
          2       binary
          String's notation
          123     decimal
          0x7F    hexadecimal
          0b101   binary
          * The notations of all the elements in a single array have to be identical
```

Return

```
int[]      The integer array formed from input string array. If notation is invalid, returns 0.
```

Note

```
ss = {"12", "ab", "cc", "dd", "10"}
value = String_ToInteger(ss)           // value = {12, 0, 0, 0, 10}
                                        // "ab","cc","dd" are invalid decimal numbers
value = String_ToInteger(ss, 2)        // value = {0, 0, 0, 0, 2}
                                        // "12","ab","cc","dd" are invalid binary numbers
value = String_ToInteger(ss, 16)       // value = {18, 171, 204, 221, 16}
value = String_ToInteger(ss, 10)       // value = {12, 0, 0, 0, 10} // "ab","cc","dd" are invalid decimal
                                        numbers
```


4.12 String_ToFloat()

Transform string to floating-point (floating-point type)

Syntax 1

```
float String_ToFloat(
    string,
    int
)
```

Parameters

string	Input string
int	Input string's notation is decimal, hexadecimal or binary format
10	decimal or auto format detecting (Default)
16	hexadecimal
2	binary
	String's notation
123	decimal
0x7F	hexadecimal
0b101	binary

Return

float The floating-point number formed from input string. If notation is invalid, returns 0.

Note

```
value = String_ToFloat("12.34", 10)    // value = 12.34
value = String_ToFloat("12.34", 16)    // value = 0          // Invalid hexadecimal format
value = String_ToFloat("12.34", 2)     // value = 0          // Invalid binary format
value = String_ToFloat("11.00", 2)     // value = 0          // Invalid binary format
value = String_ToFloat("0x1234", 10)   // value = 6.53E-42
                                         // Hexadecimal format by auto detecting
value = String_ToFloat("0x1234", 16)   // value = 6.53E-42
value = String_ToFloat("0x1234", 2)    // value = 0          // Invalid binary format
value = String_ToFloat("0b1100", 10)   // value = 1.7E-44    // Binary format by auto detecting
value = String_ToFloat("0b1100", 16)   // value = 1.016289E-39 // Valid hexadecimal format
value = String_ToFloat("0b1100", 2)    // value = 1.7E-44
value = String_ToFloat("+12.34", 10)   // value = 12.34
value = String_ToFloat("-12.34", 10)   // value = -12.34
value = String_ToFloat("-0x1234", 16)  // value = 0          // Invalid hex format
value = String_ToFloat("-0b1100", 2)   // value = 0          // Invalid format
```

Syntax 2

```
float String_ToFloat(
    string
)
```

Note

Similar to syntax1 with decimal format or auto format detection

String_ToFloat(str) => String_ToFloat(str, 10)

Syntax 3

```
float[] String_ToFloat(
```

```

    string[],
    int
)

```

Parameters

string[] Input string array

int The notation of elements in input string array is decimal, hexadecimal or binary

- 10** decimal or auto format detecting (Default)
- 16** hexadecimal
- 2** binary

String's notation

- 123** decimal
- 0x7F** hexadecimal
- 0b101** binary

* The notation of all the elements in a single array have to be identical

Return

float[] The floating-point number array formed from input string array. If notation is invalid, returns 0.

Note

```

ss = {"12.345", "ab", "cc", "dd", "10.111"}
value = String_ToFloat(ss)           // value = {12.345,0,0,0,10.111}
value = String_ToFloat(ss, 2)        // value = {0,0,0,0,0}
value = String_ToFloat(ss, 16)       // value = {0,2.4E-43,2.86E-43,3.1E-43,0}
value = String_ToFloat(ss, 10)       // value = {12.345,0,0,0,10.111}

```

4.13 String_ToDouble()

Transform string to floating-point number (double type)

Syntax 1

```
double String_ToDouble(
    string,
    int
)
```

Parameters

string	Input string
int	Input string's notation is decimal, hexadecimal or binary format
10	decimal or auto format detecting (Default)
16	hexadecimal
2	binary
	String's notation
123	decimal
0x7F	hexadecimal
0b101	binary

Return

double The floating-point number formed from input string. If notation is invalid, returns 0.

Note

```
value = String_ToDouble("12.34", 10) // value = 12.34
value = String_ToDouble("12.34", 16) // value = 0           // Invalid hexadecimal format
value = String_ToDouble("12.34", 2)  // value = 0           // Invalid binary format
value = String_ToDouble("11.00", 2)  // value = 0           // Invalid binary format
value = String_ToDouble("0x1234", 10) // value = 2.3023E-320 // Hexadecimal format by auto
                                         detecting
value = String_ToDouble("0x1234", 16) // value = 2.3023E-320
value = String_ToDouble("0x1234", 2)  // value = 0           // Invalid binary format
value = String_ToDouble("0b1100", 10) // value = 6E-323     // Binary format by auto detecting
value = String_ToDouble("0b1100", 16) // value = 3.5832E-318 // Valid hexadecimal format
value = String_ToDouble("0b1100", 2)  // value = 6E-323
value = String_ToDouble("+12.34", 10) // value = 12.34
value = String_ToDouble("-12.34", 10) // value = -12.34
value = String_ToDouble("-0x1234", 16) // value = 0           // Invalid hex format
value = String_ToDouble("-0b1100", 2) // value = 0           // Invalid binary format
```

Syntax 2

```
double String_ToDouble(
    string
)
```

Note

Similar to syntax1 with decimal format or auto format detection

String_ToDouble(str) => String_ToDouble(str, 10)

Syntax 3

```
double[] String_ToDouble(
    string[],
    int
)
```

Parameters

string[] Input string array

int The notation of elements in input string array is decimal, hexadecimal or binary

- 10** decimal or auto format detecting (Default)
- 16** hexadecimal
- 2** binary

String's notation

- 123** decimal
- 0x7F** hexadecimal
- 0b101** binary

* The notation of all the elements in a single array has to be identical

Return

double[] The floating-point number array formed from input string array. If notation is invalid, returns 0.

Note

```
ss = {"12.345", "ab", "cc", "dd", "10.111"}
value = String_ToDouble(ss) // value = {12.345, 0, 0, 0, 10.111}
value = String_ToDouble(ss, 2) // value = {0, 0, 0, 0, 0}
value = String_ToDouble(ss, 16) // value = {0, 8.45E-322, 1.01E-321, 1.09E-321, 0}
value = String_ToDouble(ss, 10) // value = {12.345, 0, 0, 0, 10.111}
```

4.14 String_ToByte()

Transform string to byte array

Syntax 1

```
byte[] String_ToByte(
    string,
    int
)
```

Parameters

string Input string
int The character encoding rules applied to input string
 0 UTF8 (Default)
 1 HEX BINARY // Stop at invalid Hex value
 2 ASCII

Return

byte[] The byte array formed from input string

Note

```
value = String_ToByte("12345", 0) // value = {0x31, 0x32, 0x33, 0x34, 0x35}
value = String_ToByte("12345", 1) // value = {0x12, 0x34, 0x50} // the insufficient part will
                                // be filled with 0
value = String_ToByte("12345", 2) // value = {0x31, 0x32, 0x33, 0x34, 0x35}
value = String_ToByte("0x12345", 0) // value = {0x30, 0x78, 0x31, 0x32, 0x33, 0x34, 0x35}
value = String_ToByte("0x12345", 1) // value = {0x00} // Only 0 be transformed, cause x is
                                // an invalid Hex value
value = String_ToByte("0x12345", 2) // value = {0x30, 0x78, 0x31, 0x32, 0x33, 0x34, 0x35}
value = String_ToByte("TM5機器人", 0) // value = {0x54, 0x4D, 0x35, 0xE6, 0xA9, 0x9F, 0xE5, 0x99, 0xA8,
                                // 0xE4, 0xBA, 0xBA}
value = String_ToByte("TM5機器人", 1) // value = {0x00} // T is an invalid Hex value
value = String_ToByte("TM5機器人", 2) // value = {0x54, 0x4D, 0x35, 0x3F, 0x3F, 0x3F}
value = String_ToByte("0123456", 1) // value = {0x01, 0x23, 0x45, 0x60}
value = String_ToByte("01234G5", 1) // value = {0x01, 0x23, 0x40} // G is an invalid Hex value
```

Syntax 2

```
byte[] String_ToByte(
    string
)
```

Note

Similar to syntax1 with UTF8 format

String_ToByte(str) => String_ToByte(str, 0)

4.15 String_IndexOf()

Search the address of the first occurrence of a specified string from left to right.

Syntax 1

```
int String_IndexOf(
    string,
    string,
    int
)
```

Parameters

string	Input string
string	The specified string to search
int	The initial address to start searching

Return

int	0..(Length of string)-1	If the specified string is found, returns the index number
	-1	Not found
	0	The specified string is "" or empty

Syntax 2

```
int String_IndexOf(
    string,
    string
)
```

Note

Same as syntax 1. It defaults 0 to the initial address of the parameter int as searching from the leftmost.

```
int index = String_IndexOf("012314", "1") // 1
index = String_IndexOf("012314", "") // 0
index = String_IndexOf("012314", empty) // 0
index = String_IndexOf("012314", "d") // -1
index = String_IndexOf("", "d") // -1
index = String_IndexOf("012314", "1", 1) // 1 // Start searching with the 1st index number
index = String_IndexOf("012314", "1", 2) // 4 // Start searching with the 2nd index number
index = String_IndexOf("012314", "1", 10) // -1
```

Syntax 3

```
int String_IndexOf(
    string[],
    string,
    int
)
```

Parameters

string	The array of the input string
string	The specified string to search
int	The initial index of the array to start searching

Return

int **0..Array Size -1** Return the index of the string array if found the string.
 -1 Not found

* The array index will start searching with the initial index from left to right.

* Will use `String_IndexOf(string, string)` to search if the array element available.

Return the index number of the array element but not the index number of the string.

Syntax 4

```
int String_IndexOf(
    string[],
    string
)
```

Note

Same as syntax 3. It defaults 0 to the initial index of the parameter `int` as searching from the foremost of the array elements.

```
string[] ss = {"012314", "ABCDEF", "123TM"}
```

```
int index = String_IndexOf(ss, "1") // 0
```

```
index = String_IndexOf(ss, "") // 0 // Since using String_IndexOf to search strings, it is available
                                   when searching for "".
```

```
index = String_IndexOf(ss, "d") // -1
```

```
index = String_IndexOf(ss, "1", 1) // 2
```

```
index = String_IndexOf(ss, "1", 10) // -1
```

4.16 String_LastIndexOf()

Search the address of the last occurrence of a specified string from right to left.

Syntax 1

```
int String_LastIndexOf(
    string,
    string,
    int
)
```

Parameters

string Input string
string The specified string to search
int The initial address to start searching

Return

int **0..(Length of string)-1** If the specified string is found, returns the index number
 -1 Not found
 (Length of string) The specified string is "" or empty

Syntax 2

```
int String_LastIndexOf(
    string,
    string
)
```

Note

Same as syntax 1. It defaults 0 to the initial address of the parameter int as searching from the rightmost.

```
int index = String_LastIndexOf("012314", "1") // 4
index = String_LastIndexOf("012314", "") // 6
index = String_LastIndexOf("012314", empty) // 6
index = String_LastIndexOf("012314", "d") // -1
index = String_LastIndexOf("", "d") // -1
index = String_LastIndexOf("012314", "1", 1) // 1
// Start searching with the 1st index number

index = String_LastIndexOf("012314", "1", 2) // 1
// Start searching with the 2nd index number

index = String_LastIndexOf("012314", "1", 10) // 4
index = String_LastIndexOf("012314", "4", 10) // 5
```

Syntax 3

```
int String_LastIndexOf(
    string[],
    string,
    int
)
```

Parameters

string The array of the input string

string The specified string to search
int The initial index of the array to start searching

Return

int **0..Array Size -1** Return the index of the string array if found the string.
 -1 Not found

* The array index will start searching with the initial index from right to left.

* Will use `String_IndexOf(string, string)` to search if the array element available.

Return the index number of the array element but not the index number of the string.

Syntax 4

```
int String_LastIndexOf(
    string[],
    string
)
```

Note

Same as syntax 3. It defaults 0 to the initial index of the parameter `int` as the array size as searching from the last of the array elements.

```
string[] ss = {"012314", "ABCDEF", "123TM"}
int index = String_LastIndexOf(ss, "1") // 2
index = String_LastIndexOf(ss, "")      // 2 // Since using String_IndexOf to search strings, it is
                                         available when searching for "".
index = String_LastIndexOf(ss, "d")     // -1
index = String_LastIndexOf(ss, "1", 1)  // 0
index = String_LastIndexOf(ss, "1", 10)// 2
```

4.17 String_DiffIndexOf()

Compare the address where the first string difference occurs starting from the start address.

Syntax 1

```
int String_DiffIndexOf(
    string,
    string,
    int
)
```

Parameters

string Input string 1
 string Input string 2
 int The initial address to start comparing

Return

int 0..(Length of string)-1

If a difference is found, returns the index number of the difference.

-1 No difference found. Namely, both strings match.

-2 The initial address exceeds the length of two strings.

Syntax 2

```
int String_DiffIndexOf(
    string,
    string
)
```

Note

Same as syntax 1. It defaults 0 to the initial address of the parameter int as searching from the leftmost.

```
string s1 = "abcdef"
string s2 = "abcDef"
string s3 = "abcdeF123"
int index = String_DiffIndexOf(s1, s2) // 3
index = String_DiffIndexOf(s1, s3) // 5
index = String_DiffIndexOf(s1, s3, 5) // 5 // Start comparing with index 5, f and F.
index = String_DiffIndexOf(s1, s3, 7) // 7 // Start comparing with index 7, '\0' and 2.
index = String_DiffIndexOf(s1, "abcdef", 7) // -1 // Both strings match.
index = String_DiffIndexOf(s1, "ABCDEF", 7) // -2 // Start comparing with index 7. Exceeded the
length of two strings.
```

4.18 String_Substring()

Retrieve a substring from input string

Syntax 1

```
string String_Substring(
    string,
    int,
    int
)
```

Parameters

string Input string
int The starting character position of sub string (0 .. (length of input string)-1)
int The length of substring

Return

string Substring
 If **starting character position** < 0, returns empty string
 If **starting character position** >= **length of input string**, returns empty string
 If **length of substring** < 0, the substring ends at the last character of the input
 string
 If the sum of starting character position and length of substring exceeds the length of input string, the substring ends at the last character of the input string

Note

```
value = String_Substring("0x12345", 2, 4)      // value = "1234"
value = String_Substring("0x12345", -1, 4)     // value = ""
value = String_Substring("0x12345", 7, 4)     // value = ""
value = String_Substring("0x12345", 2, -1)    // value = "12345"
value = String_Substring("0x12345", 2, 100)   // value = "12345"
```

Syntax 2

```
string String_Substring(
    string,
    int
)
```

Note

Similar to syntax1 with the substring ends at the last character of the input string
String_Substring(str, 2) => String_Substring(str, 2, maxlen)

Syntax 3

```
string String_Substring(
    string,
    string,
    int
)
```

Parameters

string Input string
string The target string to be searched, the substring will start at its position, if it is

found

int The length of substring

Return

string Substring

If **the target string is empty**, the substring start at index zero

If **the target string is not found**, returns empty string

If **length of substring < 0**, the substring ends at the last character of the input

string

If the sum of starting character position and length of substring exceeds the length of input string, the substring ends at the last character of the input string

Note

This syntax is the same as **String_Substring**(str, **String_IndexOf**(str1), int)

value = **String_Substring**("0x12345", "1", 4) // value = "1234"

value = **String_Substring**("0x12345", "", 4) // value = "0x12"

value = **String_Substring**("0x12345", "7", 4) // value = ""

value = **String_Substring**("0x12345", "1", -1) // value = "12345"

value = **String_Substring**("0x12345", "1", 100) // value = "12345"

Syntax 4

```
string String_Substring(
    string,
    string
)
```

Note

Similar to Syntax 3 with the substring ends at the last character of the input string

String_Substring(str, "1") => **String_Substring**(str, "1", maxlen)

Syntax 5

```
string String_Substring(
    string,
    string,
    string,
    int
)
```

Parameters

string Input string

string Prefix. The leading element of the substring

string Suffix. The trailing element of the substring

int The number of occurrence

Return

string Substring

If **prefix and suffix are empty string**, returns input string

If **the number of occurrence <= 0**, returns empty string

Note

value = **String_Substring**("0x12345", "", "", 0) // value = "0x12345"

value = **String_Substring**("0x12345", "1", "4", 1) // value = "1234"

value = **String_Substring**("0x12345", "1", "4", 2) // value = ""

value = **String_Substring**("0x12345", "1", "4", 0) // value = ""

```

value = String_Substring("0x123450x12-345", "1", "4", 1) // value = "1234"
value = String_Substring("0x123450x12-345", "1", "4", 2) // value = "12-34"
value = String_Substring("0x123450x12-345", "1", "4", 3) // value = ""
value = String_Substring("0x12345122", "1", "", 1) // value = "12345122" // Retrieves what's
// after the matched prefix
value = String_Substring("0x12345122", "1", "", 2) // value = "122" // Retrieves what's after
// the matched prefix. The matched amount
// moves from the front to the back.
value = String_Substring("0x12345122", "1", "", 4) // value = ""
value = String_Substring("0x12345433", "", "4", 1) // value = "0x123454" // Retrieves what's
// after the matched suffix
value = String_Substring("0x12345433", "", "4", 2) // value = "0x1234" // Retrieves what's
// after the matched suffix. The matched amount
// moves from the back to the front.

```

Syntax 6

```

string String_Substring(
    string,
    string,
    string
)

```

Note

Similar to Syntax 5 with the substring start at the first occurrence

String_Substring(str, prefix, suffix) => **String_Substring**(str, prefix, suffix, 1)

4.19 String_Split()

Split the string using specified separator.

Syntax 1

```
string[] String_Split(
    string,
    string,
    int
)
```

Parameters

- | | |
|--------|---|
| string | Input string |
| string | Separator (String) |
| int | Format |
| 0 | Split and keep the empty strings |
| 1 | Split and eliminate the empty strings |
| 2 | Split with the elements inside double quotation mark skipped, and keep the empty strings |
| 3 | Split with the elements inside double quotation mark skipped, and eliminate the empty strings |

Return

- string[] Split substring
- If input string is empty, it returns a substring with array. [0] = empty and deals with empty strings by separators.
- If separator is empty, it returns substring with array [0] = Input string and deals with empty strings by separators.

Note

```
value = String_Split("0x112345", "1", 0) // value = {"0x","","2345"}
value = String_Split("0x112345", "1", 1) // value = {"0x","2345"}
value = String_Split("", "1", 0)           // value = {""}    // length = 1
value = String_Split("", "1", 1)           // value = {}      // length = 0
value = String_Split("0x112345", "", 0)    // value = {"0x112345"}

s1 = "123, ""456,67"",89"
value = String_Split(s1, ",", 0)           // value = {"123", ""456", "67"", "89"} // length = 4
value = String_Split(s1, ",", 2)           // value = {"123", ""456,67"", "89"}   // length = 3
```

Syntax 2

```
string[] String_Split(
    string,
    string
)
```

Note

Similar to Syntax1 with splitting and keeping the empty strings

String_Split(str, separator) => String_Split(str, separator, 0)

4.20 String_Replace()

Return a new string in which all occurrences of a specified string in the input string are replaced with another specified string

Syntax 1

```
string String_Replace(  
    string,  
    string,  
    string  
)
```

Parameters

string	Input string
string	Old value, the string to be replaced
string	New value, the string to replace all occurrences of old value

Return

string	The string formed by replacing the old value with new value in input value. If the old value is empty, returns the input string
--------	---

Note

```
value = String_Replace("0x112345", "1", "2")    // value = "0x222345"  
value = String_Replace("0x112345", "", "2") // value = "0x112345"  
value = String_Replace("0x112345", "1", "") // value = "0x2345"
```

4.21 String_Trim()

Return a new string in which all leading and trailing occurrences of specified characters or white-space characters from the input string are removed

Syntax 1

```
string String_Trim(
    string
)
```

Parameters

string Input string

Return

string String formed by removing all leading and trailing occurrences of white-space characters

Note

```
value = String_Trim("0x112345 ")      // value = "0x112345"
value = String_Trim(" 0x112345")      // value = "0x112345"
value = String_Trim(" 0x112345  ")    // value = "0x112345"
```

White-space characters

\u0020	\u1680	\u2000	\u2001	\u2002	\u2003	\u2004
\u2005	\u2006	\u2007	\u2008	\u2009	\u200A	\u202F
\u205F	\u3000					
\u2028						
\u2029						
\u0009	\u000A	\u000B	\u000C	\u000D	\u0085	\u00A0
\u200B	\uFEFF					

Syntax 2

```
string String_Trim(
    string,
    string
)
```

Parameters

string Input string

string Specified characters to be removed from leading occurrences

Return

string String formed by removing all leading occurrences of specified characters

Syntax 3

```
string String_Trim(
    string,
    string,
    string
)
```

Parameters

string Input string

string Specified characters to be removed from leading occurrences

string Specified characters to be removed from trailing occurrences

Return

string String formed by removing all leading and trailing occurrences of the specified characters

Note

```
string s1 = "Hello Hello World Hello World"
string s2 = "HelloHelloWorldHelloWorld"
value = String_Trim(s1, "Hello")           // value = " Hello World Hello World"
value = String_Trim(s1, "World")           // value = "Hello Hello World Hello World"
value = String_Trim(s1, "", "Hello")       // value = "Hello Hello World Hello World"
value = String_Trim(s1, "", "World")       // value = "Hello Hello World Hello "
value = String_Trim(s1, "Hello", "World")  // value = " Hello World Hello "
value = String_Trim(s2, "Hello")           // value = "WorldHelloWorld"
value = String_Trim(s2, "World")           // value = "HelloHelloWorldHelloWorld"
value = String_Trim(s2, "", "Hello")       // value = "HelloHelloWorldHelloWorld"
value = String_Trim(s2, "", "World")       // value = "HelloHelloWorldHello"
value = String_Trim(s2, "Hello", "World")  // value = "WorldHello"
```

4.22 String_ToLower()

Change all the characters in a string to lower case

Syntax 1

```
string String_ToLower(  
    string  
)
```

Parameters

string Input string

Return

string The string formed by converting all the English character into lower case. Non-English character will be remained the same.

Note

```
value = String_ToLower("0x11Acz34") // value = "0x11acz34"
```

4.23 String_ToUpper()

Change all the characters in a string to upper case

Syntax 1

```
string String_ToUpper(  
    string  
)
```

Parameters

string Input string

Return

string The string formed by converting all the English character into upper case. Non-English character will remain the same.

Note

```
value = String_ToUpper("0x11AcZ34") // value = "0X11ACZ34"
```

4.24 Array_Append()

Add new data as the elements in the end of the array.

Syntax 1

```
?[] Array_Append(
    [],
    ? or ?[]
)
```

Parameters

- ?[] Parameter 1, the array to be appended. Available types: byte, int, float, double, bool, and string.
- ? or ?[] Parameter 2, the data or the array to add. The type must be the same with the type of the array to be appended.
*Both parameters must go with the same type.

Return

- ?[] The new array with the parameter 2 elements appended to the parameter 1.

Note

- ? **byte[]** n1 = {100, 200, 30}
byte[] n2 = {40, 50, 60}
n3 = **Array_Append**(n1, n2) // n3 = {100, 200, 30, 40, 50, 60}
n1 = **Array_Append**(n1, 100) // n1 = {100, 200, 30, 100}
n1 = **Array_Append**(n1, n3) // n1 = {100, 200, 30, 100, 100, 200, 30, 40, 50, 60}
- ? **float[]** n1 = {1.1, 2.2, 3.3}
float[] n2 = {0.4, 0.5}
n3 = **Array_Append**(n1, n2) // n3 = {1.1, 2.2, 3.3, 0.4, 0.5}
n4 = **Array_Append**(n3, 5.678) // n4 = {1.1, 2.2, 3.3, 0.4, 0.5, 5.678}
- ? **string[]** n1 = {"123", "ABC", "456", "DEF"}
string[] n2 = {"ABC", "123", "XYZ"}
n3 = **Array_Append**(n1, n2) // n3 = {"123", "ABC", "456", "DEF", "ABC", "123", "XYZ"}
n4 = **Array_Append**(n2, "Hello World") // n4 = {"ABC", "123", "XYZ", "Hello World"}

4.25 Array_Insert()

Insert data as the elements in the array.

Syntax1

```
?[] Array_Insert(
    [],
    int,
    ? or []
)
```

Parameters

?	Parameter 1, the array to be inserted. Available types: byte, int, float, double, bool, and string.
int	The index starting address of the parameter 1.
0	The length of the array 1 Legal value
- 1	
>=	The length of the array 1 Legal value, and will insert the value in the end of the parameter 1.
< 0	Illegal value, the project will stop by error.
? or []	Parameter 2, the data or the array to insert. The type must be the same with the type of the array to be appended.
* Both parameters must go with the same type.	

Return

?	The new array with the parameter 2 elements inserted to the index starting address of the parameter 1.
---	--

Note

```
? int[] n1 = {100, 200, 30}
int[] n2 = {40, 50, 60}
n3 = Array_Insert(n1, 0, n2) // n3 = {40, 50, 60, 100, 200, 30} // Insert to the index 0
n4 = Array_Insert(n1, 2, n2) // n4 = {100, 200, 40, 50, 60, 30} // Insert to the index 2
n5 = Array_Insert(n1, -1, n2) // n5 = {}
// The project will stop by error. Illegal index to start with

? double[] n1 = {1.4, 2.6, 3.9}
double[] n2 = {0.5, 0.7}
n3 = Array_Insert(n1, 1, n2) // n3 = {1.4, 0.5, 0.7, 2.6, 3.9}
n4 = Array_Insert(n3, 4, 1.2345) // n4 = {1.4, 0.5, 0.7, 2.6, 1.2345, 3.9}
n5 = Array_Insert(n3, 100, 9) // n5 = {1.4, 0.5, 0.7, 2.6, 3.9, 9}
// Out of the index. The value will insert in the end of the array.
```

4.26 Array_Remove()

Delete data as the elements in the array.

Syntax1

```
?[] Array_Remove(
    [],
    int,
    int
)
```

Parameters

<code>[]</code>	Parameter 1, the array to be inserted. Available types: byte, int, float, double, bool, and string.
<code>int</code>	The index starting address of the parameter 1 to remove.
<code>0</code>	The length of the parameter 1 Legal value
<code>- 1</code>	
<code>>=</code>	The length of the parameter 1 Illegal value, the project will stop by error.
<code>< 0</code>	Illegal value, the project will stop by error.
<code>int</code>	The number of the elements to remove
<code>> 0</code>	The number of the elements to remove from the index starting address or until the end of the array.
<code>< 0</code>	The number will be 0 and no element will be removed.

Return

`[]` The new array with elements removed after the index starting address.

Syntax2

```
?[] Array_Remove(
    [],
    int
)
```

Note

Same as syntax 1. The default number of the elements to remove is 1.

```
? int[] n1 = {100, 200, 30, 40, 50, 60}
n3 = Array_Remove(n1, -1) // n3 = {}
// The project will stop by error. Illegal value to start with.
n4 = Array_Remove(n1, 100) // n4 = {}
// The project will stop by error. Illegal value to start with.
n5 = Array_Remove(n1, 0) // n5 = {200, 30, 40, 50, 60} // Remove index 0
n6 = Array_Remove(n1, 1, 2) // n6 = {100, 40, 50, 60} // Remove 2 elements from index 1
n7 = Array_Remove(n1, 1, 100) // n7 = {100} // Remove 100 elements from index 1 (remove
to the end of the array)
n8 = Array_Remove(n1, Length(n1)-1) // n8 = {100,200,30,40,50,60}
// Remove from the last of index
n9 = Array_Remove(n1, Length(n1)) // n9 = {}
// The project will stop by error. Illegal value to start with.
```

4.27 Array_Equals()

Determine whether the specified two arrays are identical.

Syntax 1

```
bool Array_Equals(
    ?[],
    ?[]
)
```

Parameters

?[] Input array1 (Data type can be byte, int, float, double, bool, string)
?[] Input array2 (Data type can be byte, int, float, double, bool, string)
 * The data type of array1 and array2 must be identical.

Return

bool Two arrays are identical or not?
true two arrays are identical
false two arrays are not identical

Syntax 2

```
bool Array_Equals(
    ?[],vv
    int,
    ?[],
    int,
    int
)
```

Parameters

?[] Input array1 (Data type can be byte, int, float, double, bool, string)
int The starting index of array1 (0 .. (length of array1)-1)
?[] Input array2 (Data type can be byte, int, float, double, bool, string)
int The starting index of array2 (0 .. (length of array2)-1)
int The number of elements to be compared (0: return true, <0: return false)
 * The data type of array1 and array2 must be identical.

Return

bool The assigned elements in two arrays are identical or not?
true identical
false not identical (or parameters are not valid)

Note

```
? byte[] n1 = {100, 200, 30}
byte[] n2 = {100, 200, 30}
Array_Equals(n1, n2) // true
Array_Equals(n1, 0, n2, 0, 3) // true
Array_Equals(n1, 0, n2, 0, Length(n2)) // true

? int[] n1 = {1000, 2000, 3000}
int[] n2 = {1000, 2000, 3000, 4000}
Array_Equals(n1, n2) // false
Array_Equals(n1, 0, n2, 0, Length(n2)) // false // compare 4 elements
```

```

    Array_Equals(n1, 0, n2, 0, 3)           // true
? float[] n1 = {1.1, 2.2, 3.3}
float[] n2 = {1.1, 2.2}
    Array_Equals(n1, n2)                   // false
    Array_Equals(n1, 0, n2, 0, Length(n2)) // true    // compare 2 elements
    Array_Equals(n1, 0, n2, 0, Length(n1)) // false
? double[] n1 = {100, 200, 300, 3.3, 2.2, 1.1}
double[] n2 = {100, 200, 400, 3.3, 2.2, 4.4}
    Array_Equals(n1, n2)                   // false
    Array_Equals(n1, 0, n2, 0, Length(n2)) // false
    Array_Equals(n1, 0, n2, 0, 2)           // true
    Array_Equals(n1, 3, n2, 3, 2)           // true
? bool[] n1 = {true, false, true, true, true}
bool[] n2 = {true, false, true, false, true}
    Array_Equals(n1, n2)                   // false
    Array_Equals(n1, 0, n2, 0, -1)          // false
    Array_Equals(n1, 0, n2, 0, 0)           // true    // compare 0 element
? string[] n1 = {"123", "ABC", "456", "DEF"}
string[] n2 = {"123", "ABC", "456", "DEF"}
    Array_Equals(n1, n2)                   // true
    Array_Equals(n1, -1, n2, 0, 4)          // false    // Invalid starting index

```


4.28 Array_IndexOf()

Search for the index number of the first occurrence within array elements.

Syntax 1

```
int Array_IndexOf(
    ?[],
    ?,
    int
)
```

Parameters

?[] input array (Data type can be byte, int, float, double, bool, string)
? The target element to search (The data type needs to be the same as the input array **?[]**, but not an array.)
int The initial index of the array to start searching

Return

int **0..(length of input array)-1** If the element is found , it returns the index number.
-1 No element found

Syntax 2

```
int Array_IndexOf(
    ?[],
    ?
)
```

Note

Same as syntax 1. It defaults 0 to the initial address of the parameter int as searching from the foremost.

```
? byte[] n = {100, 200, 30, 100}
value = Array_IndexOf(n, 200) // 1
value = Array_IndexOf(n, 2000) // -1 // Since 2000 is not a byte type, it will be converted to int[],
int to search.
value = Array_IndexOf(n, 100, 1) // 3

? int[] n = {1000, 2000, 3000, 1000}
value = Array_IndexOf(n, 200) // -1
value = Array_IndexOf(n, 1000) // 0
value = Array_IndexOf(n, 1000, 1) // 3

? float[] n = {1.1, 2.2, 3.3, 1.1}
value = Array_IndexOf(n, 1.1) // 0
value = Array_IndexOf(n, 4.4) // -1
value = Array_IndexOf(n, 1.1, 1) // 3

? double[] n = {100, 200, 300, 3.3, 2.2, 1.1, 100}
value = Array_IndexOf(n, 1.1) // 5
value = Array_IndexOf(n, 500) // -1
value = Array_IndexOf(n, 100, 1) // 6
```

```
? bool[] n = {true, false, true, true, true}
value = Array_IndexOf(n, true) // 0
value = Array_IndexOf(n, false) // 1
value = Array_IndexOf(n, false, 2) // -1
value = Array_IndexOf(n, true, 2) // 2

? string[] n = {"123", "ABC", "456", "DEF", "123"}
value = Array_IndexOf(n, "456") // 2
value = Array_IndexOf(n, "789") // -1
value = Array_IndexOf(n, "123", 1) // 4
value = Array_IndexOf(n, "AB") // -1
```

4.29 Array_LastIndexOf()

Search for the index number of the last occurrence within array elements.

Syntax 1

```
int Array_LastIndexOf(
    ?[],
    ?,
    int
)
```

Parameters

?[] input array (Data type can be byte, int, float, double, bool, string)
? The target element to search (The data type needs to be the same as the input array **?[]**, but not an array.)
int The initial index of the array to start searching

Return

int **0..(length of input array)-1** If the element is found , returns the index value
-1 No element found

Syntax 2

```
int Array_LastIndexOf(
    ?[],
    ?
)
```

Note

Same as syntax 1. It defaults 0 to the initial address of the parameter int as searching from the foremost.

? **byte[]** n = {100, 200, 30}

? **byte[]** n = {100, 200, 30, 100}

value = **Array_LastIndexOf**(n, 200) // 1

value = **Array_LastIndexOf**(n, 2000) // -1 // Since 2000 is not a byte type, it will be converted to int[], int to search.

value = **Array_LastIndexOf**(n, 100, 1) // 0

? **int[]** n = {1000, 2000, 3000, 1000}

value = **Array_LastIndexOf**(n, 200) // -1

value = **Array_LastIndexOf**(n, 1000) // 3

value = **Array_LastIndexOf**(n, 1000, 1) // 0

? **float[]** n = {1.1, 2.2, 3.3, 1.1}

value = **Array_LastIndexOf**(n, 1.1) // 3

value = **Array_LastIndexOf**(n, 4.4) // -1

value = **Array_LastIndexOf**(n, 1.1, 1) // 0

? **double[]** n = {100, 200, 300, 3.3, 2.2, 1.1, 100}

value = **Array_LastIndexOf**(n, 1.1) // 5

value = **Array_LastIndexOf**(n, 500) // -1

value = **Array_LastIndexOf**(n, 100, 1) // 0

```
? bool[] n = {true, false, true, true, true}
value = Array_LastIndexOf(n, true) // 4
value = Array_LastIndexOf(n, false) // 1
value = Array_LastIndexOf(n, false, 2) // 1
value = Array_LastIndexOf(n, true, 2) // 2

? string[] n = {"123", "ABC", "456", "DEF", "123"}
value = Array_LastIndexOf(n, "456") // 2
value = Array_LastIndexOf(n, "789") // -1
value = Array_LastIndexOf(n, "123", 1) // 0
value = Array_LastIndexOf(n, "AB") // -1
```

4.30 Array_Reverse()

Reverse the sequence of the elements in the array

Syntax 1

```
?[] Array_Reverse(
    ?[]
)
```

Parameters

?[] input array (Data type can be byte, int, float, double, bool, string)

Return

?[] The reversed array

Note

```
? byte[] n = {100, 200, 30}
n = Array_Reverse(n) // n = {30, 200, 100}
? int[] n = {1000, 2000, 3000}
n = Array_Reverse(n) // n = {3000, 2000, 1000}
? float[] n = {1.1, 2.2, 3.3}
n = Array_Reverse(n) // n = {3.3, 2.2, 1.1}
? double[] n = {100, 200, 300, 3.3, 2.2, 1.1}
n = Array_Reverse(n) // n = {1.1, 2.2, 3.3, 300, 200, 100}
? bool[] n = {true, false, true, true, true}
n = Array_Reverse(n) // n = {true, true, true, false, true}
? string[] n = {"123", "ABC", "456", "DEF"}
n = Array_Reverse(n) // n = {"DEF", "456", "ABC", "123"}
```

Syntax 2

```
?[] Array_Reverse(
    ?[],
    int
)
```

Parameters

?[] input array (Data type can be byte, int, float, double, bool, string)
int the number of elements to be viewed as a section to be reversed

2 2 elements as a section

4 4 elements as a section

8 8 elements as a section

* The sequence of the elements in the same section will be reversed, but the sequence of the sections will remain the same

Return

?[] The reversed array

Note

```
? byte[] n = {100, 200, 30}
n = Array_Reverse(n, 2) // n = {200, 100, 30}
// 2 elements as a section, that is {100,200}{30}
n = Array_Reverse(n, 4) // n = {30, 200, 100}
// 4 elements as a section, that is {100,200,30}
n = Array_Reverse(n, 8) // n = {30, 200, 100}
```

```

?  int[] n = {100, 200, 300, 400}
    n = Array_Reverse(n,  2) // n = {200, 100, 400, 300}
                                // 2 elements as a section, that is {100,200}{300,400}
    n = Array_Reverse(n,  4) // n = {400, 300, 200, 100}
                                // 4 elements as a section, that is {100,200,300,400}
    n = Array_Reverse(n,  8) // n = {400, 300, 200, 100}
?  float[] n = {1.1, 2.2, 3.3, 4.4, 5.5}
    n = Array_Reverse(n,  2) // n = {2.2, 1.1, 4.4, 3.3, 5.5}
                                // 2 elements as a section, that is {1.1,2.2}{3.3,4.4}{5.5}
    n = Array_Reverse(n,  4) // n = {4.4, 3.3, 2.2, 1.1, 5.5}
                                // 4 elements as a section, that is {1.1,2.2,3.3,4.4}{5.5}
    n = Array_Reverse(n,  8) // n = {5.5, 4.4, 3.3, 2.2, 1.1}
?  double[] n = {100, 200, 300, 400, 4.4, 3.3, 2.2, 1.1, 50, 60, 70, 80}
    n = Array_Reverse(n,  2) // n = {200, 100, 400, 300, 3.3, 4.4, 1.1, 2.2, 60, 50, 80, 70}
    n = Array_Reverse(n,  4) // n = {400, 300, 200, 100, 1.1, 2.2, 3.3, 4.4, 80, 70, 60, 50}
    n = Array_Reverse(n,  8) // n = {1.1, 2.2, 3.3, 4.4, 400, 300, 200, 100, 80, 70, 60, 50}
?  bool[] n = {true, false, true, true, true, false, true, false}
    n = Array_Reverse(n,  2) // n = {false, true, true, true, false, true, false, true }
    n = Array_Reverse(n,  4) // n = {true, true, false, true, false, true, false, true}
    n = Array_Reverse(n,  8) // n = {false, true, false, true, true, true, false, true}
?  string[] n = {"123", "ABC", "456", "DEF", "000", "111"}
    n = Array_Reverse(n,  2) // n = {"ABC", "123", "DEF", "456", "111", "000"}
    n = Array_Reverse(n,  4) // n = {"DEF", "456", "ABC", "123", "111", "000"}
    n = Array_Reverse(n,  8) // n = {"111", "000", "DEF", "456", "ABC", "123"}

```

4.31 Array_Sort()

Sort the elements in a array

Syntax 1

```
?[] Array_Sort(
    [],
    int
)
```

Parameters

?[]	input array (Data type can be byte, int, float, double, bool, string)
int	Sorting direction
0	Ascending Order (Default)
1	Descending Order

Return

?[]	The array after sorting
-----	-------------------------

Syntax 2

```
?[] Array_Sort(
    []
)
```

Note

Similar to Syntax1 with sorting direction as ascending order

Array_Sort(array[]) => Array_Sort(array[], 0)

```
? int[] n = {1000, 2000, 3000}
n = Array_Sort(n) // n = {1000, 2000, 3000}
? double[] n = {100, 200, 300, 3.3, 2.2, 1.1}
n = Array_Sort(n, 1) // n = {300, 200, 100, 3.3, 2.2, 1.1}
? bool[] n = {true, false, true, true, true}
n = Array_Sort(n, 1) // n = {true, true, true, true, false}
? string[] n = {"123", "ABC", "456", "DEF"}
n = Array_Sort(n) // n = {"123", "456", "ABC", "DEF"}
```

4.32 Array_SubElements()

Retrieve the sub-elements from input array

Syntax 1

```
?[] Array_SubElements(
    [],
    int,
    int
)
```

Parameters

?[] Input array (Data type can be byte, int, float, double, bool, string)
 int The starting index of sub-elements. (0 .. (length of array)-1)
 int The number of element in sub-elements

Return

?[] The sub-elements from input arrays
 If **starting index** < 0, sub-elements equals empty array
 If **starting index** >= **length of input array**, sub-elements equals empty array
 If **sub-element number** < 0, sub-elements starts at starting index to the last element of input array
 If the sum of starting index and the number of element exceeds the length of the input array, sub-elements starts at starting index to the last element of input array

Syntax 2

```
?[] Array_SubElements(
    [],
    int
)
```

Note

Similar to Syntax 1, but the sub-elements starts at starting index to the last element of input array

Array_SubElements(array[], 2) => Array_SubElements(array[], 2, maxlen)

```
? byte[] n = {100, 200, 30}
n1 = Array_SubElements(n1, 0) // n1 = {100, 200, 30}
n1 = Array_SubElements(n1, -1) // n1 = {}
n1 = Array_SubElements(n1, 0, 3) // n1 = {100, 200, 30}
n1 = Array_SubElements(n1, 1, 3) // n1 = {200, 30}
n1 = Array_SubElements(n1, 2) // n1 = {30}
n1 = Array_SubElements(n1, 3, 3) // n1 = {}

? int[] n = {1000, 2000, 3000}
n1 = Array_SubElements(n1, 0) // n1 = {1000, 2000, 3000}
n1 = Array_SubElements(n1, -1) // n1 = {}
n1 = Array_SubElements(n1, 1, 3) // n1 = {2000, 3000}
n1 = Array_SubElements(n1, 2) // n1 = {3000}

? float[] n = {1.1, 2.2, 3.3}
n1 = Array_SubElements(n1, 0) // n1 = {1.1, 2.2, 3.3}
n1 = Array_SubElements(n1, -1) // n1 = {}
```



```

n1 = Array_SubElements(n1 , 1, 3) // n1 = {2.2, 3.3}
n1 = Array_SubElements(n1 , 2) // n1 = {3.3}
? double[] n = {100, 200, 3.3, 2.2, 1.1}
n1 = Array_SubElements(n1 , 0) // n1 = {100, 200, 3.3, 2.2, 1.1}
n1 = Array_SubElements(n1 , -1) // n1 = {}
n1 = Array_SubElements(n1 , 1, 3) // n1 = {200, 3.3, 2.2}
n1 = Array_SubElements(n1 , 2) // n1 = {3.3, 2.2, 1.1}
? bool[] n = {true, false, true, true, true}
n1 = Array_SubElements(n1 , 0) // n1 = {true, false, true, true, true}
n1 = Array_SubElements(n1 , -1) // n1 = {}
n1 = Array_SubElements(n1 , 1, 3) // n1 = {false, true, true}
n1 = Array_SubElements(n1 , 2) // n1 = {true, true, true}
? string[] n = {"123", "ABC", "456", "DEF"}
n1 = Array_SubElements(n1 , 0) // n1 = {"123", "ABC", "456", "DEF"}
n1 = Array_SubElements(n1 , -1) // n1 = {}
n1 = Array_SubElements(n1 , 1, 3) // n1 = {"ABC", "456", "DEF"}
n1 = Array_SubElements(n1 , 2) // n1 = {"456", "DEF"}

```

4.33 ValueReverse()

Reverse the sequence of byte units inside input data (int 2 bytes or 4 bytes, float 4 bytes, double 8 bytes); or reverse the sequence of character of string.

Syntax 1

```
int ValueReverse(
    int,
    int
)
```

Parameters

int Input value
int The input value follows int32 or int16 format
0 int32 (Default)
1 int16. If the data does not meets int16 format, int32 will be applied instead.
2 int16. Forced to apply int16 format. For int32 data input, there could be some bytes missing

Return

int The value formed from reversing the sequence of byte units inside the input value. For Int32 data, reverse with 4 bytes. For int16 data, reverse with 2 bytes.

Note

```
int i = 10000
value = ValueReverse(i, 0) // 10000=0x00002710 → 0x10270000 // value = 270991360
value = ValueReverse(i, 1) // 10000=0x2710 → 0x1027 // value = 4135
i = 100000 // int32 value
value = ValueReverse(i, 0) // 100000=0x000186A0 → 0xA0860100 // value = -1601830656
value = ValueReverse(i, 1) // 100000=0x000186A0 → 0xA0860100 // value = -1601830656
value = ValueReverse(i, 2) // 100000=0x000086A0 → 0xA0860000 // value = -24442
```

Syntax 2

```
int ValueReverse(
    int
)
```

Parameters

int Input value

Note

Similar to Syntax1 with int32 input format
ValueReverse(int) => ValueReverse(int, 0)

Syntax 3

```
float ValueReverse(
    float
)
```

Parameters

float Input value

Return

float The value formed from reversing the sequence of byte units inside the input value. For float data, reverse 4 bytes.

Note

```
float i = 40000
value = ValueReverse(i) // 40000=0x471C4000 → 0x00401C47 // value = 5.887616E-39
```

Syntax 4

```
double ValueReverse(
    double
)
```

Parameters

double Input value

Return

double The value formed from reversing the sequence of byte units inside the input value. For double data, reverse 8 bytes.

Note

```
double i = 80000
value = ValueReverse(i) // 80000=0x40F3880000000000 → 0x000000000088F340
// value = 4.43432217445369E-317
```

Syntax 5

```
string ValueReverse(
    string
)
```

Parameters

string Input string

Return

string The value formed from reversing the sequence of characters of input string.

Note

```
string i = "ABCDEF"
value = ValueReverse(i) // value = "FEDCBA"
```

Syntax 6

```
int[] ValueReverse(
    int[],
    int
)
```

Parameters

int[] Input array value

int The input value follows int32 or int16 format

- 0 int32 (Default)
- 1 int16. If the data does not meets int16 format, int32 will be applied instead.
- 2 int16. Forced to apply int16 format. For int32 data input, there could be some bytes missing

Return

int[] The array formed from reversing the sequence of byte units inside every element of the input array.

Note

```
int[] i = {10000, 20000, 60000, 80000}
value = ValueReverse(i, 0) // value = {270991360, 541982720, 1625948160, -2143813376}
```

```
value = ValueReverse(i, 1)    // value = {4135, 8270, 1625948160, -2143813376}
value = ValueReverse(i, 2)    // value = {4135, 8270, 24810, -32712}
```

Syntax 7

```
int[] ValueReverse(
    int[]
```

```
)
```

Parameters

```
int[] Input array value
```

Note

Similar to Syntax6 with input integer as int32

ValueReverse(int[]) => ValueReverse(int[], 0)

Syntax 8

```
float[] ValueReverse(
    float[]
```

```
)
```

Parameters

```
float[] Input array value
```

Return

```
float[] The array formed from reversing the sequence of byte units inside every
         element of the input array.
```

Note

```
float[] i = {10000, 20000}
```

```
value = ValueReverse(i) // value = {5.887614E-39, 5.933532E-39}
```

Syntax 9

```
double[] ValueReverse(
    double[]
```

```
)
```

Parameters

```
double[] Input array value
```

Return

```
double[] The array formed from reversing the sequence of byte units inside every
         element of the input array.
```

Note

```
double[] i = {10000, 20000}
```

```
value = ValueReverse(i) // value = {4.428251E-317, 4.430275E-317}
```

Syntax 10

```
string[] ValueReverse(
    string[]
```

```
)
```

Parameters

```
string[] Input string array
```

Return

```
string[] The string array formed from reversing the string inside every element of
         the input string array.
```

Note

```
string[] i = {"ABCDEFGH", "12345678"}  
value = ValueReverse(i) // value = {"GFEDCBA", "87654321"}
```

4.34 GetBytes()

Convert arbitrary data type to byte array.

Syntax 1

```
byte[] GetBytes(
    ?,
    int
)
```

Parameters

- ? The input data. Data type can be int, float, double, bool, string or array.
- int The input data as integers and floating points follows Little Endian or Big Endian
 - 0 Little Endian (Default)
 - 1 Big Endian
- The input data as string arrays separates with 0x00 0x00 for each element
 - 0 Not separate with 0x00 0x00 (Default)
 - 1 Separate with 0x00 0x00

Return

byte[] The byte array formed by input data

Syntax 2

```
byte[] GetBytes(
    ?
)
```

Note

Same as syntax 1 with Little Endian or Big Endian defaults to 0 such as returns based on Little Endian

GetBytes(?) => GetBytes(?, 0)

- ? byte n = 100
 - value = **GetBytes**(n) // value = {0x64}
 - value = **GetBytes**(n, 0) // value = {0x64}
 - value = **GetBytes**(n, 1) // value = {0x64}
- ? byte[] n = {100, 200} // Convert every element of the array to byte, 1 byte as a single unit.
 - value = **GetBytes**(n) // value = {0x64, 0xC8}
 - value = **GetBytes**(n, 0) // value = {0x64, 0xC8}
 - value = **GetBytes**(n, 1) // value = {0x64, 0xC8}
- ? int
 - value = **GetBytes**(123456) // value = {0x40, 0xE2, 0x01, 0x00}
 - value = **GetBytes**(123456, 0) // value = {0x40, 0xE2, 0x01, 0x00}
 - value = **GetBytes**(0x123456, 0) // value = {0x56, 0x34, 0x12, 0x00}
 - value = **GetBytes**(0x1234561, 1) // value = {0x01, 0x23, 0x45, 0x61}
- ? int[] n = {10000, 20000, 80000}
 - // Convert every single element of the array to byte. For int32 data, works on 4 bytes sequentially.

```

value = GetBytes(n)
// value = {0x10, 0x27, 0x00, 0x00, 0x20, 0x4E, 0x00, 0x00, 0x80, 0x38, 0x01, 0x00}
value = GetBytes(n, 0)
// value = {0x10, 0x27, 0x00, 0x00, 0x20, 0x4E, 0x00, 0x00, 0x80, 0x38, 0x01, 0x00}
value = GetBytes(n, 1)
// value = {0x00, 0x00, 0x27, 0x10, 0x00, 0x00, 0x4E, 0x20, 0x00, 0x01, 0x38, 0x80}

```

? float

```

value = GetBytes(123.456, 0) // value = {0x79, 0xE9, 0xF6, 0x42}
float n = -1.2345
value = GetBytes(n, 0)      // value = {0x19, 0x04, 0x9E, 0xBF}
value = GetBytes(n, 1)      // value = {0xBF, 0x9E, 0x04, 0x19}

```

? float[] n = {1.23, 4.56, -7.89}

```

// Convert every single element of the array to byte. For float data, works on 4 bytes sequentially.
value = GetBytes(n)
// value = {0xA4, 0x70, 0x9D, 0x3F, 0x85, 0xEB, 0x91, 0x40, 0xE1, 0x7A, 0xFC, 0xC0}
value = GetBytes(n, 0)
// value = {0xA4, 0x70, 0x9D, 0x3F, 0x85, 0xEB, 0x91, 0x40, 0xE1, 0x7A, 0xFC, 0xC0}
value = GetBytes(n, 1)
// value = {0x3F, 0x9D, 0x70, 0xA4, 0x40, 0x91, 0xEB, 0x85, 0xC0, 0xFC, 0x7A, 0xE1}

```

? double n = -1.2345

```

value = GetBytes(n, 0) // value = {0x8D, 0x97, 0x6E, 0x12, 0x83, 0xC0, 0xF3, 0xBF}
value = GetBytes(n, 1) // value = {0xBF, 0xF3, 0xC0, 0x83, 0x12, 0x6E, 0x97, 0x8D}

```

? double[] n = {1.23, -7.89}

```

// Convert every single element of the array to byte. For double data, works on 8 bytes sequentially.
value = GetBytes(n)
// value = {0xAE, 0x47, 0xE1, 0x7A, 0x14, 0xAE, 0xF3, 0x3F, 0x8F, 0xC2, 0xF5, 0x28, 0x5C, 0x8F, 0x1F, 0xC0}
value = GetBytes(n, 0)
// value = {0xAE, 0x47, 0xE1, 0x7A, 0x14, 0xAE, 0xF3, 0x3F, 0x8F, 0xC2, 0xF5, 0x28, 0x5C, 0x8F, 0x1F, 0xC0}
value = GetBytes(n, 1)
// value = {0x3F, 0xF3, 0xAE, 0x14, 0x7A, 0xE1, 0x47, 0xAE, 0xC0, 0x1F, 0x8F, 0x5C, 0x28, 0xF5, 0xC2, 0x8F}

```

? bool flag = true

```

//GetBytes converts true to 1, and false to 0.
value = GetBytes(flag) // value = {1}
value = GetBytes(flag, 0) // value = {1}
// Because bool is 1 byte, Endian Parameters are not sufficient.
value = GetBytes(flag, 1) // value = {1}

```

? bool[] flag = {true, false, true, false, false, true, true}

```

value = GetBytes(flag) // value = {1, 0, 1, 0, 0, 1, 1}
value = GetBytes(flag, 0) // value = {1, 0, 1, 0, 0, 1, 1}
value = GetBytes(flag, 1) // value = {1, 0, 1, 0, 0, 1, 1}

```

? string n = "ABCDEFGH" // string to encode in UTF8

```

value = GetBytes(n) // value = {0x41, 0x42, 0x43, 0x44, 0x45, 0x46, 0x47}

```

```
value = GetBytes(n, 0)    // value = {0x41, 0x42, 0x43, 0x44, 0x45, 0x46, 0x47}
                          // Endian Parameters not valid
value = GetBytes(n, 1)    // value = {0x41, 0x42, 0x43, 0x44, 0x45, 0x46, 0x47}
                          // Endian Parameters not valid
```

```
? string[] n = {"ABC", "DEF", "達明機器人" }
value = GetBytes(n)
// value = {0x41, 0x42, 0x43, 0x44, 0x45, 0x46,
           0xE9,0x81,0x94,0xE6,0x98,0x8E,0xE6,0xA9,0x9F,0xE5,0x99,0xA8,0xE4,0xBA,0xBA}
value = GetBytes(n, 1)
// value = {0x41, 0x42, 0x43, 0x00, 0x00, 0x44, 0x45, 0x46, 0x00, 0x00,
           0xE9,0x81,0x94,0xE6,0x98,0x8E,0xE6,0xA9,0x9F,0xE5,0x99,0xA8,0xE4,0xBA,0xBA}
```

*Conversion of `string[]` to `byte[]` can maintain the original contents without separation bytes, but it is unable to turn `byte[]` back to `string[]` effectively.

*It is effective to turn `byte[]` back to `string[]` by inserting separation bytes (2 consecutive 0x00s) between the elements in the array, but it is possible to find conversion errors if the value of the string come with 0x00 0x00.

Syntax 3

Convert integer (int type) to byte array.

```
byte[] GetBytes(
    int,
    int,
    int
)
```

Parameters

```
int    The input integer (int type)
int    The input value follows Little Endian or Big Endian
0      Little Endian (Default)
1      Big Endian
int    The input integer value's data type is int32 or int16
0      int32 (Default)
1      int16. If the data does not meets int16 format, int32 will be applied instead.
2      int16. Forced to apply int16 format. For int32 data input, there could be
        some bytes missing.
```

Return

```
byte[] The byte array formed by input integer. For int32 data, convert with 4 bytes. For
        int16 data, convert with 2 bytes.
```

Note

```
value = GetBytes(12345, 0, 0)    // value = {0x39, 0x30, 0x00, 0x00}
value = GetBytes(12345, 0, 1)    // value = {0x39, 0x30}
value = GetBytes(12345, 0, 2)    // value = {0x39, 0x30}
value = GetBytes(0x123456, 0, 0)  // value = {0x56, 0x34, 0x12, 0x00}
value = GetBytes(0x123456, 0, 1)  // value = {0x56, 0x34, 0x12, 0x00}
value = GetBytes(0x123456, 0, 2)  // value = {0x56, 0x34} // bytes missing
value = GetBytes(0x1234561, 1, 0) // value = {0x01, 0x23, 0x45, 0x61}
```



```
value = GetBytes(0x1234561, 1, 1) // value = {0x01, 0x23, 0x45, 0x61}
value = GetBytes(0x1234561, 1, 2) // value = {0x45, 0x61} // bytes missing
```

Syntax 4

Convert the integer array (int[] type) to byte array

```
byte[] GetBytes(
    int[],
    int,
    int
)
```

Parameters

int[] The input integer array (int[] type)

int The input integer array follows Little Endian or Big Endian

- 0 Little Endian (Default)
- 1 Big Endian

int The input integer array's data type is int32 or int16

- 0 int32 (Default)
- 1 int16. If the data does not meets int16 format, int32 will be applied instead
- 2 int16. Forced to apply int16 format. For int32 data input, there could be some bytes missing.

Return

byte[] The byte array formed by input integer array. Every element is converted independently and forms an array. For int32 data, convert with 4 bytes. For int16 data, convert with 2 bytes.

Note

```
i = {10000, 20000, 80000}
value = GetBytes(i, 0, 0)
// value = {0x10, 0x27, 0x00, 0x00, 0x20, 0x4E, 0x00, 0x00, 0x80, 0x38, 0x01, 0x00}
value = GetBytes(i, 0, 1) // value = {0x10, 0x27, 0x20, 0x4E, 0x80, 0x38, 0x01, 0x00}
value = GetBytes(i, 0, 2) // value = {0x10, 0x27, 0x20, 0x4E, 0x80, 0x38} // bytes missing
value = GetBytes(i, 1, 0)
// value = {0x00, 0x00, 0x27, 0x10, 0x00, 0x00, 0x4E, 0x20, 0x00, 0x01, 0x38, 0x80}
value = GetBytes(i, 1, 1) // value = {0x27, 0x10, 0x4E, 0x20, 0x00, 0x01, 0x38, 0x80}
value = GetBytes(i, 1, 2) // value = {0x27, 0x10, 0x4E, 0x20, 0x38, 0x80} // bytes missing
```

4.35 GetString()

Convert arbitrary data type to string

Syntax 1

```
string GetString(
    ?,
    int,
    int
)
```

Parameters

- ?** The input data. Data type can be int, float, double, bool, string or array.
- int** When the output string's notation is decimal, hexadecimal, or binary, the output string value is in decimal, hexadecimal, or binary.
 - 10** decimal, such as 123
 - 16** hexadecimal, such as 0x7B
 - 2** binary, such as 0b01111011
- When the input value is a string array, the output string value is in standard string format or not.
 - 0 or 10** Automatic detection. If the values in the string come with double quotations or commas, it converse to standard string format.
 - 1** Mandatory conversion to standard string format
 - Other** No conversion
- int** The output string format
 - When the output is in decimal
 - 0** Never use scientific notation
 - 1** Use scientific notation when needed.
 - When the output is in hexadecimal or binary
 - 0** Fill up digits. Add prefix 0x or 0b, e.g. 0x0C or 0b00001100
 - 1** Fill up digits. No prefix 0x or 0b, e.g. 0C or 00001100
 - 2** Don't fill up digits. Add prefix 0x or 0b, e.g. 0xC or 0b1100
 - 3** Don't fill up digits. No prefix 0x or 0b, e.g. C or 1100

Return

- string** String converted from input data. If the input data cannot be converted, returns empty string.
- If the input data is array, every element is converted respectively, and returned in "{ , , }" format

Syntax 2

```
string GetString(
    ?,
    int
)
```

Note

Similar to Syntax1 with filling up digits and adding prefix 0x or 0b.

GetString(?, 16) => GetString(?, 16, 0)

Syntax 3

```
string GetString(
    ?
)
```

Note

Same as syntax 1. The output string's notation defaults to 10 and the output string format defaults to 0.

```
GetString(?) => GetString(?, 10, 0)
GetString(?) => GetString(?, 0, 0)    // supposed ? is a string array
```

? byte n = 123

```
value = GetString(n)    // value = "123"
value = GetString(n, 10) // value = "123"
value = GetString(n, 16) // value = "0x7B"
value = GetString(n, 2)  // value = "0b01111011"
value = GetString(n, 16, 3) // value = "7B"
value = GetString(n, 2, 2) // value = "0b1111011"
```

? byte[] n = {12, 34, 56}

```
value = GetString(n)    // value = "{12,34,56}"
value = GetString(n, 10) // value = "{12,34,56}"
value = GetString(n, 16) // value = "{0x0C,0x22,0x38}"
value = GetString(n, 2)  // value = "{0b00001100,0b00100010,0b00111000}"
value = GetString(n, 16, 3) // value = "{C,22,38}"
value = GetString(n, 2, 2) // value = "{0b1100,0b100010,0b111000}"
```

? int n = 1234

```
value = GetString(n)    // value = "1234"
value = GetString(n, 10) // value = "1234"
value = GetString(n, 16) // value = "0x000004D2"
value = GetString(n, 2)  // value = "0b00000000000000000000000010011010010"
value = GetString(n, 16, 3) // value = "4D2"
value = GetString(n, 2, 2) // value = "0b10011010010"
```

? int[] n = {123, 345, -123, -456}

```
value = GetString(n)    // value = "{123,345,-123,-456}"
value = GetString(n, 10) // value = "{123,345,-123,-456}"
value = GetString(n, 16) // value = "{0x0000007B,0x00000159,0xFFFFF85,0xFFFFFE38}"
value = GetString(n, 2)  // value = "{0b00000000000000000000000000001111011,
                                0b0000000000000000000000000101011001,
                                0b11111111111111111111111110000101,
                                0b1111111111111111111111111000111000}"

value = GetString(n, 16, 3) // value = "{7B,159,FFFFFF85,FFFFFE38}"
value = GetString(n, 2, 2) // value = "{0b1111011,
                                0b101011001,
                                0b11111111111111111111111110000101,
                                0b1111111111111111111111111000111000}"
```

? float n = 12.34

```

value = GetString(n)      // value = "12.34"
value = GetString(n, 10)  // value = "12.34"
value = GetString(n, 16)  // value = "0x414570A4"
value = GetString(n, 2)   // value = "0b01000001010001010111000010100100"
value = GetString(n, 16, 3) // value = "414570A4"
value = GetString(n, 2, 2) // value = "0b1000001010001010111000010100100"

```

? **float[]** n = {123.4, 345.6, -123.4, -456.7}

```

value = GetString(n)      // value = "{123.4,345.6,-123.4,-456.7}"
value = GetString(n, 10)  // value = "{123.4,345.6,-123.4,-456.7}"
value = GetString(n, 16)  // value = "{0x42F6CCCD,0x43ACCCCD,0xC2F6CCCD,0xC3E4599A}"
value = GetString(n, 16, 3) // value = "{42F6CCCD,43ACCCCD,C2F6CCCD,C3E4599A}"

```

? **double** n = 12.34

```

value = GetString(n)      // value = "12.34"
value = GetString(n, 10)  // value = "12.34"
value = GetString(n, 16)  // value = "0x4028AE147AE147AE"
value = GetString(n, 16, 3) // value = "4028AE147AE147AE"

```

? **double[]** n = {123.45, 345.67, -123.48, -456.79}

```

value = GetString(n)      // value = "{123.45,345.67,-123.48,-456.79}"
value = GetString(n, 10)  // value = "{123.45,345.67,-123.48,-456.79}"
value = GetString(n, 16)  // value = "{0x405EDCCCCCCCCCD,0x40759AB851EB851F,
                                0xC05EDEB851EB851F,0xC07C8CA3D70A3D71}"
value = GetString(n, 16, 3) // value = "{405EDCCCCCCCCCD,40759AB851EB851F,
                                C05EDEB851EB851F,C07C8CA3D70A3D71}"

```

? **bool** n = true

```

value = GetString(n)      // value = "true"
value = GetString(n, 16)  // value = "true"
value = GetString(n, 2)   // value = "true"
value = GetString(n, 16, 3) // value = "true"

```

? **bool[]** n = {true, false, true, false, false, true}

```

value = GetString(n)      // value = "{true,false,true,false,false,true}"
value = GetString(n, 16)  // value = "{true,false,true,false,false,true}"
value = GetString(n, 2)   // value = "{true,false,true,false,false,true}"
value = GetString(n, 16, 3) // value = "{true,false,true,false,false,true}"

```

? **string** n = "1234567890"

```

value = GetString(n)      // value = "1234567890"
value = GetString(n, 16)  // value = "1234567890"
value = GetString(n, 2)   // value = "1234567890"
value = GetString(n, 16, 3) // value = "1234567890"

```

? **string[]** n = {"123.45", "345.67", "-12""3.48", "-45A6.79"}

```

value = GetString(n)      // value = "{123.45,345.67,-12""3.48,-45A6.79}"
value = GetString(n, 1)   // value = "{123.45,\"345.67\",-12""3.48,\"-45A6.79\"}"

```

```

value = GetString(n, 2)    // value = "{123.45,345.67,-12"3.48,-45A6.79}"    // -12""3.48 displayed
                           as -12"3.48
value = GetString(n, 16, 3) // value = "{123.45,345.67,-12""3.48,-45A6.79}"
value = GetString(n, 10, 3) // value = "{123.45,345.67,"-12""3.48",-45A6.79}"
                           //use automatic detection as the default

```

Syntax 4

```

string GetString(
    ?,
    string,
    int,
    int
)

```

Parameters

? The input data. Data type can be int, float, double, bool, string or array.

string Separator for output string (Only effective to array input)

int When the output string's notation is decimal, hexadecimal, or binary, the output string value is in decimal, hexadecimal, or binary.

- 10** decimal, such as 123
- 16** hexadecimal, such as 0x7B
- 2** binary, such as 0b01111011

When the input value is a string array, the output string value is in standard string format or not.

0 or 10 Automatic detection. If the values in the string come with double quotations or separation symbols, it converse to standard string format.

1 Mandatory conversion to standard string format

Other No conversion

int The output string format

When the output is in decimal

- 0** Never use scientific notation
- 1** Use scientific notation when needed.

When the output is in hexadecimal or binary

- 0** Fill up digits. Add prefix 0x or 0b, e.g. 0x0C or 0b00001100
- 1** Fill up digits. No prefix 0x or 0b, e.g. 0C or 00001100
- 2** Don't fill up digits. Add prefix 0x or 0b, e.g. 0xC or 0b1100
- 3** Don't fill up digits. No prefix 0x or 0b, e.g. C or 1100

Return

string String converted from input data. If the input data cannot be converted, returns empty string.

If the input data is array, every element is converted respectively, and returned as a string with the assigned separator

Syntax 5

```

string GetString(
    ?,
    string,
    int
)

```

Note

Same as Syntax 4 with filling up digits and adding prefix 0x or 0b

GetString(?, str, 16) => **GetString**(?, str, 16, 0)

Syntax 6

```
string GetString(
    ?,
    string
)
```

Note

Same as Syntax 4. The output string's notation defaults to 10 and the output string format defaults to 0.

GetString(?) => **GetString**(?, 10, 0)

GetString(?) => **GetString**(?, 0, 0) // supposed ? is a string array

? **byte** n = 123

value = **GetString**(n) // value = "123"

value = **GetString**(n, ";", 10) // value = "123"

value = **GetString**(n, "-", 16) // value = "0x7B"

value = **GetString**(n, "#", 2) // value = "0b01111011"

value = **GetString**(n, ",", 16, 3) // value = "7B"

value = **GetString**(n, ",", 2, 2) // value = "0b1111011"

* Separator is effective to array input only.

? **byte[]** n = {12, 34, 56}

value = **GetString**(n, "-") // value = "12-34-56"

value = **GetString**(n, Ctrl("\r\n"), 10) // value = "12\u0D0A34\u0D0A56"

value = **GetString**(n, newline, 16) // value = "0x0C\u0D0A0x22\u0D0A0x38"

value = **GetString**(n, NewLine, 2) // value =
"0b00001100\u0D0A0b00100010\u0D0A0b00111000"

value = **GetString**(n, "-", 16, 3) // value = "C-22-38"

value = **GetString**(n, "-", 2, 2) // value = "0b1100-0b100010-0b111000"

* **\u0D0A** is Newline control character, not string value.

? **string[]** n = {"123.45", "345.67", "-12""3.48", "-45A6.79"}

value = **GetString**(n, "-") // value = "123.45-345.67--12""3.48"--45A6.79""

value = **GetString**(n, "-", 1) // value = ""123.45"-345.67--12""3.48"--45A6.79""

value = **GetString**(n, "-", 2) // value = "123.45-345.67--12"3.48--45A6.79"

// Troubled for identifying the separation symbols and the negative signs.

Syntax 7

```
string GetString(
    ?,
    string,
    string,
    int,
    int
)
```

Parameters

- ?** The input data. Data type can be int, float, double, bool, string or array.
- string** The index of the output string for array input. (Only effective to ? as array type data)
 - * Support numeric format strings
- string** Separator for output string (Only effective to array input)
- int** The output string's notation is decimal, hexadecimal or binary (Can be only applied to hexadecimal or binary number)
 - 10** decimal, such as 123
 - 16** hexadecimal, such as 0x7B
 - 2** binary, such as 0b01111011

String's notation

 - 123 decimal
 - 0x7F hexadecimal
 - 0b101 binary

When the input value is a string array, the output string value is in standard string format or not.

 - 0 or 10** Automatic detection. If the values in the string come with double quotations or separation symbols, it converse to standard string format.
 - 1** Mandatory conversion to standard string format
 - Other** No conversion
- int** The output string format

When the output is in decimal

 - 0** Never use scientific notation
 - 1** Use scientific notation when needed.

When the output is in hexadecimal or binary

 - 0** Fill up digits. Add prefix 0x or 0b, e.g. 0x0C or 0b00001100
 - 1** Fill up digits. No prefix 0x or 0b, e.g. 0C or 00001100
 - 2** Don't fill up digits. Add prefix 0x or 0b, e.g. 0xC or 0b1100
 - 3** Don't fill up digits. No prefix 0x or 0b, e.g. C or 1100

Return

- string** Converse the value to the string to return. If unable to converse, it returns an empty string.

If the type is array, elements in the array will be conversed to strings with prefixes of the element index value format string separated by separation symbols to return.

There will be no braces.

Syntax 8

```
string GetString(
    ?,
    string,
    string,
    int
)
```

Note

Similar to Syntax7 with filling up digits and adding prefix.

GetString(?, str, str, 16) => GetString(?, str, str, 16, 0)

Syntax 9

```
string GetString(
    ?,
    string,
    string
)
```

Note

Similar to Syntax7 with decimal output, with filling up digits and adding prefix.

GetString(?, str, str) => **GetString**(?, str, str, 10, 0)

? **byte** n = 123

value = **GetString**(n) // value = "123"

value = **GetString**(n, "[0]=", ";", 10) // value = "123"

value = **GetString**(n, "[0]=", "-", 16) // value = "0x7B"

value = **GetString**(n, "[0]=", "#", 2) // value = "0b01111011"

* Index and separator are only effective to array input.

? **byte[]** n = {12, 34, 56}

value = **GetString**(n, "[0]=", "-") // value = "[0]=12-[1]=34-[2]=56"

value = **GetString**(n, "[0]=", Ctrl("\r\n"), 10) // value = "[0]=12\u0D0A[1]=34\u0D0A[2]=56"

value = **GetString**(n, "[0]=", newline, 16) // value = "[0]=0x0C\u0D0A[1]=0x22\u0D0A[2]=0x38"

value = **GetString**(n, "[0]=", "-", 16, 3) // value = "[0]=C-[1]=22-[2]=38"

value = **GetString**(n, "[0]=", "-", 2, 2) // value = "[0]=0b1100-[1]=0b100010-[2]=0b111000"

* "[0]=" Support numeric format strings

* **\u0D0A** is Newline control character, not string value.

4.36 GetToken()

Retrieve a substring from input string, or the sub-array from the input byte[] array

Syntax 1

```
string GetToken(
    string,
    string,
    string,
    int,
    int
)
```

Parameters

string	Input string
string	Prefix. The leading element of the substring
string	Suffix. The trailing element of the substring
int	The number of the matched substrting to retrieve
>=1	Retrieve the n th matched substring
-1	Retrieve the last matched substring
int	Remove options
0	The 1 st matched not in the start of the input string, and not remove the prefix and the suffix.
(default)	
1	The 1 st matched not in the start of the input string, and remove the prefix and the suffix.
2	The 1 st matched in the start of the input string, and not remove the prefix and the suffix.
3	The 1 st matched in the start of the input string, and remove the prefix and the suffix.

Return

string String formed by part of the input string
 If the [prefix and suffix are empty strings](#), returns the input string
 If [the number of the matched substrings <=0 or larger than the number of the total matched substrings](#), returns empty string
 If the remove option is 2 or 3, the first match retrieved must be at the start of the input string; otherwise, it returns an empty string.

Syntax 2

```
string GetToken(
    string,
    string,
    string,
    int
)
```

Note

Similar to Syntax1 with reserving prefix and suffix.

GetToken(str,str,str,1) => GetToken(str,str,str,1,0)

Syntax 3

```
string GetToken(
    string,
    string,
```

string

)

Note

Similar to Syntax1 with returning the first occurrence, and reserving prefix and suffix.

GetToken(str,str,str) => GetToken(str,str,str,1,0)

string n = "\$abcd\$1234\$ABCD\$"

value = **GetToken**(n, "", "", 0) // value = "\$abcd\$1234\$ABCD\$"

value = **GetToken**(n, "\$", "\$") // value = "\$abcd\$"

value = **GetToken**(n, "\$", "\$", 0) // value = ""

value = **GetToken**(n, "\$", "\$", 1) // value = "\$abcd\$"

value = **GetToken**(n, "\$", "\$", 2) // value = "\$ABCD\$"

value = **GetToken**(n, "\$", "\$", 3) // value = ""

value = **GetToken**(n, "\$", "\$", -1, 1) // value = "ABCD"

value = **GetToken**(n, "\$", "\$", 1, 1) // value = "abcd"

value = **GetToken**(n, "\$", "\$", 2, 1) // value = "ABCD"

value = **GetToken**(n, "\$", "", 1) // value = "\$abcd"

value = **GetToken**(n, "\$", "", 2) // value = "\$1234"

value = **GetToken**(n, "\$", "", 3) // value = "\$ABCD"

value = **GetToken**(n, "\$", "", 4) // value = "\$"

value = **GetToken**(n, "", "\$", 1) // value = "\$"

value = **GetToken**(n, "", "\$", 2) // value = "abcd\$"

value = **GetToken**(n, "", "\$", 3) // value = "1234\$"

value = **GetToken**(n, "", "\$", 4) // value = "ABCD\$"

string n = "\$abcd\$1234\$ABCD\$" + Ctrl("\r\n") + "56\r\n78\$"

value = **GetToken**(n, "\$", Ctrl("\r\n"), 1) // value = "\$abcd\$1234\$ABCD\$\u000A"

value = **GetToken**(n, "\$", newline, 2) // value = ""

value = **GetToken**(n, "\$", NewLine, 1, 1) // value = "abcd\$1234\$ABCD\$"

// Remove prefix and suffix

value = **GetToken**(n, Ctrl("\r\n"), "\$", 1) // value = "\u000A56\r\n78\$"

value = **GetToken**(n, newline, "\$", 2) // value = ""

value = **GetToken**(n, NewLine, "\$", 1, 1) // value = "56\r\n78"

* \u000A is Newline control character, not string value.

string n = "#abcd\$1234#ABCD\$5678#"

value = **GetToken**(n, "", "", 0) // value = "#abcd\$1234#ABCD\$5678#"

value = **GetToken**(n, "\$", "\$") // value = "\$1234#ABCD\$"

value = **GetToken**(n, "\$", "\$", 0) // value = ""

value = **GetToken**(n, "\$", "\$", 1) // value = "\$1234#ABCD\$"

value = **GetToken**(n, "\$", "\$", 2) // value = ""

value = **GetToken**(n, "\$", "\$", 3) // value = ""

value = **GetToken**(n, "\$", "\$", -1, 0) // value = "\$1234#ABCD\$"

value = **GetToken**(n, "\$", "\$", -1, 1) // value = "1234#ABCD"

value = **GetToken**(n, "\$", "\$", 1, 1) // value = "1234#ABCD"

value = **GetToken**(n, "\$", "\$", 2, 1) // value = ""

value = **GetToken**(n, "\$", "", 1) // value = "\$1234#ABCD"

value = **GetToken**(n, "\$", "", 2) // value = "\$5678#"

value = **GetToken**(n, "\$", "", 3) // value = ""

value = **GetToken**(n, "\$", "", 4) // value = ""

```

value = GetToken(n, "", "$", 1) // value = "#abcd$"
value = GetToken(n, "", "$", 2) // value = "1234#ABCD$"
value = GetToken(n, "", "$", 3) // value = ""
value = GetToken(n, "", "$", 4) // value = ""
value = GetToken(n, "$", "$", 1, 2) // value = ""
// The string matched $ not in the start of the input string.
value = GetToken(n, "$", "$", -1, 2) // value = ""
// The string matched $ not in the start of the input string.

value = GetToken(n, "#", "", 1, 3) // value = "abcd$1234"
value = GetToken(n, "#", "", 1, 2) // value = "#abcd$1234"
value = GetToken(n, "#", "", 2, 2) // value = "#ABCD$5678"
value = GetToken(n, "#", "", 3, 2) // value = "#"
value = GetToken(n, "#", "", 4, 2) // value = ""
value = GetToken(n, "#", "", -1, 2) // value = "#"
value = GetToken(n, "#", "", -1, 3) // value = ""
value = GetToken(n, "#", "$", 1, 2) // value = "#abcd$"
value = GetToken(n, "#", "$", 1, 3) // value = "abcd"

```

Syntax 4

```

string GetToken(
    string,
    byte[],
    byte[],
    int,
    int
)

```

Parameters

string	Input string
byte[]	Prefix. The leading element of the substring, byte[] type
byte[]	Suffix. The trailing element of the substring, byte[] type
int	The number of the matched substring to retrieve
≥1	Retrieve the n th matched substring
-1	Retrieve the last matched substring
int	Remove options
0	The 1 st matched not in the start of the input string, and not remove the prefix and the suffix.
(default)	
1	The 1 st matched not in the start of the input string, and remove the prefix and the suffix.
2	The 1 st matched in the start of the input string, and not remove the prefix and the suffix.
3	The 1 st matched in the start of the input string, and remove the prefix and the suffix.

Return

string String formed by part of the input string

If the **prefix and suffix are empty strings**, returns the input string

If the **number of the matched substrings ≤ 0 or larger than the number of the total matched substrings**, returns empty string

If the remove option is 2 or 3, the first match retrieved must be at the start of the input string; otherwise, it returns an empty string.

Syntax 5

```
string GetToken(
    string,
    byte[],
    byte[],
    int
)
```

Note

Similar to Syntax4 with reserving prefix and suffix

GetToken(str,byte[],byte[],1) => GetToken(str,byte[],byte[],1,0)

Syntax 6

```
string GetToken(
    string,
    byte[],
    byte[]
)
```

Note

Similar to Syntax 4 with the first occurrence and reserving prefix and suffix

GetToken(str,byte[],byte[]) => GetToken(str,byte[],byte[],1,0)

string n = "\$abcd\$1234\$ABCD\$"

byte[] bb0 = {}, bb1 = {0x24} // 0x24 is \$

value = **GetToken**(n, bb0, bb0, 0) // value = "\$abcd\$1234\$ABCD\$"

value = **GetToken**(n, bb1, bb1) // value = "\$abcd\$"

value = **GetToken**(n, bb1, bb1, 0) // value = ""

value = **GetToken**(n, bb1, bb1, 1) // value = "\$abcd\$"

value = **GetToken**(n, bb1, bb1, 2) // value = "\$ABCD\$"

value = **GetToken**(n, bb1, bb1, 3) // value = ""

value = **GetToken**(n, bb1, bb1, 1, 1) // value = "abcd"

value = **GetToken**(n, bb1, bb1, 2, 1) // value = "ABCD"

value = **GetToken**(n, bb1, bb0, 1) // value = "\$abcd"

value = **GetToken**(n, bb1, bb0, 2) // value = "\$1234"

value = **GetToken**(n, bb1, bb0, 3) // value = "\$ABCD"

value = **GetToken**(n, bb1, bb0, 4) // value = "\$"

value = **GetToken**(n, bb0, bb1, 1) // value = "\$"

value = **GetToken**(n, bb0, bb1, 2) // value = "abcd\$"

value = **GetToken**(n, bb0, bb1, 3) // value = "1234\$"

value = **GetToken**(n, bb0, bb1, 4) // value = "ABCD\$"

string n = "\$abcd\$1234\$ABCD\$" + Ctrl("\r\n") + "56\r\n78\$"

byte[] bb0 = {0x0D,0x0A}, bb1 = {0x24} // 0x24 is \$ // 0x0D,0x0A is \u000A

value = **GetToken**(n, bb1, bb0, 1) // value = "\$abcd\$1234\$ABCD\$\u000A"

value = **GetToken**(n, bb1, bb0, 2) // value = ""

value = **GetToken**(n, bb1, bb0, 1, 1) // value = "abcd\$1234\$ABCD\$"

// Removing the prefix and the suffix

value = **GetToken**(n, bb0, bb1, 1) // value = "\u000A56\r\n78\$"

value = **GetToken**(n, bb0, bb1, 2) // value = ""

value = **GetToken**(n, bb0, bb1, 1, 1) // value = "56\r\n78"

* \u000A is the Newline control character, not the string content.

Syntax 7

```
byte[] GetToken(
    byte[],
    string,
    string,
    int,
    int
)
```

Parameters

byte[]	The input byte[]
string	Prefix. The leading element of the output byte[], byte[] type
string	Suffix. The trailing element of the output byte[], byte[] type
int	The number of the matched substring to retrieve
>=1	Retrieve the n th matched substring
-1	Retrieve the last matched substring
int	Remove options
0	The 1 st matched not in the start of the input string, and not remove the prefix and the suffix.
(default)	
1	The 1 st matched not in the start of the input string, and remove the prefix and the suffix.
2	The 1 st matched in the start of the input string, and not remove the prefix and the suffix.
3	The 1 st matched in the start of the input string, and remove the prefix and the suffix.

Return

byte[] The byte[] formed from part of the input byte[]
 If the **prefix and suffix are empty**, returns the input array
 If **the number of the matched substrings <=0 or larger than the number of the total matched substrings**, returns empty array
 If the remove option is 2 or 3, the first match retrieved must be at the start of the input string; otherwise, it returns an empty string.

Syntax 8

```
byte[] GetToken(
    byte[],
    string,
    string,
    int
)
```

Note

Similar to Syntax7 with reserving prefix and suffix

GetToken(byte[],str,str,1) => GetToken(byte[],str,str,1,0)

Syntax 9

```
byte[] GetToken(
    byte[],
    string,
    string
)
```

Note

Similar to Syntax7 with returning the first occurrence, and reserving prefix and suffix.

GetToken(byte[],str,str) => **GetToken**(byte[],str,str,1,0)

string s="\$abcd\$1234\$ABCD\$"

byte[] n = GetBytes(s)

value = **GetToken**(n, "", "", 0)

// value = {0x24,0x61,0x62,0x63,0x64,0x24,0x31,0x32,0x33,0x34,0x24,0x41,0x42,0x43,0x44,0x24}

value = **GetToken**(n, "\$", "\$") // value = {0x24,0x61,0x62,0x63,0x64,0x24}

value = **GetToken**(n, "\$", "\$", 0) // value = {}

value = **GetToken**(n, "\$", "\$", 1) // value = {0x24,0x61,0x62,0x63,0x64,0x24}

value = **GetToken**(n, "\$", "\$", 2) // value = {0x24,0x41,0x42,0x43,0x44,0x24}

value = **GetToken**(n, "\$", "\$", 1, 1) // value = {0x61,0x62,0x63,0x64}

value = **GetToken**(n, "\$", "\$", 2, 1) // value = {0x41,0x42,0x43,0x44}

value = **GetToken**(n, "\$", "", 1) // value = {0x24,0x61,0x62,0x63,0x64}

value = **GetToken**(n, "\$", "", 2) // value = {0x24,0x31,0x32,0x33,0x34}

value = **GetToken**(n, "\$", "", 3) // value = {0x24,0x41,0x42,0x43,0x44}

value = **GetToken**(n, "\$", "", 4) // value = {0x24}

value = **GetToken**(n, "", "\$", 1) // value = {0x24}

value = **GetToken**(n, "", "\$", 2) // value = {0x61,0x62,0x63,0x64,0x24}

value = **GetToken**(n, "", "\$", 3) // value = {0x31,0x32,0x33,0x34,0x24}

value = **GetToken**(n, "", "\$", 4) // value = {0x41,0x42,0x43,0x44,0x24}

string s="\$abcd\$1234\$ABCD\$" + Ctrl("\r\n") + "56\r\n78\$"

byte[] n = GetBytes(s)

value = **GetToken**(n, "\$", Ctrl("\r\n"), 1)

// value = {0x24,0x61,0x62,0x63,0x64,0x24,0x31,0x32,0x33,0x34,0x24,0x41,0x42,0x43,0x44,0x24,0x0D,0x0A}

value = **GetToken**(n, "\$", Ctrl("\r\n"), 1, 1)

// value = {0x61,0x62,0x63,0x64,0x24,0x31,0x32,0x33,0x34,0x24,0x41,0x42,0x43,0x44,0x24}

// Removing prefix and suffix

value = **GetToken**(n, Ctrl("\r\n"), "\$", 1)

// value = {0x0D,0x0A,0x35,0x36,0x5C,0x72,0x5C,0x6E,0x37,0x38,0x24}

value = **GetToken**(n, Ctrl("\r\n"), "\$", 1, 1)

// value = {0x35,0x36,0x5C,0x72,0x5C,0x6E,0x37,0x38}

Syntax 10

byte[] **GetToken**(

byte[],

byte[],

byte[],

int,

int

)

Parameters

byte[] The input byte[] array

byte[] Prefix. The leading element of the output byte[]

byte[] Suffix. The trailing element of the output byte[]

int The number of the matched substring to retrieve

>=1 Retrieve the nth matched substring

-1 Retrieve the last matched substring

int Remove options

0 (default)	The 1 st matched not in the start of the input string, and not remove the prefix and the suffix.
1	The 1 st matched not in the start of the input string, and remove the prefix and the suffix.
2	The 1 st matched in the start of the input string, and not remove the prefix and the suffix.
3	The 1 st matched in the start of the input string, and remove the prefix and the suffix.

Return

byte[] The byte[] formed from part of the input byte[]
 If the **prefix and suffix are empty**, returns the input array
 If the **number of the matched substrings <=0 or larger than the number of total matched substrings**, returns empty array
 If the remove option is 2 or 3, the first match retrieved must be at the start of the input string; otherwise, it returns an empty string.

Syntax 11

```
byte[] GetToken(
    byte[],
    byte[],
    byte[],
    int
)
```

Note

Similar to Syntax10 with reserving the prefix and suffix

GetToken(byte[],byte[],byte[],1) => GetToken(byte[],byte[],byte[],1,0)

Syntax 12

```
byte[] GetToken(
    byte[],
    byte[],
    byte[]
)
```

Note

Similar to Syntax10 with returning the first occurrence, and reserving prefix and suffix.

GetToken(byte[],byte[],byte[]) => GetToken(byte[],byte[],byte[],1,0)

string s = "\$abcd\$1234\$ABCD\$"

byte[] n = GetBytes(s)

byte[] bb0 = {}, bb1 = {0x24} // 0x24 is \$

value = **GetToken**(n, bb0, bb0, 0)

// value = {0x24,0x61,0x62,0x63,0x64,0x24,0x31,0x32,0x33,0x34,0x24,0x41,0x42,0x43,0x44,0x24}

value = **GetToken**(n, bb1, bb1) // value = {0x24,0x61,0x62,0x63,0x64,0x24}

value = **GetToken**(n, bb1, bb1, 0) // value = {}

value = **GetToken**(n, bb1, bb1, 1) // value = {0x24,0x61,0x62,0x63,0x64,0x24}

value = **GetToken**(n, bb1, bb1, 2) // value = {0x24,0x41,0x42,0x43,0x44,0x24}

value = **GetToken**(n, bb1, bb1, 1, 1) // value = {0x61,0x62,0x63,0x64}

value = **GetToken**(n, bb1, bb1, 2, 1) // value = {0x41,0x42,0x43,0x44}

value = **GetToken**(n, bb1, bb0, 1) // value = {0x24,0x61,0x62,0x63,0x64}

value = **GetToken**(n, bb1, bb0, 2) // value = {0x24,0x31,0x32,0x33,0x34}

value = **GetToken**(n, bb1, bb0, 3) // value = {0x24,0x41,0x42,0x43,0x44}

value = **GetToken**(n, bb0, bb1, 1) // value = {0x24}

```

value = GetToken(n, bb0, bb1, 2) // value = {0x61,0x62,0x63,0x64,0x24}
value = GetToken(n, bb0, bb1, 3) // value = {0x31,0x32,0x33,0x34,0x24}
string s = "$abcd$1234$ABCD$" + Ctrl("\r\n") + "56\r\n78$"
byte[] n = GetBytes(s)
byte[] bb0 = {0x0D,0x0A}, bb1 = {0x24}
value = GetToken(n, bb1, bb0, 1)
// value = {0x24,0x61,0x62,0x63,0x64,0x24,0x31,0x32,0x33,0x34,0x24,0x41,0x42,0x43,0x44,0x24,0x0D,0x0A}
value = GetToken(n, bb1, bb0, 1, 1)
// value = {0x61,0x62,0x63,0x64,0x24,0x31,0x32,0x33,0x34,0x24,0x41,0x42,0x43,0x44,0x24}
// Remove prefix and suffix
value = GetToken(n, bb0, bb1, 1)
// value = {0x0D,0x0A,0x35,0x36,0x5C,0x72,0x5C,0x6E,0x37,0x38,0x24}
value = GetToken(n, bb0, bb1, 1, 1)
// value = {0x35,0x36,0x5C,0x72,0x5C,0x6E,0x37,0x38}

```


4.37 GetAllTokens()

Retrieve all the substrings from input string, which meets the given condition

Syntax 1

```
string[] GetAllTokens(
    string,
    string,
    string,
    int
)
```

Parameters

string	Input string
string	Prefix. The leading element of the substring
string	Suffix. The trailing element of the substring
int	Remove options
0	The 1 st matched not in the start of the input string, and not remove the prefix and the suffix.
(default)	
1	The 1 st matched not in the start of the input string, and remove the prefix and the suffix.
2	The 1 st matched in the start of the input string, and not remove the prefix and the suffix.
3	The 1 st matched in the start of the input string, and remove the prefix and the suffix.

Return

string[] String array formed from retrieving all the substrings from input string
 If the **prefix and suffix are empty**, returns the input array
 If the remove option is 2 or 3, the first match retrieved must be at the start of the input string; otherwise, it returns an empty string.

Syntax 2

```
string[] GetAllTokens(
    string,
    string,
    string
)
```

Note

Similar to Syntax1 with reserving prefix and suffix

GetAllTokens(str,str,str) => GetAllTokens(str,str,str,0)

```
string n = "$abcd$1234$ABCD$"
```

```
value = GetAllTokens(n, "", "") // value = {"$abcd$1234$ABCD$"}
value = GetAllTokens(n, "$", "$") // value = {"$abcd$", "$ABCD$"}
value = GetAllTokens(n, "$", "$", 1) // value = {"abcd", "ABCD"}
value = GetAllTokens(n, "$", "") // value = {"$abcd", "$1234", "$ABCD", "$"}
value = GetAllTokens(n, "", "$", 1) // value = {"", "abcd", "1234", "ABCD"}
```

```
string n = "#abcd$1234#ABCD$5678#"
```

```
value = GetAllTokens(n, "", "", 0) // value = {"#abcd$1234#ABCD$5678#"}
value = GetAllTokens(n, "$", "", 0) // value = {"$1234#ABCD", "$5678#"}
value = GetAllTokens(n, "$", "", 1) // value = {"1234#ABCD", "5678#"}
value = GetAllTokens(n, "$", "", 2) // value = {}
```

```

// $ is not in the start of the input string. Returns an empty array.
value = GetAllTokens(n, "$", "", 3) // value = {}
// $ is not in the start of the input string. Returns an empty array.
value = GetAllTokens(n, "$", "$", 0) // value = {"$1234#ABCD$"}
value = GetAllTokens(n, "$", "$", 1) // value = {"1234#ABCD"}
value = GetAllTokens(n, "$", "$", 2) // value = {}
// $ is not in the start of the input string. Returns an empty array.
value = GetAllTokens(n, "$", "$", 3) // value = {}
// $ is not in the start of the input string. Returns an empty array.
value = GetAllTokens(n, "#", "", 0) // value = {"#abcd$1234", "#ABCD$5678", "#"}
value = GetAllTokens(n, "#", "", 1) // value = {"abcd$1234", "ABCD$5678", ""}
value = GetAllTokens(n, "#", "", 2) // value = {"#abcd$1234", "#ABCD$5678", "#"}
value = GetAllTokens(n, "#", "", 3) // value = {"abcd$1234", "ABCD$5678", ""}
value = GetAllTokens(n, "#", "$", 0) // value = {"#abcd$", "#ABCD$"}
value = GetAllTokens(n, "#", "$", 1) // value = {"abcd", "ABCD"}
value = GetAllTokens(n, "#", "$", 2) // value = {"#abcd$", "#ABCD$"}
value = GetAllTokens(n, "#", "$", 3) // value = {"abcd", "ABCD"}
value = GetAllTokens(n, "", "$", 0) // value = {"#abcd$", "1234#ABCD$"}
value = GetAllTokens(n, "", "$", 1) // value = {"#abcd", "1234#ABCD"}
value = GetAllTokens(n, "", "$", 2) // value = {"#abcd$", "1234#ABCD$"}
value = GetAllTokens(n, "", "$", 3) // value = {"#abcd", "1234#ABCD"}
value = GetAllTokens(n, "", "#", 0) // value = {"#", "abcd$1234#", "ABCD$5678#"}
value = GetAllTokens(n, "", "#", 1) // value = {"", "abcd$1234", "ABCD$5678"}
value = GetAllTokens(n, "", "#", 2) // value = {"#", "abcd$1234#", "ABCD$5678#"}
value = GetAllTokens(n, "", "#", 3) // value = {"", "abcd$1234", "ABCD$5678"}

```

4.38 GetNow()

Get the current system time

Syntax 1

```
string GetNow(
    string
)
```

Parameters

string	The date and time format strings defining the text representation of a date and time value. The definition of each specifier is listed below. The strings not included will remains the same.
d	The day of the month, from 1 through 31.
dd	The day of the month, from 01 through 31.
ddd	The abbreviated name of the day of the week.
dddd	The full name of the day of the week.
f	The tenths of a second in a date and time value.
ff	The hundredths of a second in a date and time value.
fff	The milliseconds in a date and time value.
ffff	The ten thousandths of a second in a date and time value.
h	The hour, using a 12-hour clock from 1 to 12.
hh	The hour, using a 12-hour clock from 01 to 12.
H	The hour, using a 24-hour clock from 0 to 23.
HH	The hour, using a 24-hour clock from 00 to 23.
m	The minute, from 0 through 59.
mm	The minute, from 00 through 59.
M	The month, from 1 through 12.
MM	The month, from 01 through 12.
MMM	The abbreviated name of the month.
MMMM	The full name of the month.
s	The second, from 0 through 59.
ss	The second, from 00 through 59.
t	The first character of the AM/PM designator.
tt	The AM/PM designator.
y	The year, from 0 to 99.
yy	The year, from 00 to 99.
yyyy	The year as a four-digit number.
/	The date separator.

Return

string Current date and time. If there is errors in format setting, the default format will be applied as MM/dd/yyyy HH:mm:ss.

Note

```
value = GetNow("MM/dd/yyyy HH:mm:ss")           // value = 08/15/2017 13:40:30
value = GetNow("yyyy/MM/dd HH:mm:ss.ffff")       // value = 2017/08/15 13:40:30.1337
value = GetNow("yyyy-MM-dd hh:mm:ss tt")         // value = 2017-08-15 01:40:30 PM
```

Syntax 2

```
string GetNow(
```

)

Parameters

void No format defined. Default format "MM/dd/yyyy HH:mm:ss" will be applied

Return

string Current date and time.

Note

value = **GetNow()** // value = 08/15/2017 13:40:30

4.39 GetNowStamp()

Get the total run time or difference in total run time

Syntax 1

```
int GetNowStamp(
)
```

Parameters

void No parameter

Return

int The total run time of the current project in ms. The upper limit is 2147483647 ms
< 0 Over flow, invalid total run time

Note

```
value = GetNowStamp()           // value = 2147483647
... others ...
value = GetNowStamp()           // value = -1    // Over flow
```

Syntax 2

```
double GetNowStamp(
    bool
)
```

Parameters

bool Use double format to record project's total run time or not?
true Use double type, the upper limit is 9223372036854775807 ms
false Use int32 type, the upper limit is 2147483647 ms

Return

double The total run time of the current project
< 0 Over flow. Invalid total run time.

Note

```
value = GetNowStamp(false)       // value = 2147483647
... others ...
value = GetNowStamp(false)       // value = -1    // Over flow
value = GetNowStamp(true)        // value = 3147483647
```

Syntax 3

```
int GetNowStamp(
    int
)
```

Parameters

int Previous recorded run time in ms

Return

int The difference between the current run time and the input run time in ms.
 Run time difference = current run time – input run time
< 0 Invalid run time difference, caused by input run time larger than current run time, or over flow.

Note

```
value = GetNowStamp()           // value = 2147483546
```

```

... others ... (After 100ms)
diff = GetNowStamp(value)    // diff = 100
... others ... (After 200ms)
diff = GetNowStamp(value)    // diff = -1    // Value is over 2147483647

```

Syntax 4

```

double GetNowStamp(
    double
)

```

Parameters

double Previous recorded run time in ms

Return

double The difference between the current run time and the input run time in ms.
 Run time difference = current run time – input run time
< 0 Invalid run time difference, caused by input run time larger than current run time, or over flow.

Note

```

value = GetNowStamp()    // value = 2147483546
... others ... (After 100ms)
diff = GetNowStamp(value)    // diff = 100
... others ... (After 200ms)
diff = GetNowStamp(value)    // diff = 200

```

Syntax 5

```

bool GetNowStamp(
    int,
    int
)

```

Parameters

int Previous recorded run time in ms
int The expected run time difference

Return

bool The time difference between current run time and input run time is larger than the expected run time difference or not.
true (Current run time – input run time) >= expected run time
 Or Time difference smaller than zero or over flow
false (Current run time – input run time) < expected run time

Note

```

value = GetNowStamp()    // value = 41730494
... others ... (After 60ms)
flag = GetNowStamp(value, 100)    // diff = 60    // flag = false
... others ... (After 60ms)
flag = GetNowStamp(value, 100)    // diff = 120    // flag = true

```

Syntax 6

```

bool GetNowStamp(
    double,
    double
)

```

)

Parameters

double Previous recorded run time in ms
double The expected run time difference

Return

bool The time difference between current run time and input run time is larger than the expected run time difference or not.
true (Current run time – input run time) >= expected run time
 Or Time difference smaller than zero or over flow
false (Current run time – input run time) < expected run time

Note

```
value = GetNowStamp()           // value = 41730494
... others ... (After 60ms)
flag = GetNowStamp(value, 100)  // diff = 60      // flag = false
... others ... (After 60ms)
flag = GetNowStamp(value, 100)  // diff = 120     // flag = true
```

4.40 GetVarValue()

Retrieve the value of the variable value. Users can use the string to combine the variable names, and then retrieve the combined value of the variables.

Syntax 1

```
? GetVarValue(
    string
)
```

Parameters

string The name of the variable

Return

? Return the value of the variable. The return type goes by the definition of the variable.

Return an **error** if the variable is not existed.

Note

```
string var_s1 = "Hello World"
string var_h = " var_s1"
string var_t = " var_s"

string var_re = var_t           // var_re = " var_s"
var_re = var_t + "1"           // var_re = " var_s" + "1" = " var_s1"
var_re = GetVarValue("var_h") // var_re = " var_s1"
var_re = GetVarValue(var_h)    // var_re = "Hello World" // var_h = " var_s1"
                                // Retrieve the value of var_s1
var_re = GetVarValue(var_t + "1") // var_re = "Hello World" // var_b + "1" = " var_s1"
                                // Retrieve the value of var_s1
var_re = GetVarValue(var_t)    // Error // var_t = " var_s"
                                //Retrieve the value of var_s, but the variable is not existed.

string s
int[] var_array0 = {10, 11, 12, 13, 14}
int[] var_array1 = {20, 21, 22, 23, 24}
for (int i = 0; i < 2; i++)
{
    int[] v = GetVarValue("var_array" + i)
    for (int j = 0; j < Length(v) ; j++) s += v[j] + ", "
    s += newline
}
Display(s) // 10, 11, 12, 13, 14, \u0D0A20, 21, 22, 23, 24, \u0D0A
* \u0D0A is a written line break, not a string value.
```


4.41 SetVarValue()

For variable value settings, users can use a string to combine the variable names to set the variable value of the combined variable names.

Syntax 1

```
bool SetVarValue (
    string,
    ?
)
```

Parameters

string The name of the variable
? The variable value

Return

bool The variable setting value. Return **True** if successful; **False** if unsuccessful.
 Returns an **error** if the variable does not exist or cannot convert legitimately.

Note

```
string var_s1 = "Hello World"
SetVarValue("var_s1", "Hi World") // var_s1 = "Hi World"
int var_k = 100
SetVarValue("var_k", 200)          // var_k = 200
SetVarValue("var_k", 2.3)          // var_k = 2      // Since 2.3 converting to the int type
SetVarValue("var_k", "2.3")        // var_k = 2      // Since "2.3" converting to the int type

SetVarValue("var_k", true)         // Error
SetVarValue("var_k", "XYZ")        // Error

int[] var_array0 = {10, 11, 12, 13, 14}
int[] var_array1 = {20, 21, 22, 23, 24}
for (int i = 0; i < 2; i++)
{
    int[] v = GetVarValue("var_array" + i) // Retrieve var_array0 or var_array1
    for (int j = 0; j < Length(v); j++)
        v[j] += 100 // +100 for each element in int[] v
    SetVarValue("var_array" + i, v) // Set the value of int[] v to var_array0 or var_array1.
}
Display(var_array0) // {110,111,112,113,114}
Display(var_array1) // {120,121,122,123,124}
```

4.42 Length()

Acquire the number of byte of input data, length of string or length of array (number of elements in array)

Syntax 1

```
int Length(
    ?
)
```

Parameters

? The input data. The available data types are integer, floating-point, boolean, string, or array.

Return

int Length of data
For input as integer, floating-point number, and boolean, returns the number of byte.
For input as string, returns the length of string.
For input as array, returns the number of element in array

Note

```
? byte n = 100
value = Length(n)    // value = 1
value = Length(100)  // value = 1

? int n = 400
value = Length(n)    // value = 4
value = Length(400)  // value = 4

? float n = 1.234
value = Length(n)    // value = 4
value = Length(1.234) // value = 4

? double n = 1.234
value = Length(n)    // value = 8
value = Length(1.234) // value = 4
// float // Numbers would be stored as the smaller data type first.

? bool n = true
value = Length(n)    // value = 1
value = Length(false) // value = 1

? string n = "A""BC"
value = Length(n)    // value = 4
// The string is A"BC. Two double quotation marks represent " in string

value = Length("")    // value = 0
value = Length("123") // value = 3
value = Length(empty) // value = 0

? byte[] n = {100, 200, 30}
value = Length(n)    // value = 3

? int[] n = {}
value = Length(n)    // value = 0
n = {400, 500, 600}
value = Length(n)    // value = 3

? float[] n = {1.234}
```

```
value = Length(n)           // value = 1
? double[] n = {1.234, 200, -100, +300}
value = Length(n)           // value = 4
? bool[] n = {true, false, true, true, true, true, false}
value = Length(n)           // value = 7
? string[] n = {"A""BC", "123", "456", "ABC"}
value = Length(n)           // value = 4
```

4.43 Ctrl()

Change the integer or string to control characters

Syntax 1

```
string Ctrl(
    int
)
```

Parameters

int The input integer, which follows the Big Endian format. 4 characters could be transformed at most. 0x00 will not be transformed.

Return

string The string formed by input integer (contains the control character)

Note

```
b = 0x0D0A
value = Ctrl(b)           // value = \r\n
value = Ctrl(0x0D0A)      // value = \r\n
value = Ctrl(0x0D000A09)  // value = \r\n\t // 0x00 will not be transformed
value = Ctrl(0x0D300A09)  // value = \r0\n\t // 0x30 is transformed to 0
value = Ctrl(0x00)        // value = ""
                           // empty string does not equal to NULL. For NULL, the code is Ctrl("\0")
```

Syntax 2

```
string Ctrl(
    string
)
```

Parameters

string Input string. The following rules will be applied. For string not on the list, it will remain the same.

\0	0x00 null
\a	0x07 bell
\b	0x08 backspace
\t	0x09 horizontal tab
\r	0x0D carriage return
\v	0x0B vertical tab
\f	0x0C form feed
\n	0x0A line feed

Return

string The string formed by input integer (contains the control character)

Note

```
b = "\r\n"
value = Ctrl(b)           // value = \r\n
value = Ctrl("\r\n")      // value = \r\n
value = Ctrl("\r\n\t")    // value = \r\n\t
value = Ctrl("\r0\n\t")   // value = \r0\n\t
value = Ctrl("\0")        // value = \0 // NULL
```

Syntax 3

```
string Ctrl(
    byte[]
)
```

Parameters

byte[] The input byte array, the transfer will start from index [0] to the end of the array. (0x00 will be transferred also)

Return

string The string formed by input integer (contains the control character)

Note

```
byte[] bb1 = {0xFF,0x55,0x31,0x32,0x33,0x00,0x35,0x36,0x0D,0x0A}
value = Ctrl(bb1)           // value = U123 56\r\n
byte[] bb2 = {}
value = Ctrl(bb2)           // value = ""
```

4.44 XOR8()

Utilize XOR 8 bits algorithm to compute the checksum

Syntax 1

```
byte XOR8(
    byte[],
    int,
    int
)
```

Parameters

<code>byte[]</code>	The input byte array
<code>int</code>	The starting index
	<code>0..(array size-1)</code> Valid
	<code><0</code> Invalid. Returns the initial value 0
	<code>>=array size</code> Invalid. Returns the initial value 0
<code>int</code>	The number of elements to be computed.
	If the number of elements <code><0</code> , the calculation ends at the last element of the array
	If the sum of starting index and number of element exceeds the array size, the calculation ends at the last element of the array.

Return

`byte` Checksum.

Note

```
byte[] bb1 = {0x10, 0x20, 0x50, 0xF0, 0xFF, 0xFF, 0xFF}
value = XOR8(bb1,0,Length(bb1)) // value = 0x6F
value = XOR8(bb1,0,-1)          // value = 0x6F
value = XOR8(bb1,1,-1)          // value = 0x7F
value = XOR8(bb1,-1,-1)         // value = 0
```

Syntax 2

```
byte XOR8(
    byte[],
    int
)
```

Note

Similar to Syntax1 with computing to the last element of the array
`XOR8(byte[], int) => XOR8(byte[], int, Length(byte[]))`

Syntax 3

```
byte XOR8(
    byte[]
)
```

Note

Similar to Syntax1 with computing all the elements of the array
`XOR8(byte[]) => XOR8(byte[], 0, Length(byte[]))`

```
byte[] bb1 = {0x10, 0x20, 0x50, 0xF0, 0xFF, 0xFF, 0xFF}
```

```
value = XOR8(bb1,0,Length(bb1))    // value = 0x6F
value = XOR8(bb1,0)                // value = 0x6F
value = XOR8(bb1)                  // value = 0x6F
bb1 = Byte_Concat(bb1, XOR(bb1))    // bb1 = {0x10, 0x20, 0x50, 0xF0, 0xFF, 0xFF, 0xFF,
0x6F}
```

4.45 SUM8()

Utilize SUM 8 bits algorithm to compute the checksum

Syntax 1

```
byte SUM8(
    byte[],
    int,
    int
)
```

Parameters

byte[] The input byte array
int The starting index
0..array size-1 Valid
<0 Invalid. Returns the initial value 0
>=array size Invalid. Returns the initial value 0
int The number of elements to be computed.
 If the number of elements <0, the calculation ends at the last element of the array
 If the sum of starting index and number of element exceeds the array size, the calculation ends at the last element of the array.

Return

byte Checksum.

Note

```
byte[] bb1 = {0x10, 0x20, 0x50, 0xF0, 0xFF, 0xFF, 0xFF}
value = SUM8(bb1,0,Length(bb1)) // value = 0x6D
value = SUM8(bb1,0,-1)          // value = 0x6D
value = SUM8(bb1,1,-1)          // value = 0x5D
value = SUM8(bb1,-1,-1)         // value = 0
```

Syntax 2

```
byte SUM8(
    byte[],
    int
)
```

Note

Similar to Syntax1 with computing to the last element of the array
SUM8(byte[], int) => SUM8(byte[], int, Length(byte[]))

Syntax 3

```
byte SUM8(
    byte[]
)
```

Note

Similar to Syntax1 with computing all the elements of the array
SUM8(byte[]) => SUM8(byte[], 0, Length(byte[]))

```
byte[] bb1 = {0x10, 0x20, 0x50, 0xF0, 0xFF, 0xFF, 0xFF}
value = SUM8(bb1,0,Length(bb1)) // value = 0x6D
value = SUM8(bb1,0)             // value = 0x6D
```



```
value = SUM8(bb1)           // value = 0x6D
bb1 = Byte_Concat(bb1, SUM8(bb1)) // bb1 = {0x10, 0x20, 0x50, 0xF0, 0xFF, 0xFF, 0xFF,
0x6D}
```

4.46 SUM16()

Utilize SUM 16 bits algorithm to compute the checksum

Syntax 1

```
byte[] SUM16(
    byte[],
    int,
    int
)
```

Parameters

byte[] The input byte array

int The starting index

0..array size-1 Valid

<0 Invalid. Returns the initial value 0

>=array size Invalid. Returns the initial value 0

int The number of elements to be computed.

If the number of elements <0, the calculation ends at the last element of the array

If the sum of starting index and number of element exceeds the array size, the calculation ends at the last element of the array.

Return

byte[] Checksum. The length is 16bits 2 bytes (The Checksum follows Big Endian)

Note

```
byte[] bb1 = {0x10, 0x20, 0x50, 0xF0, 0xFF, 0xFF, 0xFF}
value = SUM16(bb1,0,Length(bb1)) // value = {0x04, 0x6D}
value = SUM16(bb1,0,-1)           // value = {0x04, 0x6D}
value = SUM16(bb1,1,-1)           // value = {0x04, 0x5D}
value = SUM16(bb1,-1,-1)          // value = {0x00, 0x00}
```

Syntax 2

```
byte[] SUM16(
    byte[],
    int
)
```

Note

Similar to Syntax1 with computing to the last element of the array

SUM16(byte[], int) => SUM16(byte[], int, Length(byte[]))

Syntax 3

```
byte[] SUM16(
    byte[]
)
```

Note

Similar to Syntax1 with computing all the elements of the array

SUM16(byte[]) => SUM16(byte[], 0, Length(byte[]))

```
byte[] bb1 = {0x10, 0x20, 0x50, 0xF0, 0xFF, 0xFF, 0xFF}
value = SUM16(bb1,0,Length(bb1)) // value = {0x04, 0x6D}
```

```
value = SUM16(bb1,0)           // value = {0x04, 0x6D}  
value = SUM16(bb1)             // value = {0x04, 0x6D}  
bb1 = Byte_Concat(bb1, SUM16(bb1)) // bb1 = {0x10, 0x20, 0x50, 0xF0, 0xFF, 0xFF, 0xFF,  
                                0x04, 0x6D}
```

4.47 SUM32()

Utilize SUM 32 bits algorithm to compute the checksum

Syntax 1

```
byte[] SUM32(
    byte[],
    int,
    int
)
```

Parameters

byte[] The input byte array

int The starting index

0..array size-1 Valid

<0 Invalid. Returns the initial value 0

>=array size Invalid. Returns the initial value 0

int The number of elements to be computed.

If the number of elements <0, the calculation ends at the last element of the array

If the sum of starting index and number of element exceeds the array size, the calculation ends at the last element of the array.

Return

byte[] Checksum. The length is 32bits 4 bytes (The Checksum follows Big Endian)

Note

```
byte[] bb1 = {0x10, 0x20, 0x50, 0xF0, 0xFF, 0xFF, 0xFF}
value = SUM32(bb1,0,Length(bb1)) // value = {0x00, 0x00, 0x04, 0x6D}
value = SUM32(bb1,0,-1)           // value = {0x00, 0x00, 0x04, 0x6D}
value = SUM32(bb1,1,-1)           // value = {0x00, 0x00, 0x04, 0x5D}
value = SUM32(bb1,-1,-1)          // value = {0x00, 0x00, 0x00, 0x00}
```

Syntax 2

```
byte[] SUM32(
    byte[],
    int
)
```

Note

Similar to Syntax1 with computing to the last element of the array

SUM32(byte[], int) => SUM32(byte[], int, Length(byte[]))

Syntax 3

```
byte[] SUM32(
    byte[]
)
```

Note

Similar to Syntax1 with computing all the elements of the array

SUM32(byte[]) => SUM32(byte[], 0, Length(byte[]))

```
byte[] bb1 = {0x10, 0x20, 0x50, 0xF0, 0xFF, 0xFF, 0xFF}
value = SUM32(bb1,0,Length(bb1)) // value = {0x00, 0x00, 0x04, 0x6D}
```

```
value = SUM32(bb1,0)           // value = {0x00, 0x00, 0x04, 0x6D}  
value = SUM32(bb1)             // value = {0x00, 0x00, 0x04, 0x6D}  
bb1 = Byte_Concat(bb1, SUM32(bb1)) // bb1 = {0x10, 0x20, 0x50, 0xF0, 0xFF, 0xFF, 0xFF, 0x00, 0x00,  
                                0x04, 0x6D}
```

4.48 CRC16()

Utilize CRC 16 bits algorithm to compute the checksum

Syntax 1

```
byte[] CRC16(
    int,
    byte[],
    int,
    int
)
```

Parameters

int	CRC16 algorithm
0	CRC16 // initial value 0x0000 // Polynomial 0xA001
1	CRC16 (Modbus) // initial value 0xFFFF // Polynomial 0xA001
2	CRC16 (Sick) // initial value 0x0000 // Polynomial 0x8005
3	CRC16-CCITT (0x1D0F) // initial value 0x1D0F // Polynomial 0x1021
4	CRC16-CCITT (0xFFFF) // initial value 0xFFFF // Polynomial 0x1021
5	CRC16-CCITT (XModem) // initial value 0x0000 // Polynomial 0x1021
6	CRC16-CCITT (Kermit) // initial value 0x0000 // Polynomial 0x8408
7	CRC16 Schunk Gripper // initial value 0xFFFF // Polynomial 0x1021
byte[]	The input byte array
int	The starting index
0..array size-1	Valid
<0	Invalid. Returns the initial value
>=array size	Invalid. Returns the initial value
int	The number of elements to be computed.
	If the number of elements <0, the calculation ends at the last element of the array
array	If the sum of starting index and number of element exceeds the array size, the calculation ends at the last element of the array.

Return

byte[] Checksum. The length is 16bits 2 bytes (The checksum follows Big Endian)

Note

```
byte[] bb1 = {0x10, 0x20, 0x50, 0xF0, 0xFF, 0xFF, 0xFF}
value = CRC16(0, bb1,0,Length(bb1)) // value = {0x2D, 0xD4}
value = CRC16(0, bb1,0,-1) // value = {0x2D, 0xD4}
value = CRC16(0, bb1,1,-1) // value = {0xEC, 0xC5}
value = CRC16(0, bb1,-1,-1) // value = {0x00, 0x00}
value = CRC16(3, bb1,0,Length(bb1)) // value = {0x42, 0x12}
value = CRC16(4, bb1,0,Length(bb1)) // value = {0xAB, 0xAE}
```

Syntax 2

```
byte[] CRC16(
    int,
    byte[],
    int
)
```

Note

Similar to Syntax1 with computing to the last element of the array
CRC16(int, byte[], int) => **CRC16**(int, byte[], int, Length(byte[]))

Syntax 3

```
byte[] CRC16(
    int,
    byte[]
)
```

Note

Similar to Syntax1 with computing all the elements of the array

CRC16(int, byte[]) => **CRC16**(int, byte[], 0, Length(byte[]))

byte[] bb1 = {0x10, 0x20, 0x50, 0xF0, 0xFF, 0xFF, 0xFF}

value = **CRC16**(0, bb1, 0, Length(bb1)) // value = {0x2D, 0xD4}

value = **CRC16**(0, bb1, 0) // value = {0x2D, 0xD4}

value = **CRC16**(0, bb1) // value = {0x2D, 0xD4}

bb1 = **Byte_Concat**(bb1, **CRC16**(0, bb1)) // bb1 = {0x10, 0x20, 0x50, 0xF0, 0xFF, 0xFF, 0xFF, 0x2D, 0xD4}

Syntax 4

```
byte[] CRC16(
    byte[],
    int,
    int
)
```

Note

Similar to Syntax1 with CRC16 algorithm as 0 CRC16

CRC16(byte[], int, int) => **CRC16**(0, byte[], int, int)

Syntax 5

```
byte[] CRC16(
    byte[],
    int
)
```

Note

Similar to Syntax1 with CRC16 algorithm as 0 CRC16 and computing to the last element of the array

CRC16(byte[], int) => **CRC16**(0, byte[], int, Length(byte[]))

Syntax 6

```
byte[] CRC16(
    byte[]
)
```

Note

Similar to Syntax1 with CRC16 algorithm as 0 CRC16 and computing all the elements of the array

CRC16(byte[]) => **CRC16**(0, byte[], 0, Length(byte[]))

4.49 CRC32()

Utilize CRC 32 bits algorithm to compute the checksum

Syntax 1

```
byte[] CRC32(
    byte[],
    int,
    int
)
```

Parameters

byte[] The input byte array

int The starting index

0..array size-1 Valid

<0 Invalid. Returns the initial value 0

>=array size Invalid. Returns the initial value 0

int The number of elements to be computed.

If the number of elements <0, the calculation ends at the last element of the array

If the sum of starting index and number of element exceeds the array size, the calculation ends at the last element of the array.

Return

byte[] Checksum. The checksum length is 32bits 4 bytes (The checksum follows Big Endian)

Note

```
byte[] bb1 = {0x10, 0x20, 0x50, 0xF0, 0xFF, 0xFF, 0xFF}
```

```
value = CRC32(bb1,0,Length(bb1)) // value = {0x43, 0xD5, 0xB9, 0xF8}
```

```
value = CRC32(bb1,0,-1) // value = {0x43, 0xD5, 0xB9, 0xF8}
```

```
value = CRC32(bb1,1,-1) // value = {0x08, 0xA5, 0x5B, 0xEB}
```

```
value = CRC32(bb1,-1,-1) // value = {0x00, 0x00, 0x00, 0x00}
```

Syntax 2

```
byte[] CRC32(
    byte[],
    int
)
```

Note

Similar to Syntax1 with computing to the last element of the array

```
CRC32(byte[], int) => CRC32(byte[], int, Length(byte[]))
```

Syntax 3

```
byte[] CRC32(
    byte[]
)
```

Note

Similar to Syntax1 with computing all the elements of the array

```
CRC32(byte[]) => CRC32(byte[], 0, Length(byte[]))
```



```
byte[] bb1 = {0x10, 0x20, 0x50, 0xF0, 0xFF, 0xFF, 0xFF}
value = CRC32(bb1,0,Length(bb1))    // value = {0x43, 0xD5, 0xB9, 0xF8}
value = CRC32(bb1,0)                // value = {0x43, 0xD5, 0xB9, 0xF8}
value = CRC32(bb1)                  // value = {0x43, 0xD5, 0xB9, 0xF8}
bb1 = Byte_Concat(bb1, CRC32(bb1))
    // bb1 = {0x10, 0x20, 0x50, 0xF0, 0xFF, 0xFF, 0xFF, 0x43, 0xD5, 0xB9, 0xF8}
```

4.50 ListenPacket()

Pack the string contents as the compatible protocol for the Listen Node (External Script Control Mode)

Syntax 1

```
string ListenPacket(
    string,
    string
)
```

Parameters

string User defined Header. For empty string, Default string "TMSCT" will be applied
string The data section in Listen Node communication format

Return

string Packed data (Including header, data length and check sum)

Note

```
string var_data1 = "1, var_i++"
string var_data2 = "Hello World"
```

```
value = ListenPacket("TMSCT", var_data1) // $TMSCT,9,1,var_i++,*06\r\n
value = ListenPacket("", var_data2)       // $TMSCT,11,Hello World,*51\r\n
                                           // Error for TMSCT

value = ListenPacket("", "2,Techman Robot") // $TMSCT,15,2,Techman Robot,*57\r\n
value = ListenPacket("TMSTA", var_data2)    // $TMSTA,11,Hello World,*53\r\n
                                           // Error for TMSTA

value = ListenPacket("TMSTA", "00")         // $TMSTA,2,00,*41\r\n
```

Syntax 2

```
string ListenPacket(
    string
)
```

Parameters

string The data section in Listen Node communication format (With TMSCT header)

Return

string Packed data (Including header, data length and check sum)

Note

```
string var_data1 = "1,var_counter++" // ScriptID, ScriptLanguage
value = ListenPacket(var_data1)      // $TMSCT,15,1, var_counter++,*26\r\n
```

4.51 ListenSend()

Send TMSTA, the communication protocol of Listen node, to the client devices connected to the Listen Server currently.

Syntax1

```
int ListenSend(
    string,
    int,
    ?
)
```

Parameters

string Target IP filtering such as 127.0.0.1 meaning to send to all client devices connecting from 127.0.0.1.

int TMSTA SubCmd numbering for sending self-defined data message only 90 .. 99

? The value to send. Available types: byte, int, float, double, bool, and string. Numeric values will be conversed in Little Endian, and string values will be converse in UTF8.

Return

int Return the result

0 sent successfully

-1 error. Listen Server is not starting.

-2 error. SubCmd must be between 90 and 99.

Syntax2

```
int ListenSend(
    int,
    ?
)
```

Parameters

int TMSTA SubCmd numbering for sending self-defined data message only 90 .. 99

? The value to send. Available types: byte, int, float, double, bool, and string. Numeric values will be conversed in Little Endian, and string values will be converse in UTF8.

Return

int Return the result

0 sent successfully

-1 error. Listen Server is not starting.

-2 error. SubCmd must be between 90 and 99.

Note

No target IP filtering will result in sending data messages to all connected client devices.

Note

string ip = "127.0.0.1"

byte b = 100

value = ListenSend(ip, 10, b)

// send 0x64 to ipfilter "127.0.0.1" // value = -2 // SubCmd must be between 90 and 99.

```

value = ListenSend(ip, 90, b)
    // send 0x64 to ipfilter "127.0.0.1"    // value = -1    // Supposedly Listen Server is not starting.
value = ListenSend(ip, 90, b)
    // send 0x64 to ipfilter "127.0.0.1"    // value = 0    // sent successfully
    // IP filtering 127.0.0.1 and send to the devices connected to Listen Server via the IP.
    // $TMSTA,4,90,d,*06                    // The value of 100 is conversed to 0x64.
value = ListenSend(ip, 90, 123456)
    // send 0x40 0xE2 0x01 0x00 to ipfilter "127.0.0.1"
    // $TMSTA,7,90,@,*,*C2
    // The value of 123456 is conversed to 0x40 0xE2 0x01 0x00 (int, Little Endian)
value = ListenSend(90, "123.456")
    // send 0x31 0x32 0x33 0x2E 0x34 0x35 0x36
    // No target IP filtering will result in sending data messages to all connected client devices.
    // $TMSTA,10,90,123.456,*7E
    // The value of "123.456" is conversed to 0x31 0x32 0x33 0x2E 0x34 0x35 0x36 (string, UTF8).
byte[] bb = {100, 200}
value = ListenSend(90, bb)
    // send 0x64 0xC8
    // $TMSTA,5,90,d,*,*CF    // The value of {100, 200} is conversed to 0x64 0xC8
string[] ss = {"T", "M", "達明機器人"}
value = ListenSend(90, ss)
    // send 0x54 0x4D 0xE9 0x81 0x94 0xE6 0x98 0x8E 0xE6 0xA9 0x9F 0xE5 0x99 0xA8 0xE4 0xBA
    // $TMSTA,20,90,TM達明機器人,*A1

```

4.52 VarSync()

Send the Variable object to TMmanager (Robot Management System)

* When performing this function, the flow will not go on until the object is sent out successfully or the maximum retry times is reached.

Syntax 1

```
int VarSync(
    int,
    int,
    ?
)
```

Parameters

- | | |
|----------------------|---|
| int | The maximum times to retry |
| <code><= 0</code> | Keep retrying as error occurred. |
| int | The time duration between two retries (millisecond) |
| <code>< 0</code> | Invalid time duration. The default value, 1000ms, will be applied |
| ? | The string or string array. The name of variables to be sent. |
| | Multiple items can be listed. If there are indefinite variables, they will be not be sent; other definite variables will be sent. |
| | * The item is the name of the variable, not what the variable equals such that it goes with "i". |
| | * If the variable is listed, the value of the variable will be used to send the matched object. |

Return

- | | |
|---------------------|--|
| int | Sending times |
| <code>> 0</code> | Send success. The return value returns the sending times |
| <code>0</code> | Send failed |
| <code>-1</code> | TM Manager function is not enabled |
| <code>-9</code> | Invalid Parameters |

Note

```
string var_s = "ABC"
string var_s1 = " var_s"
string[] var_ss = {"ABC", " var_s", " var_s1"}
value = VarSync(1, 1000, " var_s") // Send var_s variable object
value = VarSync(2, 2000, var_s)    // Send ABC variable object (Because the value of var_s is
                                   "ABC")
value = VarSync(3, 2000, var_ss)   // Send ABC, var_s, var_s1 variable object (From the value of
                                   ss string array)
value = VarSync(3, 2000, " var_ss") // Send var_ss variable object
value = VarSync(4, 2000, " var_ss", " var_s1", "ABC") // Send var_ss, var_s1, ABC variable
object
```

Syntax 2

```
int VarSync(
    int,
    ?
)
```

Note

Same as Syntax 1 with the time between two retries defaults to 1000 ms.

VarSync(int, ?) => **VarSync**(int, 1000, ?)

Syntax 3

int **VarSync**(

?

)

Note

Same as Syntax 1 with the time between two retries defaults to 1000 ms without limit of times to retry

VarSync(?) => **VarSync**(0, 1000, ?)

5. General Functions (Script)

5.1 Exit()

Stop the project running.

Syntax 1

```
void Exit(
    bool,
    int
)
```

Parameter

bool	Whether to wait for the end of the motion command to stop the project
true	Wait (default)
false	No wait
int	Ending code
> 0	Execute the closestop function.
== 0	Evaluate if the project errs. (default)
< 0	Execute the errorstop function.

Return

void No return

Syntax 2

```
void Exit(
    bool
)
```

Parameter

bool	Whether to wait for the end of the motion command to stop the project
true	Wait (default)
false	No wait

Note

Same as syntax 1. It defaults 0 to the ending code.

Syntax 3

```
void Exit(
)
```

Parameter

void No input

Note

Same as syntax 1. It defaults true to whether to wait for the end of the motion command to stop the project and 0 to the ending code.

Syntax 4

```
void Exit(
    int
)
```

Parameter

int Ending code.
 > 0 Execute the closestop funciton.
 == 0 Same as syntax 1. Evaluates if the project errs.
 < 0 Execute the errorstop funciton.

Note

Same as syntax 1. It defaults true to whether to wait for the end of the motion command to stop the project.

Exit()

// Wait for the end of the motion command, stop the project running, and assess if errors occurred.

// Run the closestop function next when there is no error.

// Run the errorstop function next when there is an error.

Exit(false) // Wait not for the end of the motion command not, stop the project running shortly, and assess if errors occurred.

Exit(false, 1) // Wait not for the end of the motion command not, stop the project running shortly, and run the closestop function.

Exit(1) // Wait for the end of the motion command, stop the project running, and run the closestop function.

Exit(0) // Wait for the end of the motion command, stop the project running, and assess if errors occurred.

Exit(-1) // Wait for the end of the motion command, stop the project running, and run the errorstop function.

5.2 Pause()

Pause the project and the motion of the robot other than non-paused threads and external script processes. Use Resume() or press the Play button on the robot stick to resume.

Syntax 1

```
bool Pause(  
)
```

Parameter

void No parameter

Return

bool True Command accepted ; False Command rejected

Note

Pause()

5.3 Resume()

Resume the project and the motion of the robot.

Syntax 1

```
bool Resume(  
)
```

Parameter

void No parameter

Return

bool True Command accepted ; False Command rejected

Note

Resume()

5.4 WaitFor()

The loop wait condition stands or time out on waiting.

Syntax 1

```
bool WaitFor(
    bool,
    int
)
```

Parameter

bool The loop wait condition. It can be true/false or a statement returns the bool value.
int wait time (millisecond)
 < 0 wait indefinitely
 >= 0 time to wait

Return

bool Return **True** for the loop wait condition stands and **False** for time out on waiting.

Note

```
int i = 0
bool flag = WaitFor(i++ > 100, 1000)
// The loop executes i++ and judges whether it is larger than 100. It exits the loop at once if the condition
// satisfies (flag = true) or it is timeout after 1000ms (flag = false).
```

Syntax 2

```
bool WaitFor(
    int
)
```

Parameter

int wait time (millisecond)
 < 0 invalid
 >= 0 time to wait

Return

bool Return **True** for the wait time stands and **False** for not. (Interrupted by the project stop)

Note

```
WaitFor(100) // Timeout after 100ms of waiting.
```

5.5 Sleep()

Stop the thread in the time specified. Same as WaitFor(int).

Syntax 1

```
bool Sleep(  
    int  
)
```

Parameter

int	wait time (millisecond)
< 0	invalid
>= 0	time to wait

Return

bool Return **True** for the wait time stands and **False** for not. (Interrupted by the project stop)

Note

```
Sleep(100)    // Timeout after 100ms of waiting.
```

5.6 Display()

Display contents on the TMflow dashboard.

Syntax 1

```
bool Display(
    string,
    string,
    string,
    string
)
```

Parameter

string	Headline text color
"Red"	Set headline background color to red.
"Green"	Set headline background color to green.
"Blue"	Set headline background color to blue.
"Yellow"	Set headline background color to yellow.
"Black"	Set headline background color to black.
"White"	Set headline background color to white.
"Gray"	Set headline background color to gray.
string	Headline background color
"Red"	Set headline text color to red.
"Green"	Set headline text color to green.
"Blue"	Set headline text color to blue.
"Yellow"	Set headline text color to yellow.
"Black"	Set headline text color to black.
"White"	Set headline text color to white.
"Gray"	Set headline text color to gray.
string	Headline text
string	Text

Return

bool Return **True** for set successfully and **False** for unsuccessfully.

Note

Display("Yellow", "Green", "Gripper Initial Finish", "Force = 30N") // Output the headline of Gripper Initial Finish with the text of Force = 30N to the TMflow dashboard and set headline text color to yellow and headline background color to green.

Syntax 2

```
bool Display(
    string,
    string
)
```

Parameter

string	Headline text
string	Text

Note

Same as Syntax 1. It sets headline text color to black and headline background color to white by default.

Syntax 3

```
bool Display(  
    string  
)
```

Parameter

string Text

Note

Same as Syntax 1. It sets headline text color to black and headline background color to white by default. The headline text is an empty string.

6. Math Functions

6.1 abs()

Return the absolute value of the designate number

Syntax 1

```
int abs(  
    int  
)
```

Parameter

int Input number in integer

Return

int Return the absolute value of the input number in integer

Note

```
int i = 10  
value = abs(i)    // 10  
i = -10  
value = abs(i)    // 10
```

Syntax 2

```
float abs(  
    float  
)
```

Parameter

float Input number in float

Return

float Return the absolute value of the input number in float

Note

```
float f = 10.1  
value = abs(f)    // 10.1  
f = -10.1  
value = abs(f)    // 10.1
```

Syntax 3

```
double abs(  
    double  
)
```

Parameter

double Input number in double

Return

double Return the absolute value of the input number in double

Note

```
double d = 10.8  
value = abs(d)    // 10.8  
d = -10.8  
value = abs(d)    // 10.8
```

6.2 pow()

Return the power of the designate base and exponent

Syntax 1

```
int pow(
    int,
    double
)
```

Parameter

int Input base in integer
double Exponent

Return

int Return the power in integer

Syntax 2

```
float pow(
    float,
    double
)
```

Parameter

float Input base in float
double Exponent

Return

float Return the power in float

Syntax 3

```
double pow(
    double,
    double
)
```

Parameter

double Input base in double
double Exponent

Return

double Return the power in double

Note

```
? int b = 100
value = pow(b, 2)    // 10000
value = pow(b, -2)   // 0        // 0.0001, but int type
value = pow(b, 0.1)  // 1        // 1.5848932, but int type
value = pow(b, 2.1)  // 15848    // 15848.932, but int type
value = pow(b, -2.1) // 0        // 6.309574E-05, but int type

? float b = -100
value = pow(b, 2)    // 10000
value = pow(b, -2)   // 0.0001
value = pow(b, 0.2)  // Error // NaN
value = pow(b, 2.2)  // Error // NaN
```



```
value = pow(b, -2.2) // Error // NaN
```

```
? double b = 100  
value = pow(b, 2) // 10000  
value = pow(b, -2) // 0.0001  
value = pow(b, 0.31) // 4.168694  
value = pow(b, 2.31) // 41686.938  
value = pow(b, -2.31) // 2.3988328E-05
```

6.3 sqrt()

Return the square root of the designate number

Syntax 1

```
float sqrt(  
    float  
)
```

Parameter

float Input number in float

Return

float Return the square root in float

Syntax 2

```
double sqrt(  
    double  
)
```

Parameter

double Input number in double

Return

double Return the square root in double

Note

```
value = sqrt(100)           // 10  
value = sqrt(100.1234)      // 10.006168  
value = sqrt(0.1234)        // 0.35128337  
value = sqrt(-100)          // Error // NaN  
value = sqrt(-100.1234)     // Error // NaN  
value = sqrt(-0.1234)       // Error // NaN
```

6.4 ceil()

Return a number rounded upward to its nearest integer.

Syntax 1

```
float ceil(  
    float  
)
```

Parameter

float input number in float

Return

float Return a number in float rounded upward to its nearest integer

Syntax 2

```
double ceil(  
    double  
)
```

Parameter

double input number in double

Return

double Return a number in double rounded upward to its nearest integer

Note

```
value = ceil(100)           // 100  
value = ceil(100.1234)      // 101  
value = ceil(0.1234)        // 1  
value = ceil(-100)          // -100  
value = ceil(-100.1234)     // -100  
value = ceil(-0.1234)       // 0
```

6.5 floor()

Return a number rounded downward to its nearest integer.

Syntax 1

```
float floor(  
    float  
)
```

Parameter

float input number in float

Return

float Return a number in float rounded downward to its nearest integer

Syntax 2

```
double floor(  
    double  
)
```

Parameter

double input number in double

Return

double Return a number in double rounded downward to its nearest integer

Note

```
value = floor(100)           // 100  
value = floor(100.1234)     // 100  
value = floor(0.1234)       // 0  
value = floor(-100)         // -100  
value = floor(-100.1234)    // -101  
value = floor(-0.1234)      // -1
```

6.6 round()

Return a number rounded to its nearest integer.

Syntax 1

```
float round(
    float,
    int
)
```

Parameter

float	input number in float
int	digits after the returned decimal point (0 by default meaning the number is rounded to integer)
0 .. 15	valid values
< 0	value invalid, will use 0 by default
> 15	value invalid, will use 0 by default

Return

float Return a number in float rounded to its nearest integer.

Syntax 2

```
float round(
    float
)
```

Note

Same as syntax 1. Obtain 0 digit after the decimal point by default.

round(float) => round(float, 0)

Syntax 3

```
double round(
    double,
    int
)
```

Parameter

double	input number in double
int	digits after the returned decimal point (0 by default meaning the number is rounded to integer)
0 .. 15	valid values
< 0	value invalid, will use 0 by default
> 15	value invalid, will use 0 by default

Return

double Return a number in double rounded to its nearest integer.

Syntax 4

```
double round(
    double
)
```

Note

Same as syntax 3. Obtain 0 digit after the decimal point by default.

round(double) => round(double, 0)

```
value = round(100)           // 100
value = round(100.456)       // 100
value = round(0.567)         // 1
value = round(-100)          // -100
value = round(-100.456)      // -100
value = round(-0.567)        // -1
value = round(100.345, 1)    // 100.3
value = round(100.345, 2)    // 100.35
value = round(-100.345, 1)   // -100.3
value = round(-100.345, 2)   // -100.35
value = round(-100.345, 16) // -100
```

6.7 random()

Return a random number in float between 0 and 1 or in integer between the lower bound and the upper bound.

Syntax 1

```
float random(
)
```

Parameter

void No parameter

Return

float Return a random number in float between 0 and 1.

Note

```
value = random()    // 0.9473762
value = random()    // 0.7764986
value = random()    // 0.9911129
```

Syntax 2

```
int random(
    int
)
```

Parameter

int The upper bound of the random number

Return

int Return a random number in integer between 0 and the upper bound

Note

```
value = random(10)    // 8
value = random(10)    // 1
value = random(10)    // 5
value = random(-10)   // 0 // The value of the upper bound must be larger than 0.
```

Syntax 3

```
int random(
    int,
    int
)
```

Parameter

int The lower bound of the random number

int The upper bound of the random number must be larger than the lower bound, or it will return the value of the lower bound in integer.

Return

int Return a random number in integer between the lower bound and the upper bound.

Note

```
value = random(5, 10) // 8
value = random(5, 10) // 8
value = random(5, 10) // 6
value = random(5, -1) // 5 // The upper bound is smaller than the lower bound. Returned the value of
```

the lower bound in integer.

6.8 sum()

Return the sum of the given numbers or the array of numbers.

Syntax 1

```
int sum(
    ?,
    ...
)
```

Parameter (variable parameter amount)

? input value, can be in the type of `byte`, `int`, `byte[]`, or `int[]`
calculate the value of each parameter or each element in the array. It returns an error and stops if it comes with a non-numeric type.

Return

`int` Return the sum, in the integer type, of the given numbers.

Syntax 2

```
double sum(
    ?,
    ...
)
```

Parameter (variable parameter amount)

? input value, can be in the type of `float`, `double`, `float[]`, or `double[]`
calculate the value of each parameter or each element in the array. It returns an error and stops if it comes with a non-numeric type.

Return

`double` Return the sum, in the double type, of the given numbers.

Note

```
int sum1 = sum(1,2,3,4,5)           // 15
int sum2 = sum(1,2,3,4,{5,6,7,8})    // 36
int sum3 = sum(1,2,3,4,{5,6,7,8},1.2) // 37    // due to putting the int type
double sum4 = sum(1,2,3,4,{5,6,7,8},1.2) // 37.2
double sum5 = sum(1,2,3,4,{5,6,7,8},9.2) // 45.2
double sum6 = sum(1,2,3,4,{5,6,7,8},9.2,sum5,{1.2,3.4}) // 95
```

6.9 average()

Return the average of the given numbers or the array of numbers.

Syntax 1

```
double average(
    ?,
    ...
)
```

Parameter (variable parameter amount)

? input value, can be in the type of `byte`, `int`, `float`, `double`, `byte[]`, `int[]`, `float[]`, or `double[]`

calculate the value of each parameter or each element in the array. It returns an error and stops if it comes with a non-numeric type.

Return

`double` Return the average, in the double type, of the given numbers.

Note

```
double avg1 = average(1,2,3,4,5)           // 3
double avg2 = average(1,2,3,4,{5,6,7,8},9.2) // 5.022222222222222
double avg3 = average(1,2,3,4,sum({5,6,7,8}),9.2) // 7.533333333333333
double avg4 = average(1,2,3,4,{5,6,7,8},9.2,{1.2,3.4}) // 4.52727272727273
```

6.10 stdevp()

Return the standard deviation $\sigma = \sqrt{\sum \left(X - \frac{\sum X}{N}\right)^2 / N}$ based on the entire population of the given numbers or the array of numbers.

Syntax 1

```
double stdevp(
    ?,
    ...
)
```

Parameter (variable parameter amount)

? input value, can be in the type of `byte`, `int`, `float`, `double`, `byte[]`, `int[]`, `float[]`, or `double[]`

calculate the value of each parameter or each element in the array. It returns an error and stops if it comes with a non-numeric type.

Return

`double` Return the standard deviation, in the double type, of the given numbers.

Note

```
double stdp1 = stdevp(1,2,3,4,5)           // 1.4142135623731
double stdp2 = stdevp(1,2,3,4,{5,6,7,8},9.2) // 2.61694384000276
double stdp3 = stdevp(1,2,3,4,sum({5,6,7,8}),9.2) // 8.66153694341958
double stdp4 = stdevp(1,2,3,4,{5,6,7,8},9.2,{1.2,3.4}) // 2.63165724111457
```

6.11 stdevs()

Return the standard deviation $s = \sqrt{\sum \left(X - \frac{\sum X}{N} \right)^2 / (N - 1)}$ based on a sample of the given numbers or the array of numbers.

Syntax 1

```
double stdevs(
    ?,
    ...
)
```

Parameter (variable parameter amount)

? input value, can be in the type of `byte`, `int`, `float`, `double`, `byte[]`, `int[]`, `float[]`, or `double[]`

calculate the value of each parameter or each element in the array. It returns an error and stops if it comes with a non-numeric type.

Return

`double` Return the standard deviation, in the double type, of the given numbers.

Note

```
double stds1 = stdevs(1,2,3,4,5)           // 1.58113883008419
double stds2 = stdevs(1,2,3,4,{5,6,7,8},9.2) // 2.77568810287547
double stds3 = stdevs(1,2,3,4,sum({5,6,7,8}),9.2) // 9.4882383331505
double stds4 = stdevs(1,2,3,4,{5,6,7,8},9.2,{1.2,3.4}) // 2.76010539983201
```

6.12 min()

Return the minimum of the given numbers or the array of numbers.

Syntax 1

```
int min(
    ?,
    ...
)
```

Parameter (variable parameter amount)

? input value, can be in the type of `byte`, `int`, `byte[]`, or `int[]`
compare the value of each parameter or each element in the array. It returns an error and stops if it comes with a non-numeric type.

Return

`int` Return the minimum, in the integer type, of the given numbers.

Syntax 2

```
double min(
    ?,
    ...
)
```

Parameter (variable parameter amount)

? input value, can be in the type of `float`, `double`, `float[]`, or `double[]`
compare the value of each parameter or each element in the array. It returns an error and stops if it comes with a non-numeric type.

Return

`double` Return the minimum, in the double type, of the given numbers.

Note

```
int min1 = min(1,2,3)           // 1
int min2 = min(1,2,3,{-4,-5.3,6.2},1.2) // -5 // due to putting the int type
double min3 = min(1,2,3,{-4,-5.3,6.2},1.2) // -5.3
double min4 = min(1,2,3,{-0.2,-0.1,0.1},9.2) // -0.2
```

6.13 max()

Return the maximum of the given numbers or the array of numbers.

Syntax 1

```
int max(
    ?,
    ...
)
```

Parameter (variable parameter amount)

? input value, can be in the type of `byte`, `int`, `byte[]`, or `int[]`
compare the value of each parameter or each element in the array. It returns an error and stops if it comes with a non-numeric type.

Return

`int` Return the maximum, in the integer type, of the given numbers.

Syntax 2

```
double max(
    ?,
    ...
)
```

Parameter (variable parameter amount)

? input value, can be in the type of `float`, `double`, `float[]`, or `double[]`
compare the value of each parameter or each element in the array. It returns an error and stops if it comes with a non-numeric type.

Return

`double` Return the maximum, in the double type, of the given numbers.

Note

```
int max1 = max(1,2,3) // 3
int max2 = max(1,2,3,{-4,-5.3,6.2},1.2) // 6 // due to putting the int type
double max3 = max(1,2,3,{-4,-5.3,6.2},1.2) // 6.2
double max4 = max(1,2,3,{-0.2,-0.1,0.1},9.2) // 9.2
```

6.14 d2r()

Convert the value of degree to radian

Syntax 1

```
float d2r(  
    float  
)
```

Parameter

float Input the value of degree in float

Return

float Return the value of radian in float

Syntax 2

```
double d2r(  
    double  
)
```

Parameter

double Input the value of degree in double

Return

double Return the value of radian in double

Note

```
value = d2r(1)           // 0.017453292
```

6.15 r2d()

Convert the value of degree to radian to degree

Syntax 1

```
float r2d(  
    float  
)
```

Parameter

float Input the value of radian in float

Return

float Return the value of degree in float

Syntax 2

```
double r2d(  
    double  
)
```

Parameter

double Input the value of radian in double

Return

double Return the value of degree in double

Note

```
value = r2d(1)           // 57.29578
```


6.16 sin()

Return the sine of the input value of degree

Syntax 1

```
float sin(  
    float  
)
```

Parameter

float Input the value of degree in float

Return

float Return the sine of the input value of degree in float

Syntax 2

```
double sin(  
    double  
)
```

Parameter

double Input the value of degree in double

Return

double Return the sine of the input value of degree in double

Note

```
value = sin(0)           // 0  
value = sin(15)          // 0.25881904  
value = sin(30)          // 0.5  
value = sin(60)          // 0.8660254  
value = sin(90)          // 1
```

6.17 cos()

Return the cosine of the input value of degree

Syntax 1

```
float cos(  
    float  
)
```

Parameter

float Input the value of degree in float

Return

float Return the cosine of the input value of degree in float

Syntax 2

```
double cos(  
    double  
)
```

Parameter

double Input the value of degree in double

Return

double Return the cosine of the input value of degree in double

Note

```
value = cos(0)           // 1  
value = cos(15)          // 0.9659258  
value = cos(30)          // 0.8660254  
value = cos(45)          // 0.70710677  
value = cos(60)          // 0.5
```

6.18 tan()

Return the tangent of the input value of degree

Syntax 1

```
float tan(  
    float  
)
```

Parameter

float Input the value of degree in float

Return

float Return the tangent of the input value of degree in float

Syntax 2

```
double tan(  
    double  
)
```

Parameter

double Input the value of degree in double

Return

double Return the tangent of the input value of degree in double

Note

```
value = tan(0)           // 0  
value = tan(15)          // 0.2679492  
value = tan(30)          // 0.57735026  
value = tan(45)          // 1  
value = tan(60)          // 1.7320508
```

6.19 asin()

Return the arcsine of the input value in degree

Syntax 1

```
float asin(  
    float  
)
```

Parameter

float Input the sine value in float between -1 and 1

Return

float Return the arcsine of the input value of degree in float

Syntax 2

```
double asin(  
    double  
)
```

Parameter

double Input the sine value in double between -1 and 1

Return

double Return the arcsine of the input value of degree in double

Note

```
value = asin(0)           // 0  
value = asin(0.258819)    // 14.999998  
value = asin(0.5)         // 30  
value = asin(0.8660254)   // 60  
value = asin(1)           // 90  
  
value = asin(sin(15))     // 15  
value = asin(sin(60))     // 60
```

6.20 acos()

Return the arccosine of the input value in degree

Syntax 1

```
float acos(  
    float  
)
```

Parameter

float Input the cosine value in float between -1 and 1

Return

float Return the degree value in float

Syntax 2

```
double acos(  
    double  
)
```

Parameter

double Input the cosine value in double between -1 and 1

Return

double Return the degree value in double

Note

```
value = acos(1)           // 0
```

```
value = acos(0.9659258)   // 15.000003
```

```
value = acos(0.8660254)   // 30.000002
```

```
value = acos(0.7071068)   // 44.999996
```

```
value = acos(0.5)         // 60
```

```
value = acos(cos(15))     // 15.000003
```

```
value = acos(cos(30))     // 30.000002
```

```
value = acos(cos(45))     // 45
```

```
value = acos(cos((double)15)) // 14.999999999999996
```

```
value = acos(cos((double)30)) // 29.999999999999993
```

6.21 atan()

Return the arctangent of the input value in degree

Syntax 1

```
float atan(  
    float  
)
```

Parameter

float Input the arctangent value in float

Return

float Return the degree value in float

Syntax 2

```
double atan(  
    double  
)
```

Parameter

double Input the arctangent value in double

Return

double Return the degree value in double

Note

```
value = atan(0)           // 0  
value = atan(0.2679492)   // 15  
value = atan(0.5773503)   // 30.000002  
value = atan(1)           // 45  
value = atan(1.732051)    // 60.000004  
  
value = atan(tan(30))     // 30  
value = atan(tan(60))     // 60
```

6.22 atan2()

Return the arctangent of the quotient of its arguments

Syntax 1

```
float atan2(
    float,
    float
)
```

Parameter

float Input a number in float representing the Y coordinate

float Input a number in float representing the X coordinate

Return

float Return the degree value in float

Syntax 2

```
double atan2(
    double,
    double
)
```

Parameter

double Input a number in double representing the Y coordinate

double Input a number in double representing the X coordinate

Return

double Return the degree value in double

Note

```
value = atan2(2, 1)           // 63.434948
value = atan2(1, 1)           // 45
value = atan2(-1, -1)         // -135
value = atan2(4, -3)          // 126.869896
```

6.23 log()

Return the natural logarithm of the input value

Syntax 1

```
float log(
    float,
    double
)
```

Parameter

float Input value in float

double The base of the logarithm

Return

float Return the logarithm of the input value and the base in float

Syntax 2

```
double log(
    double,
    double
)
```

Parameter

double Input value in double

double The base of the logarithm

Return

double Return the logarithm of the input value and the base in double

Note

```
value = log(16, 2)    // 4
value = log(16, 8)    // 1.3333334
value = log(16, 10)   // 1.20412
value = log(16, 16)   // 1
```

Syntax 3

```
float log(
    float
)
```

Parameter

float Input value in float

Return

float Return the natural logarithm of the input value and the base e in float

Syntax 4

```
double log(
    double
)
```

Parameter

double Input value in double

Return

double Return the natural logarithm of the input value and the base e in double

Note

```
value = log(16, 2)      // 4
value = log(16)         // 2.7725887
value = log(2)          // 0.6931472
value = log(16)/log(2) // 2.7725887/0.6931472 = 3.9999998557305
```

6.24 log10()

Return the logarithm of the input value with the base 10

Syntax 1

```
float log10(  
    float  
)
```

Parameter

float Input value in float

Return

float Return the logarithm of the input value with the base 10 in float

Syntax 2

```
double log10(  
    double  
)
```

Parameter

double Input value in double

Return

double Return the logarithm of the input value with the base 10 in double

Note

```
value = log(16, 10)    // 1.20412  
value = log10(16)      // 1.20412  
value = log(500, 10)   // 2.69897  
value = log10(500)     // 2.69897
```

6.25 norm2()

Return the second norm of a specified vector.

Syntax 1

```
float norm2(
    float[]
)
```

Parameter

float[] A vector whose second norm (or called Euclidean norm, vector magnitude) is to be found.

Return

float the second norm (or called Euclidean norm, vector magnitude) of a specified vector

Note

$$\|v\| = \sqrt{\sum_{i=1}^{i=N} |v_i|^2}$$

float[] vector1 = {3,4}

float[] vector2 = {3,4,5}

float[] vector3 = {3,4,5,6,8}

value = **norm2**(vector1) // 5

value = **norm2**(vector2) // 7.071068

value = **norm2**(vector3) // 12.247449

6.26 dist()

Return the distance between the two coordinates.

Syntax 1

```
float dist(  
    float[],  
    float[]  
)
```

Parameter

`float[]` The first coordinate $\{X_{1(mm)} \ Y_{1(mm)} \ Z_{1(mm)} \ RX_{1(^{\circ})} \ RY_{1(^{\circ})} \ RZ_{1(^{\circ})}\}$
`float[]` The second coordinate $\{X_{2(mm)} \ Y_{2(mm)} \ Z_{2(mm)} \ RX_{2(^{\circ})} \ RY_{2(^{\circ})} \ RZ_{2(^{\circ})}\}$

Return

`float` The distance between the two coordinates

Note

```
float[] c1 = {100,200,100,30,50,20}  
float[] c2 = {100,100,100,50,50,10}  
value = dist(c1, c2)      // 100
```

6.27 trans()

Return the displacement and rotation angle from one specified point to another point.

Syntax 1

```
float[] trans(
    float[],
    float[],
    bool
)
```

Parameter

float[]	First Point	{ $X_1(mm)$ $Y_1(mm)$ $Z_1(mm)$ $RX_1(^{\circ})$ $RY_1(^{\circ})$ $RZ_1(^{\circ})$ }
float[]	Second Point	{ $X_2(mm)$ $Y_2(mm)$ $Z_2(mm)$ $RX_2(^{\circ})$ $RY_2(^{\circ})$ $RZ_2(^{\circ})$ }
bool	The reference coordinate	
false	Refer to the robot's base(default)	
true	Refer to the first point	

Return

float[] The displacement and rotation angle from first point to second point
 { X_{trans} Y_{trans} Z_{trans} RX_{trans} RY_{trans} RZ_{trans} }
 Return an empty array if unable to calculate.

Syntax 2

```
float[] trans(
    float[],
    float[]
)
```

Note

Same as syntax 1. By default, the reference coordinate is set to false for the robot's base.

Original Transformation Matrix = $\begin{bmatrix} R & P \\ 0 & 1 \end{bmatrix}$

$$R_n = \begin{bmatrix} \cos(RZ_n)\cos(RY_n) & -\sin(RZ_n)\cos(RX_n) + \cos(RZ_n)\sin(RY_n)\sin(RX_n) & \sin(RZ_n)\sin(RX_n) + \cos(RZ_n)\sin(RY_n)\cos(RX_n) \\ \sin(RZ_n)\cos(RY_n) & \cos(RZ_n)\cos(RX_n) + \sin(RZ_n)\sin(RY_n)\sin(RX_n) & -\cos(RZ_n)\sin(RX_n) + \sin(RZ_n)\sin(RY_n)\cos(RX_n) \\ -\sin(RY_n) & \cos(RY_n)\sin(RX_n) & \cos(RY_n)\cos(RX_n) \end{bmatrix}$$

$$\text{First Point} = \begin{bmatrix} R_1 & P_1 \\ 0 & 1 \end{bmatrix}$$

$$\text{Second Point} = \begin{bmatrix} R_2 & P_2 \\ 0 & 1 \end{bmatrix}$$

If *reference coordinate* is **false** (reference coordinate is Robot Base)

$$\begin{aligned} P_{trans} &= P_2 - P_1 \\ R_{trans} &= R_2 * R_1^{-1} \end{aligned}$$

If *reference coordinate* is **true** (reference coordinate is First Point)

$$\begin{aligned} P_{trans} &= R_1^{-1} * (P_2 - P_1) \\ R_{trans} &= R_1^{-1} * R_2 \end{aligned}$$

float[] var_P1 = {100, -200, 300, 10, 20, 60}

```
float[] var_P2 = {-400, 200, 50, -20, 30, -45}

float[] var_trans_RB = trans(var_P1, var_P2)
    // {-500,400,-250,-24.615868,-15.565178,-88.613686}
float[] var_trans_i = trans(var_P1, var_P2, true)
    // {176.10095,588.32776,-308.80237,3.7459252,23.13792,-92.46916}
```

6.28 inversetrans()

Return the displacement and rotation angle {x, y, z, rx, ry, rz} opposite to the input displacement and rotation angle {x, y, z, rx, ry, rz}.

Syntax 1

```
float[] inversetrans(
    float[],
    bool
)
```

Parameter

float[] The input displacement and rotation angle {X_o Y_o Z_o RX_o RY_o RZ_o}

bool The reference coordinate

false Refer to the robot's base(default)

true Refer to the input displacement and the rotation angle

Return

float[] The displacement and rotation angle

{X_{inv} Y_{inv} Z_{inv} RX_{inv} RY_{inv} RZ_{inv}}

opposite to the input displacement and rotation angle

{X_o Y_o Z_o RX_o RY_o RZ_o}

Return an empty array if unable to calculate.

Syntax 2

```
float[] inversetrans(
    float[]
)
```

Note

Same as syntax 1. By default, the reference coordinate is set to false for the robot's base.

Original Transformation Matrix = $\begin{bmatrix} R & P \\ 0 & 1 \end{bmatrix}$

$$R_n = \begin{bmatrix} \cos(RZ_n) \cos(RY_n) & -\sin(RZ_n) \cos(RX_n) + \cos(RZ_n) \sin(RY_n) \sin(RX_n) & \sin(RZ_n) \sin(RX_n) + \cos(RZ_n) \sin(RY_n) \cos(RX_n) \\ \sin(RZ_n) \cos(RY_n) & \cos(RZ_n) \cos(RX_n) + \sin(RZ_n) \sin(RY_n) \sin(RX_n) & -\cos(RZ_n) \sin(RX_n) + \sin(RZ_n) \sin(RY_n) \cos(RX_n) \\ -\sin(RY_n) & \cos(RY_n) \sin(RX_n) & \cos(RY_n) \cos(RX_n) \end{bmatrix}$$

$$\text{Initial Point} = \begin{bmatrix} R_i & P_i \\ 0 & 1 \end{bmatrix}$$

If *reference coordinate* is **false** (reference coordinate is Robot Base)

$$P_{inv} = -P_i$$

$$R_{inv} = R_i^{-1}$$

If *reference coordinate* is **true** (reference coordinate is the input {x, y, z, rx, ry, rz}, which is equivalent to inverse of Transformation Matrix)

$$P_{inv} = R_i^{-1} * (-P_i)$$

$$R_{inv} = R_i^{-1}$$

```
float[] var_P1 = {100, -200, 300, 10, 20, 60}
```

```
float[] var_inv_RB = inversetrans(var_P1)
```

```
// {-100,200,-300,12.483133,-18.590115,-60.283367}
```

```
float[] var_inv_i = inversetrans(var_P1, true)
```

```
// {218.38095,142.13216,-268.52972,12.483133,-18.590115,-60.283367}
```

6.29 applytrans()

Return the terminal point computed by applied the displacement and rotation angle to the specified point.

Syntax 1

```
float[] applytrans(
    float[],
    float[],
    bool
)
```

Parameter

float[] Initial point $\{X_i \ Y_i \ Z_i \ RX_i \ RY_i \ RZ_i\}$
float[] the displacement and rotation angle $\{X_o \ Y_o \ Z_o \ RX_o \ RY_o \ RZ_o\}$
bool The reference coordinate
 false Refer to the robot's base(default)
 true Refer to the initial point

Return

float[] the terminal point $\{X_t \ Y_t \ Z_t \ RX_t \ RY_t \ RZ_t\}$ computed by applied the displacement and rotation angle to the initial point
 Return an empty array if unable to calculate.

Syntax 2

```
float[] applytrans(
    float[],
    float[]
)
```

Note

Same as syntax 1. By default, the reference coordinate is set to false for the robot's base.

Original Transformation Matrix = $\begin{bmatrix} R & P \\ 0 & 1 \end{bmatrix}$

$$R_n = \begin{bmatrix} \cos(RZ_n)\cos(RY_n) & -\sin(RZ_n)\cos(RX_n) + \cos(RZ_n)\sin(RY_n)\sin(RX_n) & \sin(RZ_n)\sin(RX_n) + \cos(RZ_n)\sin(RY_n)\cos(RX_n) \\ \sin(RZ_n)\cos(RY_n) & \cos(RZ_n)\cos(RX_n) + \sin(RZ_n)\sin(RY_n)\sin(RX_n) & -\cos(RZ_n)\sin(RX_n) + \sin(RZ_n)\sin(RY_n)\cos(RX_n) \\ -\sin(RY_n) & \cos(RY_n)\sin(RX_n) & \cos(RY_n)\cos(RX_n) \end{bmatrix}$$

$$\text{Initial Point} = \begin{bmatrix} R_i & P_i \\ 0 & 1 \end{bmatrix}$$

If *reference coordinate* is **false** (reference coordinate is Robot Base)

$$P_t = P_i + P_{trans}$$

$$R_t = R_{trans} * R_i$$

If *reference coordinate* is **true** (reference coordinate is Initial point)

$$P_t = P_i + R_i * P_{trans}$$

$$R_t = R_i * R_{trans}$$

```
float[] var_P1 = {100, -200, 300, 10, 20, 60}
```

```
float[] var_P2 = {-400, 200, 50, -20, 30, -45}
```

```
float[] var_trans_RB = trans(var_P1, var_P2)
```

```
// {-500,400,-250,-24.615868,-15.565178,-88.613686}
```



```
float[] var_trans_i = trans(var_P1, var_P2, true)
    // {176.10095,588.32776,-308.80237,3.7459252,23.13792,-92.46916}

float[] var_apply_RB = applytrans(var_P1, var_trans_RB)
    // {-400,200,50,-20,30,-44.999996}
float[] var_apply_i = applytrans(var_P1, var_trans_i, true)
    // {-400,200,50.000015,-20,30,-45}

float[] var_inv_RB = inversetrans(var_P1)
    // {-100,200,-300,12.483133,-18.590115,-60.283367}
float[] var_inv_i = inversetrans(var_P1, true)
    // {218.38095,142.13216,-268.52972,12.483133,-18.590115,-60.283367}

float[] var_apply_1 = applytrans(var_P1, var_inv_RB)
    // {0,0,0,-4.8045007E-07,1.4295254E-07,4.8365365E-07}
float[] var_apply_2 = applytrans(var_P1, var_inv_i, true)
    // {-3.845248E-06,6.1641267E-06,1.4917891E-06,-1.8922562E-07,-2.1415798E-07,6.352311E-07}
```

6.30 interpoint()

Return the interpolate point between two points according to the specified points and ratio

Syntax 1

```
float[] interpoint(
    float[],
    float[],
    float
)
```

Parameter

float[]	First Point {X ₁ (mm) Y ₁ (mm) Z ₁ (mm) RX ₁ (°) RY ₁ (°) RZ ₁ (°)}
float[]	Second Point {X ₂ (mm) Y ₂ (mm) Z ₂ (mm) RX ₂ (°) RY ₂ (°) RZ ₂ (°)}
float	Ratio

Return

float[] the linear interpolate point {X_i Y_i Z_i RX_i RY_i RZ_i} between initial point and endpoint according to the ratio.
Return an empty array if unable to calculate.

Note

$$\begin{aligned} \{X_i \ Y_i \ Z_i \ RX_i \ RY_i \ RZ_i\} \\ = (\{X_2 \ Y_2 \ Z_2 \ RX_2 \ RY_2 \ RZ_2\} - \{X_1 \ Y_1 \ Z_1 \ RX_1 \ RY_1 \ RZ_1\}) \times Ratio \\ + \{X_1 \ Y_1 \ Z_1 \ RX_1 \ RY_1 \ RZ_1\} \end{aligned}$$

```
float[] var_P1 = {-388.3831,-199.8061,367.0702,177.4319,1.717448,-46.02005}
float[] var_P2 = {-436.9584,115.7343,371.4378,179.4419,-42.86601,-96.91867}
float[] interp = interpoint(var_P1, var_P2, 0.5)
// {-412.67075,-42.035904,369.254,172.91898,-20.690556,-69.33843}
```

6.31 changeref()

Return the new coordinate value described with the new base converted from the original coordinate value through the base conversion. In the process of the conversion, the physical position of the original point in the world of the coordinates will remain the same, the change takes effects on its descriptions of the reference coordinates and the corresponding coordinate values.

Syntax 1

```
float[] changeref(
    float[],
    float[],
    float[]
)
```

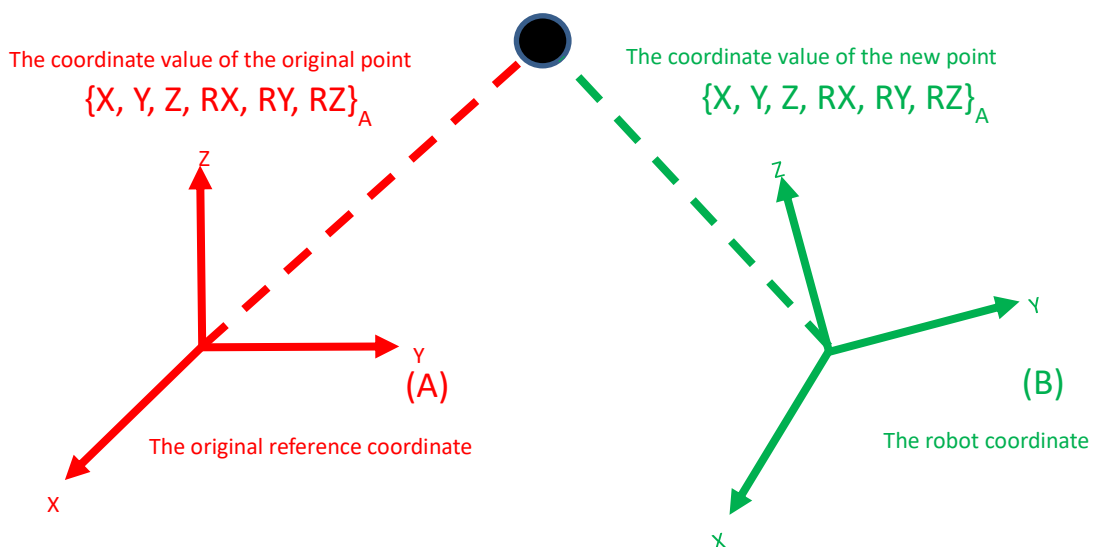
Parameter

float[] The coordinate value of the original point $\{X_o \ Y_o \ Z_o \ RX_o \ RY_o \ RZ_o\}_A$
 float[] The original reference base $\{X_{oa} \ Y_{oa} \ Z_{oa} \ RX_{oa} \ RY_{oa} \ RZ_{oa}\}_A$
 float[] The new reference base $\{X_n \ Y_n \ Z_n \ RX_n \ RY_n \ RZ_n\}_B$

Return

float[] The coordinate value of the new point
 $\{X_{nb} \ Y_{nb} \ Z_{nb} \ RX_{nb} \ RY_{nb} \ RZ_{nb}\}_B$
 Return an empty array if unable to calculate.

Note



```
P1 = {-431.927, -140.6103, 368.7306, -179.288, -0.6893783, -105.8449}
RobotBase = {0, 0, 0, 0, 0, 0}
base1 = {-431.93, -140.61, 368.73, -57.70, -44.98, 33.62}
float[] f0 = changeref(Point["P1"].Value, Base["RobotBase"].Value, Base["base1"].Value)
// f0 = {0.0020519744, 1.9731047E-05, -0.0022721738, 113.94231, 14.9346, -123.19886}
// Convert the value of "P1" in the base "RobotBase" to the value of a point in the base "base1"
```

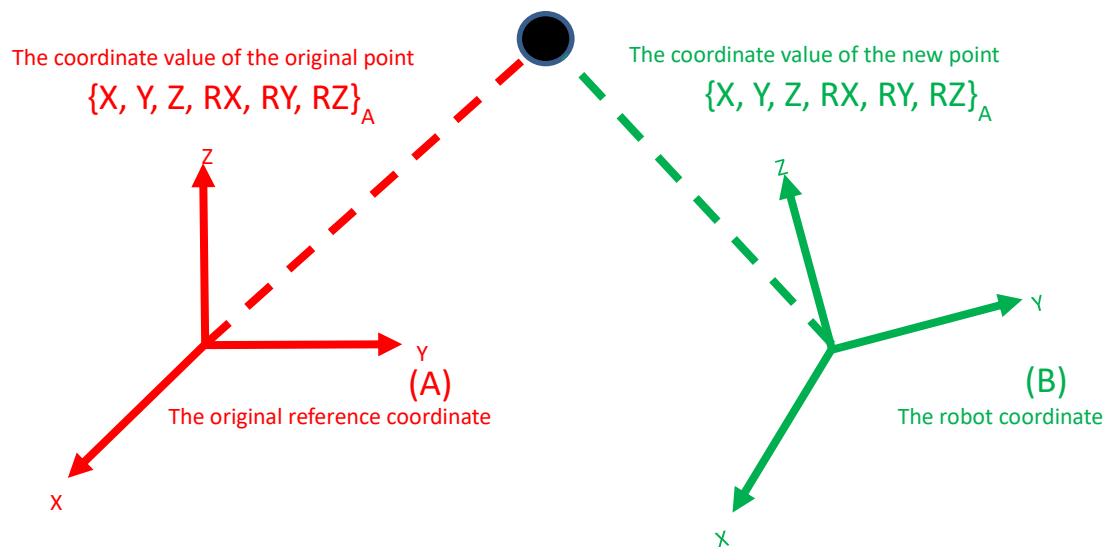
Syntax 2

```
float[] changeref(
    float[],
```

float[]
)
Parameter
 float[] The coordinate value of the original point $\{X_o \ Y_o \ Z_o \ RX_o \ RY_o \ RZ_o\}_A$
 float[] The original reference base $\{X_{oa} \ Y_{oa} \ Z_{oa} \ RX_{oa} \ RY_{oa} \ RZ_{oa}\}_A$
Return
 float[] The coordinate value of the new point
 $\{X_{nr} \ Y_{nr} \ Z_{nr} \ RX_{nr} \ RY_{nr} \ RZ_{nr}\}_R$
 Return an empty array if unable to calculate.

Note

The usage is the same as Syntax1's except assuming the robot base $\{0 \ 0 \ 0 \ 0 \ 0 \ 0\}_R$ as the default new reference base.



```
base1 = {-431.93, -140.61, 368.73, -57.70, -44.98, 33.62}
f0 = {0.002052, 0.000020, -0.002272, 113.9423, 14.9346, -123.1989}
float[] f1 = changeref(f0, Base["base1"].Value)
// f1 = {-431.927,-140.6103,368.7306,-179.288,-0.6893424,-105.84492}
```

6.32 points2coord()

Based on the input points, calculate the coordinate plane of the input points and return the values of the plane parameters converted by the three points on the plane.

Syntax 1

```
float[] points2coord(
    float[],
    float[],
    float[]
)
```

Parameters

float[] The origin coordinates in the coordinate plane
 $X_1(mm) \ Y_1(mm) \ Z_1(mm) \ RX_1(^{\circ}) \ RY_1(^{\circ}) \ RZ_1(^{\circ})$

float[] Any point on the X axis in the plane $X_2(mm) \ Y_2(mm) \ Z_2(mm) \ RX_2(^{\circ}) \ RY_2(^{\circ}) \ RZ_2(^{\circ})$

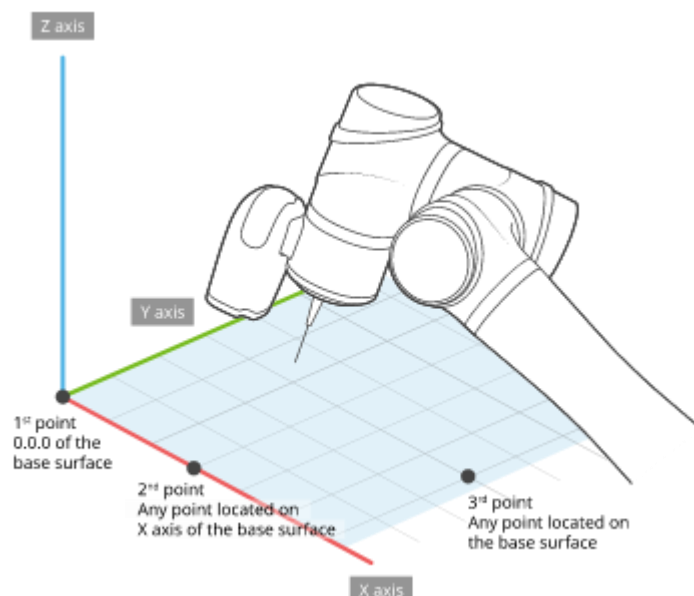
float[] Any point on the +X and +Y in the plane
 $X_3(mm) \ Y_3(mm) \ Z_3(mm) \ RX_3(^{\circ}) \ RY_3(^{\circ}) \ RZ_3(^{\circ})$

* The input points above are required to be described in the same base.

Return

float[] The value of the plane parameter formed by the three points on the plane
 $X_p(mm) \ Y_p(mm) \ Z_p(mm) \ RX_p(^{\circ}) \ RY_p(^{\circ}) \ RZ_p(^{\circ})$
 that coincided with the origin of the plane
 $X_1(mm) \ Y_1(mm) \ Z_1(mm)$
 and calculated with the corners of
 $X_2(mm) \ Y_2(mm) \ Z_2(mm) \ RX_2(^{\circ}) \ RY_2(^{\circ}) \ RZ_2(^{\circ})$
 and
 $X_3(mm) \ Y_3(mm) \ Z_3(mm) \ RX_3(^{\circ}) \ RY_3(^{\circ}) \ RZ_3(^{\circ})$

Note



Supposed there are three points: P1, P2, and P3

Point["P1"].Value = {389.9641,-4.797323,416.2175,177.3384,0.9190881,91.07789}

Point["P2"].Value = {365.0222,137.3036,423.2249,177.4598,0.9549707,112.4033}

Point["P3"].Value = {546.7307,94.02614,385.1812,176.3468,0.5975906,95.82078}

```
Base["points2coord_b1"].Value = points2coord(Point["P1"].Value, Point["P2"].Value,
Point["P3"].Value)
// {389.9641,-4.797323,416.2175,-168.6551,-2.780696,99.9553}
Point["P4"].Value = {0,0,0,0,0,0}
ChangeBase("points2coord_b1")
PTP("CPP",Point["P4"].Value,10,200,0,false)
```

Syntax 2

```
float[] points2coord(
    float, float, float,
    float, float, float,
    float, float, float
)
```

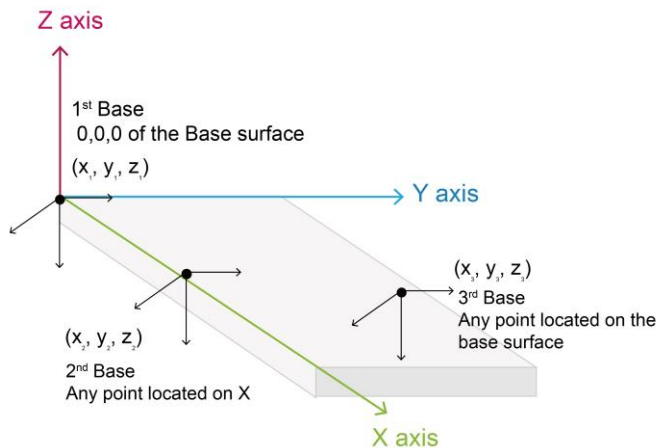
Parameter

- float, float, float The origin coordinates in the coordinate plane to calculate
 $X_1(mm) \ Y_1(mm) \ Z_1(mm)$
 - float, float, float The coordinates of any point on the X axis in the coordinate plane to calculate
 $X_2(mm) \ Y_2(mm) \ Z_2(mm)$
 - float, float, float The coordinates of any point in the coordinate plane to calculate
 $X_3(mm) \ Y_3(mm) \ Z_3(mm)$
- * The input points above are required to be described in the same base.

Return

float[] Definition parameters of the coordinate plane to calculate.
 $X_p(mm) \ Y_p(mm) \ Z_p(mm) \ RX_p(^{\circ}) \ RY_p(^{\circ}) \ RZ_p(^{\circ})$

Note



```
Base["points2coord_2"].Value = points2coord(260,0,360,260,100,360,100,0,360)
// {260,0,360,0,0,90}
Point["P1"].Value = {0,0,0,180,0,0}
ChangeBase("points2coord_2")
PTP("CPP",Point["P1"].Value,10,200,0,false)
```

6.33 intercoord()

Convert two input planes into a new plane.

Syntax 1

```
float[] intercoord(
    float[],
    float[]
)
```

Parameters

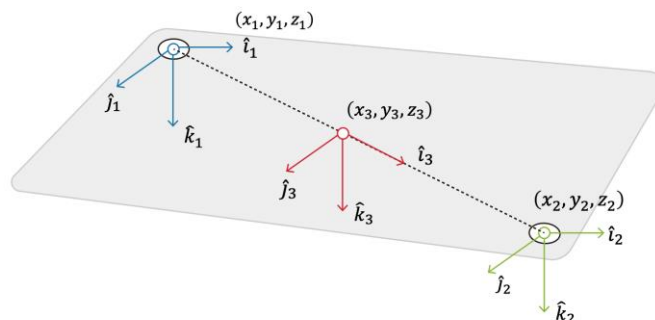
float[] The first input plane $X_1(mm)$ $Y_1(mm)$ $Z_1(mm)$ $RX_1(^{\circ})$ $RY_1(^{\circ})$ $RZ_1(^{\circ})$
float[] The second input plane $X_2(mm)$ $Y_2(mm)$ $Z_2(mm)$ $RX_2(^{\circ})$ $RY_2(^{\circ})$ $RZ_2(^{\circ})$
 * The input points above are required to be described in the same base.

Return

float[] The new plane converted by the two input planes.
 $X_3(mm)$ $Y_3(mm)$ $Z_3(mm)$ $RX_3(^{\circ})$ $RY_3(^{\circ})$ $RZ_3(^{\circ})$

Note

$$\begin{aligned}x_3 &= (x_1 + x_2)/2 \\y_3 &= (y_1 + y_2)/2 \\z_3 &= (z_1 + z_2)/2 \\i_3 &= (x_2 - x_1, y_2 - y_1, z_2 - z_1) \\j_3 &= k_1 \times i_3 \\k_3 &= i_3 \times j_3\end{aligned}$$



Supposed there are vision landmark jobs: vb_1 and vb_2

vb_1 outputs Base["vision_vb_1"].Value = {-69.73,380.02,141.79,-176.11,1.13,-121.27}

vb_2 outputs Base["vision_vb_2"].Value = {58.81,613.03,140.7,171.62,-0.89,0.8}

Base["mix_base"].Value = **intercoord**(Base["vision_vb_1"].Value,
 Base["vision_vb_2"].Value)

// {-5.460001,496.52502,141.245,176.06546,0.23468152,61.116673}

6.34 coordshift()

Calculate a new coordinate by adding a shift value to the original coordinate, where the shift depends on the position and direction of the original coordinate.

Syntax 1

```
float[] coordshift(
    float[],
    float[]
)
```

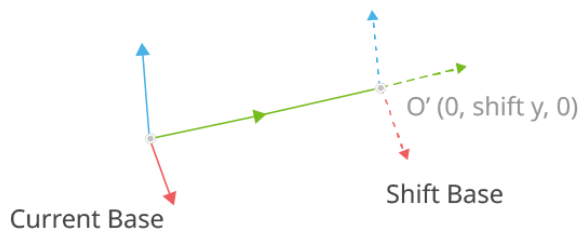
Parameters

float[]	base value	$X_{(mm)}$ $Y_{(mm)}$ $Z_{(mm)}$ $RX_{(^\circ)}$ $RY_{(^\circ)}$ $RZ_{(^\circ)}$
float[]	shift value	$X_{(mm)}$ $Y_{(mm)}$ $Z_{(mm)}$ $RX_{(^\circ)}$ $RY_{(^\circ)}$ $RZ_{(^\circ)}$

Return

float[] the new coordinate value calculated by the given position and direction of the coordinate and the shift value. $X_{(mm)}$ $Y_{(mm)}$ $Z_{(mm)}$ $RX_{(^\circ)}$ $RY_{(^\circ)}$ $RZ_{(^\circ)}$

Note



```
Base["base1"].Value = {200,100,300,0,15,0}
```

```
Base["base2"].Value = {0,0,0,0,0,-90}
```

```
float[] var_shift = {10,50,0,0,0,0}
```

```
Base["base2"].Value = coordshift(Base["base2"].Value, var_shift)
```

```
// {50, -10, 0, 0, 0, -90}
```

```
float[] var_baseNew = coordshift(Base["base1"].Value, Base["base2"].Value)
```

```
// {248.2963, 90, 287.05905, -15, 0, -90}
```


7. File Functions

- The file functions are capable of operations related to file reading, writing, or inquiry.
- File path
 1. Local path. Available in the directories named **TextFiles**, **XmlFiles**, and **TMcraft** only.

<code>FileName.txt</code> <code>.\TextFiles\FileName.txt</code> <code>.\XmlFiles\FileName.xml</code> <code>.\XmlFiles\folder\file</code> <code>..\folder</code> <code>.\TextFiles\...\folder</code>	The directory default to <code>.\TextFiles</code> (File path the same as <code>.\TextFiles\FileName.txt</code>) The file is in the local directory named TextFiles . The file is in the local directory named XmlFiles . The subdirector is in the local directory named XmlFiles . Unavailable. Unavailable.
--	---

Void for absolute paths.

<code>C:\file1</code> <code>D:\folder\file2</code> <code>\TextFiles\FileName.txt</code>	Void. Void. Void.
---	--
 2. External device path. Available to USB drives or SSD labelled **TMROBOT**.

<code>\USB\TMROBOT</code>	The root directory of the external USB drive.
---------------------------	---
 3. Remote path. Available with the Network service in TMflow.

<code>\\127.0.0.1\shared</code> <code>ftp://127.0.0.1</code>	SMB / CIFS FTP
---	-------------------
- The path is not case sensitive. For example, the paths below all point to the same file.
 - `.\TextFiles\FileName.txt`
 - `.\textfiles\fileName.txt`
 - `.\Textfiles\Filename.TXT`
- The path is available for pointing to subdirectories such as:
 - `subfolder\file`
 - `.\TextFiles\subfolder1\subfolder2\file`
 - `.\XmlFiles\subfolder\file`
 - `\USB\TMROBOT\subfolder\file`
 - `\\127.0.0.1\shared\subfolder\file`
- The maximum file size is limited to **10MB (10485760 Bytes)**.
- The type of the array to read or write depends on the definition of the array.

7.1 File_ReadBytes()

Read the file content and return in the type of byte[].

Syntax 1

```
byte[] File_ReadBytes(
    string
)
```

Parameter

string File path

Return

byte[] Return the file content in the type of byte[].

Note

```
.\TextFiles\SampleFile1.txt
1| 1Hello World!
2| 1Hello TM Robot!

byte[] var_bb1 = File_ReadBytes("sampleFile1.txt")
// {0x31,0x48,0x65,0x6C,0x6C,0x6F,0x20,0x57,0x6F,0x72,0x6C,0x64,0x21,0x0D,0x0A,
   0x31,0x48,0x65,0x6C,0x6C,0x6F,0x20,0x54,0x4D,0x20,0x52,0x6F,0x62,0x6F,0x74,0x21}
byte[] var_bb2 = File_ReadBytes(".\TextFiles\SampleFile1.txt")
// {0x31,0x48,0x65,0x6C,0x6C,0x6F,0x20,0x57,0x6F,0x72,0x6C,0x64,0x21,0x0D,0x0A,
   0x31,0x48,0x65,0x6C,0x6C,0x6F,0x20,0x54,0x4D,0x20,0x52,0x6F,0x62,0x6F,0x74,0x21}
byte[] var_bb3 = File_ReadBytes("C:\SampleFile1.txt")      // Error. Void for absolute paths.
byte[] var_bb4 = File_ReadBytes(".\SampleFile1.txt")
// Error. The file is in the local directory named TextFiles or XmlFiles.
byte[] var_bb5 = File_ReadBytes("SampleFileXX.txt")// Error. The file does not exist.
```

7.2 File_ReadText()

Read the file content and return in the type of string.

Syntax 1

```
string File_ReadText(
    string
)
```

Parameter

string File path

Return

string Return the file content in the type of string.

Note

```
.\TextFiles\SampleFile1.txt
1| 1Hello World!
2| 1Hello TM Robot!

string var_s1 = File_ReadText("sampleFile1.txt")
// "1Hello World!\u0D0A1Hello TM Robot!"
string var_s2 = File_ReadText(".\TextFiles\SampleFile1.txt")
// "1Hello World!\u0D0A1Hello TM Robot!"
* \u0D0A denotes a new line character but not a string value.

string var_s3 = File_ReadText("C:\SampleFile1.txt")
// Error. Void for absolute paths.
string var_s4 = File_ReadText(".\SampleFile1.txt")
// Error. The file is in the local directory named TextFiles or XmlFiles.
```

7.3 File_ReadLines()

Read the file content and return in the type of string separated by new line characters .

Syntax 1

```
string[] File_ReadLines(
    string
)
```

Parameter

string File path

Return

string[] Return the file content in the type of string separated by new line characters.

Syntax 2

```
string[] File_ReadLines(
    string,
    int,
    int
)
```

Parameter

string File path

int The number of the line to start to read

int The amount of the lines to read

Return

string[] Return the file content in the type of string separated by new line characters.
 If **the number of the line to start to read** ≤ 0 , it returns an empty array.
 If **the number of the line to start to read** $>$ **the total number of lines**, it returns an empty array.
 If **the amount of the lines to read** ≤ 0 , it returns from the first line to the last line.
 If **the amount of the lines to read** $>$ **the total number of lines**, it returns from the first line to star to read to the last line.

Syntax 3

```
string[] File_ReadLines(
    string,
    int
)
```

Note

Same as Syntax 2 with the parameter of amount of the lines to read defaults to 0 and returns to the last line. **File_ReadLines(string,int,int)** => **File_ReadLines(string,int,0)**

Note

```
.\TextFiles\SampleFile2.txt
1| 2Hello World!
2| 2Hello TM Robot!
3| 2Hi TM Robot!
```

```
string[] var_ss = {""}
```

```

var_ss = File_ReadLines("SampleFile2.txt")           // {"2Hello World!", "2Hello TM Robot!", "2Hi TM
                                                    Robot!"}
var_ss = File_ReadLines("SampleFile2.txt", 1, 2) // {"2Hello World!", "2Hello TM Robot!"}
var_ss = File_ReadLines("SampleFile2.txt", 2, 2) // {"2Hello TM Robot!", "2Hi TM Robot!"}
var_ss = File_ReadLines("SampleFile2.txt", 3, 2) // {"2Hi TM Robot!"} // Tops the total number of
                                                    lines. Returns to the last line.

var_ss = File_ReadLines("SampleFile2.txt", 0)        // {} // empty array
var_ss = File_ReadLines("SampleFile2.txt", 4)        // {} // empty array
int var_len = Length(var_ss)                        // 0
var_ss = File_ReadLines("SampleFile2.txt", 3, 0)    // {"2Hi TM Robot!"} // Returns from line
                                                    3 to the last line.

```

```

.\TextFiles\SampleFile3.txt
1|

```

```

var_ss = File_ReadLines("SampleFile3.txt") // {} // var_ss[0] = ""
var_len = Length(var_ss) // 1

```

7.4 File_NextLine()

Record the last read file path, and continue to read the next line of the file content or open the file to read.

Syntax 1

```
string File_NextLine(
    string
)
```

Parameter

string File path

Return

string If the same as the last read file path, it returns the next line of the file content.

If different from the last read file path, it opens the file and returns the first line of the file content. If read the end of the file, it returns an empty string.

Syntax 2

```
string File_NextLine(
    string,
    bool
)
```

Parameter

string File path

bool Whether open the file to read or not

false Try the file path. Continue to read the next line if the same. Open the file to read if different.

true Open the file and read the first line.

Return

string Whether open the file to read or not false

If the same as the last read file path, it returns the next line of the file content.

If different from the last read file path, it opens the file and returns the first line of the file content.

Whether open the file to read or not true

It opens the file and returns the first line of the file content.

If read the end of the file, it returns an empty string.

Syntax 3

```
string File_NextLine(
)
```

Parameter

void No parameter

Return

string Return the next line of the file content in the last record to read or returns an empty string if not read.

Note

```
.\TextFiles\SampleFile4.txt
1| 4Hello World!
```

```
.\TextFiles\SampleFile5.txt
1| 5Hello World!
```

```
2|
3| 4Hello TM Robot!
```

```
2| 5Hello TM Robot!
3| 5Hi TM Robot!
```

```
string var_s = ""
var_s = File_NextLine()
var_s = File_NextLine("SampleFile4.txt")
var_s = File_NextLine("SampleFile4.txt")
var_s = File_NextLine("SampleFile5.txt")

var_s = File_NextLine("SampleFile4.txt")

var_s = File_NextLine("SampleFile4.txt")
var_s = File_NextLine("SampleFile4.txt")
var_s = File_NextLine("SampleFile4.txt")

var_s = File_NextLine("SampleFile4.txt", true)

var_s = File_NextLine("SampleFile4.txt", true)

var_s = File_NextLine("SampleFile4.txt", false)

var_s = File_NextLine()
```

// "" // Not open the file to read.
// "4Hello World!"
// "" // Continue to read the next line.
// "5Hello World!"
// Different file path. Open the file to read.
// "4Hello World!"
// Different file path. Open the file to read.
// "" // Continue to read the next line
// "4Hello TM Robot!"
// "" // Continue to read the next line (to the EOF)
// "4Hello World!"
// Open the file to read the first line.
// "4Hello World!"
// Open the file to read the first line.
// ""
// Continue to read the next line
// "4Hello TM Robot!"

- * To determine a blank line or the end of the file, use syntax 4 with the size of string[].
- * Or, use File_NextEOF() to determine the end of the file.

Syntax 4

```
string[] File_NextLine(
    string,
    int
)
```

Parameter

string	File path
int	Parameters to read
0	Try the file path. Continue to read the next line if the same. Open the file to read if different.
1	Open the file and read the first line.
2	Open the file without reading. Returns a empty array.

Return

string[] Return strings in the next line of the file content in an array.
 If the array size is 1, it denotes read the strings in the next line.
 If the array size is 0, it denotes read the end of the file already.

Syntax 5

```
string[] File_NextLine(
    int
)
```

Parameter

int	Parameters to read
-----	--------------------

- 0 Try the file path. Continue to read the next line if the same. Open the file to read if different.
- 1 Open the file and read the first line.
- 2 Open the file without reading. Returns a empty array.

Return

string[]

Return strings in the next line of the file content in an array in the last record to read or an empty string array if not read.

If the array size is 1, it denotes read the strings in the next line.

If the array size is 0, it denotes read the end of the file already.

Note

.\TextFiles\SampleFile4.txt

```
1| 4Hello World!
2|
3| 4Hello TM Robot!
```

.\TextFiles\SampleFile5.txt

```
1| 5Hello World!
2| 5Hello TM Robot!
3| 5Hi TM Robot!
```

```
string[] var_ss = {}
var_ss = File_NextLine(0) // {} // Not open the file to read.
var_ss = File_NextLine("SampleFile4.txt", 0) // {"4Hello World!"}
var_ss = File_NextLine("SampleFile4.txt", 0) // {} // Continue to read the next line.
var_ss = File_NextLine("SampleFile5.txt", 0) // {"5Hello World!"}
// Different file path. Open the file to read.
var_ss = File_NextLine("SampleFile4.txt", 0) // {"4Hello World!"}
// Different file path. Open the file to read.
var_ss = File_NextLine("SampleFile4.txt", 0) // {} // Continue to read the next line.
int var_len = Length(var_ss) // 1
var_ss = File_NextLine("SampleFile4.txt", 0) // {"4Hello TM Robot!"}
var_ss = File_NextLine("SampleFile4.txt", 0) // {}
// Continue to read the next line (to the EOF)
var_len = Length(var_ss) // 0
var_ss = File_NextLine("SampleFile4.txt", 1) // {"4Hello World!"}
// Open the file and read the first line.
var_ss = File_NextLine("SampleFile4.txt", 2) // {}
// Open the file without reading.
var_len = Length(var_ss) // 0
var_ss = File_NextLine("SampleFile4.txt") // {"4Hello World!"}
// Continue to read the next line.
var_ss = File_NextLine(0) // {}
// Continue to read the next line.
var_ss = File_NextLine(0) // {"4Hello TM Robot!"}
var_ss = File_NextLine(0) // {}
```


7.5 File_NextEOF()

Try the last read file path for reading to the end of the file already.

Syntax 1

```
bool File_NextEOF(
)
```

Parameter

void No parameter but needs to use with File_NextLine()

Return

bool Return true if not read.
false Not read to the end of the file.
true Not open the file or read to the end of the file.

Note

```
.\TextFiles\SampleFile4.txt
1| 4Hello World!
2|
3| 4Hello TM Robot!

bool var_eof = File_NextEOF()           // true    // Not open the file to read.
string var_s = ""
var_s = File_NextLine("SampleFile4.txt") // "4Hello World!"
var_eof = File_NextEOF()                 // false
var_s = File_NextLine("SampleFile4.txt") // ""
var_eof = File_NextEOF()                 // false
var_s = File_NextLine("SampleFile4.txt") // 4Hello TM Robot!"
var_eof = File_NextEOF()                 // true
var_s = File_NextLine("SampleFile4.txt") // ""

File_NextLine("SampleFile4.txt", 2)      // Open the file without reading.
var_eof = File_NextEOF()                 // false
```

7.6 File_WriteBytes()

Put the data content into an array in byte and write to a file.

Syntax 1

```
bool File_WriteBytes(
    string,
    ?,
    int,
    int,
    int
)
```

Parameter

string	File path
?	Values to write. Eligible for integers, floating-point numbers, Booleans, strings, or arrays.
	Values will be converted with Little Endian, and strings will be converted with UTF8.
int	Overwrite the file or append to the file.
0	Over the file. If not existed, create a new file.
1	Append to the file. If not existed, create a new file.
int	The starting index of the value to write (eligible for strings and arrays)
0 .. length-1	Legitimate value
< 0	Illegitimate value. The starting index will be set to 0.
>= length	Illegitimate value. The starting index will be set to 0.
int	The length of the value to write (eligible for strings and arrays)
<= 0	Write from the starting index to the end of the data.
> 0	Write from the starting index for a specified number of the length up to the data ends.

Return

bool	true	Write successfully.
	false	Write unsuccessfully.

Syntax 2

```
bool File_WriteBytes(
    string,
    ?,
    int,
    int
)
```

Note

Same as Syntax 1. Set the length of the value to write to 0 writing up to the data ends.

File_WriteBytes(string,?,int,int) => File_WriteBytes(string,?,int,int,0)

Syntax 3

```
bool File_WriteBytes(
    string,
    ?,
    int
)
```

Note

Same as Syntax 1. Set the starting index and the length of the value to write to 0 writing up to the data ends.

File_WriteBytes(string,?,int) => File_WriteBytes(string,?,int,0,0)

Syntax 4

```
bool File_WriteBytes(
    string,
    ?
)
```

Note

Same as Syntax 1. Set the starting index and the length of the value to write to 0 overwriting the file up to the data ends.

File_WriteBytes(string,?) => File_WriteBytes(string,?,0,0,0)

```
byte[] var_bb1 = {0x00, 0x01, 0x02, 0x03, 0x04, 0x05, 0x06, 0x07, 0x08}
byte[] var_bb2 = {0x30, 0x31, 0x32, 0x33, 0x34, 0x35, 0x36, 0x37, 0x38}
byte[] var_bb3 = {}
bool flag = false
var_flag = File_WriteBytes("writebytes.txt", var_bb1) // Overwrite var_bb1 to the file
```

```
writebytes.txt
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
00000000 00 01 02 03 04 05 06 07 08
```

```
var_flag = File_WriteBytes("writebytes.txt", var_bb2) // Overwrite var_bb2 to the file
```

```
writebytes.txt
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
00000000 30 31 32 33 34 35 36 37 38
```

```
var_flag = File_WriteBytes("writebytes.txt", var_bb3) // Overwrite var_bb3 to the file
```

```
writebytes.txt
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
00000000
```

```
File_WriteBytes("writebytes.txt", var_bb1, 1)// Append var_bb1 to the file
```

```
writebytes.txt
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
00000000 00 01 02 03 04 05 06 07 08
```

```
File_WriteBytes("writebytes.txt", var_bb2, 1)// Append var_bb2 to the file
```

```
writebytes.txt
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
00000000 00 01 02 03 04 05 06 07 08 30 31 32 33 34 35 36
00000010 37 38
```

```
File_WriteBytes("writebytes.txt", var_bb1, 1, 3)
```

// Append var_bb1 to the file starting from index 3 to the end.

```
writebytes.txt
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
00000000 00 01 02 03 04 05 06 07 08 30 31 32 33 34 35 36
00000010 37 38 03 04 05 06 07 08
```

```
File_WriteBytes("writebytes.txt", var_bb2, 1, 3, 2)
```

// Append var_bb2 to the file starting from index 3 for the length of 2

```
writebytes.txt
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
00000000 00 01 02 03 04 05 06 07 08 30 31 32 33 34 35 36
00000010 37 38 03 04 05 06 07 08 33 34
```

```
File_WriteBytes("writebytes.txt", var_bb1, 1, -1)
```

// -1 illegitimate value // Append var_bb1 to the file starting from index 0 to the end.

```

writebytes.txt
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
00000000 00 01 02 03 04 05 06 07 08 30 31 32 33 34 35 36
00000010 37 38 03 04 05 06 07 08 33 34 00 01 02 03 04 05
00000020 06 07 08

```

File_WriteBytes("writebytes.txt", var_bb2, 1, 0, 100)

// Append var_bb2 to the file starting from index 0 for the length of 100 or to the end.

```

writebytes.txt
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
00000000 00 01 02 03 04 05 06 07 08 30 31 32 33 34 35 36
00000010 37 38 03 04 05 06 07 08 33 34 00 01 02 03 04 05
00000020 06 07 08 30 31 32 33 34 35 36 37 38

```

? **byte** var_n1 = 100 // Convert the value with Little Endian.

File_WriteBytes("writebytes2.txt", var_n1, 1)

```

writebytes2.txt
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F

```

? **byte[]** var_n2 = {100, 200} // Convert every value in the array with Little Endian one after another.

File_WriteBytes("writebytes2.txt", var_n2, 1)

```

writebytes2.txt
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
00000000 64 64 C8 10 27 00 00 10 27 00 00 20 4E 00 00 80

```

? **int** var_n3 = 10000 **File_WriteBytes("writebytes2.txt", var_n3, 1)**

```

writebytes2.txt
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
00000000 64 64 C8 10 27 00 00 10 27 00 00 20 4E 00 00 80
00000010 38 01 00

```

? **int[]** var_n4 = {10000, 20000, 80000}

File_WriteBytes("writebytes2.txt", var_n4, 1)

```

writebytes2.txt
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
00000000 64 64 C8 10 27 00 00 10 27 00 00 20 4E 00 00 80
00000010 38 01 00

```

? **float** var_n5 = 1.234

File_WriteBytes("writebytes2.txt", var_n5, 1)

```

writebytes2.txt
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
00000000 64 64 C8 10 27 00 00 10 27 00 00 20 4E 00 00 80
00000010 38 01 00 B6 F3 9D 3F

```

? **float[]** var_n6 = {1.23, 4.56, -7.89}

File_WriteBytes("writebytes2.txt", var_n6, 1)

```

writebytes2.txt
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
00000000 64 64 C8 10 27 00 00 10 27 00 00 20 4E 00 00 80
00000010 38 01 00 B6 F3 9D 3F A4 70 9D 3F 85 EB 91 40 E1
00000020 7A FC C0

```

? **double** var_n7 = -1.2345

File_WriteBytes("writebytes3.txt", var_n7, 1)

```

writebytes3.txt
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
00000000 8D 97 6E 12 83 C0 F3 BF

```

? **double[]** var_n8 = {1.23, -7.89}

File_WriteBytes("writebytes3.txt", var_n8, 1)

```

writebytes3.txt
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
00000000 8D 97 6E 12 83 C0 F3 BF AE 47 E1 7A 14 AE F3 3F
00000010 8F C2 F5 28 5C 8F 1F C0

```

? **bool** var_n9 = true // Convert true to 1 and false to 0.

File_WriteBytes("writebytes3.txt", var_n9, 1)

```

writebytes3.txt
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
00000000 8D 97 6E 12 83 C0 F3 BF AE 47 E1 7A 14 AE F3 3F
00000010 8F C2 F5 28 5C 8F 1F C0 01

```

? `bool[] var_n10 = {true, false, true, false, false, true, true}`
File_WriteBytes("writebytes3.txt", var_n10, 1)

```

writebytes3.txt
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
00000000 8D 97 6E 12 83 C0 F3 BF AE 47 E1 7A 14 AE F3 3F
00000010 8F C2 F5 28 5C 8F 1F C0 01 01 00 01 00 00 01 01

```

? `string var_n11 = "ABCDEFGH" // Convert the string into UTF8.`
File_WriteBytes("writebytes3.txt", var_n11, 1)

```

writebytes3.txt
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
00000000 8D 97 6E 12 83 C0 F3 BF AE 47 E1 7A 14 AE F3 3F
00000010 8F C2 F5 28 5C 8F 1F C0 01 01 00 01 00 00 01 01
00000020 41 42 43 44 45 46 47

```

? `string[] var_n12 = {"ABC", "DEF", "達明機器人" }`
File_WriteBytes("writebytes3.txt", var_n12, 1)

```

writebytes3.txt
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
00000000 8D 97 6E 12 83 C0 F3 BF AE 47 E1 7A 14 AE F3 3F
00000010 8F C2 F5 28 5C 8F 1F C0 01 01 00 01 00 00 01 01
00000020 41 42 43 44 45 46 47 41 42 43 44 45 46 E9 81 94
00000030 E6 98 8E E6 A9 9F E5 99 A8 E4 BA BA

```

7.7 File_WriteText()

Put the data content in a string and write to a file.

Syntax 1

```
bool File_WriteText(
    string,
    ?,
    int,
    int,
    int
)
```

Parameter

string	File path
?	Values to write. Eligible for integers, floating-point numbers, Booleans, strings, or arrays.
int	Overwrite the file or append to the file.
0	Over the file. If not existed, create a new file.
1	Append to the file. If not existed, create a new file.
int	The starting index of the value to write (eligible for strings and arrays)
0 .. length -1	Legitimate value
< 0	Illegitimate value. The starting index will be set to 0.
>= length	Illegitimate value. The starting index will be set to 0.
int	The length of the value to write (eligible for strings and arrays)
<= 0	Write from the starting index to the end of the data.
> 0	Write from the starting index for a specified number of the length up to the data ends.

Return

bool	true	Write successfully.
	false	Write unsuccessfully.

Syntax 2

```
bool File_WriteText(
    string,
    ?,
    int,
    int
)
```

Note

Same as Syntax 1. Set the length of the value to write to 0 writing up to the data ends.

File_WriteText(string,?,int,int) => File_WriteText(string,?,int,int,0)

Syntax 3

```
bool File_WriteText(
    string,
    ?,
    int
)
```

Note

Same as Syntax 1. Set the starting index and the length of the value to write to 0 writing

up to the data ends.

File_WriteText(string,?,int) => **File_WriteText**(string,?,int,0,0)

Syntax 4

File_WriteText(string,?) => **File_WriteText**(string,?,0,0,0)

string var_s1 = "Hi TM Robot"

string var_s2 = "達明機器人"

bool var_flag = false

var_flag = **File_WriteText**("writetext.txt", var_s1) // Overwrite "Hi TM Robot" to the file

writetext.txt

1| Hi TM Robot

var_flag = **File_WriteText**("writetext.txt", var_s2) // Overwrite "達明機器人" to the file

writetext.txt

1| 達明機器人

var_flag = **File_WriteText**("writetext.txt", var_s1, 1) // Append "Hi TM Robot" to the file

writetext.txt

1| 達明機器人 Hi TM Robot

var_flag = **File_WriteText**("writetext.txt", var_s2, 1, 2, 3)

// Append 3 characters from the starting index to the file

writetext.txt

1| 達明機器人 Hi TM Robot 機器人

? byte var_n1 = 100 // Use decimal for values to convert to strings

File_WriteText("writetext2.txt", var_n1, 1)

writetext2.txt

1| 100

? int[] var_n4 = {10000, 20000, 80000} // Use the "{, ,}" format for arrays

File_WriteText("writetext2.txt", var_n4, 1)

writetext2.txt

1| 100{10000,20000,80000}

// For other formats, use GetString() to convert to a string and write the string

? float var_n5 = 1.234

File_WriteText("writetext2.txt", var_n5, 1)

writetext2.txt

1| 100{10000,20000,80000}1.234

? double[] var_n8 = {1.23, -7.89}

File_WriteText("writetext2.txt", var_n8, 1)

writetext2.txt

1| 100{10000,20000,80000}1.234{1.23,-7.89}

? bool var_n9 = true

File_WriteText("writetext2.txt", var_n9, 1)

writetext2.txt

1| 100{10000,20000,80000}1.234{1.23,-7.89>true

? string var_n11 = "ABCDEFGH"

File_WriteText("writetext2.txt", var_n11, 1)

writetext2.txt

1| 100{10000,20000,80000}1.234{1.23,-7.89>trueABCDEFGH

? `string[] var_n12 = {"ABC", "DEF", "達明機器人"}`

File_WriteText("writetext2.txt", var_n12, 1)

`writetext2.txt`

1 | 100{10000,20000,80000}1.234{1.23,-7.89>trueABCDEFG{ABC,DEF,達明機器人}

7.8 File_WriteLine()

Put the data content in a string with newline characters (0x0D 0x0A) in the end and write to a file.

Syntax 1

```
bool File_WriteLine(
    string,
    ?,
    int,
    int,
    int
)
```

Parameter

string	File path
?	Values to write. Eligible for integers, floating-point numbers, Booleans, strings, or arrays.
int	Overwrite the file or append to the file.
0	Over the file. If not existed, create a new file.
1	Append to the file. If not existed, create a new file.
int	The starting index of the value to write (eligible for strings and arrays)
0 .. length -1	Legitimate value
< 0	Illegitimate value. The starting index will be set to 0.
>= length	Illegitimate value. The starting index will be set to 0.
int	The length of the value to write (eligible for strings and arrays)
<= 0	Write from the starting index to the end of the data.
> 0	Write from the starting index for a specified number of the length up to the data ends.

Return

bool	true	Write successfully.
	false	Write unsuccessfully.

Syntax 2

```
bool File_WriteLine(
    string,
    ?,
    int,
    int
)
```

Note

Same as Syntax 1. Set the length of the value to write to 0 writing up to the data ends.

File_WriteLine(string,?,int,int) => File_WriteLine(string,?,int,int,0)

Syntax 3

```
bool File_WriteLine(
    string,
    ?,
    int
)
```

Note

Same as Syntax 1. Set the starting index and the length of the value to write to 0 writing up to the data ends.

File_WriteLine(string,?,int) => **File_WriteLine**(string,?,int,0,0)

Syntax 4

```
bool File_WriteLine(
    string,
    ?
)
```

Note

Same as Syntax 1. Set the starting index and the length of the value to write to 0 overwriting the file up to the data ends.

File_WriteLine(string,?) => **File_WriteLine**(string,?,0,0,0)

```
string var_s1 = "Hi TM Robot"
string var_s2 = "達明機器人"
bool var_flag = false
var_flag = File_WriteLine("writeline.txt", var_s2)
```

// Overwrite "達明機器人\u00D0A" to the file

writeline.txt

```
1| 達明機器人
2|
```

```
var_flag = File_WriteLine("writeline.txt", var_s1)
```

// Overwrite "Hi TM Robot\u00D0A" to the file

writeline.txt

```
1| Hi TM Robot
2|
```

```
var_flag = File_WriteLine("writeline.txt", var_s2, 1)
```

// Append "達明機器人\u00D0A" to the file

writeline.txt

```
1| Hi TM Robot
2| 達明機器人
3|
```

```
var_flag = File_WriteLine("writeline.txt", var_s1, 1, 3)
```

// Append "TM Robot\u00D0A" from the starting index 3 to the end to the file.

writeline.txt

```
1| Hi TM Robot
2| 達明機器人
3| TM Robot
4|
```

? **byte[]** var_n2 = {100, 200} // The array uses the format {, ,}.

```
File_WriteLine("writeline2.txt", var_n2, 1)
```

writeline2.txt

```
1| {100,200}
2|
```

// For other formats, use GetString() to convert to string and write.

? **int** var_n3 = 10000 // Convert the value in decimal to a string.

```
File_WriteLine("writeline2.txt", var_n3, 1)
```

```
writeline2.txt
```

```
1| {100,200}
2| 10000
3|
```

? **float[]** var_n6 = {1.23, 4.56, -7.89}
File_WriteLine("writeline2.txt", var_n6, 1)

```
writeline2.txt
```

```
1| {100,200}
2| 10000
3| {1.23,4.56,-7.89}
4|
```

? **bool** var_n9 = true
File_WriteLine("writeline2.txt", var_n9, 1)

```
writeline2.txt
```

```
1| {100,200}
2| 10000
3| {1.23,4.56,-7.89}
4| true
5|
```

? **string** var_n11 = "ABCDEFGH"
File_WriteLine("writeline2.txt", var_n11, 1)

```
writeline2.txt
```

```
1| {100,200}
2| 10000
3| {1.23,4.56,-7.89}
4| true
5| ABCDEFG
6|
```

? **string[]** var_n12 = {"ABC", "DEF", "達明機器人"}
File_WriteLine("writeline2.txt", var_n12, 1)

```
writeline2.txt
```

```
1| {100,200}
2| 10000
3| {1.23,4.56,-7.89}
4| true
5| ABCDEFG
6| {ABC,DEF,達明機器人}
7|
```

7.9 File_WriteLines()

Put the data content in a string array with newline characters (0x0D 0x0A) in the end and write to a file.

Syntax 1

```
bool File_WriteLines(
    string,
    ?,
    int,
    int,
    int
)
```

Parameter

string	File path
?	Values to write. Eligible for integers, floating-point numbers, Booleans, strings, or arrays.
int	Overwrite the file or append to the file.
0	Over the file. If not existed, create a new file.
1	Append to the file. If not existed, create a new file.
int	The starting index of the value to write (eligible for strings and arrays)
0 .. length-1	Legitimate value
< 0	Illegitimate value. The starting index will be set to 0.
>= length	Illegitimate value. The starting index will be set to 0.
int	The length of the value to write (eligible for strings and arrays)
<= 0	Write from the starting index to the end of the data.
> 0	Write from the starting index for a specified number of the length up to the data ends.

Return

bool	true	Write successfully.
	false	Write unsuccessfully.

Syntax 2

```
bool File_WriteLines(
    string,
    ?,
    int,
    int
)
```

Note

Same as Syntax 1. Set the length of the value to write to 0 writing up to the data ends.

File_WriteLines(string,?,int,int) => File_WriteLines(string,?,int,int,0)

Syntax 3

```
bool File_WriteLines(
    string,
    ?,
    int
)
```

Note

Same as Syntax 1. Set the starting index and the length of the value to write to 0 writing up to the data ends.

File_WriteLines(string,?,int) => **File_WriteLines**(string,?,int,0,0)

Syntax 4

```
bool File_WriteLines(
    string,
    ?
)
```

Note

Same as Syntax 1. Set the starting index and the length of the value to write to 0 overwriting the file up to the data ends.

File_WriteLines(string,?) => **File_WriteLines**(string,?,0,0,0)

- * **File_WriteText()** : convert the data values to write to a string without adding newline characters in the end of the string.
File_WriteLine(): convert the data values to write to a string with adding newline characters in the end of the string.
File_WriteLines(): convert the data values to write to a string array with adding newline characters in the end of each element of the array.

```
string var_ss1 = {"Hi TM Robot", "達明機器人"}
bool var_flag = false
var_flag = File_WriteLines("writelines.txt", var_ss1)           // Overwrite ss1 to the file
```

```
writelines.txt
1| Hi TM Robot
2| 達明機器人
3|
```

```
var_flag = File_WriteLines("writelines.txt", var_ss1, 1, 1)
// Append ss1 from the starting index 1 to the end to the file.
```

```
writelines.txt
1| Hi TM Robot
2| 達明機器人
3| Hi TM Robot
4|
```

```
? byte[] var_n2 = {100, 200}
File_WriteLines("writelines2.txt", var_n2, 1)
```

```
writelines2.txt
1| 100
2| 200
3|
```

```
? int var_n3 = 10000
File_WriteLines("writelines2.txt", var_n3, 1)
```

```
writelines2.txt
1| 100
2| 200
3| 10000
4|
```

```
? float[] var_n6 = {1.23, 4.56, -7.89}
File_WriteLines("writelines2.txt", var_n6, 1)
```

```
writelines2.txt
```

```
1| 100
2| 200
3| 10000
4| 1.23
5| 4.56
6| -7.89
7|
```

```
? string var_n11 = "ABCDEFGH"
File_WriteLines("writelines2.txt", var_n11, 1)
```

```
writelines2.txt
```

```
1| 100
2| 200
3| 10000
4| 1.23
5| 4.56
6| -7.89
7| ABCDEFG
8|
```

```
? string[] var_n12 = {"ABC", "DEF", "達明機器人"}
File_WriteLines("writelines2.txt", var_n12, 1)
```

```
writelines2.txt
```

```
1| 100
2| 200
3| 10000
4| 1.23
5| 4.56
6| -7.89
7| ABCDEFG
8| ABC
9| DEF
10| 達明機器人
11|
```

7.10 File_Exists()

Check the file path for availability.

Syntax 1

```
bool File_Exists(
    string
)
```

Parameter

string File path

Return

bool true File path available

 false File path unavailable

*Return false file path unavailable for Voided file path without errors.

Note

```
bool var_exists = false
var_exists = File_Exists("sampleFile1.txt")      // true
var_exists = File_Exists("sampleFileX.txt")      // false      // File path unavailable
var_exists = File_Exists("C:\SampleFile1.txt")      // false      // Void for absolute paths.
```

7.11 File_Length()

Check the file size.

Syntax 1

```
int File_Length(
    string
)
```

Parameter

string File path

Return

int In int32 data type. The maximum file size is limited to 2147483647 bytes.
 -1 File path unavailable.
 -2 Exceeded the maximum file size limit.
 * Return -1 file path unavailable for void file path without errors.

Note

```
Int var_len = 0
var_len = File_Length("sampleFile1.txt")      // 31
var_len = File_Length("sampleFileX.txt")      // -1 // File path unavailable
var_len = File_Length("C:\SampleFile1.txt") // -1 // Void for absolute paths.
```


7.12 File_Delete()

Delete the file.

Syntax 1

```
bool File_Delete(
    string
    ...
)
```

Parameter

string File path
... Available for multiple strings.

Return

bool true Delete successfully. (Included unavailable or void file paths)
 false Delete unsuccessfully. (Unable to delete the file for occupied)

Syntax 2

```
bool File_Delete(
    string[]
)
```

Parameter

string[] File path

Return

bool true Delete successfully. (Included unavailable or void file paths)
 false Delete unsuccessfully. (Unable to delete the file for occupied)

Note

```
bool var_flag = false
var_flag = File_Delete("sampleFile1.txt")           // true
var_flag = File_Delete("sampleFileX.txt")           // true    // File path unavailable
var_flag = File_Delete("C:\SampleFile1.txt") // true    // Void for absolute paths.
var_flag = File_Delete("sampleFile1.txt", "sampleFileX.txt")
// Available for multiple file paths.
var_flag = File_Delete("sampleFile1.txt", "sampleFileX.txt", "C:\SampleFile1.txt")
// Available for multiple file paths.

string[] var_ss = {"sampleFile1.txt", "sampleFileX.txt", "C:\SampleFile1.txt"}
var_flag = File_Delete(var_ss)
var_flag = File_Delete(var_ss, "sampleFile2.txt") // Error. Void for syntax.
```

7.13 File_Copy()

Copy the file.

Syntax 1

```
bool File_Copy(
    string,
    string,
    string
)
```

Parameter

string Source file path
 string Target directory path
 string Target file name. Use the source file path equivalent for naming as the

default if empty.

Return

bool true Copy successfully.
 false Copy unsuccessfully.
 * Overwrite the target file if existed in the target path.

Syntax 2

```
bool File_Copy(
    string,
    string
)
```

Note

Same as Syntax 1. Set the target file name with an empty string and use the source file path equivalent for naming.

File_Copy(string,string) => File_Copy(string,string,"")

File_Copy("sampleFile1.txt", ".\TextFiles", "s1.txt")

// copy .\TextFiles\sampleFile1.txt to .\TextFiles\s1.txt

File_Copy("sampleFile1.txt", ".\XmlFiles", "")

// copy .\TextFiles\sampleFile1.txt to .\XmlFiles\sampleFile1.txt

File_Copy("sampleFile1.txt", "\USB\TMROBOT", "s1.txt")

// copy .\TextFiles\sampleFile1.txt to USB\s1.txt

File_Copy("sampleFile1.txt", "\USB\TMROBOT")

// copy .\TextFiles\sampleFile1.txt to USB\sampleFile1.txt

bool var_flag = false

var_flag = **File_Copy("sampleFile1.txt", "C:\folder")** // Error. Void for absolute paths.

var_flag = **File_Copy("sampleFile1.txt", ".")**

// Error. Neither TextFiles nor XmlFiles is in the file path.

7.14 File_CopyImage()

Copy the saved vision file.

Syntax 1

```
bool File_CopyImage(
    string,
    string,
    string,
    int
)
```

Parameters

string	Source saved vision file path
string	Target directory path
	Granted to copy and save images to external paths such as external device path directories or remote path directories
	Not granted to copy and save the image to the local path such as ".\TextFiles" or ".\XmlFiles", and an error will be reported.
string	Target file name.
	If the name is an empty string, the default files name is the same as the one in the source file path.
int	Copy options
0	No error returned when failed to copy. No relative directory of the sourced image reserved. (default)
1	No error returned when failed to copy. The relative directory of the sourced image reserved.
2	An error returned when failed to copy. No relative directory of the sourced image reserved.
3	An error returned when failed to copy. The relative directory of the sourced image reserved.

Return

bool	true	Copy successfully.
	false	Copy unsuccessfully.
		* If the file exists in the destination path, the destination file will be overwritten.

Syntax 2

```
bool File_CopyImage(
    string,
    string,
    string
)
```

Note

Same as syntax 1. Set the copy option to 0. No error returned when failed to copy. No relative directory of the sourced image reserved.

File_CopyImage(string,string,string) => File_CopyImage(string,string,string,0)

Syntax 3

```
bool File_CopyImage(
```

```
string,
string
```

```
)
```

Note

Same as syntax 1. Set the target file name to an empty string and the same as the one in the source file path. Set the copy option to 0. No error returned when failed to copy. No relative directory of the sourced image reserved.

```
File_CopyImage(string,string) => File_CopyImage(string,string,"",0)
```

Syntax 4

```
bool File_CopyImage(
    string,
    string,
    int
```

```
)
```

Note

Same as syntax 1. Set the target file name to an empty string and the same as the one in the source file path.

```
File_CopyImage(string,string,int) => File_CopyImage(string,string,"",int)
```

```
var_bool flag = false
var_flag = File_CopyImage(Job1_ImagePath_TM, ".\TextFiles", "1.png")
// false // Copy to local directories not supported.
var_flag = File_CopyImage(Job1_ImagePath_TM, ".\XmlFiles", "1.png")
// false // Copy to local directories not supported.
var_flag = File_CopyImage(Job1_ImagePath_TM, ".\XmlFiles", "1.png", 2)
// false // Copy to local directories not supported.
var_flag = File_CopyImage(Job1_ImagePath_TM, "USB\TMROBOT", "1.png")
// true // Copy Job1_ImagePath_TM (the vision AOI-only variable) to USB\1.png
var_flag = File_CopyImage(Job1_ImagePath_TM, "USB\TMROBOT", "1.png", 3)
// true // Copy Job1_ImagePath_TM (the vision AOI-only variable) to USB\ProjectName\Job1\Date\source\1.png
// Reserve the directory of the image to save.
var_flag = File_CopyImage(Job1_ImagePath_TM, "USB\TMROBOT")
// true // Copy Job1_ImagePath_TM (the vision AOI-only variable) to USB\15-16-12_423.png
// Reserve the file name of the image to save.
var_flag = File_CopyImage(Job1_ImagePath_TM, "USB\TMROBOT", 3)
// true // Copy Job1_ImagePath_TM (the vision AOI-only variable) to USB\ProjectName\Job1\Date\source\15-16-12_423.png
// Reserve the directory and the file name of the image to save.
```

7.15 File_GetImage()

After executing a task job in TMvision, the system keeps the Source image storage path and the Result image storage path in the image storage path buffer. Users can use this function to retrieve the file path stored in the buffer and proceed to other operations such as copying it externally. This function takes priority to the oldest image storage path to get images (FIFO) and removes the path automatically later.

The maximum number of stored image paths in the buffer is 60. When a new image storage path to add to the buffer, if the capacity is insufficient, the oldest image storage path will be automatically removed from the buffer, and the new image storage path will be automatically added to the buffer.

Syntax 1

```
string[] File_GetImage(
    int
)
```

Parameter

int Waiting time out to retrieve the image storage path
< 0 Wait indefinitely until retrieved the image storage path. (default)
= 0 Retrieve once.
> 0 Time out after waiting for how much time (milliseconds)
Once using waiting, the process stays at this command until retrieved the image storage path or waiting timeout and keeps on the following executions

Return

string[] If SourceImage and ResultImage are empty strings, it denotes it does not retrieve any image storage path.
[0] SourceImage Path
[1] ResultImage Path
[2] ResultData Path
[1] JobName as the name of the vision job

Syntax 2

```
string[] File_GetImage(
)
```

Note

Same as syntax 1. Set the waiting to retrieve timeout to -1 as indefinitely waiting until retrieved the image storage path.

File_GetImage() => **File_GetImage(-1)**

```
string[] var_image = File_GetImage()      // Wait until retrieved the image storage path.
bool var_flag1 = File_CopyImage(var_image[0], "USB\TMROBOT")
bool var_flag2 = File_CopyImage(var_image[1], "USB\TMROBOT")
```

Syntax 3

```
string[] File_GetImage(
    string,
    int
)
```

)

Parameter

string The name of the vision job. Retrieve the job name-matched image storage path
int Waiting time out to retrieve the image storage path
< 0 Wait indefinitely until retrieved the image storage path. (default)
= 0 Retrieve once.
> 0 Time out after waiting for how much time (milliseconds)
 Once using waiting, the process stays at this command until retrieved the image storage path or waiting timeout and keeps on the following executions

Return

string[] If SourcelImage and ResultImage are empty strings, it denotes it does not retrieve any image storag path.
[0] SourcelImage Path
[1] ResultImage Path
[2] ResultData Path
[1] JobName as the name of the vision job

Syntax 4

string[] File_GetImage(
string

)

Note

Same as syntax 3. Set the waiting to retrieve timeout to -1 as indefinitely waiting until retrieved the image storage path.

File_GetImage("Job1") => File_GetImage("Job1", -1)

```
string[] var_image = File_GetImage("Job1")           // Wait until retrieved the image storage path.
bool var_flag1 = File_CopyImage(var_image[0], "USB\TMROBOT")
bool var_flag2 = File_CopyImage(var_image[1], "USB\TMROBOT")
```

7.16 File_Replace()

Replace and overwrite the string in the file with a specified string.

Syntax 1

```
bool File_Replace(
    string,
    string,
    string
)
```

Parameter

string	File path
string	The string to be replaced
string	The string to replace

Return

bool	true	Success 1. The string to be replaced is empty. 2. The string to be replaced is absent. 3. The string to be replaced is found and overwritten in the file.
	false	Failure

Note

```
.\TextFiles\SampleFile6.txt
1| 6Hello World!
2| 6Hello TM Robot!
3| 6Hi TM Robot!

bool var_flag = false
var_flag = File_Replace("SampleFile6.txt", "Hello", "HI")

SampleFile6.txt
1| 6HI World!
2| 6HI TM Robot!
3| 6Hi TM Robot!

var_flag = File_Replace("SampleFile6.txt", "TM", "Techman")

SampleFile6.txt
1| 6HI World!
2| 6HI Techman Robot!
3| 6Hi Techman Robot!

var_flag = File_Replace("SampleFile6.txt", "6", "")

SampleFile6.txt
1| HI World!
2| HI Techman Robot!
3| Hi Techman Robot!
```

7.17 File_GetToken()

Read the file by the string pattern and retrieve the substring in the string.

Syntax 1

```
string File_GetToken(
    string,
    string,
    string,
    int,
    int
)
```

Parameter

string	File path
string	The prefix of the string to retrieve
string	The suffix of the string to retrieve
int	The number of the matched substring to retrieve
>=1	Retrieve the n th matched substring
-1	Retrieve the last matched substring
int	Remove options
0	The 1 st matched not in the start of the input string, and not remove the prefix and the suffix. (default)
1	The 1 st matched not in the start of the input string, and remove the prefix and the suffix.
2	The 1 st matched in the start of the input string, and not remove the prefix and the suffix.
3	The 1 st matched in the start of the input string, and remove the prefix and the suffix.

Return

string Return the retrieved string.
 Return the content of the string in the file if **the prefix and the suffix are empty**.
 Return an empty string if **matching number <=0**.
 If the remove option is 2 or 3, the first match retrieved must be at the start of the input string; otherwise, it returns an empty string.

Syntax 2

```
string File_GetToken(
    string,
    string,
    string,
    int
)
```

Note

Same as Syntax 1. Fill 0 for not removing the prefix and the suffix as the default.

File_GetToken(string,string,string,int) => File_GetToken(string,string,string,int,0)

Syntax 3

```
string File_GetToken(
    string,
    string,
    string
)
```


Note

Same as Syntax 1. Fill 1 for the matching and 0 for not removing the prefix and the suffix as the default.

File_GetToken(string,string,string) => File_GetToken(string,string,string,1,0)

```
.\TextFiles\SampleFile7.txt
1| $Hello World!
2| $Hello TM Robot!
3| $Hi TM Robot!$

string var_n = "SampleFile7.txt"
string var_s = ""
var_s = File_GetToken(var_n, "", "", 0) // "$Hello World!\u000A$Hello TM Robot!\u000A$Hi TM
Robot!$"

var_s = File_GetToken(var_n, "$", "$") // "$Hello World!\u000A$"
var_s = File_GetToken(var_n, "$", "$", 0) // ""
var_s = File_GetToken(var_n, "$", "$", 1) // "$Hello World!\u000A$"
var_s = File_GetToken(var_n, "$", "$", 2) // "$Hi TM Robot!$"
var_s = File_GetToken(var_n, "$", "$", 3) // ""
var_s = File_GetToken(var_n, "$", "$", 1, 1) // "Hello World!\u000A"
var_s = File_GetToken(var_n, "$", "$", 2, 1) // "Hi TM Robot!"
var_s = File_GetToken(var_n, "$", "", 1) // "$Hello World!\u000A"
var_s = File_GetToken(var_n, "$", "", 2) // "$Hello TM Robot!\u000A"
var_s = File_GetToken(var_n, "$", "", 3) // "$Hi TM Robot!"
var_s = File_GetToken(var_n, "$", "", 4) // "$"
var_s = File_GetToken(var_n, "", "$", 1) // "$"
var_s = File_GetToken(var_n, "", "$", 2) // "Hello World!\u000A$"
var_s = File_GetToken(var_n, "", "$", 3) // "Hello TM Robot!\u000A$"
var_s = File_GetToken(var_n, "", "$", 4) // "Hi TM Robot!$"
var_s = File_GetToken(var_n, "$", Ctrl("\r\n"), 1) // "$Hello World!\u000A"
var_s = File_GetToken(var_n, "$", newline, 2) // "$Hello TM Robot!\u000A"
var_s = File_GetToken(var_n, "$", NewLine, 1, 1) // "Hello World!"
// Remove the prefix and the suffix
var_s = File_GetToken(var_n, Ctrl("\r\n"), "$", 1) // "\u000A$"
var_s = File_GetToken(var_n, newline, "$", 2) // "\u000A$"
var_s = File_GetToken(var_n, NewLine, "$", 1, 1) // ""
* \u000A denotes a new line character but not a string value.

.\TextFiles\SampleFile9.txt
1| #Hello World!
2| $Hello TM Robot!
3| $Hi TM Robot!$

string var_n = "SampleFile9.txt"
string var_s = ""
var_s = File_GetToken(var_n, "", "") // "#Hello World!\u000A$Hello TM Robot!\u000A$Hi TM
Robot!$"

var_s = File_GetToken(var_n, "#", "") // "#Hello World!\u000A$Hello TM Robot!\u000A$Hi TM
Robot!$"

var_s = File_GetToken(var_n, "", "$") // "#Hello World!\u000A$"
var_s = File_GetToken(var_n, "#", newline, 1, 0) // "#Hello World!\u000A"
var_s = File_GetToken(var_n, "#", newline, 1, 1) // "Hello World!"
var_s = File_GetToken(var_n, "#", newline, 1, 2) // "#Hello World!\u000A"
```

```

var_s = File_GetToken(var_n, "#", newline, 1, 3)    // "Hello World!"
var_s = File_GetToken(var_n, "$", newline, 1, 0)    // "$Hello TM Robot!\u000A"
var_s = File_GetToken(var_n, "$", newline, 1, 2)    // ""
// $ not in the start of the file. Return an empty string.

var_s = File_GetToken(var_n, "$", "", 1, 2)          // ""
// $ not in the start of the file. Return an empty string.

var_s = File_GetToken(var_n, "#", "", 1, 2)          // "#Hello World!\u000A$Hello TM Robot!\u000A$Hi TM
Robot!$"

var_s = File_GetToken(var_n, "", "$", 1, 2)          // "#Hello World!\u000A$"
var_s = File_GetToken(var_n, "", "$", -1, 2)          // "Hi TM Robot!$"
var_s = File_GetToken(var_n, "", "$", 100, 2)        // "" // Exceeded the matching number
var_s = File_GetToken(var_n, "", "#", 1, 2)          // "#"

```

Syntax 4

```

string File_GetToken(
    string,
    byte[],
    byte[],
    int,
    int
)

```

Parameter

string	File path
byte[]	The prefix of the string to retrieve in the byte array
byte[]	The suffix of the string to retrieve in the byte array
int	The number of the matched substring to retrieve
>=1	Retrieve the n th matched substring
-1	Retrieve the last matched substring
int	Remove options
0	The 1 st matched not in the start of the input string, and not remove the prefix and the suffix. (default)
1	The 1 st matched not in the start of the input string, and remove the prefix and the suffix.
2	The 1 st matched in the start of the input string, and not remove the prefix and the suffix.
3	The 1 st matched in the start of the input string, and remove the prefix and the suffix.

Return

string Return the retrieved string.

Return the content of the string in the file if **the prefix and the suffix are empty**.

Return an empty string if **matching number <=0**.

If the remove option is 2 or 3, the first match retrieved must be at the start of the input string; otherwise, it returns an empty string.

Syntax 5

```

string File_GetToken(
    string,
    byte[],
    byte[],
    int
)

```

Note

Same as Syntax 4. Fill 0 for not removing the prefix and the suffix as the default.

File_GetToken(string,byte[],byte[],int) => File_GetToken(string,byte[],byte[],int,0)

Syntax 6

```
string File_GetToken(
    string,
    byte[],
    byte[]
)
```

Note

Same as Syntax 4. Fill 1 for the matching and 0 for not removing the prefix and the suffix as the default.

File_GetToken(string,byte[],byte[]) => File_GetToken(string,byte[],byte[],1,0)

```
.\TextFiles\SampleFile8.txt
1| $Hello World!
2| Hello$ TM Robot!
3| Hi$ TM Robot!$

string var_n = "SampleFile8.txt", var_s = ""
byte[] var_bb0 = {}, var_bb1 = {0x24}, var_bb2 = {0x0D, 0x0A}
// 0x24 is $ and 0x0D 0x0A is \u0D0A
var_s = File_GetToken(var_n, bb0, bb0, 0)
// "$Hello World\u0D0AHello$ TM Robot!\u0D0AHi$ TM Robot!$"
var_s = File_GetToken(var_n, bb1, bb1) // "$Hello World\u0D0AHello$"
var_s = File_GetToken(var_n, bb1, bb1, 0) // ""
var_s = File_GetToken(var_n, bb1, bb1, 1) // "$Hello World\u0D0AHello$"
var_s = File_GetToken(var_n, bb1, bb1, 2) // "$ TM Robot!$"
var_s = File_GetToken(var_n, bb1, bb1, 3) // ""
var_s = File_GetToken(var_n, bb1, bb1, 1, 1) // "Hello World\u0D0AHello"
var_s = File_GetToken(var_n, bb1, bb1, 2, 1) // " TM Robot!"
var_s = File_GetToken(var_n, bb1, bb0, 1) // "$Hello World\u0D0AHello"
var_s = File_GetToken(var_n, bb1, bb0, 2) // "$ TM Robot!\u0D0AHi"
var_s = File_GetToken(var_n, bb1, bb0, 3) // "$ TM Robot!"
var_s = File_GetToken(var_n, bb1, bb0, 4) // "$"
var_s = File_GetToken(var_n, bb0, bb1, 1) // "$"
var_s = File_GetToken(var_n, bb0, bb1, 2) // "Hello World\u0D0AHello$"
var_s = File_GetToken(var_n, bb0, bb1, 3) // " TM Robot!\u0D0AHi$"
var_s = File_GetToken(var_n, bb0, bb1, 4) // " TM Robot!$"
var_s = File_GetToken(var_n, bb1, bb2, 1) // "$Hello World\u0D0A"
var_s = File_GetToken(var_n, bb1, bb2, 2) // "$ TM Robot!\u0D0A"
var_s = File_GetToken(var_n, bb1, bb2, 1, 1) // "Hello World"
// Remove the prefix and the suffix
var_s = File_GetToken(var_n, bb2, bb1, 1) // "\u0D0AHello$"
var_s = File_GetToken(var_n, bb2, bb1, 2) // "\u0D0AHi$"
var_s = File_GetToken(var_n, bb2, bb1, 1, 1) // "Hello"
```

* **\u0D0A** denotes a new line character but not a string value.

7.18 File_GetAllTokens()

Read the file by the string pattern and retrieve all eligible substrings.

Syntax 1

```
string[] File_GetAllTokens(
    string,
    string,
    string,
    int
)
```

Parameter

string	File path
string	The prefix of the string to retrieve
string	The suffix of the string to retrieve
int	Remove options
0	The 1 st matched not in the start of the input string, and not remove the prefix and the suffix. (default)
1	The 1 st matched not in the start of the input string, and remove the prefix and the suffix.
2	The 1 st matched in the start of the input string, and not remove the prefix and the suffix.
3	The 1 st matched in the start of the input string, and remove the prefix and the suffix.

Return

string[] Return the eligible string in an array.
 Return the content of the string in the file as a string array if [the prefix and the suffix are empty](#).
 If the remove option is 2 or 3, the first match retrieved must be at the start of the input string; otherwise, it returns an empty string.

Syntax 2

```
string[] File_GetAllTokens(
    string,
    string,
    string
)
```

Note

Same as Syntax 1. Fill 0 for not removing the prefix and the suffix as the default.

File_GetAllTokens(string,string,string) => File_GetAllTokens(string,string,string,0)

```
.\TextFiles\SampleFile7.txt
```

```
1| $Hello World!
2| $Hello TM Robot!
3| $Hi TM Robot!$
```

```
string var_n = "SampleFile7.txt"
```

```
string[] var_ss = {}
```

```
var_ss = File_GetAllTokens(var_n, "", "") // {"$Hello World!\u000A$Hello TM Robot!\u000A$Hi TM Robot!$"}
var_ss = File_GetAllTokens(var_n, "$", "$") // {"$Hello World!\u000A$", "$Hi TM Robot!$"}
var_ss = File_GetAllTokens(var_n, "$", "$", 1) // {"Hello World!\u000A", "Hi TM Robot!"}
var_ss = File_GetAllTokens(var_n, "$", "", 1) // {"Hello World!\u000A", "Hello TM Robot!\u000A", "Hi TM Robot!", ""}
```

```

.\TextFiles\SampleFile9.txt
1| #Hello World!
2| $Hello TM Robot!
3| $Hi TM Robot!$

```

```
string var_n = "SampleFile9.txt"
```

```
string[] var_ss = {}
```

```
var_ss = File_GetAllTokens(var_n, "", "")
```

```
// {"#Hello World!\u000A$Hello TM Robot!\u000A$Hi TM Robot!$"}

```

```
var_ss = File_GetAllTokens(var_n, "$", "$", 0)
```

```
// {"$Hello TM Robot!\u000A$"}

```

```
var_ss = File_GetAllTokens(var_n, "$", "$", 1)
```

```
// {"Hello TM Robot!\u000A"}

```

```
var_ss = File_GetAllTokens(var_n, "$", "$", 2)
```

```
// {}

```

```
var_ss = File_GetAllTokens(var_n, "$", "$", 3)
```

```
// {}

```

```
var_ss = File_GetAllTokens(var_n, "$", "", 0)
```

```
// {"$Hello TM Robot!\u000A", "$Hi TM Robot!", "$"}

```

```
var_ss = File_GetAllTokens(var_n, "$", "", 2)
```

```
// {}

```

```
var_ss = File_GetAllTokens(var_n, "#", "", 0)
```

```
// {"#Hello World!\u000A$Hello TM Robot!\u000A$Hi TM Robot!$"}

```

```
var_ss = File_GetAllTokens(var_n, "#", "", 1)
```

```
// {"Hello World!\u000A$Hello TM Robot!\u000A$Hi TM Robot!$"}

```

```
var_ss = File_GetAllTokens(var_n, "#", "", 2)
```

```
// {"#Hello World!\u000A$Hello TM Robot!\u000A$Hi TM Robot!$"}

```

```
var_ss = File_GetAllTokens(var_n, "#", "", 3)
```

```
// {"Hello World!\u000A$Hello TM Robot!\u000A$Hi TM Robot!$"}

```

7.19 File_GetFiles()

Retrieve file names in the assigned directory.

Syntax 1

```
string[] File_GetFiles(
    string,
    string,
    int
)
```

Parameter

string	The assigned directory
string	The string to be searched in the file name is case-insensitive. A search is considered valid even if the string is empty.
int	Retrieving options
0	Retrieve the file names in the specified directory only.
1	Retrieve the file names in the specified directory and all its subdirectories.

Return

string[] Return the file name with the path associated with the assigned directory, such as assigned .\TextFiles as the directory.

```
[0] "Text1.txt"           // ".\TextFiles\Text1.txt"
[1] "Folder2\Text2.txt"  // ".\TextFiles\Folder2\Text2.txt"
```

Syntax 2

```
string[] File_GetFiles(
    string,
    string
)
```

Note

Same as Syntax 1, with 0 as the default retrieving option.

Syntax 3

```
string[] File_GetFiles(
    string
)
```

Note

Same as Syntax 1, with 0 as the default retrieving option and an empty string set to search in the filename.

Note

Supposed that files in the directory \TextFiles are

Text1.txt

Folder2\Text2.txt

```
File_GetFiles(".\TextFiles", "", 0)           // {"Text1.txt"}
File_GetFiles(".\TextFiles", "", 1)           // {"Text1.txt", "Folder2\Text2.txt"}
File_GetFiles(".\TextFiles", "Text1.txt", 1)  // {"Text1.txt"}
File_GetFiles(".\TextFiles", "Text2.txt") // {}
File_GetFiles(".\TextFiles", "Text2.txt", 1)  // {"Folder2\Text2.txt"}
File_GetFiles(".\TextFiles")                  // {"Text1.txt"}
```

Supposed that files in the directory [\\127.0.0.1\shared](#) are

PN-2024-07-07.log

Image1.png

Text1.txt

Folder2\Image1.png

Folder2\Text2.txt

```
File_GetFiles("\\127.0.0.1\shared", "", 0) // {"PN-2024-07-07.log", "Image1.png", "Text1.txt"}
File_GetFiles("\\127.0.0.1\shared", "", 1)
// {"PN-2024-07-07.log", "Image1.png", "Text1.txt", "Folder2\Image1.png", "Folder2\Text2.txt"}
```

```
File_GetFiles("\\127.0.0.1\shared", "image1.png", 0) // {"Image1.png"} // case-insensitive
File_GetFiles("\\127.0.0.1\shared", "image1.png", 1) // {"Image1.png", "Folder2\Image1.png"}
```

```
File_GetFiles("\\127.0.0.1\shared", "*.txt", 1) // {"Text1.txt", "Folder2\Text2.txt"}
File_GetFiles("\\127.0.0.1\shared", "*.txt") // {"Text1.txt"}
```

7.20 File_LogWrite()

Write the log file that messages are available on the view page of the flowchart.

Syntax 1

```
bool File_LogWrite(
    string,
    string,
    string,
    string, ...
)
```

Parameter

string	The device to write
string	The directory to write
string	Title
string	Content

*Titles and content can be multiple.

Return

bool	true	Write successfully.
	false	Write unsuccessfully.

Syntax 2

```
bool File_LogWrite(
    string,
    string
)
```

Parameter

string	Title
string	Content

* The system will use the device and directory settings from the previous execution of Syntax 1. No file writing if there is no previous execution of Syntax 1, but the information will still appear on the view page of the flowchart.

Return

bool	true	Write successfully.
	false	Write unsuccessfully.

Syntax 3

```
bool File_LogWrite(
    string
)
```

Parameter

string Content

* The system will use the device and directory settings from the previous execution of Syntax 1. No file writing if there is no previous execution of Syntax 1, but the information will still appear on the view page of the flowchart.

Return

bool	true	Write successfully.
	false	Write unsuccessfully.

File_LogWrite("text0") // false. Write unsuccessfully for no previous execution of Syntax 1,

File_LogWrite("\USB\TMROBOT", "aaa", "title1", "text1")

// 產出 log 檔在 \USB\TMROBOT\aaa\ProjectName-yyyy-MM-dd.log

// dd/MM/yyyy HH:mm:ss title1=text1

File_LogWrite("\USB\TMROBOT", "aaa", "title2", "text2", "title3", "text3")

// 產出 log 檔在 \USB\TMROBOT\aaa\ProjectName-yyyy-MM-dd.log

// dd/MM/yyyy HH:mm:ss title2=text2

// dd/MM/yyyy HH:mm:ss title3=text3

File_LogWrite("\USB\TMROBOT", "aaa", "title4")

// dd/MM/yyyy HH:mm:ss title4

File_LogWrite("\USB\TMROBOT", "aaa") // Will execute by syntax 2.

// dd/MM/yyyy HH:mm:ss \USB\TMROBOT=aaa

File_LogWrite("title5", "text5")

// dd/MM/yyyy HH:mm:ss title5=text5

File_LogWrite("text6")

// dd/MM/yyyy HH:mm:ss text6

8. Serial Port Functions

8.1 SerialPort Class

Use SerialPort class and declare variables to create a COM port device. The variable name will be the device name.

Construct 1

SerialPort *VariableName* = string, int, string, int, float, int, bool, bool, bool

SerialPort *VariableName* = string, int, string, int, float, int

SerialPort *VariableName* = string, int, string, int, float

SerialPort *VariableName* = string, int

Parameter

string	connection description	
int	bits per second, BaudRate	
string	parity check	"none", "odd", "even", "mark", "space" ("none" by default)
int	Data Bits	5, 6, 7, 8 (8 by default)
float	Stop Bits	1, 1.5, 2 (1 by default)
int	read/write timeout in millisecond	0 .. 10000 (10000 ms by default)
bool	DTR/DSR	true, false (false by default)
bool	RTS/CTS	true, false (false by default)
bool	XON/XOFF	true, false (false by default)

Note

SerialPort spd_c1 = "COM2",115200

// construct a device, with Baudrate 115200

SerialPort spd_c2 = "COM2",115200,"none",8,1

// construct a device with Baudrate 115200, Parity none, DataBits 8, StopBits 1

SerialPort spd_c3 = "COM2",115200,"none",8,1,10000

// construct a device with Baudrate 115200, Parity none, DataBits 8, StopBits 1, Timeout 10000ms

- In a flow project, it will create a device from the serial port list and connect to it actively.
- In a script project, after creating a device by the syntax content, it will not connect to the device. It takes the open device function to connect.
- Whether using the device to read or write, it will ask for confirmation of connecting to the device.

8.2 `com_open()`

Open a Serial Port device.

Syntax 1

```
bool com_open(  
    string  
)
```

Parameter

`string` Serial Port device name

Return

`bool True` Open successfully.

`False` Open unsuccessfully. (The project reports error.)

Note

```
SerialPort spd_dev = "COM2",115200  
com_open("spd_dev")
```

8.3 com_close()

Close a Serial Port device.

Syntax 1

```
bool com_close(  
    string  
)
```

Parameter

string Serial Port device name

Return

bool True Close successfully.

False Close unsuccessfully.

Note

```
SerialPort spd_dev = "COM2",115200  
com_open("spd_dev")  
com_close("spd_dev")
```

When the project starts running as going from the start node, it opens the serial port for connections and receives the data from the serial port consistently. For data received in the received buffer, users can use the function `com_read` to read data in the buffer.

Once opened the Serial Port, it receives data from it continuously and puts them in the receiving buffer. Users can use the function `com_read` or associated functions to get data from the buffer. When the project stops running, it closes the opened Serial Port and clears the receiving buffer.

The receiving buffer comes with a capacity limitation. If there is data coming to the buffer and the buffer is out of space, it removes the earliest data automatically for the latest data coming into the buffer.

8.4 `com_read()`

Read data in the Serial Port received buffer and return an array `byte[]`.

Syntax 1

```
byte[] com_read(
    string
)
```

Parameters

`string` The name of the device on the Serial Port

Return

`byte[]` Return all the data content. If the content is empty, it returns `byte[0]`.

Note

```
ReceivedBuffer =
{0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x20,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
byte[] value = com_read("spd")
// value byte[] = {0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x20,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
// ReceivedBuffer = {}
```

*This function reads all data in the received buffer and clears the received buffer.

Syntax 2

```
byte[] com_read(
    string,
    int,
    int
)
```

Parameters

`string` The name of the device on the Serial Port

`int` The number of the elements to read (based on the length of `byte[]`)

- `<= 0` Read all elements
- `> 0` Read a specified number of the elements (Data is available when the specified number fulfills.)

`int` The length of time to read in millisecond

- `<= 0` Read once only

> 0 Read many times until there is data or the time is up.

Return

byte[] Return the specified number of the elements with byte[]. If the elements is insufficient, it returns byte[0].

Syntax 3

```
byte[] com_read(
    string,
    int
)
```

Note

The syntax is the same as syntax 2. The default length of time to read is 0.

ReceivedBuffer =

```
{0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x20,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
```

```
value = com_read("spd", 6)
```

```
// value byte[] = {0x48,0x65,0x6C,0x6C,0x6F,0x2C}
```

```
// ReceivedBuffer = {0x20,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
```

```
value = com_read("spd", 100)
```

```
// value byte[] = {}
```

```
// Insufficient elements for no more than 100 elements in the received buffer and return byte[0].
```

```
// ReceivedBuffer = {0x20,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
```

```
value = com_read("spd", 0)
```

```
// value byte[] = {0x20,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A} // Read all elements
```

```
// ReceivedBuffer = {}
```

```
value = com_read("spd", 4, 100)
```

```
// value byte[] = {}
```

```
// Insufficient elements for no more than 4 elements in the received buffer and return byte[0].
```

```
// But the length of time to read is set to 100 ms, the process stays in the function until there is data or the time is up and exits the function.
```

```
// ReceivedBuffer = {0x31,0x32,0x33,0x34,0x35,0x36,0x37,0x38}
```

```
// Supposed it receives data after 50ms,
```

```
// value byte[] = {0x31,0x32,0x33,0x34} // it reads 4 element and exits the function.
```

```
// ReceivedBuffer = {0x35,0x36,0x37,0x38}
```

Syntax 4

```
byte[] com_read(
    string,
    byte[] or string,
    byte[] or string,
    int,
    int
)
```

Parameters

string The name of the device on the Serial Port

byte[] or string

Terms of the prefix to read. If the input is byte[0] or "", an empty string, it means no prefix terms.

byte[] or string

Terms of the suffix to read. If the input is byte[0] or "", an empty string, it means no suffix terms.

int Remove options

- 0 The 1st matched not in the start of the input string, and not remove the prefix and the suffix. (default)
- 1 The 1st matched not in the start of the input string, and remove the prefix and the suffix.
- 2 The 1st matched in the start of the input string, and not remove the prefix and the suffix.
- 3 The 1st matched in the start of the input string, and remove the prefix and the suffix.

int The length of time to read in millisecond

- ≤ 0 Read once only
- > 0 Read many times until there is data or the time is up.

Return

byte[] Return with byte[] in the first matched terms of the prefix and the suffix. It retrieves data with the content matches the first of all terms, and the rest will be reserved and not retrieved. If there is no match, it returns byte[0].

Syntax 5

```
byte[] com_read(
    string,
    byte[] or string,
    byte[] or string,
    int
)
```

Note

The syntax is the same as syntax 4. The default length of time to read is 0.

Syntax 6

```
byte[] com_read(
    string,
    byte[] or string,
    byte[] or string
)
```

Note

The syntax is the same as syntax 4. The default is not to remove the prefix and the suffix from the read content and the length of time to read is 0.

```
ReceivedBuffer =
{0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
value = com_read("spd", "He", newline) // prefix "He", suffix \u0D0A
// value byte[] = {0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A} // Hello,\u0D0A
//vReceivedBuffer = {0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
// retrieve the first match and reserve the rest
value = com_read("spd", "", newline, 1) // prefix "", suffix \u0D0A. Remove both the prefix and the
suffix.
// value byte[] = {0x57,0x6F,0x72,0x6C,0x64} // World
// ReceivedBuffer = {}
value = com_read("spd", "", newline, 1, 100)
// prefix "", suffix \u0D0A. Remove both the prefix and the suffix. The length of time to read is 100ms.
// value byte[] = {}
// No matched terms to read. Read byte[0]. Wait for 100 ms.
// ReceivedBuffer = {}
```

```
// ReceivedBuffer = {0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A}
// value byte[] = {0x48,0x65,0x6C,0x6C,0x6F,0x2C}
// ReceivedBuffer = {}
```

```
ReceivedBuffer =
{0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
value = com_read("spd", "lo", newline) // prefix "lo", suffix \u0D0A
// value byte[] = {0x6C,0x6F,0x2C,0x0D,0x0A} // lo,\u0D0A
// The data before the first matched term, {0x48,0x65,0x6C}, will be removed.
// ReceivedBuffer = {0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
// retrieve the first match and reserve the rest
byte[] bb = {}
value = com_read("spd", bb, newline) // prefix byte[0], suffix \u0D0A
// value byte[] = {0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A} // World\u0D0A
// ReceivedBuffer = {}
value = com_read("spd", bb, newline, 0, 100) // prefix byte[0], suffix \u0D0A, 100ms
// value byte[] = {}
// No matched terms to read. Read byte[0]. Wait for 100 ms.
// ReceivedBuffer = {}
// ReceivedBuffer = {0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A}
// value byte[] = {0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A}
// ReceivedBuffer = {}
```

```
ReceivedBuffer = {0x24,0x48,0x65,0x6C,0x6C,0x6F,0x0D,0x0A, // $Hello
0x23,0x48,0x69,0x0D,0x0A, // #Hi
0x24,0x54,0x4D,0x0D,0x0A, // $TM
0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A} // #Robot
value = com_read("spd", "#", newline, 2) // prefix "#" suffix \u0D0A
// value byte[] = {} // # not in the start. Return an empty array.
// ReceivedBuffer = {0x24,0x48,0x65,0x6C,0x6C,0x6F,0x0D,0x0A,0x23,0x48,0x69,0x0D,0x0A,
0x24,0x54,0x4D,0x0D,0x0A,0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A}
value = com_read("spd", "$", newline, 2) // prefix "$" suffix \u0D0A
// value byte[] = {0x24,0x48,0x65,0x6C,0x6C,0x6F,0x0D,0x0A}
// It must be in the start to retrieve the first match.
// ReceivedBuffer = {0x23,0x48,0x69,0x0D,0x0A,
0x24,0x54,0x4D,0x0D,0x0A,0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A}
value = com_read("spd", "#", newline, 2) // prefix "#" suffix \u0D0A
// value byte[] = {0x23,0x48,0x69,0x0D,0x0A} // It must be in the start to retrieve the first match.
// ReceivedBuffer = {0x24,0x54,0x4D,0x0D,0x0A,0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A}
value = com_read("spd", "$", newline, 3) // prefix "$" suffix \u0D0A
// value byte[] = {0x54,0x4D}
// ReceivedBuffer = {0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A}
value = com_read("spd", "#", newline, 3) // prefix "#" suffix \u0D0A
// value byte[] = {0x52,0x6F,0x62,0x6F,0x74}
// ReceivedBuffer = {}
```

Syntax 7

byte[] com_read(


```

string,
byte[] or string,
int,
int
)

```

Parameters

string The name of the device on the Serial Port

byte[] or string Terms of the suffix to read. If the input is byte[0] or "", an empty string, it means no suffix terms.

int Remove options

- 0** The 1st matched not in the start of the input string, and not remove the prefix and the suffix. (default)
- 1** The 1st matched not in the start of the input string, and remove the prefix and the suffix.
- 2** The 1st matched in the start of the input string, and not remove the prefix and the suffix.
- 3** The 1st matched in the start of the input string, and remove the prefix and the suffix.

int The length of time to read in millisecond

- <= 0** Read once only
- > 0** Read many times until there is data or the time is up.

* No terms of the prefix to read.

Return

byte[] Return with byte[] in the first matched terms of the prefix and the suffix. It retrieves data with the content matches the first of all terms, and the rest will be reserved and not retrieved. If there is no match, it returns byte[0].

Syntax 8

```

byte[] com_read(
    string,
    byte[] or string,
    int
)

```

Note

The syntax is the same as syntax 7. The default length of time to read is 0.

Syntax 9

```

byte[] com_read(
    string,
    byte[] or string
)

```

Note

The syntax is the same as syntax 7. The default is not to remove the prefix and the suffix from the read content and the length of time to read is 0.

Note

```

ReceivedBuffer =
{0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
value = com_read("spd", newline) // suffix \u0D0A
// value byte[] = {0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A} // Hello,\u0D0A
// ReceivedBuffer = {0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
// retrieve the first match and reserve the rest

```

```

value = com_read("spd", newline) // suffix \u0D0A
// value byte[] = {0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A} // World\u0D0A
// ReceivedBuffer = {}

```

```

ReceivedBuffer =
{0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
value = com_read("spd", newline, 1) // suffix \u0D0A
// value byte[] = {0x48,0x65,0x6C,0x6C,0x6F,0x2C} // Hello,
// ReceivedBuffer = {0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
// retrieve the first match and reserve the rest
value = com_read("spd", newline, 1) // suffix \u0D0A
// value byte[] = {0x57,0x6F,0x72,0x6C,0x64} // World
// ReceivedBuffer = {}
value = com_read("spd", newline, 1, 100) // suffix \u0D0A, 100ms
// value byte[] = {}
// No matched terms to read. Read byte[0]. Wait for 100 ms.
// ReceivedBuffer = {}
// ReceivedBuffer = {0x31,0x32,0x33,0x34,0x35,0x36,0x0D,0x0A}
// value byte[] = {0x31,0x32,0x33,0x34,0x35,0x36}
// ReceivedBuffer = {}

```

8.5 `com_read_string()`

Read the data in the Serial Port buffer, and convert the data to a UTF8 string.

Syntax 1

```
string com_read_string(
    string
)
```

Parameters

string The name of the device on the Serial Port

Return

string Return all the data content. If the content is empty, it returns an empty string.

Syntax 2

```
string com_read_string(
    string,
    int,
    int
)
```

Parameters

string The name of the device on the Serial Port

int The number of characters to read (based on the number of characters of the string)

- ≤ 0 Read all characters
- > 0 Read a specified number of the characters (Data is available when the specified number fulfills.)

int The length of time to read in millisecond

- ≤ 0 Read once only
- > 0 Read many times until there is data or the time is up.

Return

string Returns the specified number of characters as a string. If the characters are insufficient, it returns an empty string.

Syntax 3

```
string com_read_string(
    string,
    int
)
```

Note

The syntax is the same as syntax 2. The default length of time to read is 0.

Syntax 4

```
string com_read_string(
    string,
    byte[] or string,
    byte[] or string,
    int,
    int
)
```

Parameters

string The name of the device on the Serial Port

byte[] or string Terms of the prefix to read. If the input is byte[0] or "", an empty string, it means no prefix terms.

byte[] or string Terms of the suffix to read. If the input is byte[0] or "", an empty string, it means no suffix terms.

int Remove options

- 0** The 1st matched not in the start of the input string, and not remove the prefix and the suffix. (default)
- 1** The 1st matched not in the start of the input string, and remove the prefix and the suffix.
- 2** The 1st matched in the start of the input string, and not remove the prefix and the suffix.
- 3** The 1st matched in the start of the input string, and remove the prefix and the suffix.

int The length of time to read in millisecond

- <= 0** Read once only
- > 0** Read many times until there is data or the time is up.

Return

string It retrieves data with the content matches the first of all terms
It retrieves data with the content matches the first of all terms, and the rest will be reserved and not retrieved.
If there is no match, it returns an empty string.

Syntax 5

```
string com_read_string(
    string,
    byte[] or string,
    byte[] or string,
    int
)
```

Note

The syntax is the same as syntax 4. The default length of time to read is 0.

Syntax 6

```
string com_read_string(
    string,
    byte[] or string,
    byte[] or string
)
```

Note

The syntax is the same as syntax 4. The default is not to remove the prefix and the suffix from the read content and the length of time to read is 0.

Syntax 7

```
string com_read_string(
    string,
    byte[] or string,
    int,
    int
)
```

Parameters

- string** The name of the device on the Serial Port
- byte[] or string** Terms of the suffix to read. If the input is byte[0] or "", an empty string, it means no suffix terms.
- int** Remove options
- 0** The 1st matched not in the start of the input string, and not remove the prefix and the suffix. (default)
 - 1** The 1st matched not in the start of the input string, and remove the prefix and the suffix.
 - 2** The 1st matched in the start of the input string, and not remove the prefix and the suffix.
 - 3** The 1st matched in the start of the input string, and remove the prefix and the suffix.
- int** The length of time to read in millisecond
- <= 0** Read once only
 - > 0** Read many times until there is data or the time is up.
- * No terms of the prefix to read.

Return

- string** Returns as a string with the first matched terms of the prefix and the suffix. Retrieves data in the content matches to the first of all terms, and the rest will be reserved and not retrieved. If there is no match, it returns an empty string.

Syntax 8

```
string com_read_string(
    string,
    byte[] or string,
    int
)
```

Note

The syntax is the same as syntax 7. The default length of time to read is 0.

Syntax 9

```
string com_read_string(
    string,
    byte[] or string
)
```

Note

The syntax is the same as syntax 7. The default is not to remove the prefix and the suffix from the read content and the length of time to read is 0.

Note

ReceivedBuffer =

```
{0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x20,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
```

string value = **com_read_string**("spd")

```
// value string = "Hello, World\u00D0A"
```

```
// ReceivedBuffer = {}
```

ReceivedBuffer =

```
{0x54,0x4D,0xE9,0x81,0x94,0xE6,0x98,0x8E,0xE6,0xA9,0x9F,0xE5,0x99,0xA8,0xE4,0xBA,0xBA}
```

value = **com_read_string**("spd", 4)

```

// value string = "TM 達明" // {0x54,0x4D,0xE9,0x81,0x94,0xE6,0x98,0x8E}
//Retrieve 4 characters based on the length of the string.
// var _ReceivedBuffer = {0xE6,0xA9,0x9F,0xE5,0x99,0xA8,0xE4,0xBA,0xBA}
value = com_read_string("spd", 5, 100)
// value string = ""
// Insufficient characters for no more than 5 characters in the received buffer based on the length of the
string. Wait for 100 ms.
// ReceivedBuffer = {0xE6,0xA9,0x9F,0xE5,0x99,0xA8,0xE4,0xBA,0xBA}
// ReceivedBuffer = {0xE6,0xA9,0x9F,0xE5,0x99,0xA8,0xE4,0xBA,0xBA, 0x0D, 0x0A}
// value string = "機器人\u00D0A"
// ReceivedBuffer = {}

```

```

ReceivedBuffer =
{0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
value = com_read_string("spd", "He", newline) // prefix "He", suffix \u00D0A
// value string = "Hello,\u00D0A"
// ReceivedBuffer = {0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A} // retrieve the first match and reserve the
rest
value = com_read_string("spd", "", newline, 1)
// prefix "", suffix \u00D0A. Remove both the prefix and the suffix.
// value string = "World"
// ReceivedBuffer = {}
value = com_read_string("spd", "", newline, 1, 100)
// value string = "" // No matched terms to read. Read an empty string. Wait for 100 ms.
// ReceivedBuffer = {}
// ReceivedBuffer = {0xE6,0xA9,0x9F,0xE5,0x99,0xA8,0xE4,0xBA,0xBA, 0x0D, 0x0A}
// value string = "機器人"

```

```

ReceivedBuffer =
{0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
value = com_read_string("spd", "lo", newline) // prefix "lo", suffix \u00D0A
// value string = "lo,\u00D0A"
// The data before the first matched term, "Hel", will be removed.
// ReceivedBuffer = {0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
// retrieve the first match and reserve the rest
value = com_read_string("spd", newline, 1) // suffix \u00D0A
// value string = "World"
// ReceivedBuffer = {}

```

```

ReceivedBuffer = {0x24,0x48,0x65,0x6C,0x6C,0x6F,0x0D,0x0A, // $Hello
0x23,0x48,0x69,0x0D,0x0A, // #Hi
0x24,0x54,0x4D,0x0D,0x0A, // $TM
0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A} // #Robot

```

```

value = com_read_string("spd", "#", newline, 2) // prefix "#" suffix \u00D0A
// value string = "" // # not in the start. Return an empty array.
// ReceivedBuffer = {0x24,0x48,0x65,0x6C,0x6C,0x6F,0x0D,0x0A,0x23,0x48,0x69,0x0D,0x0A,
0x24,0x54,0x4D,0x0D,0x0A,0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A}

```

```

value = com_read_string("spd", "$", newline, 2)      // prefix "$" suffix \u0D0A
// value string = "$Hello\u0D0A"                    // It must be in the start to retrieve the first match.
// ReceivedBuffer = {0x23,0x48,0x69,0x0D,0x0A,
//                   0x24,0x54,0x4D,0x0D,0x0A,0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A}
value = com_read_string("spd", "#", newline, 2)      // prefix "#" suffix \u0D0A
// value string = "#Hi\u0D0A"                       // It must be in the start to retrieve the first match.
// ReceivedBuffer = {0x24,0x54,0x4D,0x0D,0x0A,0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A}

value = com_read_string("spd", "$", newline, 3)      // prefix "$" suffix \u0D0A
// value string = "TM"
// ReceivedBuffer = {0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A}
value = com_read_string("spd", "#", newline, 3)      // prefix "#" suffix \u0D0A
// value string = "Robot"
// ReceivedBuffer = {}
ReceivedBuffer =
{0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A,0x57,0x6F,0x72,0x6C,0x64}
value = com_read_string("spd", newline)             // suffix \u0D0A
// value string = "Hello,\u0D0A"
// ReceivedBuffer = {0x57,0x6F,0x72,0x6C,0x64} // retrieve the first match and reserve the rest
value = com_read_string("spd", newline, 0) // suffix \u0D0A
// value string = ""                                // No matched terms to read. Read an empty string.
// ReceivedBuffer = {0x57,0x6F,0x72,0x6C,0x64}
value = com_read_string("spd", newline, 1, 100) // suffix \u0D0A
// value string = ""
// No matched terms to read. Read an empty string. Wait for 100 ms.
// ReceivedBuffer = {0x57,0x6F,0x72,0x6C,0x64}
// ReceivedBuffer = {0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A,0x31,0x32,0x33,0x0D,0x0A}
// value string = "World"
// ReceivedBuffer = {0x31,0x32,0x33,0x0D,0x0A} // retrieve the first match and reserve the rest
value = com_read_string("spd", newline, 2) // suffix \u0D0A
// value string = "123\u0D0A"
// ReceivedBuffer = {}

```

8.6 `com_write()`

Write data to the Serial Port

Syntax 1

```
bool com_write(
    string,
    ?,
    int,
    int
)
```

Parameters

<code>string</code>	The name of the device on the Serial Port						
<code>?</code>	The value to write. Available types: int, float, , bool, string, and array. Numeric values will be conversed in Little Endian, and string values will be converse in UTF8.						
<code>int</code>	The starting index of the value to write (eligible for strings and arrays) <table border="0"> <tr> <td><code>0 .. length-1</code></td><td>Legitimate value</td></tr> <tr> <td><code>< 0</code></td><td>Illegitimate value. The starting index will be set to 0.</td></tr> <tr> <td><code>>= length</code></td><td>Illegitimate value. The starting index will be set to 0.</td></tr> </table>	<code>0 .. length-1</code>	Legitimate value	<code>< 0</code>	Illegitimate value. The starting index will be set to 0.	<code>>= length</code>	Illegitimate value. The starting index will be set to 0.
<code>0 .. length-1</code>	Legitimate value						
<code>< 0</code>	Illegitimate value. The starting index will be set to 0.						
<code>>= length</code>	Illegitimate value. The starting index will be set to 0.						
<code>int</code>	The length of the value to write (eligible for strings and arrays) <table border="0"> <tr> <td><code><= 0</code></td><td>Write from the starting index to the end of the data.</td></tr> <tr> <td><code>> 0</code></td><td>Write from the starting index for a specified number of the length up to the data ends.</td></tr> </table>	<code><= 0</code>	Write from the starting index to the end of the data.	<code>> 0</code>	Write from the starting index for a specified number of the length up to the data ends.		
<code><= 0</code>	Write from the starting index to the end of the data.						
<code>> 0</code>	Write from the starting index for a specified number of the length up to the data ends.						

Return

<code>Bool</code>	<code>True</code>	write successfully
	<code>False</code>	write unsuccessfully
		1. The value to write is an empty string or an empty array.
		2. Unable to send to the serial port correctly.

Syntax 2

```
bool com_write(
    string,
    ?,
    int,
    int
)
```

Parameters

string	The name of the device on the Serial Port	
?	The value to write. Available types: int, float, , bool, string, and array. Numeric values will be conversed in Little Endian, and string values will be converse in UTF8.	
int	The starting index of the data to write. (valid for strings or arrays)	
	0	The length of the string Legal value
	- 1	
	< 0	Illegal value, and the starting index will be 0.
	>=	The length of the string Illegal value, and the starting index will be 0.
int	The length of the data to write. (valid for strings or arrays)	

<=
0
> 0

From the start of the index to the end of the data
From the start of the index, write the specified length of the data until the data ends.

Return

bool True write successfully
False write unsuccessfully
1. The value to write is an empty string or an empty array.
2. Unable to send to the serial port correctly.

Syntax 3

```
bool com_write(
    string,
    ?
)
```

Note

The syntax is the same as syntax 2. The default length of data to write is 0.

```
flag = com_write("spd", 100)           // write 0x64
flag = com_write("spd", 1000)          // write 0xE8 0x03 0x00 0x00 (int, Little Endian)
flag = com_write("spd", (float)1.234)  // write 0xB6 0xF3 0x9D 0x3F (float, Little Endian)
flag = com_write("spd", (double)123.456)
// write 0x77 0xBE 0x9F 0x1A 0x2F 0xDD 0x5E 0x40 (double, Little Endian)
flag = com_write("spd", "Hello, World"+newline)
// write 0x48 0x65 0x6C 0x6C 0x6F 0x2C 0x20 0x57 0x6F 0x72 0x6C 0x64 0x0D 0x0A (string, UTF8)
flag = com_write("spd", 1000, 1, 2) //Invalid in the value, the starting index, and the length
// write 0xE8 0x03 0x00 0x00 (int, Little Endian)

byte[] bb = {100, 200}
flag = com_write("spd", bb)           // write 0x64 0xC8
flag = com_write("spd", bb, 1, 1)     // write 0xC8
// Array. Retrieve 1 element from the index 1. [1]=200
flag = com_write("spd", bb, -1, 1)    // write 0x64
// Array. Retrieve 1 element from the index 0. [0]=100
flag = com_write("spd", "達明機器人", 2)
// String. Retrieve from the index 2 until the index ends. "機器人"
// write 0xE6 0xA9 0x9F 0xE5 0x99 0xA8 0xE4 0xBA 0xBA (string, UTF8)
string[] ss = {"TM", "", "達明機器人"}
flag = com_write("spd", ss)
// write 0x54 0x4D 0xE9 0x81 0x94 0xE6 0x98 0x8E 0xE6 0xA9 0x9F 0xE5 0x99 0xA8 0xE4 0xBA 0xBA
flag = com_write("spd", Byte_Concat(GetBytes(ss), GetBytes(newline)))
// write 0x54 0x4D 0xE9 0x81 0x94 0xE6 0x98 0x8E 0xE6 0xA9 0x9F 0xE5 0x99 0xA8 0xE4 0xBA 0xBA
0x0D 0x0A
flag = com_write("spd", ss, 2, 100)
// Array. Retrieve 100 elements (to the end) from the index 2. [2]=達明機器人
// write 0xE9 0x81 0x94 0xE6 0x98 0x8E 0xE6 0xA9 0x9F 0xE5 0x99 0xA8 0xE4 0xBA 0xBA
```

8.7 com_writeline()

Write data to the Serial Port and add line break symbols, 0x0D 0x0A, in the end of the data automatically

Syntax 1

```
bool com_writeline(
    string,
    ?,
    int,
    int
)
```

Parameters

string	The name of the device on the Serial Port
?	The value to write. Available types: int, float, , bool, string, and array. Numeric values will be conversed in Little Endian, and string values will be converse in UTF8.
int	The starting index of the value to write (eligible for strings and arrays)
0 .. length-1	Legitimate value
< 0	Illegitimate value. The starting index will be set to 0.
>= length	Illegitimate value. The starting index will be set to 0.
int	The length of the value to write (eligible for strings and arrays)
<= 0	Write from the starting index to the end of the data.
> 0	Write from the starting index for a specified number of the length up to the data ends.

Return

bool True	write successfully
False	write unsuccessfully
	1. The value to write is an empty string or an empty array.
	2. Unable to send to the serial port correctly.

Syntax 2

```
bool com_writeline(
    string,
    ?,
    int,
)
```

Note

The syntax is the same as syntax 1. The default length of data to write is 0.

Syntax 3

```
bool com_writeline(
    string,
    ?,
)
```

Note

The syntax is the same as syntax 1. The default starting index of the data to write is 0.

The default length of data to write is 0.

```

flag = com_writeline("spd", 100)           // write 0x64 0x0D 0x0A
flag = com_writeline("spd", 1000)         // write 0xE8 0x03 0x00 0x00 0x0D 0x0A (int, Little
Little Endian)
flag = com_writeline("spd", (float)1.234) // write 0xB6 0xF3 0x9D 0x3F 0x0D 0x0A (float,
Little Endian)
flag = com_writeline("spd", (double)123.456)
// write 0x77 0xBE 0x9F 0x1A 0x2F 0xDD 0x5E 0x40 0x0D 0x0A (double, Little Endian)
flag = com_write("spd", "Hello, World"+newline)
// write 0x48 0x65 0x6C 0x6C 0x6F 0x2C 0x20 0x57 0x6F 0x72 0x6C 0x64 0x0D 0x0A (string, UTF8)
flag = com_writeline("spd", "Hello, World")
// write 0x48 0x65 0x6C 0x6C 0x6F 0x2C 0x20 0x57 0x6F 0x72 0x6C 0x64 0x0D 0x0A (string, UTF8)
flag = com_writeline("spd", 1000, 1, 2) // Invalid in the value, the starting index, and the length
// write 0xE8 0x03 0x00 0x00 0x0D 0x0A (int, Little Endian)

byte[] bb = {100, 200}
flag = com_writeline("spd", bb)           // write 0x64 0xC8 0x0D 0x0A
flag = com_writeline("spd", bb, 1, 1)     // write 0xC8 0x0D 0x0A
// Array. Retrieve 1 element from the index 1. [1]=200
flag = com_writeline("spd", bb, -1, 1)    // write 0x64 0x0D 0x0A
// Array. Retrieve 1 element from the index 0. [0]=100
flag = com_writeline("spd", "達明機器人", 2)
// String. Retrieve from the index 2 until the index ends. "機器人"
// write 0xE6 0xA9 0x9F 0xE5 0x99 0xA8 0xE4 0xBA 0xBA 0x0D 0x0A (string, UTF8)
string[] ss = {"TM", "", "達明機器人"}
flag = com_writeline("spd", ss)
// write 0x54 0x4D 0xE9 0x81 0x94 0xE6 0x98 0x8E 0xE6 0xA9 0x9F 0xE5 0x99 0xA8 0xE4 0xBA 0xBA
0x0D 0x0A
flag = com_write("spd", Byte_Concat(GetBytes(ss), GetBytes(newline)))
// write 0x54 0x4D 0xE9 0x81 0x94 0xE6 0x98 0x8E 0xE6 0xA9 0x9F 0xE5 0x99 0xA8 0xE4 0xBA 0xBA
0x0D 0x0A
flag = com_writeline("spd", ss)
// write 0x54 0x4D 0xE9 0x81 0x94 0xE6 0x98 0x8E 0xE6 0xA9 0x9F 0xE5 0x99 0xA8 0xE4 0xBA 0xBA
0x0D 0x0A
flag = com_writeline("spd", ss, 2, 100)
// Array. Retrieve 100 elements (to the end) from the index 2. [2]=達明機器人
// write 0xE9 0x81 0x94 0xE6 0x98 0x8E 0xE6 0xA9 0x9F 0xE5 0x99 0xA8 0xE4 0xBA 0xBA 0x0D
0x0A

```

9. Socket Functions

9.1 Socket Class

Use Socket class and declare variables to create a TCP/IP communication device. The variable name will be the device name.

Construct

Socket VariableName = string, int, int

Socket VariableName = string, int

Parameter

string the IP address of the remote host
int the connection port of the remote host
int read/write timeout in millisecond 0 .. 10000 (10000ms by default)

Note

Socket ntd_d1 = "192.168.1.10", 12345

// construct a device, with IP 192.168.1.10, Port 12345, Timeout 10000ms

Socket ntd_d2 = "192.168.1.11", 9999, 8000

// construct a device, with IP 192.168.1.10, Port 9999, Timeout 8000ms

- In a flow project, it will create a device from the network device list and open it.
- In a script project, after creating a device by the syntax content, it will not connect to the device. It takes the open device function to connect.
- While reading or writing with the device, it confirms if it needs to connect to the device.

9.2 socket_open()

Open a TCP/IP device.

Syntax 1

```
bool socket_open(  
    string  
)
```

Parameter

string TCP/IP device name

Return

bool True Open successfully.

False Open unsuccessfully.

Note

Socket ntd_dev = "192.168.1.10", 12345

socket_open("ntd_dev")

9.3 socket_close()

Close a TCP/IP device.

Syntax 1

```
bool socket_close(  
    string  
)
```

Parameter

string TCP/IP device name

Return

bool True Open successfully.

False Open unsuccessfully.

Note

Socket ntd_dev = "192.168.1.10", 12345

socket_open("ntd_dev")

socket_close("ntd_dev")

When a flow project starts running as the Start node initiates, it launches the TCP/IP Socket Client to connect to the specified IP and port. However, as to the script project, users have to use the `socket_open` function to open and connect to the assigned device.

After connecting to the TCP/IP device, the system keeps receiving the data in the connection, brings the received data to the Received Buffer, and uses respective functions such as `Socket_read` to read the data. When the project stops running, the existing TCP/IP Socket connection will be closed with the Received Buffer cleared.

The Received Buffer comes with a capacity limit. If the buffer capacity is insufficient when the data comes in, it automatically deletes the oldest data and adds the latest.

9.4 `socket_read()`

Read data in the Received Buffer and return in a byte array.

Syntax 1

```
byte[] socket_read(
    string
)
```

Parameters

`string` Network device name

Return

`byte[]` Return all data in the Received Buffer. Return `byte[0]` if buffer empty.

Note

```
ReceivedBuffer =
{0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x20,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
byte[] var_value = socket_read("ntd_a")
// byte[] = {0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x20,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
// ReceivedBuffer = {}
```

*This function reads all data in the Received Buffer and clear the buffer.

Syntax 2

```
byte[] socket_read(
    string,
    int,
    int
)
```

Parameters

`string` Network device name

`int` Retrieve the fixed amount of byte (by the length of `byte[]`)

`<= 0` Get all

`> 0` Get a specified amount (the specified amount required to get the data)

`int` Read time (millisecond)

`<= 0` Read once

`> 0` Read multiple times until there is data or the times fulfill.

Return

`byte[]` Return the specified amount of data in the Received Buffer with a byte array. Return `byte[0]` if insufficient amount.

Syntax 3

```
byte[] socket_read(
    string,
    int
)
```

Note

Same as syntax 2. Fill 0 as the read time by default.

```
ReceivedBuffer =
{0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x20,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
byte[] var_value = socket_read("ntd_a", 6)
    // byte[] = {0x48,0x65,0x6C,0x6C,0x6F,0x2C}
    // ReceivedBuffer = {0x20,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
var_value = socket_read("ntd_a", 100)
    // byte[] = {}           // The number is insufficient for less than 100 bytes in the Received Buffer. Byte[0] will return.
    // ReceivedBuffer = {0x20,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
var_value = socket_read("ntd_a", 0)
    // byte[] = {0x20,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}      // Retrieve all data
    // ReceivedBuffer = {}
var_value = socket_read("ntd_a", 4, 100)
    // byte[] = {} // The number is insufficient for less than 4 bytes in the Received Buffer. Byte[0] will return.
                    //But the read time is set to 100 ms, it stays in the function still waiting for data or the read times
                    fulfilled before exiting.
    // ReceivedBuffer = {0x31,0x32,0x33,0x34,0x35,0x36,0x37,0x38}
    //Supposed data received in 50 ms later
    // byte[] = {0x31,0x32,0x33,0x34}                          // Retrieve 4 bytes and exit the function.
    // ReceivedBuffer = {0x35,0x36,0x37,0x38}
```

Syntax 4

```
byte[] socket_read(
    string,
    byte[] or string,
    byte[] or string,
    int,
    int
)
```

Parameters

string Network device name

byte[] or string

The prefix condition to read. Input byte[0] or "" as the empty string to denote no prefix condition.

byte[] or string

The suffix condition to read. Input byte[0] or "" as the empty string to denote no suffix condition.

int

Fetch option

0 The 1st match not required in the beginning and not removing the prefix and the suffix (default)

- 1 The 1st match not required in the beginning and removing the prefix and the suffix
- 2 The 1st match required in the beginning and removing the prefix and the suffix
- 3 The 1st match required in the beginning and removing the prefix and the suffix

int Read time (millisecond)

<= 0 Read once

> 0 Read multiple times until there is data or the times fulfill.

Return

byte[] Return the 1st array in byte that matches the prefix condition and the suffix condition.

Fetch the first data that matches the conditions in the Received Buffer. only. The following data remains in the Received Buffer.

If the prefix condition and the suffix condition are byte[0] or an empty string, it fetches all data in the Received Buffer.

If unable to find the data matches the condition, it returns byte[0].

Syntax 5

```
byte[] socket_read(
    string,
    byte[] or string,
    byte[] or string,
    int
)
```

Note

Same as syntax 4. Fill 0 as the read time by default.

Syntax 6

```
byte[] socket_read(
    string,
    byte[] or string,
    byte[] or string
)
```

Note

Same as syntax 4. Fill 0 as the fetch option and the read time by default.

ReceivedBuffer =

```
{0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
```

```
byte[] var_value = socket_read("ntd_a", "", "") // No prefix. No suffix. Fetch all data.
```

```
// byte[] = {0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
```

```
// ReceivedBuffer = {}
```

ReceivedBuffer =

```
{0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
```

```
byte[] var_value = socket_read("ntd_a", "He", newline) // Prefix "He" Suffix \u0D0A
```

```
// byte[] = {0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A} // Hello,\u0D0A
```

```
// ReceivedBuffer = {0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
```

```

    // Fetch the 1st match. Remain the following data.
var_value = socket_read("ntd_a", "", newline, 1)    // Prefix "" Suffix \u0D0A Remove the prefix
    and the suffix.
    // byte[] = {0x57,0x6F,0x72,0x6C,0x64}    // World
    // ReceivedBuffer = {}
var_value = socket_read("ntd_a", "", newline, 1, 100)
    // Prefix "" Suffix \u0D0A Remove the prefix and the suffix. 100ms
    // byte[] = {}
    // Read byte[0] for fetch option unfulfilled. Wait for 100ms.
    // ReceivedBuffer = {}
    // ReceivedBuffer = {0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A}
    // Supposed data received in 50ms later
    // byte[] = {0x48,0x65,0x6C,0x6C,0x6F,0x2C}    // Hello,
    // ReceivedBuffer = {}

ReceivedBuffer =
{0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
byte[] var_value = socket_read("ntd_a", "lo", newline) // Prefix "lo" Suffix \u0D0A
    // byte[] = {0x6C,0x6F,0x2C,0x0D,0x0A}    // lo,\u0D0A
    // The data before the 1st match, {0x48,0x65,0x6C}, will be removed.
    // ReceivedBuffer = {0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
byte[] var_bb = {}
var_value = socket_read("ntd_a", var_bb, newline)    // Prefix byte[0] Suffix \u0D0A
    // byte[] = {0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}    // World\u0D0A
    // ReceivedBuffer = {}
var_value = socket_read("ntd_a", var_bb, newline, 0, 100)
    // Prefix byte[0] Suffix \u0D0A , 100ms
    // byte[] = {}    //Read byte[0] for fetch option unfulfilled. Wait for 100ms.
    // ReceivedBuffer = {}
    // ReceivedBuffer = {0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A}
    // Supposed data received in 50ms later
    // byte[] = {0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A}
    // ReceivedBuffer = {}

ReceivedBuffer = {0x24,0x48,0x65,0x6C,0x6C,0x6F,0x0D,0x0A,    // $Hello
    0x23,0x48,0x69,0x0D,0x0A,    // #Hi
    0x24,0x54,0x4D,0x0D,0x0A,    // $TM
    0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A}    // #Robot
byte[] var_value = socket_read("ntd_a", "#", newline, 2)    // Prefix "#" Suffix \u0D0A
    // byte[] = {}
    //Return an empty string for # not in the prefix.
    // ReceivedBuffer = {0x24,0x48,0x65,0x6C,0x6C,0x6F,0x0D,0x0A,0x23,0x48,0x69,0x0D,0x0A,
    0x24,0x54,0x4D,0x0D,0x0A,0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A}
var_value = socket_read("ntd_a", "$", newline, 2)    // Prefix "$" Suffix \u0D0A
    // byte[] = {0x24,0x48,0x65,0x6C,0x6C,0x6F,0x0D,0x0A}    // Fetch the 1st match in the prefix required.
    // ReceivedBuffer = {0x23,0x48,0x69,0x0D,0x0A,
    0x24,0x54,0x4D,0x0D,0x0A,0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A}
var_value = socket_read("ntd_a", "#", newline, 2)    // Prefix "#" Suffix \u0D0A

```

```

// byte[] = {0x23,0x48,0x69,0x0D,0x0A} // Fetch the 1st match in the prefix required.
// ReceivedBuffer = {0x24,0x54,0x4D,0x0D,0x0A,0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A}
var_value = socket_read("ntd_a", "$", newline, 3) // Prefix "$" Suffix \u0D0A
// byte[] = {0x54,0x4D}
// ReceivedBuffer = {0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A}
var_value = socket_read("ntd_a", "#", newline, 3) // Prefix "#" Suffix \u0D0A
// byte[] = {0x52,0x6F,0x62,0x6F,0x74}
// ReceivedBuffer = {}

```

Syntax 7

```

byte[] socket_read(
    string,
    byte[] or string,
    int,
    int
)

```

Parameters

string Network device name

byte[] or string The suffix condition to read. Input byte[0] or "" as the empty string to denote no suffix condition.

int Fetch option

- 0 The 1st match not required in the beginning and not removing the prefix and the suffix (default)
- 1 The 1st match not required in the beginning and removing the prefix and the suffix
- 2 The 1st match required in the beginning and removing the prefix and the suffix
- 3 The 1st match required in the beginning and removing the prefix and the suffix

int Read time (millisecond)

- <= 0 Read once
- > 0 Read multiple times until there is data or the times fulfill.

Return

byte[] Return the 1st array in byte that matches the suffix condition. (No prefix condition restricted)

Fetch the first data that matches the condition in the Received Buffer. The following data remains in the Received Buffer.

If the suffix condition is byte[0] or an empty string, it fetches all data in the Received Buffer.

If unable to find the data matches the condition, it returns byte[0].

Syntax 8

```

byte[] socket_read(
    string,
    byte[] or string,
    int
)

```

Note

Same as syntax 7. Fill 0 as the read time by default.

Syntax 9

```
byte[] socket_read(
    string,
    byte[] or string
)
```

Note

Same as syntax 7. Fill 0 as the fetch option and the read time by default.

```
ReceivedBuffer =
{0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
byte[] var_value = socket_read("ntd_a", "")           // Empty Suffix. Fetch all data.
// byte[] = {0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
// ReceivedBuffer = {}

ReceivedBuffer =
{0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
byte[] var_value = socket_read("ntd_a", newline) // Suffix \u0D0A
// byte[] = {0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A} // Hello,\u0D0A
// ReceivedBuffer = {0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A} // Fetch the 1st match. Remain the following
data.
var_value = socket_read("ntd_a", newline)           // Suffix \u0D0A
// byte[] = {0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A} // World\u0D0A
// ReceivedBuffer = {}

ReceivedBuffer = {0x24,0x48,0x65,0x6C,0x6C,0x6F,0x0D,0x0A, // $Hello
0x23,0x48,0x69,0x0D,0x0A, // #Hi
0x24,0x54,0x4D,0x0D,0x0A, // $TM
0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A} // #Robot
byte[] var_value = socket_read("ntd_a", newline, 0) // Suffix \u0D0A
// byte[] = {0x24,0x48,0x65,0x6C,0x6C,0x6F,0x0D,0x0A} // $Hello\u0D0A
// ReceivedBuffer =
{0x23,0x48,0x69,0x0D,0x0A,0x24,0x54,0x4D,0x0D,0x0A,0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A}
var_value = socket_read("ntd_a", newline, 1) // Suffix \u0D0A
// byte[] = {0x23,0x48,0x69} // #Hi
// ReceivedBuffer = {0x24,0x54,0x4D,0x0D,0x0A,0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A}
var_value = socket_read("ntd_a", newline, 2) // Suffix \u0D0A
// byte[] = {0x24,0x54,0x4D,0x0D,0x0A} // $TM\u0D0A
// ReceivedBuffer = {0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A}
var_value = socket_read("ntd_a", newline, 3) // Suffix \u0D0A
// byte[] = {0x23,0x52,0x6F,0x62,0x6F,0x74} // #Robot
// ReceivedBuffer = {}
```

9.5 socket_read_string()

Read data in the Received Buffer and convert the byte array to string text in UTF8

Syntax 1

```
string socket_read_string(
    string
)
```

Parameters

string Network device name

Return

string Return all data in the Received Buffer. Return an empty string if buffer empty.

Syntax 2

```
string socket_read_string(
    string,
    int,
    int
)
```

Parameters

string Network device name

int Retrieve the fixed amount of string (by the length of string)

<= 0 Get all

> 0 Get a specified amount (the specified amount required to get the data)

int Read time (millisecond)

<= 0 Read once

> 0 Read multiple times until there is data or the times fulfill.

Return

string Return the specified amount of data in the Received Buffer with a string. Return an empty string if insufficient amount.

Syntax 3

```
string socket_read_string(
    string,
    int
)
```

Note

Same as syntax 2. Fill 0 as the read time by default.

Syntax 4

```
string socket_read_string(
    string,
    byte[] or string,
    byte[] or string,
    int,
    int
)
```

Parameters

string Network device name

byte[] or string

The prefix condition to read. Input byte[0] or "" as the empty string to denote no prefix condition.

byte[] or string

The suffix condition to read. Input byte[0] or "" as the empty string to denote no suffix condition.

int Retrieve the fixed amount of string (by the length of string)

<= 0 Get all

> 0 Get a specified amount (the specified amount required to get the data)

int Read time (millisecond)

<= 0 Read once

> 0 Read multiple times until there is data or the times fulfill.

Return

string Return the 1st string that matches the prefix condition and the suffix condition. Fetch the first data that matches the conditions in the Received Buffer. The following data remains in the Received Buffer.

If the prefix condition and the suffix condition are byte[0] or an empty string, it fetches all data in the Received Buffer.

If unable to find the data matches the condition, it returns an empty string.

Syntax 5

```
string socket_read_string(
    string,
    byte[] or string,
    byte[] or string,
    int
```

```
)
```

Note

Same as syntax 4. Fill 0 as the read time by default.

Syntax 6

```
string socket_read_string(
    string,
    byte[] or string,
    byte[] or string
```

```
)
```

Note

Same as syntax 4. Fill 0 as the fetch option and the read time by default.

Syntax 7

```
string socket_read_string(
    string,
    byte[] or string,
    int,
    int
```

```
)
```

Parameters

string Network device name

byte[] or string

The suffix condition to read. Input byte[0] or "" as the empty string to denote no suffix condition.

int Retrieve the fixed amount of string (by the length of string)

<= 0 Get all

> 0 Get a specified amount (the specified amount required to get the data)

int Read time (millisecond)

<= 0 Read once

> 0 Read multiple times until there is data or the times fulfill.

Return

string Return the 1st string that matches the suffix condition. (No prefix condition restricted)

Fetch the first data that matches the condition in the Received Buffer. only. The following data remains in the Received Buffer.

If the suffix condition is byte[0] or an empty string, it fetches all data in the Received Buffer.

If unable to find the data matches the condition, it returns an empty string.

Syntax 8

```
string socket_read_string(
    string,
    byte[] or string,
    int
)
```

Note

Same as syntax 7. Fill 0 as the read time by default.

Syntax 9

```
string socket_read_string(
    string,
    byte[] or string
)
```

Note

Same as syntax 7. Fill 0 as the fetch option and the read time by default.

ReceivedBuffer =

```
{0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x20,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
```

```
string var_value = socket_read_string("ntd_a")
```

```
// string = "Hello, World\u0D0A"
```

```
// ReceivedBuffer = {}
```

ReceivedBuffer =

```
{0x54,0x4D,0xE9,0x81,0x94,0xE6,0x98,0x8E,0xE6,0xA9,0x9F,0xE5,0x99,0xA8,0xE4,0xBA,0xBA}
```

```
string var_value = socket_read_string("ntd_a", 4)
```

```
// string = "TM 達明" // {0x54,0x4D,0xE9,0x81,0x94,0xE6,0x98,0x8E} // Fetch 4 by the string length.
```

```
// ReceivedBuffer = {0xE6,0xA9,0x9F,0xE5,0x99,0xA8,0xE4,0xBA,0xBA}
```

```
var_value = socket_read_string("ntd_a", 5, 100)
```

```
// string = ""
```

```
// The amount to read data in the Received Buffer is less than 5. (by the string length). Wait for 100 ms.
// ReceivedBuffer = {0xE6,0xA9,0x9F,0xE5,0x99,0xA8,0xE4,0xBA,0xBA}
// ReceivedBuffer = {0xE6,0xA9,0x9F,0xE5,0x99,0xA8,0xE4,0xBA,0xBA, 0x0D, 0x0A}
// string = "機器人\u00D0A"
// ReceivedBuffer = {}
```

```
ReceivedBuffer =
{0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
string var_value = socket_read_string("ntd_a", "", "")
// string = "Hello,\u00D0AWorld\u00D0A"
// ReceivedBuffer = {}
ReceivedBuffer =
{0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
string value = socket_read_string("ntd_a", "He", newline) // Prefix "He" Suffix \u00D0A
// string = "Hello,\u00D0A"
// ReceivedBuffer = {0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
var_value = socket_read_string("ntd_a", "", newline, 1)
// Prefix "" Suffix \u00D0A Remove the prefix and the suffix.
// string = "World"
// ReceivedBuffer = {}
var_value = socket_read_string("ntd_a", "", newline, 1, 100)
// string = ""
// Read a empty string for the fetch option unfulfilled. Wait for 100 ms.
// ReceivedBuffer = {}
// ReceivedBuffer = {0xE6,0xA9,0x9F,0xE5,0x99,0xA8,0xE4,0xBA,0xBA,0x0D,0x0A}
// string = "機器人"
```

```
ReceivedBuffer =
{0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
string var_value = socket_read_string("ntd_a", "lo", newline) // Prefix "lo" Suffix \u00D0A
// string = "lo,\u00D0A"
// The data before the 1st match, "Hel", will be removed.
// ReceivedBuffer = {0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A}
var_value = socket_read_string("ntd_a", newline, 1) // Suffix \u00D0A
// string = "World"
// ReceivedBuffer = {}
```

```
ReceivedBuffer = {0x24,0x48,0x65,0x6C,0x6C,0x6F,0x0D,0x0A, // $Hello
0x23,0x48,0x69,0x0D,0x0A, // #Hi
0x24,0x54,0x4D,0x0D,0x0A, // $TM
0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A} // #Robot
string var_value = socket_read_string("ntd_a", "#", newline, 2) // Prefix "#" Suffix \u00D0A
// string = "" // Return an empty string for # not in the prefix.
// ReceivedBuffer = {0x24,0x48,0x65,0x6C,0x6C,0x6F,0x0D,0x0A,0x23,0x48,0x69,0x0D,0x0A,
0x24,0x54,0x4D,0x0D,0x0A,0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A}
var_value = socket_read_string("ntd_a", "$", newline, 2) // Prefix "$" Suffix \u00D0A
// string = "$Hello\u00D0A"
// Fetch the 1st match in the prefix required.
```



```

    // ReceivedBuffer = {0x23,0x48,0x69,0x0D,0x0A,
                        0x24,0x54,0x4D,0x0D,0x0A,0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A}
var_value = socket_read_string("ntd_a", "#", newline, 2)      // Prefix "#" Suffix \u0D0A
    // string = "#Hi\u0D0A"                                     // Fetch the 1st match in the prefix required.
    // ReceivedBuffer = {0x24,0x54,0x4D,0x0D,0x0A,0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A}

var_value = socket_read_string("ntd_a", "$", newline, 3) // Prefix "$" Suffix \u0D0A
    // string = "TM"
    // ReceivedBuffer = {0x23,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A}
var_value = socket_read_string("ntd_a", "#", newline, 3) // Prefix "#" Suffix \u0D0A
    // string = "Robot"
    // ReceivedBuffer = {}

ReceivedBuffer =
{0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x0D,0x0A,0x57,0x6F,0x72,0x6C,0x64}
string var_value = socket_read_string("ntd_a", newline) // Suffix \u0D0A
    // string = "Hello,\u0D0A"
    // ReceivedBuffer = {0x57,0x6F,0x72,0x6C,0x64}
var_value = socket_read_string("ntd_a", newline, 0)      // Suffix \u0D0A
    // string = ""
    // Read a empty string for the fetch option unfulfilled.
    // ReceivedBuffer = {0x57,0x6F,0x72,0x6C,0x64}
var_value = socket_read_string("ntd_a", newline, 1, 100) // Suffix \u0D0A
    // string = ""
    // Read a empty string for the fetch option unfulfilled. Wait for 100ms.
    // ReceivedBuffer = {0x57,0x6F,0x72,0x6C,0x64}
    // ReceivedBuffer = {0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A,0x31,0x32,0x33,0x0D,0x0A}
    // Supposed the data comes in and fulfills the fetch option.
    // string = "World"
    // ReceivedBuffer = {0x31,0x32,0x33,0x0D,0x0A}
var_value = socket_read_string("ntd_a", newline, 2)      // Suffix \u0D0A
    // string = "123\u0D0A"
    // ReceivedBuffer = {}

```

9.6 socket_send()

Send the data value to a remote device.

Syntax 1

```
int socket_send(
    string,
    ?,
    int,
    int
)
```

Parameters

string	Network device name
?	The value to write. Available types: int, float, , bool, string, and array. Numeric values will be conversed in Little Endian, and string values will be converse in UTF8.
int	The starting index of the value to write (eligible for strings and arrays)
0 .. length-1	Legitimate value
< 0	Illegitimate value. The starting index will be set to 0.
>= length	Illegitimate value. The starting index will be set to 0.
int	The length of the value to write (eligible for strings and arrays)
<= 0	Write from the starting index to the end of the data.
> 0	Write from the starting index for a specified number of the length up to the data ends.

Return

int	Send result
1	Sent successfully.
0	Unable to send the data value as an empty string or an empty array.
-1	Socket exception occurred during sending.
-2	Unable to connect to the remote device.
-3	The device name does not exist, or IP or port is incorrect.

Syntax 2

```
int socket_send(
    string,
    ?,
    int
)
```

Note

Same as syntax 1. Fill 0 as the length of the value to write by default.

Syntax 3

```
int socket_send(
    string,
    ?
)
```

Note

Same as syntax 1. Fill 0 as the starting index and the length of the value to write by

default.

```

int var_re = socket_send("ntd_a", 100)           // send 0x64
var_re = socket_send("ntd_a", 1000)             // send 0xE8,0x03,0x00,0x00 (int, Little Endian)
var_re = socket_send("ntd_a", (float)1.234)
//send 0xB6,0xF3,0x9D,0x3F (float, Little Endian)
var_re = socket_send("ntd_a", (double)123.456)
// send 0x77,0xBE,0x9F,0x1A,0x2F,0xDD,0x5E,0x40 (double, Little Endian)
var_re = socket_send("ntd_a", "Hello, World"+newline)
// send 0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x20,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A (string, UTF8)
int[] var_ii = {100, 200, 300, 400}
var_re = socket_send("ntd_a", var_ii)
// send 0x64,0x00,0x00,0x00,0xC8,0x00,0x00,0x00,0x2C,0x01,0x00,0x00,0x90,0x01,0x00,0x00 (int[],
Little Endian)
string[] var_ss = {"TM", "", "Robot"}
var_re = socket_send("ntd_a", var_ss)
// send 0x54,0x4D,0x52,0x6F,0x62,0x6F,0x74 (string[], UTF8)
// var_ss[1] is an empty string. The conversion value is still empty.

var_re = socket_send("ntd_a", 1000, 1, 2)
// Invalid in the value, the starting index, and the length
// send 0xE8,0x03,0x00,0x00 (int, Little Endian)
var_re = socket_send("ntd_a", "Hello, World"+newline, 0, 7)
// send 0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x20 (string, UTF8)
byte[] var_bb = {100, 200}
var_re = socket_send("ntd_a", var_bb)           // send 0x64,0xC8
var_re = socket_send("ntd_a", var_bb, 1, 1) // send 0xC8
// Array. Read 1 from address 1, [1]=200
var_re = socket_send("ntd_a", var_bb, -1, 1)    // send 0x64
// Array. Read 1 from address 0. [0]=100
var_re = socket_send("ntd_a", "達明機器人", 2)
// Array. Read from address 2 till the end. "機器人"
// send 0xE6,0xA9,0x9F,0xE5,0x99,0xA8,0xE4,0xBA,0xBA (string, UTF8)
var_ss = {"TM", "", "達明機器人"}
var_re = socket_send("ntd_a", var_ss)
// send 0x54,0x4D,0xE9,0x81,0x94,0xE6,0x98,0x8E,0xE6,0xA9,0x9F,0xE5,0x99,0xA8,0xE4,0xBA,0xBA
re = socket_send("ntd_a", Byte_Concat(GetBytes(var_ss), GetBytes(newline)))
// send
0x54,0x4D,0xE9,0x81,0x94,0xE6,0x98,0x8E,0xE6,0xA9,0x9F,0xE5,0x99,0xA8,0xE4,0xBA,0xBA,0x0D,0x
0A
var_re = socket_send("ntd_a", var_ss, 2, 100)
// Array. Read 100 from address 2. (till the end) [2]=達明機器人
// send 0xE9,0x81,0x94,0xE6,0x98,0x8E,0xE6,0xA9,0x9F,0xE5,0x99,0xA8,0xE4,0xBA,0xBA

```

9.7 socket_sendline()

Send the data value and add line break symbols, 0x0D 0x0A, to a remote device.

Syntax 1

```
int socket_sendline(
    string,
    ?,
    int,
    int
)
```

Parameters

string	Network device name
?	The value to write. Available types: int, float, , bool, string, and array. Numeric values will be conversed in Little Endian, and string values will be converse in UTF8.
int	The starting index of the value to write (eligible for strings and arrays)
0 .. length-1	Legitimate value
< 0	Illegitimate value. The starting index will be set to 0.
>= length	Illegitimate value. The starting index will be set to 0.
int	The length of the value to write (eligible for strings and arrays)
<= 0	Write from the starting index to the end of the data.
> 0	Write from the starting index for a specified number of the length up to the data ends.

Return

int	Send result
1	Sent successfully.
0	Unable to send the data value as an empty string or an empty array.
-1	Socket exception occurred during sending.
-2	Unable to connect to the remote device.
-3	The device name does not exist, or IP or port is incorrect.

Syntax 2

```
int socket_sendline(
    string,
    ?,
    int
)
```

Note

Same as syntax 1. Fill 0 as the length of the value to write by default.

Syntax 3

```
int socket_sendline(
    string,
    ?
)
```

Note

Same as syntax 1. Fill 0 as the starting index and the length of the value to write by

default.

```

int var_re = socket_sendline("ntd_a", 200) // send 0xC8,0x0D,0x0A
var_re = socket_sendline("ntd_a", 2000) // send 0xD0,0x07,0x00,0x00,0x0D,0x0A (int,
Little Endian)
var_re = socket_sendline("ntd_a", (float)0.234) // send 0xB2,0x9D,0x6F,0x3E,0x0D,0x0A
(float, Little Endian)
var_re = socket_sendline("ntd_a", (double)0.234)
// send 0xC1,0xCA,0xA1,0x45,0xB6,0xF3,0xCD,0x3F,0x0D,0x0A (double, Little Endian)
var_re = socket_sendline("ntd_a", "Hello, World"+newline)
// send 0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x20,0x57,0x6F,0x72,0x6C,0x64,0x0D,0x0A,0x0D,0x0A
(string, UTF8)
int[] var_ii = {100, 200, 300}
var_re = socket_sendline("ntd_a", var_ii)
// send 0x64,0x00,0x00,0x00,0xC8,0x00,0x00,0x00,0x2C,0x01,0x00,0x00,0x0D,0x0A (int[], Little
Endian)
string[] var_ss = {"TM", "", "Robot"}
var_re = socket_sendline("ntd_a", var_ss)
// send 0x54,0x4D,0x52,0x6F,0x62,0x6F,0x74,0x0D,0x0A (string[], UTF8)
//var_ss[1] is an empty string. The conversion value is still empty.

var_re = socket_sendline("ntd_a", 1000, 1, 2)
// Invalid in the value, the starting index, and the length
// send 0xE8,0x03,0x00,0x00,0x0D,0x0A (int, Little Endian)
var_re = socket_sendline("ntd_a", "Hello, World"+newline, 0, 7)
// send 0x48,0x65,0x6C,0x6C,0x6F,0x2C,0x20,0x0D,0x0A (string, UTF8)
byte[] var_bb = {123, 234}
var_re = socket_sendline("ntd_a", var_bb) // send 0x7B,0xEA,0x0D,0x0A
var_re = socket_sendline("ntd_a", var_bb, 1, 1) // send 0xEA,0x0D,0x0A
//Array. Read 1 from address 1.
var_re = socket_sendline("ntd_a", var_bb, -1, 1) // send 0x7B,0x0D,0x0A
// Array. Read 1 from address 0.
var_re = socket_sendline("ntd_a", "達明機器人", 2)
//" Array. Read from address 2 till the end. "機器人"
// send 0xE6,0xA9,0x9F,0xE5,0x99,0xA8,0xE4,0xBA,0xBA,0x0D,0x0A (string, UTF8)
var_ss = {"TM", "", "達明機器人"}
var_re = socket_sendline("ntd_a", var_ss)
// send
0x54,0x4D,0xE9,0x81,0x94,0xE6,0x98,0x8E,0xE6,0xA9,0x9F,0xE5,0x99,0xA8,0xE4,0xBA,0xBA,0x0D,
0x0A
var_re = socket_sendline("ntd_a", Byte_Concat(GetBytes(var_ss),
GetBytes(newline)))
// send
0x54,0x4D,0xE9,0x81,0x94,0xE6,0x98,0x8E,0xE6,0xA9,0x9F,0xE5,0x99,0xA8,0xE4,0xBA,0xBA,0x0D,0x0A,0x0D,
0x0A
var_re = socket_sendline("ntd_a", var_ss, 2, -1)
// Array. Read from address 2 till the end. [2]=達明機器人
// send

```

0xE9,0x81,0x94,0xE6,0x98,0x8E,0xE6,0xA9,0x9F,0xE5,0x99,0xA8,0xE4,0xBA,0xBA,0x0D,0x0A

10. Manual Decision Functions

10.1 MDecision Class

Use the MDecision class with variable declaration to allow users to configure M-Decision nodes for conditional judgments or to display pop-up messages.

Construct 1

MDecision *VariableName*

MDecision *VariableName* = string, string

Parameter

string Headline

string Text

Note

MDecision md1 // Headline and text are empty strings.

MDecision md2 = "Test1", "MDecision Test"

// The headline is "Test1", and the text, "MDecision Test".

Member Methods

Name	Default	Description
Reset()	-	Reset all MDecision parameters to the default.
Title()	""	The pop-up headline
Description()	""	The pop-up text
Timeout()	-1	Timeout
TimeoutDefaultCase()	0	The default index after the timeout
Case()	-	Add, delete, or modify the conditional judgement
Show()		Check the conditional judgement

Reset()

Reset all MDecision parameters to the default.

Syntax 1

```
void Reset(
)
```

Parameters

void No input value

Return

void No return

10.1.2 Title()

Set the pop-up headline

Syntax 1

```
void Title(
    string
)
```

Parameters

string The headline

Return

void No return

10.1.3 Description()

Set the pop-up text

Syntax 1

```
void Description(
    string
)
```

Parameters

string The text

Return

void No return

10.1.4 Timeout()

Set the stop condition of timeout.

Syntax 1

```
void Timeout(
    int,
    int
)
```

Parameters

int Timeout in millisecond

`< 0` Disable
`>= 0` Timeout duration
`int` The default index after timeout
Return
`void` No return

Syntax 2

```
void Timeout(
  int
)
```

Note

Same as Syntax 1. For setting the timeout duration.

Syntax 3

```
void Timeout(
)
```

Parameters

`void` No input value for cancelling the timeout.

Return

`void` No return

10.1.5 TimeoutDefaultCase()

Set the default index after the timeout

Syntax 1

```
void TimeoutDefaultCase(
  int
)
```

Parameters

`int` The default index after the timeout.

Return

`void` No return

10.1.6 Case()

Add, delete, or modify the conditional judgment. It creates a new index if none exists, updates an existing index, or deletes an index if only the index parameter is available.

Syntax 1

```
void Case(
  int,
  string,
  string,
  string,
  bool
)
```

Parameters

`int` The condition or the index of the button in the pop-up
`< 0` Error

	<code>>= 0</code>	Legal index value
<code>string</code>		Text to display of the button
<code>string</code>		Text color of the button
	<code>"Red"</code>	Set the text color to red.
	<code>"Green"</code>	Set the text color to green.
	<code>"Blue"</code>	Set the text color to blue.
	<code>"Yellow"</code>	Set the text color to yellow.
	<code>"Black"</code>	Set the text color to black.
	<code>"White"</code>	Set the text color to white.
	<code>"Gray"</code>	Set the text color to gray.
<code>string</code>		Background color of the button
	<code>"Red"</code>	Set the background color to red.
	<code>"Green"</code>	Set the background color to green.
	<code>"Blue"</code>	Set the background color to blue.
	<code>"Yellow"</code>	Set the background color to yellow.
	<code>"Black"</code>	Set the background color to black.
	<code>"White"</code>	Set the background color to white.
	<code>"Gray"</code>	Set the background color to gray.
<code>bool</code>		The conditional can be true/false or a bool return of the statement.
Return		
<code>void</code>		No return

Syntax 2

```
void Case(
    int,
    string,
    string,
    string
)
```

Parameters

<code>int</code>	The condition or the index of the button in the pop-up
<code>string</code>	Text to display of the button
<code>string</code>	Text color of the button
<code>string</code>	Background color of the button

Note

Same as syntax 1. The condition is set to false meaning the condition of the index does not meet.

Syntax 3

```
void Case(
    int,
    string
)
```

Parameters

<code>int</code>	The condition or the index of the button in the pop-up
<code>string</code>	Text to display of the button

Note

Same as syntax 1. The condition is set to false, with the default text color as white and the

default background color as blue.

Syntax 4

```
void Case(
    int,
    string,
    bool
)
```

Parameters

int	The condition or the index of the button in the pop-up
string	Text to display of the button
bool	The conditional can be true/false or a bool return of the statement.

Note

Same as syntax 1 with the default text color as white and the default background color as blue.

Syntax 5

```
void Case(
    int
)
```

Parameters

int	The condition or the index of the button in the pop-up
< 0	Error
>= 0	Legal index value

Note

Delete the set conditional judgment.

10.1.7 Show()

Check the condition. If it is true, it returns the index value without prompting a pop-up window. If the condition is false, it prompts a pop-up window and continues to loop it concurrently. It returns the respective index value if the condition meets, or the button index value when a user clicks it. If there is a set timeout, it returns the default index value after the timeout.

Syntax 1

```
int Show(
)
```

Parameters

void	No input value
------	----------------

Return

int	The index value to return
-----	---------------------------

Note

```
int var_count = 0
MDecision md1 = "TM", "Techman Robot?"
md1.Case(1, "Yes")           // Add index 1, button text "Yes"
md1.Case(2, "OK")           // Add index 2
md1.Case(3, "No")           // Add index 3
md1.Case(4, "If", var_count > 100)
// Add index 4, button text "If", and a condition of var_count > 100 for judgment.
```

```
md1.Timeout(10000)           // Timeout for 10000ms
md1.TimeoutDefaultCase(2)
md1.Case(2)                  // Delete index 2
int re = md1.Show()          // Return the index that meets the condition or the index where the
                             // button was clicked. If there is a timeout, it returns index 2.
```

Display(re)

MDecision md2 = "TM", "Techman Robot?"

```
md2.Case(6, "YES")           // Add index 6, button text "YES"
md2.Case(2, "OK")            // Add index 2
md2.Case(1, "NO")            // Add index 1
md2.Case(2, "OK-2")          // Modify index 2
int re2 = md2.Show()
Display(re2)
```

11. Parameterized objects

Using parameterized objects is the same as using user defined variables. Parameterized objects can be used without declarations to get or modify point data through the syntaxes in the project operations and make the robot go with more flexibility. The expression comes with 3 parts, item, index, and attribute, and the syntax is shown as below.

`parameterized item[index].attribute`

The supported parameterized items include:

1. Point
2. Base
3. TCP
4. VPoint
5. IO
6. Robot
7. FT

Definitions of the indexes and the attributes vary from parameterized items.

Take the reading and writing of the `coordinate` (attribute) of the `Point` (item) `"P1"` (index) as an example. The index is defined as the name of the point, and the attribute, as the data type of float (the same usage as the array's) with modes of reading and writing.

Value	<code>float[]</code>	R/W	point coordinate{X,Y,Z,RX,RY,RZ}
-------	----------------------	-----	----------------------------------

Read values

<code>float[] f = Point["P1"].Value</code>	<code>// In the item Point, the index is defined as the name of the point and the data type of string.</code>
<code>float f1 = Point["P1"].Value[0]</code>	<code>// The x value of "P1" can be obtained solely</code>

Write values

<code>Point["P1"].Value = {0, 0, 90, 0, 90, 0}</code>	<code>// Replace to the coordinate of "P1" with {0,0,90,0,90,0}</code>
<code>Point["P1"].Value[2] = 120</code>	<code>// or replace the z value of "P1" with 12 solely</code>

11.1 Point

Syntax

Base

`Point[string].attribute`

Item

`Point`

Index

`string` The name of the point in the point manager

Attribute

Name	Type	Mode	Description	Format
Value	float[]	R/W	The coordinate of the point	{X, Y, Z, RX, RY, RZ}, Size = 6
Joint	float[]	R/W	The joint angle	{J1, J2, J3, J4, J5, J6}, Size = 6
Pose	int[]	R/W	The pose of the robot	{Config1, Config2, Config3}, Size = 3
Flange	float[]	R	The coordinate of the flange's center	{X, Y, Z, RX, RY, RZ}, Size = 6
BaseName	string	R	The name of the base	"Base Name"
TCPName	string	R	The name of the TCP	"TCP Name"
TeachValue	float[]	R	The original coordinate of the teaching point	{X, Y, Z, RX, RY, RZ}, Size = 6
TeachJoint	float[]	R	The joint angle (the original angle at the teach point)	{J1, J2, J3, J4, J5, J6}, Size = 6
TeachPose	int[]	R	The original pose of the robot on the teaching point	{Config1, Config2, Config3}, Size = 3

*It recalculates the the Joint and Flange when setting the Value, and it recalculates the Value and Flange when setting the Joint. Therefore, it reports an error if it cannot calculate the set value.

Note

// Read values

`float[] f = Point["P1"].Value` / Obtain the coordinate {X, Y, Z, RX, RY, RZ} of "P1"

`float f1 = Point["P1"].Value[0]` // or retrieve the x value of "P1" solely

`float f1 = Point["P1"].Value[6]` // Return error, exceeding the array's access range

`string s = Point["P1"].BaseName` // s ="RobotBase"

// Write values

`Point["P1"].Value = {0, 0, 90, 0, 90, 0}` // Replace the coordinate of "P1" with {0,0,90,0,90,0}

`Point["P1"].Value[2] = 120` // or replace the z value of "P1" with 120 solely

`Point["P1"].Flange = {0, 0, 90, 0, 90, 0}` // Read only, invalid operation

`Point["P1"].Value = {0, 0, 90, 0, 90}` // Return error, writing elements to the array do not match to 6 (writing 5 elements)

`Point["P1"].Pose = {1, 2, 4, 0}` // Return error, writing elements to the array do not match to 3 (writing 4 elements)

11.2 Base

Syntax

Base

Base[string].attribute or
Base[string, int].attribute

Item

Base

Index

string The name of the base in the base manager
*The name of the base comes with the attribute of the mode in reading without writing only.
"RobotBase"

int The index of the base, available to assign with multiple bases built by vision one shot get all, ranging from 0 as the default to N.

Attribute

Name	Type	Mode	Description	Format
Value	float[]	R/W	The base value	{X, Y, Z, RX, RY, RZ}, Size = 6
Type	string	R	The type of the base	"R": Robot Base "V": Vision Base "C": Custom Base
TeachValue	float[]	R	The original teach base value	{X, Y, Z, RX, RY, RZ}, Size = 6

Note

// Read values

```
float[] f = Base["RobotBase"].Value // Obtain the base value {0,0,0,0,0,0} of the base "RobotBase"
```

```
float f1 = Base["base1"].Value[0] // or retrieve the x value of "base1" solely
```

```
string s = Base["base1"].Type // s = "C"
```

```
s = Base[Point["P1"].BaseName].Type // s = "R" // Given the type of "P1" is "RobotBase"
```

```
float[] f = Base["vision_osga", 1].Value // Obtain the 2nd value of the "vision_osga"
```

// Write values

```
Base["RobotBase"].Value = {0, 0, 90, 0, 90, 0} // Read only, invalid operation, because "RobotBase" is the system base
```

```
Base["base1"].Value = {0, 90, 0, 0, 90, 0} // Replace the value of "base1" with {0,90,0,0,90,0}
```

```
Base["base1"].Value[4] = 120 // or replace the RY value of "base1" with 120 solely
```

```
Base["base1"].Value[6] = 120 // Return error, exceeding the array's access range
```

```
Base["base1"].Type = "C" // Read only, invalid operation
```

```
Base["base1"].Value = {0, 0, 90, 0, 90} // Return error, writing elements to the array do not match to 6 (writing 5 elements)
```

```
Base["base1"].Value = {0, 0, 90, 0, 90, 0, 100} // Return error, writing elements to the array do not match to 6 (writing 7 elements)
```

11.3 TCP

Syntax

TCP

`TCP[string].attribute`

Item

`TCP`

Index

`string`

The name of the TCP in the TCP list

*The name of the TCP comes with the attribute of the mode in reading without writing only.

"RobotEndFlange"

"HandCamera"

"HandCamera2"

Attribute

Name	Type	Mode	Description	Format
Value	float[]	R/W	The value of the TCP	{X, Y, Z, RX, RY, RZ}, Size = 6
Mass	float	R/W	The value of mass	Mass in kg
MOI	float[]	R/W	The value of the Principal Moments of Inertia	{Ixx, Iyy, Izz}, Size = 3
MCF	float[]	R/W	The value of Mass center frame with principle axes w.r.t tool frame	{X, Y, Z, RX, RY, RZ}, Size = 6
TeachValue	float[]	R	The original value of the TCP	{X, Y, Z, RX, RY, RZ}, Size = 6
TeachMass	float	R	The original value of mass	Mass in kg
TeachMOI	float[]	R	The original value of the Principal Moments of Inertia	{Ixx, Iyy, Izz}, Size = 3
TeachMCF	float[]	R	The original value of Mass center frame with principle axes w.r.t tool frame	{X, Y, Z, RX, RY, RZ}, Size = 6

Note

// Read values

`float[] f = TCP["RobotEndFlange"].Value`

// Obtain the value {0,0,0,0,0,0} of the TCP " RobotEndFlange "

`float f1 = TCP["RobotEndFlange"].Value[0]`

// or retrieve the x value of " RobotEndFlange " solely

`float mass = TCP["T1"].Mass` // mass = 2.0

`float[] moi = TCP["T1"].MOI` // moi = {0,0,0}

`float[] mcf = TCP["T1"].MCF` // mcf = {0,0,0,0,0,0}

// Write values

`TCP["RobotEndFlange"].Value = {0, -10, 0, 0, 0, 0}`

// Read only, invalid operation, because " RobotEndFlange " is the system TCP

`TCP["T1"].Value = {0, -10, 0, 0, 0, 0}` // Replace the value of "T1" with {0,-10,0,0,0,0}

`TCP["T1"].Value[0] = 10` // or replace the X value of "T1" with 10 solely

`TCP["T1"].Mass = 2.4` // Replace the mass value of "T1" with 2.4 kg

`TCP["T1"].MOI = {0, 0, 0, 1, 2}` // Return error, writing elements to the array do not match to 3 (writing 5 elements)


```
TCP["T1"].MCF = {0, -20, 0, 0, 0, 0, 0} // Return error, writing elements to the array do not match  
to 6 (writing 7 elements)
```

11.4 VPoint

Syntax

VPoint

VPoint[string].attribute

Item

VPoint Initial position of the vision job

Index

string The name of the VPoint

Attribute

Name	Type	Mode	Description	Format
Value	float[]	R/W	The initial coordinate of VPoint	{X, Y, Z, RX, RY, RZ}, Size = 6
BaseName	string	R	The name of the VPoint	"Base Name"
TeachValue	float[]	R	The original job initial coordinate of VPoint	{X, Y, Z, RX, RY, RZ}, Size = 6

Note

// Read values

float[] f = VPoint["Job1"].Value // Obtain the initial coordinate {X, Y, Z, RX, RY, RZ} of VPoint "Job1"

float f1 = VPoint["Job1"].Value[0] // or retrieve the x value of "Job1"

float f1 = VPoint["Job1"].Value[6] // Return error, exceeding the array's access range

string s = VPoint["Job1"].BaseName // s="RobotBase"

// Write values

VPoint["Job1"].Value = {0, 0, 90, 0, 90, 0} // Replace the initial coordinate of VPoint "Job1" with {0,0,90,0,90,0}

VPoint["Job1"].Value[2] = 120 // or replace the Z value of "Job1" with 120 solely

VPoint["Job1"].BaseName = "base1" // Read only, invalid operation

VPoint["Job1"].Value = {0, 0, 90, 0, 90} // Return error, writing elements to the array do not match to 6 (writing 5 elements)

VPoint["Job1"].Value = {0, 0, 90, 0, 90, 0, 100} // Return error, writing elements to the array do not match to 6 (writing 7 elements)

11.5 IO

Syntax

IO

IO[string].attribute

Item

IO Input/Output

Index

string The name of the control module
 ControlBox the control box
 EndModule the end module
 ExtModuleN the external module (N = 0 .. n)
 Safety the safety module

Attribute

ControlBox / EndModule / ExtModuleN

Name	Type	Mode	Description	Format
DI	byte[]	R	Digital input	[0] = DI0 0: Low, 1: High [1] = DI1 [n] = DI n
DO	byte[]	R/W	Digital output	[0] = DO0 0: Low, 1: High [1] = DO1 [n] = DO n
AI	float[]	R	Analog input	[0] = AI0 -10.24V .. +10.24V (Voltage) [1] = AI1 [n] = AI n
AO	float[]	R/W	Analog output	[0] = AO0 -10.00V .. + 10.00V (Voltage) [1] = AO1 [n] = AO n
InstantDI	byte[]	R	Digital input (Instant Command)	[0] = DI0 0: Low, 1: High [1] = DI1 [n] = DI n
InstantDO	byte[]	R/W	Digital output (Instant Command)	[0] = DO0 0: Low, 1: High [1] = DO1 [n] = DO n
InstantAI	float[]	R	Analog input (Instant Command)	[0] = AI0 -10.24V .. +10.24V (Voltage) [1] = AI1 [n] = AI n
InstantAO	float[]	R/W	Analog output (Instant Command)	[0] = AO0 -10.00V .. + 10.00V (Voltage) [1] = AO1 [n] = AO n

* The sets of DI[n]/DO[n]/AI[n]/AO[n] vary from the actual hardware device identification.

Safety

Name	Type	Mode	Description	Format
SI	byte[]	R	Safety function input	0: Low, 1: High SI[0] = SF1 User Connected ESTOP input

Name	Type	Mode	Description	Format
				SI[1] = SF3 User Connected External Safeguard Input SI[2] According to Safety Input Ports Assign SI[3] According to Safety Input Ports Assign .. SI[7] According to Safety Input Ports Assign
SO	byte[]	R	Safety function output	0: Low, 1: High SO[0] According to Safety Output Ports Assign SO[1] According to Safety Output Ports Assign SO[2] According to Safety Output Ports Assign SO[3] According to Safety Output Ports Assign .. SO[7] According to Safety Output Ports Assign

The differences between DI/DO/AI/AO and InstantDI/InstantDO/InstantAI/InstantAO

DI/DO/AI/AO is the queue command with reservations in the main flow of the project. If a DI/DO/AI/AO is after the robot motion function such as a point node with the mixture of trajectories, the DI/DO/AI/AO will be operated after the point node is finished. If using an InstantDI/InstantDO/InstantAI/InstantAO command, It will be operated while the point node is on the run and without waiting for the point node to finish.

In addition, if DI/DO/AI/AO turns to an instant command automatically in the thread page of a project as going without waiting for the point node to finish before running along, the result is the same as using InstantDI/InstantDO/InstantAI/InstantAO.

Note

// Read values

```
byte[] di = IO["ControlBox"].DI // Obtain the digital input status of ControlBox
int dlen = Length(di)           // Obtain the amount of digital PINs with the size of the array
byte di0 = IO["ControlBox"].DI[0] // Obtain the status of ControlBox DI[0]
byte di32 = IO["ControlBox"].DI[32] // Return error, exceeding the array's access range (given DI is
                                     an array with the length of 16 where the indexes start with 0
                                     and end with 15.

float[] ai = IO["ControlBox"].AI // Obtain the analog input status of ControlBox
float[] ao = IO["ControlBox"].AO // Obtain the analog output status of ControlBox
byte si0 = IO["Safety"].SI[0]    // Obtain the safety input status of Safety SI[0]
byte so4 = IO["Safety"].SO[4]    // Obtain the safety output status of Safety SO[4]
byte si1 = IO["ControlBox"].SI[1] // Return error, ControlBox does not support SI attribute.
byte di2 = IO["Safety"].DI[2]    // Return error, Safety does not support DI attribute.
byte di7 = IO["ControlBox"].InstantDI[7] // Obtain the status of ControlBox" DI[7] (instant execution)
```

// Write values

```
IO["ControlBox"].DI = {1,1,0,0} // Read only, invalid operation
IO["ControlBox"].DI[0] = 0       // Read only, invalid operation
IO["ControlBox"].DO[2] = 1       // Set DO 2 to High
IO["ControlBox"].AO[0] = 3.3     // Set AO0 to 3.3V
IO["ControlBox"].DO = {1,1,0,0} // Return error, elements to write mismatch to array's size
                                     (given DI is an array with the length of 16 which covers 16
                                     elements)
```

```
IO["ControlBox"].InstantDO[0] = 1    // Set DO 0 to High (Instant Execution)
IO["Safety"].SI[0] = 0                // Read only, invalid operation
IO["Safety"].SO[4] = 1                // Read only, invalid operation
IO["ControlBox"].SO[1] = 1            // Return error, ControlBox does not support SO attribute.
IO["Safety"].DO[2] = 1                // Return error, Safety does not support DO attribute.
```

11.6 Robot

Syntax

Robot

Robot[int].attribute

Item

Robot

Index

int The index of the robot fixed at 0

Attribute

Name	Type	Mode	Description	Format
CoordRobot	float[]	R	The TCP coordinate of the robot end point opposite to the RobotBase of the robot	{X, Y, Z, RX, RY, RZ}, Size = 6
CoordBase	float[]	R	The TCP coordinate of the robot end point opposite to the current base of the robot.	{X, Y, Z, RX, RY, RZ}, Size = 6
Joint	float[]	R	The current robot joint angle	{J1, J2, J3, J4, J5, J6}, Size = 6
BaseName	string	R	The name of the current base	"Base Name"
TCPName	string	R	The name of the current TCP	"TCP Name"
Payload	float	R	The current compensatory payload	kg
CameraLight	byte	R/W	The lighting of the robot's camera	0: Low (Off), 1: High (On)
InstantCameraLight	byte	R/W	The lighting of the robot's camera (Instant Command)	0: Low (Off), 1: High (On)
TCPForce3D	float	R	The current TCP force as the composite force of the robot base x, y, and z.	N
TCPSpeed3D	float	R	The current TCP speed as a composite speed of the robot base x, y, and z.	mm/s

Note

// Read values

```
float[] rtool = Robot[0].CoordRobot // Obtain the current TCP coordinate of the robot end
                                     // point opposite to the RobotBase of the robot
float[] ftool = Robot[0].CoordBase  // Obtain the current TCP coordinate of the robot end point
                                     // opposite to the current base of the robot.
float f = Robot[0].CoordBase[0]     // or retrieve the X value of the current TCP coordinate of the
                                     // robot end point opposite to the current base of the robot solely.

f = Robot[0].CoordBase[6]           // Return error, exceeding the array's access range
float[] joint = Robot[0].Joint       // Obtain the current robot joint angle
float j = Robot[0].Joint[0]         // or retrieve the current angle of the robot's 1st joint solely
string b = Robot[0].BaseName        // b = "RobotBase"
string t = Robot[0].TCPName         // t = "RobotEndFlange"
float load = Robot[0].Payload       // 0
```

```
byte light = Robot[0].CameraLight      // light = 0 (OFF)
float tf3d = Robot[0].TCPForce3D        // tf3d = 1.234
float ts3d = Robot[0].TCPSpeed3D        // ts3d = 1.234

// Write values
Robot[0].CoordRobot = {0, 90, 0, 0, 0, 0} // Read only, invalid operation
Robot[0].CoordBase = {0, 0, 90, 0, 90, 0}  // Read only, invalid operation
Robot[0].BaseName = "Base1"                // Read only, invalid operation
Robot[0].TCPName = "Tool1"                 // Read only, invalid operation
Robot[0].Payload= 3.4                      // Read only, invalid operation
Robot[0].CameraLight = 1                   // Turn on the lighting of the robot's camera
Robot[0].CameraLight = 0                   // Turn off the lighting of the robot's camera
Robot[0].TCPSpeed3D = 1.234                // Read only, invalid operation
```

11.7 FT

Syntax

FT

FT[string].attribute

Item

FT Force Torque sensor status

Index

string The name of F/T sensor in the F/T sensor list

Attribute

Name	Type	Mode	Description	Format
X	float	R	The force value of the X axis	
Y	float	R	The force value of the y axis	
Z	float	R	The force value of the z axis	
TX	float	R	The torque value of the X axis	
TY	float	R	The torque value of the y axis	
TZ	float	R	The torque value of the z axis	
F3D	float	R	The XYZ resultant force value	
T3D	float	R	The XYZ torque value	
Value	float[]	R	The XYZ resultant force value and torque value array.	{X, Y, Z, TX, TY, TZ}, Size = 6
ForceValue	float[]	R	The XYZ resultant force value array	{X, Y, Z}, Size = 3
TorqueValue	float[]	R	The XYZ torque value array	{TX, TY, TZ}, Size = 3
RefCoordX	float	R	The X-axis force value measured based on the reference base set in the node	
RefCoordY	float	R	The Y-axis force value measured based on the reference base set in the node	
RefCoordZ	float	R	The Z-axis force value measured based on the reference base set in the node	
RefCoordTX	float	R	The X-axis torque value measured based on the reference base set in the node	
RefCoordTY	float	R	The Y-axis torque value measured based on the reference base set in the node	
RefCoordTZ	float	R	The Z-axis torque value measured based on the reference base set in the node	
RefCoordF3D	float	R	The XYZ resultant force value measured based on the reference base set in the node	

Name	Type	Mode	Description	Format
RefCoordT3D	float	R	The XYZ torque measured based on the reference base set in the node	
RefCoordForceValue	float[]	R	The XYZ resultant force value matrix measured based on the reference base set in the node	{RefCoordX, RefCoordY, RefCoordZ}, Size = 3
RefCoordTorqueValue	float[]	R	The XYZ torque value matrix measured based on the reference base set in the node	{RefCoordTX, RefCoordTY, RefCoordTZ}, Size = 3
Model	string	R	The Model name of the F/T sensor	
Zero	byte	R/W	Turn on or off F/T sensor offset	0: Zero OFF, 1: Zero ON

Attributes associated with RefCoord come with values when in Force Control.

Note

// Read values

```
float x = FT["fts1"].X           // Obtain the current X-axis force value of F/T sensor "fts1"
float tx = FT["fts1"].TX        // Obtain the current X-axis torque value of F/T sensor "fts1"
float f3d = FT["fts1"].F3D      // Obtain the current XYZ resultant force value of F/T sensor "fts1"
float[] force = FT["fts1"].ForceValue // Obtain the current XYZ resultant force value array of F/T sensor "fts1"

string mode = FT["fts1"].Model  // Obtain the model name of F/T sensor "fts1"
```

// Write values

```
FT["fts1"].Y = 3.14              // Read only, invalid operation
FT["fts1"].TY = 1.34             // Read only, invalid operation
FT["fts1"].T3D = 4.13           // Read only, invalid operation
FT["fts1"].TorqueValue = {1.1, 2.2, 3.3} // Read only, invalid operation
FT["fts1"].Zero = 1             // Book the current offset of F/T sensor
```

12. Robot Teach Class

12.1 TPoint Class

Use TPoint Class and declare variables to create the point names and the point values. The variable name will be the point name. *It takes calculations for the values of the coordinates and the angles to construct points. It returns an error if it fails to calculate the values of the coordinates and the angles with the parameter values.

Construct 1

TPoint VariableName = float[]

TPoint VariableName = float, float, float, float, float, float

Parameters

float[] The Robot End TCP Coordinate: X(mm), Y(mm), Z(mm), RX(°), RY(°), RZ(°) recorded by RobotBase and RobotEndFlange TCP.

Note

TPoint p1 = {0,-282.75,1094.9,90,0,0}

// the coordinate value 0,-282.75,1094.9,90,0,0 by RobotBase and RobotEndFlange

TPoint p2 = 517.5,-147.8,442.45,180,0,90

// the coordinate value 517.5,-147.8,442.45,180,0,90 by RobotBase and RobotEndFlange

Construct 2

TPoint VariableName = string, float[]

TPoint VariableName = string, float, float, float, float, float, float

Parameters

string Point Definitions. Default to "C".

"C" Point coordinate. When the project operation ends, it does not write the values back to the record file. It uses the declared values with the next project operation.(Same as Construct 1)

"D" Point coordinate. When the project operation ends, it writes the values back to the record file. It prioritizes using the record file with the next project operation. If the record file comes empty, it uses the declared values.

"J" Joint angle. When the project operation ends, it does not the values back to the record file. It uses the declared values with the next project operation.

"JD" Joint angle. When the project operation ends, it writes the values back to the record file. It prioritizes using the record file with the next project operation. If the record file comes empty, it uses the declared values.

float[] If expressed in coordinates, this is the six elements of the TCP coordinates at the robot end: X(mm), Y(mm), Z(mm), RX(°), RY(°), RZ(°); if expressed in angles, the six elements of the robot joints: Joint 1(°), Joint 2(°), Joint 3(°), Joint 4(°), Joint 5(°), Joint 6(°) by the RobotBase base and RobotEndFlange TCP records.

Note

```

TPoint p3 = "D",{522.07,130.75,442.45,180,0,120}
TPoint p4 = "D",134.95,-147.8,1094.9,90,0,90    // Return error , unable to calculate the angle
                                                value by the coordinate value

TPoint p5 = "J",0,0,90,0,90,0
// the angle value 0,0,90,0,90,0 by RobotBase and RobotEndFlange
TPoint p6 = "JD",{30,0,90,0,90,0}
// the angle value 30,0,90,0,90,0 by RobotBase and RobotEndFlange

```

Construct 3

```

TPoint VariableName = string, float[], string, string
TPoint VariableName = string, float, float, float, float, float, float, string, string
TPoint VariableName = float[], string, string
TPoint VariableName = float, float, float, float, float, float, string, string

```

Parameters

string	Point Definitions. Default to "C".
"C"	Point coordinate. When the project operation ends, it does not write the values back to the record file. It uses the declared values with the next project operation.
"D"	Point coordinate. When the project operation ends, it writes the values back to the record file. It prioritizes using the record file with the next project operation. If the record file comes empty, it uses the declared values.
"J"	Joint angle. When the project operation ends, it does not the values back to the record file. It uses the declared values with the next project operation.
"JD"	Joint angle. When the project operation ends, it writes the values back to the record file. It prioritizes using the record file with the next project operation. If the record file comes empty, it uses the declared values.
float[]	If expressed in coordinates, this is the six elements of the TCP coordinates at the robot end: X(mm), Y(mm), Z(mm), RX(°), RY(°), RZ(°); if expressed in angles, the six elements of the robot joints: Joint 1(°), Joint 2(°), Joint 3(°), Joint 4(°), Joint 5(°), Joint 6(°). Record by the base name and the TCP name.
string	Base name. Use the current base name in the record if an empty string.
string	TCP name. Use the current TCP name in the record if an empty string.

Note

```

TBase base1 = 0,0,90,0,0,0
TTCP tcp1 = 0,0,10,0,0,0
TPoint p7 = "D",{0,-282.75,1094.9,90,0,0},"RobotBase"," RobotEndFlange "
TPoint p8 = "J",0,0,90,0,90,0,"base1"," RobotEndFlange" // by base1 and RobotEndFlange
TPoint p9 = "JD",{30,0,90,0,90,0},"RobotBase","tcp1"    // by RobotBase and tcp1
TPoint p0 = "C",{517.5,-147.8,342.45,180,0,90},"base1","tcp1" // by base1 and tcp1
// The syntax below is in the non-define section.
ChangeBase("base1")
TPoint p00 = {517.5,-147.8,342.45,180,0,90},"" ,"tcp1"    // by the current base1 and the
                                                         assigned tcp1

```

Construct 4

TPoint VariableName = string, float[], float[], string, string

TPoint VariableName = string, float[], float[]

TPoint VariableName = float[], float[], string, string

TPoint VariableName = float[], float[]

Parameters

string	Point Definitions. Default to "C".
"C"	When the project operation ends, it does not write the values back to the record file. It uses the declared values with the next project operation.
"D"	When the project operation ends, it writes the values back to the record file. It prioritizes using the record file with the next project operation. If the record file comes empty, it uses the declared values.
float[]	Expressed in coordinates, this is the six elements of the TCP coordinates at the robot end: X(mm), Y(mm), Z(mm), RX(°), RY(°), RZ(°).
float[]	Expressed in angles, the six elements of the robot joints: Joint 1(°), Joint 2(°), Joint 3(°), Joint 4(°), Joint 5(°), Joint 6(°). Record by the base name and the TCP name.
string	Base name. Use the current base name in the record if an empty string.
string	TCP name. Use the current TCP name in the record if an empty string.

Note

TBase base1 = 0,0,90,0,0,0

TTCP tcp1 = 0,0,10,0,0,0

TPoint tp1 = "C",{0,-282.75,1094.9,90,0,0},{0,0,0,0,0,0},"RobotBase"," RobotEndFlange

TPoint tp2 = "D",{517.5,-147.8,442.45,180,0,90},{0,0,90,0,90,0}

// by RobotBase and RobotEndFlange

TPoint tp3 = {0,-292.75,1004.9,90,0,0},{0,0,0,0,0,0},"base1","tcp1" // by base1 and tcp1

TPoint tp4 = {134.95,-147.8,1094.9,90,0,90},{0,0,0,0,90,0}

// by RobotBase and RobotEndFlange

// The syntax below is in the non-define section.

ChangeBase("base1")

ChangeTCP("tcp1")

TPoint tp5 = {0,-292.75,1004.9,90,0,0},{0,0,0,0,0,0},"","" // by the current base1 and the current tcp1

Member Attributes

Name	Type	Mode	Description	Format
Value	float[]	R/W	Point coordinate	{X, Y, Z, RX, RY, RZ}, Size = 6
Joint	float[]	R/W	Joint angle	{J1, J2, J3, J4, J5, J6}, Size = 6
Pose	int[]	R/W	Robot pose	{Config1, Config2, Config3}, Size = 3
Flange	float[]	R	Flange center coordinate	{X, Y, Z, RX, RY, RZ}, Size = 6
BaseName	string	R	Base name	"Base Name"
TCPName	string	R	TCP name	"TCP Name"
TeachValue	float[]	R	Point coordinate (the original teach point coordinate)	{X, Y, Z, RX, RY, RZ}, Size = 6

Name	Type	Mode	Description	Format
TeachJoint	float[]	R	Joint angle (the original teach point angle)	{J1, J2, J3, J4, J5, J6}, Size = 6
TeachPose	int[]	R	Robot pose (the original teach point pose)	{Config1, Config2, Config3}, Size = 3

* It recalculates the Joint and the Flange while setting the Value, and it recalculates the Value and the Flange while setting the Joint. Therefore, it returns an error if failed to calculate.

Note

TPoint P1 = {0,-282.75,1094.9,90,0,0}

// "P1"Coordinate value 0,-282.75,1094.9,90,0,0 by RobotBase and RobotEndFlange

// Read values

float[] f = P1.Value // Get the point coordinate of the point "P1" {0,-282.75,1094.9,90,0,0}

float f1 = P1.Value[1] // or get the Y value along of point "P1"

float f2 = P1.Value[6] // Return error. Exceeding the array's access range

float[] f3 = P1.Joint // Get the angle value of the point "P1" {0,0,0,0,0,0}

string s = P1.BaseName // s = "RobotBase"

// Write values

P1.Value = {517.5,-147.8,442.45,180,0,90} // Modify the point coordinate of the point "P1" to {517.5,-147.8,442.45,180,0,90}

P1.Value[2] = 450 // or modify the z value along of point "P1" to 450

P1.Flange = {0,0,90,0,90,0} // Read only, invalid operation

P1.Value = {0,0,90,0,90} // Return error. The amount of the array elements to write does not match to 6. (5)

P1.Pose = {1,2,4,0} // Return error. The amount of the array elements to write does not match to 3. (4)

12.2 TBase Class

Use TBase Class and declare variables to create the point names and the point values. The variable name will be the point name.

*The system base name comes with the attribute of the read-only mode and without the write mode.

"RobotBase"

Once the variable name is the system base name, it is for variable declaration only. The input value is invalid in the construct.

Construct 1

TBase VariableName = float[]

TBase VariableName = float, float, float, float, float, float

Parameters

float[] Base Value: X(mm), Y(mm), Z(mm), RX(°), RY(°), RZ(°)

Note

TBase base1 = {0,0,90,0,0,0} // Name: base1, Type: C, Value: 0,0,90,0,0,0

TBase base2 = 0,90,90,0,0,0 // Name: base2, Type: C, Value: 0,90,90,0,0,0

TBase RobotBase = 0,0,90,0,0,90

// Since RobotBase is the system base, the input values are void but variable declarations.

Construct 2

TBase VariableName = string, float[]

TBase VariableName = string, float, float, float, float, float, float

Parameters

string Base Types. Default to "C".

"C" When the project operation ends, it does not write the values back to the record file. It uses the declared values with the next project operation.(Same as Construct 1)

"D" When the project operation ends, it writes the values back to the record file. It prioritizes using the record file with the next project operation. If the record file comes empty, it uses the declared values.

"V" Defined the same as type "D." This type goes with vision jobs mainly. The name must comply with the naming rule beginning with *vision_*.

* Only in the define variable section will the variable declaration come with the write back to the record file mechanism.

float[] Base Value: X(mm), Y(mm), Z(mm), RX(°), RY(°), RZ(°)

Note

TBase base5 = "C",{0,0,90,0,0,0} // Name: base5, Type: C, Value: 0,0,90,0,0,0

TBase base6 = "D",0,90,90,0,0,0 // Name: base6, Type: D, Value: 0,90,90,0,0,0

TBase vision_base7 = "V",90,90,90,0,0,0 // Name: vision_base7, Type: V, Value: 90,90,90,0,0,0

TBase base8 = "V",90,90,90,0,0,0

// Return error. The name, base8 , does not comply the naming rule of the type V.

Member Attributes

Name	Type	Mode	Description	Format
Value	float[]	R/W	Base value	{X, Y, Z, RX, RY, RZ}, Size = 6

Name	Type	Mode	Description	Format
Type	string	R	Base type	"R": Robot Base "C": Custom Base "D": Custom Base with write-back "V": Vision Base
TeachValue	float[]	R	Base value (the original teach point base value)	{X, Y, Z, RX, RY, RZ}, Size = 6

Member Methods

Name	Description
GetValue()	Retrieve the base value
SetValue()	Set the base value
ConShift()	Convert a new base value (the original teach value) with the base value and the shift value.

12.2.1 GetValue()

Retrieve the base value. (applicable to multiple bases created in the vision jobs)

Syntax 1

```
float[] GetValue(
    int
)
```

Parameters

int

Base Index. Users can assign from multiple bases created by vision one shot get all. The index is 0 to N and defaulted to 0.

Return

float[]

Base Value: X(mm), Y(mm), Z(mm), RX(°), RY(°), RZ(°)
If the base index value is larger than the number of multiple bases, it returns an empty array.

12.2.2 SetValue()

Set the base value. (applicable to multiple bases created in the vision jobs)

Syntax 1

```
bool SetValue(
    int,
    float[]
)
```

Parameters

int

Base Index. Users can assign from multiple bases created by vision one shot get all. The index is 0 to N and defaulted to 0.

float[]

Base Value: X(mm), Y(mm), Z(mm), RX(°), RY(°), RZ(°)

Return

bool Return **True** if successful. Return **False** if failed.
If the base index value is larger than the number of multiple bases, it sets failed.

Note

```

TBase base1 = {0,0,90,0,0,0}           // Name: base1, Type: C, Value: 0,0,90,0,0,0
TBase RobotBase = 0,0,90,0,0,90
// Since RobotBase is the system base, the input values are void but variable declarations.
TBase base5 = "C",{0,0,90,0,0,0}        // Name: base5, Type: C, Value: 0,0,90,0,0,0
TBase base6 = "D",0,90,90,0,0,0        // Name: base6, Type: D, Value: 0,90,90,0,0,0
TBase vision_base8 = "V",90,90,90,0,0,0
// Name: vision_base8, Type: V, Value: 90,90,90,0,0,0

// Read values
float[] b1 = base1.Value                // Base: "base1", Value: {0,0,90,0,0,0}
float[] b2 = RobotBase.Value            // Base: "RobotBase", Value {0,0,0,0,0,0}
float b3 = base1.Value[2]               // Z value of Base "base1": 90
string t1 = base5.Type                  // "C"
string t2 = base6.Type                  // "D"
string t3 = vision_base8.Type           // "V"
float[] b50 = base5.GetValue(0) // {0,0,90,0,0,0} Obtain the 1st base value in the base "base5".
float[] b51 = base5.GetValue(1)        // {} empty array Obtain the 2nd base value in the base "base5".
float[] vb = vision_base8.GetValue(1)   // Obtain the 2nd base value in the base "vision_base8".
// Suppose there is a vision job, it creates and updates the base
// value of vision_base8.

// Write values
RobotBase.Value = {0,0,90,0,90,0} // Read only, invalid operation for "RobotBase" being the
// system base.
base1.Value = {0,90,0,0,90,0}       // Modify the base "base1" to {0,90,0,0,90,0}
base1.Value[4] = 120                // or modify the the RY value of the base "base1" alone to 120.
base1.Value[6] = 120                // Return error. Exceeding the array's access range.
base1.Type = "C"                    // Read only, invalid operation
base1.Value = {0,0,90,0,90}
// Return error. The amount of the array elements to write does not match to 6. (5)
base1.Value = {0,0,90,0,90,0,100}
// Return error. The amount of the array elements to write does not match to 6. (7)
base6.Value = {30,90,90,0,0,0}      // Write back to the record file since "base6" is type D.
base5.SetValue(0, {90,90,90,0,0,0}) // True. Set the 1st coordinate value in the base "base5".
base5.SetValue(1, {0,0,0,90,90,90}) // False. Failed to set.

```

12.2.3 ConvShift()

Convert a new base value with the base value (the original base teach value) and the shift value, where the shift depends on the position and direction of the original coordinate.

Syntax 1

```

float[] ConvShift(
    float[]
)

```


Parameters

`float[]` Shift value $X_{(mm)}$ $Y_{(mm)}$ $Z_{(mm)}$ $RX(^{\circ})$ $RY(^{\circ})$ $RZ(^{\circ})$

Return

`float[]` the new base value calculated by the given position and direction of the original base teach value and the shift value.
 $X_{(mm)}$ $Y_{(mm)}$ $Z_{(mm)}$ $RX(^{\circ})$ $RY(^{\circ})$ $RZ(^{\circ})$

Syntax 2

```
float[] ConvShift(
    float, float, float, float, float, float
)
```

Note

Same as Syntax 1. It replaces the `float[]` type with `float, float, float, float, float, float` parameters.

Syntax 3

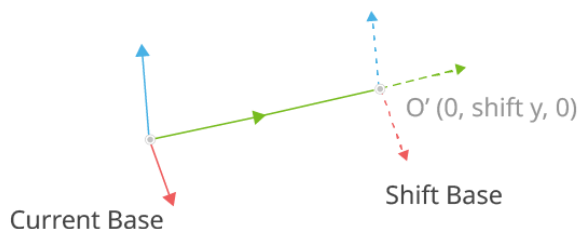
```
float[] ConvShift(
    float[],
    float[]
)
```

Parameters

`float[]` Base value $X_{(mm)}$ $Y_{(mm)}$ $Z_{(mm)}$ $RX(^{\circ})$ $RY(^{\circ})$ $RZ(^{\circ})$
`float[]` Shift value $X_{(mm)}$ $Y_{(mm)}$ $Z_{(mm)}$ $RX(^{\circ})$ $RY(^{\circ})$ $RZ(^{\circ})$

Return

`float[]` the new base value calculated by the given position and direction of the original base teach value and the shift value.
 $X_{(mm)}$ $Y_{(mm)}$ $Z_{(mm)}$ $RX(^{\circ})$ $RY(^{\circ})$ $RZ(^{\circ})$

Note

```
TBase base1 = {200,200,0,0,0,90} // Name: base1, Type: C, Vaule: 200,200,0,0,0,90
```

```
float[] base_shift = {10,20,30,0,0,0}
```

```
float[] new_base = {0,0,0,0,0,0}
```

```
base1.Value = base1.ConvShift(base_shift) // {180,210,30,0,0,90}
```

```
new_base = base1.ConvShift(10,20,30,0,0,0) // {180,210,30,0,0,90}
```

```
new_base = base1.ConvShift(base1.Value, {10,20,30,0,0,0}) // {160,220,60,0,0,90}
```

```
new_base = base1.ConvShift(base1.TeachValue, {10,20,30,0,0,0}) // {180,210,30,0,0,90}
```

12.3 TTCP Class

Use TTCP Class and declare variables to create the point names and the point values. The variable name will be the point name.

*The system TCP name comes with the attribute of the read-only mode and without the write mode.

"RobotEndFlange"

"HandCamera"

"HandCamera2"

Once the variable name is the system TCP name or is in the tool setting of TMflow, it is for variable declaration only. The input value is invalid in the construct.

Construct 1

TTCP VariableName = float[], float

TTCP VariableName = float[]

TTCP VariableName = float, float, float, float, float, float, float

TTCP VariableName = float, float, float, float, float, float

Parameters

float[] TCP value: X(mm), Y(mm), Z(mm), RX(°), RY(°), RZ(°)

float Tool Mass (kg). Default to 0.

Note

TTCP tcp1 = {0,0,90,0,0,0} // Name: tcp1, Type: C, Value: 0,0,90,0,0,0, Mass: 0 kg

TTCP tcp2 = {0,0,90,0,0,0},2 // Name: tcp2, Type: C, Value: 0,0,90,0,0,0, Mass: 2 kg

TTCP tcp3 = 0,0,90,0,0,0 // Name: tcp3, Type: C, Value: 0,0,90,0,0,0, Mass: 0 kg

TTCP tcp4 = 0,0,90,0,0,0,2 // Name: tcp4, Type: C, Value: 0,0,90,0,0,0, Mass: 2 kg

TTCP RobotEndFlange = 0,0,90,0,0,90

// Since RobotEndFlange is the system tool name, the input values are void but variable declarations.

Construct 2

TTCP VariableName = string, float[], float

TTCP VariableName = string, float[]

TTCP VariableName = string, float, float, float, float, float, float, float

TTCP VariableName = string, float, float, float, float, float, float

Parameters

string Tool Types. Default to "C".

"C" When the project operation ends, it does not write the values back to the record file. It uses the declared values with the next project operation.(Same as Construct 1)

"D" When the project operation ends, it writes the values back to the record file. It prioritizes using the record file with the next project operation. If the record file comes empty, it uses the declared values.

"V" Defined the same as type "D." This type goes with vision jobs mainly. The name must comply with the naming rule beginning with *vision_*.

* Only in the define variable section will the variable declaration come with the write back to the record file mechanism.

float[] TCP value: X(mm), Y(mm), Z(mm), RX(°), RY(°), RZ(°)

float Tool Mass (kg). Default to 0.

Note

```

TTCP tcp5 = "C",{0,0,90,0,0,0} // Name: tcp5, Type: C, Value: 0,0,90,0,0,0, Mass: 0 kg
TTCP tcp6 = "D",{0,90,90,0,0,0},2 // Name: tcp6, Type: D, Value: 0,90,90,0,0,0, Mass: 2 kg
TTCP visionTCP_tcp7 = "V",90,90,90,0,0,0 // Name: visionTCP_tcp7, Type: V, Value:
                                           90,90,90,0,0,0

TTCP tcp7v = "V",90,90,90,0,0,0
                                           // Return error. The name, tcp7v, does not comply the naming rule of the
                                           type V.

TTCP tcp8 = "D",90,90,90,0,0,0,2 // Name: tcp8, Type: D, Value: 90,90,90,0,0,0, Mass: 2 kg

```

Construct 3

TTCP VariableName = string, float[], float, float[], float[]

TTCP VariableName = float[], float, float[], float[]

Parameters

string Tool Types. Default to "C".

"C" When the project operation ends, it does not write the values back to the record file. It uses the declared values with the next project operation.(Same as Construct 1)

"D" When the project operation ends, it writes the values back to the record file. It prioritizes using the record file with the next project operation. If the record file comes empty, it uses the declared values.

"V" Defined the same as type "D." This type goes with vision jobs mainly. The name must comply with the naming rule beginning with *vision_*.
* Only in the define variable section will the variable declaration come with the write back to the record file mechanism.

float[] TCP value: X(mm), Y(mm), Z(mm), RX(°), RY(°), RZ(°)

float Tool Mass (kg). Default to 0.

float[] Principal Moments of Inertia: lxx(kg-mm²), lyy(kg-mm²), lzz(kg-mm²)

float[] Mass Center Frame with Principle Axes w.r.t Tool Frame: X(mm), Y(mm), Z(mm), RX(°), RY(°), RZ(°)

Note

```

TTCP tcp9 = {0,0,90,0,0,0},2,{2,0.5,0.5},{0,0,-80,0,0,0}
// Name: tcp9, Type: C, Value: 0,0,90,0,0,0, Mass: 2 kg, Inertia: 2,0.5,0.5, The Reference Coordinate:
0,0,-80,0,0,0

TTCP tcp0 = "D",{0,0,150,0,0,90},1,{1,0.5,0.5},{0,0,-80,0,0,0}
// Name: tcp0, Type: D, Value: 0,0,150,0,0,90, Mass: 1 kg, Inertia: 1,0.5,0.5, The Reference Coordinate:
0,0,-80,0,0,0

```

Member Attributes

Name	Type	Mode	Description	Format
Value	float[]	R/W	TCP value	{X, Y, Z, RX, RY, RZ}, Size = 6
Mass	float	R/W	Mass	Mass in kg
MOI	float[]	R/W	Principal Moments of Inertia	{lxx, lyy, lzz}, Size = 3
MCF	float[]	R/W	Mass center frame with principle axes w.r.t tool frame	{X, Y, Z, RX, RY, RZ}, Size = 6
TeachValue	float[]	R	TCP value (the original TCP setting value)	{X, Y, Z, RX, RY, RZ}, Size = 6

Name	Type	Mode	Description	Format
TeachMass	float	R	Mass (the original TCP setting value)	Mass in kg
TeachMOI	float[]	R	Principal Moments of Inertia (the original TCP setting value)	{Ixx, Iyy, Izz}, Size = 3
TeachMCF	float[]	R	Mass center frame with principle axes w.r.t tool frame (the original TCP setting value)	{X, Y, Z, RX, RY, RZ}, Size = 6

Note

```

TTCP tcp1 = {0,0,90,0,0,0},3           // Name: tcp1, Type: C, Value: 0,0,90,0,0,0, Mass: 3 kg
TTCP RobotEndFlange = 0,0,90,0,0,90
           // Since RobotEndFlange is the system tool name, the input values are void but variable
           declarations.
TTCP tcp9 = "D",{0,0,90,0,0,0},2,{2,0.5,0.5},{0,0,-80,0,0,0}
// Read values
float[] t0 = RobotEndFlange.Value
// Obtain the TCP value of the tool named RobotEndFlange: {0,0,0,0,0,0}
float[] t1 = tcp1.Value           // Obtain the TCP value of the tool named tcp1: {0,0,90,0,0,0}
float t2 = tcp1.Value[2]         // or obtain the Z value of the tool named tcp1 alone.
float mass = tcp1.Mass           // mass = 3
float[] mcf = tcp1.MCF           // mcf = {0,0,0,0,0,0}
float mass9 = tcp9.Mass          // mass9 = 2 // Obtain 2.1 kg if operate after tcp9.Mass = 2.1
float[] moi9 = tcp9.MOI          // moi9 = {2,0.5,0.5}
// Write values
RobotEndFlange.Value = {0, -10, 0, 0, 0, 0}
           // Read only, invalid operation for RobotEndFlange being the system TCP.
tcp1.Value = {0, -10, 0, 0, 0, 0} // Modify the TCP named tcp1 to {0,-10,0,0,0,0}
tcp1.Value[0] = 10                // or modify the X value of tcp1 alone to 10
tcp1.Mass = 2.4                   // Modify the the mass of the TCP named tcp1 to 2.4 kg.
tcp9.Mass = 2.1                  // Modify the the mass of the TCP named tcp9 to 2.1 kg.
tcp1.MOI = {0, 0, 0, 1, 2}        // Return error. The amount of the array elements to wirte does not
                                // match to 3. (5)
tcp1.MCF = {0, -20, 0, 0, 0, 0, 0}
                                // Return error. The amount of the array elements to wirte does not
                                // match to 6. (7)

```

13. Robot Motion & Vision Job Function

For robot motion functions and vision job functions to be available if using flow projects, it requires the External Script to have the project flow enter the Listen node (external script control mode) or the script node. If using script project programming, they are available in the project programming directly.

Calling in the main thread, namely, the main flows or subflows in the flow projects, or the main functions and the functions to call in the script projects, is mandatory for robot motion functions and vision job functions moving to the initial position. Calling motion functions is not allowed in other threads. All the motion processes will be queued temporarily and processed in sequence. Users can use the queue tag numbers to understand the current motion command process if necessary.

13.1 QueueTag()

Set robot motions with Queue Tag Numbers to denote the current robot motion in process. The status of each queue tag can be monitored using TMSTA SubCmd 01.

Syntax 1

```
bool QueueTag(
    int,
    int
)
```

Parameters

int	The tag number. Valid for integers between 1 and 15.
int	Wait for the tagging to continue processing or not.
0	Not wait (default)
1	Wait
	When the value is set to 1, the process stays in the function and waits for the tagging to complete and continue processing.

Return

bool Return **True** when tagged successfully. Return **False** when tagged unsuccessfully.

Syntax 2

```
bool QueueTag(
    int
)
```

Note

Same as syntax 1. It default to 0 not waiting for the tagging to continue processing.

13.2 WaitQueueTag()

Wait for the Queue Tag Number of the robot motion to complete.

Syntax 1

```
int WaitQueueTag(
    int,
    int
)
```

Parameters

int The tag number
 1..15 Valid tag numbers.
 0 Invalid tag number, but waits for the timeout. (No waiting for the time out if set the timeout to infinite.)
 <0 Unavailable tag number. No waiting for the time out.
 >15 Unavailable tag number. No waiting for the time out.

int Set the time to the timeout
 < 0 Wait infinitely. Valid when the tag number is between 1 to 15 legitimately. (default) **= 0** Wait to check once
 > 0 Wait in milliseconds before timeout
 When using waiting in queue, the process stays in the function until the tagging is completed, the tagging is not existed, or timeout, and then continues processing.

Return

int Return the result of waiting
 1 The tagging is completed
 0 The tagging is incomplete or timeout
 -1 The tagging is not existed

* Tag numbers can be reused.

* The tag number will retain the status of the final four tags, and if the tag number did not occur or exceeds them, it returns not existed.

Syntax 2

```
int WaitQueueTag(
    int
)
```

Note

The syntax is the same as syntax 1. The default is no timeout and required to wait for the tagging to complete (or not existed)

WaitQueueTag(int, int) => WaitQueueTag(int, 0)

Syntax 3

```
void WaitQueueTag(
)
```

Parameters

void No input value

Return

void No return

Note

No tag number is required. It waits for all the robot motions in the current queue to complete before going forward execution.

13.3 CheckQueueTag()

Check the Queue Tag Number of the robot motion to complete.

Syntax 1

```
int CheckQueueTag(
    int
)
```

Parameters

int The tag number
1..15 Valid tag numbers.

Return

int Return the result of checking
1 The tagging is completed.
0 The tagging is incomplete.
-1 The tagging is not existed.

* Tag numbers can be reused.

* The tag number will retain the status of the final four tags, and if the tag number did not occur or exceeds them, it returns not existed.

* Same as WaitQueueTag(int, 0) and denotes the result of one time checking only tag number.

● Motion Function Queue Tag

Motion function queue tags are used to cooperate with the robot motion functions. Since all motion functions are queued in the buffer and executed in order, use the cooperative queue tags, it is possible to know which motion function is in execution currently.

```
1. < $TMSCT,172,2,float[] targetP1= {0,0,90,0,90,0}\r\n
    PTP("JPP",targetP1,10,200,0,false)\r\n
    QueueTag(1)\r\n           // QueueTag(1) not wait and continue processing
    float[] targetP2 = {0,0,90,90,0,0}\r\n
    PTP("JPP",targetP2,10,200,10,false)\r\n
    QueueTag(2)\r\n           // QueueTag(2) not wait and continue processing
    ,*49\r\n
```

When executed the script content, since QueueTag() did not wait, after execution, the process returned

```
> $TMSCT,4,2,OK,*5F\r\n
When robot motion executed PTP() targetP1, because of QueueTag(1), it will return
> $TMSTA,10,01,01,true,*64\r\n           // TMSTA SubCmd 01, TagNumber 01, completed
When robot motion executed PTP() targetP2, because of QueueTag(2) , it will return
> $TMSTA,10,01,02,true,*67\r\n           // TMSTA SubCmd 01, TagNumber 02, completed
```

```
2. < $TMSCT,174,2,float[] targetP3= {0,0,90,0,90,0}\r\n
    PTP("JPP",targetP3,10,200,0,false)\r\n
    QueueTag(3,1)\r\n       // QueueTag(3) wait and stay in the function until the tagging completed
    float[] targetP4 = {0,0,90,90,0,0}\r\n
    PTP("JPP",targetP4,10,200,10,false)\r\n
    QueueTag(4)\r\n           // QueueTag(4) not wait and continue processing
    ,*56\r\n
```

When executed the script content, since QueueTag(3,1) is set to wait, after tagging completed, the process returned

```
> $TMSTA,10,01,03,true,*66\r\n           // TMSTA SubCmd 01, TagNumber 03, completed
When QueueTag(3) completed, the process continues, since QueueTag(4) is not set to wait,
after execution, the process returned
> $TMSCT,4,2,OK,*5F\r\n
When robot motion executed PTP() targetP2, because of QueueTag(4) , it will return
> $TMSTA,10,01,04,true,*61\r\n           // TMSTA SubCmd 01, TagNumber 04, completed
```

* \$TMSCT,4,2,OK is returned when the process executed the script. Therefore, if using QueueTag to wait or WaitQueueTag to wait, it will return after the execution as well.

13.4 StopAndClearBuffer()

Stop the motion of the robot and clear existing commands of the robot in the buffer.

Syntax

```
bool StopAndClearBuffer(  
)
```

Parameter

void No parameter

Return

bool True Command accepted ; False Command rejected

Note

StopAndClearBuffer()

13.5 ChangeBase()

Set the reference base for the consecutive motion.

Syntax 1

```
bool ChangeBase(
    string,
    int
)
```

Parameters

string	The base name
int	The base index. Users can specify the bases created by the vision task of one shot get all. The value is 0 N, and the default is 0.

Return

bool	True	Change successfully.
	False	Change unsuccessfully.

Syntax 2

```
bool ChangeBase(
    string
)
```

Note

Same as syntax 1 and fill 0 as the default base index.

ChangeBase("RobotBase")	//change the reference base for motion to "RobotBase"
ChangeBase("vision_job1", 1)	//change the reference base to the 2 nd base value of "vision_job1". If the name or the index specified of the base does exist, it reports an error.
ChangeBase("RobotBase", 10)	//change the reference base for motion to "RobotBase". Since it is the name of the base, the base index is invalid.

Syntax 3

```
bool ChangeBase(
    float[]
)
```

Parameters

float[] Base parameters X, Y, Z, RX, RY, and RZ

Return

bool	True	Change successfully.
	False	Change unsuccessfully.

Note

```
float[] Base1 = {20,30,10,0,0,90}      //declare a floating point array for the base values.
ChangeBase(Base1)                      //change the reference base for motion to the set base value
```

Syntax 4

```
bool ChangeBase(
    float, float, float, float, float, float
)
```

Parameters

float, float, float, float, float, float
Base parameters X, Y, Z, RX, RY, and RZ

Return

bool	True	Change successfully.
	False	Change unsuccessfully.

Note

```
ChangeBase(20,30,10,0,0,90)      //change the reference base to {20,30,10,0,0,90}
```

13.6 ChangeTCP()

Set the end tool parameters for the consecutive motion.

Syntax 1

```
bool ChangeTCP(
    string
```

```
)
```

Parameters

string TCP name

Return

bool True Change successfully.
 False Change unsuccessfully.

Note

```
ChangeTCP("RobotEndFlange")            //change the end tool to "RobotEndFlange"
```

Syntax 2

```
bool ChangeTCP(
    float[]
```

```
)
```

Parameters

float[] TCP parameters X, Y, Z, RX, RY, and RZ

Return

bool True Change successfully.
 False Change unsuccessfully.

Note

```
float[] Tool1 = {0,0,150,0,0,90}            // declare a floating point array for the end tool values.
ChangeTCP(Tool1)                            // change the end tool value to the set end tool value
```

Syntax 3

```
bool ChangeTCP(
    float[],
    float
```

```
)
```

Parameters

float[] TCP parameters X, Y, Z, RX, RY, and RZ
float TCP mass

Return

bool True Change successfully.
 False Change unsuccessfully.

Note

```
float[] Tool1 = {0,0,150,0,0,90}            // declare a floating point array for the end tool values.
ChangeTCP(Tool1,2)                            // change the end tool value to the set end tool value.
The weight of the end tool is 2kg.
```

Syntax 4

```
bool ChangeTCP(
    float[],
    float,
    float[]
```

```
)
```

Parameters

float[] TCP parameters X, Y, Z, RX, RY, and RZ
float TCP mass
float[] TCP rotation inertia includes nine values: (1) lxx, (2) lyy, (3) lzz, along with the mass center coordinates relative to the tool coordinates, which are (4) X, (5) Y, (6) Z, (7) RX, (8) RY, (9) RZ.

Return

bool True Change successfully.
 False Change unsuccessfully.

Note

```
float[] Tool1 = {0,0,150,0,0,90}           // declare a floating point array for the end tool values.
float[] COM1 = {2,0.5,0.5,0,0,-80,0,0,0}    // declare a floating point array for the rotation inertia and
                                             // the reference coordinate relative to the tool coordinates.

ChangeTCP(Tool1,2,COM1)                   // change the end tool value to the set end tool value. The
                                             // weight of the end tool is 2kg,.and it applies the rotation
                                             // inertia and its reference base.
```

Syntax 5

```
bool ChangeTCP(
    float, float, float, float, float, float
)
```

Parameters

float, float, float, float, float, float
TCP parameters X, Y, Z, RX, RY, and RZ

Return

bool True Change successfully.
 False Change unsuccessfully.

Note

```
ChangeTCP(0,0,150,0,0,90)                 // change the end tool value to {0,0,150,0,0,90}
```

Syntax 6

```
bool ChangeTCP(
    float, float, float, float, float, float,
    float
)
```

Parameters

float, float, float, float, float, float
TCP parameters X, Y, Z, RX, RY, and RZ

float
TCP mass

Return

bool True Change successfully.
 False Change unsuccessfully.

Note

ChangeTCP(0,0,150,0,0,90,2) // change the end tool value to {0,0,150,0,0,90}. The weight of the end tool is 2kg.

Syntax 7

```
bool ChangeTCP(
    float, float, float, float, float, float,
    float,
    float, float, float, float, float, float, float, float
)
```

Parameters

float, float, float, float, float, float
TCP parameters X, Y, Z, RX, RY, and RZ

float
TCP mass

float, float, float, float, float, float, float, float, float
TCP rotation inertia includes nine values: (1) lxx, (2) lyy, (3) lzz, along with the mass center coordinates relative to the tool coordinates, which are (4) X, (5) Y, (6) Z, (7) RX, (8) RY, (9) RZ.

Return

bool True Change successfully.
 False Change unsuccessfully.

Note

ChangeTCP(0,0,150,0,0,90,2, 2,0.5,0.5,0,0,-80,0,0,0) // change the end tool value to {0,0,150,0,0,90}. The weight of the end tool is 2kg. The rotation inertia is {2,0.5,0.5} and its reference coordinate is {0,0,-80,0,0,0}.

13.7 ChangeLoad()

Set the load weight to compensate.

Syntax 1

```
bool ChangeLoad(
    float
)
```

Parameters

float Load weight in kilograms

Return

bool	True	Change successfully.
	False	Change unsuccessfully.

Note

```
ChangeLoad(2.3) // Set the load weight to 2.3 kg.
```

Syntax 2

```
bool ChangeLoad(
    float,
    float
)
```

Parameters

float	Load weight in kilograms
float	Weight Conversion Estimated Distance in millimeters

Return

bool	True	Change successfully.
	False	Change unsuccessfully.

Note

```
ChangeLoad(2.3, 100) // Set the load weight to 2.3 kg.
```

Syntax 3

```
bool ChangeLoad(
    string,
    int,
    float
)
```

Parameters

string	the name of the sensor
int	the sensing condition
0	Picking up unknown load
1	Releasing unknown load
float	Weight Conversion Estimated Distance in millimeters

Return

bool	True	Change successfully.
	False	Change unsuccessfully.

Note

FTSensor fts1 = "TMFT300"

ChangeLoad("fts1", 0, 100) // Use sensor fts1 to pick up an unknown load and change the load weight by the detected weight.

13.8 PTP()

Define and send PTP motion command into buffer for execution.

Syntax 1

```
bool PTP(
    string,
    float[],
    int,
    int,
    int,
    bool
)
```

Parameters

string	Definition of data format, combines three letters #1: Motion target format: "C" expressed in Cartesian coordinate "J" expressed in joint angles #2: Speed format: "P" expressed as a percentage #3: Blending format "P" expressed as a percentage
float[]	Motion target. If defined with coordinate, it includes the coordinate of tool center point at the robot end: X (mm), Y (mm), Z (mm), RX(°), RY(°), RZ(°). If defined with joint angle, it includes the angles of six joints: Joint1(°), Joint 2(°), Joint 3(°), Joint 4(°), Joint 5(°), Joint 6(°).
int	The robot end moving speed setting, expressed as a percentage (%) ≥0 Percentage of speed (%) -1 Using the Acceleration Table (supported on TM Robot S series models) <0 Invalid value
int	The time interval to accelerate to top speed (ms). Invalid when using the Acceleration Table.
int	Blending value, expressed as a percentage (%)
bool	Disable precise positioning true Disable precise positioning false Enable precise positioning

Return

bool **True** Command accepted; **False** Command rejected (format error)

Note

Data format parameter includes: (1) "CPP", (2) "JPP"

```
float[] target1 = {0,0,90,0,90,0}
PTP("JPP",target1,10,200,0,false)    // variable ingress available
PTP("JPP",{0,0,90,0,90,0},10,200,0,false) // array constant ingress available
PTP("CPP",Point["P2"].Value,10,200,0,false) // point coordinate retrieving with point
                                              parameterization available (robot posture attention
                                              required)
PTP("JPP",Point["P2"].Joint,10,200,0,false) // joint angle retrieving with point parameterization
                                              available
// move to target with PTP, speed = 10%, time to top speed = 200ms
```

Syntax 2

```
bool PTP(
    string,
    float[],
    int,
    int,
    int,
    bool,
    int[]
)
```

Parameters

string	Definition of data format, combines three letters #1: Motion target format: "C" expressed in Cartesian coordinate #2: Speed format: "P" expressed as a percentage #3: Blending format "P" expressed as a percentage
float[]	Motion target. If defined with coordinate, it includes the coordinate of tool center point at the robot end: X (mm), Y (mm), Z (mm), RX(°), RY(°), RZ(°).
int	The robot end moving speed setting, expressed as a percentage (%) ≥0 Percentage of speed (%) -1 Using the Acceleration Table (supported on TM Robot S series models) <0 Invalid value
int	The time interval to accelerate to top speed (ms). Invalid when using the Acceleration Table.
int	Blending value, expressed as a percentage (%)
bool	Disable precise positioning true Disable precise positioning false Enable precise positioning
int[]	The pose of robot : [Config1, Config2, Config3], please find more information in appendix

Return

bool True Command accepted; **False** Command rejected (format error)

Note

Data format parameter includes: (1) "CPP"

```
float[] targetP1 = {417.50,-122.30,343.90,180.00,0.00,90.00}
```

```
float[] pose = {0,2,4}
```

```
PTP("CPP", target1,50,200,0,false, pose)                   // variable ingress available
```

```
PTP("CPP",{417.50,-122.30,343.90,180.00,0.00,90.00},10,200,0,false,pose)
```

```
                                                          // array constant ingress available
```

```
PTP("CPP",{417.50,-122.30,343.90,180.00,0.00,90.00},10,200,0,false,{0,2,4})
```

```
                                                          // array constant ingress available
```

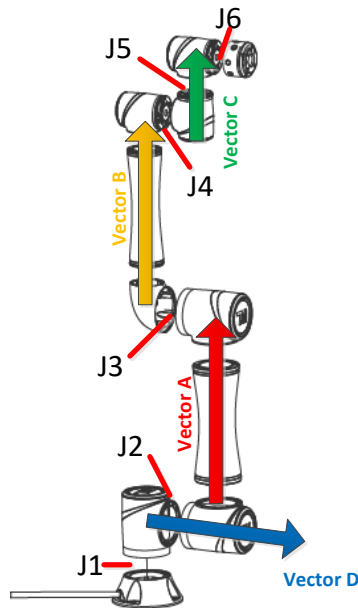
```
PTP("CPP",Point["P2"].Value,10,200,0,false,{0,2,4}) // point coordinate retrieving with point  
                                                          parameterization available
```

```
                                                          // move to target with PTP, speed = 10%, time
```

to top speed = 200ms, and pose configuration = 024.

● Pose Configuration Parameters: [Config1, Config2, Config3]

Config: config1, config2, config3



config1=0:

if [(Vector A + Vector B + Vector C) projects on X-Y plane] cross [Vector D projects on X-Y plane] is on negative-Z

config1=1:

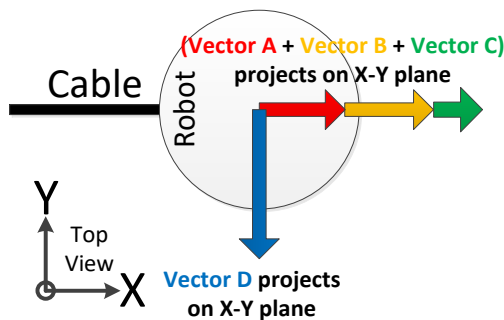
if [(Vector A + Vector B + Vector C) projects on X-Y plane] cross [Vector D projects on X-Y plane] is on positive-Z

config2=2:

if (config1=0 and J3 is positive) or (config1=1 and J3 is negative)

config2=3:

if (config1=0 and J3 is negative) or (config1=1 and J3 is positive)



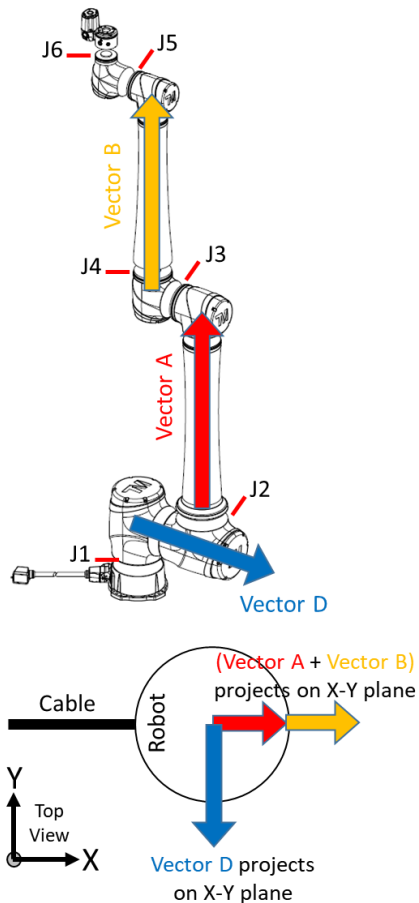
config3=4:

if (config1=0 and J5 is positive) or (config1=1 and J5 is negative)

config3=5:

if (config1=0 and J5 is negative) or (config1=1 and J5 is positive)

TM25S/TM30S



config1=0:

if [(Vector A + Vector B) projects on X-Y plane] cross [Vector D projects on X-Y plane] is on negative-Z

config1=1:

if [(Vector A + Vector B) projects on X-Y plane] cross [Vector D projects on X-Y plane] is on positive-Z

config2=2:

if (config1=0 and J3 is positive) or (config1=1 and J3 is negative)

config2=3:

if (config1=0 and J3 is negative) or (config1=1 and J3 is positive)

config3=4:

if (config1=0 and J5 is positive) or (config1=1 and J5 is negative)

config3=5:

if (config1=0 and J5 is negative) or (config1=1 and J5 is positive)

Syntax 3

```
bool PTP(
    string,
    float[],
    int,
    int,
    int,
    bool,
    float[],
)
```

Parameters

- | | |
|----------------|--|
| string | Definition of data format, combines three letters
#1: Motion target format:
"C" expressed in coordinate
#2: Speed format:
"P" expressed as a percentage
#3: Blending format:
"P" expressed as a percentage |
| float[] | Motion target expressed in coordinate, It is six elements of the TCP coordinate at the robot end: X (mm), Y (mm), Z (mm), RX(°), RY(°), RZ(°). |
| int | The robot end moving speed setting, expressed as a percentage (%)
≥0 Percentage of speed (%) |

-1 Using the Acceleration Table (supported on TM Robot S series models)
 <0 Invalid value

int The time interval to accelerate to top speed (ms). Invalid when using the Acceleration Table.

int Blending value, expressed as a percentage (%)

bool Disable precise positioning
 true Disable precise positioning
 false Enable precise positioning

float[] Motion target expressed in reference joint angles. It is six elements of the robot joints: Joint 1(°), Joint 2(°), Joint 3(°), Joint 4(°), Joint 5(°), Joint 6(°).

Return

bool **True** Command accepted; **False** Command rejected

Note

Data format parameter includes: (1) "CPP"

`float[] target1 = {417.50,-122.30,343.90,180.00,0.00,90.00}`

`float[] joint6 = {3.53,-13.26,116.3,-13.04,90,3.53}`

`PTP("CPP",target1,10,200,0,false,joint6)` *// variable ingress available*

`PTP("CPP",{417.50,-122.30,343.90,180.00,0.00,90.00},10,200,0,false,joint6)` *// array constant ingress available*

`PTP("CPP",{417.50,-122.30,343.90,180.00,0.00,90.00},10,200,0,false,{3.53,-13.26,116.3,-13.04,90,3.53})`

`PTP("CPP",Point["P2"].Value,10,200,0,false,Point["P2"].Joint)`
*// point coordinate retrieving with point
 parameterization available
 // move to target with PTP, speed = 10%,
 time to top speed = 200ms, and attempt to
 reach the closest assigned joint target.*

13.9 Move_PTP()

Define and send PTP relative motion commands for execution.

Syntax 1

```
bool Move_PTP(
    string,
    float[],
    int,
    int,
    int,
    bool
)
```

Parameters

string	Definition of data format made of three letters #1: Relative motion target format: "C": expressed with the current robot base "T": expressed with the tool coordinate "J": expressed in joint angles #2: Speed format: "P": expressed as a percentage #3: Blending format: "P": expressed as a percentage
float[]	relative motion parameters. If expressed in coordinate, it includes the robot end TCP relative motion value with respect to the specified coordinate: X (mm), Y (mm), Z (mm), RX(°), RY(°), RZ(°); If defined with joint angle, it includes the angles of six joints: Joint1(°), Joint 2(°), Joint 3(°), Joint 4(°), Joint 5(°), Joint 6(°)
int	The robot end moving speed setting, expressed as a percentage (%) ≥0 Percentage of speed (%) -1 Using the Acceleration Table (supported on TM Robot S series models) <0 Invalid value
int	The time interval to accelerate to top speed (ms). Invalid when using the Acceleration Table.
int	Blending value, expressed as a percentage (%)
bool	Disable precise positioning true Disable precise positioning false Enable

Return

bool **True** Command accepted; **False** Command rejected (format error)

Note

Motion command parameter includes: (1) "CPP", (2) "TPP" or (3) "JPP"

```
float[] relmove = {0,0,10,45,0,0}
```

```
Move_PTP("TPP",var_rel,10,200,0,false)           // variable ingress available
```

```
Move_PTP("TPP",{0,0,10,45,0,0},10,200,0,false)   // array constant ingress available
```

```
// Move 0,0,10,45,0,0, with respect to tool coordinate, with PTP, velocity = 10%, time to top speed = 200ms
```

13.10 Line()

Define and send Line motion command into buffer for execution.

Syntax 1

```
bool Line(
    string,
    float[],
    int,
    int,
    int,
    bool
)
```

Parameters

string	Definition of data format, combines three letters #1: Motion target format: "C" expressed with Cartesian coordinate #2: Speed format: "P" expressed by percentage "A" expressed with absolute speed and in synchronization with the project speed "D" expressed with absolute speed and not in synchronization with the project speed #3: Blending format: "P" expressed by percentage "R" expressed by radius
float[]	Motion target. It includes the coordinate of tool center point at the robot end: X (mm), Y (mm), Z (mm), RX(°), RY(°), RZ(°)
int	The robot end moving speed setting, expressed as a percentage (%) or an absolute speed (mm/s) ≥0 Percentage of speed (%) or absolute speed (mm/s) -1 Using the Acceleration Table (supported on TM Robot S series models) <0 Invalid value
int	The time interval to accelerate to top speed (ms). Invalid when using the Acceleration Table.
int	Blending value, expressed as a percentage (%) or in radius (mm)
bool	Disable precise positioning true Disable precise positioning false Enable precise positioning

Return

bool **True** Command accepted; **False** Command rejected (format error)

Note

Data format parameter includes:

```
float[] target1 = {417.50,-122.30,343.90,180.00,0.00,90.00}
Line("CAR",target1,100,200,50,false) // variable ingress available
Line("CAR",{417.50,-122.30,343.90,180.00,0.00,90.00},100,200,50,false)
// array constant ingress available
Line("CAR",Point["P2"].Value,100,200,50,false)
// point coordinate retrieving with point parameterization available
```

```
// move to target with Line, speed = 10%, time to top speed = 200ms, and blending value = 50

// Opt for the singularity handling mechanism set by LineSingularity().
LineSingularity(0) // Disable singularity handling.
PTP("JPP", {45,-20,125,-100,-15,0}, 100, 500, 0, false)
Line("CPP", {307.8192,-100.8191,436.7351,90.1633,-0.4788,0.4848}, 100, 500, 0, false)
// Disable
Line("CPP", {138.0930,-411.4636,438.1152,89.5760,-0.3520,0.4868}, 100, 500, 0, false)
// Disable, through singularity

LineSingularity(1) // Enable singularity handling: path avoidance.
PTP("JPP", {45,-20,125,-100,-15,0}, 100, 500, 0, false)
Line("CPP", {307.8192,-100.8191,436.7351,90.1633,-0.4788,0.4848}, 100, 500, 0, false)
// Enable
Line("CPP", {138.0930,-411.4636,438.1152,89.5760,-0.3520,0.4868}, 100, 500, 0, false)
// Enable, path avoidance
```

Syntax 2

```
bool Line(
    string,
    float[],
    int,
    int,
    int,
    bool,
    int
)
```

Parameters

string	Definition of data format, combines three letters #1: Motion target format: "C" expressed with Cartesian coordinate #2: Speed format: "P" expressed by percentage "A" expressed with absolute speed and in synchronization with the project speed "D" expressed with absolute speed and not in synchronization with the project speed #3: Blending format: "P" expressed by percentage "R" expressed by radius
float[]	Motion target. It includes the coordinate of tool center point at the robot end: X (mm), Y (mm), Z (mm), RX(°), RY(°), RZ(°)
int	The robot end moving speed setting, expressed as a percentage (%) or an absolute speed (mm/s) ≥0 Percentage of speed (%) or absolute speed (mm/s) -1 Using the Acceleration Table (supported on TM Robot S series models) <0 Invalid value
int	The time interval to accelerate to top speed (ms). Invalid when using the Acceleration Table.
int	Blending value, expressed as a percentage (%) or in radius (mm)
bool	Disable precise positioning

true Disable precise positioning
false Enable precise positioning
int Singularity handling.
 0 Disbale
 1 Path avoidance
 2 Speed change(TM25S, TM30S)
 * After exiting from the function, the singularity handling mechanism returns to what it was before entering.

Return

bool **True** Command accepted; **False** Command rejected (format error)

Note

Data format parameter includes:

(1) "CPP", (2) "CPR", (3) "CAP", (4) "CAR", (5) "CDP", (6) "CDR"

Line("CAR", 417.50,-122.30,343.90,180.00,0.00,90.00,100,200,50,false)

// Move to 417.50,-122.30,343.90,180.00,0.00,90.00 with Line, velocity = 100mm/s, time to top speed = 200ms, blending radius = 50mm

13.11 Move_Line()

Define and send Line relative motion commands for execution.

Syntax 1

```
bool Move_Line(
    string,
    float[],
    int,
    int,
    int,
    bool
)
```

Parameters

string	Definition of data format made of three letters #1: Relative motion target format: "C": expressed with w.r.t. current base "T": expressed with w.r.t. tool coordinate #2: Speed format: "P": expressed by percentage "A" expressed with absolute speed and in synchronization with the project speed "D" expressed with absolute speed and not in synchronization with the project speed #3: Blending format: "P": expressed by percentage "R": expressed by radius
float[]	Relative motion parameters. It includes the coordinate of tool center point at the robot end: X (mm), Y (mm), Z (mm), RX(°), RY(°), RZ(°)
int	The robot end moving speed setting, expressed as a percentage (%) or an absolute speed (mm/s) ≥0 Percentage of speed (%) or absolute speed (mm/s) -1 Using the Acceleration Table (supported on TM Robot S series models) <0 Invalid value
int	The time interval to accelerate to top speed (ms). Invalid when using the Acceleration Table.
int	Blending value, expressed as a percentage (%) or in radius (mm)
bool	Disable precise positioning true Disable precise positioning false Enable

Return

bool True Command accepted; **False** Command rejected (format error)

Note

Motion command parameter includes: (1) "CPP", (2) "CPR", (3) "CAP", (4) "CAR", (5) "CDP", (6) "CDR", (7) "TPP", (8) "TPR", (9) "TAP", (10) "TAR", (11) "TDP", (12) "TDR"

```
float[] var_rel= {0, 0, 10, 25, 0, 0}
```

```
Move_Line("TAP",var_rel,125,200,0,false)     // variable ingress available
```

```
Move_Line("TAP",{0,0,10,25,0,0},125,200,0,false)     // array constant ingress available  
// move to target with Line, velocity = 125mm/s, time to top speed = 200ms
```

```
// Opt for the singularity handling mechanism set by LineSingularity().
LineSingularity(0) // Disable singularity handling.
PTP("JPP", {45,-20,125,-100,-15,0}, 100, 500, 0, false)
Line("CPP", {307.8192,-100.8191,436.7351,90.1633,-0.4788,0.4848}, 100, 500, 0, false) // Disable
Move_Line("CPP", {0,-300,0,0,0,0}, 100, 500, 0, false) // Disable, through singularity

LineSingularity(1) // Enable singularity handling: path avoidance.
PTP("JPP", {45,-20,125,-100,-15,0}, 100, 500, 0, false)
Line("CPP", {307.8192,-100.8191,436.7351,90.1633,-0.4788,0.4848}, 100, 500, 0, false)
// Enable
Move_Line("CPP", {0,-300,0,0,0,0}, 100, 500, 0, false) // Enable, path avoidance
```

Syntax 2

```
bool Move_Line(
    string,
    float[],
    int,
    int,
    int,
    bool,
    int
)
```

Parameters

string	Definition of data format made of three letters #1: Relative motion target format: "C": expressed with w.r.t. current base "T": expressed with w.r.t. tool coordinate #2: Speed format: "P": expressed by percentage "A" expressed with absolute speed and in synchronization with the project speed "D" expressed with absolute speed and not in synchronization with the project speed #3: Blending format: "P": expressed by percentage "R": expressed by radius
float[]	Motion target. It includes the coordinate of tool center point at the robot end: X (mm), Y (mm), Z (mm), RX(°), RY(°), RZ(°)
int	The robot end moving speed setting, expressed as a percentage (%) or an absolute speed (mm/s) ≥0 Percentage of speed (%) or absolute speed (mm/s) -1 Using the Acceleration Table (supported on TM Robot S series models) <0 Invalid value
int	The time interval to accelerate to top speed (ms). Invalid when using the Acceleration Table.
int	Blending value, expressed as a percentage (%) or in radius (mm)
bool	Disable precise positioning true Disable precise positioning false Enable
int	Singularity handling.

- 0 Disbale
- 1 Path avoidance
- 2 Speed change(TM25S, TM30S)

* After exiting from the function, the singularity handling mechanism returns to what it was before entering.

Return

bool True Command accepted ; False Command rejected (format error)

Note

Motion command parameter includes: (1) "CPP", (2) "CPR", (3) "CAP", (4) "CAR", (5) "CDP", (6) "CDR", (7) "TPP", (8) "TPR", (9) "TAP", (10) "TAR", (11) "TDP", (12) "TDR"

// Opt for the singularity handling mechanism set by LineSingularity().

LineSingularity(0) // Disable singularity handling.

PTP("JPP", {45,-20,125,-100,-15,0}, 100, 500, 0, false)

Line("CPP", {307.8192,-100.8191,436.7351,90.1633,-0.4788,0.4848}, 100, 500, 0, false) // Disable

Move_Line("CPP", {0,-300,0,0,0,0}, 100, 500, 0, false) // Disable, through singularity

LineSingularity(1) // Enable singularity handling: path avoidance.

PTP("JPP", {45,-20,125,-100,-15,0}, 100, 500, 0, false)

Line("CPP", {307.8192,-100.8191,436.7351,90.1633,-0.4788,0.4848}, 100, 500, 0, false)

// Enable

Move_Line("CPP", {0,-300,0,0,0,0}, 100, 500, 0, false) // Enable, path avoidance

13.12 Circle()

Define and send Circle motion command into buffer for execution.

Syntax 1

```
bool Circle(
    string,
    float[],
    float[],
    int,
    int,
    int,
    int,
    bool
)
```

Parameters

string	Definition of data format, combines three letters #1: Motion target format: "C" expressed with Cartesian coordinate #2: Speed format: "P" expressed by percentage "A" expressed with absolute speed and in synchronization with the project speed "D" expressed with absolute speed and not in synchronization with the project speed #3: Blending format: "P" expressed by percentage
float[]	A point on arc. It includes the coordinate of tool center point at the robot end: X (mm), Y (mm), Z (mm), RX(°), RY(°), RZ(°)
float[]	The end point of arc. It includes the coordinate of tool center point at the robot end: X (mm), Y (mm), Z (mm), RX(°), RY(°), RZ(°)
int	The robot end moving speed setting, expressed as a percentage (%) or an absolute speed (mm/s) ≥0 Percentage of speed (%) or absolute speed (mm/s) -1 Using the Acceleration Table (supported on TM Robot S series models) <0 Invalid value
int	The time interval to accelerate to top speed (ms). Invalid when using the Acceleration Table.
int	Blending value, expressed as a percentage (%)
int	Arc angle(°), If non-zero value is given, the TCP will keep the same pose and move from current point to the assigned arc angle via the given point and end point on arc; If zero is given, the TCP will move from current point and pose to end point and pose via the point on arc with linear interpolation on pose.
bool	Disable precise positioning true Disable precise positioning false Enable precise positioning

Return

bool **True** Command accepted; **False** Command rejected (format error)

Note

Data format parameter includes: (1) "CPP", (2) "CAP", (3) "CDP"

PTP("JPP", {0, 0, 90, 0, 90, 0}, -1, 500, 0, false)

float[] targetPass= {417.50,-122.30,343.90,180.00,0.00,90.00}

float[] targetEnd= {381.70,208.74,343.90,180.00,0.00,135.00}

Circle("CAP",targetPass,targetEnd,100,200,50,270,false) // variable ingress available

Circle("CAP",targetPass,{381.7,208.74,343.9,180,0,135},100,200,50,270,false)

// array constant ingress available

Circle("CAP",Point["P1"].Value,{381.7,208.74,343.9,180,0,135},100,200,50,270,false)

// point coordinate retrieving with point parameterization available

// Move on 270° arc, velocity = 100mm/s, time to top speed = 200ms, blending value = 50%

Syntax 2

```
bool Circle(
    string,
    float[],
    float[],
    int,
    int,
    int,
    int,
    int,
    bool
)
```

Parameters

string	Definition of data format, combines three letters #1: Motion target format: "C" expressed with Cartesian coordinate #2: Speed format: "P" expressed by percentage "A" expressed with absolute speed and in synchronization with the project speed "D" expressed with absolute speed and not in synchronization with the project speed #3: Blending format: "P" expressed by percentage
float[]	A point on arc. It includes the coordinate of tool center point at the robot end: X (mm), Y (mm), Z (mm), RX(°), RY(°), RZ(°)
float[]	The end point of arc. It includes the coordinate of tool center point at the robot end: X (mm), Y (mm), Z (mm), RX(°), RY(°), RZ(°)
int	The robot end moving speed setting, expressed as a percentage (%) or an absolute speed (mm/s) ≥0 Percentage of speed (%) or absolute speed (mm/s) -1 Using the Acceleration Table (supported on TM Robot S series models) <0 Invalid value
int	The time interval to accelerate to top speed (ms). Invalid when using the Acceleration Table.
int	Blending value, expressed as a percentage (%)
int	Arc angle(°), If non-zero value is given, the TCP will keep the same pose and move

from current point to the assigned arc angle via the given point and end point on arc; If zero is given, the TCP will move from current point and pose to end point and pose via the point on arc with linear interpolation on pose.

int Rotation
 0 Linear Interpolation
 1 Keep Rotation
bool Disable precise positioning
 true Disable precise positioning
 false Enable precise positioning

Return

bool True Command accepted; False Command rejected (format error)

Note

Data format parameter includes: (1) "CPP", (2) "CAP", (3) "CDP"

```
float[] targetPass= {417.50,-122.30,343.90,180.00,0.00,90.00}
float[] targetEnd= {381.70,208.74,343.90,180.00,0.00,135.00}
```

```
PTP("JPP", {0, 0, 90, 0, 90, 0}, -1, 500, 0, false)
Circle("CAP",targetPass,targetEnd,100,200,50,270,false)
Circle("CAP",targetPass,targetEnd,100,200,50,270,1,false)
// Move on 270° arc, rotation = keep rotation
```

```
PTP("JPP", {0, 0, 90, 0, 90, 0}, -1, 500, 0, false)
Circle("CAP",targetPass,targetEnd,100,200,50,0,false)
Circle("CAP",targetPass,targetEnd,100,200,50,0,0,false)
// Move on the reach end point arc, rotation = linear interpolation
```

```
PTP("JPP", {0, 0, 90, 0, 90, 0}, -1, 500, 0, false)
Circle("CAP",targetPass,targetEnd,100,200,50,270,0,false)
// Move on 270° arc, rotation = linear interpolation
```

```
PTP("JPP", {0, 0, 90, 0, 90, 0}, -1, 500, 0, false)
Circle("CAP",targetPass,targetEnd,100,200,50,0,1,false)
// Move on the reach end point arc, rotation = keep rotation
```

13.13 PLine()

Define and send PLine motion command into buffer for execution.

Syntax 1

```
bool PLine(
    string,
    float[],
    int,
    int,
    int
)
```

Parameters

string	Definition of data format, combines three letters #1: Motion target format: "J": expressed with joint angles "C": expressed with Cartesian coordinate #2: Speed format: "A" expressed with absolute speed and in synchronization with the project speed "D" expressed with absolute speed and not in synchronization with the project speed #3: Blending format: "P": expressed by percentage
float[]	Motion target. If expressed in joint angles, it includes the angles of six joints: Joint1(°), Joint 2(°), Joint 3(°), Joint 4(°), Joint 5(°), Joint 6(°) ; If expressed in coordinate, it includes the the coordinate of tool center point at the robot end: X (mm), Y (mm), Z (mm), RX(°), RY(°), RZ(°)
int	The robot end moving speed setting, expressed as an absolute speed (mm/s)
int	The time interval to accelerate to top speed (ms)
int	Blending value, expressed as a percentage (%)

Return

bool True Command accepted ; False Command rejected (format error)

Note

Data format parameter includes: (1) "CAP", (2) "CDP", (3) "JAP", (4) "JDP"

```
float[] targetP1 = {417.50,-122.30,343.90,180.00,0.00,90.00}
```

```
PLine("CAP",targetP1,100,200,50) // variable ingress available
```

```
PLine("CAP",{417.50,-122.30,343.90,180.00,0.00,120.00},100,200,50)
```

```
// array constant ingress available
```

```
PLine("CAP",Point["P1"].Value,100,200,50)
```

```
// point coordinate retrieving with point parameterization available
```

```
// move to target with PLine, speed = 100mm/s, time to top speed = 200ms, and blending value = 50
```


13.14 Move_PLine()

Define and send PLine relative motion commands for execution.

Syntax 1

```
bool Move_PLine(
    string,
    float[],
    int,
    int,
    int
)
```

Parameters

- string** Definition of data format made of three letters
 - #1: Relative motion target format:
 - "C": expressed with w.r.t. current base
 - "T": expressed with w.r.t. tool coordinate
 - "J": expressed with joint angles
 - #2: Speed format:
 - "A" expressed with absolute speed and in synchronization with the project speed
 - "D" expressed with absolute speed and not in synchronization with the project speed
 - #3: Blending format:
 - "P": expressed by percentage
- float[]** Relative motion parameters expressed in coordinate. It includes the coordinate of tool center point at the robot end: X (mm), Y (mm), Z (mm), RX(°), RY(°), RZ(°)
- int** The robot end moving speed setting, expressed as an absolute speed (mm/s)
- int** The time interval to accelerate to top speed (ms)
- int** Blending value, expressed as a percentage (%)

Return

bool True Command accepted; False Command rejected (format error)

Note

Motion command parameter includes:

(1) "CAP", (2) "CDP", (3) "TAP", (4) "TDP", (5) "JAP", (6) "JDP"

```
float[] var_rel= {0, 0, 10, 25, 0, 0}
```

```
Move_PLine("CAP",var_rel,125,200,0) // variable ingress available
```

```
Move_PLine("CAP",{0,0,10,25,0,0},125,200,0) // array constant ingress available
```

```
// move relatively to target with PLine, speed = 125mm/s, time to top speed = 200ms
```

13.15 LineSingularity()

Set the line motion control with singularity handling or not.

Syntax 1

```
bool LineSingularity(
    int
```

```
)
```

Parameters

int	Singularity handling.
0	Disbale
1	Path avoidance
2	Speed change(TM25S, TM30S)

Return

bool True Command accepted; False Command rejected

Note

When the project is running, it defaults to 0 to disable. Once configured, it will affect the line motion control used later.

LineSingularity(0) // Disable singularity handling.

PTP("JPP", {45,-20,125,-100,-15,0}, 100, 500, 0, false)

Line("CPP", {307.8192,-100.8191,436.7351,90.1633,-0.4788,0.4848}, 100, 500, 0, false)

// Disable

Line("CPP", {138.0930,-411.4636,438.1152,89.5760,-0.3520,0.4868}, 100, 500, 0, false)

// Disable, through singularity

LineSingularity(1) // Enable singularity handling: path avoidance.

PTP("JPP", {45,-20,125,-100,-15,0}, 100, 500, 0, false)

Line("CPP", {307.8192,-100.8191,436.7351,90.1633,-0.4788,0.4848}, 100, 500, 0, false) // Enable

Line("CPP", {138.0930,-411.4636,438.1152,89.5760,-0.3520,0.4868}, 100, 500, 0, false)

// Enable, path avoidance

13.16 CollisionCheck()

Pre-collision checks help prevent robots from interfering with self-testing and avoid collisions between the gripper and the box. By adjusting and modifying the exported scene from TMstudio or using the built-in box container in TMvision, users can simulate robot motions using robot parts, tools, and virtual boxes. This simulation allows for early detection of potential issues during motions, reducing collisions in applications like box picking, using large hardware, and executing critical poses.

Syntax 1

```
string CollisionCheck(
    float,
    string
)
```

Parameter

float	Safety distance (mm)
string	The subflow name. For use in flow projects only. Returns errors if using in script projects.

Return

string	the result of checking
== ""	The value returned is an empty string, and there was no collision detected.
!= ""	The value returned is a non-empty string, and there was a collision detected. The contest will indicate which node.

Syntax 2

```
string CollisionCheck(
    float,
    ?
)
```

Parameter

float	Safety distance (mm)
?	The target motion to check. It can be a statement or a self-defined function.

Return

string	the result of checking
== ""	The value returned is an empty string, and there was no collision detected.
!= ""	The value returned is a non-empty string, and there was a collision detected. The contest will indicate which line number.

- * Check motion functions PTP(), Move_PTP(), Line(), Move_Line(), and Circle() only.
- * Support variable operations, general functions, math functions, and parameterized objects (read only) only. Does not operate with other functions.
- * The result of the variable operation is valid only during the pre-checking. The values revert to the default after checking.
- * All statements within the conditional expression will be checked.
- * All statements within the loop expression will be checked only once and will not repeat.

Note

- (1)


```
string result = CollisionCheck(15, "CCheck0")
// Check subflow CCheck0 (use only in Flow projects)
```
- (2)


```
define
{
```

```

TPoint P2 = {516.65,-147.75,445.37,179.44,-0.32,89.91},{0,0,90,0,90,0}
TPoint P3 = {516.14,156.17,443.43,179.59,0.30,111.24},{32.92,0.26,89.95,0,89.30,11.58}
TPoint P4 = {446.79,-313.56,435.31,-179.58,0.24,90.22},{-19.17,1.39,89.79,-0.32,89.31,-
19.49}
ModbusTCP localhost = 127.0.0.1,502
int count = 0
}
main
{
    Display(count)                // count = 0
    Pause()
    string re = CollisionCheck(15, TargetMotion())
    // Set safety distance to 15 mm for TargetMotion() function to check
    if (re == "") TargetMotion()  // No collision detected. Call TargetMotion() for the real operation.
    else Display("Collision: " + re) // Collision detected.
    Display(count)                // count = 0 or 1
    // +1 during pre-checking. Revert to the default, 0, after checking. If no collision detected, it calls the function
    TargetMotion(), and the the count value is 1. If collision detected, it does not call the function TargetMotion(), and
    the the count value is 0.
}
void TargetMotion()
{
    count++                      // count = 1 // Support variable operation during pre-checking.
    while (true)                 // The loop expression will be checked once only during pre-checking.
    {
        PTP("CPP",Point["P2"].Value,50,200,100,false) // check
        // check only during pre-checking without motion
        PTP("JPP",Point["P3"].Joint,50,200,100,false) // check
        Move_PTP("JPP",{0,0,0,0,0,60},50,200,100,false) // check
        Line("CPP",Point["P4"].Value,50,200,100,false) // check
        Move_Line("CPP", {20,20,20,0,0,0},50,200,100,false) // check
        Circle("CPP", Point["P2"].Value, Point["P3"].Value, 50, 200, 100, 100, false) // check
        WaitQueueTag()          // ignore // Do not support to this function during pre-checking.
        byte[] bb = modbus_read("localhost", 1, "DO", 0, 10)
        // ignore // Do not support to this function during pre-checking.
        string ss = GetString(bb) // General function. The operation result is ss = "{}".
    }
}

```

13.17 PVTEnter()

Set PVT mode to start with Joint/Cartesian command

Syntax 1

```
bool PVTEnter(
    int
)
```

Parameter

int	PVT Mode
0	Joint
1	Cartesian

Return

bool True Command accepted; False Command rejected

Note

Syntax 2

```
bool PVTEnter(
)
```

Parameter

Void	No parameter. Use PVT mode with Joint Command by default.
------	---

Return

bool True Command accepted; False Command rejected

Note

PVTEnter(1)

- * Before switching to PVT mode, it waits for other robot motion instructions to complete.
- * When the PVT mode begins, it only supports to functions tied to PVT, QueueTag, IO, and Payload.

13.18 PVTExit()

Set PVT mode motion to exit

Syntax 1

```
bool PVTExit(  
)
```

Parameter

void No parameter

Return

bool True Command accepted; False Command rejected

Note

PVTExit()

- * After exiting PVT mode, it waits for PVT function motion instructions to complete.

13.19 PVTPoint()

Set the PVT mode parameters of motion in position, velocity, and duration.

Syntax 1

```
bool PVTPoint(
    float[],
    float[],
    float,
    bool
)
```

Parameter

float[]	Target position {J1, J2, J3, J4, J5, J6} in PVT mode with Joint {X, Y, Z, RX, RY, RZ} in PVT mode with Cartesian
float[]	Target velocity {J1, J2, J3, J4, J5, J6}' in PVT mode with Joint {X, Y, Z, RX, RY, RZ}' in PVT mode with Cartesian
float	Duration (second), it requires ≥ 0.01 seconds.
bool	Synchronize to the project speed or not. true Synchronize to the project speed false Not synchronize to the project speed

Return

bool True Command accepted; False Command rejected

Note

Syntax 2

```
bool PVTPoint(
    float[],
    float[],
    float
)
```

Note

Same as Syntax 1. It default true to synchronize to the project speed or not.

Syntax 3

```
bool PVTPoint(
    float, float, float, float, float, float,
    float, float, float, float, float, float,
    float,
    bool
)
```

Parameter

float, float, float, float, float, float,	Target Position. {J1, J2, J3, J4, J5, J6} in PVT mode with Joint
---	---

{X, Y, Z, RX, RY, RZ} in PVT mode with Cartesian
 float, float, float, float, float, float,
 Target Velocity.
 {J1, J2, J3, J4, J5, J6}' in PVT mode with Joint
 {X, Y, Z, RX, RY, RZ}' in PVT mode with Cartesian
 float Duration (second), it requires ≥ 0.01 seconds.
 bool Synchronize to the project speed or not.
 true Synchronize to the project speed
 false Not synchronize to the project speed

Return

bool True Command accepted; False Command rejected

Note

```

PVTEnter(1)
PVTPoint(467.5,-122.2,359.7,180,0,90,50,50,0,0,0,0,0.5)
PVTPoint(467.5,-72.2,359.7,180,0,90,-50,50,0,0,0,0,0.5)
PVTPoint(417.5,-72.2,359.7,180,0,90,0,0,0,0,0,0,0.5)
PVTPoint(417.5,-122.2,359.7,180,0,90,50,50,0,0,0,0,0.5)
PVTPoint(417.5,-122.2,359.7,180,0,60,50,50,0,0,0,0,3)
PVTPoint(417.5,-122.2,359.7,180,0,90,50,50,0,0,0,0,3)
PVTExit()
  
```

Syntax 4

```

bool PVTPoint(
    float, float, float, float, float, float,
    float, float, float, float, float, float,
    float
)
  
```

Note

Same as Syntax 3. It default true to synchronize to the project speed or not.

```

PVTEnter(1)
PVTPoint(467.5,-122.2,359.7,180,0,90,50,50,0,0,0,0,0.5)
PVTPoint(467.5,-72.2,359.7,180,0,90,-50,50,0,0,0,0,0.5)
PVTPoint(417.5,-72.2,359.7,180,0,90,0,0,0,0,0,0,0.5, true)
PVTPoint(417.5,-122.2,359.7,180,0,90,50,50,0,0,0,0,0.5, false)
PVTPoint(417.5,-122.2,359.7,180,0,60,50,50,0,0,0,0,3, false)
PVTPoint(417.5,-122.2,359.7,180,0,90,50,50,0,0,0,0,3)
PVTExit()
  
```


13.20 PVTPause()

Set PVT mode motion to pause

Syntax 1

```
bool PVTPause(  
)
```

Parameter

void No parameter

Return

bool True Command accepted; False Command rejected

Note

PVTPause()

13.21 PVTResume()

Set PVT mode motion to resume

Syntax 1

```
bool PVTResume(  
)
```

Parameter

void No parameter

Return

bool True Command accepted; False Command rejected

Note

PVTResume()

13.22 PathOffset_Set()

Set Path Offset parameters

Syntax 1

```
bool PathOffset_Set(
    bool,
    float[],
    int,
    float
)
```

Parameter

bool	Enable Path Offset or not.
false	Disable
true	Enable
float[]	Offset values in coordinate for the six elements of the robot coordinate: X(mm), Y(mm), Z(mm), RX(°), RY(°), RZ(°)
int	Offset reference base
0	the robot base
1	tool (default)
2	the current base
3	trajectory
float	Alpha filter coefficient value ranging from 0 to 1. The lesser the value is, the slower the offset compensate speed is. In practical use, it is best to start testing at a lower setting, such as 0.05, and gradually increase it to about 0.1 to determine the most suitable value. Also, monitor closely for excessively rapid compensation increases once the setting exceeds 0.5.

* Calling in the flow required.

Return

bool True Command accepted; **False** Command rejected

Syntax 2

```
bool PathOffset_Set(
    bool,
    float[],
    int
)
```

Note

Parameter definition same as Syntax 1. Setting the enable path offset, the offset values, and the offset reference base without setting the alpha filter coefficient value.

* Calling in the flow required.

Syntax 3

```
bool PathOffset_Set(
    bool,
    float[]
)
```

Note

Parameter definition same as Syntax 1. Setting the enable path offset and the offset values by the tool reference base without setting the alpha filter coefficient value.

* Calling in the flow required.

Syntax 4

```
bool PathOffset_Set(
    bool,
    float
)
```

Note

Parameter definition same as Syntax 1. Setting the enable path offset and the alpha coefficient without setting the offset values and the alpha filter coefficient value.

* Calling in process required.

Syntax 5

```
bool PathOffset_Set(
    bool
)
```

Note

Parameter definition same as Syntax 1. Setting the enable path offset only without setting the offset values and the alpha filter coefficient value.

* Calling in process required.

Suppose in the Flow page

// Enable Path Offset with setting {10,0,0,0,0,0} as the offset value by the tool base and 0.005 to the alpha coefficient.

PathOffset_Set(true, {10,0,0,0,0,0}, 1, 0.005)

PathOffset_Set(false) // Disable Path Offsetting

// Enable Path Offset with setting {10,0,0,0,0,0} as the offset value by the tool base and without setting the alpha coefficient.

PathOffset_Set(true, {10,0,0,0,0,0}, 1)

PathOffset_Set(false) // Disable Path Offsetting

// Enable Path Offset with setting {10,0,0,0,0,0} as the offset value by the tool base and without setting the alpha coefficient.

PathOffset_Set(true, {10,0,0,0,0,0})

PathOffset_Set(false) // Disable Path Offsetting

// Enable Path Offset with setting the offset value and the alpha coefficient.

PathOffset_Set(true)

Suppose in the Thread page

PathOffset_Set(true, {10,0,0,0,0,0}) // Return Error. Must be in the Flow page to enable or disable Path Offset.

PathOffset_Set(false) // Return Error. Must be in the Flow page to enable or disable Path Offset.

Syntax 6

```
bool PathOffset_Set(
```

```
float[],
int
)
```

Parameter

float[]	Offset values in coordinate for the six elements of the robot coordinate: X(mm), Y(mm), Z(mm), RX(°), RY(°), RZ(°)
int	Offset reference base
0	the robot base
1	tool (default)
2	the current base
3	trajectory

* Calling and setting the offset value and the reference base in the flow or the thread available.

Return

bool True Command accepted; False Command rejected

Syntax 7

```
bool PathOffset_Set(
float[]
)
```

Note

Parameter definition same as Syntax 6. Setting the offset value by the reference tool base.

Suppose in the Flow page

```
PathOffset_Set(true)     // Enable Path Offset without setting the offset value and the
                           alpha coefficient.
```

Suppose in the Thread page

```
PathOffset_Set({10,0,0,0,0,0}, 0)     // Set {10,0,0,0,0,0} as the offset value by the robot base.
PathOffset_Set({5,0,0,0,0,0})     // Set {5,0,0,0,0,0} as the offset value by the tool base.
```

13.23 PathOffset_Get()

Get the Path Offset values

Syntax 1

```
float[] PathOffset_Get(
    int
)
```

Parameter

int	Get the current offset value.
0	Get by the robot base.
1	Get by the offset reference base. (Default)

Return

float[]	Offset values in coordinate for the six elements of the robot coordinate: X(mm), Y(mm), Z(mm), RX(°), RY(°), RZ(°)
---------	---

Syntax 2

```
float[] PathOffset_Get(
)
```

Parameter

void	No input value
------	----------------

Return

float[]	Offset values in coordinate for the six elements of the robot coordinate: X(mm), Y(mm), Z(mm), RX(°), RY(°), RZ(°)
---------	---

Note

Same as Syntax 1. Get the offset value by the current offset reference base.

// Enable Path Offset with setting {10,0,0,0,0,0} as the offset value by the tool base and 0.005 to the alpha coefficient.

PathOffset_Set(true, {10,0,0,0,0,0}, 1, 0.005)

// It proceeds with offset compensation, whether the robot is moving, once Path Offset is enabled.

// So the current offset value to read tends to the setting offset value continuously.

PathOffset_Get() // {1.1,0,0,0,0,0}

PathOffset_Get() // {2.4,0,0,0,0,0}

...

PathOffset_Get() // {9.999952,0,0,0,0,0}

13.24 PathOffset_IsEnabled()

Check if the Path Offset enabled.

Syntax 1

```
bool PathOffset_IsEnabled(  
)
```

Parameter

void No input value

Return

bool if the Path Offset enabled
false Disabled
true Enabled

Note

```
PathOffset_IsEnabled()            // false  
PathOffset_Set(true, {10,0,0,0,0,0}, 1, 0.005)  
PathOffset_IsEnabled()            // true
```

13.25 PathOffset_AlphaFilter()

Set the Alpha Filter coefficient to avoid over-acceleration.

Syntax 1

```
bool PathOffset_AlphaFilter(
    float
)
```

Parameter

Float Alpha filter coefficient value ranging from 0 to 1. The lesser the value is, the slower the offset compensate speed is. In practical use, it is best to start testing at a lower setting, such as 0.05, and gradually increase it to about 0.1 to determine the most suitable value. Also, monitor closely for excessively rapid compensation increases once the setting exceeds 0.5.

Return

bool **True** Command accepted; **False** Command rejected

Note

```
PathOffset_Set(true, {10,0,0,0,0,0}, 1, 0.005)
PathOffset_AlphaFilter(0.01)
PathOffset_AlphaFilter(1.2)           // Return false for out of range.
```


13.26 PathOffset_MaxOffset()

Set the upper bounds of the distance and the rotation.

Syntax 1

```
bool PathOffset_MaxOffset(
    float,
    float
)
```

Parameter

float the distance upper bound in mm
float the rotation upper bound in degree

Return

bool **True** Command accepted; **False** Command rejected

Note

// Set the offset compensation upper bound.

PathOffset_MaxOffset(10, 3)

// Enable Path Offset with setting {20,0,0,0,0,0} as the offset value by the tool base and 0.005 to the alpha coefficient.

PathOffset_Set(true, {20,0,0,0,0,0}, 1, 0.005)

// It proceeds with offset compensation, whether the robot is moving, once Path Offset is enabled.

// So the current offset value to read tends to the setting offset value continuously.

PathOffset_Get() // {1.1,0,0,0,0,0}

PathOffset_Get() // {2.4,0,0,0,0,0}

...

PathOffset_Get() // {9.999952,0,0,0,0,0}

...

PathOffset_Get() // {9.999952,0,0,0,0,0} // Set the distance upper bound to 10mm

(1) In the Flow project

Flow	Thread
Point Node: P1 Point Node: P2 Script Node: Script1 PathOffset_Set(true) <i>// Enable Path Offset</i> Point Node: P3 Script Node: Script2 PathOffset_Set(false) <i>// Disable Path Offsetting</i> Point Node: P4 ... Link To P1 ...	Script Node: Script3 <i>// Set the alpha coefficient</i> PathOffset_AlphaFilter(0.005) <i>// Set the offset compensation upper bound</i> PathOffset_MaxOffset(20, 5) <i>// Read the required offset compensation</i> <i>value from mbus1</i> var_X = modbus_read ("mbus1", "offsetX") <i>// Offset to the X coordinate</i> var_Offset[0] = var_X <i>// Update the offset value by the tool base</i> PathOffset_Set(var_Offset, 1) Goto Node: Goto1 ... Link to Script3 ...
● It enables Path Offset when it is on the path of P2 toward P3. ● It disables Path Offset after moving to P3.	● Since the enable and the disable of the Path Offset come in the Flow, it is required to reset the AlphaFilter and the MaxOffset in the Thread. ● Keep updating the offset value in the Thread.

(2) In the Script project

```

define
{
    float var_X = 0
    float[] var_Offset = {0,0,0,0,0,0}
    ModbusTCP mbus1 = "127.0.0.1"
}
main
{
    ThreadRun(Offsetting())
    while (true)
    {
        PTP("JPP",{15,0,90,0,90,0},50,200,100,false)
        PTP("JPP",{15,0,75,0,90,0},50,200,100,false)
        PathOffset_Set(true)           // Enable Path Offset
        PTP("JPP",{ -15,0,75,0,90,0},50,200,100,false)
        PathOffset_Set(false)        // Disable Path Offset
        PTP("JPP",{ -15,0,90,0,90,0},50,200,100,false)

        Sleep(10)
    }
}
void Offsetting()
{
    mbus1.Preset("offsetX", 1, "RO", 9000, "float")
}

```

```
while (true)
{
    PathOffset_AlphaFilter(0.005)    // Set the alpha coefficient
    PathOffset_MaxOffset(20, 5)      // Set the offset compensation upper bound

    // Read the required offset compensation value from mbus1.
    var_X = modbus_read("mbus1", "offsetX")

    var_Offset[0] = var_X            // Offset to the X coordinate.
    PathOffset_Set(var_Offset, 1)    // Update the offset value by the tool base.

    Sleep(10)
}
}
```

13.27 Velocity()

Define and activate continuous speed control mode. Under this mode, robot will change motion instantaneously once the target speed vector changes.

- * Syntax 1/2/3/4 is to define the motion parameters, while syntax 5/6, the target speed vector.
- * When the robot is in continuous speed control mode, sending any other types of motion control commands (such as PTP(), Line(), etc.) will trigger an error stop.
- * This function is not affected by the project override speed (%), and the robot will run at the speed specified in the function within the reachable speed range.
- * Users must carefully assess the risk of the trajectory generated by Velocity() before sending the command. In other words, users or the motion control program must constantly check whether the robot is moving beyond its working range or nearing a singularity.

Syntax 1

```
bool Velocity(
    bool,
    string,
    int,
    int
)
```

Parameter

bool	whether to enable the adaptive speed control mode or not
false	disable
true	enable
string	control mode
	"C": linear speed control with the current base
	"T": linear speed control with the tool base
	"J": angular speed control with joint angles
int	The time interval to accelerate to top speed (ms). Valid range: 150 between 2000
int	The control value protection time (millisecond). Valid value ≥ 0 . If users do not update and it is beyond the protection time, the Velocity(float[]) value will be adjusted to 0 automatically.

* While disable can be called in the process or threads, enable must be called in the process.

Return

bool True Command accepted; False Command rejected (format error)

Syntax 2

```
bool Velocity(
    bool,
    string,
    int
)
```

Parameter

bool	whether to enable the adaptive speed control mode or not
false	disable
true	enable
string	control mode
	"C": linear speed control with the current base
	"T": linear speed control with the tool base
	"J": angular speed control with joint angles
int	The time interval to accelerate to top speed (ms). Valid range: 150 between 2000

* While disable can be called in the process or threads, enable must be called in the process.

Note

Same as syntax 1. It sets a 1000ms to the control value protection time.

Syntax 3

```
bool Velocity(
    bool,
    string
)
```

Parameter

bool	whether to enable the adaptive speed control mode or not
false	disable
true	enable
string	control mode
	"C": linear speed control with the current base
	"T": linear speed control with the tool base
	"J": angular speed control with joint angles

* While disable can be called in the process or threads, enable must be called in the process.

Note

Same as syntax 1. It sets a 500 ms to the time interval to accelerate to the top speed and a 1000ms to the control value protection time.

Syntax 4

```
bool Velocity(
    bool
)
```

Parameter

bool	whether to enable the adaptive speed control mode or not
false	disable
true	enable

* Disable can be called in the process or threads.

* For disable only.

Note

If on the Flow page

```
Velocity(true, "C", 300, 3000)
```

// Enable velocity control for the linear speed by the current base, setting a 500 ms time interval to accelerate to the top speed and a 3000 ms control value protection time.

```
Velocity(true, "T", 300)
```

// If continuing from the previous line, the "C" control will be turned off first.

// Enable velocity control for the linear speed by the tool base, setting a 500 ms time interval to accelerate to the top speed and a 1000 ms control value protection time.

```
Velocity(true, "J")
```

// If continuing from the previous line, the "T" control will be turned off first.

// Enable velocity control for the angular speed by the joint angle, setting a 500 ms time interval to accelerate to the top speed and a 1000 ms control value protection time.

```
Position(true, "J")
```

If continuing from the previous line, the "T" control will be turned off first.

// Enable position control // Only one control, Position or Velocity, can be active at a time.

```
Velocity(false)                      // Disable velocity control
```

```
Velocity(true)                      // Return error. Control Mode Parameters Required to enable
```

If on the Thread page

```
Velocity(true, "J")                      // Return error. Must be in the Flow page to enable velocity control
```

```
Velocity(false)                      // Disable velocity control
```

Syntax 5

```
bool Velocity(
    float[]
)
```

Parameter

float[] Speed control value for the associated applicable control mode
"C" or **"T"**: linear speed control for X,Y,Z (mm/s) RX,RY,RZ (degree/s)
"J": angular speed control for J1,J2,J3,J4,J5,J6 (degree/s)

* Calling available in the process or threads.

Return

bool **True** Command accepted; **False** Command rejected

Syntax 6

```
bool Velocity(
    float, float, float, float, float, float
)
```

Note

Same as syntax 5. It replaces float[] with parameters float,float,float,float,float,float.

```
PTP("JPP", {90,0,90,0,90,0}, 10, 500, 0, true)
Velocity(true, "J", 150, 5000)
// Enable adaptive speed control for the angular speed by the joint angle

Sleep(1000)
for (int i = 0; i < 30; i++)
{
    float[] AngleSpeed = {0, 0, i, 0, 0, 0} // Adjust the speed for angle J3 along with the loop.
    Velocity(AngleSpeed) // Set the speed for angle J30
    Sleep(100)
}
Velocity(false)
```

13.28 Position()

Define and activate continuous position control mode. Under this mode, the robot will change motion instantaneously once the target position vector changes.

- * Syntax 1/2/3/4 is to define the motion parameters, while syntax 5/6, the target position vector.
- * When the robot is in continuous position control mode, sending any other types of motion control commands (such as PTP(), Line(), etc.) will trigger an error stop.
- * Users must carefully assess the risk of the trajectory generated by Position() before sending the command. In other words, users or the motion control program must constantly check whether the robot is moving beyond its working range or nearing a singularity.

Syntax 1

```
bool Position(
    bool,
    string,
    int,
    float,
    int
)
```

Parameter

bool	whether to enable the position adaptive speed control mode or not
false	disable
true	enable
string	control mode
	"C": target coordinate control with the current base
	"T": target coordinate control with the tool base
	"J": target angle control with the joint angle
int	The time interval to accelerate to top speed (ms). Valid range: 150 between 2000
float	The Motion Control Gain value. Valid range: 1 between 100. The larger the value, the faster the response to the new control value.
int	The expected control value protection time interval (millisecond). Valid value >= 20. In adaptive control mode, Position (float[]) requires updating regularly. If the difference from the expected time interval varies significantly (less or more than), it will affect the control speed algorithm.

* While disable can be called in the process or threads, enable must be called in the process.

Return

bool True Command accepted; False Command rejected (format error)

Syntax 2

```
bool Position(
    bool,
    string,
    int,
    float
)
```

Parameter

bool whether to enable the position adaptive speed control mode or not

false disable
 true enable
 string control mode
 "C": target coordinate control with the current base
 "T": target coordinate control with the tool base
 "J": target angle control with the joint angle
 int The time interval to accelerate to top speed (ms). Valid range: 150 between 2000
 float The Motion Control Gain value. Valid range: 1 between 100. The larger the value,
 the faster the response to the new control value.
 * While disable can be called in the process or threads, enable must be called in the process.

Note

Same as syntax 1. It sets a 1000ms to the expected control value protection time interval.

Syntax 3

```
bool Position(
    bool,
    string
)
```

Parameter

bool whether to enable the position adaptive speed control mode or not
 false disable
 true enable
 string control mode
 "C": target coordinate control with the current base
 "T": target coordinate control with the tool base
 "J": target angle control with the joint angle
 * While disable can be called in the process or threads, enable must be called in the process.

Note

Same as syntax 1. It sets a 500 ms to the time interval to accelerate to the top speed, a 4.0 to the motion control gain value, and a 1000ms to the expected control value protection time.

Syntax 4

```
bool Position(
    bool
)
```

Parameter

```
bool          whether to enable the adaptive speed control mode or not
              false      disable
              true       enable
```

* Disable can be called in the process or threads.

* For disable only.

Note

If on the Flow page

```
Position(true, "C", 300, 3, 500)
```

// Enable position control for the target coordinate control by the current base, setting a 300 ms to the time interval to accelerate to the top speed, a 3.0 to the motion control gain value, and a 500ms to the expected control value protection time.

```
Position(true, "T", 300, 3)
```

// If continuing from the previous line, the "C" control will be turned off first.

// Enable position control for the target coordinate control by the tool base, setting a 300 ms to the time interval to accelerate to the top speed, a 3.0 to the motion control gain value, and a 1000ms to the expected control value protection time.

```
Position(true, "J")
```

// If continuing from the previous line, the "T" control will be turned off first.

// Enable position control for the target angle control by the joint angle, setting a 500ms to the time interval to accelerate to the top speed, a 4.0 to the motion control gain value, and a 1000ms to the expected control value protection time.

```
Velocity(true, "J")
```

// If continuing from the previous line, the Position "J" control will be turned off first.

// Enable velocity control // Only one control, Position or Velocity, can be active at a time.

```
Position(false)          // Disable position control
```

```
Position(true)           // Return error. Control Mode Parameters Required to enable
```

If on the Thread page

```
Position(true, "J")       // Return error. Must be in the Flow page to enable position control
```

```
Position(false)          // Disable position control
```

Syntax 5

```
bool Position(
    float[]
)
```

Parameter

float[] Position control value for the associated applicable control mode
 "C" or "T": target coordinate control for X,Y,Z (mm) RX,RY,RZ (degree)
 "J": target angle control for J1,J2,J3,J4,J5,J6 (degree)
 * Calling available in the process or threads.

Return

bool **True** Command accepted; **False** Command rejected

Syntax 6

```
bool Position(
    float, float, float, float, float, float
)
```

Note

Same as syntax 5. It replaces float[] with parameters float,float,float,float,float,float.

```
PTP("JPP", {90,0,90,0,90,0}, 10, 500, 0, true)
Position(true, "J", 150, 1, 500)            // Enable position control by the angle
Sleep(1000)
for (int i = 0; i < 30; i++)
{
    float[] AngleTarget = {90, 0, 90+i, 0, 90, 0}
    // Adjust the angle for axis-angle J3 along with the loop.
    Position(AngleTarget)    // Set the speed for angle J30
    Sleep(100)
}
Position(false)
```

13.29 SetTCPSPeedLimit()

Set Linear Speed Limit and Rotation Speed Limit for Line Motion.

Syntax 1

```
bool SetTCPSPeedLimit(
    bool,
    int,
    int
)
```

Parameter

bool	whether to enable the line motion speed limit
false	disable. Limited by the system default.
true	enable. Limited by the input parameters for linear and rotation.
int	Linear speed limit, expressed as mm/s. Maximum limit if <= 0.
int	Rotation speed limit, expressed as deg/s. Maximum limit if <= 0.

Return

bool	True	Command accepted;	False	Command rejected.
------	------	-------------------	-------	-------------------

Syntax 2

```
bool SetTCPSPeedLimit(
    int,
    int
)
```

Parameter

int	Linear speed limit, expressed as mm/s. Maximum limit if <= 0.
int	Rotation speed limit, expressed as deg/s. Maximum limit if <= 0.

Note

Same as syntax 1. It sets true to whether to enable the line motion speed limit.

Syntax 3

```
bool SetTCPSPeedLimit(
    bool
)
```

Parameter

bool	whether to enable the line motion speed limit
false	disable. Limited by the system default.
true	enable. Limited by the input parameters for linear and rotation.

Return

bool	True	Command accepted;	False	Command rejected.
------	------	-------------------	-------	-------------------

Note

When activated, the line motion speed limit uses the existing settings for the Linear Speed Limit and the Rotation Speed Limit. It defaults the limits to the maximum if never set.

```
SetTCPSPeedLimit(true, 0, 0)
Display("Default Speed Limit")
Line("CPP",{517.67212,-331.18498,343.19177,179.75342,-0.32395,89.91306},100,600,0,false)
// Point1
Line("CPP",{517.67505,-131.19205,343.19144,89.83959,89.9993,0},100,600,0,false)
// Point2
Line("CPP",{517.67212,-331.18498,343.19177,179.75342,-0.32395,89.91306},100,600,0,false)
```

```
// Point1
```

```
SetTCPSpeedLimit(true, 0, 30)
```

```
Display("Rotational Speed Limit = 30 deg/s")
```

```
Line("CPP",{517.67212,-331.18498,343.19177,179.75342,-0.32395,89.91306},100,600,0,false)
```

```
// Point1
```

```
Line("CPP",{517.67505,-131.19205,343.19144,89.83959,89.9993,0},100,600,0,false)
```

```
// Point2
```

```
Line("CPP",{517.67212,-331.18498,343.19177,179.75342,-0.32395,89.91306},100,600,0,false)
```

```
// Point1
```

```
SetTCPSpeedLimit(false)    // Disable. Limited by the system default.
```

13.30 SetAccTable()

Set the acceleration table, and the system will automatically fetch the speed and the time to top speed from the table by the distance when the robot is in motion control. Currently, it is available for the TM Robot S series models.

The acceleration table comes with five groups at most and with some rules.

1. There must be two groups at least, and the first and the last group cannot be disabled.
2. The maximum distance of each group must be greater than or equal to the maximum distance of the previous group.
3. If Speed 2 (%) gets more than Speed 1 (%), it means speed up with distance; if less, it means speed down with distance.
4. The default values vary from the robot models. The table below exemplifies one of the default values in the acceleration table.

Distance Min (mm)	Distance Max (mm)	Speed 1 (%)	Speed 2 (%)	Time to Top Speed 1 (ms)	Time to Top Speed 2 (ms)
0	50	5	15	172.5	217.5
50	300	15	45	217.5	352.5
300	700	45	65	352.5	442.5
700	1000	65	75	442.5	487.5
1000	1300	75	100	487.5	600

Syntax 1

```
bool SetAccTable(
)
```

Parameter

void No input

Return

bool True Command accepted; False Command rejected.

Note

Reset the acceleration table to defaults.

Syntax 2

```
bool SetAccTable(
    int,
    float,
    float,
    float,
    float
)
```

Parameter

int The acceleration table group. Valid from 1 to 5.
float Distance Max (mm). A value of <= 0 means the group configuration is disabled.
 (Configuration disabling is available for intermediate groups.)
float Speed 1 (%)
float Speed 2 (%)
float Time to Top Speed 2 (ms)

Return

bool True Command accepted; False Command rejected.

Note

Time to top speed 1 (ms) is calculated using the following equation:

Time to Top Speed 1

$$= \text{Accelerate Time Min} + \frac{\text{Speed 1} \times (\text{Time to Top Speed 2} - \text{Accelerate Time Min})}{\text{Speed 2}}$$

* Where Accelerate Time Min is defined based on the robot model.

Syntax 3

```
bool SetAccTable(
    int,
    float,
    float,
    float
)
```

Parameter

int The acceleration table group. Valid from 1 to 5.
float Distance Max (mm). A value of <= 0 means the group configuration is disabled. (Configuration disabling is available for intermediate groups.)
float Speed 1 (%)
float Speed 2 (%)

Note

Time to top speed 1 (ms) and Time to top speed 2 (ms) are calculated using the following equation:

Time to Top Speed 1

$$= \text{Accelerate Time Min} + \frac{\text{Speed 1} \times (\text{Accelerate Time Default} - \text{Accelerate Time Min})}{100}$$

Time to Top Speed 2

$$= \text{Accelerate Time Min} + \frac{\text{Speed 2} \times (\text{Accelerate Time Default} - \text{Accelerate Time Min})}{100}$$

* Where Accelerate Time Min and Accelerate Time Default are defined based on the robot model.

Syntax 4

```
bool SetAccTable(
    int,
    float,
    float
)
```

Parameter

int The acceleration table group. Valid from 1 to 5.
float Distance Max (mm). A value of <= 0 means the group configuration is disabled. (Configuration disabling is available for intermediate groups.)
float Speed 2 (%)

Note

Same as syntax 3. It sets Speed 1 (%) the same as Speed 2 (%).

Syntax 5

```
bool SetAccTable(
    int,
    bool
)
```

Parameter

int The acceleration table group. Valid from 2 to 4. Groups 1 and 5 are permanently enabled.
bool Enable or disable the configuration of the group.

Return

`bool` `True` Command accepted; `False` Command rejected.

Note

The acceleration table requires at least two groups (groups 1 and 5), so the sequence of groups to be enabled or disabled is 2 to 4.

13.31 GetAccTable()

Retrieve the contents of the acceleration table. Available for TM Robot S series models currently.

Syntax 1

```
float[] GetAccTable(
    int
)
```

Parameter

int The acceleration table group. Valid from 1 to 5.

Return

float[] Retrieve the setting of the acceleration table index associated group. The array length is six and goes by Distance Min, Distance Max, Speed 1, Speed 2, Time to Top Speed 1, and Time to Top Speed 2. If all the values in the array are 0, the group configuration is closed.

Note

```
SetAccTable() // Reset the acceleration table to defaults.
```

```
string s1
```

```
for (int i = 1; i <= 5; i++)
```

```
    s1 += GetString(GetAccTable(i)) + newline
```

```
Display(s1)
```

```
// {0,50,5,15,172.5,217.5}\u00D0A
```

```
    {50,300,15,45,217.5,352.5}\u00D0A
```

```
    {300,700,45,65,352.5,442.5}\u00D0A
```

```
    {700,1000,65,75,442.5,487.5}\u00D0A
```

```
    {1000,1300,75,100,487.5,600}
```

```
Pause()
```

```
SetAccTable(1, 50, 10, 20, 220)
```

```
SetAccTable(2, 320, 20, 50)
```

```
SetAccTable(3, 750, 70)
```

```
SetAccTable(4, false)
```

```
SetAccTable(5, 1300, 100)
```

```
string s2
```

```
for (int i = 1; i <= 5; i++)
```

```
    s2 += GetString(GetAccTable(i)) + newline
```

```
Display(s2)
```

```
// {0,50,10,20,185,220}\u00D0A
```

```
    {50,320,20,50,240,375}\u00D0A
```

```
    {320,750,70,70,465,465}\u00D0A
```

```
    {0,0,0,0,0}\u00D0A
```

```
    {750,1300,100,100,600,600}
```

13.32 Vision_DoJob()

Execute the existing vision jobs in the project but not the ones with vision capture points.

- * Must create vision jobs in the project first without checking **Vision Capture Point**.
- * It is required to codify the respective variable definitions in the define segment if using script projects.
- * The respective variable definition values update after the vision job execution.

Syntax 1

```
bool Vision_DoJob(
    string
)
```

Parameter

string Vision job name.

Return

bool	True	Vision job completed and the result is pass.
	False	Vision job failed. 1. The result is fail.
		2. Fail to execute.
		3. Vision job comes with the initial point.

Note

```
bool var_result = Vision_DoJob("job1")
```

13.33 Vision_DoJob_PTP()

Execute the existing vision jobs in the project, move to the vision capture point by the PTP motion, and queue with the motion commands.

- * Must create vision jobs in the project first. Check **Vision Capture Point** for moving to the initial position and uncheck for not.
- * It is required to codify the respective variable definitions in the define segment if using script projects.
- * The respective variable definition values update after the vision job execution.

Syntax 1

```
bool Vision_DoJob_PTP(
    string,
    int,
    int,
    bool
)
```

Parameters

string	Vision job name.
int	The robot end moving speed setting, expressed as a percentage (%) ≥0 Percentage of speed (%) -1 Using the Acceleration Table (supported on TM Robot S series models) <0 Invalid value
int	The time interval to accelerate to top speed (ms). Invalid when using the Acceleration Table.
bool	Whether to use the smart pose of robot selection mode true Use the pose of robot by the system. false Use the pose of robot recorded teaching the vision job.

Return

bool	True	Vision job completed and the result is pass.
	False	Vision job failed. 1. The result is fail. 2. Fail to execute.

Syntax 2

```
bool Vision_DoJob_PTP(
    string,
    int,
    int
)
```

Note

Same as syntax 1. It defaults false to whether to use the smart pose of robot selection mode.

```
bool var_result = Vision_DoJob_PTP("job1", 100, 500)
var_result = Vision_DoJob_PTP("job1", 100, 500, true)
```

13.34 Vision_DoJob_Line()

Execute the existing vision jobs in the project, move to the initial position by the line motion, and queue with the motion commands.

*Must create vision jobs in the project first. Check **Start at Initial Position** for moving to the initial position and uncheck for not.

*It is required to codify the respective variable definitions in the define segment if using script projects.

* The respective variable definition values update after the vision job execution.

Syntax 1

```
bool Vision_DoJob_Line(
    string,
    int
)
```

Parameters

string	Vision job name
int	The robot end moving speed setting, expressed as a percentage (%)
	≥0 Percentage of speed (%)
	-1 Using the Acceleration Table (supported on TM Robot S series models)
	<0 Invalid value

Return

bool	True	Vision job completed and the result is pass.
	False	Vision job failed. 1. The result is fail.
		2. Fail to execute.

Note

```
bool var_result = Vision_DoJob_Line("job1", 100)
```

Syntax 2

```
bool Vision_DoJob_Line(
    string,
    int,
    int,
    bool
)
```

Parameters

string	Vision job name
int	The robot end moving speed setting, expressed as an absolute speed (mm/s)
int	The time interval to accelerate to top speed (ms)
bool	Whether to link to the project speed.
	True Link to the project speed.
	false Unlink to the project speed.

Return

bool	True	Vision job completed and the result is pass.
	False	Vision job failed. 1. The result is fail.

2. Fail to execute.

Syntax 3

```
bool Vision_DoJob_Line(  
    string,  
    int,  
    int  
)
```

Note

Same as syntax 2. It defaults true to whether to link to the project speed.

```
bool var_result = Vision_DoJob_Line("job1", 100, 500)  
var_result = Vision_DoJob_Line("job1", 100, 500, false)
```

14. Vision Functions

14.1 Vision_IsJobAvailable()

Check if the vision job in the current project present and valid

Syntax 1

```
bool Vision_IsJobAvailable(  
    string  
)
```

Parameter

string Vision job name

Return

bool True the vision present and valid
 False the vision absent and invalid

Note

Use this function to check the presence and validity of the visual job before calling to avoid errors caused by absence or invalidity while calling.

```
bool var_result = Vision_IsJobAvailable("job1")
```

14.2 Multi-Object Result Output and Flying Trigger/Inspection Tasks

Once the parameters in the output result of the vision job come with more than one dimensions, or the visual job is created by Flying Trigger/Inspection, it requires function calls for the relevant information retrieval. The parameter should lead in the variable name, which indicates the result data of the desired vision job and the module to retrieve. For example, "Job1_Count_Blob_1_DetectObjectX_TM" is the output variable of the vision job Job1, which is output by the Counting (Blobs) module 1. If there are multiple object outputs in the vision job, the output variable values of the Counting (Blobs) module will be a two-dimensional array. Each row of the array denotes the result of an object (find), and each column signifies the multiple results of that object.

14.2.1 Vision_GetOutputArraySize()

Retrieve the array size of the 2D output variables after the vision job execution.

- For vision jobs not created by Flying Trigger or Flying Inspection.

Syntax 1

```
int Vision_GetOutputArraySize(
    string
)
```

Parameter

string The variable name as the output by the vision job

Return

int the object count. (the row count of the 2D array)

Syntax 2

```
int Vision_GetOutputArraySize(
    string,
    int
)
```

Parameter

string The variable name as the output by the vision job
int the object index. (the row index of the 2D array)

Return

int The length of the result array under the object. (row index).

- For vision jobs created by Flying Trigger or Flying Inspection.

Syntax 1

```
int Vision_GetOutputArraySize(
    string
    int or string
)
```

Parameter

string	The variable name as the output by the vision job
int or string	The index of the same visual job name or the name of the Flying point in in Flying Trigger/Inspection.

Return

int	the object count. (the row count of the 2D array)
-----	---

Syntax 2

```
int Vision_GetOutputArraySize(
    string,
    int or string,
    int
)
```

Parameter

string	The variable name as the output by the vision job
int or string	The index of the same visual job name or the name of the Flying point in in Flying Trigger/Inspection.
int	the find object index. (the row index of the 2D array)

Return

int	The length of the result array under the object. (row index).
-----	---

14.2.2 Vision_GetOutputArrayValue()

Retrieve the array values of the 2D output variables after the vision job execution.

- For vision jobs not created by Flying Trigger or Flying Inspection.

Syntax 1

```
string[] Vision_GetOutputArrayValue(
    string,
    int
)
```

Parameter

string	The variable name as the output by the vision job with the data type of string array.
int	the object index. (the row index of the 2D array)

Return

?[]	The result array under the object. (row index).
-----	---

Syntax 2

```
string Vision_GetOutputArrayValue(
    string,
    int,
    int
)
```

Parameter

string	The variable name as the output by the vision job with the data type of string array.
int	the object index. (the row index of the 2D array)
int	The output result array value under the object. (column index)

Return

?[]	The result value of the column index in the result array under the object. (row index).
-----	---

- For vision jobs created by Flying Trigger or Flying Inspection.

Syntax 1

```
? Vision_GetOutputArrayValue(
    string
    int or string
)
```

Parameter

string	The variable name as the output by the vision job
int or string	The index of the same visual job name or the name of the Flying point in in Flying Trigger/Inspection.

Return

?	The result output value
---	-------------------------

Syntax 2

```
? Vision_GetOutputArrayValue(
    string,
    int or string,
    int
)
```

Parameter

string	The variable name as the output by the vision job
int or string	The index of the same visual job name or the name of the Flying point in in Flying Trigger/Inspection.
int	The index value. It can be a result index in a single-object multi-result value or an object index in a multi-object single-result value.

Return

?	The result output value
---	-------------------------

Syntax 3

? **Vision_GetOutputArrayValue**(

string,
int or string,
int,
int

)

Parameter

string The variable name as the output by the vision job
int or string The index of the same visual job name or the name of the Flying point in
in Flying Trigger/Inspection.
int the object index. (the row index of the 2D array)
int The output result array value under the object. (column index)

Return

? The result value of the column index in the result array under the object.
(row index).

● For vision jobs not created by Flying Trigger or Flying Inspection.

Suppose that the visual job Job1 has four objects (find). The Counting Blobs result of the first object is "0", "1", and "2". The Counting Blobs result of the second object is "3" and "4". The Counting Blobs result of the third object is "5", and the Counting Blobs result of the fourth object is "6" and "7". Then, the variable Job1_Count_Blob_1_DetectObjectX_TM represents the result as an array: { {"0", "1", "2"}, {"3", "4"}, {"5"}, {"6", "7"} }.

Row \ Column	0	1	2
0	"0"	"1"	"2"
1	"3"	"4"	
2	"5"		
3	"6"	"7"	

● Get the object (find) count

Vision_GetOutputArraySize("Job1_Count_Blob_1_DetectObjectX_TM")

// Retrieve the row count of the variable *Job1_Count_Blob_1_DetectObjectX_TM* (i.e. the object (find) count)

// 4 // Row count

Vision_GetOutputArraySize("Job1_Count_Blob_1_DetectObjectX_TM", 1)

// Retrieve the array length of the output result from row 1, the 2nd object, of the variable

Job1_Count_Blob_1_DetectObjectX_TM

// 2 // Column count of Row[1]

● Get the object (find) result value

Vision_GetOutputArrayValue("Job1_Count_Blob_1_DetectObjectX_TM", 0)

// Retrieve the array of the output result from row 0, the 1st object, of the variable

Job1_Count_Blob_1_DetectObjectX_TM

// {"0", "1", "2"}

Vision_GetOutputArrayValue("Job1_Count_Blob_1_DetectObjectX_TM", 2, 0)

// Retrieve the output result from row 2 and column 0, the 3rd object, of the variable

Job1_Count_Blob_1_DetectObjectX_TM

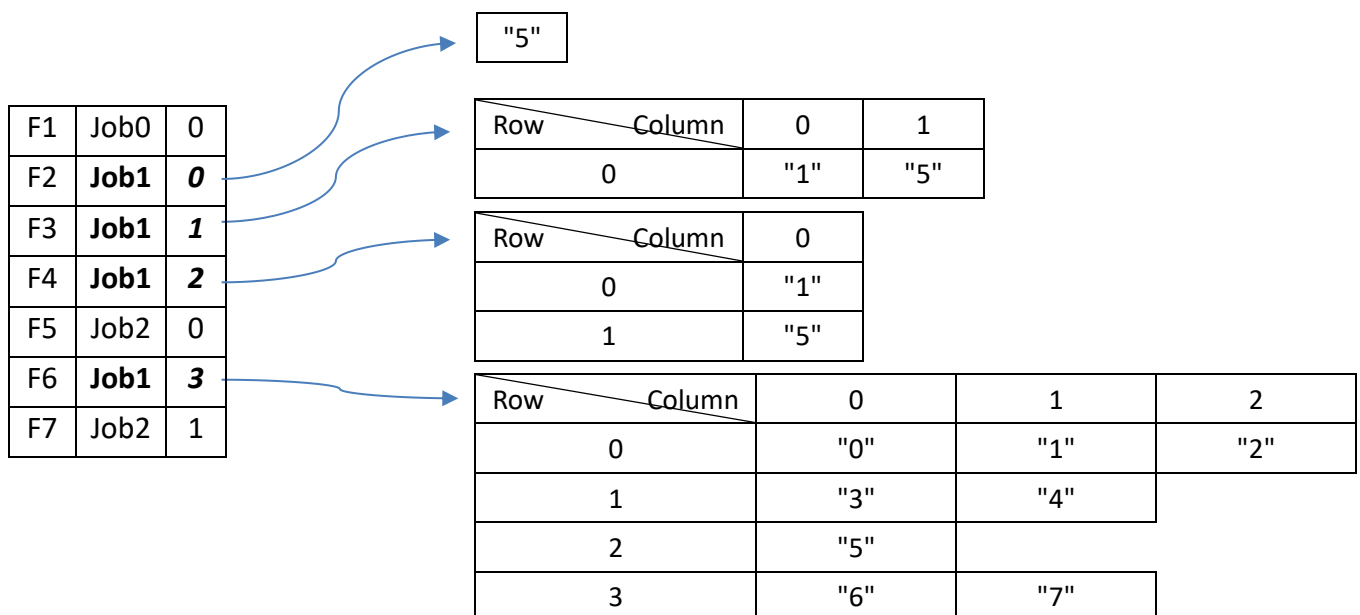
```
// "5"
```

```
Vision_GetOutputArrayValue("Job1_Count_Blob_1_DetectObjectX_TM", 4, 0)
```

```
// Return error. Exceeding the access range. (the object (find) count is 4, and the valid index is 0 to 3.
```

● For vision jobs not created by Flying Trigger or Flying Inspection.

In Flying Trigger or Flying Inspection, users can select the same vision job for repetitive execution. Therefore, the added parameter is the index or the flying point name that uses the same vision job name in Flying Trigger/Inspection. Assume the list of the vision job selected in Flying Trigger/Inspection is {"Job0", "Job1", "Job1", "Job1", "Job1", "Job2", "Job1", "Job2"}, and assume the count of its output objects and the result value is as follows.0



- (1) Suppose the output of the first Job1 is a **single object** and a **single result value**.

F1	Job0	0
F2	Job1	0
F3	Job1	1
F4	Job1	2
F5	Job2	0
F6	Job1	3
F7	Job2	1



- *Get the object (find) count*

```
Vision_GetOutputArraySize("Job1_Count_Blob_1_DetectObjectX_TM", 0)
```

```
Vision_GetOutputArraySize("Job1_Count_Blob_1_DetectObjectX_TM", "F2")
```

// Retrieve the row count (i.e. the object (find) count) of the variable

Job1_Count_Blob_1_DetectObjectX_TM in the first Job1 (the flying point name is F2.)

```
// 1 // Row count
```

```
Vision_GetOutputArraySize("Job1_Count_Blob_1_DetectObjectX_TM", 0, 0)
```

```
Vision_GetOutputArraySize("Job1_Count_Blob_1_DetectObjectX_TM", "F2", 0)
```

// Retrieve the array length of the output result from row 0, the 1st object, of the variable

Job1_Count_Blob_1_DetectObjectX_TM in the first Job1 (the flying point name is F2.)

```
// 1 // Column count of Row[1]
```

- *Get the object (find) result value*

```
Vision_GetOutputArrayValue("Job1_Count_Blob_1_DetectObjectX_TM", 0)
```

```
Vision_GetOutputArrayValue("Job1_Count_Blob_1_DetectObjectX_TM", "F2")
```

// Retrieve the output result of the variable *Job1_Count_Blob_1_DetectObjectX_TM* in the first Job1 (the flying point name is F2.)

```
// "5"
```

- (2) Suppose the output of the second Job1 is a **single object** and a **multiple result value**.

F1	Job0	0
F2	Job1	0
F3	Job1	1
F4	Job1	2
F5	Job2	0
F6	Job1	3
F7	Job2	1

Row \ Column	0	1
0	"1"	"5"

- *Get the object (find) count*

```
Vision_GetOutputArraySize("Job1_Count_Blob_1_DetectObjectX_TM", 1)
Vision_GetOutputArraySize("Job1_Count_Blob_1_DetectObjectX_TM", "F3")
// Retrieve the row count (i.e. the object (find) count) of the variable
Job1_Count_Blob_1_DetectObjectX_TM in the second Job1 (the flying point name is F3.)
// 1      // Row count

Vision_GetOutputArraySize("Job1_Count_Blob_1_DetectObjectX_TM", 1, 0)
Vision_GetOutputArraySize("Job1_Count_Blob_1_DetectObjectX_TM", "F3", 0)
// Retrieve the array length of the output result from row 0, the 1st object, of the variable
Job1_Count_Blob_1_DetectObjectX_TM in the second Job1 (the flying point name is F3.)
// 2      // Column count of Row[1]
```

- *Get the object (find) result value*

```
Vision_GetOutputArrayValue("Job1_Count_Blob_1_DetectObjectX_TM", 1, 0)
Vision_GetOutputArrayValue("Job1_Count_Blob_1_DetectObjectX_TM", "F3", 0)
// Retrieve the index 0 output result of the variable Job1_Count_Blob_1_DetectObjectX_TM in the second
Job1 (the flying point name is F3.)
// "1"
```

(3) Suppose the output of the third Job1 is a multiple object and a single result value.

F1	Job0	0
F2	Job1	0
F3	Job1	1
F4	Job1	2
F5	Job2	0
F6	Job1	3
F7	Job2	1



Row \ Column	0
0	"1"
1	"5"

- *Get the object (find) count*

```

Vision_GetOutputArraySize("Job1_Count_Blob_1_DetectObjectX_TM", 2)
Vision_GetOutputArraySize("Job1_Count_Blob_1_DetectObjectX_TM", "F4")
// Retrieve the row count (i.e. the object (find) count) of the variable
Job1_Count_Blob_1_DetectObjectX_TM in the third Job1 (the flying point name is F4.)
// 2      // Row count
Vision_GetOutputArraySize("Job1_Count_Blob_1_DetectObjectX_TM", 2, 1)
Vision_GetOutputArraySize("Job1_Count_Blob_1_DetectObjectX_TM", "F4", 1)
// Retrieve the array length of the output result from row 1, the 2nd object, of the variable
Job1_Count_Blob_1_DetectObjectX_TM in the third Job1 (the flying point name is F4.)
// 1      // Column count of Row[1]

```

- *Get the object (find) result value*

```

Vision_GetOutputArrayValue("Job1_Count_Blob_1_DetectObjectX_TM", 2, 1)
Vision_GetOutputArrayValue("Job1_Count_Blob_1_DetectObjectX_TM", "F4", 1)
// Retrieve the index 1 output result of the variable Job1_Count_Blob_1_DetectObjectX_TM in the third
Job1 (the flying point name is F4.)
// "5"

```

(4) Suppose the output of the fourth Job1 is a multiple object and a multiple result value.

F1	Job0	0
F2	Job1	0
F3	Job1	1
F4	Job1	2
F5	Job2	0
F6	Job1	3
F7	Job2	1

Row \ Column	0	1	2
0	"0"	"1"	"2"
1	"3"	"4"	
2	"5"		
3	"6"	"7"	

- *Get the object (find) count*

```

Vision_GetOutputArraySize("Job1_Count_Blob_1_DetectObjectX_TM", 3)
Vision_GetOutputArraySize("Job1_Count_Blob_1_DetectObjectX_TM", "F6")
// Retrieve the row count (i.e. the object (find) count) of the variable
Job1_Count_Blob_1_DetectObjectX_TM in the fourth Job1 (the flying point name is F6.)
// 4      // Row count
Vision_GetOutputArraySize("Job1_Count_Blob_1_DetectObjectX_TM", 3, 1)
Vision_GetOutputArraySize("Job1_Count_Blob_1_DetectObjectX_TM", "F6", 1)
// Retrieve the array length of the output result from row 1, the 2nd object, of the variable
Job1_Count_Blob_1_DetectObjectX_TM in the fourth Job1 (the flying point name is F6.)
// 2      // Column count of Row[1]

```

- *Get the object (find) result value*

```

Vision_GetOutputArrayValue("Job1_Count_Blob_1_DetectObjectX_TM", 3, 0, 2)
Vision_GetOutputArrayValue("Job1_Count_Blob_1_DetectObjectX_TM", "F6", 0, 2)
// Retrieve the output result in row 0, the 1st object, and column 2, item 3, of the variable
Job1_Count_Blob_1_DetectObjectX_TM in the fourth Job1 (the flying point name is F4.)
// "2"
Vision_GetOutputArrayValue("Job1_Count_Blob_1_DetectObjectX_TM", 3, 4, 0)
// Return error. Exceeding the access range. (the object (find) count is 4, and the valid index is 0 to 3.)
Vision_GetOutputArrayValue("Job1_Count_Blob_1_DetectObjectX_TM", 4, 0, 0)
// Return error. Exceeding the access range. (Job1 repeats four times, and the valid index is 0 to 3.)
Vision_GetOutputArrayValue("Job1_Count_Blob_1_DetectObjectX_TM", "F8", 0, 0)
// Return error. The Flying point named "F8" does not exist.
Vision_GetOutputArrayValue("Job1_Count_Blob_1_DetectObjectX_TM", "F7", 0, 0)
// Return error. The vision job associated to the Flying point named "F7" is Job2

```


14.3 Vision_GetTriggerJobOutputCount()

Retrieve the output variable count in the buffer after the vision IO Trigger Job execution.

Syntax 1

```
int Vision_GetTriggerJobOutputCount(  
    string  
)
```

Parameter

string	The variable name as output by the IO Trigger Job with the data type of string array.
--------	---

Return

int	The output variable count of the designated variable in the buffer.
-----	---

14.4 Vision_GetTriggerJobOutputValue()

Retrieve the output variable values after the vision IO Trigger Job execution.. (Retrieve and remove from the buffer.)

Syntax 1

```
string[] Vision_GetTriggerJobOutputValue(
    string
)
```

Parameter

string The variable name as output by the IO Trigger Job with the data type of string array.

Return

string[] The output values of the designated variables in the buffer.

Note

After the vision IO Trigger Job execution (Enabling the IO trigger mode is a must during vision editing.), the result updates to the variable after each IO trigger. Since the output of the IO Trigger Job is multiple outputs, to prevent the output data from being overwritten because of out-of-process, it stores the output data in the buffer one by one (first in, first out), and users can use the function calls to retrieve the count or the values in the buffer. For example, job1_Count_Blob_1_DetectObjectX_TM is an output variable of the vision job job1 to be output by the Counting (Blobs) module 1. After the vision execution, it updates and stores the result in the buffer at each IO trigger, so users can call the functions to retrieve the count or the values of the variable job1_Count_Blob_1_DetectObjectX_TM in the buffer.

The buffer comes with a capacity limit. When results are about to enter, if the buffer capacity is insufficient, it removes the oldest data automatically and adds the latest to the buffer.

Suppose the current content in the buffer goes by { {"0", "1", "2"}, {"3", "4"}, {"5"} }

Retrieve the output variable count of the vision trigger job in the buffer.

```
int var_count = Vision_GetTriggerJobOutputCount
("job1_Count_Blob_1_DetectObjectX_TM")
```

```
// var_count = 3
```

Retrieve the output variable values in the buffer.

```
string[] var_s = Vision_GetTriggerJobOutputValue
("job1_Count_Blob_1_DetectObjectX_TM")
```

```
// var_s = {"0", "1", "2"}
```

```
// After retrieving, the content in the buffer turns to { {"3", "4"}, {"5"} }.
```

```
var_count = Vision_GetTriggerJobOutputCount
("job1_Count_Blob_1_DetectObjectX_TM")
```

```
// var_count = 2
```

```
var_s = Vision_GetTriggerJobOutputValue ("job1_Count_Blob_1_DetectObjectX_TM")
// var_s = {"3", "4"}
```

```
// After retrieving, the content in the buffer turns to { {"5"} }.
```

```
var_s = Vision_GetTriggerJobOutputValue ("job1_Count_Blob_1_DetectObjectX_TM")
// var_s = {"5"}
```

```
// After retrieving, the content in the buffer turns to { }.  
var_s = Vision_GetTriggerJobOutputValue ("job1_Count_Blob_1_DetectObjectX_TM")  
// var_s = {}  
var_count = Vision_GetTriggerJobOutputCount  
("job1_Count_Blob_1_DetectObjectX_TM")  
// var_count = 0
```

15. External Script

15.1 Listen Node

Users can establish a socket TCPListener (server site) in the listen node to connect to external devices and communicate based on the packet format. All features available in TM_Robot_Function can also be operated in the listen node.

1. **Send Message:** When entering this node, it will initiate a message
2. **Print Log:** Enable Communication Log (shown on the right)
3. **Connection Timeout:** When entering this node, if more than the time (milliseconds) is not connected, it will be overtime.
If ≤ 0 , no timeout
4. **Data Timeout:** When connected, the timeout will be exceeded when there is no communication packet
If ≤ 0 , no timeout

Socket TCPListener is started up after the project being executed, and closed as the project stopped. The IP and listen port will be shown on the Notice Log window on the right, after the Socket TCPListener is started up.

IP HMI → System → Network → IP Address
Port 5890

When entering the Listen Node, the flow will keep at Listen Node until either of the two exit conditions is fulfilled.

Pass: *ScriptExit()* is executed or the project is stopped

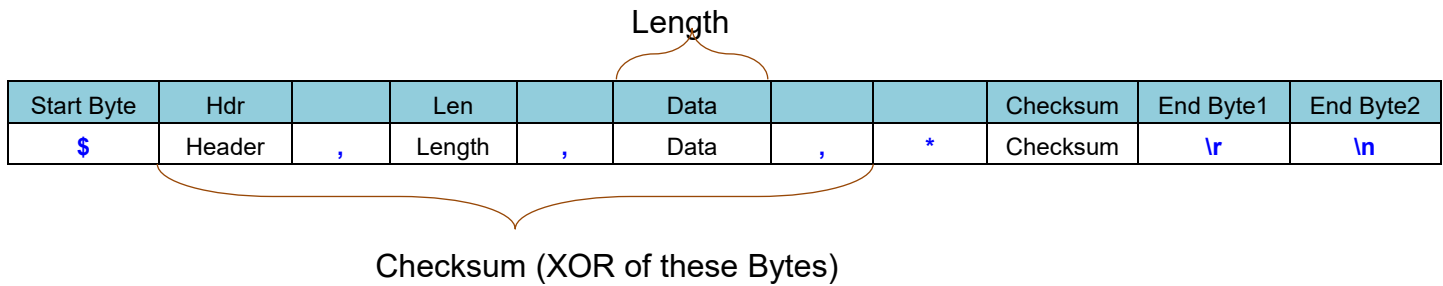
Fail:

1. Connection Timeout
2. Data Timeout
3. Before the TCP Listener is started up, the flow has entered this Listen Node

The command received by listen node will be executed in order. If the command is not valid, an error message will be returned carrying the line number with errors. If the command is valid, it will be executed.

The command can be divided into two categories. The first category is commands which can be accomplished in instance, like assigning variable value. The second category is commands needs to be executed in sequence, like motion command and IO value assigning. The second category command will be placed in queue and executed in order.

15.2 Communication Protocol



Name	Size	ASCII	HEX	Description
Start Byte	1	\$	0x24	Start Byte for Communication
Header	X			Header for Communication
Separator	1	,	0x2C	Separator between Header and Length
Length	Y			Length of Data
Separator	1	,	0x2C	Separator between Length and Data
Data	Z			Communication Data
Separator	1	,	0x2C	Separator between Data and Checksum
Sign	1	*	0x2A	Begin Sign of Checksum
Checksum	2			Checksum of Communication
End Byte 1	1	\r	0x0D	
End Byte 2	1	\n	0x0A	End Byte of Communication

1. Header

Defines the purpose of the communication package. The data definition could be different with different Header.

- **TMSCT** External Script
- **TMSTA** Acquiring status or properties
- **CPERR** Communication data error (E.g. Packet error, checksum error, header error, etc.)

2. Length

Length defines the length in UTF8 byte. It can be represented in decimal, hexadecimal or binary, the upper limit is int 32bits

Example:

```
$TMSCT,100,Data,*CS\r\n // Decimal 100, that is the data length is 100 bytes
$TMSCT,0x100,Data,*CS\r\n // Hexadecimal 0x100, that is the data length is 256 bytes
$TMSCT,0b100,Data,*CS\r\n // Binary 0b100, that is the data length is 4 bytes
$TMSCT,8,1,達明,*58\r\n // The Data length 1,達明 is 8 bytes (UTF8)
```

3. Data

The content of the communication package. Arbitrary characters are supported (including 0x00 .. 0xFF in UTF8). The data length is defined in Length and the purpose is defined in Header

4. Checksum

The checksum of the communication package. The checksum is calculated with XOR(exclusive OR), and the range for checksum computation starts from \$ to * (\$ and * are

excluded) as shown below:

\$TMSCT,100,Data*CS\r\n

Checksum = Byte[1] ^ Byte[2] ... ^ Byte[N-6]

The representation of checksum is fixed to 2 bytes in hexadecimal format (without 0x).

For example:

\$TMSCT,5,10,OK,*6D

CS = 0x54 ^ 0x4D ^ 0x53 ^ 0x43 ^ 0x54 ^ 0x2C ^ 0x35 ^ 0x2C ^ 0x31 ^ 0x30 ^ 0x2C ^ 0x4F ^
0x4B ^ 0x2C = 0x6D

CS = 6D (0x36 0x44)

15.3 TMSCT

Start Byte	Hdr		Len		Data			Checksum	End Byte1	End Byte2
\$	TMSCT	,	Length	,	Data	,	*	Checksum	\r	\n

ID		SCRIPT
Script ID	,	Script Language

TMSCT defines the communication package as External Script Language. In External Script Language, the data contains two parts and is separated by comma. One is ID and the other is SCRIPT

- ID** Script ID, can be arbitrary English character or number (a CPERR 04 error will be reported when encountering non-alphanumeric byte). The ID is used as specifying the target SCRIPT of return message.
- ,** Separator
- SCRIPT** The content defined in Script Language. In a communication package, multi-line scripts can fit into the SCRIPT section with separator (0x0D 0x0A)

Note

TMSCT is available only when in the external script control mode, otherwise CPEER error packets will be replied.

Return (Robot→External Device)

- When it enters Listen Node, the robot will send a message to all the connected device. The ID is set to 0.
\$TMSCT,9,0,Listen1,*4C\r\n
9 The length of *0,Listen1* is 9 bytes
0 The Script ID is 0
, Separator
Listen1 The message to send
- The OK or ERROR message is replied according to the Script's content. For message with **;N**, **;N** represents the number of line with error or warning. **After the message is received, robot will execute the message, then send back the return message, if the Script is valid. For invalid Script, the return message will be sent back immediately without executed.**
\$TMSCT,4,1,OK,*5C\r\n // Response to ID 1
 // OK means valid Script.
\$TMSCT,8,2,OK;2;3,*52\r\n // Response to ID 2
 // OK;2;3 means valid Script with warnings in line 2 and 3.
\$TMSCT,13,3,ERROR;1;2;3,*3F\r\n // Response to ID 3
 // ERROR;1;2;3 means invalid Script with errors in line 1, 2 and 3.

Receive (Robot←External Device)

- When it enters the listen node, the robot will start to receive, check, and execute the external script. If the robot did not enter the listen node (not in the external script control mode), the Script received will be disposed and CPEER error packets will be replied.
- The message from external device should define the Script ID as a ID used in messages replied by robot.

< **\$TMSCT,25,1,ChangeBase("RobotBase"),*08\r\n** // Defined as ID 1

> \$TMSCT,4,1,OK,*5C\r\n // Response to ID 1

3. In a communication package, multi-line scripts can fit into the SCRIPT section with separator \r\n

```
< $TMSCT,72,2,ChangeBase("RobotBase")\r\n
ChangeTCP("RobotEndFlange")\r\n
ChangeLoad(10.1),*5A\r\n // Three lines Script in a communication package (Lines
are separated by \r\n)
> $TMSCT,4,2,OK,*5F\r\n
```

4. In Listen Node, local variables are supported and valid before quitting the Listen Node.

```
< $TMSCT,40,3,int var_i = 100\r\n
var_i = 1000\r\n
var_i++,*5A\r\n
> $TMSCT,4,3,OK,*5E\r\n

< $TMSCT,42,4,int var_i = 100\r\n
var_i = 1000\r\n
var_i++\r\n
,*58\r\n
> $TMSCT,9,4,ERROR;1,*02\r\n // Because int var_i has been declared, an error
occurred.
```

5. In the listen node, it is possible to access or modify the project's variables, but no new variable can be declared since the variables created in the listen node are local variables.

15.4 TMSTA

Start Byte	Hdr		Len		Data			Checksum	End Byte1	End Byte2
\$	TMSTA	,	Length	,	Data	,	*	Checksum	\r	\n

SubCmd		
SubCmd	...	...

(Based on SubCmd)

TMSTA defines the communication package as acquiring status or properties. The data section of the package contains different sub command (SubCmd). The package format could be different according to different SubCmd. The definitions are listed below.

SubCmd

- 00 In external script control mode or not
- 01 Complete the configured QueueTag numbering or not
- 90..99 Date message to send (the format of data is self-definable)

Note

TMSTA could be executed without entering the Listen Node

SubCmd 00 In external script control mode or not

Format

Response (Robot→External Device)

SubCmd		Entry		Message
00	,	false	,	
00	,	true	,	message

Receive (Robot←External Device)

SubCmd
00

Response (Robot→External Device)

- If not in external script control mode, it will reply false.

\$TMSTA,9,00,false,,*37\r\n

- 9 Indicates the length of 00,false, is 9 bytes
- 00 Indicates SubCmd as 00
- ,
- false The flow has not entered Listen Node
- ,
- Empty string (Have not entered Listen Node)

- If in external script control mode, it will reply true.

\$TMSTA,15,00,true,Listen1,*79\r\n

- 15 Indicates the length of 00,true,Listen1 is 15 bytes
- 00 Indicates SubCmd as 00
- ,
- true The flow has entered the Listen Node
- ,
- Listen1 The message to be sent as in Listen Node (It indicates the flow is in Listen1)

Receive (Robot←External Device)

1. Send to the robot from the external device

\$**TMSTA**,**2**,**00**,*41\r\n

2 Indicates the length of 00 is 2 bytes.

00 Indicates the SubCmd is 00 whether in external script control mode or not.

SubCmd 01 Complete the configured QueueTag numbering or not

Format

Send (Robot→External Device)

SubCmd		Tag Number		Status
01	,	01 .. 15	,	true/false/none

Receive (Robot←External Device)

SubCmd		Tag Number
01	,	01 .. 15

Note

When inquiring with TMSTA 01, users can look up to the status of the last 4 tag numbers.

Send (Robot→External Device)

1. Send to the external device from the robot. Spontaneously sending after QueueTag numbering completed.

\$**TMSTA**,**10**,**01**,**08**,**true**,*6D\r\n

10 Indicates the length of 01,00,true is 10 bytes

01 Indicates SubCmd as 01 to send the status of Tag Number

, Separation symbol

08 Tag Number 08

, Separation symbol

true **true** Indicates Tag Number complete

false Indicates Tag Number incomplete

none Indicates Tag Number not existed

Receive (Robot←External Device)

1. Send from the external device to the robot. Users can look up to the status of the last 4 tag numbers.

\$**TMSTA**,**5**,**01**,**15**,*6F\r\n

5 Indicates the length of 01,88 is 5 bytes

01 Indicates SubCmd as 01 to send the status of Tag Number

, Separation symbol

15 Tag Number 15

> \$**TMSTA**,**10**,**01**,**15**,**none**,*7D\r\n // TagNumber 15 not existed

2. Tag Number uses the value of integers between 1 and 15. If the value is invalid, it relies none for not existed.

\$**TMSTA**,**5**,**01**,**88**,*6B\r\n

> \$**TMSTA**,**10**,**01**,**88**,**none**,*79\r\n // TagNumber 88 not existed

SubCmd 90 .. 99 Send data message

Format

Send (Robot→External Device)

SubCmd		Data
90 .. 99	,	...

Receive (Robot←External Device)

None

Note

1. When sending with TMSTA 90 .. 99, users can use their self-defined formats.
2. Self-defined formats denote the formats are defined by both the project flow and the external device.
3. To enhance the flexibility of usages, users can various SubCmd of 90 .. 99 to define different formats to send such as

SubCmd 90 defined as string ;

SubCmd 91 defined as float[] ;

SubCmd 92 defined as byte[]

...

and so on for the external device to analyze and resolve based on the SubCmd with different methods.

Send (Robot→External Device)

1. Send to the external device from the robot. When the external script executes the ListenSend() function, it will send data.

string var_s = "Hello World"

float[] var_f = {1,2,3,4}

byte[] var_b = {0x10, 0x11, 0x12, 0x13}

ListenSend(90, var_s)

// the content of communication \$TMSTA,14,90,Hello World,*73\r\n

// 0x39,0x30,0x2C,0x48,0x65,0x6C,0x6C,0x6F,0x20,0x57,0x6F,0x72,0x6C,0x64

ListenSend(91, var_f)

// the content of communication \$TMSTA,19,91,...,*60\r\n

//

0x39,0x31,0x2C,0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00,0x40,0x40,0x00,0x00,0x80,0x40

ListenSend(92, var_b)

// the content of communication \$TMSTA,7,92,...,*63\r\n

// 0x39,0x32,0x2C,0x10,0x11,0x12,0x13

15.5 CPERR

Start Byte	Hdr		Len		Data			Checksum	End Byte1	End Byte2
\$	CPERR	,	Length	,	Data	,	*	Checksum	\r	\n

Error Code
Code (00 .. FF)

CPERR defines the communication package as sending the Communication Protocol Error. The data section is defined as Error Code.

Error Code	Error code, presented in 2 bytes hexadecimal format (without 0x)
00	Packet correct. No error. (The return message usually reply to the content of packet instead of returning no error)
01	Packet Error.
02	Checksum Error.
03	Header Error.
04	Packet Data Error.
F1	Have not entered Listen Node

Note

Used by robot to response to external device

Response (Robot→External Device)

- 01 Packet Error
- ```
< $TMSCT,-100,1,ChangeBase("RobotBase"),*13\r\n // Length cannot be negative
> $CPERR,2,01,*49\r\n // CPERR Error Code 01
```
- 01 Packet Error
- ```
< $TMSCT,24,1,ChangeBase("RobotBase"),*09\r\n
// Length value is incorrect (should be 25)
> $CPERR,2,01,*49\r\n // CPERR Error Code 01
```
- 01 Packet Error
- ```
< $TMSCT,26,1,ChangeBase("RobotBase"),*0B\r\n
// When sending only the packet mentioned above, it will not bring out a response because its Length
// of 26 exceeds the actual data amount, indicating an insufficient number of packets.
< $TMSCT,26,1,ChangeBase("RobotBase"),*0B\r\n
// Length value is incorrect (should be 25)
> $CPERR,2,01,*49\r\n // CPERR Error Code 01
```
- 01 Packet Error
- ```
< $TMSCT,25,1,ChangeBase("RobotBase"),*088\r\n
// 088 is not a correct packet ending format.
> $CPERR,2,01,*49\r\n // CPERR Error Code 01
```
- 01 Packet Error
- ```
< $TMSCT,25,1,ChangeBase("RobotBase"),*8\r\n
// When sending only the packet mentioned above, it will not bring out a response because of a
// Checksum Byte coming short.
```

```
< $TMSCT,25,1,ChangeBase("RobotBase"),*8\r\n
// 8 is not a correct packet ending format.
> $CPERR,2,01,*49\r\n // CPERR Error Code 01
```

#### 01 Packet Error

```
< $TMSCT,25,1,ChangeBase("RobotBase"),*08\n
// When sending only the packet mentioned above, it will not bring out a response because of an End
short.
< $TMSCT,25,1,ChangeBase("RobotBase"),*08\n
// \n is not a correct packet ending format.
> $CPERR,2,01,*49\r\n // CPERR Error Code 01
```

#### 02 Checksum Error

```
< $TMSCT,25,1,ChangeBase("RobotBase"),*09\r\n // 09 is not a correct Checksum
> $CPERR,2,02,*4A\r\n // CPERR Error Code 02
```

#### 03 Header Error

```
< $TMsct,25,1,ChangeBase("RobotBase"),*28\r\n // TMscT is not a correct
Header
> $CPERR,2,03,*4B\r\n // CPERR Error Code 03
```

#### 04 Packet Data Error

```
< $TMSTA,4,XXXX,*47\r\n // There is no XXXX SubCmd under
TMSTA
> $CPERR,2,04,*4C\r\n // CPERR Error Code 04
```

#### F1 No External Script Mode

```
< $TMSCT,25,1,ChangeBase("RobotBase"),*0D\r\n
// Suppose currently not in external script control mode
> $CPERR,2,F1,*3F\r\n // CPERR Error Code F1
```

## 15.6 ScriptListen()

Enter the external script control mode.

### Syntax 1

```
bool ScriptListen(
 string,
 bool,
 int,
 int
)
```

#### Parameters

- |               |                                                                                                                                                                                                                |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>string</b> | The message string to send while in the external script control mode.                                                                                                                                          |
| <b>bool</b>   | Whether to display the receiving external messages in the notice window.                                                                                                                                       |
| <b>int</b>    | After entering external Script control mode, it will time out if not connected within a specific timeframe (measured in milliseconds).<br>If <code>&lt;= 0</code> No timeout                                   |
| <b>Int</b>    | After entering external Script control mode and connected, it will time out if no communication packets received within a specific timeframe (measured in milliseconds).<br>If <code>&lt;= 0</code> No timeout |

#### Return

- |             |                                                                                                                                                                                                                                                                                                                                                            |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>bool</b> | When entering the external Script control mode, it will remain there until meeting certain conditions and exit by the conditions.<br><b>True</b> execute <a href="#">ScriptExit()</a> or stop the project<br><b>False</b> 1. Connection Timeout occurred<br>2. Data Timeout occurred<br>3. If the TCP Listener fails to set up, ScriptListen() is invoked. |
| *           | Accessible only through the Listen node, the Script node, or Script project programming.                                                                                                                                                                                                                                                                   |
| *           | Calling in the flow required.                                                                                                                                                                                                                                                                                                                              |

### Syntax 2

```
bool ScriptListen(
 string,
 bool
)
```

#### Note

Same as Syntax 1. Set the message strings to send and whether to show in the notice window. There is no timeout by default.

### Syntax 3

```
bool ScriptListen(
 string
)
```

#### Note

Same as Syntax 1. Set the message strings to send. There is no show in the notice window and no timeout by default.

```
if (ScriptListen("Listen1", false, 0, 0))
 Display("PASS")
else
```

Display("FAIL")

## 15.7 ScriptExit()

Exit the external script control mode.

### Syntax 1

```
bool ScriptExit(
)
```

#### Parameters

void No parameter

#### Return

bool True Command accepted; False Command rejected (format error)

#### Note

Exit the external script control mode and wait for the command to finish, and then quit the listen node and move on with the pass route, or quit ScriptListen() and return True.

- \* Execute via TMSCT communication packets required.
- \* Functions after ScriptExit() will not be executed such as

```
< $TMSCT,86,2,ChangeBase("RobotBase")\r\n
 ChangeTCP("RobotEndFlange")\r\n
 ScriptExit()\r\n // Exit the external script control mode.
 ChangeLoad(10.1),*58\r\n // ChangeLoad will not be executed.
```

\* After exiting the script mode, it is required to wait for all the commands and the functions to complete executions until quitting the listen node and moving on with the pass route. At the time being of waiting for quitting the listen node, it is not in the external script control mode, so no more external commands will be accepted and CPEER error packets will be replied.



## 15.8 Priority Commands

Due to the serial execution nature of the TMsyntax, if using the queue syntax such as QueueTag(1, 1) or WaitQueueTag(1) to wait for the tag number to arrive, the program will stay put until the conditions are comprehended before going on execution. Therefore, if received the ScriptExit() syntax sent from the outside while waiting, it is impossible to exit the external Script control mode since the program is still waiting for the condition to meet.

When the program comes to the Listen node (the external Script control mode), other than the serial execution TMsyntax, there are also priority commands to use. In Listen nodes, the priority commands go with a higher execution priority than the syntax execution, and the priority commands will run at once as defined below.

### 1. ScriptExit(0)

Stop the robot motion immediately, clear the robot motion instructions in the buffer, and exit the external Script control mode. Go to the Fail path after leaving the Listen node.

### 2. ScriptExit(1)

Stop the robot motion immediately, clear the robot motion instructions in the buffer, and exit the external Script control mode. Go to the Pass path after leaving the Listen node.

### 3. StopAndClearBuffer(0)

Stop the robot motion immediately and clear the robot motion instructions in the buffer.

### 4. StopAndClearBuffer(1)

Stop the robot motion immediately, clear the robot motion instructions in the buffer, exit the current Script program in execution, and continue to the next Script program.

### 5. StopAndClearBuffer(2)

Stop the robot motion immediately, clear the robot motion instructions in the buffer, exit the current Script program in execution, and clear all the script programs in the Received Script buffer.

- Priority commands support the general use of the command definitions only but not the use with variables and functions such as

```
< $TMSCT,42,1,int var_st=2\r\n
 StopAndClearBuffer(var_st),*3A\r\n
> $TMSCT,9,1,ERROR;2,*04\r\n // Invalid syntax StopAndClearBuffer(var_st)
```

- Using priority commands with TMsyntax leads to priority command executions only but not syntax executions.

```
< $TMSCT,94,2,float[] targetP1= {0,0,90,0,90,0}\r\n // Will not execute.
 PTP("JPP",targetP1,10,200,0,false)\r\n // Will not execute.
 StopAndClearBuffer(0),*75\r\n // Priority execution StopAndClearBuffer(0)
> $TMSCT,4,2,OK,*5F\r\n
```

- The system will handle one priority command only in one external Script packet. If the packet comes with numerous priority commands, the system will handle ScriptExit(0/1) before StopAndClearBuffer(0/1/2). If there are numerous ScriptExit and

StopAndClearBuffer respectively, the system will handle the first only.

```
< $TMSCT,46,3,StopAndClearBuffer(2)\r\n // Will execute.
// There are many of StopAndClearBuffer and the system will
// handle the 1st of them.
StopAndClearBuffer(1),*68\r\n // Will not execute.
> $TMSCT,4,3,OK,*5E\r\n
< $TMSCT,61,4,StopAndClearBuffer(2)\r\n // Will not execute.
StopAndClearBuffer(1)\r\n // Will not execute.
ScriptExit(1),*52\r\n // Will execute.
// Due to the higher priority, the system handles ScriptExit
// before StopAndClearBuffer.
> $TMSCT,4,4,OK,*59\r\n
```

```
1. < $TMSCT,86,1,float[] targetP1= {0,0,90,0,90,0}\r\n
PTP("JPP",targetP1,10,200,0,false)\r\n
QueueTag(1,1),*60\r\n // QueueTag(1,1) waits.
// The program will stay put with the command until tag 1 finishes.
< $TMSCT,15,2,ScriptExit(0),*55\r\n
// When receiving the command ScriptExit(0), if tag 1 does not finish,
> $TMSCT,4,1,OK,*5C\r\n // Respond 1 OK
// It does not send out TMSTA tag number for finishing due to the clearing of the
// motion command and QueueTag().
> $TMSCT,4,2,OK,*5F\r\n // Respond 2 OK
```

It will exit the external Script control mode and go to the Fail path after leaving the Listen node.

```
2. < $TMSCT,86,1,float[] targetP1= {0,0,90,0,90,0}\r\n
PTP("JPP",targetP1,10,200,0,false)\r\n
QueueTag(1,1),*60\r\n // QueueTag(1,1) waits.
// The program will stay put with the command until tag 1 finishes.
< $TMSCT,23,2,StopAndClearBuffer(0),*55\r\n
// When receiving the command StopAndClearBuffer(0), if tag 1 does not
// finish
> $TMSCT,4,1,OK,*5C\r\n // Respond 1 OK
// It does not send out TMSTA tag number for finishing due to the clearing of the
// motion command and QueueTag().
> $TMSCT,4,2,OK,*5F\r\n // Respond 2 OK
It does not exit the external Script control mode and is still in the Listen node.
```

```
3. < $TMSCT,145,1,float[] targetP1= {0,0,90,0,90,0}\r\n
PTP("JPP",targetP1,10,200,0,false)\r\n
QueueTag(1,0)\r\n
WaitQueueTag(0,60000)\r\n // Tag 0 will wait for the 60-seconds timeout.
PTP("JPP",targetP1,40,200,0,false),*64\r\n // It takes a 60-seconds wait before the execution.
< $TMSCT,23,2,StopAndClearBuffer(0),*55\r\n
// If receiving the command StopAndClearBuffer(0) in 60 seconds, it will clear the 1st PTP() function.
> $TMSCT,4,2,OK,*5F\r\n // Respond 2 OK
It waits for the 60-seconds timeout before the 2nd PTP() function execution and responds 1
```

OK.

```
> $TMSCT,4,1,OK,*5C\r\n // Respond 1 OK
```

\* StopAndClearBuffer(0) clears the motion commands only but not the function syntax or the logic operations. While WaitQueueTag(0) sets the number 0, it waits for the timeout and is impossible to exit with the motion command clearing. It is required to use the command StopAndClearBuffer(1/2) to exit from the current Script program in execution.

```
4. < $TMSCT,86,1,float[] targetP1= {0,0,90,0,90,0}\r\n
 PTP("JPP",targetP1,10,200,0,false)\r\n
 QueueTag(1,1),*60\r\n // QueueTag(1,1) waits.
 The program will stay put with the command until tag 1 finishes.
< $TMSCT,86,2,float[] targetP2= {90,0,0,90,0,0}\r\n
 PTP("JPP",targetP2,10,200,0,false)\r\n
 QueueTag(2,1),*60\r\n // The last packet still waits, and this packet does not execute.
< $TMSCT,23,3,StopAndClearBuffer(1),*55\r\n
 // When receiving the command StopAndClearBuffer(1), if tag 1 does not finish
> $TMSCT,4,1,OK,*5C\r\n // Respond 1 OK
 // It does not send out TMSTA tag number for finishing due to the clearing of the
 motion command and QueueTag().
> $TMSCT,4,3,OK,*5E\r\n // Respond 3 OK
It clears and exits the current Script 1 in execution, and it continues to the next, Script 2.
> $TMSCT,4,2,OK,*5F\r\n // Respond 2 OK
> $TMSTA,10,01,02,true,*67\r\n // Tag 2 finished
```

```
5. < $TMSCT,86,1,float[] targetP1= {0,0,90,0,90,0}\r\n
 PTP("JPP",targetP1,10,200,0,false)\r\n
 QueueTag(1,1),*60\r\n // QueueTag(1,1) waits.
 The program will stay put with the command until tag 1 finishes.
< $TMSCT,86,2,float[] targetP2= {90,0,0,90,0,0}\r\n
 PTP("JPP",targetP2,10,200,0,false)\r\n
 QueueTag(2,1),*60\r\n // The last packet still waits, and this packet does not execute.
< $TMSCT,23,3,StopAndClearBuffer(2),*56\r\n
 // When receiving the command StopAndClearBuffer(2), if tag 1 does not
 finish
> $TMSCT,4,1,OK,*5C\r\n // Respond 1 OK
 // It does not send out TMSTA tag number for finishing due to the clearing of the
 motion command and QueueTag().
> $TMSCT,4,3,OK,*5E\r\n // Respond 3 OK
It clears and exits the current Script 1 in execution, and clears all the Script programs in the
Received Script buffer. Therefore, it clears Script 2 without the response to Script 2.
```

## 16. Modbus Functions

### 16.1 ModbusTCP Class

Use Modbus TCP class and declare variables to create a Modbus TCP device. The variable name will be the device name.

#### Construct 1

**ModbusTCP** *VariableName* = string, int, int

**ModbusTCP** *VariableName* = string, int

**ModbusTCP** *VariableName* = string

#### Parameters

string remote host IP address  
 int remote host connection port (502 by default)  
 int read/write timeout in millisecond 0 .. 10000 (10000ms by default)

#### Note

**ModbusTCP** mtcp1 = "192.168.1.10"

// construct a device, with IP 192.168.1.10

**ModbusTCP** mtcp2 = "192.168.1.10", 502

// construct a device, with IP 192.168.1.10, Port 502

**ModbusTCP** mtcp3 = "192.168.1.10", 502, 8000

// construct a device, with IP 192.168.1.10, Port 502, Timeout 8000ms

\* After construction, either in flow projects or script projects, the device will not connect actively until proceeding to read or write.

#### Member Methods

| Name         | Description                                                   |
|--------------|---------------------------------------------------------------|
| Preset()     | Configure the preset ModbusTCP parameters.                    |
| IODDPreset() | Read IODD file and configure the preset ModbusTCP parameters. |

#### 16.1.1 Preset()

Configure the preset ModbusTCP parameters.

#### Syntax 1

```
bool Preset(
 string,
 byte,
 string,
 int,
 string,
 int
)
```

#### Parameters

string preset device name  
 byte Slave ID  
 string signal type "DO", "DI", "RO", "RI"  
 int starting address

**string** type "bool", "byte", "int16", "int32", "float", "double", "string"  
**int** suffix parameter, when type is  
 "int32" 0 Little Endian (CD AB) 1 Big Endian (AB CD) (default)  
 "float" 0 Little Endian (CD AB) 1 Big Endian (AB CD) (default)  
 "double" 0 Little Endian (CD AB) 1 Big Endian (AB CD) (default)  
 "string" address count (0 by default)  
 This is invalid for other types.

**Return**

**bool** preset successfully **True**, preset unsuccessfully **False**

**Syntax 2**

```
bool Preset(
 string,
 byte,
 string,
 int,
 string
)
```

**Note**

Same as syntax 1. Fill default to suffix parameter by default.

**Note**

**ModbusTCP** mbus1 = "127.0.0.1"

// construct a device, with IP 127.0.0.1, Port 502, Timeout 8000ms

mbus1.Preset("light", 1, "DO", 7206, "bool") // set preset device name to "light"

mbus1.Preset("9000", 1, "RO", 9000, "string") // set preset device name to "9000"

// Return error, naming must be in alphanumeric combination.

mbus1.Preset("preset\_9000", 1, "RO", 9000, "int", 0)

// set preset device name to "preset\_9000" // Little Endian

mbus1.Preset("preset\_9000", 1, "RO", 9000, "int", 1)

// set preset device name to "preset\_9000" // Big Endian

mbus1.Preset("preset\_9000", 1, "RO", 9000, "int")

// set preset device name to "preset\_9000" // Big Endian

mbus1.Preset("preset\_9000", 1, "RO", 9000, "string", 5)

// set preset device name to "preset\_9000" // string type

// Once the name, preset\_9000, exists, it overwrites the content to the configuration of the same name by the processing sequence.

bool flag = modbus\_read("mbus1", "light") // flag = false // suppose camera light is off

modbus\_write("mbus1", "light", true) // write true // camera light is on

flag = modbus\_read("mbus1", "light") // flag = true

modbus\_write("mbus1 ", "preset\_9000", Ctrl("\0\0\0\0\0\0\0\0\0"))

// clears preset\_9000 in string type and occupies five addresses (5 RO = 10 bytes)

modbus\_write("mbus1", "preset\_9000", 1234) // write "1234" // because preset\_9000 is in string type

```

int var_i = modbus_read("mbus1", "preset_9000") // var_i = 1234 // it tries to convert
 "1234" to integer.

// because preset_9000 is in string type and occupies five addresses, it then reads 10 bytes and converts by strings.
// (ends when encountered 0x00.)

modbus_write("mbus1", "preset_9000", "HelloWorld") // write "HelloWorld"
string var_s = modbus_read("mbus1", "preset_9000") // var_s = "HelloWorld"
var_i = modbus_read("mbus1", "preset_9000")
// Return error, unable to convert HelloWorld to integer.

modbus_write("mbus1", "preset_9000", 1234) // write "1234"
// continue the last piece of data and write "HelloWorld" // therefore, the current data in the address 9000 is
 "1234oWorld"

var_i = modbus_read("mbus1", "preset_9000")
// Return error, unable to convert 1234oWorld to integer.

```

### 16.1.2 IODDPreset()

Read IODD file and configure the preset ModbusTCP parameters.

#### Syntax 1

```

bool IODDPreset(
 string,
 byte,
 int,
 int
)

```

#### Parameters

**string** IODD file name (read the IODD file stored in the directory `.\XmlFiles\IODD` at local host)

**byte** Slave ID

**int** starting address In

**int** starting address Out

#### Return

**bool** preset successfully **True**, preset unsuccessfully **False**

#### Note

**ModbusTCP** mbus1 = "192.168.1.10"

// construct a device, with IP 192.168.1.10, Port 502, Timeout 10000ms

```

mbus1.IODDPreset("OMRON-E2EQ-X3B4-IL2-20170301-IODD1.1.xml", 1, 100, 200)
// load the file, .\XmlFiles\IODD\OMRON-E2EQ-X3B4-IL2-20170301-IODD1.1.xml, and add the preset
configuration parameters.
// define preset names in the same way as flow projects rules.

```

## 16.2 ModbusRTU Class

Use Modbus TCP class and declare variables to create a Modbus TCP device. The variable name will be the device name.

### Construct

**ModbusRTU** *VariableName* = string, int, string, int, float, int, bool, bool, bool

**ModbusRTU** *VariableName* = string, int, string, int, float, int

**ModbusRTU** *VariableName* = string, int, string, int, float

**ModbusRTU** *VariableName* = string, int

### Parameters

|        |                                   |                                                            |
|--------|-----------------------------------|------------------------------------------------------------|
| string | connection description            |                                                            |
| int    | bits per second, BaudRate         |                                                            |
| string | parity check                      | "none", "odd", "even", "mark", "space" ("none" by default) |
| int    | Data Bits                         | 5, 6, 7, 8 (8 by default)                                  |
| float  | Stop Bits                         | 1, 1.5, 2 (1 by default)                                   |
| int    | read/write timeout in millisecond | 0 .. 10000 (10000 ms by default)                           |
| bool   | DTR/DSR                           | true, false (false by default)                             |
| bool   | RTS/CTS                           | true, false (false by default)                             |
| bool   | XON/XOFF                          | true, false (false by default)                             |

### Note

**ModbusRTU** mrtu1 = "COM2",115200

// construct a device with Baudrate 115200

**ModbusRTU** mrtu2 = "COM2",115200,"none",8,1

// construct a device with Baudrate 115200, Parity none, DataBits 8, StopBits 1

**ModbusRTU** mrtu3 = "COM2",115200,"none",8,1,10000

// construct a device with Baudrate 115200, Parity none, DataBits 8, StopBits 1, Timeout 10000ms

\* After construction, either in flow projects or script projects, the device will not connect actively until proceeding to read or write.

### Member Methods

| Name                         | Description                                                   |
|------------------------------|---------------------------------------------------------------|
| <a href="#">Preset()</a>     | Configure the preset ModbusTCP parameters.                    |
| <a href="#">IODDPreset()</a> | Read IODD file and configure the preset ModbusTCP parameters. |

### 16.2.1 Preset()

Configure the preset ModbusTCP parameters.

#### Syntax 1

```
bool Preset(
 string,
 byte,
 string,
 int,
 string,
```



```
int
```

```
)
```

### Parameters

```
string preset device name
```

```
byte Slave ID
```

```
string signal type "DO", "DI", "RO", "RI"
```

```
int starting address
```

```
string type "bool", "byte", "int16", "int32", "float", "double", "string"
```

```
int suffix parameter, when type is
```

```
"int32" 0 Little Endian (CD AB) 1 Big Endian (AB CD) (default)
```

```
"float" 0 Little Endian (CD AB) 1 Big Endian (AB CD) (default)
```

```
"double" 0 Little Endian (CD AB) 1 Big Endian (AB CD) (default)
```

```
"string" address count (0 by default)
```

This is invalid for other types.

### Return

```
bool preset successfully True , preset unsuccessfully False
```

### Syntax 2

```
bool Preset(
```

```
string,
```

```
byte,
```

```
string,
```

```
int,
```

```
string
```

```
)
```

### Note

Same as syntax 1. Fill default to suffix parameter by default.

### Note

```
ModbusRTU mbus1 = "COM2",115200
```

```
// construct a device with Baudrate 115200, Parity none, DataBits 8, StopBits 1, Timeout 10000ms
```

```
mbus1.Preset("light", 1, "DO", 7206, "bool") // set preset device name to "light"
```

```
mbus1.Preset("9000", 1, "RO", 9000, "string") // set preset device name to "9000"
```

```
// Return error, naming must be in alphanumeric combination.
```

```
mbus1.Preset("preset_9000", 1, "RO", 9000, "int", 0)
```

```
// set preset device name to "preset_9000" // Little Endian
```

```
mbus1.Preset("preset_9000", 1, "RO", 9000, "int", 1)
```

```
// set preset device name to "preset_9000" // Big Endian
```

```
mbus1.Preset("preset_9000", 1, "RO", 9000, "int")
```

```
// set preset device name to "preset_9000" // Big Endian
```

```
mbus1.Preset("preset_9000", 1, "RO", 9000, "string", 5)
```

```
// set preset device name to "preset_9000" // string type
```

```
// Once the name, preset_9000, exists, it overwrites the content to the configuration of the same name by the processing sequence.
```

```
bool flag = modbus_read("mbus1", "light")// flag = false // suppose camera light is off
```



```

modbus_write("mbus1", "light", true) // write true // camera light is on
flag = modbus_read("mbus1", "light") // flag = true

modbus_write("mbus1 ", "preset_9000", Ctrl("\0\0\0\0\0\0\0\0\0\0"))
// clears preset_9000 in string type and occupies five addresses (5 RO = 10 bytes)

modbus_write("mbus1", "preset_9000", 1234) // write "1234" // because preset_9000 is in
 string type
int var_i = modbus_read("mbus1", "preset_9000") // var_i = 1234 // it tries to convert
 "1234" to integer.

// because preset_9000 is in string type and occupies five addresses, it then reads 10 bytes and converts by strings.
(ends when encountered 0x00.)

modbus_write("mbus1", "preset_9000", "HelloWorld") // write "HelloWorld"
string var_s = modbus_read("mbus1", "preset_9000") // var_s = "HelloWorld"
var_i = modbus_read("mbus1", "preset_9000")
// Return error, unable to convert HelloWorld to integer.

modbus_write("mbus1", "preset_9000", 1234) // write "1234"
// contiune the last piece of data and write "HelloWorld" // therefore, the current data in the address 9000 is
 "1234oWorld"

var_i = modbus_read("mbus1", "preset_9000")
// Return error, unable to convert 1234oWorld to integer.

```

## 16.2.2 IODDPreset()

Read IODD file and configure the preset ModbusTCP parameters.

### Syntax 1

```

bool IODDPreset(
 string,
 byte,
 int,
 int
)

```

#### Parameters

**string** IODD file name (read the IODD file stored in the directory [.XmlFiles\IODD](#) at local host)  
**byte** Slave ID  
**int** starting address In  
**int** starting address Out

#### Return

**bool** preset successfully **True**, preset unsuccessfully **False**

#### Note

```

ModbusRTU mbus1 = "COM2",115200
// construct a device with Baudrate 115200, Parity none, DataBits 8, StopBits 1, Timeout 10000ms

mbus1.IODDPreset("OMRON-E2EQ-X3B4-IL2-20170301-IODD1.1.xml", 1, 100, 200)
// load the file, .XmlFiles\IODD\OMRON-E2EQ-X3B4-IL2-20170301-IODD1.1.xml, and add the preset

```

configuration parameters.

// define preset names in the same way as flow projects rules.

## 16.3 modbus\_open()

Open the connection to the Modbus TCP/RTU device.

### Syntax 1

```
bool modbus_open(
 string
)
```

#### Parameters

string TCP/RTU device name

#### Return

bool True Open successfully.

False Open unsuccessfully.

#### Note

```
ModbusTCP mbus1 = "127.0.0.1" // construct a device, with IP 127.0.0.1, Port 502, Timeout
 10000ms
```

```
modbus_open("mbus1") // connect to the device with IP: 127.0.0.1 and port: 502.
```

## 16.4 modbus\_close()

Close the connection from the Modbus TCP/RTU device.

### Syntax 1

```
bool modbus_close(
 string
)
```

#### Parameters

string TCP/RTU device name

#### Return

bool True close successfully

False close unsuccessfully

#### Note

**ModbusTCP** mbus1 = "127.0.0.1" // construct a device, with IP 127.0.0.1, Port 502, Timeout 10000ms

**modbus\_open**("mbus1") // connect to the device with IP: 127.0.0.1 and port: 502.

**modbus\_close**("mbus1") // close the connection.

## 16.5 modbus\_read()

Modbus TCP/RTU read function

### Syntax 1 (TCP/RTU)

```
? modbus_read(
 string,
 string
)
```

#### Parameters

**string** TCP/RTU device name  
**string** The predefined parameters belong to TCP/RTU device

#### Return

? The return data type is decided by the predefined parameters

| Signal Type     | Function Code | Type   | Num Of Addr | Return data type         |
|-----------------|---------------|--------|-------------|--------------------------|
| Digital Output  | 01            | byte   | 1           | byte (H: 1)(L: 0)        |
|                 |               | bool   | 1           | bool (H: true)(L: false) |
| Digital Input   | 02            | byte   | 1           | byte (H: 1)(L: 0)        |
|                 |               | bool   | 1           | bool (H: true)(L: false) |
| Register Output | 03            | byte   | 1           | byte                     |
|                 |               | int16  | 1           | int                      |
|                 |               | int32  | 2           | int                      |
|                 |               | float  | 2           | float                    |
|                 |               | double | 4           | double                   |
|                 |               | string | ?           | string                   |
| Register Input  | 04            | bool   | 1           | bool                     |
|                 |               | byte   | 1           | byte                     |
|                 |               | int16  | 1           | int                      |
|                 |               | int32  | 2           | int                      |
|                 |               | float  | 2           | float                    |
|                 |               | double | 4           | double                   |
|                 |               | string | ?           | string                   |
|                 |               | bool   | 1           | bool                     |

\* According to the Little Endian (CD AB) or Big Endian (AB CD) setting, the int32, float, double data will transformed automatically.

\* string will follows the UTF8 data format transformation (Stop at 0x00)

#### Note

Modbus Address data size

Digital 1 address = 1 bit size  
 Register 1 address = 2 bytes size

If the default values are applied in Preset Setting

preset\_800 DO 800 byte  
 preset\_7202 DI 7202 bool

|             |    |      |        |                    |
|-------------|----|------|--------|--------------------|
| preset_9000 | RO | 9000 | string | 4                  |
| preset_7001 | RI | 7001 | float  | Big-Endian (AB CD) |

```

value = modbus_read("TCP_1", "preset_800") // value = 1 // DO 1 address = 1 bit
value = modbus_read("TCP_1", "preset_7202") // value = true // DI 1 address = 1 bit
value = modbus_read("TCP_1", "preset_9000") // value = ab1234cd // RO 4 address = 8
 bytes size
value = modbus_read("TCP_1", "preset_7001") // value = -314.1593 // RI 2 address = 4 bytes
 size (float)

```

## Syntax 2 (TCP/RTU)

```

byte[] modbus_read(
 string,
 byte,
 string,
 int,
 int
)

```

### Parameters

|        |                     |                                |  |
|--------|---------------------|--------------------------------|--|
| string | TCP/RTU Device Name |                                |  |
| byte   | Slave ID            |                                |  |
| string | Read type           |                                |  |
| DO     | Digital Output      | (FC 01 Read Coil Status)       |  |
| DI     | Digital Input       | (FC 02 Read Input Status)      |  |
| RO     | Register Output     | (FC 03 Read Holding Registers) |  |
| RI     | Register Input      | (FC 04 Read Input Registers)   |  |
| int    | Starting address    |                                |  |
| int    | Data length         |                                |  |

### Return

byte[] The returned byte array from Modbus server  
 \*User defined modbus\_read only follows **Big-Endian (AB CD)** format to read

byte[]

### Note

Modbus Address data size

|          |                          |
|----------|--------------------------|
| Digital  | 1 address = 1 bit size   |
| Register | 1 address = 2 bytes size |

If the user defined values are applied to User Setting as

|            |   |    |      |    |
|------------|---|----|------|----|
| TCP device | 0 | DO | 800  | 4  |
| TCP device | 0 | DI | 7202 | 3  |
| TCP device | 0 | RO | 9000 | 6  |
| TCP device | 0 | RI | 7001 | 12 |
| TCP device | 0 | RI | 7301 | 6  |

```

value = modbus_read("TCP_1", 0, "DO", 800, 4)
// value = {0,0,0,0} // DO 4 address = 4 bit to byte array

```

```
value = modbus_read("TCP_1", 0, "DI", 7202, 3)
 // value = {1,0,0} // DI 3 address = 3 bit to byte array
value = modbus_read("TCP_1", 0, "RO", 9000, 6)
 // value = {0x54,0x65,0x63,0x68,0x6D,0x61,0x6E,0xE9,0x81,0x94,0xE6,0x98}
 // RO 6 address = 12 bytes size
value = modbus_read("TCP_1", 0, "RI", 7001, 12)
 // value =
{0x29,0x30,0x9F,0x4C,0xC3,0x7C,0x99,0x9A,0x44,0x5E,0xEC,0xCD,0x42,0xB4,0x00,0x00,
 0x80,0x00,0x00,0x00,0x00,0x00,0x00,0x00} // RI 12 address = 24 bytes size
value = modbus_read("TCP_1", 0, "RI", 7301, 6)
 // value = {0x07,0xE2,0x00,0x05,0x00,0x12,0x00,0x0F,0x00,0x0A,0x00,0x39}
 // RI 6 address = 12 bytes size
```

## 16.6 modbus\_read\_int16()

Modbus TCP/RTU read function, and transform Modbus address data array to int16 array

### Syntax 1 (TCP/RTU)

```
int[] modbus_read_int16(
 string,
 byte,
 string,
 int,
 int,
 int
)
```

#### Parameters

string TCP/RTU Device Name

byte Slave ID

string Read type

DO Digital Output (FC 01 Read Coil Status)

DI Digital Input (FC 02 Read Input Status)

RO Register Output (FC 03 Read Holding Registers)

RI Register Input (FC 04 Read Input Registers)

int Starting address

int Data length

int Follows Little Endian (CD AB) or Big Endian (AB CD) to transform the address data to int16 array. \*Invalid Parameter. Only support int32, float, double

0 Little Endian

1 Big Endian (Default)

#### Return

int[] The returned int array from Modbus server

### Syntax 2 (TCP/RTU)

```
int[] modbus_read_int16(
 string,
 byte,
 string,
 int,
 int
)
```

#### Note

Similar to Syntax1 with Big Endian (AB CD) setting

**modbus\_read\_int16("TCP\_1", 0, "DI", 7200, 2) => modbus\_read\_int16("TCP\_1", 0, "DI", 7200, 2, 1)**

Modbus Address data size

Digital 1 address = 1 bit size

Register 1 address = 2 bytes size

If the user defined values are applied to User Setting as



|            |   |    |      |    |
|------------|---|----|------|----|
| TCP device | 0 | DO | 800  | 4  |
| TCP device | 0 | DI | 7202 | 3  |
| TCP device | 0 | RO | 9000 | 6  |
| TCP device | 0 | RI | 7001 | 12 |
| TCP device | 0 | RI | 7301 | 6  |

```

value = modbus_read_int16("TCP_1", 0, "DO", 800, 4)
 // byte[] = {0,0,0,0} to int16[] value = {0,0} // byte[0][1] , byte[2][3]
value = modbus_read_int16("TCP_1", 0, "DI", 7202, 3)
 // byte[] = {1,0,0} to int16[] value = {256,0}
 // byte[0][1] , byte[2][3] // Fill up to [3] automatically
value = modbus_read_int16("TCP_1", 0, "RO", 9000, 6)
 // byte[] = {0x54,0x65,0x63,0x68,0x6D,0x61,0x6E,0xE9,0x81,0x94,0xE6,0x98}
 // to int16[] value = {21605,25448,28001,28393,-32364,-6504}
value = modbus_read_int16("TCP_1", 0, "RI", 7001, 12)
 // byte[] =
{0x29,0x30,0x9F,0x4C,0xC3,0x7C,0x99,0x9A,0x44,0x5E,0xEC,0xCD,0x42,0xB4,0x00,0x00,
 0x80,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // to int16[] value = {10544,-24756,-15492,-26214,17502,-4915,17076,0,-32768,0,0,0}
value = modbus_read_int16("TCP_1", 0, "RI", 7301, 6)
 // byte[] = {0x07,0xE2,0x00,0x05,0x00,0x12,0x00,0x0F,0x00,0x31,0x00,0x23}
 // to int16[] value = {2018,5,18,15,49,35}
value = modbus_read_int16("TCP_1", 0, "RI", 7301, 6, 0)
 // byte[] = {0x07,0xE2,0x00,0x05,0x00,0x12,0x00,0x0F,0x00,0x31,0x00,0x23}
 // to int16[] value = {2018,5,18,15,49,35}

```

## 16.7 modbus\_read\_int32()

Modbus TCP/RTU read function, and transform Modbus address data array to int32 array

### Syntax 1 (TCP/RTU)

```
int[] modbus_read_int32(
 string,
 byte,
 string,
 int,
 int,
 int
)
```

#### Parameters

|        |                                                                                                   |                                |  |
|--------|---------------------------------------------------------------------------------------------------|--------------------------------|--|
| string | TCP/RTU DEVICE NAME                                                                               |                                |  |
| byte   | Slave ID                                                                                          |                                |  |
| string | Read type                                                                                         |                                |  |
| DO     | Digital Output                                                                                    | (FC 01 Read Coil Status)       |  |
| DI     | Digital Input                                                                                     | (FC 02 Read Input Status)      |  |
| RO     | Register Output                                                                                   | (FC 03 Read Holding Registers) |  |
| RI     | Register Input                                                                                    | (FC 04 Read Input Registers)   |  |
| int    | Starting address                                                                                  |                                |  |
| int    | Data length                                                                                       |                                |  |
| int    | Follows Little Endian (CD AB) or Big Endian (AB CD) to transform the address data to int32 array. |                                |  |
| 0      | Little Endian                                                                                     |                                |  |
| 1      | Big Endian (Default)                                                                              |                                |  |

#### Return

int[] The returned int array from Modbus server

### Syntax 2 (TCP/RTU)

```
int[] modbus_read_int32(
 string,
 byte,
 string,
 int,
 int
)
```

#### Note

Similar to Syntax1 with Big Endian (AB CD) setting.

**modbus\_read\_int32("TCP\_1", 0, "DI", 7200, 2) => modbus\_read\_int32("TCP\_1", 0, "DI", 7200, 2, 1)**

Modbus Address data size

|          |                          |
|----------|--------------------------|
| Digital  | 1 address = 1 bit size   |
| Register | 1 address = 2 bytes size |

If the user defined values are applied to User Setting as

|            |   |    |     |   |
|------------|---|----|-----|---|
| TCP device | 0 | DO | 800 | 4 |
|------------|---|----|-----|---|

|            |   |    |      |    |
|------------|---|----|------|----|
| TCP device | 0 | DI | 7202 | 3  |
| TCP device | 0 | RO | 9000 | 6  |
| TCP device | 0 | RI | 7001 | 12 |
| TCP device | 0 | RI | 7301 | 6  |

```

value = modbus_read_int32("TCP_1", 0, "DO", 800, 4)
 // byte[] = {0,0,0,0} to int32[] value = {0} // byte[0][1][2][3]
value = modbus_read_int32("TCP_1", 0, "DI", 7202, 3)
 // byte[] = {1,0,0} to int32[] value = {16777216}
 // byte[0][1][2][3] // Fill up to [3] automatically.
value = modbus_read_int32("TCP_1", 0, "RO", 9000, 6)
 // byte[] = {0x54,0x65,0x63,0x68,0x6D,0x61,0x6E,0xE9,0x81,0x94,0xE6,0x98}
 // to int32[] value = {1415930728,1835101929,-2120948072}
value = modbus_read_int32("TCP_1", 0, "RI", 7001, 12)
 // byte[] =
 {0x29,0x30,0x9F,0x4C,0xC3,0x7C,0x99,0x9A,0x44,0x5E,0xEC,0xCD,0x42,0xB4,0x00,0x00,
 0x80,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // to int32[] value = {691052364,-1015244390,1147071693,1119092736,-2147483648,0}
value = modbus_read_int32("TCP_1", 0, "RI", 7301, 6)
 // byte[] = {0x07,0xE2,0x00,0x05,0x00,0x12,0x00,0x0F,0x00,0x31,0x00,0x23}
 // to int32[] value = {132251653,1179663,3211299}
value = modbus_read_int32("TCP_1", 0, "RI", 7301, 6, 0) // byte[2][3][0][1]
 // byte[] = {0x07,0xE2,0x00,0x05,0x00,0x12,0x00,0x0F,0x00,0x31,0x00,0x23}
 // to int32[] value = {0x000507E2,0x000F0012,0x00230031} = {329698,983058,2293809}

```

## 16.8 modbus\_read\_float()

Modbus TCP/RTU read function, and transform Modbus address data array to float array

### Syntax 1 (TCP/RTU)

```
float[] modbus_read_float(
 string,
 byte,
 string,
 int,
 int,
 int
)
```

#### Parameters

|        |                                                                                                   |                                |  |
|--------|---------------------------------------------------------------------------------------------------|--------------------------------|--|
| string | TCP/RTU DEVICE NAME                                                                               |                                |  |
| byte   | Slave ID                                                                                          |                                |  |
| string | Read type                                                                                         |                                |  |
| DO     | Digital Output                                                                                    | (FC 01 Read Coil Status)       |  |
| DI     | Digital Input                                                                                     | (FC 02 Read Input Status)      |  |
| RO     | Register Output                                                                                   | (FC 03 Read Holding Registers) |  |
| RI     | Register Input                                                                                    | (FC 04 Read Input Registers)   |  |
| int    | Starting address                                                                                  |                                |  |
| int    | Data length                                                                                       |                                |  |
| int    | Follows Little Endian (CD AB) or Big Endian (AB CD) to transform the address data to float array. |                                |  |
| 0      | Little Endian                                                                                     |                                |  |
| 1      | Big Endian (Default)                                                                              |                                |  |

#### Return

float[]      The returned float array from Modbus server

### Syntax 2 (TCP/RTU)

```
float[] modbus_read_float(
 string,
 byte,
 string,
 int,
 int
)
```

#### Note

Similar to Syntax1 with Big Endian (AB CD) setting.

**modbus\_read\_float("TCP\_1", 0, "DI", 7200, 2) => modbus\_read\_float("TCP\_1", 0, "DI", 7200, 2, 1)**

Modbus Address data size

|          |                          |
|----------|--------------------------|
| Digital  | 1 address = 1 bit size   |
| Register | 1 address = 2 bytes size |

If the user defined values are applied to User Setting as

|            |   |    |      |    |
|------------|---|----|------|----|
| TCP device | 0 | DO | 800  | 4  |
| TCP device | 0 | DI | 7202 | 3  |
| TCP device | 0 | RO | 9000 | 6  |
| TCP device | 0 | RI | 7001 | 12 |
| TCP device | 0 | RI | 7301 | 6  |

```

value = modbus_read_float("TCP_1", 0, "DO", 800, 4)
 // byte[] = {0,0,0,0} to float[] value = {0} // byte[0][1][2][3]
value = modbus_read_float("TCP_1", 0, "DI", 7202, 3)
 // byte[] = {1,0,0} to float[] value = {2.350989E-38} // byte[0][1][2][3]
 // Fill up to [3] automatically.
value = modbus_read_float("TCP_1", 0, "RO", 9000, 6)
 // byte[] = {0x54,0x65,0x63,0x68,0x6D,0x61,0x6E,0xE9,0x81,0x94,0xE6,0x98}
 // to float[] value = {3.940861E+12,4.360513E+27,-5.46975E-38}
value = modbus_read_float("TCP_1", 0, "RI", 7001, 12)
 // byte[] =
 {0x29,0x30,0x9F,0x4C,0xC3,0x7C,0x99,0x9A,0x44,0x5E,0xEC,0xCD,0x42,0xB4,0x00,0x00,
 0x80,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // to float[] value = {3.921802E-14,-252.6,891.7,90,0,0}
value = modbus_read_float("TCP_1", 0, "RI", 7001, 12, 0) // byte[2][3][0][1]
 // to float[] value =
 {0x9F4C2930,0x999AC37C,0xECCD445E,0x000042B4,0x00008000,0x00000000}
 = {-4.323275E-20,-1.600218E-23,-1.985221E+27,2.392857E-41,4.591775E-
 41,0}
value = modbus_read_float("TCP_1", 0, "RI", 7301, 6)
 // byte[] = {0x07,0xE2,0x00,0x05,0x00,0x12,0x00,0x0F,0x00,0x3A,0x00,0x26}
 // to float[] value = {3.400471E-34,1.65306E-39,5.326512E-39}

```

## 16.9 modbus\_read\_double()

Modbus TCP/RTU read function, and transform Modbus address data array to double array.

### Syntax 1 (TCP/RTU)

```
double[] modbus_read_double(
 string,
 byte,
 string,
 int,
 int,
 int
)
```

#### Parameters

|        |                                                                                                    |                                |  |
|--------|----------------------------------------------------------------------------------------------------|--------------------------------|--|
| string | TCP/RTU DEVICE NAME                                                                                |                                |  |
| byte   | Slave ID                                                                                           |                                |  |
| string | Read type                                                                                          |                                |  |
| DO     | Digital Output                                                                                     | (FC 01 Read Coil Status)       |  |
| DI     | Digital Input                                                                                      | (FC 02 Read Input Status)      |  |
| RO     | Register Output                                                                                    | (FC 03 Read Holding Registers) |  |
| RI     | Register Input                                                                                     | (FC 04 Read Input Registers)   |  |
| int    | Starting address                                                                                   |                                |  |
| int    | Data length                                                                                        |                                |  |
| int    | Follows Little Endian (CD AB) or Big Endian (AB CD) to transform the address data to double array. |                                |  |
| 0      | Little Endian                                                                                      |                                |  |
| 1      | Big Endian (Default)                                                                               |                                |  |

#### Return

double[] The returned double array from Modbus server

### Syntax 2 (TCP/RTU)

```
double[] modbus_read_double(
 string,
 byte,
 string,
 int,
 int
)
```

#### Note

Similar to Syntax1 with Big Endian (AB CD) setting.

**modbus\_read\_double("TCP\_1", 0, "DI", 7200, 2) => modbus\_read\_double("TCP\_1", 0, "DI", 7200, 2, 1)**

Modbus Address data size

|          |                          |
|----------|--------------------------|
| Digital  | 1 address = 1 bit size   |
| Register | 1 address = 2 bytes size |

If the user defined values are applied to User Setting as

|            |   |    |      |    |
|------------|---|----|------|----|
| TCP device | 0 | DO | 800  | 4  |
| TCP device | 0 | DI | 7202 | 3  |
| TCP device | 0 | RO | 9000 | 6  |
| TCP device | 0 | RI | 7001 | 12 |
| TCP device | 0 | RI | 7301 | 6  |

```

value = modbus_read_double("TCP_1", 0, "DO", 800, 4)
 // byte[] = {0,0,0,0} to double[] value = {0} // byte[0][1][2][3][4][5][6][7]
value = modbus_read_double("TCP_1", 0, "DI", 7202, 3)
 // byte[] = {1,0,0} to double[] value = {7.2911220195564E-304}
 // byte[0][1][2][3][4][5][6][7]
value = modbus_read_double("TCP_1", 0, "RO", 9000, 6)
 // byte[] = {0x54,0x65,0x63,0x68,0x6D,0x61,0x6E,0xE9,0x81,0x94,0xE6,0x98}
 // to double[] value = {3.65481260356117E+98,-4.87647898854073E-301}
value = modbus_read_double("TCP_1", 0, "RI", 7001, 12)
 // byte[] =
 {0x29,0x30,0x9F,0x4C,0xC3,0x7C,0x99,0x9A,0x44,0x5E,0xEC,0xCD,0x42,0xB4,0x00,0x00,
 0x80,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // to double[] value = {2.76472410615396E-110,2.2818627604613E+21,0}
value = modbus_read_double("TCP_1", 0, "RI", 7001, 12, 0) // byte[6][7][4][5][2][3][0][1]
 // to double[] value = {0x999AC37C9F4C2930,0x000042B4ECCD445E,0x00000000000008000}
 = {-2.4604103205376E-185,3.62371629877526E-310,1.6189543082926E-319}
value = modbus_read_double("TCP_1", 0, "RI", 7301, 6)
 // byte[] = {0x07,0xE2,0x00,0x05,0x00,0x12,0x00,0x10,0x00,0x0B,0x00,0x29}
 // to double[] value = {1.06475148078395E-270,1.52982527955113E-308}

```

## 16.10 modbus\_read\_string()

Modbus TCP/RTU read function, and convert Modbus address data array to string text in UTF8

### Syntax 1 (TCP/RTU)

```
string modbus_read_string(
 string,
 byte,
 string,
 int,
 int,
 int
)
```

#### Parameters

|        |                                                                                                                                                                                                                               |                 |                                |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------------------------------|
| string | TCP/RTU DEVICE NAME                                                                                                                                                                                                           |                 |                                |
| byte   | Slave ID                                                                                                                                                                                                                      |                 |                                |
| string | Read type                                                                                                                                                                                                                     |                 |                                |
|        | DO                                                                                                                                                                                                                            | Digital Output  | (FC 01 Read Coil Status)       |
|        | DI                                                                                                                                                                                                                            | Digital Input   | (FC 02 Read Input Status)      |
|        | RO                                                                                                                                                                                                                            | Register Output | (FC 03 Read Holding Registers) |
|        | RI                                                                                                                                                                                                                            | Register Input  | (FC 04 Read Input Registers)   |
| int    | Starting address                                                                                                                                                                                                              |                 |                                |
| int    | Data length                                                                                                                                                                                                                   |                 |                                |
| int    | Follows Little Endian (CD AB) or Big Endian (AB CD) to transform the address data to string. *Invalid Parameter. Only support int32, float, double. String follows UTF8 and is sequentially transferred according to address. |                 |                                |
| 0      | Little Endian                                                                                                                                                                                                                 |                 |                                |
| 1      | Big Endian (Default)                                                                                                                                                                                                          |                 |                                |

#### Return

string The returned UTF8 string from Modbus server (Stop at 0x00)

### Syntax 2 (TCP/RTU)

```
string modbus_read_string(
 string,
 byte,
 string,
 int,
 int
)
```

#### Note

Similar to Syntax1 with Big Endian (AB CD) setting.

**modbus\_read\_string**("TCP\_1", 0, "RO", 9000, 2) => **modbus\_read\_string**("TCP\_1", 0, "RO", 9000, 2, 1)

Modbus Address data size

|          |                          |
|----------|--------------------------|
| Digital  | 1 address = 1 bit size   |
| Register | 1 address = 2 bytes size |



If the user defined values are applied to User Setting as

|            |   |    |      |   |
|------------|---|----|------|---|
| TCP device | 0 | RO | 9000 | 1 |
|------------|---|----|------|---|

## 16.11 modbus\_write()

Modbus TCP/RTU write function

### Syntax 1 (TCP/RTU)

```
bool modbus_write(
 string,
 string,
 ?,
 int
)
```

#### Parameters

**string** TCP/RTU Device Name

**string** TCP/RTU The predefined parameters belong to TCP/RTU device

**?** The input data. The predefined parameters will be applied according to the table below.

| Signal Type     | Function Code | Type   | Input type | Input value         |
|-----------------|---------------|--------|------------|---------------------|
| Digital Output  | 05            | byte   | byte       | (H: 1)(L: 0)        |
|                 |               | bool   | bool       | (H: true)(L: false) |
| Register Output | 06            | byte   | byte       |                     |
|                 |               | bool   | bool       |                     |
|                 |               | int16  | int        |                     |
| Register Output | 16            | int32  | int        |                     |
|                 |               | float  | float      |                     |
|                 |               | double | double     |                     |
|                 |               | string | string     |                     |

\* int32, float, double will be transferred with Little Endian (CD AB) or Big Endian (AB CD) according to user's setting.

\* string will be transferred with UTF8 format

\* Writing array value is not supported with predefined parameters. To write with the array value, user defined method should be applied (Syntax 3/4)

**int** The maximum number of addresses to be write, only effective to string type data

**> 0** Valid address length. Write with defined address length

**<= 0** Invalid address length. Write all the data

When this parameter is skipped (As shown in Syntax2), the predefined address length will be applied.

#### Return

**bool True** Write success

**False** Write failed

1. If the input data ? is empty string or array
2. If an error occurred in Modbus communication

### Syntax 2 (TCP/RTU)

```
bool modbus_write(
```

```

string,
string,
?,
)

```

### Note

Similar to Syntax1 with predefined address length to write. If the predefined address length  $\leq 0$ , it will write all the data.

Modbus Address data size

|          |                          |
|----------|--------------------------|
| Digital  | 1 address = 1 bit size   |
| Register | 1 address = 2 bytes size |

If the user defined values are applied to User Setting as

|             |    |      |        |   |
|-------------|----|------|--------|---|
| preset_800  | DO | 800  | bool   |   |
| preset_9000 | RO | 9000 | string | 4 |

```

modbus_write("TCP_1", "preset_800", 1) // write 1 (true)
modbus_write("TCP_1", "preset_800", 0) // write 0 (false)
bool flag = true
modbus_write("TCP_1", "preset_800", flag) // write 1 (true)
modbus_write("TCP_1", "preset_800", false) // write 0 (false)

string ss = "ABCDEFGHJKLMNOPQRST" // With no number of address, the predefined
 // address length, 4, is applied. That is 4 RO = 8 bytes size
 // can be written.
modbus_write("TCP_1", "preset_9000", ss) // write ABCDEFGH // The exceeding part will be
 // skipped
 // With no number of address, the predefined address
 // length, 4, is applied. That is 4 RO = 8 bytes size can be
 // written.
modbus_write("TCP_1", "preset_9000", "1234567") // write 1234567\0 // Use 0x00 to fill up
 // 4 address
 // With address length = 0, write all the data.
 // "09AB123" needs 4 addresses.
modbus_write("TCP_1", "preset_9000", "09AB123", 0) // write 09AB123\0 // Use 0x00 to
 // fill up 4 address
 // With address length = 5, write data in 5
 // addresses. That is 5 RO = 10 bytes size can
 // be wrote.
modbus_write("TCP_1", "preset_9000", "09AB1234", 5) // write 09AB1234 // The input
 // data needs only 4 addresses.

```

### Syntax 3 (TCP/RTU)

```

bool modbus_write(
string,
byte,
string,
int,

```

?,  
int

)

### Parameters

string TCP/RTU DEVICE NAME

byte Slave ID

string Write type

DO Digital Output (FC 05/15 Write Single/Multiple Coil(s))

RO Register Output (FC 06/16 Write Single/Multiple Register(s))

int Starting address

? Input data

| Signal Type     | Function Code | Input ? type | Input value         |
|-----------------|---------------|--------------|---------------------|
| Digital Output  | 05            | byte         | (H: 1)(L: 0)        |
|                 |               | bool         | (H: true)(L: false) |
| Digital Output  | 15            | byte[]       | (H: 1)(L: 0)        |
|                 |               | bool[]       | (H: true)(L: false) |
| Register Output | 06            | byte         |                     |
|                 |               | bool         |                     |
| Register Output | 16            | int          |                     |
|                 |               | float        |                     |
|                 |               | double       |                     |
|                 |               | string       |                     |
|                 |               | byte[]       |                     |
|                 |               | int[]        |                     |
|                 |               | float[]      |                     |
|                 |               | double[]     |                     |
|                 |               | string[]     |                     |
|                 |               | bool[]       |                     |

\*User defined modbus\_write will follows **Big-Endian (AB CD)** format to write

\* Here int means int32. For int16 type data, GetBytes() needs to be applied first to change int16 to byte[]

int The maximum number of addresses to be write, only effective to string type data

> 0 Valid address length. Write with defined address length

<= 0 Invalid address length. Write all the data

### Return

bool True Write success

False Write failed

1. If the input data ? is empty string or array

2. If an error occurred in Modbus communication

### Syntax 4 (TCP/RTU)

bool modbus\_write(

string,

byte,  
string,  
int,  
?

)

### Note

Similar to Syntax3 with address length  $\leq 0$ , it will write all the data.

**modbus\_write**("TCP\_1", 0, "RO", 9000, bb) => **modbus\_write**("TCP\_1", 0, "RO", 9000, bb, 0)

Modbus Address data size

|          |                          |
|----------|--------------------------|
| Digital  | 1 address = 1 bit size   |
| Register | 1 address = 2 bytes size |

If the user defined values are applied to User Setting as

|            |   |    |      |    |
|------------|---|----|------|----|
| TCP device | 0 | DO | 800  | 4  |
| TCP device | 0 | RO | 9000 | 12 |

byte[] bb = {10, 20, 30}

**modbus\_write**("TCP\_1", 0, "DO", 800, bb) // write 1,1,1

// Zero value, write 0. Non-zero value, write 1.

**modbus\_write**("TCP\_1", 0, "DO", 800, bb, 2) // write 1,1

// Address number = 2, only write 2 addresses.

**modbus\_write**("TCP\_1", 0, "DO", 800, true) // write 1

int i = 10000

**modbus\_write**("TCP\_1", 0, "RO", 9000, i) // write 0x00,0x00,0x27,0x10

// with int32 BigEndian (AB CD) default

bb = **GetBytes**(i, 0, 1)

// bb = {0x10,0x27}

// transfer to int16 LittleEndian (CD AB)

**modbus\_write**("TCP\_1", 0, "RO", 9000, bb) // write 0x10,0x27

string[] n = {"ABC", "12", "34"}

**modbus\_write**("TCP\_1", 0, "RO", 9000, n, 2) // write ABC1

// Only 2 addresses available, the exceeding values cannot be applied.

**modbus\_write**("TCP\_1", 0, "RO", 9000, n, 5) // write ABC12340

// The data needs 4 addresses (0xAB 0xC1 0x23 - 0x40)

## 17. TM Ethernet Slave

Ethernet Slave comes with functions established with Socket TCP based on the framework of client/server connections. Once enabled, the robot establishes a Socket TCP Listener Serve to send the robot status and data to all of the connected clients or receive the contents from the clients to execute the respective instructions and update the respective information periodically and promptly without the real-time guarantee.

Like the Modbus Slave, the Ethernet Slave will automatically start on its own after power cycling if it was previously set to **Enable**. The established IP and Port will be shown in the notice window.

IP            TMflow → System → Network → IP Address  
Port        5891

### 17.1 GUI Setting

Modbus Slave Status Enable

Ethernet Slave

Data Table Setting

IP Filter:

|     |     |   |     |   |     |                                     |                  |  |
|-----|-----|---|-----|---|-----|-------------------------------------|------------------|--|
|     |     |   |     | ~ |     | <input type="checkbox"/>            | Write Permission |  |
| 192 | 168 | 1 | 100 | ~ | 200 | <input checked="" type="checkbox"/> | Write Permission |  |
| 192 | 168 | 2 | 100 | ~ | 200 | <input type="checkbox"/>            | Write Permission |  |

**Enable/Disable** Enable or disable Ethernet Slave

**IP Filter** IP whitelist

Sets ranges for eligible IP addresses that are allowed to connect to the Ethernet Slave. If no filters are set, all devices on the network can connect to the Ethernet Slave.

**Write Permission** If checked, allows devices within the corresponding IP range to write to the Ethernet Server with TMSVR commands.

For example, setting IP Filter.

Group 1 192.168.1.100 ~ 200 denotes IP 192.168.1.100, 192.168.1.101, ... , and 192.168.1.200 are available for connections.

Group 2 192.168.2.100 ~ 200 denotes IP 192.168.2.100, 192.168.2.101, ... , and 192.168.2.200 are available for connections.

If the IP address of the client is not in the range of the IPs listed above, it rejects the client to connect.

Group 1, 192.168.1.100 ~ 200, has permission to write, so clients connecting within this group range sending data to Ethernet Slave makes Ethernet Slave write data. Group 2, 192.168.2.100 ~ 200, does not have permission to write. When sending data to Ethernet Slave, it does not write data and will respond with the error code of write permission.

## 17.2 Data Table

Users can use the items listed the Data Table to customize the required data content as well as configure the communication protocol to transmit between the Ethernet Slave and clients, and save the settings as a communication file. When the Ethernet Slave is enabled, the data items in the communication file will be established with the relevant data content to the item to send to the connected clients periodically (**no real-time guarantee**). The types of the data format is defined by the settings in the communication file. The client can send data to the server with any type of the supported data formats.

In the protocol, the types of the supported data format are:

|        |                                                              |
|--------|--------------------------------------------------------------|
| BINARY | Binary format, converse in Byte array (Little Endian / UTF8) |
| STRING | String format, similar to the external command format        |
| JSON   | JSON string format                                           |

Receive/Send Data Table Setting

The interface consists of a main table on the left and a 'Transmit File (Data Table)' window on the right. The main table has three tabs: 'Predefined', 'User defined', and 'Global Variable'. The 'Predefined' tab is active, showing a table with columns 'Item', 'Description', 'Data Type', and 'Data'. The 'Transmit File (Data Table)' window has a 'Setting Transmit File Name' field with 'TransTable' and an 'Open' button. It also has a 'Communicate Mode' dropdown set to 'STRING'. Below this is a table with columns 'Item', 'Description', and 'Data Type'. The table contains items like 'TCP\_Value' (float), 'Ctrl\_DO0' (byte), 'Ctrl\_DO1' (byte), and 'g\_ss' (string[]). At the bottom of the window are 'Delete All' and 'Save' buttons.

The configuration interface is a left-to-right mechanism. Users can add items at the left to the communication data table at the right and adjust the arrangement order of each item in the communication data table at the right. In the content to send, there will always be an item, **Robot\_Link**, predefined in Ethernet Slave as a type of byte with the attribute of read-only to denote whether to connect to the robot.

### 1. Predefined

Items and settings in this section are defined by TMflow, and the data content of the items is updated by TMflow. The defined items are the general statuses of the robot, such as the coordinates of the robot, the state of the project, the state of the electrical control box, or the IO related statuses, such as digital input / digital output, analog input / analog output.

### 2. User defined

Items and settings in this section are defined by TMflow users for project programs to read / write item data through the Expression Editor or for external users to read / write item data through the TMSVR commands over a TCP/IP connection. With the user defined tab, the project programs can work with external communication devices as a data exchange protocol. The item list in the user-defined tab can be saved as a custom-defined file to be edited or exchanged data in the future.

### 3. Global Variable

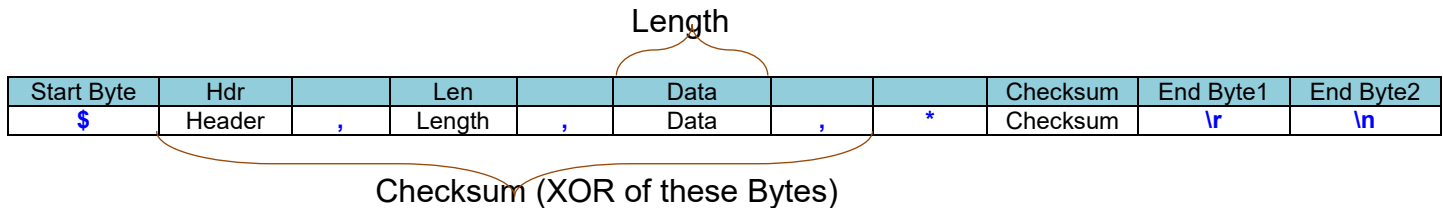
In the global variable tab, the variable list created by the TMflow users provides a way to directly use the variable name for read / write operations in the project programming, and the external communication devices can read / write global variables with the communication protocol.

#### 4. **TransmitFile**

When the Ethernet Slave starts up, it will create the data content of the associated item by the list of items in the selected communication file and send the content to the connected client according to a fixed cycle. The Data section format will follow the specifications defined in the communication file. The Ext?\_DO\_Mask and Ext?\_AO\_Mask in the Predefined System Definition Area are primarily used by the client to transmit data to the server along with their associated Ext?\_DO and Ext?\_AO. Since they are intended solely for client-to-server communication, Ext?\_DO\_Mask and Ext?\_AO\_Mask are not in the periodic communication data table.



## 17.3 Communication Protocol



| Name       | Size | ASCII | HEX  | Description                         |
|------------|------|-------|------|-------------------------------------|
| Start Byte | 1    | \$    | 0x24 | Start Byte for Communication        |
| Header     | X    |       |      | Header for Communication            |
| Separator  | 1    | ,     | 0x2C | Separator between Header and Length |
| Length     | Y    |       |      | Length of Data                      |
| Separator  | 1    | ,     | 0x2C | Separator between Length and Data   |
| Data       | Z    |       |      | Communication Data                  |
| Separator  | 1    | ,     | 0x2C | Separator between Data and Checksum |
| Sign       | 1    | *     | 0x2A | Begin Sign of Checksum              |
| Checksum   | 2    |       |      | Checksum of Communication           |
| End Byte 1 | 1    | \r    | 0x0D |                                     |
| End Byte 2 | 1    | \n    | 0x0A | End Byte of Communication           |

\*Using the same communication protocol with external commands.

### 1. Header

Defines the purpose of communication packets. Different headers come with different definitions of communication packets and data.

- **TMSVR** Defines the function of TM Ethernet Slave
- **CPERR** Defines the errors of the communication packets such as packer errors, checksum errors, header errors, and so on.

\*Using the same content definitions with CPERR in external commands.

### 2. Length

The length indicates the length in the UTF8 bytes occupied by Data. Users can use decimal, hexadecimal, or binary format. The maximum length is 32 bits.

For example,

```
$TMSVR,100,Data,*CS\r\n // 100 in decimal indicates the data length is 100 bytes
$TMSVR,0x100,Data,*CS\r\n // 0x100 in hexadecimal indicates the data length is 256 bytes
$TMSVR,0b100,Data,*CS\r\n // 0b100 in binary indicates the data length is 4 bytes
$TMSVR,8,1,達明,*CS\r\n // indicates the length of Data, 1,達明, is 8 bytes (UTF8)
```

### 3. Data

The content of the communication packet can support any character (including 0x00 .. 0xFF and uses UTF8 encoding), and the data length determines by Length. The purpose and description defined in Data must be defined by the header.

#### 4. Checksum

The checksum of the communication packet. The calculation method is XOR (exclusive OR). The calculation range is all Bytes between \$ and \* (excluding \$ and \*) as shown below.

\$TMSVR,100,Data,\*CS\r\n

Checksum = Byte[1] ^ Byte[2] ... ^ Byte[N-6]

The checksum format is set to 2 bytes in hexadecimal (but not 0x), such as

\$TMSVR,5,10,OK,\*7E

CS = 0x54 ^ 0x4D ^ 0x53 ^ 0x56 ^ 0x52 ^ 0x2C ^ 0x35 ^ 0x2C ^ 0x31 ^ 0x30 ^ 0x2C ^ 0x4F ^ 0x4B ^ 0x2C = 0x7E

CS = 7E (0x37 0x45)

## 17.4 TMSVR

| Start Byte | Hdr   |   | Len    |   | Data |   |   | Checksum | End Byte1 | End Byte2 |
|------------|-------|---|--------|---|------|---|---|----------|-----------|-----------|
| \$         | TMSVR | , | Length | , | Data | , | * | Checksum | \r        | \n        |

| ID             |   | Mode                 |   | Content        |
|----------------|---|----------------------|---|----------------|
| Transaction ID | , | 0/1/2/3/<br>11/12/13 | , | Item and Value |

TMSVR is defined as the TM Ethernet Slave protocol. The Data section of the packet is further divided into three segments, ID (Transaction ID), Mode (Content Mode), and Content (Item and Value), separated with commas and described below.

- ID** The transaction number expressed in any alphanumeric characters. (Reports the CPERR 04 error if a non-alphanumeric byte is encountered.) When used as a communication packet response, it is a transaction number that identifies which group of commands to respond.
- ,
- Mode** The mode as the format of the data content
  - 0 Indicates the server responds to the client command in string format.
  - 1 Indicates the content data type in binary format
  - 2 Indicates the content data type in string format
  - 3 Indicates the content data type in JSON format
  - 11 Indicates the content data type in binary format (Request read)
  - 12 Indicates the content data type in string format (Request read)
  - 13 Indicates the content data type in JSON format (Request read)
  - 1/2/3 are for client write to server and client read from server. The client read from server is that the server sends contents to the connected client periodically.
  - 11/12/13 are for client read from server with request read, which is the client sends read request for the item and the server responds with the item value to the client.
- ,
- Content** The data content. Formatted by the mode definition.

### Note

TMSVR command is for the client and the server to communicate in both directions. Under normal circumstances, the server will broadcast the data items from the Transmit and User Defined communication files to the connected clients periodically. When the server sends to the client, the data is sent to the client to read from the server with no response to the server required. When the client sends to the server, the data is received by the server from the client to write with response to the client required.

- ID Transaction Number

When the server sends data, cycles from 0 to 9 with each iteration. When the client sends data to the server, the transaction number can be in any alphanumeric characters customized at the client side. If the communication packet format is checked and correct, the server will reply the client with the command processing status by the transaction number in the packet.

- Mode Format and Mode for the Data Content

When the Mode is 1/2/3, it can be the function of client write to server or client read from server, in which client read from server means the server sends content to the connected client periodically. If client write to server, it means the clients sends data writing to the server. When the client sends an item to write, not only can one item be written, but multiple items at a time, and the server will send the command processing status according to the format of Mode 0.

When the Mode is 11/12/13, the request read method is used. The client sends an item request and the server responds to the item value. The purpose is to get the other item data not in the communication data table sent periodically. When the client sends a request to read, not only can one item be retrieved, but multiple items at a time. When the request succeeds, the server will send the item value to the client associated to the format of Mode 11/12/13. However, if the request fails, for example, the item name does not exist, the server will send the command processing status according to the format of Mode 0.

Whether data to write or request to read, the items can be all in the Predefined area but not limited to saving in the data table. However, in the Userdefined area and the GlobalVariable area, for the sake of custom definitions, it still need to be saved in the data table to proceed with writing or reading.

- Writing or Reading Confirmation

When the client sends data writing or reading requests to the server, no matter what format the data table used to proceed sending periodically, the servers supports to the data format of Mode 1/2/3/11/12/13 concurrently and **checks whether all the criteria are correct before performing the request. If there is any error with the write command, no request will be performed.** The criteria to write for inspection are:

1. The validity of the mode as the format of the data content
2. The connected client's write permissions based on the IP Filter.
3. The data content matches to the mode.
4. The item to write or read exists.
5. The attribute of the item to write is not read only.
6. The robot is in the appropriate mode (M/A).
7. The written data matches the data type of each item.

### 17.4.1 Mode = 0 (the status the server responds to the client command processing)

After the server receives and processes a write command from a client, it will respond with another TMSVR command with Mode 0. The details for Mode 0 are as follows.

Data

| ID             |   | Mode |   | Error Code |   | Error Description |
|----------------|---|------|---|------------|---|-------------------|
| Transaction ID | , | 0    | , | 00 .. 07   | , |                   |

Transaction ID    Defined while the client sends the command for the server to reply with.

Mode              0 for the server to respond to the client

Error Code    Error code definitions. **Fixed as 2 bytes and in hexadecimal (but not 0x)**

|                   |                                                                                                                                                                                         |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 00                | Correct writing. No error.                                                                                                                                                              |
| 01                | The communication format or mode is not supported. (Ex. Mode = 99)                                                                                                                      |
| 02                | The connected client is not permitted to write. (IP filter without write permission)                                                                                                    |
| 03                | The communication format and the data content format are mismatched.<br>(Ex. Mode = 3, but the data content is not in JSON format)                                                      |
| 04                | Item to write or read does not exist.                                                                                                                                                   |
| 05                | Unable to write to read-only items.                                                                                                                                                     |
| 06                | Incorrect M/A mode while writing.                                                                                                                                                       |
| 07                | Values to write mismatches with the configured type or the size.                                                                                                                        |
| Error Description | Error description, following the error code.                                                                                                                                            |
| 00                | OK                                                                                                                                                                                      |
| 01                | NotSupport                                                                                                                                                                              |
| 02                | WritePermission                                                                                                                                                                         |
| 03                | InvalidData                                                                                                                                                                             |
| 04                | NotExist;XXX // ;XXX denotes which data item                                                                                                                                            |
| 05                | ReadOnly;XXX                                                                                                                                                                            |
| 06                | ModeError;XXX                                                                                                                                                                           |
| 07                | ValueError;XXX                                                                                                                                                                          |
| <                 | \$TMSVR,15,S0,2,Ctrl_DO0=1,*76\r\n // transaction ID S0, string format, set Ctrl_DO0=1                                                                                                  |
| >                 | \$TMSVR,10,S0,0,00,OK,*18\r\n<br>// server responds transaction ID S0, mode 0, error code 00, correct writing                                                                           |
| <                 | \$TMSVR,16,S1,99,Ctrl_DO0=1,*46\r\n // transaction ID S1, mode 99                                                                                                                       |
| >                 | \$TMSVR,18,S1,0,01,NotSupport,*0E\r\n<br>// server responds transaction ID S1, mode 0, error code 01, mode not support                                                                  |
| <                 | \$TMSVR,15,S2,2,Ctrl_DO0=1,*74\r\n // transaction ID S2, string format, set Ctrl_DO0=1                                                                                                  |
| >                 | \$TMSVR,23,S2,0,02,WritePermission,*6A\r\n<br>// server responds transaction ID S2, mode 0, error code 02, the connected client is not granted with write permission.                   |
| <                 | \$TMSVR,15,S3,3,Ctrl_DO0=1,*74\r\n // transaction ID S3, JSON format, set Ctrl_DO0=1                                                                                                    |
| >                 | \$TMSVR,19,S3,0,03,InvalidData,*74\r\n<br>// server responds transaction ID S3, mode 0, error code 03, JSON format, data format (JSON) mismatched with the content data format (STRING) |
| <                 | \$TMSVR,16,S4,2,Ctrl_DO32=1,*40\r\n // transaction ID S4, string format, set Ctrl_DO32=1                                                                                                |
| >                 | \$TMSVR,26,S4,0,04,NotExist;Ctrl_DO32,*58\r\n<br>// server responds transaction ID S4, mode 0, error code 04, item Ctrl_DO32 does not exist.                                            |
| <                 | \$TMSVR,17,S5,2,Robot_Link=1,*07\r\n<br>// transaction ID S5, string format, set Robot_Link=1                                                                                           |
| >                 | \$TMSVR,27,S5,0,05,ReadOnly;Robot_Link,*1E\r\n<br>// server responds transaction ID S5, mode 0, error code 05, the item Robot_Link is read only.                                        |

Supposed the user defined Item: adata, Type: int, Size: 4, and Write: Auto.

```
< $TMSVR,20,S6,2,adata={1,2,3,4},*55\r\n
 // transaction ID S6, string format, set adata={1,2,3,4}
> $TMSVR,23,S6,0,06,ModeError;adata,*2D\r\n
 // server responds transaction ID S6, mode 0, error code 06, M/A mode mismatched while writing
 (suppose it Manual Mode while writing).

< $TMSVR,18,S7,2,adata={1,2,3},*47\r\n // transaction ID S7, string format, set adata={1,2,3}
> $TMSVR,24,S7,0,07,ValueError;adata,*42\r\n
 // server responds transaction ID S7, mode 0, error code 07, writing values and data size or type
 mismatched. (the configured size is 4, but there is only 3 to write.)
```

## 17.4.2 Mode = 1 BINARY

The data content is transmitted in binary mode by converting the data item name with the Little Endian value and the value with UTF8 to a byte array accordingly. The format is shown as below.

Data

| ID             |   | Mode |   | Content        |
|----------------|---|------|---|----------------|
| Transaction ID | , | 1    | , | Item and Value |

|                                                |                     |                                                 |                                      |     |
|------------------------------------------------|---------------------|-------------------------------------------------|--------------------------------------|-----|
| <b>Length of Item</b><br>2 bytes Little Endian | <b>Item</b><br>UTF8 | <b>Length of Value</b><br>2 bytes Little Endian | <b>Value</b><br>Little Endian / UTF8 | ... |
|------------------------------------------------|---------------------|-------------------------------------------------|--------------------------------------|-----|

Length of Item      2 bytes in Little Endian, value from 0 to 65535 indicating the length of the item that follows

Item                item name

Length of Value    2 bytes in Little Endian), value from 0 to 65535 indicating the length of the data value that follows

Value                data value

Suppose taking Check TCP\_Value float[] and Ctrl\_DO0 byte as the communication data and transmitting in binary mode.

| Predefined     |                       |           |             | User defined |             |           |             | Global Variable |             |           |             |
|----------------|-----------------------|-----------|-------------|--------------|-------------|-----------|-------------|-----------------|-------------|-----------|-------------|
| Item           | Description           | Data Type | Data Length | Item         | Description | Data Type | Data Length | Item            | Description | Data Type | Data Length |
| Get_UI_Control | Get UI Control or Not | bool      | 1           |              |             |           |             |                 |             |           |             |
| Get_Control    | Get UI Control or Not | bool      | 1           |              |             |           |             |                 |             |           |             |
| Robot_Error    | Error or Not          | bool      | 1           |              |             |           |             |                 |             |           |             |
| Error_Code     | Last Error Code       | int       | 1           |              |             |           |             |                 |             |           |             |

| Transmit File (Data Table) |                  |                  |             |
|----------------------------|------------------|------------------|-------------|
| Setting Transmit File Name |                  | Communicate Mode |             |
| TransTable                 | Open             | BINARY           |             |
| Item                       | Description      | Data Type        | Data Length |
| TCP_Value                  | TCP Value        | float            | 6           |
| Ctrl_DO0                   | Digital Output 0 | byte             | 1           |

```
> 24 54 4D 53 56 52 2C // $TMSVR, // Header
 36 39 2C // 69, // Length
 30 2C 31 2C // 0,1, // transaction ID 0, mode 1, binary
 0A 00 52 6F 62 6F 74 5F 4C 69 6E 6B 01 00 00 // Robot_Link=0
 // The name occupied 10 bytes, the value, 1 byte
 09 00 54 43 50 5F 56 61 6C 75 65 18 00 00 00 80 3F 00 00 80 3F 00 00 80 3F CD
```

```

CC CC 3D CD CC 4C 3E CD CC CC 3D // TCP_Value={1,1,1,0.1,0.2,0.1}
// The name occupied 9 bytes, the value, 24 bytes
08 00 43 74 72 6C 5F 44 4F 30 01 00 00 // Ctrl_DO0=0
// The name occupied 8 bytes, the value, 1 byte
2C 2A 39 36 0D 0A // ,*96\r\n // Checksum

< 24 54 4D 53 56 52 2C // $TMSVR, // Header
31 38 2C // 18, // Length
54 31 2C 31 2C // T1,1, // transaction ID T1, mode 1, binary
08 00 43 74 72 6C 5F 44 4F 30 01 00 01 // Ctrl_DO0=1
// The name occupied 8 bytes, the value, 1 byte
2C 2A 37 41 0D 0A // ,*7A\r\n // Checksum
> $TMSVR,10,T1,0,00,OK,*1E\r\n
// server responds to ID T1, mode 1, error code 00, correct writing

```

Once the data type of the item to send is string [], two bytes, 0x00 0x00, are inserted between the string elements as the separators.

```

> 24 54 4D 53 56 52 2C // $TMSVR, // Header
39 30 2C // 90, // Length
30 2C 31 2C // 0,1 // transaction ID 0, mode 1, binary
0A 00 52 6F 62 6F 74 5F 4C 69 6E 6B 01 00 00 // Robot_Link=0
//The name occupied 10 bytes, the value, 1 byte
09 00 54 43 50 5F 56 61 6C 75 65 18 00 00 00 80 3F 00 00 80 3F 00 00 80 3F CD
CC CC 3D CD CC 4C 3E CD CC CC 3D // TCP_Value={1,1,1,0.1,0.2,0.1}
//The name occupied 9 bytes, the value, 24 bytes
08 00 43 74 72 6C 5F 44 4F 30 01 00 01 // Ctrl_DO0=1
// The name occupied 8 bytes, the value, 1 byte
04 00 67 5F 73 73 0D 00 48 69 00 00 54 4D 00 00 52 6F 62 6F 74
// g_ss={"Hi", "TM", "Robot"}
// The name occupied 4 bytes, the value, 13 bytes
2C 2A 44 43 0D 0A // ,*DC\r\n // Checksum

```

Also, if the data type of the item to receive is string [], when converting to a byte array, two bytes, 00 00, are inserted between the string elements as the separators.

```

< 24 54 4D 53 56 52 2C // $TMSVR, // Header
32 35 2C // 25, // Length
54 32 2C 31 2C // T2,1, // transaction ID T2, mode 1, binary
04 00 67 5F 73 73 0C 00 48 65 6C 6C 6F 00 00 57 6F 72 6C 64
// g_ss={"Hello", "World"} // The name occupied 4 bytes, the value, 12 bytes
2C 2A 30 32 0D 0A // ,*02\r\n // Checksum
> $TMSVR,10,T2,0,00,OK,*1D\r\n
//server responds to ID T2, mode 0, error code 00, correct writing

```

## 17.4.3 Mode = 2 STRING

The data content is transmitted as a string with the name and value of the data item in the Script string of an external command. The format is shown as below.

Data

| ID             |   | Mode |   | Content        |
|----------------|---|------|---|----------------|
| Transaction ID | , | 2    | , | Item and Value |

|             |   |              |      |     |
|-------------|---|--------------|------|-----|
| <i>Item</i> | = | <i>Value</i> | \r\n | ... |
|-------------|---|--------------|------|-----|

Item                      item name  
 =                          equal  
 Value                    data value  
 \r\n                      symbol of carriage return as required if there is an item up next for separation.

Suppose taking Check TCP\_Value float[] and Ctrl\_DO0 byte, Ctrl\_DO1 byte, g\_ss string[] as the communication data and transmitting in string mode.

| Predefined     |                       | User defined |             | Global Variable |  |
|----------------|-----------------------|--------------|-------------|-----------------|--|
| Item           | Description           | Data Type    | Data Length |                 |  |
| Get_UI_Control | Get UI Control or Not | bool         | 1           |                 |  |
| Get_Control    | Get UI Control or Not | bool         | 1           |                 |  |
| Robot_Error    | Error or Not          | bool         | 1           |                 |  |
| Error_Code     | Last Error Code       | int          | 1           |                 |  |
| Error_Time     | Last Error Time       | string       | 1           |                 |  |
| Camera_Light   | Light                 | byte         | 1           |                 |  |

Transmit File (Data Table)

Setting Transmit File Name: TransTable Open Communicate Mode: STRING

| Item      | Description      | Data Type | Data Length |
|-----------|------------------|-----------|-------------|
| TCP_Value | TCP Value        | float     | 6           |
| Ctrl_DO0  | Digital Output 0 | byte      | 1           |
| Ctrl_DO1  | Digital Output 1 | byte      | 1           |
| g_ss      |                  | string[]  | 0           |

```

> $TMSVR,9,2,Robot_Link=0\r\n // Robot_Link=0 // transaction ID 9, mode 2, string
TCP_Value={1,1,1,0.1,0.2,0.1}\r\n // TCP_Value={1,1,1,0.1,0.2,0.1}
Ctrl_DO0=1\r\n // Ctrl_DO0=1
Ctrl_DO1=0\r\n // Ctrl_DO1=0
g_ss={"Hi","TM","Robot"},*77\r\n // g_ss={"Hi","TM","Robot"}

< $TMSVR,15,T2,2,Ctrl_DO0=0\r\n // set Ctrl_DO0=0 // transaction ID T2, mode 2, string
Ctrl_DO1=1,*34\r\n // set Ctrl_DO1=1

> $TMSVR,10,T2,0,00,OK,*1D\r\n
// server responds to ID T2, mode 0, error code 00, correct writing

```



## 17.4.4 Mode = 3 JSON

The data content is transmitted as a JSON string with the name and value of the data item serialized in the JSON format as shown below.

Data

| ID             |   | Mode |   | Content        |
|----------------|---|------|---|----------------|
| Transaction ID | , | 3    | , | Item and Value |

Item  
Value

item name  
data value

```
public class TMSVRJsonData
{
 public string Item;
 public object Value;
}
```

\*[] array is in use when it comes it multiple items.

Suppose taking TCP\_Value float[] and Ctrl\_DO0 byte, Ctrl\_DO1 byte, g\_ss string[] as the communication data and transmitting in JSON mode.

| Predefined     | User defined          | Global Variable |
|----------------|-----------------------|-----------------|
| Item           | Description           | Data Type       |
| Get_UI_Control | Get UI Control or Not | bool            |
| Get_Control    | Get UI Control or Not | bool            |
| Robot_Error    | Error or Not          | bool            |
| Error_Code     | Last Error Code       | int             |
| Error_Time     | Last Error Time       | string          |
| Camera_Light   | Light                 | byte            |

Transmit File (Data Table)

Setting Transmit File Name

TransTable

Open

Communicate Mode

JSON

| Item      | Description      | Data Type | Data Le |
|-----------|------------------|-----------|---------|
| TCP_Value | TCP Value        | float     | 6       |
| Ctrl_DO0  | Digital Output 0 | byte      | 1       |
| Ctrl_DO1  | Digital Output 1 | byte      | 1       |
| g_ss      |                  | string[]  | 0       |

```
> $TMSVR,196,5,3,{"Item":"Robot_Link","Value":0}, // Robot_Link=0
// transaction ID 5, mode 3, JSON
{"Item":"TCP_Value","Value":[1.0,1.0,1.0,0.1,0.2,0.1]},
// TCP_Value={1,1,1,0.1,0.2,0.1}
{"Item":"Ctrl_DO0","Value":0}, // Ctrl_DO0=0
{"Item":"Ctrl_DO1","Value":0}, // Ctrl_DO1=0
{"Item":"g_ss","Value":["Hi","TM","Robot"]},*3A\r\n
// g_ss={"Hi","TM","Robot"}

< $TMSVR,113,T9,3,{"Item":"Ctrl_DO0","Value":1}, // Ctrl_DO0=1
{"Item":"Ctrl_DO1","Value":0}, // Ctrl_DO1=0
{"Item":"g_ss","Value":["Hello","TM","Robot"]},*7C\r\n
// g_ss={"Hello","TM","Robot"}

> $TMSVR,10,T9,0,00,OK,*16\r\n
// server responds to ID T9, mode 0, error code 0, correct writing
```

### 17.4.5 Mode = 11 BINARY (Request read)

The data content is transmitted in binary mode by converting the data item name with the Little Endian value and the value with UTF8 to a byte array accordingly. The format is shown as below.

Data (client to server)

| ID             |   | Mode |   | Content |
|----------------|---|------|---|---------|
| Transaction ID | , | 11   | , | Item    |

|                                                |                     |     |                                                                        |
|------------------------------------------------|---------------------|-----|------------------------------------------------------------------------|
| <b>Length of Item</b><br>2 bytes Little Endian | <b>Item</b><br>UTF8 | ... | The difference of the read request from Mode = 1 is no value required. |
|------------------------------------------------|---------------------|-----|------------------------------------------------------------------------|

Length of Item      2 bytes in Little Endian, value from 0 to 65535 indicating the length of the item that follows

Item                      Item name

Suppose taking TCP\_Value float[] and Ctrl\_DO0 byte as the communication data and transmitting in binary mode.

| Predefined     |                       | User defined |             | Global Variable |  |
|----------------|-----------------------|--------------|-------------|-----------------|--|
| Item           | Description           | Data Type    | Data Length |                 |  |
| Get_UI_Control | Get UI Control or Not | bool         | 1           |                 |  |
| Get_Control    | Get UI Control or Not | bool         | 1           |                 |  |
| Robot_Error    | Error or Not          | bool         | 1           |                 |  |
| Error_Code     | Last Error Code       | int          | 1           |                 |  |

| Transmit File (Data Table) |                  |           |             | Communicate Mode |
|----------------------------|------------------|-----------|-------------|------------------|
| Setting Transmit File Name |                  |           |             |                  |
| TransTable                 |                  | Open      |             | BINARY           |
| Item                       | Description      | Data Type | Data Length |                  |
| TCP_Value                  | TCP Value        | float     | 6           |                  |
| Ctrl_DO0                   | Digital Output 0 | byte      | 1           |                  |

server periodical delivery

```
> 24 54 4D 53 56 52 2C // $TMSVR, // Header
 36 39 2C // 69, // Length
 30 2C 31 2C // 0,1, // transaction ID 0, mode 1, binary
 0A 00 52 6F 62 6F 74 5F 4C 69 6E 6B 01 00 00 // Robot_Link=0
// The name occupied 10 bytes, the value, 1 byte
 09 00 54 43 50 5F 56 61 6C 75 65 18 00 00 00 80 3F 00 00 80 3F 00 00 80 3F CD
 CC CC 3D CD CC 4C 3E CD CC CC 3D // TCP_Value={1,1,1,0.1,0.2,0.1}
// The name occupied 9 bytes, the value, 24 bytes
 08 00 43 74 72 6C 5F 44 4F 30 01 00 00 // Ctrl_DO0=0
// The name occupied 8 bytes, the value, 1 byte
 2C 2A 39 36 0D 0A //,*96\r\n // Checksum
```

client requested to read

```
< 24 54 4D 53 56 52 2C // $TMSVR, // Header
 32 36 2C // 26, // Length
 51 31 2C 31 31 2C // Q1,11, // transaction ID Q1, mode 11 binary (Request read)
 08 00 43 74 72 6C 5F 44 4F 30 // Ctrl_DO0 // The name occupied 8 bytes
 08 00 54 43 50 5F 4D 61 73 73 // TCP_Mass // The name occupied 8 bytes
 2C 2A 37 46 0D 0A //,*7F\r\n // Checksum
```

server replied with the item value

```
> 24 54 4D 53 56 52 2C // $TMSVR, // Header
```

```

33 35 2C // 35, // Length
51 31 2C 31 31 2C // Q1,11, // server responds to ID Q1, mode 11 binary
08 00 43 74 72 6C 5F 44 4F 30 01 00 00
 // Ctrl_DO0=0 // The name occupied 8 bytes, the value, 1 byte
08 00 54 43 50 5F 4D 61 73 73 04 00 00 00 00
 // TCP_Mass=0 // The name occupied 8 bytes, the value, 4 byte
2C 2A 37 38 0D 0A // ,*78\r\n // Checksum

```

\* server replied the same content format as Mode = 1 BINARY

client requested to read

```

< 24 54 4D 53 56 52 2C // $TMSVR, // Header
 32 36 2C // 26, // Length
 51 32 2C 31 31 2C // Q2,11, // transaction ID Q1, mode 11 binary (Request read)
 08 00 43 74 72 6C 5F 44 4F 30 // Ctrl_DO0 // The name occupied 8 bytes
 08 00 54 43 50 5F 4D 61 58 58 // TCP_MaXX // The name occupied 8 bytes
 2C 2A 37 43 0D 0A // ,*7C\r\n // Checksum

```

server replied with the item value

```

> $TMSVR,25,Q2,0,04,NotExist;TCP_MaXX,*17\r\n
 // server responds to ID Q2, mode 0, error code 04, item not existed

```

## 17.4.6 Mode = 12 STRING (Request read)

The data content is transmitted as a string with the name and value of the data item in the Script string of an external command. The format is shown as below.

Data (client to server)

| ID             |   | Mode |   | Content        |
|----------------|---|------|---|----------------|
| Transaction ID | , | 12   | , | Item and Value |

|             |             |     |                    |
|-------------|-------------|-----|--------------------|
| <i>Item</i> | <i>\r\n</i> | ... | No Value required. |
|-------------|-------------|-----|--------------------|

Item                      Item name  
\r\n                      The newline characters. Required only as a delimiter if the next item comes.

Suppose taking TCP\_Value float[] and Ctrl\_DO0 byte, Ctrl\_DO1 byte, g\_ss string[] as the communication data and transmitting in STRING mode.

| Predefined     |                       | User defined |             | Global Variable |  |
|----------------|-----------------------|--------------|-------------|-----------------|--|
| Item           | Description           | Data Type    | Data Length |                 |  |
| Get_UI_Control | Get UI Control or Not | bool         | 1           |                 |  |
| Get_Control    | Get UI Control or Not | bool         | 1           |                 |  |
| Robot_Error    | Error or Not          | bool         | 1           |                 |  |
| Error_Code     | Last Error Code       | int          | 1           |                 |  |
| Error_Time     | Last Error Time       | string       | 1           |                 |  |
| Camera_Light   | Light                 | byte         | 1           |                 |  |

Transmit File (Data Table)

Setting Transmit File Name:

Communicate Mode: STRING

| Item      | Description      | Data Type | Data Length |
|-----------|------------------|-----------|-------------|
| TCP_Value | TCP Value        | float     | 6           |
| Ctrl_DO0  | Digital Output 0 | byte      | 1           |
| Ctrl_DO1  | Digital Output 1 | byte      | 1           |
| g_ss      |                  | string[]  | 0           |

server periodical delivery

```
> $TMSVR,97,9,2,Robot_Link=0\r\n // Robot_Link=0 // transaction ID 9, mode 2
TCP_Value={1,1,1,0.1,0.2,0.1}\r\n // TCP_Value={1,1,1,0.1,0.2,0.1}
Ctrl_DO0=1\r\n // Ctrl_DO0=1
Ctrl_DO1=0\r\n // Ctrl_DO1=0
g_ss={"Hi","TM","Robot"},*77\r\n // g_ss={"Hi","TM","Robot"}
```

client requested to read

```
< $TMSVR,28,Q2,12,Robot_Link\r\n // Item Robot_Link
// transaction ID Q2, mode 12 JSON (Request read)
TCP_Mass,*0E\r\n // Item TCP_Mass
```

server replied with the item value

```
> $TMSVR,30,Q2,12,Robot_Link=0\r\n // server responds to ID Q2, mode 12
TCP_Mass=0,*09\r\n
```

\* server replied the same content format as Mode = 2 STRING

### 17.4.7 Mode = 13 JSON (Request read)

The data content is transmitted as a JSON string with the name and value of the data item serialized in the JSON format as shown below.

Data (client to server)

| ID             |   | Mode |   | Content        |
|----------------|---|------|---|----------------|
| Transaction ID | , | 13   | , | Item and Value |

Item Value      Item name Data value

```
public class TMSVRJsonData
{
 public string Item;
 public object Value;
}
```

\* [] array is in use when it comes to multiple items.

\* Shared with Mode = 3 JSON for using the same class for serialization / deserialization, but the Value attribute may not exist

Suppose taking TCP\_Value float[] and Ctrl\_DO0 byte, Ctrl\_DO1 byte, g\_ss string[] as the communication data and transmitting in JSON mode.

| Predefined     |                       | User defined |             | Global Variable |             |
|----------------|-----------------------|--------------|-------------|-----------------|-------------|
| Item           | Description           | Data Type    | Data Length | Item            | Description |
| Get_UI_Control | Get UI Control or Not | bool         | 1           |                 |             |
| Get_Control    | Get UI Control or Not | bool         | 1           |                 |             |
| Robot_Error    | Error or Not          | bool         | 1           |                 |             |
| Error_Code     | Last Error Code       | int          | 1           |                 |             |
| Error_Time     | Last Error Time       | string       | 1           |                 |             |
| Camera_Light   | Light                 | byte         | 1           |                 |             |

Transmit File (Data Table)

Setting Transmit File Name: TransTable  Communicate Mode: JSON

| Item      | Description      | Data Type | Data Length |
|-----------|------------------|-----------|-------------|
| TCP_Value | TCP Value        | float     | 6           |
| Ctrl_DO0  | Digital Output 0 | byte      | 1           |
| Ctrl_DO1  | Digital Output 1 | byte      | 1           |
| g_ss      |                  | string[]  | 0           |

server periodical delivery

```
> $TMSVR,196,5,3,{"Item":"Robot_Link","Value":0}, // Robot_Link=0
// transaction ID 5, mode 3
{"Item":"TCP_Value","Value":[1.0,1.0,1.0,0.1,0.2,0.1]}, // TCP_Value={1,1,1,0.1,0.2,0.1}
{"Item":"Ctrl_DO0","Value":0}, // Ctrl_DO0=0
{"Item":"Ctrl_DO1","Value":0}, // Ctrl_DO1=0
{"Item":"g_ss","Value":["Hi","TM","Robot"]},*3A\r\n // g_ss={"Hi","TM","Robot"}
```

client requested to read

```
< $TMSVR,27,Q3,13,{"Item":"TCP_Mass"},*3C\r\n
// transaction ID Q3, mode 13 JSON (Request read)
```

server replied with the item value

```
> $TMSVR,39,Q3,13,{"Item":"TCP_Mass","Value":0.0},*40\r\n
// server responds to ID Q3, mode 13
```

\* server replied the same content format as Mode = 3 JSON

## 17.5 svr\_read()

Read the item value in the communication data table of Ethernet Slave in the Connection Tab of Robot Setting at the local host.

### Syntax 1

```
? svr_read(
 string
)
```

#### Parameter

string Item name

#### Return

? Return value by set data type

#### Note

Suppose taking TCP\_Value float[], Ctrl\_DO0 byte, Ctrl\_DO1 byte, and g\_ss string[] as the communication data table.

| Predefined     | User defined          | Global Variable |
|----------------|-----------------------|-----------------|
| Item           | Description           | Data Type       |
| Get_UI_Control | Get UI Control or Not | bool            |
| Get_Control    | Get UI Control or Not | bool            |
| Robot_Error    | Error or Not          | bool            |
| Error_Code     | Last Error Code       | int             |
| Error_Time     | Last Error Time       | string          |
| Camera_Light   | Light                 | byte            |

>>

| Transmit File (Data Table) |                  |                  |             |
|----------------------------|------------------|------------------|-------------|
| Setting Transmit File Name |                  | Communicate Mode |             |
| TransTable                 | Open             | STRING           | ▼           |
| Item                       | Description      | Data Type        | Data Length |
| TCP_Value                  | TCP Value        | float            | 6           |
| Ctrl_DO0                   | Digital Output 0 | byte             | 1           |
| Ctrl_DO1                   | Digital Output 1 | byte             | 1           |
| g_ss                       |                  | string[]         | 0           |

```
float[] fva0= svr_read("TCP_Value") // {1, 1, 1, 0.1, 0.2, 0.1}
byte b0 = svr_read("Ctrl_DO0") // 0
byte b1 = svr_read("Ctrl_DO1") // 1
string[]ss = svr_read("g_ss") // {"Hi","TM","Robot"}
ss = g_ss // make the variable to take, g_ss, a variable name directly
byte st = svr_read("Robot_Link") // 0 (Robot disconnected) 1 (Robot connected)
```

```
float[] fva1 = svr_read("TCP_Value") // Report error. Suppose Ethernet Slave is not launched.
float[] fva2 = svr_read("TCP_Value1") // Report error. Item name TCP_Value1 does not exist.
```

```
float[] fva3 = svr_read("g_ff") // Report error.. Item name g_ff does not exist in the
 communication table.
```

// If g\_ff is assumed to exist among the global variables, it cannot be accessed because not it does not load all global variables.

```
float[] fva4 = svr_read("Coord_Base_Flange") // {0.01,-252.6,891.7,90,0,0}
// Although added as the communication data, it's accessible for item name Coord_Base_Flange is in the
system definitions.
```

## 17.6 svr\_write()

Write the item value into the communication data table of Ethernet Slave in the Connection Tab of Robot Setting at the local host.

### Syntax 1

```
bool svr_write(
 string,
 ?
)
```

#### Parameters

string    Item name  
?        Item value

#### Return

bool True      Write successfully  
False      Write failed    Possible causes

1. Item name does not exist.
2. Unable to write the read-only item name.
3. Item value to write mismatched with item data type.

### Note

Suppose taking TCP\_Value float[], Ctrl\_DO0 byte, Ctrl\_DO1 byte, and g\_ss string[] in the communication data table.

| Predefined     |                       | User defined | Global Variable |
|----------------|-----------------------|--------------|-----------------|
| Item           | Description           | Data Type    | Data Length     |
| Get_UI_Control | Get UI Control or Not | bool         | 1               |
| Get_Control    | Get UI Control or Not | bool         | 1               |
| Robot_Error    | Error or Not          | bool         | 1               |
| Error_Code     | Last Error Code       | int          | 1               |
| Error_Time     | Last Error Time       | string       | 1               |
| Camera_Light   | Light                 | byte         | 1               |

&gt;&gt;

| Transmit File (Data Table) |                                     |                  |             |
|----------------------------|-------------------------------------|------------------|-------------|
| Setting Transmit File Name |                                     | Communicate Mode |             |
| TransTable                 | <input type="button" value="Open"/> | STRING           | ▼           |
| Item                       | Description                         | Data Type        | Data Length |
| TCP_Value                  | TCP Value                           | float            | 6           |
| Ctrl_DO0                   | Digital Output 0                    | byte             | 1           |
| Ctrl_DO1                   | Digital Output 1                    | byte             | 1           |
| g_ss                       |                                     | string[]         | 0           |

```
float[] tvalue = {1,2,3,0.1,0.2,0.3}
```

```
bool flag = false
```

```
flag = svr_write("TCP_Value", tvalue) // flag = false read-only, invalid process (not an error)
```

```
flag = svr_write("Ctrl_DO0", 1) // flag = true , Ctrl_DO0 = 1
```

```
flag = svr_write("Ctrl_DO1", 0) // flag = true , Ctrl_DO1 = 0
```

```
flag = svr_write("TCP_Value", tvalue) // Error. Suppose Ethernet Slave is not launched.
```

```
flag = svr_write("TCP_Value1", tvalue) // Error. Item name TCP_Value1 does not exist.
```

```
flag = svr_write("Ctrl_DO0", "True") // Error. Item name Ctrl_DO0 writes value as string (the data type is set to byte)
```

## 18. Profinet Functions

The robot communicates with external controllers via the Profinet communication protocol. In the mechanism of the Profinet communication protocol, the robot works as a Profinet IO device for external devices to read and write the robot data. Meanwhile, TMflow monitors the table of data receiving from external devices and the table of data sending to external devices with Profinet functions as well as changes the custom definition section in the table of data sending to external devices.

### Communication Data Table

The data table is composed of the input data and the output data. Input Data Table is for external devices posting on the robot, and Output Data Table is for the robot sending to external devices. Both of the data tables come with System Definition Section and Custom Definition Section for data.

1. System Definition Section: Items and settings are defined by the robot, and the data contents are updated by the robot or external devices. The defined items are robot status relevant such as robot bases, project status, control box status, or input/output status relevant such as digital I/Os and analog I/Os. Users can use Profinet functions to read the input data table and the output data table in the system definition section.
2. Custom Definition Section: Items and settings are defined by users, and the data contents are updated by users or external devices. In the meantime of the project editing, users can use Profinet functions to read and write the output data table in the custom definition section or read input data table in the custom definition section as well as use the custom definition section as a data exchange register between the project and external devices.

| Communication Data Table<br>(at the robot's viewpoint) | Data Section              | TMflow Profinet Function Permissions | External Device Permissions |
|--------------------------------------------------------|---------------------------|--------------------------------------|-----------------------------|
| Input Data Table                                       | System Definition Section | Read                                 | Write                       |
|                                                        | Custom Definition Section | Read                                 | Write                       |
| Output Data Table                                      | System Definition Section | Read                                 | Read                        |
|                                                        | Custom Definition Section | Read/Write                           | Read                        |



## 18.1 profinet\_read\_input()

Read the input table content.

### Syntax 1

```
byte[] profinet_read_input(
 int,
 int
)
```

#### Parameters

**int** Starting address  
**int** The address amount to read

#### Return

**byte[]** Return data in a byte array.

#### Note

```
byte[] var_ba = profinet_read_input(148,16)
// {0x30,0x31,0x30,0x36,0x30,0x31,0x31,0x32,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
```

### Syntax 2

```
byte profinet_read_input(
 int,
)
```

#### Parameters

**int** Starting address

#### Return

**byte** Return data in byte.

#### Note

```
byte var_b = profinet_read_input(148)
// 0x30
```

### Syntax 3

```
? profinet_read_input(
 string,
 int,
 int
)
```

#### Parameters

**string** Item name  
**int** The starting shifted address of the item  
**int** The amount of the addresses to read

#### Return

**?** The data type returned by the item definition in the communication data table.  
 \* Data type includes **byte,byte[],int,int[],float,float[],string**

### Syntax 4

```
? profinet_read_input(
 string,
 int,
```

)

## Parameters

|        |                                          |
|--------|------------------------------------------|
| string | Item name                                |
| int    | The starting shifted address of the item |

## Return

? The data type returned by the item definition in the communication data table.  
\* Data type includes **byte,byte[],int,int[],float,float[],string**

## Note

- \* Same as syntax 3. Read to the end of the item by default.
- \* Reading data based on the configuration file Little Endian (DCBA) or Big Endian (ABCD).

## Syntax 5

? profinet\_read\_input(  
string

)

## Parameter

string    Item name

## Return

? The data type returned by the item definition in the communication data table.  
\* Data type includes **byte,byte[],int,int[],float,float[],string**

### Note

- \* Same as syntax 3. Fill 0 as the starting shifted address of the item and read to the end of the item by default.
- \* Reading data based on the configuration file Little Endian (DCBA) or Big Endian (ABCD).

```
byte var_b = profinet_read_input("StickStatus",0,1)
 // 0x02
var_b = profinet_read_input("StickStatus",0)
 // 0x02
var_b = profinet_read_input("StickStatus")
 // 0x02
byte[] var_ba = profinet_read_input("CtrlBox_DO",0,2)
 // {0x00,0x04}
var_ba = profinet_read_input("CtrlBox_DO")
 // {0x00,0x04}
int[] var_ia = profinet_read_input("Register_Int",0,12)
 // byte[] = {0x00,0x00,0x00,0x01,0x00,0x00,0x00,0x02,0x00,0x00,0x00,0x03} (Little Endian)
 to int[]
 // int[] = {0x00000001,0x00000002,0x00000003} (Little Endian)
 // int[] = {1,2,3}
var_ia = profinet_read_input("Register_Int",12)
 // byte[] = {0x00,0x00,0x00,0x04,0x00,0x00,0x00,0x05,0x00,0x00,0x00,0x06,0x00,0x00,0x00,0x07,
 0x00,0x00,0x00,0x08,0x00,0x00,0x00,0x09,0x00,0x00,0x00,0x0A,0x00,0x00,0x00,0x0B,
 0x00,0x00,0x00,0x0C,0x00,0x00,0x00,0x0D,0x00,0x00,0x00,0x0E,0x00,0x00,0x00,0x0F,
 0x00,0x00,0x00,0x10,0x00,0x00,0x00,0x11,0x00,0x00,0x00,0x12,0x00,0x00,0x00,0x13,
 0x00,0x00,0x00,0x14,0x00,0x00,0x00,0x15,0x00,0x00,0x00,0x16,0x00,0x00,0x00,0x17,
 0x00,0x00,0x00,0x18,0x00,0x00,0x00,0x19,0x00,0x00,0x00,0x1A,0x00,0x00,0x00,0x1B,
 0x00,0x00,0x00,0x1C,0x00,0x00,0x00,0x1D,0x00,0x00,0x00,0x1E,0x00,0x00,0x00,0x1F} (Little Endian) to int[]
```

[illegible]

```
// float[] = {0x3F8CCCCD,0x3F99999A,0x3FA66666,0x40066666,0x400CCCCD,0x40133333,
 0x00000000,0x00000000,0x00000000,0x00000000,0x00000000,0x00000000,
 0x00000000,0x00000000,0x00000000,0x00000000,0x00000000,0x00000000,
 0x00000000,0x00000000,0x00000000,0x00000000,0x00000000,0x00000000,
 0x00000000,0x00000000,0x00000000,0x00000000,0x00000000,0x00000000} (Big Endian)
// float[] = {1.1,1.2,1.3,2.1,2.2,2.3,0}
```

## 18.2 profinet\_read\_input\_int()

Read the input table content and convert the data to the 32-bit integer.

### Syntax 1

```
int[] profinet_read_input_int(
 int,
 int,
 int
)
```

#### Parameters

|     |                                                                                                     |
|-----|-----------------------------------------------------------------------------------------------------|
| int | Starting address                                                                                    |
| int | The address amount to read                                                                          |
| int | The conversion of the read data to an int array based on Little Endian (DCBA) or Big Endian (ABCD). |
| 0   | Little-Endian                                                                                       |
| 1   | Big-Endian                                                                                          |
| 2   | Based on the configuration file.                                                                    |

#### Return

int[] Return data in an integer array.

### Syntax 2

```
int[] profinet_read_input_int(
 int,
 int
)
```

#### Parameters

|     |                            |
|-----|----------------------------|
| int | Starting address           |
| int | The address amount to read |

#### Note

Same as Syntax 1 with the parameter of the conversion of the read data defaults to **2**.

\* Convert the read data to an int array based on Little Endian (DCBA) or Big Endian (ABCD).

```
int[] var_ia = profinet_read_input_int(164,12,0)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian)
 to int[]
 // int[] = {0x00007FFF,0x0001869F,0xFFFF8000} (Little Endian)
 // int[] = {32767,99999,-32768}
```

```
var_ia = profinet_read_input_int(164,11,0)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF} (Little Endian) to int[]
 // int[] = {0x00007FFF,0x0001869F,0x00FF8000} (Little Endian)
 // int[] = {32767,99999,16744448}
```

```
var_ia = profinet_read_input_int(164,10,0)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80} (Little Endian)to int[]
 // int[] = {0x00007FFF,0x0001869F,0x00008000} (Little Endian)
 // int[] = {32767,99999,32768}
```

```
var_ia = profinet_read_input_int(164,12,1)
// byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Big Endian) to int[]
// int[] = {0xFF7F0000,0x9F860100,0x0080FFFF} (Big Endian)
// int[] = {-8454144,-1618607872,8454143}
```

```
var_ia = profinet_read_input_int(164,12,2)
// byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian)
to int[]
// int[] = {0x00007FFF,0x0001869F,0xFFFF8000} (Little Endian)
// int[] = {32767,99999,-32768}
```

```
var_ia = profinet_read_input_int(164,12)
// byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian)
to int[]
// int[] = {0x00007FFF,0x0001869F,0xFFFF8000} (Little Endian)
// int[] = {32767,99999,-32768}
```

### Syntax 3

```
int profinet_read_input_int(
 int
)
```

#### Parameters

**int** Starting address  
 \* Convert the read data to an int array based on Little Endian (DCBA) or Big Endian (ABCD).

#### Return

**int** Return data in an integer

#### Note

```
int var_i = profinet_read_input_int(164)
// byte[] = {0xE4,0x07,0x00,0x00} (Little Endian) to int
// int = 0x000007E4 (Little Endian)
// int = 2020
var_i = profinet_read_input_int(164)
// byte[] = {0x00,0x00,0x07,0xE4} (Big Endian) to int
// int = 0x000007E4 (Big Endian)
// int = 2020
```

## 18.3 profinet\_read\_input\_float()

Read the input table content and convert the data to the 32-bit floating-point number.

### Syntax 1

```
float[] profinet_read_input_float(
 int,
 int,
 int
)
```

#### Parameters

|     |                                                                                                      |
|-----|------------------------------------------------------------------------------------------------------|
| int | Starting address                                                                                     |
| int | The address amount to read                                                                           |
| int | The conversion of the read data to a float array based on Little Endian (DCBA) or Big Endian (ABCD). |
| 0   | Little-Endian                                                                                        |
| 1   | Big-Endian                                                                                           |
| 2   | Based on the configuration file.                                                                     |

#### Return

float[] Return data in a floating-point number array.

### Syntax 2

```
float[] profinet_read_input_float(
 int,
 int
)
```

#### Note

Same as Syntax 1 with the parameter of the conversion of the read data defaults to 2.

\* Convert the read data to a float array based on Little Endian (DCBA) or Big Endian (ABCD).

```
float[] var_fa = profinet_read_input_float(284,12,0)
 // byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00,0x40,0x40} (Little Endian)
 to float[]
 // float[] = {0x3F800000,0x40000000,0x40400000} (Little Endian)
 // float[] = {1.0,2.0,3.0}
var_fa = profinet_read_input_float(284,11,0)
 // byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00,0x40} (Little Endian) to float[]
 // float[] = {0x3F800000,0x40000000,0x00400000} (Little Endian)
 // float[] = {1.0,2.0,5.877472E-39}
var_fa = profinet_read_input_float(284,10,0)
 // byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00} (Little Endian) to float[]
 // float[] = {0x3F800000,0x40000000,0x00000000} (Little Endian)
 // float[] = {1.0,2.0,0.0}

var_fa = profinet_read_input_float(284,12,1)
 // byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00,0x40,0x40} (Little Endian)
 to float[]
```

```

// float[] = {0x0000803F,0x00000040,0x00004040} (Big Endian)
// float[] = {4.600603E-41,8.96831E-44,2.304856E-41}

var_fa = profinet_read_input_float(284,12,2)
// byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00,0x40,0x40} (Little Endian)
to float[]
// float[] = {0x3F800000,0x40000000,0x40400000} (Little Endian)
// float[] = {1.0,2.0,3.0}

var_fa = profinet_read_input_float(284,12)
// byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00,0x40,0x40} (Little Endian)
to float[]
// float[] = {0x3F800000,0x40000000,0x40400000} (Little Endian)
// float[] = {1.0,2.0,3.0}

```

### Syntax 3

```
float profinet_read_input_float(
 int
)
```

#### Parameters

**int** Starting address  
 \* Convert the read data to a floating-point number based on Little Endian (DCBA) or Big Endian (ABCD).

#### Return

**float** Return data in floating-point number.

#### Note

```

float var_f = profinet_read_input_float(284)
// byte[] = {0x00,0x00,0x80,0x3F} (Little Endian) to float
// float = 0x3F800000 (Little Endian)
// float = 1.0

var_f = profinet_read_input_float(284)
// byte[] = {0x3F,0x80,0x00,0x00} (Big Endian) to float
// float = {0x3F800000} (Big Endian)
// float = {1.0}

```



## 18.4 profinet\_read\_input\_string()

Read the input table content and convert the data to the string encoded in UTF8.

### Syntax 1

```
string profinet_read_input_string(
 int,
 int
)
```

#### Parameters

**int** Starting address  
**int** The address amount to read

#### Return

**string** Return data in a UTF8 string (ending with 0x00 encountered).

#### Note

```
string var_s = profinet_read_input_string(148,16)
 // byte[] = {0x54,0x4D,0x35,0x2D,0x37,0x30,0x30,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // string = "TM5-700"
var_s = profinet_read_input_string(148,32)
 // byte[] = {0x30,0x31,0x30,0x36,0x30,0x31,0x31,0x32,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
 // 0x54,0x4D,0x35,0x2D,0x37,0x30,0x30,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // string = "01060112"
var_s = profinet_read_input_string(148,32)
 // byte[] = {0x61,0x62,0x63,0x64,0xE9,0x81,0x94,0xE6,0x98,0x8E,0xE6,0xA9,0x9F,0xE5,0x99,0xA8,
 // 0xE4,0xBA,0xBA,0x31,0x32,0x33,0x34,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // string = "abcd 達明機器人 1234"
var_s = profinet_read_input_string(148,10)
 // byte[] = {0x61,0x62,0x63,0x64,0xE9,0x81,0x94,0xE6,0x98,0x8E}
 // string = "abcd 達明"
var_s = profinet_read_input_string(148,8)
 // byte[] = {0x61,0x62,0x63,0x64,0xE9,0x81,0x94,0xE6}
 // string = "abcd 達◆"
```

## 18.5 profinet\_read\_input\_bit()

Read the input table content and retrieve the  $n^{\text{th}}$  bit value of the data byte.

### Syntax 1

```
byte profinet_read_input_bit(
 int,
 int
)
```

#### Parameters

**int** Starting address  
**int** The  $n^{\text{th}}$  bit value in the data byte

#### Return

**byte** Return data in byte.  
 Return 1 for bit value == 1.  
 Return 0 for bit value == 0.

#### Note

```
byte var_b = profinet_read_input_bit(148,0)
// 0x30 get bit: "0"
// 0
var_b = profinet_read_input_bit(148,5)
// 0x30 get bit: "5"
// 1
```

### Syntax 2

```
byte profinet_read_input_bit(
 string,
 int
)
```

#### Parameters

**string** Item name  
**int** The  $n^{\text{th}}$  bit value

#### Return

**byte** Return data in byte.  
 Return 1 for bit value == 1.  
 Return 0 for bit value == 0.  
 \*Data in bit will return in byte.

#### Note

```
byte var_b = profinet_read_input_bit("Register_Bit",0)
// byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1,...} // total
// 1 get bit: "0"
var_b = profinet_read_input_bit("Register_Bit",17)
// byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1,...} // total
// 0 get bit: "17"
```

### Syntax 3

```
byte[] profinet_read_input_bit(
 int,
```

```

 int,
 int
)

```

#### Parameters

```

 int Starting address
 int Starting bit
 int The amount of bit to read

```

#### Return

```

 byte[] Return data in byte[].
 Return 1 for bit value == 1.
 Return 0 for bit value == 0.
 *Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].

```

#### Note

```

byte[] var_ba = profinet_read_input_bit(148,0,20)
 // byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1,...} // total
 // byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
var_ba = profinet_read_input_bit(148,12,8)
 // byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1,...} // total
 // byte[] = {1,1,0,1,0,0,1,1}

```

#### Syntax 4

```

byte[] profinet_read_input_bit(
 string,
 int,
 int
)

```

#### Parameters

```

 string Item name
 int Starting bit
 int The amount of bit to read

```

#### Return

```

 byte[] Return data in byte[].
 Return 1 for bit value == 1.
 Return 0 for bit value == 0.
 *Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].

```

#### Note

```

byte[] var_ba = profinet_read_input_bit("Register_Bit",0,20)
 // byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1,...} // total
 // byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
var_ba = profinet_read_input_bit("Register_Bit",12,8)
 // byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1,...} // total
 // byte[] = {1,1,0,1,0,0,1,1}

```

## 18.6 profinet\_read\_output()

Read the output table content.

### Syntax 1

```
byte[] profinet_read_output(
 int,
 int
)
```

#### Parameters

**int** Starting address  
**int** The address length to read

#### Return

**byte[]** Return data in a byte array.

#### Note

```
byte[] var_ba = profinet_read_output(540,16)
// {0x30,0x31,0x30,0x36,0x30,0x31,0x31,0x32,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
```

### Syntax 2

```
byte profinet_read_output(
 int
)
```

#### Parameters

**int** Starting address

#### Return

**byte** Return data in byte

#### Note

```
byte var_b = profinet_read_output(540)
// 0x30
```

### Syntax 3

```
? profinet_read_output(
 string,
 int,
 int
)
```

#### Parameters

**string** Item name  
**int** The starting shifted address of the item  
**int** The amount of the addresses to read

#### Return

**?** The data type returned by the item defined in the communication data table.  
 \* Data type includes **byte,byte[],int,int[],float,float[],string**

### Syntax 4

```
? profinet_read_output(
 string,
 int,
)
```

**Parameters**

**string** Item name  
**int** The starting shifted address of the item

**Return**

**?** The data type returned by the item defined in the communication data table.  
 \* Data type includes **byte,byte[],int,int[],float,float[],string**

**Note**

- \* Same as syntax 3. Read to the end of the item by default.
- \* Reading data based on the configuration file Little Endian (DCBA) or Big Endian (ABCD).

**Syntax 5**

**? profinet\_read\_output(**  
**string**  
**)**

**Parameter**

**string** Item name

**Return**

**?** The data type returned by the item defined in the communication data table.  
 \* Data type includes **byte,byte[],int,int[],float,float[],string**

**Note**

- \* Same as syntax 3. Fill 0 as the starting shifted address of the item and read to the end of the item by default.
- \* Reading data based on the configuration file Little Endian (DCBA) or Big Endian (ABCD).

```
byte var_b = profinet_read_output("ManualAuto",0,1)
// 0x02
var_b = profinet_read_output("ManualAuto",0)
// 0x02
var_b = profinet_read_output("ManualAuto")
// 0x02
```

```
byte[] var_ba = profinet_read_output("Error_Code",0,4)
// {0x00,0x04,0x80,0x0C}
var_ba = profinet_read_output("Error_Code",0,2)
// {0x00,0x04}
var_ba = profinet_read_output("Error_Code")
// {0x00,0x04,0x80,0x0C}
```

```
var_int i = profinet_read_output("Current_Time_YY")
// byte[] = {0x00,0x00,0x07,0xE4} (Little Endian) to int
// int = 0x000007E4 (Little Endian)
// int = 2020
```

```
int[] var_ia = profinet_read_output("Register_Int",0,12)
// byte[] = {0x00,0x00,0x00,0x01,0x00,0x00,0x00,0x02,0x00,0x00,0x00,0x03} (Little Endian)
// to int[]
// int[] = { 0x00000001,0x00000002,0x00000003} (Little Endian)
// int[] = {1,2,3}
```

```

var_ia = profinet_read_output("Register_Int",12)
 // byte[] = {0x00,0x00,0x00,0x04,0x00,0x00,0x00,0x05,0x00,0x00,0x00,0x06,0x00,0x00,0x00,0x07,
 // 0x00,0x00,0x00,0x08,0x00,0x00,0x00,0x09,0x00,0x00,0x00,0x0A,0x00,0x00,0x00,0x0B,
 // 0x00,0x00,0x00,0x0C,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
 // 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
 // 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
 // 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian) to int[]
 // int[] = {0x00000004,0x00000005,0x00000006,0x00000007,0x00000008,0x00000009,
 // 0x0000000A,0x0000000B,0x0000000C,0x00000000,0x00000000,0x00000000,
 // 0x00000000,0x00000000,0x00000000,0x00000000,0x00000000,0x00000000,
 // 0x00000000,0x00000000,0x00000000,0x00000000,0x00000000,0x00000000,
 // 0x00000000,0x00000000,0x00000000} (Little Endian)
 // int[] = {4,5,6,7,8,9,10,11,12,0}
var_ia = profinet_read_output("Register_Int")
 // byte[] = {0x00,0x00,0x00,0x01,0x00,0x00,0x00,0x02,0x00,0x00,0x00,0x03,0x00,0x00,0x00,0x04,
 // 0x00,0x00,0x00,0x05,0x00,0x00,0x00,0x06,0x00,0x00,0x00,0x07,0x00,0x00,0x00,0x08,
 // 0x00,0x00,0x00,0x09,0x00,0x00,0x00,0x0A,0x00,0x00,0x00,0x0B,0x00,0x00,0x00,0x0C,
 // 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
 // 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
 // 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
 // 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
 // 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian) to int[]
 // int[] = {0x00000001,0x00000002,0x00000003,0x00000004,0x00000005,0x00000006,
 // 0x00000007,0x00000008,0x00000009,0x0000000A,0x0000000B,0x0000000C,
 // 0x00000000,0x00000000,0x00000000,0x00000000,0x00000000,0x00000000,
 // 0x00000000,0x00000000,0x00000000,0x00000000,0x00000000,0x00000000,
 // 0x00000000,0x00000000,0x00000000,0x00000000,0x00000000,0x00000000} (Little Endian)
 // int[] = {1,2,3,4,5,6,7,8,9,10,11,12,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0}

float var_f = profinet_read_output("Current_TCP_Mass")
 // byte[] = {0x40,0x40,0x00,0x00} (Big Endian) to float
 // float = 0x40400000 (Big Endian)
 // float = 3.0

var_fa = profinet_read_output("Current_TCP_Value",4,12)
 // byte[] = {0x3F,0x99,0x99,0x9A,0x3F,0xA6,0x66,0x66,0x40,0x06,0x66,0x66} (Big Endian) to float[]
 // float[] = {0x3F99999A,0x3FA66666,0x40066666} (Big Endian)
 // float[] = {1.2,1.3,2.1}
var_fa = profinet_read_output("Current_TCP_Value",12)
 // byte[] = { 0x40,0x06,0x66,0x66,0x40,0x0C,0xCC,0xCD,0x40,0x13,0x33,0x33} (Big Endian)
 // to float[]
 // float[] = {0x40066666,0x400CCCCD,0x40133333} (Big Endian)
 // float[] = {2.1,2.2,2.3}
var_fa = profinet_read_output("Current_TCP_Value")
 // byte[] = {0x3F,0x8C,0xCC,0xCD,0x3F,0x99,0x99,0x9A,0x3F,0xA6,0x66,0x66,0x40,0x06,0x66,0x66,
 // 0x40,0x0C,0xCC,0xCD,0x40,0x13,0x33,0x33} (Big Endian) to float[]
 // float[] = {0x3F8CCCCD,0x3F99999A,0x3FA66666,0x40066666,0x400CCCCD,0x40133333}

```

```
(Big Endian)
// float[] = { 1.1,1.2,1.3,2.1,2.2,2.3}

string var_s = profinet_read_output ("RobotModel",0,16)
// byte[] = {0x54,0x4D,0x35,0x2D,0x37,0x30,0x30,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
// string = "TM5-700"
var_s = profinet_read_output ("RobotModel",4,3)
// byte[] = {0x37,0x30,0x30}
// string = "700"
var_s = profinet_read_output ("RobotModel")
// byte[] = {0x54,0x4D,0x35,0x2D,0x37,0x30,0x30,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
// string = "TM5-700"
```

## 18.7 profinet\_read\_output\_int()

Read the output table content and convert the data to the 32-bit integer.

### Syntax 1

```
int[] profinet_read_output_int(
 int,
 int,
 int
)
```

#### Parameters

|     |                                                                                                     |
|-----|-----------------------------------------------------------------------------------------------------|
| int | Starting address                                                                                    |
| int | The address amount to read                                                                          |
| int | The conversion of the read data to an int array based on Little Endian (DCBA) or Big Endian (ABCD). |
| 0   | Little-Endian                                                                                       |
| 1   | Big-Endian                                                                                          |
| 2   | Based on the configuration file.                                                                    |

#### Return

int[] Return data in an integer array.

### Syntax 2

```
int[] profinet_read_output_int(
 int,
 int
)
```

#### Note

Same as Syntax 1 with the parameter of the conversion of the read data defaults to **2**.

\*Convert the read data to an int array based on Little Endian (DCBA) or Big Endian (ABCD).

```
int[] var_ia = profinet_read_output_int(556,12,0)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian)
 to int[]
 // int[] = {0x00007FFF,0x0001869F,0xFFFF8000} (Little Endian)
 // int[] = {32767,99999,-32768}
var_ia = profinet_read_output_int(556,11,0)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF} (Little Endian) to int[]
 // int[] = {0x00007FFF,0x0001869F,0x00FF8000} (Little Endian)
 // int[] = {32767,99999,16744448}
var_ia = profinet_read_output_int(556,10,0)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80} (Little Endian)to int[]
 // int[] = {0x00007FFF,0x0001869F,0x00008000} (Little Endian)
 // int[] = {32767,99999,32768}
var_ia = profinet_read_output_int(556,12,1)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian)
 to int[]
 // int[] = {0xFF7F0000,0x9F860100,0x0080FFFF} (Big Endian)
```



```

// int[] = {-8454144,-1618607872,8454143}

var_ia = profinet_read_output_int(556,12,2)
// byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian)
to int[]
// int[] = {0x00007FFF,0x0001869F,0xFFFF8000} (Little Endian)
// int[] = {32767,99999,-32768}

var_ia = profinet_read_output_int(556,12)
// byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian)
to int[]
// int[] = {0x00007FFF,0x0001869F,0xFFFF8000} (Little Endian)
// int[] = {32767,99999,-32768}

```

### Syntax 3

```

int profinet_read_output_int(
 int
)

```

#### Parameters

**int** Starting address  
 \* Convert the read data to an integer based on Little Endian (DCBA) or Big Endian (ABCD).

#### Return

**int** Return data in an integer

#### Note

```

int var_i = profinet_read_output_int(556)
// byte[] = {0xE4,0x07,0x00,0x00} (Little Endian) to int
// int = 0x000007E4 (Little Endian)
// int = 2020

var_i = profinet_read_output_int(556)
// byte[] = {0x00,0x00,0x07,0xE4} (Big Endian) to int
// int = 0x000007E4 (Big Endian)
// int = 2020

```

## 18.8 profinet\_read\_output\_float()

Read the output table content and convert the data to the 32-bit floating-point number.

### Syntax 1

```
float[] profinet_read_output_float(
 int,
 int,
 int
)
```

#### Parameters

|     |                                                                                                      |
|-----|------------------------------------------------------------------------------------------------------|
| int | Starting address                                                                                     |
| int | The address amount to read                                                                           |
| int | The conversion of the read data to a float array based on Little Endian (DCBA) or Big Endian (ABCD). |
| 0   | Little-Endian                                                                                        |
| 1   | Big-Endian                                                                                           |
| 2   | Based on the configuration file.                                                                     |

#### Return

float[] Return data in a floating-point number array.

### Syntax 2

```
float[] profinet_read_output_float(
 int,
 int
)
```

#### Note

Same as Syntax 1 with the parameter of the conversion of the read data defaults to 2.  
 \*Convert the read data to a float array based on Little Endian (DCBA) or Big Endian (ABCD).

```
float[] var_fa = profinet_read_output_float(676,12,0)
 // byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00,0x40,0x40} (Little Endian)
 // to float[]
 // float[] = {0x3F800000,0x40000000,0x40400000} (Little Endian)
 // float[] = {1.0,2.0,3.0}
var_fa = profinet_read_output_float(676,11,0)
 // byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00,0x40} (Little Endian) to float[]
 // float[] = {0x3F800000,0x40000000,0x00400000} (Little Endian)
 // float[] = {1.0,2.0,5.877472E-39}
var_fa = profinet_read_output_float(676,10,0)
 // byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00} (Little Endian) to float[]
 // float[] = {0x3F800000,0x40000000,0x00000000} (Little Endian)
 // float[] = {1.0,2.0,0.0}

var_fa = profinet_read_output_float(676,12,1)
 // byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00,0x40,0x40} (Little Endian)
```

```

to float[]
// float[] = {0x0000803F,0x00000040,0x00004040} (Big Endian)
// float[] = {4.600603E-41,8.96831E-44,2.304856E-41}

```

```

var_fa = profinet_read_output_float(676,12,2)
// byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00,0x40,0x40} (Little Endian)
to float[]
// float[] = {0x3F800000,0x40000000,0x40400000} (Little Endian)
// float[] = {1.0,2.0,3.0}

```

```

var_fa = profinet_read_output_float(676,12)
// byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00,0x40,0x40} (Little Endian)
to float[]
// float[] = {0x3F800000,0x40000000,0x40400000} (Little Endian)
// float[] = {1.0,2.0,3.0}

```

### Syntax 3

```

float profinet_read_output_float(
 int
)

```

#### Parameters

**int** Starting address  
 \* Convert the read data to a floating-point number based on Little Endian (DCBA) or Big Endian (ABCD).

#### Return

**float** Return data in a floating-point number.

#### Note

```

float var_f = profinet_read_output_float(676)
// byte[] = {0x00,0x00,0x80,0x3F} (Little Endian) to float
// float = 0x3F800000 (Little Endian)
// float = 1.0
var_f = profinet_read_output_float(676)
// byte[] = {0x3F,0x80,0x00,0x00} (Big Endian) to float
// float = 0x3F800000 (Big Endian)
// float = 1.0

```

## 18.9 profinet\_read\_output\_string()

Read the output table content and convert the data to the string encoded in UTF8.

### Syntax 1

```
string profinet_read_output_string(
 int,
 int
)
```

#### Parameters

**int** Starting address  
**int** The address amount to read

#### Return

**string** Return data in a UTF8 string (ending with 0x00 encountered).

#### Note

```
string var_s = profinet_read_output_string(540,16)
 // byte[] = {0x54,0x4D,0x35,0x2D,0x37,0x30,0x30,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // string = "TM5-700"

var_s = profinet_read_output_string(540,32)
 // byte[] = {0x30,0x31,0x30,0x36,0x30,0x31,0x31,0x32,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
 // 0x54,0x4D,0x35,0x2D,0x37,0x30,0x30,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // string = "01060112"

var_s = profinet_read_output_string(540,32)
 // byte[] = {0x61,0x62,0x63,0x64,0xE9,0x81,0x94,0xE6,0x98,0x8E,0xE6,0xA9,0x9F,0xE5,0x99,0xA8,
 // 0xE4,0xBA,0xBA,0x31,0x32,0x33,0x34,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // string = "abcd 達明機器人 1234"

var_s = profinet_read_output_string(540,10)
 // byte[] = {0x61,0x62,0x63,0x64,0xE9,0x81,0x94,0xE6,0x98,0x8E}
 // string = "abcd 達明"

var_s = profinet_read_output_string(540,8)
 // byte[] = {0x61,0x62,0x63,0x64,0xE9,0x81,0x94,0xE6}
 // string = "abcd 達◆"
```

## 18.10 profinet\_read\_output\_bit()

Read the output table content and retrieve the n<sup>th</sup> bit value of the data byte.

### Syntax 1

```
byte profinet_read_output_bit(
 int,
 int
)
```

#### Parameters

**int** Starting address  
**int** The n<sup>th</sup> bit value in the data byte

#### Return

**byte** Return data in byte.  
 Return 1 for bit value == 1.  
 Return 0 for bit value == 0.

#### Note

```
byte var_b = profinet_read_output_bit(540,0)
// 0x30 get bit: "0"
// 0
var_b = profinet_read_output_bit(540,5)
// 0x30 get bit: "5"
// 1
```

### Syntax 2

```
byte profinet_read_output_bit(
 string,
 int
)
```

#### Parameters

**string** Item name  
**int** The n<sup>th</sup> bit value

#### Return

**byte** Return data in byte.  
 Return 1 for bit value == 1.  
 Return 0 for bit value == 0.  
 \*Data in bit will return in byte.

#### Note

```
byte[] var_data = {57,184,12}
profinet_write_output(540,var_data,3)
// {00111001,10111000,00001100} (binary)
byte var_b = profinet_read_output_bit("Register_Bit",0)
// 1
var_b = profinet_read_output_bit("Register_Bit",17)
// 0
```

### Syntax 3

```
byte[] profinet_read_output_bit(
 int,
```

```
int,
int
)
```

#### Parameters

```
int Starting address
int Starting bit
int The amount of bit to read
```

#### Return

```
byte[] Return data in byte[].
 Return 1 for bit value == 1.
 Return 0 for bit value == 0.
 *Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].
```

#### Note

```
byte[] var_data = {57,184,12}
profinet_write_output(540,var_data,3)
// {00111001,10111000,00001100} (binary)
byte[] var_ba = profinet_read_output_bit(540,0,20)
// byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
var_ba = profinet_read_output_bit(540,12,8)
// byte[] = {1,1,0,1,0,0,1,1}
```

#### Syntax 4

```
byte[] profinet_read_output_bit(
 string,
 int,
 int
)
```

#### Parameters

```
string Item name
int Starting bit
int The amount of bit to read
```

#### Return

```
byte[] Return data in byte[].
 Return 1 for bit value == 1.
 Return 0 for bit value == 0.
 *Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].
```

#### Note

```
byte[] var_data = {57,184,12}
profinet_write_output(540,var_data,3)
// {00111001,10111000,00001100} (binary)
byte[] var_ba = profinet_read_output_bit("Register_Bit",0,20)
// byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
var_ba = profinet_read_output_bit("Register_Bit",12,8)
// byte[] = {1,1,0,1,0,0,1,1}
```

## 18.11 profinet\_write\_output()

Write data to the output table.

### Syntax 1

```
bool profinet_write_output (
 int,
 ?,
 int
)
```

#### Parameters

|      |                                                                                        |
|------|----------------------------------------------------------------------------------------|
| int  | Starting address                                                                       |
| ?    | The data to write                                                                      |
|      | * Available data types include <b>byte, byte[], int, int[], float, float[], string</b> |
| int  | The maximum amount of the address to write                                             |
| > 0  | Legitimate data length. Write by the amount of the address.                            |
| <= 0 | Illegitimate data length. Write by the complete length of the data to write.           |

#### Return

|           |                                                              |
|-----------|--------------------------------------------------------------|
| bool True | Write successfully.                                          |
| False     | Write unsuccessfully.                                        |
|           | 1. If the data to write is an empty string or an empty array |
|           | 2. Unable to send and receive correctly.                     |

#### Note

\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

### Syntax 2

```
bool profinet_write_output(
 int,
 ?
)
```

#### Parameters

|     |                                                                                        |
|-----|----------------------------------------------------------------------------------------|
| int | Starting address                                                                       |
| ?   | The data to write                                                                      |
|     | * Available data types include <b>byte, byte[], int, int[], float, float[], string</b> |

#### Return

|      |       |                                                              |
|------|-------|--------------------------------------------------------------|
| Bool | True  | Write successfully.                                          |
|      | False | Write unsuccessfully.                                        |
|      |       | 1. If the data to write is an empty string or an empty array |
|      |       | 2. Unable to send and receive correctly.                     |

#### Note

Same as syntax 1. Write with the full length of the data to write by default.

\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

### Syntax 3

```
bool profinet_write_output(
 int,
 ?,

```

```
int,
int
)
```

### Parameters

|     |                                                                                  |
|-----|----------------------------------------------------------------------------------|
| int | Starting address                                                                 |
| ?   | The data to write                                                                |
|     | * Available data types include <b>byte,byte[],int,int[],float,float[],string</b> |
| int | Starting address of the data to write                                            |
| int | The amount of the address to write                                               |

### Return

|           |                       |
|-----------|-----------------------|
| bool True | Write successfully.   |
| False     | Write unsuccessfully. |

1. If the data to write is an empty string or an empty array
2. Unable to send and receive correctly.

### Note

\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

```
byte var_data = 255
profinet_write_output(540,var_data,1)
byte var_b = profinet_read_output(540)
// 0xFF

byte[] var_data = {1,127,255}
profinet_write_output(540,var_data,3)
byte[] var_ba = profinet_read_output(540,3)
// {0x01,0x7F,0xFF}
profinet_write_output(540,var_data,2)
var_ba = profinet_read_output(540,3)
// {0x01,0x7F,0x00}
profinet_write_output(540,var_data,-1)
var_ba = profinet_read_output(540,3)
// {0x00,0x7F,0xFF}

int var_data = 32767
profinet_write_output(556,var_data,4)
int var_i = profinet_read_output_int(556)
// byte[] = {0xFF,0x7F,0x00,0x00} (Little Endian) to int
// int = 0x00007FFF (Little Endian)
// int = 32767
profinet_write_output(556,var_data,1)
var_i = profinet_read_output_int(556)
// byte[] = {0xFF,0x00,0x00,0x00} (Little Endian) to int
// int = 0x000000FF (Little Endian)
// int = 255

int[] var_data = {32767,99999,-32768}
profinet_write_output(556,var_data,12)
```



```

int[] var_ia = profinet_read_output_int(556,12)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian)
 to int[]
 // int[] = {0x00007FFF,0x0001869F,0xFFFF8000} (Little Endian)
 // int[] = {32767,99999,-32768}
profinet_write_output(556,var_data,3)
var_ia = profinet_read_output_int(556,12)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian)
 to int[]
 // int[] = {0x00007FFF,0x00000000,0x00000000} (Little Endian)
 // int[] = {32767,0,0}
profinet_write_output(556,var_data,11)
var_ia = profinet_read_output_int(556,12)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0x00} (Little Endian)
 to int[]
 // int[] = {0x00007FFF,0x0001869F,0x00FF8000} (Little Endian)
 // int[] = {32767,99999,16744448}
profinet_write_output(556,var_data,4,4)
var_ia = profinet_read_output_int(556,12)
 // byte[] = {0x9F,0x86,0x01,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian)
 to int[]
 // int[] = {0x0001869F,0x00000000,0x00000000} (Little Endian)
 // int[] = {99999,0,0}

float var_data = -10.0
profinet_write_output(676,var_data,4)
float var_f = profinet_read_output_float(676)
 // byte[] = {0x00,0x00,0x20,0xC1} (Little Endian) to float
 // float = 0xC1200000 (Little Endian)
 // float = -10.0
profinet_write_output(676,var_data,1)
var_f = profinet_read_output_float(676)
 // byte[] = {0x00,0x00,0x00,0x00} (Little Endian) to float
 // float = 0x00000000 (Little Endian)
 // float = 0

float[] var_data = {-10.0,3.3,123.45}
profinet_write_output(676,var_data,12)
float[] var_fa = profinet_read_output_float(676,12)
 // byte[] = {0x00,0x00,0x20,0xC1,0x33,0x33,0x53,0x40,0x66,0xE6,0xF6,0x42} (Little Endian)
 to float[]
 // float[] = {0xC1200000,0x40533333,0x42F6E666} (Little Endian)
 // float[] = {-10,3.3,123.45}
profinet_write_output(676,var_data,3)
var_fa = profinet_read_output_float(676,12)
 // byte[] = {0x00,0x00,0x20,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian)
 to float[]
 // float[] = {0x00200000,0x00000000,0x00000000} (Little Endian)

```

```

 // float[] = {2.938736E-39,0,0}
 profinet_write_output(676,var_data,11)
 var_fa = profinet_read_output_float(676,12)
 // byte[] = {0x00,0x00,0x20,0xC1,0x33,0x33,0x53,0x40,0x66,0xE6,0xF6,0x00} (Little Endian)
 to float[]
 // float[] = {0xC1200000,0x40533333,0x00F6E666} (Little Endian)
 // float[] = {-10,3.3,2.267418E-38}
 profinet_write_output(676,var_data,4,4)
 var_fa = profinet_read_output_float(676,12)
 // byte[] = {0x33,0x33,0x53,0x40,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian)
 to float[]
 // float[] = {0x40533333,0x00000000,0x00000000} (Little Endian)
 // float[] = {3.3,0,0}

 string var_data = "abcd 達明機器人 1234"
 profinet_write_output(540,var_data,32)
 string var_s = profinet_read_output_string(540,32)
 // byte[] = {0x61,0x62,0x63,0x64,0xE9,0x81,0x94,0xE6,0x98,0x8E,0xE6,0xA9,0x9F,0xE5,0x99,0xA8,
 // 0xE4,0xBA,0xBA,0x31,0x32,0x33,0x34,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // string = "abcd 達明機器人 1234"
 profinet_write_output(540,var_data,10)
 var_s = profinet_read_output_string(540,32)
 // byte[] = { 0x61,0x62,0x63,0x64,0xE9,0x81,0x94,0xE6,0x98,0x8E,0x00,0x00,0x00,0x00,0x00,0x00,
 // 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // string = "abcd 達明"
 profinet_write_output(540,var_data,8)
 var_s = profinet_read_output_string(540,32)
 // byte[] = {0x61,0x62,0x63,0x64,0xE9,0x81,0x94,0xE6,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
 // 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // string = "abcd 達💎"
 profinet_write_output(540,var_data,4,15)
 var_s = profinet_read_output_string(540,15)
 // byte[] = {0xE9,0x81,0x94,0xE6,0x98,0x8E,0xE6,0xA9,0x9F,0xE5,0x99,0xA8,0xE4,0xBA,0xBA}
 // string = "達明機器人"

```

#### Syntax 4

```

bool profinet_write_output(
 string,
 int,
 ?,
 int,
 int
)

```

#### Parameters

|                                                                                                                                        |                                          |
|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| string                                                                                                                                 | Item name                                |
| int                                                                                                                                    | The starting shifted address of the item |
| ?                                                                                                                                      | The data to write                        |
| * Available data types include <b>byte</b> , <b>byte[]</b> , <b>int</b> , <b>int[]</b> , <b>float</b> , <b>float[]</b> , <b>string</b> |                                          |

`int` Starting address of the data to write  
`int` The amount of the address to write

#### Return

`bool True` Write successfully.  
`False` Write unsuccessfully.  
 1. If the data to write is an empty string or an empty array  
 2. Unable to send and receive correctly.

#### Note

\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

#### Syntax 5

```
bool profinet_write_output(
 string,
 int,
 ?
 int
)
```

#### Parameters

`string` Item name  
`int` The starting shifted address of the item  
`?` The data to write  
 \* Available data types include `byte, byte[], int, int[], float, float[], string`  
`int` Starting address of the data to write

#### Return

`bool True` Write successfully.  
`False` Write unsuccessfully.  
 1. If the data to write is an empty string or an empty array  
 2. Unable to send and receive correctly.

#### Note

Same as syntax 4. Write with the full length of the data to write by default.  
 \* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

#### Syntax 6

```
bool profinet_write_output(
 string,
 int,
 ?
)
```

#### Parameters

`string` Item name  
`int` The starting shifted address of the item  
`?` The data to write  
 \* Available data types include `byte, byte[], int, int[], float, float[], string`

#### Return

`bool True` Write successfully.

**False** Write unsuccessfully.

1. If the data to write is an empty string or an empty array
2. Unable to send and receive correctly.

#### Note

Same as syntax 4. Fill 0 as the starting address to write and write with the full length of the data to write by default.

\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

#### Syntax 7

```
bool profinet_write_output(
 string,
 ?
)
```

#### Parameters

**string** Item name  
**?** The data to write  
 \* Available data types include **byte,byte[],int,int[],float,float[],string**

#### Return

**bool True** Write successfully.  
**False** Write unsuccessfully.  

1. If the data to write is an empty string or an empty array
2. Unable to send and receive correctly.

#### Note

Same as syntax 4. Fill 0 as the starting shifted address and the starting address to write as well as write with the full length of the data to write by default.

\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

```
int[] var_data = {32767,99999,-32768}
profinet_write_output("Register_Int",0,var_data,0,12)
int[] var_ia = profinet_read_output_int(556,12)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian)
 to int[]
 // int[] = {0x00007FFF,0x0001869F,0xFFFF8000} (Little Endian)
 // int[] = {32767,99999,-32768}
profinet_write_output("Register_Int",4,var_data,4,4)
var_ia = profinet_read_output_int(556,12)
 // byte[] = {0x00,0x00,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x00,0x00,0x00} (Little Endian)
 to int[]
 // int[] = {0x00000000,0x0001869F,0x00000000} (Little Endian)
 // int[] = {0,99999,0}
profinet_write_output("Register_Int",4,var_data)
var_ia = profinet_read_output_int(556,20)
 // byte[] = {0x00,0x00,0x00,0x00,0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF,
 0x00,0x00,0x00,0x00} (Little Endian) to int[]
 // int[] = {0x00000000,0x00007FFF,0x0001869F,0xFFFF8000,0x00000000} (Little Endian)
 // int[] = {0,32767,99999,-32768,0}
```

```

float[] var_data = {-10.0,3.3,123.45}
profinet_write_output("Register_Float",0,var_data,0,12)
float[] var_fa = profinet_read_output_float(676,12)
 // byte[] = {0x00,0x00,0x20,0xC1,0x33,0x33,0x53,0x40,0x66,0xE6,0xF6,0x42} (Little Endian)
 to float[]
 // float[] = {0xC1200000,0x40533333,0x42F6E666} (Little Endian)
 // float[] = {-10,3.3,123.45}
profinet_write_output("Register_Float",4,var_data,4,8)
var_fa = profinet_read_output_float(676,12)
 // byte[] = {0x00,0x00,0x00,0x00,0x33,0x33,0x53,0x40,0x66,0xE6,0xF6,0x42} (Little Endian)
 to float[]
 // float[] = {0x00000000,0x40533333,0x42F6E666} (Little Endian)
 // float[] = {0,3.3,123.45}
profinet_write_output("Register_Float",8,var_data)
var_fa = profinet_read_output_float(676,20)
 // byte[] = {0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x20,0xC1,0x33,0x33,0x53,0x40,
 0x66,0xE6,0xF6,0x42} (Little Endian) to float[]
 // float[] = {0x00000000,0x00000000,0xC1200000,0x40533333,0x42F6E666} (Little Endian)
 // float[] = {0,0,-10,3.3,123.45}

```

## 18.12 profinet\_write\_output\_bit()

Write content to the n<sup>th</sup> bit value of the data byte in the output table.

### Syntax 1

```
bool profinet_write_output_bit(
 int,
 int,
 int
)
```

#### Parameters

|     |                                                |
|-----|------------------------------------------------|
| int | Starting address                               |
| int | The n <sup>th</sup> bit value in the data byte |
| int | The data to write                              |

#### Return

bool True Write successfully.

False Write unsuccessfully.

1. Unable to send correctly and receive .

#### Note

```
byte var_data = 240
profinet_write_output(540,var_data)
byte var_b = profinet_read_output(540)
// 0xF0
profinet_write_output_bit(540,1,1)
var_b = profinet_read_output_bit(540,1)
// 0xF2 get bit: "1"
// 1
profinet_write_output_bit(540,7,0)
var_b = profinet_read_output_bit(540,7)
// 0x72 get bit: "7"
// 0
```

### Syntax 2

```
bool profient_write_output_bit(
 string,
 int,
 int
)
```

#### Parameter

|     |                               |
|-----|-------------------------------|
| int | Item name                     |
| int | The n <sup>th</sup> bit value |
| int | The data to write             |

\*Data in bit will write in int.

#### Return

bool True Write successfully.

False Write unsuccessfully.

1. Unable to send correctly and receive .

#### Note

```
byte var_data = 240
profinet_write_output(540,var_data)
```

```

byte var_b = profinet_read_output(540)
// 0xF0
profinet_write_output_bit("Register_Bit",1,1)
var_b = profinet_read_output_bit(540,1)
// 0xF2 get bit: "1"
// 1
profinet_write_output_bit("Register_Bit",7,0)
var_b = profinet_read_output_bit(540,7)
// 0x72 get bit: "7"
// 0

```

### Syntax 3

```

bool profinet_write_output_bit(
 int,
 int,
 byte[],
 int,
 int
)

```

#### Parameters

|        |                                                                                                      |
|--------|------------------------------------------------------------------------------------------------------|
| int    | Starting address                                                                                     |
| int    | Starting bit                                                                                         |
| byte[] | Data to write.<br>*Data in bit will write in byte such as bit[0] for byte[0] and bit[1] for byte[1]. |
| int    | Starting bit to write data                                                                           |
| int    | The amount of bit to write data                                                                      |

#### Return

|           |                       |                                           |
|-----------|-----------------------|-------------------------------------------|
| bool True | Write successfully.   |                                           |
| False     | Write unsuccessfully. | 1. Unable to send correctly and receive . |

#### Note

Bit value = 1 for byte value >=1  
 Bit value = 0 for byte value ==0

### Syntax 4

```

bool profinet_write_output_bit(
 int,
 int,
 byte[],
 int
)

```

#### Parameters

|        |                                                                                                          |
|--------|----------------------------------------------------------------------------------------------------------|
| int    | Starting address                                                                                         |
| int    | The n <sup>th</sup> bit value                                                                            |
| byte[] | The data to write<br>*Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1]. |
| int    | The starting bit of the data to write                                                                    |

#### Return

|           |                     |
|-----------|---------------------|
| bool True | Write successfully. |
|-----------|---------------------|

**False** Write unsuccessfully. 1. Unable to send correctly and receive .

#### Note

\*Same as syntax 3. Write with the full length of the rest data to write.

Bit value = 1 for byte value >=1

Bit value = 0 for byte value ==0

#### Syntax 5

```
bool profinet_write_output_bit(
 int,
 int,
 byte[]
)
```

#### Parameters

**int** Starting address

**int** The n<sup>th</sup> bit value

**byte[]** The data to write

\*Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].

#### Return

**bool True** Write successfully.

**False** Write unsuccessfully. 1. Unable to send correctly and receive .

#### Note

\*Same as syntax 3. Fill 0 as the starting bit to write as well as write with the full length of the data to write by default.

Bit value = 1 for byte value >=1

Bit value = 0 for byte value ==0

```
byte[] var_data = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
```

```
profinet_write_output_bit(540,0,var_data,0,20)
```

```
byte[] var_ba = profinet_read_output (540,0,3)
```

```
// byte[] = {0x39,0xB8,0x0C}
```

```
var_ba = profinet_read_output_bit(540,0,20)
```

```
// byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
```

```
profinet_write_output_bit(540,3,var_data,5,10)
```

```
var_ba = profinet_read_output (540,0,3)
```

```
// byte[] = {0x08,0x0E,0x00}
```

```
var_ba = profinet_read_output_bit(540,0,20)
```

```
// byte[] = {0,0,0,1,0,0,0,0,0,1,1,1,0,0,0,0,0,0,0,0}
```

#### Syntax 6

```
bool profinet_write_output_bit(
 string,
 int,
 byte[],
 int,
 int
)
```

#### Parameters

**string** Item name

**int** Starting bit



**byte[]** The data to write  
 \*Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].  
**int** The starting bit to write data  
**int** The bit amount of the data to write

**Return**

**bool True** Write successfully.  
**False** Write unsuccessfully. 1. Unable to send correctly and receive .

**Note**

Bit value = 1 for byte value >=1  
 Bit value = 0 for byte value ==0

**Syntax 7**

```
bool profinet_write_output_bit(
 string,
 int,
 byte[],
 int
)
```

**Parameters**

**string** Item name  
**int** Starting bit  
**byte[]** The data to write  
 \*Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].  
**int** The starting bit to write data

**Return**

**bool True** Write successfully.  
**False** Write unsuccessfully. 1. Unable to send correctly and receive .

**Note**

\*Same as syntax 6. Write with the full length of the rest data to write.  
 Bit value = 1 for byte value >=1  
 Bit value = 0 for byte value ==0

**Syntax 8**

```
bool profinet_write_output_bit(
 string,
 int,
 byte[]
)
```

**Parameters**

**string** Item name  
**int** Starting bit  
**byte[]** The data to write  
 \*Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].

**Return**

**bool True** Write successfully.  
**False** Write unsuccessfully. 1. Unable to send correctly and receive .

**Note**

\*Same as syntax 6. Fill 0 as the starting bit to write as well as write with the full length of

the data to write by default.

Bit value = 1 for byte value >=1

Bit value = 0 for byte value ==0

```
byte[] var_data = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
profinet_write_output_bit("Register_Bit",0,var_data,0,20)
byte[] var_ba = profinet_read_output (540,3)
 // byte[] = {0x39,0xB8,0x0C}
var_ba = profinet_read_output_bit(540,0,20)
 // byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
profinet_write_output_bit("Register_Bit",3,var_data,5,10)
var_ba = profinet_read_output (540,3)
 // byte[] = {0x08,0x0E,0x00}
var_ba = profinet_read_output_bit(540,0,20)
 // byte[] = {0,0,0,1,0,0,0,0,0,1,1,1,0,0,0,0,0,0,0,0}
```

## 19. EtherNet/IP Functions

The robot communicates with external controllers via the EtherNet/IP communication protocol. In the mechanism of the EtherNet/IP communication protocol, the robot works as an EtherNet/IP IO device for external devices to read and write the robot data. Meanwhile, TMflow monitors the table of data receiving from external devices and the table of data sending to external devices with EtherNet/IP functions as well as changes the custom definition section in the table of data sending to external devices.

### Communication Data Table

The data table is composed of the input data and the output data. Input Data Table is for external devices posting on the robot, and Output Data Table is for the robot sending to external devices. Both of the data tables come with System Definition Section and Custom Definition Section for data.

1. System Definition Section: Items and settings are defined by the robot, and the data contents are updated by the robot or external devices. The defined items are robot status relevant such as robot bases, project status, control box status, or input/output status relevant such as digital I/Os and analog I/Os. Users can use EtherNet/IP functions to read the input data table and the output data table in the system definition section.
2. Custom Definition Section: Items and settings are defined by users, and the data contents are updated by users or external devices. In the meantime of the project editing, users can use EtherNet/IP functions to read and write the output data table in the custom definition section or read input data table in the custom definition section as well as use the custom definition section as a data exchange register between the project and external devices.

| Communication Data Table<br>(at the robot's viewpoint) | Data Section              | TMflow EtherNet/IP Function Permissions | External Device Permissions |
|--------------------------------------------------------|---------------------------|-----------------------------------------|-----------------------------|
| Input Data Table                                       | System Definition Section | Read                                    | Write                       |
|                                                        | Custom Definition Section | Read                                    | Write                       |
| Output Data Table                                      | System Definition Section | Read                                    | Read                        |
|                                                        | Custom Definition Section | Read/Write                              | Read                        |

## 19.1 eip\_read\_input()

Read the input table content.

### Syntax 1

```
byte[] eip_read_input(
 int,
 int
)
```

#### Parameters

**int** Starting address  
**int** The address amount to read

#### Return

**byte[]** Return data in a byte array.

#### Note

```
byte[] var_ba = eip_read_input(104,8)
// {0x30,0x31,0x30,0x36,0x30,0x31,0x31,0x32}
```

### Syntax 2

```
byte eip_read_input(
 int,
)
```

#### Parameters

**int** Starting address

#### Return

**byte** Return data in byte.

#### Note

```
byte var_b = eip_read_input(104)
// 0x30
```

### Syntax 3

```
? eip_read_input(
 string,
 int,
 int
)
```

#### Parameters

**string** Item name  
**int** The starting shifted address of the item  
**int** The amount of the addresses to read

#### Return

**?** The data type returned by the item definition in the communication data table.  
 \* Data type includes **byte,byte[],int,int[],float,float[],string**

### Syntax 4

```
? eip_read_input(
 string,
 int,
```

)

**Parameters**

**string** Item name  
**int** The starting shifted address of the item

**Return**

**?** The data type returned by the item definition in the communication data table.  
 \* Data type includes **byte,byte[],int,int[],float,float[],string**

**Note**

- \* Same as syntax 3. Read to the end of the item by default.
- \* Reading data based on the configuration file Little Endian (DCBA) or Big Endian (ABCD).

**Syntax 5**

**? eip\_read\_input(**  
**string**

)

**Parameters**

**string** Item name

**Return**

**?** The data type returned by the item definition in the communication data table.  
 \* Data type includes **byte,byte[],int,int[],float,float[],string**

**Note**

- \* Same as syntax 3. Fill 0 as the starting shifted address of the item and read to the end of the item by default.
- \* Reading data based on the configuration file Little Endian (DCBA) or Big Endian (ABCD).

```
byte var_b = eip_read_input("O2T_StickStatus",0,1)
```

```
// 0x02
```

```
var_b = eip_read_input("O2T_StickStatus",0)
```

```
// 0x02
```

```
var_b = eip_read_input("O2T_StickStatus")
```

```
// 0x02
```

```
byte[] var_ba = eip_read_input("O2T_CtrlBox_DO",0,2)
```

```
// {0x00,0x04}
```

```
var_ba = eip_read_input("O2T_CtrlBox_DO")
```

```
// {0x00,0x04}
```

```
int[] var_ia = eip_read_input("O2T_Register_Int",0,12)
```

```
// byte[] = {0x00,0x00,0x00,0x01,0x00,0x00,0x00,0x02,0x00,0x00,0x00,0x03} (Little Endian) to int[]
```

```
// int[] = {0x00000001,0x00000002,0x00000003} (Little Endian)
```

```
// int[] = {1,2,3}
```

```
var_ia = eip_read_input("O2T_Register_Int",12)
```

```
// byte[] = {0x00,0x00,0x00,0x04,0x00,0x00,0x00,0x05,0x00,0x00,0x00,0x06,0x00,0x00,0x00,0x07,
0x00,0x00,0x00,0x08,0x00,0x00,0x00,0x09,0x00,0x00,0x00,0x0A,0x00,0x00,0x00,0x0B,
0x00,0x00,0x00,0x0C,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little
Endian)
```

```
to int[]
```

```
// int[] = {0x00000004,0x00000005,0x00000006,0x00000007,0x00000008,0x00000009,
```

```

 0x0000000A,0x0000000B,0x0000000C,0x00000000,0x00000000,0x00000000} (Little Endian)
// int[] = {4,5,6,7,8,9,10,11,12,0,0,0}

var_fa = eip_read_input("O2T_Register_Int")
// byte[] = {0x00,0x00,0x00,0x01,0x00,0x00,0x00,0x02,0x00,0x00,0x00,0x03,0x00,0x00,0x00,0x04,
// 0x00,0x00,0x00,0x05,0x00,0x00,0x00,0x06,0x00,0x00,0x00,0x07,0x00,0x00,0x00,0x08,
// 0x00,0x00,0x00,0x09,0x00,0x00,0x00,0x0A,0x00,0x00,0x00,0x0B,0x00,0x00,0x00,0x0C,
// 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian) to int[]
// int[] = {0x00000001,0x00000002,0x00000003,0x00000004,0x00000005,0x00000006,
// 0x00000007,0x00000008,0x00000009,0x0000000A,0x0000000B,0x0000000C,
// 0x00000000,0x00000000,0x00000000} (Little Endian)
// int[] = {1,2,3,4,5,6,7,8,9,10,11,12,0,0,0}

float[] var_fa = eip_read_input("O2T_Register_Float",4,12)
// byte[] = {0x3F,0x99,0x99,0x9A,0x3F,0xA6,0x66,0x66,0x40,0x06,0x66,0x66} (Big Endian) to float[]
// float[] = {0x3F99999A,0x3FA66666,0x40066666} (Big Endian)
// float[] = {1.2,1.3,2.1}

var_fa = eip_read_input("O2T_Register_Float",12)
// byte[] = {0x40,0x06,0x66,0x66,0x40,0x0C,0xCC,0xCD,0x40,0x13,0x33,0x33,0x00,0x00,0x00,0x00,
// 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
// 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
// (Big Endian) to float[]
// float[] = {0x40066666,0x400CCCD,0x40133333,0x00000000,0x00000000,0x00000000,
// 0x00000000,0x00000000,0x00000000,0x00000000,0x00000000,0x00000000} (Big Endian)
// float[] = {2.1,2.2,2.3,0,0,0,0,0,0,0,0,0}

var_fa = eip_read_input("O2T_Register_Float")
// byte[] = {0x3F,0x8C,0xCC,0xCD,0x3F,0x99,0x99,0x9A,0x3F,0xA6,0x66,0x66,0x40,0x06,0x66,0x66,
// 0x40,0x0C,0xCC,0xCD,0x40,0x13,0x33,0x33,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
// 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
// 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
// (Big Endian) to float[]
// float[] = {0x3F8CCCD,0x3F99999A,0x3FA66666,0x40066666,0x400CCCD,0x40133333,
// 0x00000000,0x00000000,0x00000000,0x00000000,0x00000000,0x00000000,
// 0x00000000,0x00000000,0x00000000} (Big Endian)
// float[] = { 1.1,1.2,1.3,2.1,2.2,2.3,0,0,0,0,0,0,0,0,0,0}

```

## 19.2 eip\_read\_input\_int()

Read the input table content and convert the data to the 32-bit integer.

### Syntax 1

```
int[] eip_read_input_int(
 int,
 int,
 int
)
```

#### Parameters

|     |                                                                                                     |
|-----|-----------------------------------------------------------------------------------------------------|
| int | Starting address                                                                                    |
| int | The address amount to read                                                                          |
| int | The conversion of the read data to an int array based on Little Endian (DCBA) or Big Endian (ABCD). |
| 0   | Little-Endian                                                                                       |
| 1   | Big-Endian                                                                                          |
| 2   | Based on the configuration file.                                                                    |

#### Return

int[] Return data in an integer array.

### Syntax 2

```
int[] eip_read_input_int(
 int,
 int
)
```

#### Parameters

|     |                            |
|-----|----------------------------|
| int | Starting address           |
| int | The address amount to read |

#### Note

Same as Syntax 1 with the parameter of the conversion of the read data defaults to **2**.  
 \*Convert the read data to an int array based on Little Endian (DCBA) or Big Endian.

```
int[] var_ia = eip_read_input_int(112,12,0)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian)
 to int[]
 // int[] = {0x00007FFF,0x0001869F,0xFFFF8000} (Little Endian)
 // int[] = {32767,99999,-32768}
var_ia = eip_read_input_int(112,11,0)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF} (Little Endian) to int[]
 // int[] = {0x00007FFF,0x0001869F,0x00FF8000} (Little Endian)
 // int[] = {32767,99999,16744448}
var_ia = eip_read_input_int(112,10,0)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80} (Little Endian)to int[]
 // int[] = {0x00007FFF,0x0001869F,0x00008000} (Little Endian)
 // int[] = {32767,99999,32768}

var_ia = eip_read_input_int(112,12,1)
```

```
// byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Big Endian) to int[]
// int[] = {0xFF7F0000,0x9F860100,0x0080FFFF} (Big Endian)
// int[] = {-8454144,-1618607872,8454143}
```

```
var_ia = eip_read_input_int(112,12,2)
// byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian)
to int[]
// int[] = {0x00007FFF,0x0001869F,0xFFFF8000} (Little Endian)
// int[] = {32767,99999,-32768}
```

```
var_ia = eip_read_input_int(112,12)
// byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian)
to int[]
// int[] = {0x00007FFF,0x0001869F,0xFFFF8000} (Little Endian)
// int[] = {32767,99999,-32768}
```

### Syntax 3

```
int eip_read_input_int(
 int
```

```
)
```

#### Parameter

**int** Starting address  
 \*Convert the read data to an integer based on Little Endian (DCBA) or Big Endian (ABCD).

#### Return

**int** Return data in an integer

#### Note

```
int var_i = eip_read_input_int(112)
// byte[] = {0xE4,0x07,0x00,0x00} (Little Endian) to int
// int = 0x000007E4 (Little Endian)
// int = 2020
var_i = eip_read_input_int(112)
// byte[] = {0x00,0x00,0x07,0xE4} (Big Endian) to int
// int = 0x000007E4 (Big Endian)
// int = 2020
```



## 19.3 eip\_read\_input\_float()

Read the input table content and convert the data to the 32-bit floating-point number.

### Syntax 1

```
float[] eip_read_input_float(
 int,
 int,
 int
)
```

#### Parameter

|     |                                                                                                      |
|-----|------------------------------------------------------------------------------------------------------|
| int | Starting address                                                                                     |
| int | The address amount to read                                                                           |
| int | The conversion of the read data to a float array based on Little Endian (DCBA) or Big Endian (ABCD). |
| 0   | Little-Endian                                                                                        |
| 1   | Big-Endian                                                                                           |
| 2   | Based on the configuration file.                                                                     |

#### Return

float[] Return data in a floating-point number array.

### Syntax 2

```
float[] eip_read_input_float(
 int,
 int
)
```

#### Parameter

|     |                            |
|-----|----------------------------|
| int | Starting address           |
| int | The address amount to read |

#### Note

Same as Syntax 1 with the parameter of the conversion of the read data defaults to 2.  
 \*Convert the read data to a float array based on Little Endian (DCBA) or Big Endian (ABCD).

```
float[] var_fa = eip_read_input_float(172,12,0)
 // byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00,0x40,0x40} (Little Endian)
 // to float[]
 // float[] = {0x3F800000,0x40000000,0x40400000} (Little Endian)
 // float[] = {1.0,2.0,3.0}
var_fa = eip_read_input_float(172,11,0)
 // byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00,0x40} (Little Endian) to float[]
 // float[] = {0x3F800000,0x40000000,0x00400000} (Little Endian)
 // float[] = {1.0,2.0,5.877472E-39}
var_fa = eip_read_input_float(172,10,0)
 // byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00} (Little Endian) to float[]
 // float[] = {0x3F800000,0x40000000,0x00000000} (Little Endian)
 // float[] = {1.0,2.0,0.0}
```

```
var_fa = eip_read_input_float(172,12,1)
// byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00,0x40,0x40} (Little Endian)
to float[]
// float[] = {0x0000803F,0x00000040,0x00004040} (Big Endian)
// float[] = {4.600603E-41,8.96831E-44,2.304856E-41}
```

```
var_fa = eip_read_input_float(172,12,2)
// byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00,0x40,0x40} (Little Endian)
to float[]
// float[] = {0x3F800000,0x40000000,0x40400000} (Little Endian)
// float[] = {1.0,2.0,3.0}
```

```
var_fa = eip_read_input_float(172,12)
// byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00,0x40,0x40} (Little Endian)
to float[]
// float[] = {0x3F800000,0x40000000,0x40400000} (Little Endian)
// float[] = {1.0,2.0,3.0}
```

### Syntax 3

```
float eip_read_input_float(
 int
)
```

#### Parameter

**int** Starting address  
 \*Convert the read data to a floating-point number based on Little Endian (DCBA) or Big Endian (ABCD).

#### Return

**Float** Return data in a floating-point number.

#### Note

```
float var_f = eip_read_input_float(172)
// byte[] = {0x00,0x00,0x80,0x3F} (Little Endian) to float
// float = 0x3F800000 (Little Endian)
// float = 1.0
var_f = eip_read_input_float(172)
// byte[] = {0x3F,0x80,0x00,0x00} (Big Endian) to float
// float = 0x3F800000 (Big Endian)
// float = 1.0
```

## 19.4 eip\_read\_input\_string()

Read the input table content and convert the data to the string encoded in UTF8.

### Syntax 1

```
string eip_read_input_string(
 int,
 int
)
```

#### Parameter

**int** Starting address  
**int** The address amount to read

#### Return

**string** Return data in a UTF8 string (ending with 0x00 encountered).

#### Note

```
string var_s = eip_read_input_string(104,16)
 // byte[] = {0x54,0x4D,0x35,0x2D,0x37,0x30,0x30,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // string = "TM5-700"

var_s = eip_read_input_string(104,32)
 // byte[] = {0x30,0x31,0x30,0x36,0x30,0x31,0x31,0x32,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
 // 0x54,0x4D,0x35,0x2D,0x37,0x30,0x30,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // string = "01060112"

var_s = eip_read_input_string(104,32)
 // byte[] = {0x61,0x62,0x63,0x64,0xE9,0x81,0x94,0xE6,0x98,0x8E,0xE6,0xA9,0x9F,0xE5,0x99,0xA8,
 // 0xE4,0xBA,0xBA,0x31,0x32,0x33,0x34,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // string = "abcd 達明機器人 1234"

var_s = eip_read_input_string(104,10)
 // byte[] = {0x61,0x62,0x63,0x64,0xE9,0x81,0x94,0xE6,0x98,0x8E}
 // string = "abcd 達明"

var_s = eip_read_input_string(104,8)
 // byte[] = {0x61,0x62,0x63,0x64,0xE9,0x81,0x94,0xE6}
 // string = "abcd 達◆"
```

## 19.5 eip\_read\_input\_bit()

Read the input table content and retrieve the  $n^{\text{th}}$  bit value of the data byte.

### Syntax 1

```
byte eip_read_input_bit(
 int,
 int
)
```

#### Parameter

**int** Starting address  
**int** The  $n^{\text{th}}$  bit value in the data byte

#### Return

**byte** Return data in byte.  
 Return 1 for bit value == 1.  
 Return 0 for bit value == 0.  
 \*Data in bit will return in byte.

#### Note

```
byte var_b = eip_read_input_bit(104,0)
// 0x30 get bit: "0"
// 0
var_b = eip_read_input_bit(104,5)
// 0x30 get bit: "5"
// 1
```

### Syntax 2

```
byte eip_read_input_bit(
 string,
 int
)
```

#### Parameter

**string** Item name  
**int** The  $n^{\text{th}}$  bit value

#### Return

**byte** Return data in byte.  
 Return 1 for bit value == 1.  
 Return 0 for bit value == 0.  
 \*Data in bit will return in byte.

#### Note

```
byte var_b = eip_read_input_bit("O2T_Register_Bit",0)
// byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,0,1,0,0,1,1,...} // total
// 1 get bit: "0"
var_b = eip_read_input_bit("O2T_Register_Bit",17)
// byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,0,1,0,0,1,1,...} // total
// 0 get bit: "17"
```

### Syntax 3

```
byte[] eip_read_input_bit(
```

```
int,
int,
int
```

```
)
```

#### Parameter

```
int Starting address
int Starting bit
int The amount of bit to read
```

#### Return

```
byte[] Return data in byte[].
 Return 1 for bit value == 1.
 Return 0 for bit value == 0.
 *Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].
```

#### Note

```
byte[] var_ba = eip_read_input_bit(104,0,20)
 // byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1,...} // total
 // byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
var_ba = eip_read_input_bit(104,12,8)
 // byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1,...} // total
 // byte[] = {1,1,0,1,0,0,1,1}
```

#### Syntax 4

```
byte[] eip_read_input_bit(
 string,
 int,
 int
```

```
)
```

#### Parameter

```
string Item name
int Starting bit
int The amount of bit to read
```

#### Return

```
byte[] Return data in byte[].
 Return 1 for bit value == 1.
 Return 0 for bit value == 0.
 *Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].
```

#### Note

```
byte[] var_ba = eip_read_input_bit("O2T_Register_Bit",0,20)
 // byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1,...} // total
 // byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
var_ba = eip_read_input_bit("O2T_Register_Bit",12,8)
 // byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1,...} // total
 // byte[] = {1,1,0,1,0,0,1,1}
```

## 19.6 eip\_read\_output()

Read the output table content.

### Syntax 1

```
byte[] eip_read_output(
 int,
 int
)
```

#### Parameter

**int** Starting address  
**int** The address length to read

#### Return

**byte[]** Return data in a byte array.

#### Note

```
byte[] var_ba = eip_read_output(300,8)
// {0x30,0x31,0x30,0x36,0x30,0x31,0x31,0x32}
```

### Syntax 2

```
byte eip_read_output(
 int
)
```

#### Parameter

**int** Starting address

#### Return

**Byte** Return data in byte.

#### Note

```
byte var_b = eip_read_output(300)
// 0x30
```

### Syntax 3

```
? eip_read_output(
 string,
 int,
 int
)
```

#### Parameter

**string** Item name  
**int** The starting shifted address of the item  
**int** The amount of the addresses to read

#### Return

**?** The data type returned by the item defined in the communication data table.  
 \* Data type includes **byte,byte[],int,int[],float,float[],string**

### Syntax 4

```
? eip_read_output(
 string,
 int,
```

)

**Parameter**

**string** Item name  
**int** The starting shifted address of the item

**Return**

**?** The data type returned by the item defined in the communication data table.  
 \* Data type includes **byte,byte[],int,int[],float,float[],string**

**Note**

- \* Same as syntax 3. Read to the end of the item by default.
- \* Reading data based on the configuration file Little Endian (DCBA) or Big Endian (ABCD).

**Syntax 5**

**? eip\_read\_output(**  
**string**

)

**Parameter**

**string** Item name

**Return**

**?** The data type returned by the item defined in the communication data table.  
 \* Data type includes **byte,byte[],int,int[],float,float[],string**

**Note**

- \* Same as syntax 3. Fill 0 as the starting shifted address of the item and read to the end of the item by default.
- \* Reading data based on the configuration file Little Endian (DCBA) or Big Endian (ABCD).

```
byte var_b = eip_read_output("ManualAuto",0,1)
```

```
// 0x02
```

```
var_b = eip_read_output("ManualAuto",0)
```

```
// 0x02
```

```
var_b = eip_read_output("ManualAuto")
```

```
// 0x02
```

```
byte[] var_ba = eip_read_output("Error_Code",0,4)
```

```
// {0x00,0x04,0x80,0x0C}
```

```
var_ba = eip_read_output("Error_Code",0,2)
```

```
// {0x00,0x04}
```

```
var_ba = eip_read_output("Error_Code")
```

```
// {0x00,0x04,0x80,0x0C}
```

```
int var_i = eip_read_output("Current_Time_Year")
```

```
// byte[] = {0x00,0x00,0x07,0xE4} (Little Endian) to int
```

```
// int = 0x000007E4 (Little Endian)
```

```
// int = 2020
```

```
int[] var_ia = eip_read_output("T2O_Register_Int",0,12)
```

```
// byte[] = {0x00,0x00,0x00,0x01,0x00,0x00,0x00,0x02,0x00,0x00,0x00,0x03} (Little Endian)
```

```
to int[]
```

```
// int[] = { 0x00000001,0x00000002,0x00000003} (Little Endian)
```

```

 // int[] = {1,2,3}
var_ia = eip_read_output("T2O_Register_Int",12)
 // byte[] = {0x00,0x00,0x00,0x04,0x00,0x00,0x00,0x05,0x00,0x00,0x00,0x06,0x00,0x00,0x00,0x07,
 // 0x00,0x00,0x00,0x08,0x00,0x00,0x00,0x09,0x00,0x00,0x00,0x0A,0x00,0x00,0x00,0x0B,
 // 0x00,0x00,0x00,0x0C,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // (Little Endian) to int[]
 // int[] = {0x00000004,0x00000005,0x00000006,0x00000007,0x00000008,0x00000009,
 // 0x0000000A,0x0000000B,0x0000000C,0x00000000,0x00000000,0x00000000} (Little Endian)
 // int[] = {4,5,6,7,8,9,10,11,12,0,0,0}
var_ia = eip_read_output("T2O_Register_Int")
 // byte[] = {0x00,0x00,0x00,0x01,0x00,0x00,0x00,0x02,0x00,0x00,0x00,0x03,0x00,0x00,0x00,0x04,
 // 0x00,0x00,0x00,0x05,0x00,0x00,0x00,0x06,0x00,0x00,0x00,0x07,0x00,0x00,0x00,0x08,
 // 0x00,0x00,0x00,0x09,0x00,0x00,0x00,0x0A,0x00,0x00,0x00,0x0B,0x00,0x00,0x00,0x0C,
 // 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian) to int[]
 // int[] = {0x00000001,0x00000002,0x00000003,0x00000004,0x00000005,0x00000006,
 // 0x00000007,0x00000008,0x00000009,0x0000000A,0x0000000B,0x0000000C,
 // 0x00000000,0x00000000,0x00000000} (Little Endian)
 // int[] = {1,2,3,4,5,6,7,8,9,10,11,12,0,0,0}

float var_f = eip_read_output("Current_TCP_Mass")
 // byte[] = {0x40,0x40,0x00,0x00} (Big Endian) to float
 // float = 0x40400000 (Big Endian)
 // float = 3.0

float[] var_fa = eip_read_output("Current_TCP_Value",4,12)
 // byte[] = {0x3F,0x99,0x99,0x9A,0x3F,0xA6,0x66,0x66,0x40,0x06,0x66,0x66} (Big Endian) to float[]
 // float[] = {0x3F99999A,0x3FA66666,0x40066666} (Big Endian)
 // float[] = {1.2,1.3,2.1}

var_fa = eip_read_output("Current_TCP_Value",12)
 // byte[] = { 0x40,0x06,0x66,0x66,0x40,0x0C,0xCC,0xCD,0x40,0x13,0x33,0x33} (Big Endian)
 // to float[]
 // float[] = {0x40066666,0x400CCCCD,0x40133333} (Big Endian)
 // float[] = {2.1,2.2,2.3}
var_fa = eip_read_output("Current_TCP_Value")
 // byte[] = {0x3F,0x8C,0xCC,0xCD,0x3F,0x99,0x99,0x9A,0x3F,0xA6,0x66,0x66,0x40,0x06,0x66,0x66,
 // 0x40,0x0C,0xCC,0xCD,0x40,0x13,0x33,0x33} (Big Endian) to float[]
 // float[] = {0x3F8CCCCD,0x3F99999A,0x3FA66666,0x40066666,0x400CCCCD,0x40133333}
 // (Big Endian)
 // float[] = { 1.1,1.2,1.3,2.1,2.2,2.3}

string var_s = eip_read_output ("ControlBoxID",0,16)
 // byte[] = {0x30,0x31,0x30,0x36,0x30,0x31,0x31,0x32,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // string = "01060112"
var_s = eip_read_output ("ControlBoxID",4,3)
 // byte[] = {0x30,0x31,0x31}
 // string = "011"
var_s = eip_read_output ("ControlBoxID")

```



```
// byte[] = {0x30,0x31,0x30,0x36,0x30,0x31,0x31,0x32,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
// string = "01060112"
```

## 19.7 eip\_read\_output\_int()

Read the output table content and convert the data to the 32-bit integer.

### Syntax 1

```
int[] eip_read_output_int(
 int,
 int,
 int
)
```

#### Parameter

|     |                                                                                                     |
|-----|-----------------------------------------------------------------------------------------------------|
| int | Starting address                                                                                    |
| int | The address amount to read                                                                          |
| int | The conversion of the read data to an int array based on Little Endian (DCBA) or Big Endian (ABCD). |
| 0   | Little-Endian                                                                                       |
| 1   | Big-Endian                                                                                          |
| 2   | Based on the configuration file.                                                                    |

#### Return

int[] Return data in an integer array.

### Syntax 2

```
int[] eip_read_output_int(
 int,
 int
)
```

#### Parameter

|     |                            |
|-----|----------------------------|
| int | Starting address           |
| int | The address amount to read |

#### Note

Same as Syntax 1 with the parameter of the conversion of the read data defaults to **2**.

\* Convert the read data to an int array based on Little Endian (DCBA) or Big Endian (ABCD).

```
int[] var_ia = eip_read_output_int(308,12,0)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian)
 to int[]
 // int[] = {0x00007FFF,0x0001869F,0xFFFF8000} (Little Endian)
 // int[] = {32767,99999,-32768}
var_ia = eip_read_output_int(308,11,0)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF} (Little Endian) to int[]
 // int[] = {0x00007FFF,0x0001869F,0x00FF8000} (Little Endian)
 // int[] = {32767,99999,16744448}
var_ia = eip_read_output_int(308,10,0)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80} (Little Endian)to int[]
 // int[] = {0x00007FFF,0x0001869F,0x00008000} (Little Endian)
 // int[] = {32767,99999,32768}

var_ia = eip_read_output_int(308,12,1)
```

```
// byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian)
to int[]
// int[] = {0xFF7F0000,0x9F860100,0x0080FFFF} (Big Endian)
// int[] = {-8454144,-1618607872,8454143}
```

```
var_ia = eip_read_output_int(308,12,2)
// byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian)
to int[]
// int[] = {0x00007FFF,0x0001869F,0xFFFF8000} (Little Endian)
// int[] = {32767,99999,-32768}
```

```
var_ia = eip_read_output_int(308,12)
// byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian)
to int[]
// int[] = {0x00007FFF,0x0001869F,0xFFFF8000} (Little Endian)
// int[] = {32767,99999,-32768}
```

### Syntax 3

```
int eip_read_output_int(
 int
)
```

#### Parameter

**int** Starting address  
 \* Convert the read data to an integer based on Little Endian (DCBA) or Big Endian (ABCD).

#### Return

**int** Return data in an integer

#### Note

```
int var_i = eip_read_output_int(308)
// byte[] = {0xE4,0x07,0x00,0x00} (Little Endian) to int
// int = 0x000007E4 (Little Endian)
// int = 2020
var_i = eip_read_output_int(308)
// byte[] = {0x00,0x00,0x07,0xE4} (Big Endian) to int
// int = 0x000007E4 (Big Endian)
// int = 2020
```

## 19.8 eip\_read\_output\_float()

Read the output table content and convert the data to the 32-bit floating-point number.

### Syntax 1

```
float[] eip_read_output_float(
 int,
 int,
 int
)
```

#### Parameter

|     |                                                                                                      |
|-----|------------------------------------------------------------------------------------------------------|
| int | Starting address                                                                                     |
| int | The address amount to read                                                                           |
| int | The conversion of the read data to a float array based on Little Endian (DCBA) or Big Endian (ABCD). |
| 0   | Little-Endian                                                                                        |
| 1   | Big-Endian                                                                                           |
| 2   | Based on the configuration file.                                                                     |

#### Return

float[] Return data in a floating-point number array.

### Syntax 2

```
float[] eip_read_output_float(
 int,
 int
)
```

#### Parameter

|     |                            |
|-----|----------------------------|
| int | Starting address           |
| int | The address amount to read |

#### Note

Same as Syntax 1 with the parameter of the conversion of the read data defaults to 2.

\*Convert the read data to a float array based on Little Endian (DCBA) or Big Endian (ABCD).

```
float[] var_fa = eip_read_output_float(368,12,0)
 // byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00,0x40,0x40} (Little Endian)
 // to float[]
 // float[] = {0x3F800000,0x40000000,0x40400000} (Little Endian)
 // float[] = {1.0,2.0,3.0}
var_fa = eip_read_output_float(368,11,0)
 // byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00,0x40} (Little Endian) to float[]
 // float[] = {0x3F800000,0x40000000,0x00400000} (Little Endian)
 // float[] = {1.0,2.0,5.877472E-39}
var_fa = eip_read_output_float(368,10,0)
 // byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00} (Little Endian) to float[]
 // float[] = {0x3F800000,0x40000000,0x00000000} (Little Endian)
 // float[] = {1.0,2.0,0.0}
```

```
var_fa = eip_read_output_float(368,12,1)
// byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00,0x40,0x40} (Little Endian)
to float[]
// float[] = {0x0000803F,0x00000040,0x00004040} (Big Endian)
// float[] = {4.600603E-41,8.96831E-44,2.304856E-41}
```

```
var_fa = eip_read_output_float(368,12,2)
// byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00,0x40,0x40} (Little Endian)
to float[]
// float[] = {0x3F800000,0x40000000,0x40400000} (Little Endian)
// float[] = {1.0,2.0,3.0}
```

```
var_fa = eip_read_output_float(368,12)
// byte[] = {0x00,0x00,0x80,0x3F,0x00,0x00,0x00,0x40,0x00,0x00,0x40,0x40} (Little Endian)
to float[]
// float[] = {0x3F800000,0x40000000,0x40400000} (Little Endian)
// float[] = {1.0,2.0,3.0}
```

### Syntax 3

```
float eip_read_output_float(
 int
)
```

#### Parameter

**int** Starting address  
 \*Convert the read data to a floating-point number based on Little Endian (DCBA) or Big

#### Return

**float** Return data in a floating-point number

#### Note

```
float var_f = eip_read_output_float(368)
// byte[] = {0x00,0x00,0x80,0x3F} (Little Endian) to float
// float = 0x3F800000 (Little Endian)
// float = 1.0
var_f = eip_read_output_float(368)
// byte[] = {0x3F,0x80,0x00,0x00} (Big Endian) to float
// float = 0x3F800000 (Big Endian)
// float = 1.0
```

## 19.9 eip\_read\_output\_string()

Read the output table content and convert the data to the string encoded in UTF8.

### Syntax 1

```
string eip_read_output_string(
 int,
 int
)
```

#### Parameter

**int** Starting address  
**int** The address amount to read

#### Return

**string** Return data in a UTF8 string (ending with 0x00 encountered).

#### Note

```
string var_s = eip_read_output_string(300,16)
 // byte[] = {0x54,0x4D,0x35,0x2D,0x37,0x30,0x30,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // string = "TM5-700"

var_s = eip_read_output_string(300,32)
 // byte[] = {0x30,0x31,0x30,0x36,0x30,0x31,0x31,0x32,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
 // 0x54,0x4D,0x35,0x2D,0x37,0x30,0x30,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // string = "01060112"

var_s = eip_read_output_string(300,32)
 // byte[] = {0x61,0x62,0x63,0x64,0xE9,0x81,0x94,0xE6,0x98,0x8E,0xE6,0xA9,0x9F,0xE5,0x99,0xA8,
 // 0xE4,0xBA,0xBA,0x31,0x32,0x33,0x34,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // string = "abcd 達明機器人 1234"

var_s = eip_read_output_string(300,10)
 // byte[] = {0x61,0x62,0x63,0x64,0xE9,0x81,0x94,0xE6,0x98,0x8E}
 // string = "abcd 達明"

var_s = eip_read_output_string(300,8)
 // byte[] = {0x61,0x62,0x63,0x64,0xE9,0x81,0x94,0xE6}
 // string = "abcd 達◆"
```

## 19.10 eip\_read\_output\_bit()

Read the output table content and retrieve the  $n^{\text{th}}$  bit value of the data byte.

### Syntax 1

```
byte eip_read_output_bit(
 int,
 int
)
```

#### Parameter

**int** Starting address  
**int** The  $n^{\text{th}}$  bit value in the data byte

#### Return

**Byte** Return data in byte.  
 Return 1 for bit value == 1.  
 Return 0 for bit value == 0.  
 \*Data in bit will return in byte.

#### Note

```
byte var_b = eip_read_output_bit(300,0)
// 0x30 get bit: "0"
// 0
var_b = eip_read_output_bit(300,5)
// 0x30 get bit: "5"
// 1
```

### Syntax 2

```
byte eip_read_output_bit(
 string,
 int
)
```

#### Parameter

**string** Item name  
**int** The  $n^{\text{th}}$  bit value

#### Return

**byte** Return data in byte.  
 Return 1 for bit value == 1.  
 Return 0 for bit value == 0.  
 \*Data in bit will return in byte.

#### Note

```
byte[] var_data = {57,184,12}
eip_write_output(300,var_data,3)
// {00111001,10111000,00001100} (binary)
byte var_b = eip_read_output_bit("T2O_Register_Bit",0)
// 1
var_b = eip_read_output_bit("T2O_Register_Bit",17)
// 0
```

### Syntax 3

```
byte[] eip_read_output_bit(
 int,
 int,
 int
)
```

#### Parameter

int Starting address  
 int Starting bit  
 int The amount of bit to read

#### Return

byte[] Return data in byte[].  
 Return 1 for bit value == 1.  
 Return 0 for bit value == 0.  
 \*Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].

#### Note

```
byte[] var_data = {57,184,12}
eip_write_output(300,var_data,3)
// {00111001,10111000,00001100} (binary)
byte[] var_ba = eip_read_output_bit(300,0,20)
// byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
var_ba = eip_read_output_bit(300,12,8)
// byte[] = {1,1,0,1,0,0,1,1}
```

#### Syntax 4

```
byte[] eip_read_output_bit(
 string,
 int,
 int
)
```

#### Parameter

string Item name  
 int Starting bit  
 int The amount of bit to read

#### Return

byte[] Return data in byte[].  
 Return 1 for bit value == 1.  
 Return 0 for bit value == 0.  
 \*Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].

#### Note

```
byte[] var_data = {57,184,12}
eip_write_output(300,var_data,3)
// {00111001,10111000,00001100} (binary)
byte[] var_ba = eip_read_output_bit("T2O_Register_Bit",0,20)
// byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
var_ba = eip_read_output_bit("T2O_Register_Bit",12,8)
// byte[] = {1,1,0,1,0,0,1,1}
```



## 19.11 eip\_write\_output()

Write data to the output table.

### Syntax 1

```
bool eip_write_output(
 int,
 ?,
 int
)
```

#### Parameter

|      |                                                                                        |
|------|----------------------------------------------------------------------------------------|
| int  | Starting address                                                                       |
| ?    | The data to write                                                                      |
|      | * Available data types include <b>byte, byte[], int, int[], float, float[], string</b> |
| int  | The maximum amount of the address to write                                             |
| > 0  | Valid data length. Write by the amount of the address.                                 |
| <= 0 | Invalid data length. Write by the complete length of the data to write.                |

#### Return

|           |                                                              |
|-----------|--------------------------------------------------------------|
| bool True | Write successfully.                                          |
| False     | Write unsuccessfully.                                        |
|           | 1. If the data to write is an empty string or an empty array |
|           | 2. Unable to send and receive correctly.                     |

#### Note

\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

### Syntax 2

```
bool eip_write_output(
 int,
 ?
)
```

#### Parameter

|     |                                                                                        |
|-----|----------------------------------------------------------------------------------------|
| int | Starting address                                                                       |
| ?   | The data to write                                                                      |
|     | * Available data types include <b>byte, byte[], int, int[], float, float[], string</b> |

#### Return

|           |                                                              |
|-----------|--------------------------------------------------------------|
| bool True | Write successfully.                                          |
| False     | Write unsuccessfully.                                        |
|           | 1. If the data to write is an empty string or an empty array |
|           | 2. Unable to send and receive correctly.                     |

#### Note

Same as syntax 1. Write with the full length of the data to write by default.

\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

### Syntax 3

```
bool eip_write_output(
 int,
```

```
?,
int,
int
)
```

### Parameter

|     |                                                                                                                                                             |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| int | Starting address                                                                                                                                            |
| ?   | The data to write<br>* Available data types include <b>byte</b> , <b>byte[]</b> , <b>int</b> , <b>int[]</b> , <b>float</b> , <b>float[]</b> , <b>string</b> |
| int | Starting address of the data to write                                                                                                                       |
| int | The amount of the address to write                                                                                                                          |

### Return

|           |                       |
|-----------|-----------------------|
| bool True | Write successfully.   |
| False     | Write unsuccessfully. |

1. If the data to write is an empty string or an empty array
2. Unable to send and receive correctly.

### Note

\*\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

```
byte var_data = 255
eip_write_output(300,var_data,1)
byte var_b = eip_read_output(300)
// 0xFF

byte[] var_data = {1,127,255}
eip_write_output(300,var_data,3)
byte[] var_ba = eip_read_output(300,3)
// {0x01,0x7F,0xFF}
eip_write_output(300,var_data,2)
var_ba = eip_read_output(300,3)
// {0x01,0x7F,0x00}
eip_write_output(300,var_data,-1)
var_ba = eip_read_output(300,3)
// {0x00,0x7F,0xFF}

int var_data = 32767
eip_write_output(308,var_data,4)
int var_i = eip_read_output_int(308)
// byte[] = {0xFF,0x7F,0x00,0x00} (Little Endian) to int
// int = 0x00007FFF (Little Endian)
// int = 32767
eip_write_output(308,var_data,1)
var_i = eip_read_output_int(308)
// byte[] = {0xFF,0x00,0x00,0x00} (Little Endian) to int
// int = 0x000000FF (Little Endian)
// int = 255

int[] var_data = {32767,99999,-32768}
```

```

eip_write_output(308,var_data,12)
int[] var_ia = eip_read_output_int(308,12)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian)
 to int[]
 // int[] = {0x00007FFF,0x0001869F,0xFFFF8000} (Little Endian)
 // int[] = {32767,99999,-32768}
eip_write_output(308,var_data,3)
var_ia = eip_read_output_int(308,12)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian)
 to int[]
 // int[] = {0x00007FFF,0x00000000,0x00000000} (Little Endian)
 // int[] = {32767,0,0}
eip_write_output(308,var_data,11)
var_ia = eip_read_output_int(308,12)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0x00} (Little Endian)
 to int[]
 // int[] = {0x00007FFF,0x0001869F,0x00FF8000} (Little Endian)
 // int[] = {32767,99999,16744448}
eip_write_output(308,var_data,4,4)
var_ia = eip_read_output_int(308,12)
 // byte[] = {0x9F,0x86,0x01,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian)
 to int[]
 // int[] = {0x0001869F,0x00000000,0x00000000} (Little Endian)
 // int[] = {99999,0,0}

float var_data = -10.0
eip_write_output(368,var_data,4)
float var_f = eip_read_output_float(368)
 // byte[] = {0x00,0x00,0x20,0xC1} (Little Endian) to float
 // float = 0xC1200000 (Little Endian)
 // float = -10.0
eip_write_output(368,var_data,1)
var_f = eip_read_output_float(368)
 // byte[] = {0x00,0x00,0x00,0x00} (Little Endian) to float
 // float = 0x00000000 (Little Endian)
 // float = 0

float[] var_data = {-10.0,3.3,123.45}
eip_write_output(368,var_data,12)
float[] var_fa = eip_read_output_float(368,12)
 // byte[] = {0x00,0x00,0x20,0xC1,0x33,0x33,0x53,0x40,0x66,0xE6,0xF6,0x42} (Little Endian)
 to float[]
 // float[] = {0xC1200000,0x40533333,0x42F6E666} (Little Endian)
 // float[] = {-10,3.3,123.45}
eip_write_output(368,var_data,3)
var_fa = eip_read_output_float(368,12)
 // byte[] = {0x00,0x00,0x20,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian)
 to float[]

```

```

 // float[] = {0x00200000,0x00000000,0x00000000} (Little Endian)
 // float[] = {2.938736E-39,0,0}
 eip_write_output(368,var_data,11)
 var_fa = eip_read_output_float(368,12)
 // byte[] = {0x00,0x00,0x20,0xC1,0x33,0x33,0x53,0x40,0x66,0xE6,0xF6,0x00} (Little Endian)
 to float[]
 // float[] = {0xC1200000,0x40533333,0x00F6E666} (Little Endian)
 // float[] = {-10,3.3,2.267418E-38}
 eip_write_output(368,var_data,4,4)
 var_fa = eip_read_output_float(368,12)
 // byte[] = {0x33,0x33,0x53,0x40,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian)
 to float[]
 // float[] = {0x40533333,0x00000000,0x00000000} (Little Endian)
 // float[] = {3.3,0,0}

 string var_data = "abcd 達明機器人 1234"
 eip_write_output(300,var_data,32)
 string var_s = eip_read_output_string(300,32)
 // byte[] = {0x61,0x62,0x63,0x64,0xE9,0x81,0x94,0xE6,0x98,0x8E,0xE6,0xA9,0x9F,0xE5,0x99,0xA8,
 // 0xE4,0xBA,0xBA,0x31,0x32,0x33,0x34,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // string = "abcd 達明機器人 1234"
 eip_write_output(300,var_data,10)
 var_s = eip_read_output_string(300,32)
 // byte[] = { 0x61,0x62,0x63,0x64,0xE9,0x81,0x94,0xE6,0x98,0x8E,0x00,0x00,0x00,0x00,0x00,0x00,
 // 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // string = "abcd 達明"
 eip_write_output(300,var_data,8)
 var_s = eip_read_output_string(300,32)
 // byte[] = {0x61,0x62,0x63,0x64,0xE9,0x81,0x94,0xE6,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
 // 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // string = "abcd 達💎"
 eip_write_output(300,var_data,4,15)
 var_s = eip_read_output_string(300,15)
 // byte[] = {0xE9,0x81,0x94,0xE6,0x98,0x8E,0xE6,0xA9,0x9F,0xE5,0x99,0xA8,0xE4,0xBA,0xBA}
 // string = "達明機器人"

```

#### Syntax 4

```

bool eip_write_output(
 string,
 int,
 ?,
 int,
 int
)

```

##### Parameter

|        |                                          |
|--------|------------------------------------------|
| string | Item name                                |
| int    | The starting shifted address of the item |
| ?      | The data to write                        |

\* Available data types include **byte,byte[],int,int[],float,float[],string**

**int** Starting address of the data to write  
**int** The amount of the address to write

#### Return

**bool True** Write successfully.

**False** Write unsuccessfully.

1. If the data to write is an empty string or an empty array
2. Unable to send and receive correctly.

#### Note

\*\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

### Syntax 5

```
bool eip_write_output(
 string,
 int,
 ?
 int
)
```

#### Parameter

**string** Item name

**int** The starting shifted address of the item

**?** The data to write

\* Available data types include **byte,byte[],int,int[],float,float[],string**

**int** Starting address of the data to write

#### Return

**bool True** Write successfully.

**False** Write unsuccessfully.

1. If the data to write is an empty string or an empty array
2. Unable to send and receive correctly.

#### Note

Same as syntax 4. Write with the full length of the data to write by default.

\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

### Syntax 6

```
bool eip_write_output(
 string,
 int,
 ?
)
```

#### Parameter

**string** Item name

**int** The starting shifted address of the item

**?** The data to write

\* Available data types include **byte,byte[],int,int[],float,float[],string**

#### Return

**bool True** Write successfully.

**False** Write unsuccessfully.

1. If the data to write is an empty string or an empty array
2. Unable to send and receive correctly.

### Note

Same as syntax 4. Fill 0 as the starting address to write and write with the full length of the data to write by default.

\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

### Syntax 7

```
bool eip_write_output(
 string,
 ?
)
```

#### Parameter

**string** Item name  
**?** The data to write  
 \* Available data types include **byte, byte[], int, int[], float, float[], string**

#### Return

**bool True** Write successfully.  
**False** Write unsuccessfully.  
 1. If the data to write is an empty string or an empty array  
 2. Unable to send and receive correctly.

### Note

Same as syntax 4. Fill 0 as the starting shifted address and the starting address to write as well as write with the full length of the data to write by default.

\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

```
int[] var_data = {32767,99999,-32768}
eip_write_output("T2O_Register_Int",0,var_data,0,12)
int[] var_ia = eip_read_output_int(308,12)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian)
 to int[]
 // int[] = {0x00007FFF,0x0001869F,0xFFFF8000} (Little Endian)
 // int[] = {32767,99999,-32768}
eip_write_output("T2O_Register_Int",4,var_data,4,4)
var_ia = eip_read_output_int(308,12)
 // byte[] = {0x00,0x00,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x00,0x00,0x00} (Little Endian)
 to int[]
 // int[] = {0x00000000,0x0001869F,0x00000000} (Little Endian)
 // int[] = {0,99999,0}
eip_write_output("T2O_Register_Int",4,var_data)
var_ia = eip_read_output_int(308,20)
 // byte[] = {0x00,0x00,0x00,0x00,0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF,
 0x00,0x00,0x00,0x00} (Little Endian) to int[]
 // int[] = {0x00000000,0x00007FFF,0x0001869F,0xFFFF8000,0x00000000} (Little Endian)
 // int[] = {0,32767,99999,-32768,0}
```

```

float[] var_data = {-10.0,3.3,123.45}
eip_write_output("T2O_Register_Float",0,var_data,0,12)
float[] var_fa = eip_read_output_float(368,12)
 // byte[] = {0x00,0x00,0x20,0xC1,0x33,0x33,0x53,0x40,0x66,0xE6,0xF6,0x42} (Little Endian)
 to float[]
 // float[] = {0xC1200000,0x40533333,0x42F6E666} (Little Endian)
 // float[] = {-10,3.3,123.45}
eip_write_output("T2O_Register_Float",4,var_data,4,8)
var_fa = eip_read_output_float(368,12)
 // byte[] = {0x00,0x00,0x00,0x00,0x33,0x33,0x53,0x40,0x66,0xE6,0xF6,0x42} (Little Endian)
 to float[]
 // float[] = {0x00000000,0x40533333,0x42F6E666} (Little Endian)
 // float[] = {0,3.3,123.45}
eip_write_output("T2O_Register_Float",8,var_data)
var_fa = eip_read_output_float(368,20)
 // byte[] = {0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x20,0xC1,0x33,0x33,0x53,0x40,
 0x66,0xE6,0xF6,0x42} (Little Endian) to float[]
 // float[] = {0x00000000,0x00000000,0xC1200000,0x40533333,0x42F6E666} (Little Endian)
 // float[] = {0,0,-10,3.3,123.45}

```

## 19.12 eip\_write\_output\_bit()

Write content to the  $n^{\text{th}}$  bit value of the data byte in the output table.

### Syntax 1

```
bool eip_write_output_bit(
 int,
 int,
 int
)
```

#### Parameter

|     |                                 |
|-----|---------------------------------|
| int | Starting address                |
| int | The $n^{\text{th}}$ bit value   |
| int | The data to write               |
|     | *Data in bit will write in int. |

#### Return

|      |       |                       |
|------|-------|-----------------------|
| Bool | True  | Write successfully.   |
|      | False | Write unsuccessfully. |

1. Unable to send correctly and receive .

#### Note

```
byte var_data = 240
eip_write_output(300,var_data)
byte var_b = eip_read_output(300)
// 0xF0
eip_write_output_bit(300,1,1)
var_b = eip_read_output_bit(300,1)
// 0xF2 get bit: "1"
// 1
eip_write_output_bit(300,7,0)
var_b = eip_read_output_bit(300,7)
// 0x72 get bit: "7"
// 0
```

### Syntax 2

```
bool eip_write_output_bit(
 string,
 int,
 int
)
```

#### Parameter

|        |                                 |
|--------|---------------------------------|
| string | Item name                       |
| int    | The $n^{\text{th}}$ bit value   |
| int    | The data to write               |
|        | *Data in bit will write in int. |

#### Return

|      |       |                       |
|------|-------|-----------------------|
| bool | True  | Write successfully.   |
|      | False | Write unsuccessfully. |

1. Unable to send correctly and receive .

#### Note

```
byte var_data = 240
```



```

eip_write_output(300,var_data)
byte var_b = eip_read_output(300)
 // 0xF0
eip_write_output_bit("T2O_Register_Bit",1,1)
var_b = eip_read_output_bit(300,1)
 // 0xF2 get bit: "1"
 // 1
eip_write_output_bit("T2O_Register_Bit",7,0)
var_b = eip_read_output_bit(300,7)
 // 0x72 get bit: "7"
 // 0

```

### Syntax 3

```

bool eip_write_output_bit(
 int,
 int,
 byte[],
 int,
 int
)

```

#### Parameter

|        |                                                                                     |
|--------|-------------------------------------------------------------------------------------|
| int    | Starting address                                                                    |
| int    | The n <sup>th</sup> bit value                                                       |
| byte[] | The data to write                                                                   |
|        | *Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1]. |
| int    | The starting bit of the data to write                                               |
| int    | The bit amount of the data to write                                                 |

#### Return

|           |                       |                                           |
|-----------|-----------------------|-------------------------------------------|
| bool True | Write successfully.   |                                           |
| False     | Write unsuccessfully. | 1. Unable to send correctly and receive . |

#### Note

Bit value = 1 for byte value >=1  
 Bit value = 0 for byte value ==0

### Syntax 4

```

bool eip_write_output_bit(
 int,
 int,
 byte[],
 int
)

```

#### Parameter

|        |                                                                                     |
|--------|-------------------------------------------------------------------------------------|
| int    | Starting address                                                                    |
| int    | The n <sup>th</sup> bit value                                                       |
| byte[] | The data to write                                                                   |
|        | *Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1]. |
| int    | The starting bit of the data to write                                               |

#### Return

`bool True` Write successfully.  
`False` Write unsuccessfully. 1. Unable to send correctly and receive .

#### Note

\*Same as syntax 3. Write with the full length of the rest data to write.

Bit value = 1 for byte value >=1

Bit value = 0 for byte value ==0

#### Syntax 5

```
bool eip_write_output_bit(
 int,
 int,
 byte[]
)
```

#### Parameter

`int` Starting address

`int` The n<sup>th</sup> bit value

`byte[]` The data to write

\*Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].

#### Return

`bool True` Write successfully.

`False` Write unsuccessfully. 1. Unable to send correctly and receive .

#### Note

\*Same as syntax 3. Fill 0 as the starting bit to write as well as write with the full length of the data to write by default.

Bit value = 1 for byte value >=1

Bit value = 0 for byte value ==0

```
byte[] var_data = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
```

```
eip_write_output_bit(300,0,var_data,0,20)
```

```
byte[] var_ba = eip_read_output (300,0,3)
```

```
// byte[] = {0x39,0xB8,0x0C}
```

```
var_ba = eip_read_output_bit(300,0,20)
```

```
// byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
```

```
eip_write_output_bit(300,3,var_data,5,10)
```

```
var_ba = eip_read_output (300,0,3)
```

```
// byte[] = {0x08,0x0E,0x00}
```

```
var_ba = eip_read_output_bit(300,0,20)
```

```
// byte[] = {0,0,0,1,0,0,0,0,0,1,1,1,0,0,0,0,0,0,0,0}
```

#### Syntax 6

```
bool eip_write_output_bit(
 string,
 int,
 byte[],
 int,
 int
)
```

#### Parameter

**string**    Item name  
**int**       Starting bit  
**byte[]**    The data to write  
             \*Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].  
**int**       The starting bit to write data  
**int**       The bit amount of the data to write

**Return**

**bool True**    Write successfully.  
**False**       Write unsuccessfully.       1. Unable to send correctly and receive .

**Note**

Bit value = 1 for byte value >=1  
 Bit value = 0 for byte value ==0

**Syntax 7**

```

bool eip_write_output_bit(
 string,
 int,
 byte[],
 int
)

```

**Parameter**

**string**    Item name  
**int**       Starting bit  
**byte[]**    The data to write  
             \*Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].  
**int**       The starting bit to write data

**Return**

**bool True**    Write successfully.  
**False**       Write unsuccessfully.       1. Unable to send correctly and receive .

**Note**

\*Same as syntax 6. Write with the full length of the rest data to write.  
 Bit value = 1 for byte value >=1  
 Bit value = 0 for byte value ==0

**Syntax 8**

```

bool eip_write_output_bit(
 string,
 int,
 byte[]
)

```

**Parameter**

**string**    Item name  
**int**       Starting bit  
**byte[]**    The data to write  
             \*Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].

**Return**

**bool True**    Write successfully.  
**False**       Write unsuccessfully.       1. Unable to send correctly and receive .

**Note**

\*Same as syntax 6. Fill 0 as the starting bit to write as well as write with the full length of the data to write by default.

Bit value = 1 for byte value >=1

Bit value = 0 for byte value ==0

```
byte[] var_data = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
eip_write_output_bit("T2O_Register_Bit",0,var_data,0,20)
byte[] var_ba = eip_read_output (300,3)
 // byte[] = {0x39,0xB8,0x0C}
var_ba = eip_read_output_bit(300,0,20)
 // byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
eip_write_output_bit("T2O_Register_Bit",3,var_data,5,10)
var_ba = eip_read_output (300,3)
 // byte[] = {0x08,0x0E,0x00}
var_ba = eip_read_output_bit(300,0,20)
 // byte[] = {0,0,0,1,0,0,0,0,0,1,1,1,0,0,0,0,0,0,0,0}
```

## 20. EtherCAT Functions

The robot communicates with external controllers via the EtherCAT communication protocol. In the mechanism of the EtherCAT communication protocol, the robot works as an EtherCAT IO device for external devices to read and write the robot data. At the same time, TMflow can monitor the robot's "data table received from external devices" and "data table sent to external devices" via EtherCAT functions. Additionally, TMflow can modify the user-defined areas within the "data table sent to external devices."

### Communication Data Table

The data table is composed of input data and output data. The Input Data Table is for external devices posting on the robot, and the Output Data Table is for the robot sending to external devices. Both data tables come with the System Definition Section and the Custom Definition Section for data.

1. System Definition Section: Items and settings are defined by the robot, and the data contents are updated by the robot or external devices. The defined items are robot status relevant such as robot bases, project status, control box status, or input/output status relevant such as digital I/Os and analog I/Os. Users can use EtherCAT functions to read the input data table and the output data table in the system definition section.
2. Custom Definition Section: Items and settings are defined by users, and the data contents are updated by users or external devices. While the project editing, users can use EtherCAT functions to read and write the output data table in the custom definition section or read input data table in the custom definition section. They can also use the custom definition section as a data exchange register between the project and external devices.

| Communication Data Table<br>(at the robot's viewpoint) | Data Section              | TMflow EtherCAT Function Permissions | External Device Permissions |
|--------------------------------------------------------|---------------------------|--------------------------------------|-----------------------------|
| Input Data Table                                       | System Definition Section | Read                                 | Write                       |
|                                                        | Custom Definition Section | Read                                 | Write                       |
| Output Data Table                                      | System Definition Section | Read                                 | Read                        |
|                                                        | Custom Definition Section | Read/Write                           | Read                        |

## 20.1 ethercat\_read\_input()

Read the input table content.

### Syntax 1

```
byte[] ethercat_read_input(
 int,
 int
)
```

#### Parameter

**int** Starting address  
**int** The address amount to read

#### Return

**byte[]** Return data in a byte array.

#### Note

```
byte[] var_ba = ethercat_read_input(4,20)
// {0x45,0x74,0x68,0x65,0x72,0x43,0x41,0x54,0x5F,0x54,0x65,0x73,0x74,0x5F,0x31,0x00,0x00,0x00,0x00,0x00}
```

### Syntax 2

```
byte ethercat_read_input(
 int
)
```

#### Parameter

**int** Starting address

#### Return

**byte** Return data in byte.

#### Note

```
byte var_b = ethercat_read_input(4)
// 0x45
```

**Syntax 3**

```
? ethercat_read_input(
 string,
 int,
 int
)
```

**Parameter**

|        |                                          |
|--------|------------------------------------------|
| string | Item name                                |
| int    | The starting shifted address of the item |
| int    | The amount of the addresses to read      |

**Return**

? The data type returned by the item definition in the communication data table.  
 \* Data type includes **byte,byte[],int,int[],float,float[],string**

**Syntax 4**

```
? ethercat_read_input(
 string,
 int
)
```

**Parameter**

|        |                                          |
|--------|------------------------------------------|
| string | Item name                                |
| int    | The starting shifted address of the item |

**Return**

? The data type returned by the item definition in the communication data table.  
 \* Data type includes **byte,byte[],int,int[],float,float[],string**

**Note**

- \* Same as syntax 3. Read to the end of the item by default.
- \* Reading data based on the configuration file Little Endian (DCBA) or Big Endian (ABCD).

**Syntax 5**

```
? ethercat_read_input(
 string
)
```

**Parameter**

**string**    Item name

**Return**

**?**    The data type returned by the item definition in the communication data table.  
 \* Data type includes **byte,byte[],int,int[],float,float[],string**

**Note**

- \* Same as syntax 3. Fill **0** as the starting shifted address of the item and read to the end of the item by default.
- \* Reading data based on the configuration file Little Endian (DCBA) or Big Endian (ABCD).

```
byte var_b = ethercat_read_input("CameraLightMask")
// 0x01
```

```
byte[] var_ba = ethercat_read_input("CtrlBox_DO",0,2)
// {0x99,0x66}
var_ba = ethercat_read_input("CtrlBox_DO",1,1)
// {0x66}
var_ba = ethercat_read_input("CtrlBox_DO")
// {0x99,0x66}
```

```
float var_f = ethercat_read_input("CtrlBox_AO")
// byte[] = {0xA4,0x70,0x9D,0x3F} (Little Endian) to float
// float = 0x3F9D70A4 (Little Endian)
// float = 1.23
```

```
string var_s = ethercat_read_input("AutoRun_ProjectName",0,10)
// byte[] = {0x45,0x74,0x68,0x65,0x72,0x43,0x41,0x54,0x5F,0x54}
// string = EtherCAT_T
```

```
var_s = ethercat_read_input("AutoRun_ProjectName",3,3)
// byte[] = {0x65,0x72,0x43}
// string = erC
```

```
var_s = ethercat_read_input("AutoRun_ProjectName")
// byte[] =
0x45,0x74,0x68,0x65,0x72,0x43,0x41,0x54,0x5F,0x54,0x65,0x73,0x74,0x5F,0x31,0x00,0x00,0x00,0x00,0x00}
// string = EtherCAT_Test_1
```



## 20.2 ethercat\_read\_input\_int()

Read the input table content and convert the data to the 32-bit integer.

### Syntax 1

```
int[] ethercat_read_input_int(
 int,
 int,
 int
)
```

#### Parameters

|     |                                                                                                     |
|-----|-----------------------------------------------------------------------------------------------------|
| int | Starting address                                                                                    |
| int | The address amount to read                                                                          |
| int | The conversion of the read data to an int array based on Little Endian (DCBA) or Big Endian (ABCD). |
| 0   | Little-Endian                                                                                       |
| 1   | Big-Endian                                                                                          |
| 2   | Based on the configuration file.                                                                    |

#### Return

int[] Return data in an integer array.

### Syntax 2

```
int[] ethercat_read_input_int(
 int,
 int
)
```

#### Parameters

|     |                            |
|-----|----------------------------|
| int | Starting address           |
| int | The address amount to read |

#### Note

Same as Syntax 1 with the parameter of the conversion of the read data defaults to 2.

\*Convert the read data to an int array based on Little Endian (DCBA) or Big Endian.

```

int[] var_ia = ethercat_read_input_int(72,4,0)
 // byte[] = {0xB1,0x68,0xDE,0x3A} (Little Endian)to int[]
 // int[] = {0x3ADE68B1} (Little Endian)
 // int[] = {987654321}
var_ia = ethercat_read_input_int(72,1,0)
 // byte[] = {0xB1} (Little Endian) to int[]
 // int[] = {0x000000B1} (Little Endian)
 // int[] = {177}
var_ia = ethercat_read_input_int(72,4)
 // byte[] = {0xB1,0x68,0xDE,0x3A} (Little Endian)to int[]
 // int[] = {0x3ADE68B1} (Little Endian)
 // int[] = {987654321}

```

### Syntax 3

```

int ethercat_read_input_int(
 int
)

```

#### Parameter

**int** Starting address  
 \*Convert the read data to an integer based on Little Endian (DCBA) or Big Endian (ABCD).

#### Return

**int** Return data in an integer

#### Note

```

int var_i = ethercat_read_input_int(72)
 // byte[] = {0xB1,0x68,0xDE,0x3A} (Little Endian)to int
 // int = 0x3ADE68B1 (Little Endian)
 // int = 987654321

```

## 20.3 ethercat\_read\_input\_float()

Read the input table content and convert the data to the 32-bit floating-point number.

### Syntax 1

```
float[] ethercat_read_input_float(
 int,
 int,
 int
)
```

#### Parameter

|     |                                                                                                      |
|-----|------------------------------------------------------------------------------------------------------|
| int | Starting address                                                                                     |
| int | The address amount to read                                                                           |
| int | The conversion of the read data to a float array based on Little Endian (DCBA) or Big Endian (ABCD). |
| 0   | Little-Endian                                                                                        |
| 1   | Big-Endian                                                                                           |
| 2   | Based on the configuration file.                                                                     |

#### Return

float[] Return data in a floating-point number array.

### Syntax 2

```
float[] ethercat_read_input_float(
 int,
 int
)
```

#### Parameter

|     |                            |
|-----|----------------------------|
| int | Starting address           |
| int | The address amount to read |

#### Note

Same as Syntax 1 with the parameter of the conversion of the read data defaults to 2.  
 \*Convert the read data to a float array based on Little Endian (DCBA) or Big Endian (ABCD).

```
float[] var_fa = ethercat_read_input_float(56,8,0)
 // byte[] = {0xA4,0x70,0x9D,0x3F,0x85,0xEB,0x91,0xC0} (Little Endian) to float[]
 // float[] = {0x3F9D70A4,0xC091EB85} (Little Endian)
 // float[] = {1.23,-4.56}
var_fa = ethercat_read_input_float(56,4,0)
 // byte[] = {0xA4,0x70,0x9D,0x3F} (Little Endian) to float[]
 // float[] = {0x3F9D70A4} (Little Endian)
 // float[] = {1.23}
var_fa = ethercat_read_input_float(56,2,0)
 // byte[] = {0xA4,0x70} (Little Endian) to float[]
 // float[] = {0x000070A4} (Little Endian)
 // float[] = {0}
var_fa = ethercat_read_input_float(56,8)
 // byte[] = {0xA4,0x70,0x9D,0x3F,0x85,0xEB,0x91,0xC0} (Little Endian) to float[]
 // float[] = {0x3F9D70A4,0xC091EB85} (Little Endian)
```

```
// float[] = {1.23,-4.56}
```

### Syntax 3

```
float ethercat_read_input_float(
 int
)
```

#### Parameter

**int** Starting address  
 \*Convert the read data to a floating-point number based on Little Endian (DCBA) or Big Endian (ABCD).

#### Return

**Float** Return data in a floating-point number.

#### Note

```
int var_f = ethercat_read_input_float(132)
// byte[] = {0xE1,0x7A,0xFC,0x40} (Little Endian) to float
// float = 0x40FC7AE1 (Little Endian)
// float = 7.89
```

## 20.4 ethercat\_read\_input\_string()

Read the input table content and convert the data to the string encoded in UTF8.

### Syntax 1

```
string eip_read_input_string(
 int,
 int
)
```

#### Parameter

**int** Starting address  
**int** The address amount to read

#### Return

**string** Return data in a UTF8 string (ending with 0x00 encountered).

#### Note

```
string var_s = ethercat_read_input_string(4,20)
// byte[] =
// {0x45,0x74,0x68,0x65,0x72,0x43,0x41,0x54,0x5F,0x54,0x65,0x73,0x74,0x5F,0x31,0x00,0x00,0x00,0x00,0x00}
// string = EtherCAT_Test_1
var_s = ethercat_read_input_string (4,10)
// byte[] = {0x45,0x74,0x68,0x65,0x72,0x43,0x41,0x54,0x5F,0x54}
// string = EtherCAT_T
```

## 20.5 ethercat\_read\_input\_bit()

Read the input table content and retrieve the  $n^{\text{th}}$  bit value of the data byte.

### Syntax 1

```
byte ethercat_read_input_bit(
 int,
 int
)
```

#### Parameter

**int** Starting address  
**int** The  $n^{\text{th}}$  bit value in the data byte

#### Return

**byte** Return data in byte.  
 Return 1 for bit value == 1.  
 Return 0 for bit value == 0.  
 \*Data in bit will return in byte.

#### Note

```
byte var_b = ethercat_read_input_bit(50,0)
// byte[] = {0x99}
// 1
var_b = ethercat_read_input_bit(50,7)
// byte[] = {0x99}
// 1
```

### Syntax 2

```
byte ethercat_read_input_bit(
 string,
 int
)
```

#### Parameter

**string** Item name  
**int** The  $n^{\text{th}}$  bit value

#### Return

**byte** Return data in byte.  
 Return 1 for bit value == 1.  
 Return 0 for bit value == 0.  
 \*Data in bit will return in byte.

#### Note

```
byte var_b = ethercat_read_input_bit("Register_Bit",0)
// byte[] = {0x99}
// 1
var_b = ethercat_read_input_bit("Register_Bit",7)
// byte[] = {0x99}
// 1
```

### Syntax 3

```
byte[] ethercat_read_input_bit(
```

```
int,
int,
int
```

```
)
```

#### Parameter

```
int Starting address
int Starting bit
int The amount of bit to read
```

#### Return

```
byte[] Return data in byte[].
 Return 1 for bit value == 1.
 Return 0 for bit value == 0.
 *Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].
```

#### Note

```
byte[] var_ba = ethercat_read_input_bit(50,0,16)
 // byte[] = {0x99,0x66}
 // byte[] = {1,0,0,1,1,0,0,1,0,1,1,0,0,1,1,0}
var_ba = ethercat_read_input_bit (50,8,4)
 // byte[] = {0x66}
 // byte[] = {0,1,1,0}
var_ba = ethercat_read_input_bit (50,6,4)
 // byte[] = {0x99,0x66}
 // byte[] = {0,1,0,1}
```

#### Syntax 4

```
byte[] ethercat_read_input_bit(
 string,
 int,
 int
```

```
)
```

#### Parameter

```
string Item name
int Starting bit
int The amount of bit to read
```

#### Return

```
byte[] Return data in byte[].
 Return 1 for bit value == 1.
 Return 0 for bit value == 0.
 *Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].
```

#### Note

```
byte[] var_ba = ethercat_read_input_bit("Register_Bit",0,16)
 // byte[] = {0xEE,0x66}
 // byte[] = {0,1,1,1,0,1,1,1,0,1,1,0,0,1,1,0}
var_ba = ethercat_read_input_bit ("Register_Bit",8,4)
 // byte[] = {0x66}
 // byte[] = {0,1,1,0}
var_ba = ethercat_read_input_bit ("Register_Bit",6,4)
 // byte[] = {0xEE,0x66}
 // byte[] = {1,1,0,1}
```

## 20.6 ethercat\_read\_output()

Read the output table content.

### Syntax 1

```
byte[] ethercat_read_output(
 int,
 int
)
```

#### Parameter

**int** Starting address  
**int** The address length to read

#### Return

**byte[]** Return data in a byte array.

#### Note

```
byte[] var_ba = ethercat_read_output(40,12)
// {0x45,0x74,0x68,0x65,0x72,0x43,0x41,0x54,0x5F,0x54,0x65,0x73}
```

### Syntax 2

```
byte ethercat_read_output(
 int
)
```

#### Parameter

**int** Starting address

#### Return

**byte** Return data in byte.

#### Note

Same as syntax 1. Fill 1 as the last parameter int by default.

```
byte var_b = ethercat_read_output(40)
// 0x45
```



**Syntax 3**

```
? ethercat_read_output(
 string,
 int,
 int
)
```

**Parameter**

string    Item name  
 int      The starting shifted address of the item  
 int      The amount of the addresses to read

**Return**

?        The data type returned by the item defined in the communication data table.  
 \* Data type includes **byte,byte[],int,int[],float,float[],string**

**Syntax 4**

```
? ethercat_read_output(
 string,
 int
)
```

**Parameter**

string    Item name  
 int      The starting shifted address of the item

**Return**

?        The data type returned by the item defined in the communication data table.  
 \* Data type includes **byte,byte[],int,int[],float,float[],string**

**Note**

- \* Same as syntax 3. Read to the end of the item by default.
- \* Reading data based on the configuration file Little Endian (DCBA) or Big Endian (ABCD).

**Syntax 5**

```
? ethercat_read_output(
 string
)
```

**Parameter**

**string**    Item name

**Return**

**?**            The data type returned by the item defined in the communication data table.  
 \* Data type includes **byte,byte[],int,int[],float,float[],string**

**Note**

- \* Same as syntax 3. Fill 0 as the starting shifted address of the item and read to the end of the item by default.
- \* Reading data based on the configuration file Little Endian (DCBA) or Big Endian (ABCD).

```
byte var_b = ethercat_read_output("ManualAuto",0,1)
 // 0x01
```

```
var_b = ethercat_read_output("ManualAuto",0)
 // 0x01
```

```
var_b = ethercat_read_output("ManualAuto")
 // 0x01
```

```
byte[] var_ba = ethercat_read_output("Error_Code",0,4)
 // {0x04,0x44,0x04,0x00}
```

```
var_ba = ethercat_read_output("Error_Code",0,2)
 // {0x04,0x44}
```

```
var_ba = ethercat_read_output("Error_Code")
 // {0x04,0x44,0x04,0x00}
```

```
int var_i = ethercat_read_output("Current_Time_Year")
 // byte[] = {0xE8,0x07,0x00,0x00} (Little Endian) to int
 // int = 0x000007E8 (Little Endian)
 // int = 2024
```

```
int[] var_ia = ethercat_read_output("Current_Time_Year")
```

```
// byte[] = {0xE8,0x07,0x00,0x00} (Little Endian) to int[]
// int16[] = {0x000007E8} (Little Endian)
// int[] = {2024}
```

```
float var_f = ethercat_read_output("CtrlBox_AO")
```

```
// byte[] = {0x9D,0x67,0x9D,0x3F} (Little Endian) to float[]
// float = 0x3F9D679D (Little Endian)
// float = 1.23
```

```
float[] var_fa = ethercat_read_output("Coord_CurrBase_Tool",4,12)
```

```
// byte[] = {0x9A,0x99,0x7C,0xC3,0xCD,0xEC,0x5E,0x44,0x00,0x00,0xB4,0x42} (Little Endian) to float[]
// float[] = {0xC37C999A,0x445EECCD,0x42B40000} (Little Endian)
// float[] = {-252.6,891.7,90}
```

```
var_fa = ethercat_read_output("Coord_CurrBase_Tool",4)
```

```
// byte[] =
{0x9A,0x99,0x7C,0xC3,0xCD,0xEC,0x5E,0x44,0x00,0x00,0xB4,0x42,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian) to float[]
// float[] = {0xC37C999A,0x445EECCD,0x42B40000,0x00000000,0x00000000} (Little Endian)
// float[] = {-252.6,891.7,90,0,0}
```

```
var_fa = ethercat_read_output("Coord_CurrBase_Tool")
```

```
// byte[] = {0x0A,0xD7,0x23,0x3C,0x9A,0x99,0x7C,0xC3,0xCD,0xEC,0x5E,0x44,0x00,0x00,0xB4,0x42,
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Big Endian) to float[]
// float[] = {0x3C23D70A,0xC37C999A,0x445EECCD,0x42B40000,0x00000000,0x00000000} (Little Endian)
// float[] = {0.01,-252.6,891.7,90,0,0}
```

## 20.7 ethercat\_read\_output\_int()

Read the output table content and convert the data to the 32-bit integer.

### Syntax 1

```
int[] ethercat_read_output_int(
 int,
 int,
 int
)
```

#### Parameter

|     |                                                                                                     |
|-----|-----------------------------------------------------------------------------------------------------|
| int | Starting address                                                                                    |
| int | The address amount to read                                                                          |
| int | The conversion of the read data to an int array based on Little Endian (DCBA) or Big Endian (ABCD). |
| 0   | Little-Endian                                                                                       |
| 1   | Big-Endian                                                                                          |
| 2   | Based on the configuration file.                                                                    |

#### Return

int[] Return data in an integer array.

### Syntax 2

```
int[] ethercat_read_output_int(
 int,
 int
)
```

#### Parameter

|     |                            |
|-----|----------------------------|
| int | Starting address           |
| int | The address amount to read |

#### Note

Same as syntax 1. Fill **2** as the last parameter int by default.

\* Convert the read data to an int array based on Little Endian (DCBA) or Big Endian (ABCD).

```

int[] var_ia = ethercat_read_output_int(216,12,0)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian) to
 int[]
 // int[] = {0x00007FFF,0x0001869F,0xFFFF8000} (Little Endian)
 // int[] = {32767,99999,-32768}

var_ia = ethercat_read_output_int(216,6,0)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86} (Little Endian) to int[]
 // int[] = {0x00007FFF,0x0000869F} (Little Endian)
 // int[] = {32767,34463}

var_ia = ethercat_read_output_int(216,1,0)
 // byte[] = {0xFF} (Little Endian) to int[]
 // int[] = {0x000000FF} (Little Endian)
 // int[] = {255}

var_ia = ethercat_read_output_int(216,12)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian) to
 int[]
 // int[] = {0x00007FFF,0x0001869F,0xFFFF8000} (Little Endian)
 // int[] = {32767,99999,-32768}

```

### Syntax 3

```

int ethercat_read_output_int(
 int
)

```

#### Parameter

**int** Starting address  
 \* Convert the read data to an integer based on Little Endian (DCBA) or Big Endian (ABCD).

#### Return

**int** Return data in an integer

#### Note

```

int var_i = ethercat_read_output_int(216)
 // byte[] = {0xFF,0x7F,0x00,0x00} (Little Endian) to int[]
 // int[] = 0x00007FFF (Little Endian)
 // int = 32767

```

## 20.8 ethercat\_read\_output\_float()

Read the output table content and convert the data to the 32-bit floating-point number.

### Syntax 1

```
float[] ethercat_read_output_float(
 int,
 int,
 int
)
```

#### Parameter

|     |                                                                                                      |
|-----|------------------------------------------------------------------------------------------------------|
| int | Starting address                                                                                     |
| int | The address amount to read                                                                           |
| int | The conversion of the read data to a float array based on Little Endian (DCBA) or Big Endian (ABCD). |
| 0   | Little-Endian                                                                                        |
| 1   | Big-Endian                                                                                           |
| 2   | Based on the configuration file.                                                                     |

#### Return

float[] Return data in a floating-point number array.

### Syntax 2

```
float[] ethercat_read_output_float(
 int,
 int
)
```

#### Parameter

|     |                            |
|-----|----------------------------|
| int | Starting address           |
| int | The address amount to read |

#### Note

Same as syntax 1. Fill **2** as the last parameter int by default.

\*Convert the read data to a float array based on Little Endian (DCBA) or Big Endian (ABCD).

```
float[] var_fa = ethercat_read_output_float(120,24,0)
// byte[] =
// {0x0A,0xD7,0x23,0x3C,0x9A,0x99,0x7C,0xC3,0xCD,0xEC,0x5E,0x44,0x00,0x00,0xB4,0x42,
// 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian) to float[]
// float[] = {0x3C23D70A,0xC37C999A,0x445EECCD,0x42B40000,0x00000000,0x00000000} (Little
// Endian)
// float[] = {0.01,-252.6,891.7,90,0,0}
var_fa = ethercat_read_output_float(120,12,0)
// byte[] = {0x0A,0xD7,0x23,0x3C,0x9A,0x99,0x7C,0xC3,0xCD,0xEC,0x5E,0x44} (Little Endian) to
// float[]
// float[] = {0x3C23D70A,0xC37C999A,0x445EECCD} (Little Endian)
// float[] = {0.01,-252.6,891.7}
var_fa = ethercat_read_output_float(120,2,0)
// byte[] = {0x0A,0xD7} (Little Endian) to float[]
// float[] = {0x0000D70A} (Little Endian)
```

```

 // float[] = {0}
var_fa = ethercat_read_output_float(120,24)
 // byte[] =
 {0x0A,0xD7,0x23,0x3C,0x9A,0x99,0x7C,0xC3,0xCD,0xEC,0x5E,0x44,0x00,0x00,0xB4,0x42,
 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian) to float[]
 // float[] = {0x3C23D70A,0xC37C999A,0x445EECCD,0x42B40000,0x00000000,0x00000000} (Little
 Endian)
 // float[] = {0.01,-252.6,891.7,90,0,0}

```

### Syntax 3

```

float ethercat_read_output_float(
 int
)

```

#### Parameter

**int** Starting address  
 \*Convert the read data to a floating-point number based on Little Endian (DCBA) or Big

#### Return

**float** Return data in a floating-point number

#### Note

```

float var_f = ethercat_read_output_float(120)
 // byte[] = {0x0A,0xD7,0x23,0x3C} (Little Endian) to float[]
 // float[] = 0x3C23D70A (Little Endian)
 // float = 0.01

```

## 20.9 ethercat\_read\_output\_string()

Read the output table content and convert the data to the string encoded in UTF8.

### Syntax 1

```
string ethercat_read_output_string(
 int,
 int
)
```

#### Parameter

**int** Starting address  
**int** The address amount to read

#### Return

**string** Return data in a UTF8 string (ending with 0x00 encountered).

#### Note

```
string var_s = ethercat_read_output_string(40,20)
 // byte[] =
 {0x45,0x74,0x68,0x65,0x72,0x43,0x41,0x54,0x5F,0x54,0x65,0x73,0x74,0x5F,0x31,0x00,0x00,0x00,0x00,0x00}
 // string = "EtherCAT_Test_1"
var_s = ethercat_read_output_string(40,3)
 // byte[] = {0x45,0x74,0x68}
 // string = "Eth"
```



## 20.10 ethercat\_read\_output\_bit()

Read the output table content and retrieve the n<sup>th</sup> bit value of the data byte.

### Syntax 1

```
byte ethercat_read_output_bit(
 int,
 int
)
```

#### Parameter

**int** Starting address  
**int** The n<sup>th</sup> bit value in the data byte

#### Return

**Byte** Return data in byte.  
 Return 1 for bit value == 1.  
 Return 0 for bit value == 0.  
 \*Data in bit will return in byte.

#### Note

```
byte var_b = ethercat_read_output_bit(186,0)
// byte[] = {0x99}
// 1
var_b = ethercat_read_output_bit(186,7)
// byte[] = {0x99}
// 1
```

### Syntax 2

```
byte ethercat_read_output_bit(
 string,
 int
)
```

#### Parameter

**string** Item name  
**int** The n<sup>th</sup> bit value

#### Return

**byte** Return data in byte.  
 Return 1 for bit value == 1.  
 Return 0 for bit value == 0.  
 \*Data in bit will return in byte.

#### Note

```
byte var_b = ethercat_read_output_bit("Register_Bit",0)
// byte[] = {0x84}
// 0
var_b = ethercat_read_output_bit("Register_Bit",7)
// byte[] = {0x84}
// 1
```

### Syntax 3

```
byte[] ethercat_read_output_bit(
```

```
int,
int,
int
)
```

#### Parameter

```
int Starting address
int Starting bit
int The amount of bit to read
```

#### Return

```
byte[] Return data in byte[].
 Return 1 for bit value == 1.
 Return 0 for bit value == 0.
 *Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].
```

#### Note

```
byte[] var_ba = ethercat_read_output_bit(186,0,16)
// byte[] = {0x99,0x66}
// byte[] = {1,0,0,1,1,0,0,1,0,1,1,0,0,1,1,0}
var_ba = ethercat_read_output_bit (186,8,4)
// byte[] = {0x66}
// byte[] = {0,1,1,0}
var_ba = ethercat_read_output_bit (186,6,4)
// byte[] = {0x99,0x66}
// byte[] = {0,1,0,1}
```

#### Syntax 4

```
byte[] ethercat_read_output_bit(
 string,
 int,
 int
)
```

#### Parameter

```
string Item name
int Starting bit
int The amount of bit to read
```

#### Return

```
byte[] Return data in byte[].
 Return 1 for bit value == 1.
 Return 0 for bit value == 0.
 *Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].
```

#### Note

```
byte[] var_ba = ethercat_read_output_bit("Register_Bit",0,16)
// byte[] = {0x84,0xB7}
// byte[] = {0,0,1,0,0,0,0,1,1,1,1,0,1,1,0,1}
var_ba = ethercat_read_output_bit ("Register_Bit",8,4)
// byte[] = {0xB7}
// byte[] = {1,1,1,0}
var_ba = ethercat_read_output_bit ("Register_Bit",6,4)
// byte[] = {0x84,0xB7}
// byte[] = {0,1,1,1}
```

## 20.11 ethercat\_write\_output()

Write data to the output table.

### Syntax 1

```
bool ethercat_write_output(
 int,
 ?,
 int
)
```

#### Parameter

|      |                                                                                        |
|------|----------------------------------------------------------------------------------------|
| int  | Starting address                                                                       |
| ?    | The data to write                                                                      |
|      | * Available data types include <b>byte, byte[], int, int[], float, float[], string</b> |
| int  | The maximum amount of the address to write                                             |
| > 0  | Valid data length. Write by the amount of the address.                                 |
| <= 0 | Invalid data length. Write by the complete length of the data to write.                |

#### Return

|      |       |                                                              |
|------|-------|--------------------------------------------------------------|
| Bool | True  | Write successfully.                                          |
|      | False | Write unsuccessfully.                                        |
|      |       | 1. If the data to write is an empty string or an empty array |
|      |       | 2. Unable to send and receive correctly.                     |

#### Note

\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

### Syntax 2

```
bool ethercat_write_output(
 int,
 ?
)
```

#### Parameter

|     |                                                                                        |
|-----|----------------------------------------------------------------------------------------|
| int | Starting address                                                                       |
| ?   | The data to write                                                                      |
|     | * Available data types include <b>byte, byte[], int, int[], float, float[], string</b> |

#### Return

|      |       |                                                              |
|------|-------|--------------------------------------------------------------|
| Bool | True  | Write successfully.                                          |
|      | False | Write unsuccessfully.                                        |
|      |       | 1. If the data to write is an empty string or an empty array |
|      |       | 2. Unable to send and receive correctly.                     |

#### Note

Same as syntax 1. Write with the full length of the data to write by default.

\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

### Syntax 3

```
bool ethercat_write_output(
 int,
```

```
?,
int,
int
)
```

### Parameter

|     |                                                                                                                                                             |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| int | Starting address                                                                                                                                            |
| ?   | The data to write<br>* Available data types include <b>byte</b> , <b>byte[]</b> , <b>int</b> , <b>int[]</b> , <b>float</b> , <b>float[]</b> , <b>string</b> |
| int | Starting address of the data to write                                                                                                                       |
| int | The amount of the address to write                                                                                                                          |

### Return

|      |       |                       |
|------|-------|-----------------------|
| bool | True  | Write successfully.   |
|      | False | Write unsuccessfully. |

1. If the data to write is an empty string or an empty array
2. Unable to send and receive correctly.

### Note

\*\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

```
byte var_data = 255
ethercat_write_output(208,var_data,1)
byte var_b = ethercat_read_output(208)
// 0xFF

byte[] var_data = {1,127,255}
ethercat_write_output(208,var_data,2)
byte[] var_ba = ethercat_read_output(208,3)
// {0x01,0x7F,0x00}
ethercat_write_output(208,var_data,3)
var_ba = ethercat_read_output(208,3)
// {0x01,0x7F,0xFF}
ethercat_write_output(208,var_data,-1)
var_ba = ethercat_read_output(208,3)
// {0x01,0x7F,0xFF}

int var_data = 32767
ethercat_write_output(216,var_data,4)
var_i = ethercat_read_output_int(216)
// byte[] = {0xFF,0x7F,0x00,0x00} (Little Endian) to int
// int = 0x00007FFF (Little Endian)
// int = 32767
ethercat_write_output(216,var_data,1)
var_i = ethercat_read_output_int(216)
// byte[] = {0xFF,0x00,0x00,0x00} (Little Endian) to int
// int = 0x000000FF (Little Endian)
// int = 255
```

```

int[] var_data = {32767,99999,-32768}
ethercat_write_output(216,var_data,12)
int[] var_ia = ethercat_read_output_int(216,12)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian) to int[]
 // int[] = {0x00007FFF,0x0001869F,0xFFFF8000} (Little Endian)
 // int[] = {32767,99999,-32768}
ethercat_write_output(216,var_data,3)
var_ia = ethercat_read_output_int(216,12)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian) to int[]
 // int[] = {0x00007FFF,0x00000000,0x00000000} (Little Endian)
 // int[] = {32767,0,0}
ethercat_write_output(216,var_data,4,4)
var_ia = ethercat_read_output_int(216,12)
 // byte[] = {0x9F,0x86,0x01,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian) to int[]
 // int[] = {0x0001869F,0x00000000,0x00000000} (Little Endian)
 // int[] = {99999,0,0 }

float var_data = -10.0
ethercat_write_output(232,var_data,4)
float var_f = ethercat_read_output_float(232)
 // byte[] = {0x00,0x00,0x20,0xC1} (Little Endian) to float
 // float = 0xC1200000 (Little Endian)
 // float = -10.0
ethercat_write_output(232,var_data,1)
var_f = ethercat_read_output_int(232)
 // byte[] = {0x00,0x00,0x00,0x00} (Little Endian) to float
 // float = 0x00000000 (Little Endian)
 // float = 0

float[] var_data = {-10.0,3.3,123.45}
ethercat_write_output(232,var_data,12)
float[] var_fa = ethercat_read_output_float(232,12)
 // byte[] = {0x00,0x00,0x20,0xC1,0x33,0x33,0x53,0x40,0x66,0xE6,0xF6,0x42} (Little Endian) to float[]
 // float[] = {0xC1200000,0x40533333,0x42F6E666} (Little Endian)
 // float[] = {-10.0,3.3,123.45}
ethercat_write_output(232,var_data,3)
var_fa = ethercat_read_output_float(232,12)
 // byte[] = {0x00,0x00,0x20,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian) to float[]
 // float[] = {0x00200000,0x00000000,0x00000000} (Little Endian)
 // float[] = {0,0,0}
ethercat_write_output(232,var_data,4,4)
var_fa = ethercat_read_output_float(232,12)
 // byte[] = {0x33,0x33,0x53,0x40,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian) to float[]
 // float[] = {0x40533333,0x00000000,0x00000000} (Little Endian)
 // float[] = {3.3,0,0}

string var_data = "abcdefg1234567"
ethercat_write_output(208,var_data,16)
string var_s = ethercat_read_output_string(208,20)

```

```

// byte[] =
{0x61,0x62,0x63,0x64,0x65,0x66,0x67,0x31,0x32,0x33,0x34,0x35,0x36,0x37,0x00,0x00,0x00,0x00,0x00,0x00}
// string = abcdefg1234567

ethercat_write_output(208,var_data,8)
var_s = ethercat_read_output_string(208,20)
// byte[] =
{0x61,0x62,0x63,0x64,0x65,0x66,0x67,0x31,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
// string = abcdefg1

ethercat_write_output(208,var_data,3,3)
var_s = ethercat_read_output_string(208,20)
// byte[] =
{0x64,0x65,0x66,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
// string = def

```

#### Syntax 4

```

bool ethercat_write_output(
 string,
 int,
 ?,
 int,
 int
)

```

##### Parameter

|        |                                                                                                                                                             |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| string | Item name                                                                                                                                                   |
| int    | The starting shifted address of the item                                                                                                                    |
| ?      | The data to write<br>* Available data types include <b>byte</b> , <b>byte[]</b> , <b>int</b> , <b>int[]</b> , <b>float</b> , <b>float[]</b> , <b>string</b> |
| int    | Starting address of the data to write                                                                                                                       |
| int    | The amount of the address to write                                                                                                                          |

##### Return

|      |              |                       |
|------|--------------|-----------------------|
| bool | <b>True</b>  | Write successfully.   |
|      | <b>False</b> | Write unsuccessfully. |

1. If the data to write is an empty string or an empty array
2. Unable to send and receive correctly.

##### Note

\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

#### Syntax 5

```

bool ethercat_write_output(
 string,
 int,
 ?,
 int
)

```

##### Parameter

|        |                                                                                                                                                             |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| string | Item name                                                                                                                                                   |
| int    | The starting shifted address of the item                                                                                                                    |
| ?      | The data to write<br>* Available data types include <b>byte</b> , <b>byte[]</b> , <b>int</b> , <b>int[]</b> , <b>float</b> , <b>float[]</b> , <b>string</b> |

**int** Starting address of the data to write

### Return

**bool**     **True**     Write successfully.  
              **False**    Write unsuccessfully.

1. If the data to write is an empty string or an empty array
2. Unable to send and receive correctly.

### Note

Same as syntax 4. Write with the full length of the data to write by default.

\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

### Syntax 6

```
bool ethercat_write_output(
 string,
 int,
 ?
)
```

### Parameter

**string**     Item name  
**int**        The starting shifted address of the item  
**?**         The data to write

\* Available data types include **byte,byte[],int,int[],float,float[],string**

### Return

**bool**     **True**     Write successfully.  
              **False**    Write unsuccessfully.

1. If the data to write is an empty string or an empty array
2. Unable to send and receive correctly.

### Note

Same as syntax 4. Fill **0** as **the starting address to write** and write with the full length of the data to write by default.

\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

### Syntax 7

```
bool ethercat_write_output(
 string,
 ?
)
```

### Parameter

**string**     Item name  
**?**         The data to write

\* Available data types include **byte,byte[],int,int[],float,float[],string**

### Return

**bool**     **True**     Write successfully.  
              **False**    Write unsuccessfully.

1. If the data to write is an empty string or an empty array
2. Unable to send and receive correctly.

## Note

Same as syntax 4. Fill 0 as the starting shifted address and the starting address to write as well as write with the full length of the data to write by default.

\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

```
int[] var_data = {32767,99999,-32768}
ethercat_write_output("Register_Int",0,var_data,0,12)
int[] var_ia = ethercat_read_output_int(216,12)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian) to int[]
 // int[] = {0x00007FFF,0x0001869F,0xFFFF8000} (Little Endian)
 // int[] = {32767,99999,-32768}
ethercat_write_output("Register_Int",4,var_data,4,4)
var_ia = ethercat_read_output_int(216,12)
 // byte[] = {0x00,0x00,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x00,0x00,0x00} (Little Endian) to int[]
 // int[] = {0x00000000,0x0001869F,0x00000000} (Little Endian)
 // int[] = {0,99999,0}
ethercat_write_output("Register_Int",4,var_data)
var_ia = ethercat_read_output_int(216,12)
 // byte[] = {0x00,0x00,0x00,0x00,0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00} (Little Endian) to int[]
 // int[] = {0x00000000,0x00007FFF,0x0001869F} (Little Endian)
 // int[] = {0,32767,99999}

float[] var_data = {-10.0,3.3,123.45}
ethercat_write_output("Register_Float",0,var_data,0,12)
float[] var_fa = ethercat_read_output_float(232,12)
 // byte[] = {0x00,0x00,0x20,0xC1,0x33,0x33,0x53,0x40,0x66,0xE6,0xF6,0x42} (Little Endian) to float[]
 // float[] = {0xC1200000,0x40533333,0x42F6E666} (Little Endian)
 // float[] = {-10.0,3.3,123.45}
ethercat_write_output("Register_Float",4,var_data,4,8)
var_fa = ethercat_read_output_float(232,12)
 // byte[] = {0x00,0x00,0x00,0x00,0x33,0x33,0x53,0x40,0x66,0xE6,0xF6,0x42} (Little Endian) to float[]
 // float[] = {0x00000000,0x40533333,0x42F6E666} (Little Endian)
 // float[] = {0,3.3,123.45}
ethercat_write_output("Register_Float",4,var_data)
var_fa = ethercat_read_output_float(232,12)
 // byte[] = {0x00,0x00,0x00,0x00,0x00,0x00,0x20,0xC1,0x33,0x33,0x53,0x40} (Little Endian) to float[]
 // float[] = {0x00000000,0xC1200000,0x40533333} (Little Endian)
 // float[] = {0,-10,3.3}
```



## 20.12 ethercat\_write\_output\_bit()

Write content to the nth bit value of the data byte in the output table.

### Syntax 1

```
bool ethercat_write_output_bit(
 int,
 int,
 int
)
```

#### Parameter

|     |                                 |
|-----|---------------------------------|
| int | Starting address                |
| int | The n <sup>th</sup> bit value   |
| int | The data to write               |
|     | *Data in bit will write in int. |

#### Return

|      |       |                       |
|------|-------|-----------------------|
| Bool | True  | Write successfully.   |
|      | False | Write unsuccessfully. |

1. Unable to send correctly and receive .

#### Note

```
byte var_data = 240
ethercat_write_output(208,var_data)
byte var_b = ethercat_read_output(208)
// 0xF0
ethercat_write_output_bit(208,1,1)
var_b = ethercat_read_output_bit(208,1)
// 0xF2 get bit: "1"
// 1
ethercat_write_output_bit(208,7,0)
var_b = ethercat_read_output_bit(208,7)
// 0x72 get bit: "7"
// 0
```

**Syntax 2**

```
bool ethercat_write_output_bit(
 string,
 int,
 int
)
```

**Parameter**

|        |                                 |
|--------|---------------------------------|
| string | Item name                       |
| int    | The n <sup>th</sup> bit value   |
| int    | The data to write               |
|        | *Data in bit will write in int. |

**Return**

|      |       |                       |                                          |
|------|-------|-----------------------|------------------------------------------|
| bool | True  | Write successfully.   |                                          |
|      | False | Write unsuccessfully. | 1. Unable to send correctly and receive. |

**Note**

```
byte var_data = 240
ethercat_write_output(208,var_data)
byte var_b = ethercat_read_output(208)
// 0xF0
ethercat_write_output_bit("Register_Bit",1,1)
var_b = ethercat_read_output_bit(208,1)
// 0xF2 get bit: "1"
// 1
ethercat_write_output_bit("Register_Bit",7,0)
var_b = ethercat_read_output_bit(208,7)
// 0x72 get bit: "7"
// 0
```

**Syntax 3**

```
bool ethercat_write_output_bit(
 int,
 int,
 byte[],
 int,
 int
)
```

**Parameter**

|        |                                                                                     |
|--------|-------------------------------------------------------------------------------------|
| int    | Starting address                                                                    |
| int    | The n <sup>th</sup> bit value                                                       |
| byte[] | The data to write                                                                   |
|        | *Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1]. |
| int    | The starting bit of the data to write                                               |
| int    | The bit amount of the data to write                                                 |

**Return**

|      |       |                       |                                          |
|------|-------|-----------------------|------------------------------------------|
| bool | True  | Write successfully.   |                                          |
|      | False | Write unsuccessfully. | 1. Unable to send correctly and receive. |

**Note**

Bit value = 1 for byte value >=1  
 Bit value = 0 for byte value ==0

**Syntax 4**

```
bool ethercat_write_output_bit(
 int,
 int,
 byte[],
 int
)
```

**Parameter**

|        |                                                                                     |
|--------|-------------------------------------------------------------------------------------|
| int    | Starting address                                                                    |
| int    | The n <sup>th</sup> bit value                                                       |
| byte[] | The data to write                                                                   |
|        | *Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1]. |
| int    | The starting bit of the data to write                                               |

**Return**

|      |       |                       |                                          |
|------|-------|-----------------------|------------------------------------------|
| Bool | True  | Write successfully.   |                                          |
|      | False | Write unsuccessfully. | 1. Unable to send correctly and receive. |

**Note**

\*Same as syntax 3. Write with the full length of the rest data to write.  
 Bit value = 1 for byte value >=1  
 Bit value = 0 for byte value ==0

**Syntax 5**

```
bool ethercat_write_output_bit(
 int,
 int,
 byte[]
)
```

**Parameter**

**int** Starting address  
**int** The n<sup>th</sup> bit value  
**byte[]** The data to write  
 \*Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].

**Return**

**bool**     **True**     Write successfully.  
           **False**    Write unsuccessfully.     1. Unable to send correctly and receive.

**Note**

\*Same as syntax 3. Fill 0 as the starting bit to write as well as write with the full length of the data to write by default.

Bit value = 1 for byte value >=1

Bit value = 0 for byte value ==0

```

byte[] var_data = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
ethercat_write_output_bit(208,0,var_data,0,20)
byte[] var_ba = ethercat_read_output(208,3)
// byte[] = {0x39,0xB8,0x0C}
var_ba = ethercat_read_output_bit (208,0,20)
// byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
ethercat_write_output_bit(208,3,var_data,5,10)
var_ba = ethercat_read_output (208,3)
// byte[] = {0x08,0x0E,0x00}
var_ba = ethercat_read_output_bit (208,0,20)
// byte[] = {0,0,0,1,0,0,0,0,0,1,1,1,0,0,0,0,0,0,0,0}

```

**Syntax 6**

```

bool ethercat_write_output_bit(
 string,
 int,
 byte[],
 int,
 int
)

```

**Parameter**

**string** Item name  
**int** Starting bit  
**byte[]** The data to write  
 \*Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].  
**int** The starting bit to write data  
**int** The bit amount of the data to write

**Return**

**bool**     **True**     Write successfully.  
           **False**    Write unsuccessfully.     1. Unable to send correctly and receive.

**Note**

Bit value = 1 for byte value >=1

Bit value = 0 for byte value ==0

**Syntax 7**

```
bool ethercat_write_output_bit(
 string,
 int,
 byte[],
 int
)
```

**Parameter**

|        |                                                                                     |
|--------|-------------------------------------------------------------------------------------|
| string | Item name                                                                           |
| int    | Starting bit                                                                        |
| byte[] | The data to write                                                                   |
|        | *Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1]. |
| int    | The starting bit to write data                                                      |

**Return**

|      |       |                       |                                          |
|------|-------|-----------------------|------------------------------------------|
| bool | True  | Write successfully.   |                                          |
|      | False | Write unsuccessfully. | 1. Unable to send correctly and receive. |

**Note**

\*Same as syntax 6. Write with the full length of the rest data to write.

Bit value = 1 for byte value >=1

Bit value = 0 for byte value ==0

**Syntax 8**

```
bool ethercat_write_output_bit(
 string,
 int,
 byte[]
)
```

**Parameter**

|        |                                                                                     |
|--------|-------------------------------------------------------------------------------------|
| string | Item name                                                                           |
| int    | Starting bit                                                                        |
| byte[] | The data to write                                                                   |
|        | *Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1]. |

**Return**

|      |       |                       |                                          |
|------|-------|-----------------------|------------------------------------------|
| bool | True  | Write successfully.   |                                          |
|      | False | Write unsuccessfully. | 1. Unable to send correctly and receive. |

**Note**

\*Same as syntax 6. Fill 0 as the starting bit to write as well as write with the full length of the data to write by default.

Bit value = 1 for byte value >=1

Bit value = 0 for byte value ==0

```
byte[] var_data = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
ethercat_write_output_bit("Register_Bit",0,var_data,0,20)
byte[] var_ba = ethercat_read_output(208,3)
// byte[] = {0x39,0xB8,0x0C}
var_ba = ethercat_read_output_bit (208,0,20)
// byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
```

```
ethercat_write_output_bit("Register_Bit ",3,var_data,5,10)
var_ba = ethercat_read_output (208,3)
 // byte[] = {0x08,0x0E,0x00}
var_ba = ethercat_read_output_bit (208,0,20)
 // byte[] = {0,0,0,1,0,0,0,0,0,1,1,1,0,0,0,0,0,0,0}
```

## 21. CC-Link Functions

The robot communicates with external controllers via the CC-Link communication protocol. In the mechanism of the CC-Link communication protocol, the robot works as an CC-Link IO device for external devices to read and write the robot data. Meanwhile, TMflow monitors the table of data receiving from external devices and the table of data sending to external devices with CC-Link functions as well as changes the custom definition section in the table of data sending to external devices.

### Communication Data Table

The data table is composed of the input data and the output data. Input Data Table is for external devices posting on the robot, and Output Data Table is for the robot sending to external devices. Both of the data tables come with System Definition Section and Custom Definition Section for data.

1. System Definition Section: Items and settings are defined by the robot, and the data contents are updated by the robot or external devices. The defined items are robot status relevant such as robot bases, project status, control box status, or input/output status relevant such as digital I/Os and analog I/Os. Users can use CC-Link functions to read the input data table and the output data table in the system definition section.
2. Custom Definition Section: Items and settings are defined by users, and the data contents are updated by users or external devices. In the meantime of the project editing, users can use CC-Link functions to read and write the output data table in the custom definition section or read input data table in the custom definition section as well as use the custom definition section as a data exchange register between the project and external devices.

| Communication Data Table<br>(at the robot's viewpoint) | Data Section              | TMflow CC Link Function Permissions | External Device Permissions |
|--------------------------------------------------------|---------------------------|-------------------------------------|-----------------------------|
| Input Data Table                                       | System Definition Section | Read                                | Write                       |
|                                                        | Custom Definition Section | Read                                | Write                       |
| Output Data Table                                      | System Definition Section | Read                                | Read                        |
|                                                        | Custom Definition Section | Read/Write                          | Read                        |

## 21.1 cclink\_read\_input()

Read the input table content.

### Syntax 1

```
byte[] cclink_read_input(
 int,
 int
)
```

#### Parameter

int      Starting address  
int      The address amount to read

#### Return

byte[]    Return data in a byte array.

#### Note

```
byte[] var_ba = cclink_read_input(112,32)
// {0x61,0x61,0x62,0x62,0x63,0x63,0x64,0x64,0x65,0x65,0x66,0x66,0x00,0x00,0x00,0x00,
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
```

### Syntax 2

```
byte cclink_read_input(
 int
)
```

#### Parameter

int      Starting address

#### Return

byte      Return data in byte.

#### Note

```
byte var_b = cclink_read_input(112)
// 0x61
```



**Syntax 3**

```
? cclink_read_input(
 string,
 int,
 int
)
```

**Parameter**

|        |                                          |
|--------|------------------------------------------|
| string | Item name                                |
| int    | The starting shifted address of the item |
| int    | The amount of the addresses to read      |

**Return**

? The data type returned by the item definition in the communication data table.  
 \* Data type includes **byte,byte[],int,int[],float,float[],string**

**Syntax 4**

```
? cclink_read_input(
 string,
 int
)
```

**Parameter**

|        |                                          |
|--------|------------------------------------------|
| string | Item name                                |
| int    | The starting shifted address of the item |

**Return**

? The data type returned by the item definition in the communication data table.  
 \* Data type includes **byte,byte[],int,int[],float,float[],string**

**Note**

- \* Same as syntax 3. Read to the end of the item by default.
- \* Reading data based on the configuration file Little Endian (DCBA) or Big Endian (ABCD).

**Syntax 5**

```
? cclink_read_input(
 string
)
```

**Parameter**

**string**    Item name

**Return**

**?**    The data type returned by the item definition in the communication data table.  
 \* Data type includes **byte,byte[],int,int[],float,float[],string**

**Note**

- \* Same as syntax 3. Fill **0** as the starting shifted address of the item and read to the end of the item by default.
- \* Reading data based on the configuration file Little Endian (DCBA) or Big Endian (ABCD).

```
byte var_b = cclink_read_input("CameraLightMask")
// 0x00
```

```
byte[] var_ba = cclink_read_input("CtrlBox_DO",0,2)
// {0xEE,0x66}
```

```
var_ba = cclink_read_input("CtrlBox_DO",1,1)
// {0x66}
```

```
var_ba = cclink_read_input("CtrlBox_DO")
// {0xEE,0x66}
```

```
float var_f = cclink_read_input("CtrlBox_AO")
// byte[] = {0x71,0x3D,0x8A,0x40} (Little Endian) to float
// float = 0x408A3D71 (Little Endian)
// float = 4.32
```

```
string var_s = cclink_read_input("AutoRun_ProjectName",0,10)
// byte[] = {0x61,0x61,0x62,0x62,0x63,0x63,0x64,0x64,0x65,0x65}
// string = aabbccdde
```

```
var_s = cclink_read_input("AutoRun_ProjectName",3,3)
// byte[] = {0x62,0x63,0x63}
// string = bcc
```

```
var_s = cclink_read_input("AutoRun_ProjectName")
// byte[] = {0x61,0x61,0x62,0x62,0x63,0x63,0x64,0x64,0x65,0x65,0x66,0x66,0x00,0x00,0x00,0x00,
// 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
// string = aabbccddeeff
```

## 21.2 cclink\_read\_input\_int16()

Read the input table content and convert the data to the 16-bit integer.

### Syntax 1

```
int[] cclink_read_input_int16(
 int,
 int,
 int
)
```

#### Parameter

- int Starting address
- int The address amount to read
- int The conversion of the read data to an int array based on Little Endian (DCBA) or Big Endian (ABCD).
  - 3 Little-Endian
  - 4 Big-Endian
  - 5 Based on the configuration file.

#### Return

int[] Return data in an integer array.

### Syntax 2

```
int[] cclink_read_input_int16(
 int,
 int
)
```

#### Parameter

- int Starting address
- int The address amount to read

#### Note

Same as Syntax 1 with the parameter of the conversion of the read data defaults to **2**.

\*Convert the read data to an int array based on Little Endian (DCBA) or Big Endian (ABCD).

```

int[] var_ia = cclink_read_input_int16(6,2,0)
 // byte[] = {0xEE,0x66} (Little Endian) to int16[]
 // int16[] = {0x000066EE} (Little Endian)
 // int[] = {26350}
var_ia = cclink_read_input_int16(6,1,0)
 // byte[] = {0xEE} (Little Endian) to int16[]
 // int16[] = {0x000000EE} (Little Endian)
 // int[] = {238}
var_ia = cclink_read_input_int16(6,2)
 // byte[] = {0xEE,0x66} (Little Endian) to int16[]
 // int16[] = {0x000066EE} (Little Endian)
 // int[] = {26350}

```

### Syntax 3

```

int cclink_read_input_int16(
 int
)

```

#### Parameter

**int** Starting address  
 \*Convert the read data to an integer based on Little Endian (DCBA) or Big Endian (ABCD).

#### Return

**int** Return data in an integer

#### Note

```

int var_i = cclink_read_input_int16(6)
 // byte[] = {0xEE,0x66} (Little Endian) to int16
 // int16 = 0x000066EE (Little Endian)
 // int = 26350

```

## 21.3 cclink\_read\_input\_int()

Read the input table content and convert the data to the 32-bit integer.

### Syntax 1

```
int[] cclink_read_input_int(
 int,
 int,
 int
)
```

#### Parameters

- int Starting address
- int The address amount to read
- int The conversion of the read data to an int array based on Little Endian (DCBA) or Big Endian (ABCD).
  - 0 Little-Endian
  - 1 Big-Endian
  - 2 Based on the configuration file.

#### Return

- int[] Return data in an integer array.

### Syntax 2

```
int[] cclink_read_input_int(
 int,
 int
)
```

#### Parameters

- int Starting address
- int The address amount to read

#### Note

Same as Syntax 1 with the parameter of the conversion of the read data defaults to **2**.

\*Convert the read data to an int array based on Little Endian (DCBA) or Big Endian (ABCD).

```

int[] var_ia = cclink_read_input_int(6,4,0)
 // byte[] = {0xEE,0x66,0x0F,0x09} (Little Endian) to int[]
 // int[] = {0x090F66EE} (Little Endian)
 // int[] = {152004334}
var_ia = cclink_read_input_int(6,1,0)
 // byte[] = {0xEE} (Little Endian) to int[]
 // int[] = {0x000000EE} (Little Endian)
 // int[] = {238}
var_ia = cclink_read_input_int(6,4)
 // byte[] = {0xEE,0x66,0x0F,0x09} (Little Endian) to int[]
 // int[] = {0x090F66EE} (Little Endian)
 // int[] = {152004334}

```

### Syntax 3

```

int cclink_read_input_int(
 int
)

```

#### Parameter

**int** Starting address  
 \*Convert the read data to an integer based on Little Endian (DCBA) or Big Endian (ABCD).

#### Return

**int** Return data in an integer

#### Note

```

int var_i = cclink_read_input_int(6)
 // byte[] = {0xEE,0x66,0x0F,0x09} (Little Endian) to int
 // int = 0x090F66EE (Little Endian)
 // int = 152004334

```

## 21.4 cclink\_read\_input\_float()

Read the input table content and convert the data to the 32-bit floating-point number.

### Syntax 1

```
float[] cclink_read_input_float(
 int,
 int,
 int
)
```

#### Parameter

|     |                                                                                                      |
|-----|------------------------------------------------------------------------------------------------------|
| int | Starting address                                                                                     |
| int | The address amount to read                                                                           |
| int | The conversion of the read data to a float array based on Little Endian (DCBA) or Big Endian (ABCD). |
| 3   | Little-Endian                                                                                        |
| 4   | Big-Endian                                                                                           |
| 5   | Based on the configuration file.                                                                     |

#### Return

float[] Return data in a floating-point number array.

### Syntax 2

```
float[] cclink_read_input_float(
 int,
 int
)
```

#### Parameter

|     |                            |
|-----|----------------------------|
| int | Starting address           |
| int | The address amount to read |

#### Note

Same as Syntax 1 with the parameter of the conversion of the read data defaults to 2.

\*Convert the read data to a float array based on Little Endian (DCBA) or Big Endian (ABCD).

```
float[] var_fa = cclink_read_input_float(144,8,0)
 // byte[] = {0x71,0x3D,0x8A,0x40,0xA4,0x70,0x9D,0xBF} (Little Endian) to float[]
 // float[] = {0x408A3D71,0xBF9D70A4} (Little Endian)
 // float[] = {4.32,-1.23}
var_fa = cclink_read_input_float(144,4,0)
 // byte[] = {0x71,0x3D,0x8A,0x40} (Little Endian) to float[]
 // float[] = {0x408A3D71} (Little Endian)
 // float[] = {4.32}
var_fa = cclink_read_input_float(144,2,0)
 // byte[] = {0x71,0x3D} (Little Endian) to float[]
 // float[] = {0x00003D71} (Little Endian)
 // float[] = {0}
var_fa = cclink_read_input_float(144,8)
 // byte[] = {0x71,0x3D,0x8A,0x40,0xA4,0x70,0x9D,0xBF} (Little Endian) to float[]
 // float[] = {0x408A3D71,0xBF9D70A4} (Little Endian)
```

```
// float[] = {4.32,-1.23}
```

### Syntax 3

```
float cclink_read_input_float(
 int
)
```

#### Parameter

**int** Starting address  
\*Convert the read data to a floating-point number based on Little Endian (DCBA) or Big Endian (ABCD).

#### Return

**Float** Return data in a floating-point number.

#### Note

```
int var_f = cclink_read_input_float(144)
 // byte[] = {0x71,0x3D,0x8A,0x40} (Little Endian) to float
 // float = 0x408A3D71 (Little Endian)
 // float = 4.32
```



## 21.5 cclink\_read\_input\_string()

Read the input table content and convert the data to the string encoded in UTF8.

### Syntax 1

```
string cclink_read_input_string(
 int,
 int
)
```

#### Parameter

int Starting address  
int The address amount to read

#### Return

string Return data in a UTF8 string (ending with 0x00 encountered).

#### Note

```
string var_s = cclink_read_input_string(112,32)
 // byte[] = {0x61,0x61,0x62,0x62,0x63,0x63,0x64,0x64,0x65,0x65,0x66,0x66,0x00,0x00,0x00,0x00,
 // 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // string = aabbccddeeff
var_s = cclink_read_input_string (112,10)
 // byte[] = {0x61,0x61,0x62,0x62,0x63,0x63,0x64,0x64,0x65,0x65}
 // string = aabbccdde
```

## 21.6 cclink\_read\_input\_bit()

Read the input table content and retrieve the  $n^{\text{th}}$  bit value of the data byte.

### Syntax 1

```
byte cclink_read_input_bit(
 int,
 int
)
```

#### Parameter

**int** Starting address  
**int** The  $n^{\text{th}}$  bit value in the data byte

#### Return

**byte** Return data in byte.  
 Return 1 for bit value == 1.  
 Return 0 for bit value == 0.  
 \*Data in bit will return in byte.

#### Note

```
byte var_b = cclink_read_input_bit(6,0)
// byte[] = {0xEE}
// 0
var_b = cclink_read_input_bit(6,7)
// byte[] = {0xEE}
// 1
var_b = cclink_read_input_bit(6,15)
// byte[] = {0x66}
// 0
```

### Syntax 2

```
byte cclink_read_input_bit(
 string,
 int
)
```

#### Parameter

**string** Item name  
**int** The  $n^{\text{th}}$  bit value

#### Return

**byte** Return data in byte.  
 Return 1 for bit value == 1.  
 Return 0 for bit value == 0.  
 \*Data in bit will return in byte.

#### Note

```
byte var_b = cclink_read_input_bit("CtrlBox_DO",0)
// byte[] = {0xEE}
// 0
var_b = cclink_read_input_bit("CtrlBox_DO",7)
// byte[] = {0xEE}
// 1
```

```
var_b = cclink_read_input_bit("CtrlBox_DO",15)
 // byte[] = {0x66}
 // 0
```

### Syntax 3

```
byte[] cclink_read_input_bit(
 int,
 int,
 int
)
```

#### Parameter

|     |                           |
|-----|---------------------------|
| int | Starting address          |
| int | Starting bit              |
| int | The amount of bit to read |

#### Return

byte[] Return data in byte[].  
 Return 1 for bit value == 1.  
 Return 0 for bit value == 0.  
 \*Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].

#### Note

```
byte[] var_ba = cclink_read_input_bit(6,0,16)
 // byte[] = {0xEE,0x66}
 // byte[] = {0,1,1,1,0,1,1,1,0,1,1,0,0,1,1,0}
var_ba = cclink_read_input_bit (6,8,4)
 // byte[] = {0x66}
 // byte[] = {0,1,1,0}
var_ba = cclink_read_input_bit (6,6,4)
 // byte[] = {0xEE,0x66}
 // byte[] = {1,1,0,1}
```

### Syntax 4

```
byte[] cclink_read_input_bit(
 string,
 int,
 int
)
```

#### Parameter

|        |                          |
|--------|--------------------------|
| string | Item name                |
| int    | Starting bit             |
| int    | he amount of bit to read |

#### Return

byte[] Return data in byte[].  
 Return 1 for bit value == 1.  
 Return 0 for bit value == 0.  
 \*Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].

#### Note

```
byte[] var_ba = cclink_read_input_bit("CtrlBox_DO",0,16)
 // byte[] = {0xEE,0x66}
```

```
// byte[] = {0,1,1,1,0,1,1,1,0,1,1,0,0,1,1,0}
var_ba = cclink_read_input_bit ("CtrlBox_DO",8,4)
// byte[] = {0x66}
// byte[] = {0,1,1,0}
var_ba = cclink_read_input_bit ("CtrlBox_DO",6,4)
// byte[] = {0xEE,0x66}
// byte[] = {1,1,0,1}
```

## 21.7 cclink\_read\_output()

Read the output table content.

### Syntax 1

```
byte[] cclink_read_output(
 int,
 int
)
```

#### Parameter

int Starting address  
int The address length to read

#### Return

byte[] Return data in a byte array.

#### Note

```
byte[] var_ba = cclink_read_output(160,12)
// {0xE8,0x07,0x06,0x00,0x18,0x00,0x11,0x00,0x1A,0x00,0x23,0x00}
```

### Syntax 2

```
byte cclink_read_output(
 int
)
```

#### Parameter

int Starting address

#### Return

byte Return data in byte.

#### Note

Same as syntax 1. Fill 1 as the last parameter int by default.

```
byte var_b = cclink_read_output(160)
// 0xE8
```

**Syntax 3**

```
? cclink_read_output(
 string,
 int,
 int
)
```

**Parameter**

string    Item name  
int        The starting shifted address of the item  
int        The amount of the addresses to read

**Return**

?        The data type returned by the item defined in the communication data table.  
\* Data type includes **byte,byte[],int,int[],float,float[],string**

**Syntax 4**

```
? cclink_read_output(
 string,
 int
)
```

**Parameter**

string    Item name  
int        The starting shifted address of the item

**Return**

?        The data type returned by the item defined in the communication data table.  
\* Data type includes **byte,byte[],int,int[],float,float[],string**

**Note**

- \* Same as syntax 3. Read to the end of the item by default.
- \* Reading data based on the configuration file Little Endian (DCBA) or Big Endian (ABCD).

**Syntax 5**

```
? cclink_read_output(
 string
)
```

**Parameter**

**string**    Item name

**Return**

**?**    The data type returned by the item defined in the communication data table.  
 \* Data type includes **byte,byte[],int,int[],float,float[],string**

**Note**

- \* Same as syntax 3. Fill 0 as the starting shifted address of the item and read to the end of the item by default.
- \* Reading data based on the configuration file Little Endian (DCBA) or Big Endian (ABCD).

```
byte var_b = cclink_read_output("ManualAuto",0,1)
 // 0x01
var_b = cclink_read_output("ManualAuto",0)
 // 0x01
var_b = cclink_read_output("ManualAuto")
 // 0x01
```

```
byte[] var_ba = cclink_read_output("Error_Code",0,4)
 // {0x0D,0x82,0x04,0x00}
var_ba = cclink_read_output("Error_Code",0,2)
 // {0x0D,0x82}
var_ba = cclink_read_output("Error_Code")
 // {0x0D,0x82,0x04,0x00}
```

```
int var_i = cclink_read_output("Current_Time_YY")
 // byte[] = {0xE8,0x07} (Little Endian) to int
 // int = 0x000007E8 (Little Endian)
 // int = 2024
```

```
int[] var_ia = cclink_read_output("Current_Time_YY")
```

```
// byte[] = {0xE8,0x07} (Little Endian) to int16[]
// int16[] = {0x000007E8} (Little Endian)
// int[] = {2024}
```

```
float var_f = cclink_read_output("CtrlBox_AO")
```

```
// byte[] = {0x93,0x35,0xD3,0xC0} (Little Endian) to float[]
// float = 0xC0D33593 (Little Endian)
// float = -6.60
```

```
float[] var_fa = cclink_read_output("Coord_CurrBase_Tool",4,12)
```

```
// byte[] = {0x9A,0x99,0x7C,0xC3,0xCD,0xEC,0x5E,0x44,0x00,0x00,0xB4,0x42} (Little Endian) to float[]
// float[] = {0xC37C999A,0x445EECCD,0x42B40000} (Little Endian)
// float[] = {-252.6,891.7,90}
```

```
var_fa = cclink_read_output("Coord_CurrBase_Tool",4)
```

```
// byte[] = {0x9A,0x99,0x7C,0xC3,0xCD,0xEC,0x5E,0x44,0x00,0x00,0xB4,0x42,
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian) to float[]
// float[] = {0xC37C999A,0x445EECCD,0x42B40000,0x00000000,0x00000000} (Little Endian)
// float[] = {-252.6,891.7,90,0,0}
```

```
var_fa = cclink_read_output("Coord_CurrBase_Tool")
```

```
// byte[] =
{0x0A,0xD7,0x23,0x3C,0x9A,0x99,0x7C,0xC3,0xCD,0xEC,0x5E,0x44,0x00,0x00,0xB4,0x42,
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Big Endian) to float[]
// float[] = {0x3C23D70A,0xC37C999A,0x445EECCD,0x42B40000,0x00000000,0x00000000} (Little Endian)
// float[] = {0.01,-252.6,891.7,90,0,0}
```



## 21.8 cclink\_read\_output\_int16()

Read the output table content and convert the data to the 16-bit integer.

### Syntax 1

```
int[] cclink_read_output_int16(
 int,
 int,
 int
)
```

#### Parameter

- int Starting address
- int The address amount to read
- int The conversion of the read data to an int array based on Little Endian (DCBA) or Big Endian (ABCD).
  - 0 Little-Endian
  - 1 Big-Endian
  - 2 Based on the configuration file.

#### Return

- int[] Return data in an integer array.

### Syntax 2

```
int[] cclink_read_output_int16(
 int,
 int
)
```

#### Parameter

- int Starting address
- int The address amount to read

#### Note

Same as syntax 1. Fill 2 as the last parameter int by default.

\* Convert the read data to an int array based on Little Endian (DCBA) or Big Endian (ABCD).

```

int[] var_ia = cclink_read_output_int16(160,12,0)
 // byte[] = {0xE8,0x07,0x06,0x00,0x19,0x00,0x0B,0x00,0x20,0x00,0x0D,0x00} (Little Endian) to
 int16[]
 // int16[] = {0x000007E8,0x00000006,0x00000019,0x0000000B,0x00000020,0x0000000D} (Little
 Endian)
 // int[] = {2024,6,25,11,32,13}
var_ia = cclink_read_output_int16(160,6,0)
 // byte[] = {0xE8,0x07,0x06,0x00,0x19,0x00} (Little Endian) to int16[]
 // int16[] = {0x000007E8,0x00000006,0x00000019} (Little Endian)
 // int[] = {2024,6,25}
var_ia = cclink_read_output_int16(160,1,0)
 // byte[] = {0xE8} (Little Endian) to int16[]
 // int16[] = {0x000000E8} (Little Endian)
 // int[] = {232}
var_ia = cclink_read_output_int16(160,12)
 // byte[] = {0xE8,0x07,0x06,0x00,0x19,0x00,0x0B,0x00,0x2B,0x00,0x38,0x00} (Little Endian) to
 int16[]
 // int16[] = {0x000007E8,0x00000006,0x00000019,0x0000000B,0x0000002B,0x00000038} (Little
 Endian)
 // int[] = {2024,6,25,11,43,56}

```

### Syntax 3

```

int cclink_read_output_int16(
 int
)

```

#### Parameter

**int** Starting address  
 \* Convert the read data to an integer based on Little Endian (DCBA) or Big Endian (ABCD).

#### Return

**int** Return data in an integer

#### Note

```

int var_i = cclink_read_output_int16(160)
 // byte[] = {0xE8,0x07} (Little Endian) to int16
 // int16 = 0x000007E8 (Little Endian)
 // int = 2024

```

## 21.9 cclink\_read\_output\_int()

Read the output table content and convert the data to the 32-bit integer.

### Syntax 1

```
int[] cclink_read_output_int(
 int,
 int,
 int
)
```

#### Parameter

- int Starting address
- int The address amount to read
- int The conversion of the read data to an int array based on Little Endian (DCBA) or Big Endian (ABCD).
  - 0 Little-Endian
  - 1 Big-Endian
  - 2 Based on the configuration file.

#### Return

- int[] Return data in an integer array.

### Syntax 2

```
int[] cclink_read_output_int(
 int,
 int
)
```

#### Parameter

- int Starting address
- int The address amount to read

#### Note

Same as syntax 1. Fill **2** as the last parameter int by default.

\* Convert the read data to an int array based on Little Endian (DCBA) or Big Endian (ABCD).

```

int[] var_ia = cclink_read_output_int(160,12,0)
 // byte[] = {0xE8,0x07,0x06,0x00,0x19,0x00,0x0D,0x00,0x39,0x00,0x24,0x00} (Little Endian) to
 int[]
 // int[] = {0x000607E8,0x000D0019,0x00240039} (Little Endian)
 // int[] = {395240,851993,2359353}
var_ia = cclink_read_output_int(160,6,0)
 // byte[] = {0xE8,0x07,0x06,0x00,0x19,0x00} (Little Endian) to int[]
 // int[] = {0x000607E8,0x00000019} (Little Endian)
 // int[] = {395240,25}
var_ia = cclink_read_output_int(160,1,0)
 // byte[] = {0xE8} (Little Endian) to int[]
 // int[] = {0x000000E8} (Little Endian)
 // int[] = {232}
var_ia = cclink_read_output_int(160,12)
 // byte[] = {0xE8,0x07,0x06,0x00,0x19,0x00,0x0E,0x00,0x02,0x00,0x08,0x00} (Little Endian) to int[]
 // int[] = {0x000607E8,0x000E0019,0x00080002} (Little Endian)
 // int[] = {395240,917529,524290}

```

### Syntax 3

```

int cclink_read_output_int(
 int
)

```

#### Parameter

**int** Starting address  
 \* Convert the read data to an integer based on Little Endian (DCBA) or Big Endian (ABCD).

#### Return

**int** Return data in an integer

#### Note

```

int var_i = cclink_read_output_int(160)
 // byte[] = {0xE8,0x07,0x06,0x00} (Little Endian) to int[]
 // int[] = 0x000607E8 (Little Endian)
 // int = 395240

```

## 21.10 cclink\_read\_output\_float()

Read the output table content and convert the data to the 32-bit floating-point number.

### Syntax 1

```
float[] cclink_read_output_float(
 int,
 int,
 int
)
```

#### Parameter

|     |                                                                                                      |
|-----|------------------------------------------------------------------------------------------------------|
| int | Starting address                                                                                     |
| int | The address amount to read                                                                           |
| int | The conversion of the read data to a float array based on Little Endian (DCBA) or Big Endian (ABCD). |
| 0   | Little-Endian                                                                                        |
| 1   | Big-Endian                                                                                           |
| 2   | Based on the configuration file.                                                                     |

#### Return

float[] Return data in a floating-point number array.

### Syntax 2

```
float[] cclink_read_output_float(
 int,
 int
)
```

#### Parameter

|     |                            |
|-----|----------------------------|
| int | Starting address           |
| int | The address amount to read |

#### Note

Same as syntax 1. Fill 2 as the last parameter int by default.

\*Convert the read data to a float array based on Little Endian (DCBA) or Big Endian (ABCD).

```
float[] var_fa = cclink_read_output_float(256,24,0)
// byte[] =
// {0x0A,0xD7,0x23,0x3C,0x9A,0x99,0x7C,0xC3,0xCD,0xEC,0x5E,0x44,0x00,0x00,0xB4,0x42,
// 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian) to float[]
// float[] = {0x3C23D70A,0xC37C999A,0x445EECCD,0x42B40000,0x00000000,0x00000000} (Little
// Endian)
// float[] = {0.01,-252.6,891.7,90,0,0}
var_fa = cclink_read_output_float(256,12,0)
// byte[] = {0x0A,0xD7,0x23,0x3C,0x9A,0x99,0x7C,0xC3,0xCD,0xEC,0x5E,0x44 } (Little Endian) to
// float[]
// float[] = {0x3C23D70A,0xC37C999A,0x445EECCD} (Little Endian)
// float[] = {0.01,-252.6,891.7}
var_fa = cclink_read_output_float(256,2,0)
// byte[] = {0x0A,0xD7} (Little Endian) to float[]
```

```

 // float[] = {0x0000D70A} (Little Endian)
 // float[] = {0}
var_fa = cclink_read_output_float(256,24)
 // byte[] =
 {0x0A,0xD7,0x23,0x3C,0x9A,0x99,0x7C,0xC3,0xCD,0xEC,0x5E,0x44,0x00,0x00,0xB4,0x42,
 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian) to float[]
 // float[] = {0x3C23D70A,0xC37C999A,0x445EECCD,0x42B40000,0x00000000,0x00000000} (Little
 Endian)
 // float[] = {0.01,-252.6,891.7,90,0,0}

```

### Syntax 3

```

float cclink_read_output_float(
 int
)

```

#### Parameter

**int** Starting address  
 \*Convert the read data to a floating-point number based on Little Endian (DCBA) or Big

#### Return

**float** Return data in a floating-point number

#### Note

```

float var_f = cclink_read_output_float(256)
 // byte[] = {0x0A,0xD7,0x23,0x3C} (Little Endian) to float[]
 // float[] = 0x3C23D70A (Little Endian)
 // float = 0.01

```

## 21.11 cclink\_read\_output\_string()

Read the output table content and convert the data to the string encoded in UTF8.

### Syntax 1

```
string cclink_read_output_string(
 int,
 int
)
```

#### Parameter

int Starting address  
int The address amount to read

#### Return

string Return data in a UTF8 string (ending with 0x00 encountered).

#### Note

```
string var_s = cclink_read_output_string(172,32)
// byte[] = {0x63,0x63,0x6C,0x69,0x6E,0x6B,0x5F,0x66,0x75,0x6E,0x63,0x00,0x00,0x00,0x00,0x00,
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
// string = "cclink_func"
var_s = cclink_read_output_string(172,3)
// byte[] = {0x0A,0xD7,0x23,0x3C,0x9A,0x99,0x7C,0xC3,0xCD,0xEC,0x5E,0x44 }
// string = " ccl"
```

## 21.12 cclink\_read\_output\_bit()

Read the output table content and retrieve the n<sup>th</sup> bit value of the data byte.

### Syntax 1

```
byte cclink_read_output_bit(
 int,
 int
)
```

#### Parameter

**int** Starting address  
**int** The n<sup>th</sup> bit value in the data byte

#### Return

**Byte** Return data in byte.  
 Return 1 for bit value == 1.  
 Return 0 for bit value == 0.  
 \*Data in bit will return in byte.

#### Note

```
byte var_b = cclink_read_output_bit(6,0)
// byte[] = {0xEE}
// 0
var_b = cclink_read_output_bit(6,7)
// byte[] = {0xEE}
// 1
var_b = cclink_read_output_bit(6,15)
// byte[] = {0x66}
// 0
```

### Syntax 2

```
byte cclink_read_output_bit(
 string,
 int
)
```

#### Parameter

**string** Item name  
**int** The n<sup>th</sup> bit value

#### Return

**byte** Return data in byte.  
 Return 1 for bit value == 1.  
 Return 0 for bit value == 0.  
 \*Data in bit will return in byte.

#### Note

```
byte var_b = cclink_read_output_bit("CtrlBox_DO",0)
// byte[] = {0xEE}
// 0
var_b = cclink_read_output_bit("CtrlBox_DO",7)
// byte[] = {0xEE}
// 1
```



```
var_b = cclink_read_output_bit("CtrlBox_DO",15)
 // byte[] = {0x66}
 // 0
```

### Syntax 3

```
byte[] cclink_read_output_bit(
 int,
 int,
 int
)
```

#### Parameter

|     |                           |
|-----|---------------------------|
| int | Starting address          |
| int | Starting bit              |
| int | The amount of bit to read |

#### Return

byte[] Return data in byte[].  
 Return 1 for bit value == 1.  
 Return 0 for bit value == 0.  
 \*Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].

#### Note

```
byte[] var_ba = cclink_read_output_bit(6,0,16)
 // byte[] = {0xEE,0x66}
 // byte[] = {0,1,1,1,0,1,1,1,0,1,1,0,0,1,1,0}
var_ba = cclink_read_output_bit (6,8,4)
 // byte[] = {0x66}
 // byte[] = {0,1,1,0}
var_ba = cclink_read_output_bit (6,6,4)
 // byte[] = {0xEE,0x66}
 // byte[] = {1,1,0,1}
```

### Syntax 4

```
byte[] cclink_read_output_bit(
 string,
 int,
 int
)
```

#### Parameter

|        |                           |
|--------|---------------------------|
| string | Item name                 |
| int    | Starting bit              |
| int    | The amount of bit to read |

#### Return

byte[] Return data in byte[].  
 Return 1 for bit value == 1.  
 Return 0 for bit value == 0.  
 \*Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].

#### Note

```
byte[] var_ba = cclink_read_output_bit("CtrlBox_DO",0,16)
 // byte[] = {0xEE,0x66}
```

```
// byte[] = {0,1,1,1,0,1,1,1,0,1,1,0,0,1,1,0}
var_ba = cclink_read_output_bit ("CtrlBox_DO",8,4)
// byte[] = {0x66}
// byte[] = {0,1,1,0}
var_ba = cclink_read_output_bit ("CtrlBox_DO",6,4)
// byte[] = {0xEE,0x66}
// byte[] = {1,1,0,1}
```

## 21.13 cclink\_write\_output()

Write data to the output table.

### Syntax 1

```
bool cclink_write_output(
 int,
 ?,
 int
)
```

#### Parameter

|      |                                                                                        |
|------|----------------------------------------------------------------------------------------|
| int  | Starting address                                                                       |
| ?    | The data to write                                                                      |
|      | * Available data types include <b>byte, byte[], int, int[], float, float[], string</b> |
| int  | The maximum amount of the address to write                                             |
| > 0  | Valid data length. Write by the amount of the address.                                 |
| <= 0 | Invalid data length. Write by the complete length of the data to write.                |

#### Return

|      |       |                       |
|------|-------|-----------------------|
| Bool | True  | Write successfully.   |
|      | False | Write unsuccessfully. |

1. If the data to write is an empty string or an empty array
2. Unable to send and receive correctly.

#### Note

\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

### Syntax 2

```
bool cclink_write_output(
 int,
 ?
)
```

#### Parameter

|     |                                                                                        |
|-----|----------------------------------------------------------------------------------------|
| int | Starting address                                                                       |
| ?   | The data to write                                                                      |
|     | * Available data types include <b>byte, byte[], int, int[], float, float[], string</b> |

#### Return

|      |       |                       |
|------|-------|-----------------------|
| Bool | True  | Write successfully.   |
|      | False | Write unsuccessfully. |

1. If the data to write is an empty string or an empty array
2. Unable to send and receive correctly.

#### Note

Same as syntax 1. Write with the full length of the data to write by default.

\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

### Syntax 3

```
bool cclink_write_output(
```

```
int,
?,
int,
int
```

```
)
```

### Parameter

|     |                                                                                                                                                             |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| int | Starting address                                                                                                                                            |
| ?   | The data to write<br>* Available data types include <b>byte</b> , <b>byte[]</b> , <b>int</b> , <b>int[]</b> , <b>float</b> , <b>float[]</b> , <b>string</b> |
| int | Starting address of the data to write                                                                                                                       |
| int | The amount of the address to write                                                                                                                          |

### Return

|      |       |                       |
|------|-------|-----------------------|
| bool | True  | Write successfully.   |
|      | False | Write unsuccessfully. |

1. If the data to write is an empty string or an empty array
2. Unable to send and receive correctly.

### Note

\*\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

```
byte var_data = 255
```

```
cclink_write_output(12,var_data,1)
```

```
byte var_b = cclink_read_output(12)
```

```
// 0xFF
```

```
byte[] var_data = {1,127,255}
```

```
cclink_write_output(12,var_data,2)
```

```
byte[] var_ba = cclink_read_output(12,3)
```

```
// {0x01,0x7F,0x00}
```

```
cclink_write_output(12,var_data,3)
```

```
var_ba = cclink_read_output(12,3)
```

```
// {0x01,0x7F,0xFF}
```

```
cclink_write_output(12,var_data,-1)
```

```
var_ba = cclink_read_output(12,3)
```

```
// {0x01,0x7F,0xFF}
```

```
int var_data = 32767
```

```
cclink_write_output(12,var_data,4)
```

```
var_i = cclink_read_output_int(12)
```

```
// byte[] = {0xFF,0x7F,0x00,0x00} (Little Endian) to int
```

```
// int = 0x00007FFF (Little Endian)
```

```
// int = 32767
```

```
cclink_write_output(12,var_data,1)
```

```
var_i = cclink_read_output_int(12)
```

```
// byte[] = {0xFF,0x00,0x00,0x00} (Little Endian) to int
```

```
// int = 0x000000FF (Little Endian)
```

```
// int = 255
```

```

int[] var_data = {32767,99999,-32768}
cclink_write_output(12,var_data,12)
int[] var_ia = cclink_read_output_int(12,12)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian) to int[]
 // int[] = {0x00007FFF,0x0001869F,0xFFFF8000} (Little Endian)
 // int[] = {32767,99999,-32768}
cclink_write_output(12,var_data,3)
var_ia = cclink_read_output_int(12,12)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian) to int[]
 // int[] = {0x00007FFF,0x00000000,0x00000000} (Little Endian)
 // int[] = {32767,0,0}
cclink_write_output(12,var_data,4,4)
var_ia = cclink_read_output_int(12,12)
 // byte[] = {0x9F,0x86,0x01,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian) to int[]
 // int[] = {0x0001869F,0x00000000,0x00000000} (Little Endian)
 // int[] = {99999,0,0 }

float var_data = -10.0
cclink_write_output(12,var_data,4)
float var_f = cclink_read_output_float(12)
 // byte[] = {0x00,0x00,0x20,0xC1} (Little Endian) to float
 // float = 0xC1200000 (Little Endian)
 // float = -10.0
cclink_write_output(12,var_data,1)
var_f = cclink_read_output_int(12)
 // byte[] = {0x00,0x00,0x00,0x00} (Little Endian) to float
 // float = 0x00000000 (Little Endian)
 // float = 0

float[] var_data = {-10.0,3.3,123.45}
cclink_write_output(12,var_data,12)
float[] var_fa = cclink_read_output_float(12,12)
 // byte[] = {0x00,0x00,0x20,0xC1,0x33,0x33,0x53,0x40,0x66,0xE6,0xF6,0x42} (Little Endian) to float[]
 // float[] = {0xC1200000,0x40533333,0x42F6E666} (Little Endian)
 // float[] = {-10.0,3.3,123.45}
cclink_write_output(12,var_data,3)
var_fa = cclink_read_output_float(12,12)
 // byte[] = {0x00,0x00,0x20,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian) to float[]
 // float[] = {0x00200000,0x00000000,0x00000000} (Little Endian)
 // float[] = {0,0,0}
cclink_write_output(12,var_data,4,4)
var_fa = cclink_read_output_float(12,12)
 // byte[] = {0x33,0x33,0x53,0x40,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00} (Little Endian) to float[]
 // float[] = {0x40533333,0x00000000,0x00000000} (Little Endian)
 // float[] = {3.3,0,0}

string var_data = "abcdefg1234567"
cclink_write_output(12,var_data,16)

```

```

string var_s = cclink_read_output_string(12,32)
 // byte[] = {0x61,0x62,0x63,0x64,0x65,0x66,0x67,0x31,0x32,0x33,0x34,0x35,0x36,0x37,0x00,0x00,
 // 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // string = abcdefg1234567
cclink_write_output(12,var_data,8)
var_s = cclink_read_output_string(12,32)
 // byte[] = {0x61,0x62,0x63,0x64,0x65,0x66,0x67,0x31,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
 // 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // string = abcdefg1
cclink_write_output(12,var_data,3,3)
var_s = cclink_read_output_string(12,32)
 // byte[] = {0x64,0x65,0x66,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
 // 0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}
 // string = def

```

#### Syntax 4

```

bool cclink_write_output(
 string,
 int,
 ?,
 int,
 int
)

```

##### Parameter

|        |                                                                                                                                                             |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| string | Item name                                                                                                                                                   |
| int    | The starting shifted address of the item                                                                                                                    |
| ?      | The data to write<br>* Available data types include <b>byte</b> , <b>byte[]</b> , <b>int</b> , <b>int[]</b> , <b>float</b> , <b>float[]</b> , <b>string</b> |
| int    | Starting address of the data to write                                                                                                                       |
| int    | The amount of the address to write                                                                                                                          |

##### Return

|      |              |                       |
|------|--------------|-----------------------|
| bool | <b>True</b>  | Write successfully.   |
|      | <b>False</b> | Write unsuccessfully. |

1. If the data to write is an empty string or an empty array
2. Unable to send and receive correctly.

##### Note

\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

#### Syntax 5

```

bool cclink_write_output(
 string,
 int,
 ?,
 int
)

```

##### Parameter

|        |                                          |
|--------|------------------------------------------|
| string | Item name                                |
| int    | The starting shifted address of the item |

? The data to write  
 \* Available data types include **byte,byte[],int,int[],float,float[],string**  
 int Starting address of the data to write

**Return**

bool      True      Write successfully.  
           False      Write unsuccessfully.

1. If the data to write is an empty string or an empty array
2. Unable to send and receive correctly.

**Note**

Same as syntax 4. Write with the full length of the data to write by default.  
 \* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

**Syntax 6**

```
bool cclink_write_output(
 string,
 int,
 ?
)
```

**Parameter**

string      Item name  
 int          The starting shifted address of the item  
 ?            The data to write  
 \* Available data types include **byte,byte[],int,int[],float,float[],string**

**Return**

bool      True      Write successfully.  
           False      Write unsuccessfully.

1. If the data to write is an empty string or an empty array
2. Unable to send and receive correctly.

**Note**

Same as syntax 4. Fill 0 as the starting address to write and write with the full length of the data to write by default.  
 \* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

**Syntax 7**

```
bool cclink_write_output(
 string,
 ?
)
```

**Parameter**

string      Item name  
 ?            The data to write  
 \* Available data types include **byte,byte[],int,int[],float,float[],string**

**Return**

bool      True      Write successfully.  
           False      Write unsuccessfully.

1. If the data to write is an empty string or an empty array
2. Unable to send and receive correctly.

### Note

Same as syntax 4. Fill 0 as the starting shifted address and the starting address to write as well as write with the full length of the data to write by default.

\* Write data based on Little Endian (DCBA) or Big Endian (ABCD) in the configuration file.

```
int[] var_data = {32767,99999,-32768}
cclink_write_output("Register_1",0,var_data,0,12)
int[] var_ia = cclink_read_output_int(12,12)
 // byte[] = {0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x80,0xFF,0xFF} (Little Endian) to int[]
 // int[] = {0x00007FFF,0x0001869F,0xFFFF8000} (Little Endian)
 // int[] = {32767,99999,-32768}
cclink_write_output("Register_1",4,var_data,4,4)
var_ia = cclink_read_output_int(12,12)
 // byte[] = {0x00,0x00,0x00,0x00,0x9F,0x86,0x01,0x00,0x00,0x00,0x00,0x00} (Little Endian) to int[]
 // int[] = {0x00000000,0x0001869F,0x00000000} (Little Endian)
 // int[] = {0,99999,0}
cclink_write_output("Register_1",4,var_data)
var_ia = cclink_read_output_int(12,12)
 // byte[] = {0x00,0x00,0x00,0x00,0xFF,0x7F,0x00,0x00,0x9F,0x86,0x01,0x00} (Little Endian) to int[]
 // int[] = {0x00000000,0x00007FFF,0x0001869F} (Little Endian)
 // int[] = {0,32767,99999}

float[] var_data = {-10.0,3.3,123.45}
cclink_write_output("Register_1",0,var_data,0,12)
float[] var_fa = cclink_read_output_float(12,12)
 // byte[] = {0x00,0x00,0x20,0xC1,0x33,0x33,0x53,0x40,0x66,0xE6,0xF6,0x42} (Little Endian) to float[]
 // float[] = {0xC1200000,0x40533333,0x42F6E666} (Little Endian)
 // float[] = {-10.0,3.3,123.45}
cclink_write_output("Register_1",4,var_data,4,8)
var_fa = cclink_read_output_float(12,12)
 // byte[] = {0x00,0x00,0x00,0x00,0x33,0x33,0x53,0x40,0x66,0xE6,0xF6,0x42} (Little Endian) to float[]
 // float[] = {0x00000000,0x40533333,0x42F6E666} (Little Endian)
 // float[] = {0,3.3,123.45}
cclink_write_output("Register_1",8,var_data)
var_fa = cclink_read_output_float(12,12)
 // byte[] = {0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x20,0xC1} (Little Endian) to float[]
 // float[] = {0x00000000,0x00000000,0xC1200000} (Little Endian)
 // float[] = {0,0,-10}
```



## 21.14 cclink\_write\_output\_bit()

Write content to the  $n^{\text{th}}$  bit value of the data byte in the output table.

### Syntax 1

```
bool cclink_write_output_bit(
 int,
 int,
 int
)
```

#### Parameter

|     |                                 |
|-----|---------------------------------|
| int | Starting address                |
| int | The $n^{\text{th}}$ bit value   |
| int | The data to write               |
|     | *Data in bit will write in int. |

#### Return

|      |       |                       |
|------|-------|-----------------------|
| Bool | True  | Write successfully.   |
|      | False | Write unsuccessfully. |

1. Unable to send correctly and receive .

#### Note

```
byte var_data = 240
cclink_write_output(12,var_data)
byte var_b = cclink_read_output(12)
// 0xF0
cclink_write_output_bit(12,1,1)
var_b = cclink_read_output_bit(12,1)
// 0xF2 get bit: "1"
// 1
cclink_write_output_bit(12,7,0)
var_b = cclink_read_output_bit(12,7)
// 0x72 get bit: "7"
// 0
```

**Syntax 2**

```
bool cclink_write_output_bit(
 string,
 int,
 int
)
```

**Parameter**

|        |                                 |
|--------|---------------------------------|
| string | Item name                       |
| int    | The n <sup>th</sup> bit value   |
| int    | The data to write               |
|        | *Data in bit will write in int. |

**Return**

|      |       |                       |
|------|-------|-----------------------|
| bool | True  | Write successfully.   |
|      | False | Write unsuccessfully. |

1. Unable to send correctly and receive.

**Note**

```
byte var_data = 240
cclink_write_output(12,var_data)
byte var_b = cclink_read_output(12)
// 0xF0
cclink_write_output_bit("Register_1",1,1)
var_b = cclink_read_output_bit(12,1)
// 0xF2 get bit: "1"
// 1
cclink_write_output_bit("Register_1",7,0)
var_b = cclink_read_output_bit(12,7)
// 0x72 get bit: "7"
// 0
```

**Syntax 3**

```
bool cclink_write_output_bit(
 int,
 int,
 byte[],
 int,
 int
)
```

**Parameter**

|        |                                                                                     |
|--------|-------------------------------------------------------------------------------------|
| int    | Starting address                                                                    |
| int    | The n <sup>th</sup> bit value                                                       |
| byte[] | The data to write                                                                   |
|        | *Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1]. |
| int    | The starting bit of the data to write                                               |
| int    | The bit amount of the data to write                                                 |

**Return**

|      |       |                       |                                          |
|------|-------|-----------------------|------------------------------------------|
| bool | True  | Write successfully.   |                                          |
|      | False | Write unsuccessfully. | 1. Unable to send correctly and receive. |

**Note**

Bit value = 1 for byte value >=1  
 Bit value = 0 for byte value ==0

**Syntax 4**

```
bool cclink_write_output_bit(
 int,
 int,
 byte[],
 int
)
```

**Parameter**

|        |                                                                                     |
|--------|-------------------------------------------------------------------------------------|
| int    | Starting address                                                                    |
| int    | The n <sup>th</sup> bit value                                                       |
| byte[] | The data to write                                                                   |
|        | *Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1]. |
| int    | The starting bit of the data to write                                               |

**Return**

|      |       |                       |                                          |
|------|-------|-----------------------|------------------------------------------|
| Bool | True  | Write successfully.   |                                          |
|      | False | Write unsuccessfully. | 1. Unable to send correctly and receive. |

**Note**

\*Same as syntax 3. Write with the full length of the rest data to write.  
 Bit value = 1 for byte value >=1  
 Bit value = 0 for byte value ==0

**Syntax 5**

```
bool cclink_write_output_bit(
 int,
 int,
 byte[]
)
```

**Parameter**

**int** Starting address  
**int** The n<sup>th</sup> bit value  
**byte[]** The data to write  
 \*Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].

**Return**

**bool** **True** Write successfully.  
**False** Write unsuccessfully. 1. Unable to send correctly and receive.

**Note**

\*Same as syntax 3. Fill 0 as the starting bit to write as well as write with the full length of the data to write by default.

Bit value = 1 for byte value >=1

Bit value = 0 for byte value ==0

```

byte[] var_data = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
cclink_write_output_bit(12,0,var_data,0,20)
byte[] var_ba = cclink_read_output(12,3)
// byte[] = {0x39,0xB8,0x0C}
var_ba = cclink_read_output_bit (12,0,20)
// byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
cclink_write_output_bit(12,3,var_data,5,10)
var_ba = cclink_read_output (12,3)
// byte[] = {0x08,0x0E,0x00}
var_ba = cclink_read_output_bit (12,0,20)
// byte[] = {0,0,0,1,0,0,0,0,0,1,1,1,0,0,0,0,0,0,0,0}

```

**Syntax 6**

```

bool cclink_write_output_bit(
 string,
 int,
 byte[],
 int,
 int
)

```

**Parameter**

**string** Item name  
**int** Starting bit  
**byte[]** The data to write  
 \*Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1].  
**int** The starting bit to write data  
**int** The bit amount of the data to write

**Return**

**bool** **True** Write successfully.  
**False** Write unsuccessfully. 1. Unable to send correctly and receive.

**Note**

Bit value = 1 for byte value >=1

Bit value = 0 for byte value ==0

**Syntax 7**

```
bool cclink_write_output_bit(
 string,
 int,
 byte[],
 int
)
```

**Parameter**

|        |                                                                                                          |
|--------|----------------------------------------------------------------------------------------------------------|
| string | Item name                                                                                                |
| int    | Starting bit                                                                                             |
| byte[] | The data to write<br>*Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1]. |
| int    | The starting bit to write data                                                                           |

**Return**

|      |       |                       |                                          |
|------|-------|-----------------------|------------------------------------------|
| bool | True  | Write successfully.   |                                          |
|      | False | Write unsuccessfully. | 1. Unable to send correctly and receive. |

**Note**

\*Same as syntax 6. Write with the full length of the rest data to write.

Bit value = 1 for byte value >=1

Bit value = 0 for byte value ==0

**Syntax 8**

```
bool cclink_write_output_bit(
 string,
 int,
 byte[]
)
```

**Parameter**

|        |                                                                                                          |
|--------|----------------------------------------------------------------------------------------------------------|
| string | Item name                                                                                                |
| int    | Starting bit                                                                                             |
| byte[] | The data to write<br>*Data in bit will return in byte such as bit[0] for byte[0] and bit[1] for byte[1]. |

**Return**

|      |       |                       |                                          |
|------|-------|-----------------------|------------------------------------------|
| bool | True  | Write successfully.   |                                          |
|      | False | Write unsuccessfully. | 1. Unable to send correctly and receive. |

**Note**

\*Same as syntax 6. Fill 0 as the starting bit to write as well as write with the full length of the data to write by default.

Bit value = 1 for byte value >=1

Bit value = 0 for byte value ==0

```
byte[] var_data = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
```

```
cclink_write_output_bit("Register_1",0,var_data,0,20)
```

```
byte[] var_ba = cclink_read_output(12,3)
```

```
// byte[] = {0x39,0xB8,0x0C}
```

```
var_ba = cclink_read_output_bit (12,0,20)
```

```
// byte[] = {1,0,0,1,1,1,0,0,0,0,0,1,1,1,0,1,0,0,1,1}
```

```
cclink_write_output_bit("Register_1",3,var_data,5,10)
```

```
var_ba = cclink_read_output (12,3)
 // byte[] = {0x08,0x0E,0x00}
var_ba = cclink_read_output_bit (12,0,20)
 // byte[] = {0,0,0,1,0,0,0,0,1,1,1,0,0,0,0,0,0,0}
```

## 22. Real Time Remote Server

Real Time Remote Server (RTRS) comes with functions established with Socket TCP based on the framework of client/server connections. Once enabled, the robot establishes a Real time Socket TCP Listener Serve to send the robot status and data to all of the connected clients or receive the contents from the clients to execute the respective instructions and update the respective information periodically and promptly without the real-time guarantee.

When the TMflow program starts, the Real-Time Remote Server either launches automatically or stays inactive, depending on the configured settings. It shuts down only when the TMflow program is closed or when explicitly configured to stop. The server's IP address and port number appear in the Notice Window.

IP        TMflow → Configuration → Remote Control Settings →  
              Command Interface → RTRS → Static IP  
 Port     5891

### 22.1 GUI Setting

Real Time Remote Server

Status ☒ Enable 

☐ Motion Command 

☒ Static IP

IP Address

Subnet Mask

Default Gateway

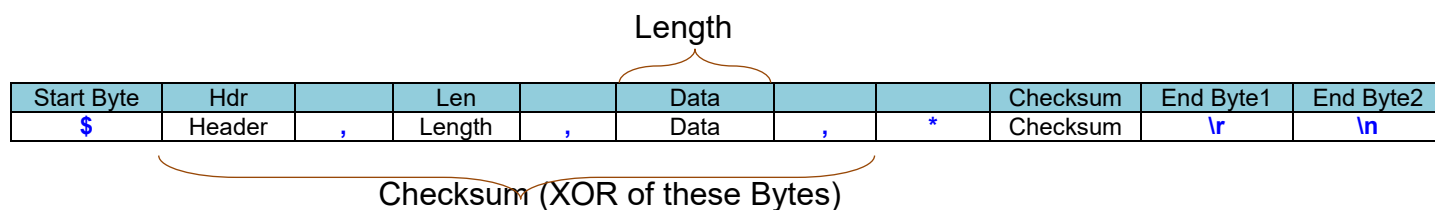
**Enable/Disable** Enable or disable Real Time Remote Server. Once enabled, a system restart is required for the changes to take effect.

**Motion Control Instructions** Motion control commands become available once enabled, and the robot will switch to external control mode.

**TMRTS** Packet header for data communication manages data communication. The system activates without requiring the selection of the Motion Command option in the interface. The connection becomes available after a system restart. For detailed packet specifications, refer to the TMRTS documentation.

**TMRTC** Packet Header for Data Communication manages data communication. The system activates by requiring the selection of the Motion Command option in the interface. The connection becomes available after a system restart. This packet format is used solely for motion control. To obtain robot status for coordinated control, use it together with TMRTS packets. For motion control packet details, refer to the TMRTC documentation.

## 22.2 Communication Protocol



| Name       | Size | ASCII | HEX  | Description                         |
|------------|------|-------|------|-------------------------------------|
| Start Byte | 1    | \$    | 0x24 | Start Byte for Communication        |
| Header     | X    |       |      | Header for Communication            |
| Separator  | 1    | ,     | 0x2C | Separator between Header and Length |
| Length     | Y    |       |      | Length of Data                      |
| Separator  | 1    | ,     | 0x2C | Separator between Length and Data   |
| Data       | Z    |       |      | Communication Data                  |
| Separator  | 1    | ,     | 0x2C | Separator between Data and Checksum |
| Sign       | 1    | *     | 0x2A | Begin Sign of Checksum              |
| Checksum   | 2    |       |      | Checksum of Communication           |
| End Byte 1 | 1    | \r    | 0x0D |                                     |
| End Byte 2 | 1    | \n    | 0x0A | End Byte of Communication           |

\*Using the same communication protocol with external commands.

### 1. Header

Defines the purpose of communication packets. Different headers come with different definitions of communication packets and data.

- **TMSVR** Defines the data transmission functions of TM Real Time Remote Server
- **TMRTC** Defines the motion control functions of TM Real Time Remote Server
- **CPERR** Defines the errors of the communication packets such as packer errors, checksum errors, header errors, and so on.

\*Using the same content definitions with CPERR in external commands.



## 2. Length

The length indicates the length in the UTF8 bytes occupied by Data. Users can use decimal, hexadecimal, or binary format. The maximum length is 32 bits.

For example,

|                            |                                                                |
|----------------------------|----------------------------------------------------------------|
| \$TMRTS,100,Data,*CS\r\n   | // 100 in decimal indicates the data length is 100 bytes       |
| \$TMRTS,0x100,Data,*CS\r\n | // 0x100 in hexadecimal indicates the data length is 256 bytes |
| \$TMRTS,0b100,Data,*CS\r\n | // 0b100 in binary indicates the data length is 4 bytes        |
| \$TMRTS,8,1,達明,*CS\r\n     | // indicates the length of Data, 1,達明, is 8 bytes (UTF8)       |

## 3. Data

The content of the communication packet can support any character (including 0x00 .. 0xFF and uses UTF8 encoding), and the data length determines by Length. The purpose and description defined in Data must be defined by the header.

## 4. Checksum

The checksum of the communication packet. The calculation method is XOR (exclusive OR). The calculation range is all Bytes between \$ and \* (excluding \$ and \*) as shown below.

\$TMRTS,100,Data,\*CS\r\n

Checksum = Byte[1] ^ Byte[2] ... ^ Byte[N-6]

The checksum format is set to 2 bytes in hexadecimal (but not 0x), such as

\$TMRTS,5,10,OK,\*7C

CS = 0x54 ^ 0x4D ^ 0x52 ^ 0x54 ^ 0x53 ^ 0x2C ^ 0x35 ^ 0x2C ^ 0x31 ^ 0x30 ^ 0x2C ^ 0x4F ^ 0x4B ^ 0x2C = 0x7C

CS = 7C (0x37 0x43)

## 22.3 TMRTS

| Start Byte | Hdr   |   | Len    |   | Data |   |   | Checksum | End Byte1 | End Byte2 |
|------------|-------|---|--------|---|------|---|---|----------|-----------|-----------|
| \$         | TMRTS | , | Length | , | Data | , | * | Checksum | \r        | \n        |

| ID             |  |   | Mode          |   | Content       |
|----------------|--|---|---------------|---|---------------|
| Transaction ID |  | , | 0/1/<br>7/8/9 | , | Item or Value |

TMRTS is defined as the communication protocol used on port 5895 in the Real Time Remote Server (RTRS). The Data section of the packet is further divided into three segments, ID (Transaction ID), Mode (Content Mode), and Content (Item and Value), separated with commas and described below.

- ID** The transaction number expressed in any alphanumeric characters. (Reports the CPERR 04 error if a non-alphanumeric byte is encountered.) When used as a communication packet response, it is a transaction number that identifies which group of commands to respond.
- ,
- Mode** The data format and structure define the type of content transmitted.
- 4 Indicates the server responds to a client command; data format is string.
  - 5 Indicates either that the server sends data to the client at fixed intervals or the client sends a command to the server; data format is binary.
  - 7 Initiates or stops data transmission from the server to the client.
  - 8 Sets the transmission frequency (in Hz) for data sent from the server to the client.
  - 9 Configures the data content transmitted from the server to the client
- ,
- Content** The data content. Formatted by the mode definition.

### Note

TMSVR command is for the client and the server to communicate in both directions. Under normal circumstances, the server periodically sends data to the client based on the item list configured by the client using Mode 9. When the server sends to the client, the data is sent to the client to read from the server with no response to the server required. When the client sends to the server, the data is received by the server from the client to write with response to the client required.

#### ● ID Transaction Number

When the server sends data, cycles from 0 to 9 with each iteration. When the client sends data to the server, the transaction number can be in any alphanumeric characters customized at the client side. If the communication packet format is checked and correct, the server will reply the client with the command processing status by the transaction number in the packet.

#### ● Mode Format and Mode for the Data Content

When Mode = 1, the client writes data to the server. The client can write multiple items in

a single transmission, not limited to one. The server will send the command processing status according to the format of Mode 0.

When Mode = 8 or 9, the client configures the data transmission frequency and content. When Mode = 7, the client starts or stops data transmission. The typical usage flow is as follows:

1. Enable the RTRS service.
2. Configure the transmission frequency and data content using the TMRTS protocol, Mode 8 and Mode 9, respectively.
3. Start data transmission using TMRTS Mode 7.
4. To write data, send a command using TMRTS Mode 1.

- Writing or Reading Confirmation

When the client sends data writing to the server, the servers **checks whether all the criteria are correct before performing the request. If there is any error with the write command, no request will be performed.** The criteria to write for inspection are:

1. The validity of the mode as the format of the data content
2. The connected client's write permissions based on the IP Filter.
3. The data content matches to the mode.
4. The item to write or read exists.
5. The attribute of the item to write is not read only.
6. The written data matches the data type of each item.
7. Data transmission settings (Mode 9) successfully configured.

### 22.3.1 Mode = 0 (the status the server responds to the client command processing)

After the server receives and processes a write command from a client, it will respond with another command with Mode 0. The details for Mode 0 are as follows.

Data

| ID             |   | Mode |   | Error Code |   | Error Description |
|----------------|---|------|---|------------|---|-------------------|
| Transaction ID | , | 0    | , | 00 .. 08   | , |                   |

Transaction ID      Defined while the client sends the command for the server to reply with.

Mode      0 for the server to respond to the client

Error Code      Error code definitions. Fixed as 2 bytes and in hexadecimal (but not 0x)

- 00      Correct writing. No error.
- 01      The communication format or mode is not supported. (Ex. Mode = 99)
- 02      The connected client is not permitted to write. (IP filer without write permission)
- 03      The communication format and the data content format are mismatched. (Ex Not in binary format)
- 04      Item to write does not exist.
- 05      Unable to write to read-only items.
- 07      Values to write mismatches with the configured type or the size.
- 08      Incomplete data transmission configuration.

Error Description      Error description, following the error code

- 00      OK
- 01      NotSupport;XXX
- 02      WritePermission
- 03      InvalidData
- 04      NotExist;XXX      // ;XXX denotes which the incorrect data item
- 05      ReadOnly;XXX
- 07      ValueError;XXX
- 08      SettingIncomplete

### When input is correct and the server responds

```

< 24 54 4D 52 54 53 2C // $TMRTS, // Header
 35 2C // 5, // Length
 32 2C 37 2C // 2,7, // transaction ID 2, mode 7
 30 // 0 // Close data transmission
 2C 2A 36 30 0D 0A // ,*60\r\n // Checksum
> 24 54 4D 52 54 53 2C // $TMRTS, // Header
 39 2C // 9, // Length
 32 2C 30 2C // 2,0, // server responds to ID 2, mode 0
 30 30 2C // 00, // Error Code 00
 4F 4B // OK // Correct writing
 2C 2A 37 33 0D 0A // ,*73\r\n // Checksum

```

### When using a non-existent Mode

```

< 24 54 4D 52 54 53 2C // $TMRTS, // Header
 35 2C // 5, // Length
 33 2C 35 2C // 3,5, // transaction ID 3, mode 5
 30 // 0 // Close data transmission
 2C 2A 36 33 0D 0A // ,*63\r\n // Checksum
> 24 54 4D 52 54 53 2C // $TMRTS, // Header
 32 30 2C // 20, // Length
 33 2C 30 2C // 3,0, // server responds to ID 30, mode 0
 30 31 2C // 01, // Error Code 01
 4E 6F 74 53 75 70 70 6F 72 74 3B 20 35 // NotSupport; 5 // mode 5 not supported
 2C 2A 37 38 0D 0A // ,*78\r\n // Checksum

```

When multiple clients write to the same item simultaneously, all writes will occur on the first client if a client without write permission issues a write command.

```

< 24 54 4D 52 54 53 2C // $TMRTS, // Header
 31 37 2C // 17, // Length
 31 2C 31 2C // 1,1, // transaction ID 1, mode 1
 08 00 43 74 72 6C 5F 44 4F 31 01 00 01 // Ctrl_DO1=1 // The name occupied 8 bytes, the value, 1
 byte
 2C 2A 32 32 0D 0A // ,*22\r\n // Checksum
> 24 54 4D 52 54 53 2C // $TMRTS, // Header
 32 32 2C // 22, // Length
 31 2C 30 2C // 1,0, // server responds to ID 1, mode 0
 30 32 2C // 02, // Error Code 02
 57 72 69 74 65 50 65 72 6D 69 73 73 69 6F 6E // WritePermission // No writing permission
 2C 2A 33 39 0D 0A // ,*39\r\n // Checksum

```

### When the data content is invalid

```

< 24 54 4D 52 54 53 2C // $TMRTS, // Header
 31 37 2C // 17, // Length
 33 32 2C 39 2C // 32,9, // transaction ID 32, mode 9
 0B 00 4A 6F 69 6E 74 41 6E 67 6C 65 // JointAngle

```

```

// The name occupied 11 bytes, and 10 bytes, actually
2C 2A 34 32 0D 0A // ,*42\r\n // Checksum
> 24 54 4D 52 54 53 2C // $TMRTS, // Header
31 39 2C // 19, // Length
33 32 2C 30 2C // 32,0, // server responds to ID 32, mode 0
30 33 2C // 03, // Error Code 03
49 6E 76 61 6C 69 64 44 61 74 61 // InvalidData // Incorrect number of bytes used for
// the item name.
2C 2A 31 37 0D 0A // ,*17\r\n // Checksum

```

When attempting to configure a non-existent item as a data exchange target

```

< 24 54 4D 52 54 53 2C // $TMRTS, // Header
37 2C // 7, // Length
30 2C 39 2C // 0,9, // transaction ID 0, mode 9
01 00 30 // 0 // The name occupied 1 byte
2C 2A 36 46 0D 0A // ,*6F\r\n // Checksum
> 24 54 4D 52 54 53 2C // $TMRTS, // Header
31 37 2C // 17, // Length
30 2C 30 2C // 0,0, // server responds to ID 0, mode 0
30 34 2C // 04, // Error Code 04
4E 6F 74 45 78 69 73 74 3B 30 // NotExist;0 // Item 0 does not exist.
2C 2A 34 33 0D 0A // ,*43\r\n // Checksum

```

When the client writes a value to a read-only item

```

< 24 54 4D 52 54 53 2C // $TMRTS, // Header
31 37 2C // 17, // Length
32 37 2C 31 2C // 27,1, // transaction ID 27, mode 1
07 00 45 6E 64 5F 44 49 30 01 00 01 // End_DI0=1 // The name occupied 7 bytes, the value, 1
// byte
2C 2A 37 38 0D 0A // ,*78\r\n // Checksum
> 24 54 4D 52 54 53 2C // $TMRTS, // Header
32 34 2C // 24, // Length
32 37 2C 30 2C // 27,0, // server responds to ID 27, mode 0
30 35 2C // 05, // Error Code 05
52 65 61 64 4F 6E 6C 79 3B 45 6E 64 5F 44 49 30 // ReadOnly;End_DI0//Read-only
// access only.
2C 2A 36 41 0D 0A // ,*6A\r\n // Checksum

```

When the value in the client's command exceeds the allowable range

```

< 24 54 4D 52 54 53 2C // $TMRTS, // Header
39 2C // 9, // Length
33 30 2C 38 2C // 30,8, // transaction ID 30, mode 8
00 00 16 44 // 600 // Floating-point number
2C 2A 33 30 0D 0A // ,*30\r\n // Checksum
> 24 54 4D 52 54 53 2C // $TMRTS, // Header
34 34 2C // 44, // Length

```

```

33 30 2C 30 2C // 30,0, // server responds to ID 30, mode 0
30 37 2C // 07, // Error Code 07
56 61 6C 75 65 45 72 72 6F 72 3B 54 68 65 20 76 61 6C 75 65 20 69 73 20 6F 75 74 20 6F
66 20 72 61 6E 67 65 // ValueError;The value is out of range // Value out of range
2C 2A 34 30 0D 0A // ,*40\r\n // Checksum

```

When the client attempts to start data transmission without completing the data exchange item configuration

```

< 24 54 4D 52 54 53 2C // $TMRTS, // Header
 35 2C // 5, // Length
 31 2C 37 2C // 1,7, // transaction ID 1, mode 7
 31 // 1 // Open data transmission
 2C 2A 36 32 0D 0A // ,*62\r\n // Checksum
> 24 54 4D 52 54 53 2C // $TMRTS, // Header
 32 34 2C // 24, // Length
 31 2C 30 2C // 1,0, // server responds to ID 1, mode 0
 30 38 2C // 08, // Error Code 08
 53 65 74 74 69 6E 67 49 6E 63 6F 6D 70 6C 65 74 65 // SettingIncomplete
 // Configuration incomplete.
 2C 2A 33 42 0D 0A // ,*3B\r\n // Checksum

```

### 22.3.2 Mode = 1 BINARY

The data content is transmitted in binary mode by converting the data item name with the Little Endian value and the value with UTF8 to a byte array accordingly. The format is shown as below. Mode 1 serves as the server regularly sends predefined data items to the client concurrently, or the client writes values to writable data items on the server. When multiple clients are connected, write access is granted to the first client that writes to a given data exchange item. Due to the potential for high-frequency communication, the server does not send a response when a client command is correct.

Data

| ID             |   | Mode |   | Content        |
|----------------|---|------|---|----------------|
| Transaction ID | , | 1    | , | Item and Value |

| <i>Length of Item</i><br>2 bytes Little Endian | <i>Item</i><br>UTF8 | <i>Length of Value</i><br>2 bytes Little Endian | <i>Value</i><br>Little Endian / UTF8 | ... |
|------------------------------------------------|---------------------|-------------------------------------------------|--------------------------------------|-----|
|------------------------------------------------|---------------------|-------------------------------------------------|--------------------------------------|-----|

Length of Item      2 bytes in Little Endian, value from 0 to 65535 indicating the length of the item that follows

Item                item name

Length of Value    2 bytes in Little Endian, value from 0 to 65535 indicating the length of the data value that follows

Value               data value

Assume the communication data includes Joint\_Angle (type: float[]) transmitted from the Server to the Client.

```
> 24 54 4D 52 54 53 2C // $TMRTS, // Header
 34 33 2C // 43, // Length
 30 2C 31 2C // 0,1, // transaction ID 0, mode 1, binary
 0B 00 4A 6F 69 6E 74 5F 41 6E 67 6C 65 18 00 3C BD 2D 41 3F F7 14 C3 77 ED 36 42 AB
 FE 0F 43 78 FA 95 41 4F 6F 07 42 // Joint_Angle={10.8587, -148.9658, 45.7319,
 143.9948, 18.7473, 33.8587}
 // The name occupied 11 bytes, the value, 24 bytes
 2C 2A 30 42 0D 0A // ,*0B\r\n // Checksum
```

Assume the Client sets the values of Ctrl\_DO2 (type: byte) and Ctrl\_AO0 (type: float). Note that in this case, the Server will not respond to the Client.

```
< 24 54 4D 52 54 53 2C // $TMRTS, // Header
 34 31 2C // 41, // Length
 31 2C 31 2C // 1,1, // transaction ID 1, mode 1, binary
 08 00 43 74 72 6C 5F 44 4F 32 01 00 01 // Ctrl_DO2=1
 // The name occupied 8 bytes, the value, 1 byte
 08 00 43 74 72 6C 5F 41 4F 30 04 00 00 00 80 3F // Ctrl_AO0=1
 // The name occupied 8 bytes, the value, 4 bytes
 2C 2A 44 39 0D 0A // ,*D9\r\n // Checksum
```



### 22.3.3 Mode = 7 START/STOP Data Streaming

The system transmits content in UTF-8 format to configure the start or stop of data transmission.

The table below provides a detailed outline of the format.

Data

| ID             |   | Mode |   | Content |
|----------------|---|------|---|---------|
| Transaction ID | , | 7    | , | Value   |

|             |
|-------------|
| <b>UTF8</b> |
| '0': STOP   |
| '1': START  |

To start data transmission

```

< 24 54 4D 52 54 53 2C // $TMRTS, // Header
 35 2C // 5, // Length
 32 2C 37 2C // 2,7, // transaction ID 2,7
 31 // 1 // Start data transmission
 2C 2A 36 31 0D 0A // ,*61\r\n // Checksum
> 24 54 4D 52 54 53 2C // $TMRTS, // Header
 39 2C // 9, // Length
 32 2C 30 2C // 2,0, // server responds to ID 2, mode 0
 30 30 2C // 00, // Error Code 00
 4F 4B // OK // Correct writing
 2C 2A 37 33 0D 0A // ,*73\r\n // Checksum

```

To enable the streaming function before completing the data transmission settings (Mode 9)

```

< 24 54 4D 52 54 53 2C // $TMRTS, // Header
 35 2C // 5, // Length
 31 2C 37 2C // 1,7, // transaction ID 1, mode 7, binary
 31 // 1 // Activate data transmission
 2C 2A 36 32 0D 0A // ,*62\r\n // Checksum
> 24 54 4D 52 54 53 2C // $TMRTS, // Header
 32 34 2C // 24, // Length
 31 2C 30 2C // 1,0, // server responds to ID 1, mode 0
 30 38 2C // 08, // Error Code 08
 53 65 74 74 69 6E 67 49 6E 63 6F 6D 70 6C 65 74 65
 // SettingIncomplete // Response for incomplete data transmission
 settings (Mode 9)
 2C 2A 33 42 0D 0A // ,*3B\r\n // Checksum

```

### 22.3.4 Mode = 8 SET Streaming Frequency

The system transmits data in binary format at a default frequency of 50 Hz. The system converts the frequency and rounds it to the nearest whole millisecond. The server then transmits the data at the specified frequency. The table below provides a detailed outline of the format.

Data

| ID             |   | Mode |   | Content |
|----------------|---|------|---|---------|
| Transaction ID | , | 8    | , | Value   |

**float Little Endian**  
 50-500 (Hz)

To configure the streaming transmission frequency

```

< 24 54 4D 52 54 53 2C // $TMRTS, // Header
 38 2C // 8, // Length
 32 2C 38 2C // 2,8, // transaction ID 2, mode 8
 00 00 48 42 // 50 // Set transmission frequency to 50 Hz.
 2C 2A 35 38 0D 0A // ,*58\r\n // Checksum
> 24 54 4D 52 54 53 2C // $TMRTS, // Header
 39 2C // 9, // Length
 32 2C 30 2C // 2,0, // server responds to ID 2, mode 0
 30 30 2C 4F 4B // 00,OK // Error Code 00 , Correct writing
 2C 2A 37 33 0D 0A // ,*73\r\n // Checksum

< 24 54 4D 52 54 53 2C // $TMRTS, // Header
 38 2C // 8, // Length
 33 2C 38 2C // 3,8, // transaction ID 3, mode 8
 00 00 7A 44 // 1000 // Set transmission frequency to 1000Hz
 2C 2A 36 44 0D 0A // ,*6D\r\n // Checksum
> 24 54 4D 52 54 53 2C // $TMRTS, // Header
 34 33 2C // 43, // Length
 33 2C 30 2C // 3,0, // transaction ID 3mode 0
 30 37 2C // 07, // Error Code 07
 56 61 6C 75 65 45 72 72 6F 72 3B 54 68 65 20 76 61 6C 75 65 20 69 73 20 6F 75 74 20 6F
 66 20 72 61 6E 67 65 // ValueError;The value is out of range
 // The configured value exceeds the limit.
 2C 2A 37 37 0D 0A // ,*77\r\n // Checksum

```

### 22.3.5 Mode = 9 SET Streaming Data

The system transmits data content using a binary format. Configure the names of the data items to be transmitted periodically using the format shown in the table below.

Data (client to server)

| ID             |   | Mode |   | Content |
|----------------|---|------|---|---------|
| Transaction ID | , | 9    | , | Item    |

|                                                |                     |     |                                                                           |
|------------------------------------------------|---------------------|-----|---------------------------------------------------------------------------|
| <b>Length of Item</b><br>2 bytes Little Endian | <b>Item</b><br>UTF8 | ... | Configure the data item without requiring a Value field, unlike Mode = 1. |
|------------------------------------------------|---------------------|-----|---------------------------------------------------------------------------|

**Length of Item** Present the item name length using 2 bytes in Little Endian format. The value ranges from 0 to 65535 and indicates the length of the following item name.

**Item** The name of the item.

To configure Joint\_Angle as the data content that the server periodically sends to the client,

```

< 24 54 4D 52 54 53 2C // $TMRTS, // Header
 31 37 2C // 17, // Length
 31 2C 39 2C // 1,9, // transaction ID 1, mode 9
 0B 00 4A 6F 69 6E 74 5F 41 6E 67 6C 65 // Joint_Angle // The name occupied 11 bytes
 2C 2A 32 44 0D 0A // ,*2D\r\n // Checksum
> 24 54 4D 52 54 53 2C // $TMRTS, // Header
 39 2C // 9, // Length
 31 2C 30 2C // 1,0, // transaction ID 1, mode 0
 30 30 2C 4F 4B // 00,OK // Error Code 00 , Correct writing
 2C 2A 37 30 0D 0A // ,*70\r\n // Checksum

```

## 22.4 TMRTC

| Start Byte | Hdr   |   | Len    |   | Data |   |   | Checksum | End Byte1 | End Byte2 |
|------------|-------|---|--------|---|------|---|---|----------|-----------|-----------|
| \$         | TMRTC | , | Length | , | Data | , | * | Checksum | lr        | ln        |

| ID             |   | Mode        |   | Content        |
|----------------|---|-------------|---|----------------|
| Transaction ID | , | 0/1/<br>7/8 | , | Item and Value |

TMRTC is defined as the communication protocol used on port 5896 in the Real Time Remote Server (RTRS). The Data section of the packet is further divided into three segments, ID (Transaction ID), Mode (Content Mode), and Content (Item and Value), separated with commas and described below.

- ID** The transaction number expressed in any alphanumeric characters. (Reports the CPERR 04 error if a non-alphanumeric byte is encountered.) When used as a communication packet response, it is a transaction number that identifies which group of commands to respond.
- ,
- Mode** The data format and structure define the type of content transmitted.
  - 0 Indicates that the server responds to a command from the client as a string format
  - 1 Indicates that the client sends a motion control command to the server as a binary format
  - 7 Indicates to set the start or stop of the server's motion control mode
  - 8 Indicates to configure the type and parameters of motion control
- ,
- Content** The data content. Formatted by the mode definition.

### Note

TMRTC command is for the client and the server to communicate in both directions. When the client sends a motion control setting (Mode 8) to the server, the server responds with Mode 0. However, when the client sends a motion control command (Mode 1), the server does not respond to the client.

TMRTC only allows a single client connection at a time. During motion execution, the server continuously receives motion commands from the client. If an unexpected disconnection occurs during this process, the server may remain in a state of waiting for further commands. In such cases, rebooting the robot arm is required.

#### ● ID Transaction Number

TMRTC, unlike TMRTS, does not perform periodic data transmission. It only responds to the client when receiving a motion control setting command (Mode 8). Therefore, all transmission IDs are user-defined alphanumeric strings assigned by the client. After verifying that the packet format is correct, the server uses the transmission ID in the packet to reply with the command processing status.

#### ● Mode Format and Mode for the Data Content

When Mode = 8, the client configures the server's motion control settings, and the server must respond using Mode 0 to acknowledge the configuration. When Mode = 1, the client sends motion control values to the server, and the server does not respond using Mode 0.

When Mode = 7, the client sets the server to start or stop motion control mode. The typical usage flow is as follows:

1. Enable the RTRS service and check motion control instructions.
2. Connect to port 5895 (TMRTS) and retrieve the robot status.
3. Connect to port 5896 (TMRTC) and configure the motion control settings using Mode 8.
4. Enable motion control Mode 7.
5. Update the motion control target values using Mode 1.

- Writing or Reading Confirmation

When the client sends data writing to the server, the servers **checks whether all the criteria are correct before performing the request. If there is any error with the write command, no request will be performed.** The criteria to write for inspection are:

1. The validity of the mode as the format of the data content
2. The connected client's write permissions based on the IP Filter.
3. The data content matches to the mode.
4. The written data matches the data type of each item.
5. Motion control settings (Mode 8) successfully configured.
6. Whether the system is in a motion-controllable state.
7. Check motion control for activation.

### 22.4.1 Mode = 0 the status the server responds to the client command processing)

After the server receives and processes a write command from a client, it will respond with another command with Mode 0. The details for Mode 0 are as follows.

Data

| ID             |   | Mode |   | Error Code                     |   | Error Description |
|----------------|---|------|---|--------------------------------|---|-------------------|
| Transaction ID | , | 0    | , | 00, 01, 02, 03, 07, 08, 09, 10 | , |                   |

Transaction ID

Defined while the client sends the command for the server to reply

with

Mode 0 for the server to respond to the client

Error Code Error code definitions. Fixed as 2 bytes and in hexadecimal (but not 0x)

00 Correct writing. No error.

01 The communication format or mode is not supported. (Ex. Mode = 99)

02 The connected client is not permitted to write. (IP filer without write permission)

03 The communication format and the data content format are mismatched

07 Values to write mismatches with the configured type or the size.

08 Incomplete motion control configuration.

09 Motion command is not permitted.

10 Motion has already started

Error Description Error description, following the error code.

00 OK

01 NotSupport;XXX

02 WritePermission

03 InvalidData

07 ValueError;XXX

08 SettingIncomplete

09 MotionNotAllowed

10 MotionStarted

When the setting command is correct

```

< 24 54 4D 52 54 43 2C // $TMRTC, // Header
 35 2C // 5, // Length
 31 2C 37 2C // 1,7, // transaction ID 1, mode 7
 30 // 0 // Stop motion control
 2C 2A 37 33 0D 0A // ,*73\r\n // Checksum
> 24 54 4D 52 54 43 2C // $TMRTC, // Header
 39 2C // 9, // Length
 31 2C 30 2C // 1,0, // server responds to ID 1, mode 0
 30 30 2C // 00, // Error Code 00
 4F 4B // OK // Correct writing
 2C 2A 36 30 0D 0A // ,*60\r\n // Checksum

```

When using a non-existent Mode

```

< 24 54 4D 52 54 43 2C // $TMRTC, // Header
 35 2C // 5, // Length
 30 2C 35 2C // 1,7, // transaction ID 0, mode 5
 30 // 0 //
 2C 2A 37 30 0D 0A // ,*70\r\n // Checksum
> 24 54 4D 52 54 43 2C // $TMRTC, // Header
 32 30 2C // 20, // Length
 30 2C 30 2C // 0,0, // server responds to ID 0, mode 0
 30 31 2C // 01, // Error Code 01
 4E 6F 74 53 75 70 70 6F 72 74 3B 20 35 // NotSupport; 5 // mode 5 not supported
 2C 2A 36 42 0D 0A // ,*6B\r\n // Checksum

```

When Mode 7 is activated and a Mode 8 command is subsequently issued

```

< 24 54 4D 52 54 43 2C // $TMRTC, // Header
 31 39 2C // 19, // Length
 31 31 2C 38 2C // 11,8, // transaction ID 11, mode 8
 53 4A F4 01 00 00 00 00 80 3F F4 01 00 00 // S J 500 1 500
 // Position mode, Joint angle mode, Acceleration-to-max-speed time:
 // 500 ms, Gain: 1, Expected command interval: 500 ms
 2C 2A 45 36 0D 0A // ,*E6\r\n // Checksum
> 24 54 4D 52 54 43 2C // $TMRTC, // Header
 32 33 2C // 23, // Length
 31 31 2C 30 2C // 11,0, // server responds to ID 11, mode 0
 30 32 2C // 02, // Error Code 02
 57 72 69 74 65 50 65 72 6D 69 73 69 6F 6E // WritePermission
 // When Mode 7 is active, it is not possible to write Mode 8 settings.
 2C 2A 31 39 0D 0A // ,*19\r\n // Checksum

```

When the Mode is not specified

```

< 24 54 4D 52 54 43 2C // $TMRTC, // Header
 34 2C // 4, // Length

```

```

31 2C __ 2C // 1,, // transaction ID 1, no mode
31 // 1
2C 2A 34 34 0D 0A // ,*44\r\n // Checksum
> 24 54 4D 52 54 43 2C // $TMRTC, // Header
31 38 2C // 18, // Length
31 2C 30 2C // 1,0, // server responds to ID 1, mode 0
30 33 2C // 03, // Error Code 03
49 6E 76 61 6C 69 64 44 61 74 61 // InvalidData // Invalid content
2C 2A 33 36 0D 0A // ,*36\r\n // Checksum

```

If the value in the command is invalid (blank)

```

< 24 54 4D 52 54 43 2C // $TMRTC, // Header
34 2C // 4, // Length
30 2C 38 2C // 0,8, // transaction ID 0, mode 8
 //
2C 2A 34 43 0D 0A // ,*4C\r\n // Checksum
> 24 54 4D 52 54 43 2C // $TMRTC, // Header
34 32 2C // 42, // Length
30 2C 30 2C // 0,0, // server responds to ID 0, mode 0
30 37 2C // 07, // Error Code 07
56 61 6C 75 65 45 72 72 6F 72 3B 54 68 65 20 76 61 6C 75 65 20 69 73 20 6E 6F 74 20 73
75 70 70 6F 72 74 // ValueError;The value is not support // Invalid value
2C 2A 34 37 0D 0A // ,*47\r\n // Checksum

```

If the value in the command is invalid (incorrect value)

```

< 24 54 4D 52 54 43 2C // $TMRTC, // Header
31 38 2C // 18, // Length
30 2C 38 2C // 0,8, // transaction ID 0, mode 8
53 56 96 00 00 00 00 00 80 3F E8 03 00 00 // S V 150 1 1000
 // Position mode, V mode (not supported), Acceleration-to-max-speed
 // time: 150 ms, Gain: 1, Expected command interval: 1000 ms
2C 2A 42 36 0D 0A // ,*B6\r\n // Checksum
> 24 54 4D 52 54 43 2C // $TMRTC, // Header
34 32 2C // 42, // Length
30 2C 30 2C // 0,0, // transaction ID 0, mode 0
30 37 2C // 07, // Error Code 07
56 61 6C 75 65 45 72 72 6F 72 3B 54 68 65 20 76 61 6C 75 65 20 69 73 20 6E 6F 74 20 73
75 70 70 6F 72 74 // ValueError;The value is not support
 // Unsupported value error
2C 2A 34 37 0D 0A // ,*47\r\n // Checksum

```

If the motion setting is incomplete and the client issues a start motion command (Mode 7)

```

< 24 54 4D 52 54 43 2C // $TMRTC, // Header
35 2C // 5, // Length
33 2C 37 2C // 3,7, // transaction ID 3, mode 7
31 // 1 // Start motion control

```



```

2C 2A 37 30 0D 0A //,*70\r\n // Checksum
> 24 54 4D 52 54 43 2C // $TMRTC, // Header
32 34 2C // 24, // Length
33 2C 30 2C // 3,0, // server responds to ID 3, mode 0
30 38 2C // 08, // Error Code 08
53 65 74 74 69 6E 67 49 6E 63 6F 6D 70 6C 65 74 65 // SettingIncomplete
// Motion control configuration incomplete
2C 2A 32 39 0D 0A //,*29\r\n // Checksum

```

If a motion control command is issued before motion control is enabled

```

< 24 54 4D 52 54 43 2C // $TMRTC, // Header
32 38 2C // 28, // Length
30 2C 31 2C // 0,1, // transaction ID 0, mode 1
CD CC CC 3D CD CC CC 3D CD CC CC 3D CD CC CC 3D CD CC CC 3D CD CC CC 3D
// {0.1.0.1,0.1,0.1,0.1,0.1}
2C 2A 37 42 0D 0A //,*7B\r\n // Checksum
> 24 54 4D 52 54 43 2C // $TMRTC, // Header
32 33 2C // 23, // Length
30 2C 30 2C // 0,0, // server responds to ID 0, mode 0
30 39 2C // 09, // Error Code 09
4D 6F 74 69 6F 6E 4E 6F 74 41 6C 6C 6F 77 65 64 // MotionNotAllowed
// Motion command not permitted
2C 2A 36 37 0D 0A //,*67\r\n // Checksum

```

If motion control has already started and the client issues a start motion command (Mode 7)

```

< 24 54 4D 52 54 43 2C // $TMRTC, // Header
35 2C // 5, // Length
32 2C 37 2C // 2,7, // transaction ID 2, mode 7
31 // 1 // Start motion control
2C 2A 37 30 0D 0A //,*71\r\n // Checksum
> 24 54 4D 52 54 43 2C // $TMRTC, // Header
32 30 2C // 20, // Length
32 2C 30 2C // 2,0, // server responds to ID 2, mode 0
31 30 2C // 10, // Error Code 10
4D 6F 74 69 6F 6E 53 74 61 72 74 65 64 // MotionStarted // Motion control already started
2C 2A 32 32 0D 0A //,*22\r\n // Checksum

```

## 22.4.2 Mode = 1 BINARY

The data content is transmitted in binary mode by converting the data item name with the Little Endian value and the value with UTF8 to a byte array accordingly. The format is shown as below.

Data

| ID             |   | Mode |   | Content |
|----------------|---|------|---|---------|
| Transaction ID | , | 1    | , | Value   |

6 \* float Little Endian

Configure motion control command values

```
< 24 54 4D 52 54 43 2C // $TMRTC, // Header
 32 38 2C // 28, // Length
 30 2C 31 2C // 0,1, // transaction ID 0, mode 1
 3C BD 2D 41 3F F7 14 C3 77 ED 36 42 AB FE 0F 43 78 FA 95 41 4F 6F 07 42
 // {10.8587,-148.9658,45.7319,143.9948,18.7473,33.8587}
 2C 2A 34 44 0D 0A // ,*4D\r\n // Checksum
```

If the command value format is incorrect-

```
< 24 54 4D 52 54 43 2C // $TMRTC, // Header
 32 35 2C // 25, // Length
 31 32 2C 31 2C // 12,1, // transaction ID 12, mode 1
 00 00 C8 42 00 00 C8 42 00 00 C8 42 00 00 C8 42 00 00 C8 42
 // {100,100,100,100,100} five values only
 2C 2A 43 46 0D 0A // ,*CF\r\n // Checksum

> 24 54 4D 52 54 43 2C // $TMRTC, // Header
 34 30 2C // 40, // Length
 31 32 2C 30 2C // 12,0, // transaction ID 12, mode 0
 30 37 2C // 07, // Error Code 07
 56 61 6C 75 65 45 72 72 6F 72 3B 43 6F 6E 74 65 6E 74 4C 65 6E 67 74 68 20 69 6C 6C 65
 67 61 6C
 // ValueError;ContentLength illegal // Invalid content length
 2C 2A 37 37 0D 0A // ,*77\r\n // Checksum
```

### 22.4.3 Mode = 7 START/STOP Motion Control

The system transmits content in UTF-8 format to configure the start or stop of motion control.

The table below provides a detailed outline of the format.

Data

| ID             |   | Mode |   | Content |
|----------------|---|------|---|---------|
| Transaction ID | , | 7    | , | Value   |

|             |
|-------------|
| <b>UTF8</b> |
| '0': STOP   |
| '1': START  |

To activate motion control, use the example below.

```

< 24 54 4D 52 54 43 2C // $TMRTC, // Header
 35 2C // 5, // Length
 38 2C 37 2C // 8,7, // transaction ID 2, mode 7, binary
 31 // 1 // Start motion control
 2C 2A 37 42 0D 0A // ,*7B\r\n // Checksum
> 24 54 4D 52 54 43 2C // $TMRTC, // Header
 39 2C // 9, // Length
 38 2C 30 2C // 8,0, // transaction ID 8, mode 0
 30 30 2C // 00, // Error Code 0
 4F 4B // OK // Correct writing
 2C 2A 36 39 0D 0A // ,*69\r\n // Checksum

< 24 54 4D 52 54 43 2C // $TMRTC, // Header
 35 2C // 5, // Length
 32 2C 37 2C // 2,7, // transaction ID 2, mode 7, binary
 31 // 1 // Start motion control
 2C 2A 37 31 0D 0A // ,*71\r\n // Checksum
> 24 54 4D 52 54 43 2C // $TMRTC, // Header
 32 34 2C // 24, // Length
 32 2C 30 2C // 2,0, // transaction ID 2, mode 0
 30 38 2C // 08 // Error Code 08
 53 65 74 74 69 6E 67 49 6E 63 6F 6D 70 6C 65 74 65 // SettingIncomplete
 // Configuration incomplete.
 2C 2A 32 38 0D 0A // ,*28\r\n // Checksum

```

## 22.4.4 Mode = 8 SET Motion Control Settings

The system transmits data in binary format to define the motion control type and its parameters.

The table below provides a detailed outline of the format.

Data

| ID             |   | Mode |   | Content        |
|----------------|---|------|---|----------------|
| Transaction ID | , | 8    | , | Item and Value |

|                      |                                         |                                                           |                                                     |                                       |                                               |
|----------------------|-----------------------------------------|-----------------------------------------------------------|-----------------------------------------------------|---------------------------------------|-----------------------------------------------|
| <b>Position(S) →</b> | <b>Position</b><br>1 byte string<br>'S' | <b>Control Mode</b><br>1 byte string<br>'C' or 'T' or 'J' | <b>Acceleration Time</b><br>4 bytes int<br>150-2000 | <b>Gain</b><br>4 bytes float<br>1-100 | <b>Control Interval</b><br>4 bytes int<br>≥ 2 |
|----------------------|-----------------------------------------|-----------------------------------------------------------|-----------------------------------------------------|---------------------------------------|-----------------------------------------------|

|                      |                                         |                                                           |                                                     |                                               |
|----------------------|-----------------------------------------|-----------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------|
| <b>Velocity(V) →</b> | <b>Velocity</b><br>1 byte string<br>'V' | <b>Control Mode</b><br>1 byte string<br>'C' or 'T' or 'J' | <b>Acceleration Time</b><br>4 bytes int<br>150-2000 | <b>Control Interval</b><br>4 bytes int<br>≥ 2 |
|----------------------|-----------------------------------------|-----------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------|

To set the motion control to Position mode and Joint angle mode, acceleration time to max-speed time: 150 ms, gain value of 1, and the expected command interval of 1000 ms, use the following example:

```
< 24 54 4D 52 54 43 2C // $TMRTC, // Header
31 38 2C // 18, // Length
35 2C 38 2C // 5,8, // transaction ID 5, mode 8
53 4A 96 00 00 00 00 00 80 3F E8 03 00 00 // S J 150 1 1000
// Position mode, Joint angle mode, Acceleration-to-max-speed time:
// 150 ms, Gain: 1, Expected command interval: 1000 ms
```

```
2C 2A 41 46 0D 0A // ,*AF\r\n // Checksum
> 24 54 4D 52 54 43 2C // $TMRTC, // Header
39 2C // 9, // Length
35 2C 30 2C // 5,0, // transaction ID 5, mode 0
30 30 2C // 00, // Error Code 00
4F 4B // OK // Correct writing
2C 2A 36 34 0D 0A // ,*64\r\n // Checksum
```

If the command contains invalid values

```
< 24 54 4D 52 54 43 2C // $TMRTC, // Header
31 38 2C // 18, // Length
30 2C 38 2C // 0,8, // transaction ID 0, mode 8
53 56 96 00 00 00 00 00 80 3F E8 03 00 00 // S V 150 1 1000
// Position mode, V mode (not supported), Acceleration-to-max-speed time:
// 150 ms, Gain: 1, Expected command interval: 1000 ms
2C 2A 42 36 0D 0A // ,*B6\r\n // Checksum
> 24 54 4D 52 54 43 2C // $TMRTC, // Header
34 32 2C // 42, // Length
30 2C 30 2C // 0,0, // transaction ID 0, mode 0
30 37 2C // 07, // Error Code 07
56 61 6C 75 65 45 72 72 6F 72 3B 54 68 65 20 76 61 6C 75 65 20 69 73 20 6E 6F 74 20 73
75 70 70 6F 72 74 // ValueError;The value is not support
```

```

// Unsupported value error
2C 2A 34 37 0D 0A //,*47\r\n // Checksum

```

If the command contains invalid values (e.g., incorrect number of parameters)

```

< 24 54 4D 52 54 43 2C // $TMRTC, // Header
31 38 2C // 18, // Length
33 2C 38 2C // 3,8, // transaction ID 3, mode 8
56 4A 96 00 00 00 02 00 00 00 E8 03 00 00 // V J 150 2 1000
// Velocity mode, J mode, acceleration to maximum speed in 150 ms,
// expected control command interval: 2 ms, unknown parameter: 1000

2C 2A 31 31 0D 0A //,*11\r\n // Checksum
> 24 54 4D 52 54 43 2C // $TMRTC, // Header
34 32 2C // 42, // Length
33 2C 30 2C // 3,0, // transaction ID 3, mode 0
30 37 2C // 07, // Error Code 07
56 61 6C 75 65 45 72 72 6F 72 3B 54 68 65 20 76 61 6C 75 65 20 69 73 20 6E 6F 74 20 73
75 70 70 6F 72 74 // ValueError;The value is not support
// Unsupported value error
2C 2A 34 34 0D 0A //,*44\r\n // Checksum

```

If the command contains invalid values (e.g., value out of range)

```

< 24 54 4D 52 54 43 2C // $TMRTC, // Header
31 34 2C // 14, // Length
30 2C 38 2C // 0,8, // transaction ID 0, mode 8
56 4A 64 00 00 00 02 00 00 00 // V J 100 2
// Velocity mode, J mode, acceleration to maximum speed in 100 ms
// (exceeds the range), expected control command interval: 2 ms

2C 2A 30 37 0D 0A //,*07\r\n // Checksum
> 24 54 4D 52 54 43 2C // $TMRTC, // Header
34 32 2C // 42, // Length
30 2C 30 2C // 0,0, // transaction ID 0, mode 0
30 37 2C // 07, // Error Code 07
56 61 6C 75 65 45 72 72 6F 72 3B 54 68 65 20 76 61 6C 75 65 20 69 73 20 6E 6F 74 20 73
75 70 70 6F 72 74 // ValueError;The value is not support
// Unsupported value error
2C 2A 34 34 0D 0A //,*47\r\n // Checksum

```

## 23. Compliance Functions

### 23.1 Compliance Class

Use the Compliance class and declare variables to provide users with the robot compliance control setting and the stop conditions.

#### Construct 1

**Compliance** *VariableName*

**Parameters** No input parameter

**Note**

**Compliance** cp1      *// Declare compliance control variable.*

#### Member Methods

| Name                             | Default | Description                                                        |
|----------------------------------|---------|--------------------------------------------------------------------|
| <a href="#">Reset()</a>          |         | Reset all the compliance control motion parameters to the default. |
| <a href="#">Frame()</a>          | 1       | Set the reference coordinate of the compliance control motion.     |
| <a href="#">HighResistance()</a> | false   | Resistance on the non-motion direction (X,Y,Z).                    |
| <a href="#">Single()</a>         |         | The single axis parameter of the compliance control motion         |
| <a href="#">Teach()</a>          |         | The teach parameter of the compliance control motion               |
| <a href="#">Multiple()</a>       |         | The multiple axes parameter of the compliance control motion       |
| <a href="#">Impedance</a>        |         | The impedance parameter of the compliance control motion           |
| <a href="#">Timeout()</a>        | -       | The stop condition of timeout                                      |
| <a href="#">DInput()</a>         | -       | The stop condition of digital input                                |
| <a href="#">AInput()</a>         | -       | The stop condition of analog input                                 |
| <a href="#">Condition()</a>      | -       | The stop condition of the conditional expression                   |
|                                  |         |                                                                    |
| <a href="#">Start()</a>          | true    | Start the compliance control motion.                               |
| <a href="#">Stop()</a>           |         | Stop the compliance control motion.                                |

#### 23.1.1 Reset()

Reset all the compliance control motion parameters to the default.

#### Syntax 1

```
void Reset(
)
```

**Parameters**

**void**      No input value

**Return**

**void**      No return

### 23.1.2 Frame()

Set the reference base of the compliance control motion.

#### Syntax 1

```
void Frame(
 int
)
```

#### Parameters

|     |                                             |
|-----|---------------------------------------------|
| int | The base associated with the control motion |
| 1   | Tool                                        |
| 2   | The current base                            |

#### Return

void No return

### 23.1.3 HighResistance()

Set the resistance on the non-motion direction (X,Y,Z).

#### Syntax 1

```
void HighResistance(
 bool
)
```

#### Parameters

|       |                                                    |
|-------|----------------------------------------------------|
| bool  | The resistance on the non-motion direction (X,Y,Z) |
| false | Default                                            |
| true  | High resistance                                    |

#### Return

void No return

### 23.1.4 Single()

Set the single axis parameters of compliance control motion.

#### Syntax 1

```
void Single(
 int or string,
 int
)
```

#### Parameters

int or string

Direction

|   |    |     |   |   |    |      |    |
|---|----|-----|---|---|----|------|----|
| 0 | or | "X" | X | 3 | or | "RX" | RX |
| 1 | or | "Y" | Y | 4 | or | "RY" | RY |
| 2 | or | "Z" | Z | 5 | or | "RZ" | RZ |

int Distance X,Y,Z (mm) RX,RY,RZ (degree)

#### Return

void No return

### 23.1.5 Teach()

Set the teach parameters of compliance control motion.

#### Syntax 1

```
void Teach(
 int or string,
 float[] or string,
 float[] or string,
 int
)
```

#### Parameters

int or string

Linear direction or rotating direction

0 or "Linear" Linear direction

1 or "Rotation" Rotating direction

float[] or string

The 1<sup>st</sup> point  $\{X_1(mm) \ Y_1(mm) \ Z_1(mm) \ RX_1(^{\circ}) \ RY_1(^{\circ}) \ RZ_1(^{\circ})\}$  or

The name of the 1<sup>st</sup> point (use the robot TCP coordinate)

float[] or string

The 2<sup>nd</sup> point  $\{X_2(mm) \ Y_2(mm) \ Z_2(mm) \ RX_2(^{\circ}) \ RY_2(^{\circ}) \ RZ_2(^{\circ})\}$  或

The name of the 2<sup>nd</sup> point (use the robot TCP coordinate)

int

Adjustment range Linear (mm) Rotation (degree)

#### Return

void No return

### 23.1.6 Multiple()

Set the Multiple Axes parameters of compliance control motion.

#### Syntax 1

```
void Multiple(
 int or string,
 bool,
 int,
 int,
)
```

#### Parameters

int or string

Direction

0 or "X" X 3 or "RX" RX

1 or "Y" Y 4 or "RY" RY

2 or "Z" Z 5 or "RZ" RZ

bool

Enable the control to the assigned direction or not.

false Disable

true Enable

int

Distance restriction upper bound X,Y,Z (mm) RX,RY,RZ (degree)



**int** Distance restriction lower bound X,Y,Z (mm) RX,RY,RZ (degree)  
**Return**  
**void** No return

### Syntax 2

**void Multiple**(  
**int or string**,  
**bool**  
 )

### Note

Same as Syntax 1 parameter definitions for setting whether to enable the control to the assigned direction.

## 23.1.7 Impedance()

Set the impedance parameters of compliance control motion.

### Syntax 1

**void Impedance**(  
**int or string**,  
**int**  
 )

### Parameters

**int or string**

Axes to release

0 or "All" Release all axes

1 or "XYZ" Release X, Y, Z

2 or "RXYZ" Release RX, RY, RZ

**int**

Rigidity level

0 Soft

1 Intermediate

2 Stiff

### Return

**void** No return

## 23.1.8 Timeout()

Set the stop condition of timeout.

### Syntax 1

**void Timeout**(  
**int**  
 )

### Parameters

**int**

Timeout in milliseconds

< 0 Disable

>= 0 Timeout duration

### Return

**void**      No return

### Syntax 2

```
void Timeout(
)
```

#### Parameters

**void**      No parameter for cancelling the stop condition.

#### Return

**void**      No return

## 23.1.9 DInput()

Set the stop condition of the digital input.

### Syntax 1

```
void DInput(
 string,
 int,
 int or string
)
```

#### Parameters

|                      |                                     |
|----------------------|-------------------------------------|
| <b>string</b>        | Control module name                 |
| <b>ControlBox</b>    | The control box                     |
| <b>EndModule</b>     | The end module                      |
| <b>ExtModuleN</b>    | The external module (N = 0 .. n)    |
| <b>int</b>           | Input channel 0 .. n                |
| <b>int or string</b> | Set the stop condition to Low/High. |
| <b>0    or   "L"</b> | Low                                 |
| <b>1    or   "H"</b> | High                                |

#### Return

**void**      No return

### Syntax 2

```
void DInput(
 int
)
```

#### Parameters

**int**      The stroke ratio to start receiving digital input (valid when the operation mode is " Compliance ")

#### Return

**void**      No return

### Syntax 3

```
void DInput(
)
```

#### Parameters

**void** No parameter for cancelling the stop condition.  
**Return**  
**void** No return

### 23.1.10 AInput()

Set the stop condition of the analog input.

#### Syntax 1

```
void AInput(
 string,
 int,
 int or string,
 float
)
```

#### Parameters

|                      |                                                                                                     |
|----------------------|-----------------------------------------------------------------------------------------------------|
| <b>string</b>        | Control module name                                                                                 |
| <b>ControlBox</b>    | The control box                                                                                     |
| <b>EndModule</b>     | The end module                                                                                      |
| <b>ExtModuleN</b>    | The external module (N = 0 .. n)                                                                    |
| <b>int</b>           | Input channel 0 .. n                                                                                |
| <b>int or string</b> | Set the condition to judge                                                                          |
| 0 or ">"             | Greater than                                                                                        |
| 1 or ">="            | Greater than or equal to                                                                            |
| 2 or "=="            | Equal to (Recommend not to use since it is not easy to hold the equal condition with analog input.) |
| 3 or "<="            | Less than or equal to                                                                               |
| 4 or "<"             | Less than                                                                                           |
| <b>float</b>         | Condition value                                                                                     |
| <b>Return</b>        |                                                                                                     |
| <b>void</b>          | No return                                                                                           |

#### Syntax 2

```
void AInput(
)
```

#### Parameters

**void** No parameter for cancelling the stop condition.

#### Return

**void** No return

### 23.1.11 Condition()

Set the stop condition of the conditional expression.

#### Syntax 1

```
void Condition(

```

bool or ?

)

#### Parameters

bool or ? The conditional that can be true/false or a bool return of the statement.

#### Return

void No return

### Syntax 2

void Condition(

)

#### Parameters

void No parameter for cancelling the stop condition.

#### Return

void No return

## 23.1.12 Start()

Start the compliance control motion.

### Syntax 1

int Start(

)

#### Parameters

void No input value

#### Return

int Return the result value after the force control motion stops.

|     |                               |
|-----|-------------------------------|
| 0   | Not Working                   |
| 1   | Working                       |
| 2   | Timeout                       |
| 3   | Distance reached              |
| 4   | IO triggered                  |
| 5   | Resisted                      |
| 6   | Error                         |
| 14  | Over Speed                    |
| 201 | Digital IO triggered          |
| 202 | Analog IO triggered           |
| 203 | Variable                      |
| 204 | Force is comprehended         |
| 205 | Allowable Position Tolerances |
| 206 | Motion Finish                 |

## 23.1.13 Stop()

Stop the compliance control motion.

### Syntax 1

int Stop(

)

**Parameters**

**void** No input value

**Return**

**int** Return the result value after the compliance control motion stops.

- **Parameter Settings**

```

Compliance cp1 // Declare the compliance control variable.
(1)
 cp1.Frame(1) // Set the compliance control base as tool.
 cp1.Frame(2) // Set the compliance control base as the current base (will overwrite the
 previous setting).
 cp1.Reset() // Reset all parameters
(2)
 cp1.Frame(1)
 cp1.Single("X", 30) // Set direction X with distance of 30mm
 cp1.Single("Z", 40) // Set direction Z with distance of 40mm (will overwrite the previous Single
 setting)
 cp1.Reset() // Reset all parameters
(3)
 cp1.Frame(1)
 cp1.HighResistance(true) // Set the resistance on the non-motion direction to High Resistance
 // Control
 cp1.Single("Z", 40) // Set direction Z with distance of 40mm (use Single mode)
 cp1.Teach("Linear", "P1", "P2", 0) // Set the teach point P1 and P2 (will use Teach
 mode instead)
 cp1.AdvSet ("X", true, 100, -100) // Set direction X with advanced parameters (will use
 Multiple mode instead)
 cp1.AdvSet ("Z", true, 100, -10030) // Set direction Z
 cp1.AdvSet ("X", true, 10, -10) // Set direction X with advanced parameters (will overwrite
 the previous Multiple X parameter)
 cp1.Impedance("All", 1) // Set the impedance parameter (will use Impedance mode instead)
 // Stop Criteria
 cp1.Timeout(10000) // Set timeout to 10000ms
 cp1.DInput("ControlBox", 0, "H") // When ControlBox DI0 is High
 cp1.AInput("ControlBox", 0, ">=", 3.3) // When ControlBox AI0 is greater than or equal to 3.3V
 int count = 0
 cp1.Condition(count > 1000) // conditional expression
 cp1.Reset() // Reset all parameters

```

- **Compliance Control**

**Compliance** cp1

**(1) Single Axis**

```

PTP("JPP",{0,0,90,0,90,0},50,200,0,false)
cp1.Single("Z", 100) // Set direction Z with distance of 100mm
int re = cp1.Start() // Start the compliance control motion
// Set direction Z with a moving distance of 100mm, so it will move toward direction Z. Once the moving
distance is comprehended, it stops the control motion, exits the function, and returns the result
values.

```

## (2) Teach

```

TPoint P1 = {516.655,-147.754,445.381,179.444,-0.324,89.912},{0,0,90,0,90,0}
TPoint P2 = {517.419,-147.175,345.184,179.533,-0.319,89.918},{0.057,1.805,103.178,-
14.896,89.998,0.085}
TPoint P3 = {516.655,-147.740,445.388,-162.754,-0.281,89.819},{0.017,6.570,77.958,23.270,89.971,-
0.043}
PTP("JPP",{0,0,90,0,90,0},50,200,0,false)
cp1.Teach("Linear", "P1", "P2", 0) // Set the teach point P1 and P2 in the linear direction
cp1.Timeout(5000) // Set timeout to 5000ms
int re = cp1.Start() // Start the compliance control motion
// Teach the two points in the linear direction. Once the moving distance and the teach linear direction
// are comprehended, it stops the control motion, exits the function, and return the result values. Since
// the timeout is set, if it does not comprehend the moving distance in time, it stops the control motion,
// exits the function, and returns the result values.
PTP("JPP",{0,0,90,0,90,0},50,200,0,false)
cp1.Teach("Rotation", "P1", "P3", 0)
// Set the teach point P1 and P3 in the rotating direction
re = cp1.Start()
// Teach the two points in the rotating direction. Once the moving distance and the teach rotating
// direction are comprehended, it stops the control motion, exits the function, and return the result
// values. Since the timeout is set (without resetting the timeout of CP1), if it does not comprehend the
// moving distance in time, it stops the control motion, exits the function, and returns the result values.

```

## (3) Multiple Axis

```

PTP("JPP",{0,0,90,0,90,0},50,200,0,false)
cp1.Multiple("X", true, 100, -100) // Set direction X with multiple axis parameters
cp1.Multiple("Z", true, 100, -100) // Set direction Z with multiple axis parameters
cp1.Multiple("X", true, 100, -100) // Set direction X with multiple axis parameters (will overwrite the
// previous multiple axis parameters).
int re = cp1.Start() // Start the compliance control motion
// Set direction X and Z for compliance control. Once the moving distance is comprehended, it stop the
// control motion, exits the function, and returns the result values.

```

## (4) Impedance

```

PTP("JPP",{0,0,90,0,90,0},50,200,0,false)
cp1.Impedance("all", 1) // Release all axes
cp1.DInput("ControlBox", 0, "H") // When ControlBox DI0 is High
int re = cp1.Start() // Start the compliance control motion
// Release all axes. The robot stays put in the current position and complies with external forces to
// lessen the torque on the joints. When the stop criterion, ControlBox DI0 is High, is comprehended, it
// stop the control motion, exits the function, and returns the result values.

```

## 24. TouchStop Functions

### 24.1 TouchStop class

Use the TouchStop class and declare variables to provide users with the robot touch stop setting and the stop conditions to fulfill the touch stop function.

#### Construct1

**TouchStop** *VariableName* = string

**TouchStop** *VariableName* = string, string

#### Parameters

string      Operation mode  
                  "Compliance"  
                  "Line"  
                  "Force"

string      Device name of the sensor (valid when the operation mode is "Force")

#### note

**TouchStop** ts1 = "Compliance"      // Declare a touchstop control variable with the function of compliance

**TouchStop** ts2 = "Line"      // Declare a touchstop control variable with the function of line

**FTSensor** fts3 = "TMFT300"      // Build a device named TMFT300.

**TouchStop** ts3 = "Force", "fts3"      // Declare a touchstop control variable with the function of force tethered with the sensor device fts3.

#### Member Methods

| Name                             | Default | Description                                                                                                        |
|----------------------------------|---------|--------------------------------------------------------------------------------------------------------------------|
| <a href="#">Reset()</a>          |         | Reset all the compliance control motion parameters to the default except the operation mode and the sensor device. |
| <a href="#">Frame()</a>          | 1       | Set the reference coordinate of the touchstop control motion.                                                      |
| <a href="#">HighResistance()</a> | false   | Resistance on the non-motion direction (X,Y,Z). (Valid when the operation mode is "Compliance").                   |
| <a href="#">BrakeDistance()</a>  | 0       | (Valid when the operation mode is "Line").                                                                         |
| <a href="#">RecordPosPoint()</a> | -       | Set the record touchstop coordinate value to the point name.                                                       |
| <a href="#">Single()</a>         |         | The single axis parameter of the compliance control motion                                                         |
| <a href="#">Teach()</a>          |         | The teach parameter of the compliance control motion                                                               |
| <a href="#">AdvSet()</a>         |         | The advance setting parameter of the compliance control motion. (Valid when the operation mode is "Compliance").   |
| <a href="#">Timeout()</a>        | -       | The stop condition of timeout                                                                                      |
| <a href="#">DInput()</a>         | -       | The stop condition of digital input                                                                                |
| <a href="#">AInput()</a>         | -       | The stop condition of analog input                                                                                 |
| <a href="#">Condition()</a>      | -       | The stop condition of the conditional expression                                                                   |
| <a href="#">Resisted()</a>       | -       | Stop condition of the external resistance detection. (Valid when the operation mode is "Compliance").              |

| Name                             | Default       | Description                                                                                                                            |
|----------------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <code>FTReached()</code>         | -             | The stop condition of monitoring the values of force, torque, or resultant force acquired. (Valid when the operation mode is "Force"). |
| <code>Start()</code>             | true          | Start the compliance control motion.                                                                                                   |
| <code>Stop()</code>              |               | Stop the compliance control motion.                                                                                                    |
| <code>GetStoppedPos()</code>     | {0,0,0,0,0,0} | Get the stopped position values after the robot touch-stopped.                                                                         |
| <code>GetTriggeredPos()</code>   | {0,0,0,0,0,0} | Get the triggered position values after the robot touch-stopped.                                                                       |
| <code>GetMovingDistance()</code> | 0             | Get the moving distance of the touchstop motion.                                                                                       |

### 24.1.1 Reset()

Reset all the touchstop control motion parameters to the default except the operation mode and the sensor device

#### Syntax 1

```
void Reset(
)
```

#### Parameters

`void` No input value

#### Return

`void` No return

### 24.1.2 Frame()

Set the reference base of the touchtop control motion.

#### Syntax 1

```
void Frame(
 int
)
```

#### Parameters

`int` The base associated with the control motion

- 1 Tool
- 2 The current base

#### Return

`void` No return

### 24.1.3 HighResistance()

Set the resistance on the non-motion direction (X,Y,Z). (Valid Valid when the operation mode is "Compliance".)

#### Syntax 1

```
void HighResistance(
 bool
)
```



)

**Parameters**

**bool**      The resistance on the non-motion direction (X,Y,Z)  
                  **false**      Default  
                  **true**      High resistance

**Return**

**void**      No return

**24.1.4 BrakeDistance()**

Set the braking distance. (Valid Valid when the operation mode is "Line".)

**Syntax 1**

```
void BrakeDistance(
 int
```

)

**Parameters**

**int**      Braking distance in mm

**Return**

**void**      No return

**24.1.5 RecordPosPoint()**

Set the record touchstop coordinate value to the point name.

\* If in Flow projects, it requires to create the associated point name in the point manager. (Use the TouchStop node to create.)

\* If in Script projects, it requires to specify the associated point name in the define section. (Define the point as "D" or "JD".)

**Syntax 1**

```
void RecordPosPoint(
 string,
 bool
```

)

**Parameters**

**string**      The point name. Denote record canceled if empty string.  
**bool** Record the touchstop stop point or the trigger point  
                  **false**      The stop point  
                  **true**      The trigger point

**Return**

**void**      No return

**Syntax 2**

```
void RecordPosPoint(
 string
```

)

**Note**

Same as Syntax 1. Fill false to the parameter of the recording the touchstop stop point or

the trigger point

### Syntax 3

```
void RecordPosPoint(
)
```

#### Parameters

**void** No input value. Use to cancel recording the touchstop stop point or the trigger point

#### Return

**void** No return

## 24.1.6 Single()

Set the single axis parameters of touchstop control motion.

### Syntax 1

```
void Single(
 int or string,
 int,
 int,
 int
)
```

#### Parameters

**int or string**

Direction

|   |    |     |   |   |    |      |    |
|---|----|-----|---|---|----|------|----|
| 0 | or | "X" | X | 3 | or | "RX" | RX |
| 1 | or | "Y" | Y | 4 | or | "RY" | RY |
| 2 | or | "Z" | Z | 5 | or | "RZ" | RZ |

**int** Distance X,Y,Z (mm) RX,RY,RZ (degree)

**If operation mode is Compliance**

**If operation mode is Line or Force**

**int** Force X,Y,Z (N) RX,RY,RZ (mNm) (degree/s)

Speed X,Y,Z (mm/s) RX,RY,RZ

**int** Target speed X,Y,Z (mm/s) RX,RY,RZ (degree/s) Accelerate time in

millisecond

#### Return

**void** No return

### Syntax 2

```
void Single(
 int or string,
 int,
 int,
 int,
 bool,
 bool
)
```

#### Parameters

**int or string**

|      |                                         |                                            |     |   |   |    |      |    |
|------|-----------------------------------------|--------------------------------------------|-----|---|---|----|------|----|
|      | Direction                               |                                            |     |   |   |    |      |    |
|      | 0                                       | or                                         | "X" | X | 3 | or | "RX" | RX |
|      | 1                                       | or                                         | "Y" | Y | 4 | or | "RY" | RY |
|      | 2                                       | or                                         | "Z" | Z | 5 | or | "RZ" | RZ |
| int  | Distance X,Y,Z (mm) RX,RY,RZ (degree)   |                                            |     |   |   |    |      |    |
| int  | Speed X,Y,Z (mm/s) RX,RY,RZ (degree/s)  |                                            |     |   |   |    |      |    |
| int  | Accelerate time in millisecond          |                                            |     |   |   |    |      |    |
| bool | Cancel precision positioning            |                                            |     |   |   |    |      |    |
|      | true                                    | Cancel precision positioning               |     |   |   |    |      |    |
|      | false                                   | Precision positioning (default)            |     |   |   |    |      |    |
| bool | Synchronize to the project speed or not |                                            |     |   |   |    |      |    |
|      | true                                    | Synchronize to the project speed (default) |     |   |   |    |      |    |
|      | false                                   | Synchronize to the project speed not       |     |   |   |    |      |    |
| urn  |                                         |                                            |     |   |   |    |      |    |
| void | No return                               |                                            |     |   |   |    |      |    |

### 24.1.7 Teach()

Set the teach parameters of compliance control motion.

#### Syntax 1

```
void Teach(
 int or string,
 float[] or string,
 float[] or string,
 int,
 int,
 int
)
```

#### Parameters

|                   |                                                                                                                                                                                           |                                        |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| int or string     | Linear direction or rotating direction                                                                                                                                                    |                                        |
| 0                 | or "Linear"                                                                                                                                                                               | Linear direction                       |
| 1                 | or "Rotation"                                                                                                                                                                             | Rotating direction                     |
| float[] or string | The 1 <sup>st</sup> point { $X_1(mm)$ $Y_1(mm)$ $Z_1(mm)$ $RX_1(^{\circ})$ $RY_1(^{\circ})$ $RZ_1(^{\circ})$ } or<br>The name of the 1 <sup>st</sup> point (use the robot TCP coordinate) |                                        |
| float[] or string | The 2 <sup>nd</sup> point { $X_2(mm)$ $Y_2(mm)$ $Z_2(mm)$ $RX_2(^{\circ})$ $RY_2(^{\circ})$ $RZ_2(^{\circ})$ } 或<br>The name of the 2 <sup>nd</sup> point (use the robot TCP coordinate)  |                                        |
| int               | Adjustment range Linear (mm) Rotation (degree)                                                                                                                                            |                                        |
|                   | If operation mode is Compliance                                                                                                                                                           | If operation mode is Line or Force     |
| int               | Force X,Y,Z (N) RX,RY,RZ (mNm)                                                                                                                                                            | Speed X,Y,Z (mm/s) RX,RY,RZ (degree/s) |
| int               | Target speed X,Y,Z (mm/s) RX,RY,RZ (degree/s) Accelerate time in millisecond                                                                                                              |                                        |

#### Return

void No return

**Syntax 2**

```
void Teach(
 int or string,
 float[] or string,
 float[] or string,
 int,
 int,
 int,
 bool,
 bool
)
```

**Parameters**

int or string

Linear direction or rotating direction

0 or "Linear" Linear direction

1 or "Rotation" Rotating direction

float[] or string

The 1<sup>st</sup> point { $X_{1(mm)}$   $Y_{1(mm)}$   $Z_{1(mm)}$   $RX_{1(^\circ)}$   $RY_{1(^\circ)}$   $RZ_{1(^\circ)}$ } or

The name of the 1<sup>st</sup> point (use the robot TCP coordinate)

float[] or string

The 2<sup>nd</sup> point { $X_{2(mm)}$   $Y_{2(mm)}$   $Z_{2(mm)}$   $RX_{2(^\circ)}$   $RY_{2(^\circ)}$   $RZ_{2(^\circ)}$ } 或

The name of the 2<sup>nd</sup> point (use the robot TCP coordinate)

int

Adjustment range Linear (mm) Rotation (degree)

int

Speed X,Y,Z (mm/s) RX,RY,RZ (degree/s)

int

Accelerate time in millisecond

bool

Cancel precision positioning

true Cancel precision positioning

false Precision positioning (default)

bool

Synchronize to the project speed or not

true Synchronize to the project speed (default)

false Synchronize to the project speed not

**Return**

void No return

**24.1.8 AdvSet()**

Set the advanced parameters of touchstop control motion. (Valid when the operation mode is "Compliance".)

**Syntax 1**

```
void AdvSet(
 int or string,
 bool,
 int,
 int,
 int,
 int
)
```

)

### Parameters

int or string

Direction

|   |    |     |   |   |    |      |    |
|---|----|-----|---|---|----|------|----|
| 0 | or | "X" | X | 3 | or | "RX" | RX |
| 1 | or | "Y" | Y | 4 | or | "RY" | RY |
| 2 | or | "Z" | Z | 5 | or | "RZ" | RZ |

bool Enable the control to the assigned direction or not.

false Disable

true Enable

int Distance restriction upper bound X,Y,Z (mm) RX,RY,RZ (degree)

int Distance restriction lower bound X,Y,Z (mm) RX,RY,RZ (degree)

int Force X,Y,Z (N) RX,RY,RZ (mNm)

int Target speed X,Y,Z (mm/s) RX,RY,RZ (degree/s)

### Return

void No return

### Syntax 2

```
void AdvSet(
 int or string,
 bool
```

)

### note

Same as Syntax 1 parameter definitions for setting whether to enable the control to the assigned direction.

## 24.1.9 Timeout()

Set the stop condition of timeout.

### Syntax 1

```
void Timeout(
 int
```

)

### Parameters

int Timeout in milliseconds

< 0 Disable

>= 0 Timeout duration

### Return

void No return

### Syntax 2

```
void Timeout(
)
```

### Parameters

void No parameter for cancelling the stop condition.

### Return

void No return

### 24.1.10 DInput()

Set the stop condition of the digital input.

#### Syntax 1

```
void DInput(
 string,
 int,
 int or string
)
```

#### Parameters

|               |                                     |
|---------------|-------------------------------------|
| string        | Control module name                 |
| ControlBox    | The control box                     |
| EndModule     | The end module                      |
| ExtModuleN    | The external module (N = 0 .. n)    |
| int           | Input channel 0 .. n                |
| int or string | Set the stop condition to Low/High. |
| 0 or "L"      | Low                                 |
| 1 or "H"      | High                                |

#### Return

void No return

#### Syntax 2

```
void DInput(
 int
)
```

#### Parameters

|     |                                                                                                     |
|-----|-----------------------------------------------------------------------------------------------------|
| int | The stroke ratio to start receiving digital input (valid when the operation mode is " Compliance ") |
|-----|-----------------------------------------------------------------------------------------------------|

#### Return

void No return

#### Syntax 3

```
void DInput(
)
```

#### Parameters

|      |                                                 |
|------|-------------------------------------------------|
| void | No parameter for cancelling the stop condition. |
|------|-------------------------------------------------|

#### Return

void No return

### 24.1.11 AInput()

Set the stop condition of the analog input.

#### Syntax 1

```
void AInput(
```

```

 string,
 int,
 int or string,
 float
)
Parameters
 string Control module name
 ControlBox The control box
 EndModule The end module
 ExtModuleN The external module (N = 0 .. n)
 int Input channel 0 .. n
 int or string
 Set the condition to judge
 0 or ">" Greater than
 1 or ">=" Greater than or equal to
 2 or "==" Equal to (Recommend not to use since it is not easy to
 hold the equal condition with analog input.)
 3 or "<=" Less than or equal to
 4 or "<" Less than
 float Condition value
Return
 void No return

```

## Syntax 2

```

void AInput(
)
Parameters
 void No parameter for cancelling the stop condition.
Return
 void No return

```

## 24.1.12 Condition()

Set the stop condition of the conditional expression.

## Syntax 1

```

void Condition(
 bool or ?
)
Parameters
 bool or ? The conditional that can be true/false or a bool return of the statement.
Return
 void No return

```

## Syntax 2

```

void Condition(

```

)

**Parameters****void** No parameter for cancelling the stop condition.**Return****void** No return**24.1.13 Resisted()**

Set the stop condition of the external resistance detection. (valid when the operation mode is " Compliance ")

**Syntax 1****void Resisted(****bool**

)

**Parameters****bool** External resistance detection**false** Disable**true** Enable**Return****void** No return**Syntax 2****void Resisted(**

)

**Parameters****void** No parameter for cancelling the stop condition.**Return****void** No return**24.1.14 FTReached()**

Set the stop condition of monitoring the values of force, torque, or resultant force acquired. (valid when the operation mode is " Force")

**Syntax 1****void FTReached(****int or string,****bool,****float**

)

**Parameters****int** The values of force, torque, or resultant force.**6 or "F3D"** F3D (N)**7 or "T3D"** T3D (Nm)**8 or "Force"** Force (N) (default)

Whether to enable monitoring the assigned the values of force, torque, or resultant force.

**false** Disable



**Return**

|                    |                      |
|--------------------|----------------------|
| <code>true</code>  | Enable               |
| <code>float</code> | The monitoring value |
| <code>void</code>  | No return            |

**Syntax 2**

```
void FTReached(
 int or string,
 bool
)
```

**Note**

Same as syntax 1. Use to whether to enable monitoring the assigned the values of force, torque, or resultant force.

**Syntax 3**

```
void FTReached(
 bool
)
```

**Parameters**

|                    |                                                   |
|--------------------|---------------------------------------------------|
| <code>bool</code>  | Whether to enable the absolute values monitoring. |
| <code>false</code> | Disable                                           |
| <code>true</code>  | Enable                                            |

**Syntax 4**

```
void FTReached(
)
```

**Parameters**

|                   |                                                 |
|-------------------|-------------------------------------------------|
| <code>void</code> | No parameter for cancelling the stop condition. |
|-------------------|-------------------------------------------------|

**Return**

|                   |           |
|-------------------|-----------|
| <code>void</code> | No return |
|-------------------|-----------|

**24.1.15 Start()**

Start the touchstop control motion.

**Syntax 1**

```
int Start(
 bool
)
```

**Parameters**

|                    |                                                                                    |
|--------------------|------------------------------------------------------------------------------------|
| <code>bool</code>  | Zero out force sensor before execution. (valid when the operation mode is "Force") |
| <code>true</code>  | Enable (default)                                                                   |
| <code>false</code> | Disable                                                                            |

**Return**

|                  |                                                                   |
|------------------|-------------------------------------------------------------------|
| <code>int</code> | Return the result value after the touchstop control motion stops. |
| <code>0</code>   | Not Working                                                       |

|     |                               |
|-----|-------------------------------|
| 1   | Working                       |
| 2   | Timeout                       |
| 3   | Distance reached              |
| 4   | IO triggered                  |
| 5   | Resisted                      |
| 6   | Error                         |
| 14  | Over Speed                    |
| 201 | Digital IO triggered          |
| 202 | Analog IO triggered           |
| 203 | Variable                      |
| 204 | Force is Comprehended         |
| 205 | Allowable Position Tolerances |
| 206 | Motion Finish                 |

**Syntax 2**

```
int Start(
)
```

**note**

Same as syntax 1. Fill true to zero out force sensor before execution.

**24.1.16 Stop()**

Stop the touchstop control motion.

**Syntax 1**

```
int Stop(
)
```

**Parameters**

void No input value

**Return**

int Return the result value after the touchstop control motion stops.

**24.1.17 GetStoppedPos()**

Get the stopped position values after the robot touch-stopped.

**Syntax 1**

```
void GetStoppedPos(
 int
)
```

**Parameters**

int Get the coordinate value or the angle value of the stop point of the touch-stop control motion.

|   |                                                          |
|---|----------------------------------------------------------|
| 0 | the robot flange center coordinate (by the current base) |
| 1 | the robot joint angle                                    |
| 2 | the robot end TCP coordinate (by the current base)       |
| 3 | the robot TCP coordinate (by the RobotBase base)         |

4 the robot end TCP coordinate (by the RobotBase base)

### Return

**float[]** If in coordinates, it is the six elements of the robot coordinates: X(mm), Y(mm), Z(mm), RX(°), RY(°), and RZ(°); if in angle, the six elements of the robot joint: Joint 1(°), Joint 2(°), Joint 3(°), Joint 4(°), Joint 5(°), and Joint 6(°)

## 24.1.18 GetTriggeredPos()

Get the triggered position values after the robot touch-stopped.

### Syntax 1

```
void GetTriggeredPos(
 int
)
```

### Parameters

**int** Get the coordinate value or the angle value of the stop point of the touch-stop control motion.

- 0 the robot flange center coordinate (by the current base)
- 1 the robot joint angle
- 2 the robot end TCP coordinate (by the current base)
- 3 the robot TCP coordinate (by the RobotBase base)
- 4 the robot end TCP coordinate (by the RobotBase base)

### Return

**float[]** If in coordinates, it is the six elements of the robot coordinates: X(mm), Y(mm), Z(mm), RX(°), RY(°), and RZ(°); if in angle, the six elements of the robot joint: Joint 1(°), Joint 2(°), Joint 3(°), Joint 4(°), Joint 5(°), and Joint 6(°)

## 24.1.19 GetMovingDistance()

Get the moving distance of the touchstop motion.

### Syntax 1

```
float GetMovingDistance(
)
```

### Parameters

**void** No input value

### Return

**float** Return the moving distance of the touchstop motion in mm.

## ● Parameter Settings

```
TouchStop ts1 = "Compliance" // Declare the touchstop control variable functioning as Compliance.
TouchStop ts2 = "Line" // Declare the touchstop control variable functioning as Line.
FTSensor fts3 = "TMFT300" // Construct the device TMFT300.
```

```

TouchStop ts3 = "Force", "fts3" // Declare the touchstop control variable and tether fts3 as the
 sensor name.

TPoint P0 = "JD",{0,0,0,0,0,0}

(1)
 ts1.Frame(1) // Set the touchstop control base as tool.
 ts1.Frame(2) // Set the touchstop control base as the current base (will overwrite the
 previous setting).
 ts1.Reset() // Reset all parameters

(2)
 ts1.Frame(1)
 ts1.Single("X", 30, 30, 30) // Set direction X with distance of 30mm, force of 30N, and target speed of
 30 mm/s
 ts1.Single("Z", 40, 30, 30) // Set direction Z with distance of 40mm (will overwrite the previous Single
 setting)
 ts1.Reset() // Reset all parameters
 ts2.Single("Z", 30, 40, 500, false, true) // Set direction Z with distance of 30mm, target of
 40mm/s, and accelerate time of 500ms.

(3)
 ts1.Frame(1)
 ts1.HighResistance(true) // Set the resistance on the non-motion direction to High Resistance
 ts2.BrakeDistance(20) // Set the braking distance to 20mm
 ts2.RecordPosPoint("P0", true) // Set the record touchstop coordinate value to P0.

// Control
 ts1.Single("Z", 40, 30, 30) // Set direction Z with distance of 40mm (use Single
 mode)
 ts1.Teach("Linear", "P1", "P2", 0, 30, 30) // Set the teach point P1 and P2 (will use Teach
 mode instead)
 ts1.AdvSet ("X", true, 100, -100, 30, 30) // Set direction X with advanced parameters (will use
 Advanced mode instead)
 ts1.AdvSet ("Z", true, 100, -100, 30, 30) // Set direction Z with advanced parameters
 ts1.AdvSet ("X", true, 10, -10, 30, 30) // Set direction X with advanced parameters (will overwrite
 the previous AdvSet X parameter)

// Stop Criteria
 ts1.Timeout(10000) // Set timeout to 10000ms
 ts1.DInput("ControlBox", 0, "H") // When ControlBox DI0 is High
 ts1.DInput(100) // Start receiving digital input with stroke ratio to 100%
 ts1.AInput("ControlBox", 0, ">=", 3.3) // When ControlBox AI0 is greater than or equal to 3.3V
 int count = 0
 ts1.Condition(count > 1000) // conditional expression
 ts1.Resisted(true) // external resistance detection
 ts1.Reset() // Reset all parameters (reserve tethering "fts1" as sensor
 name)
 ts3.FTReached("Force", true, 10) // Enable monitoring of the resultant force of the control
 direction to satisfy 10N.
 ts3.FTReached(true) // Enable monitoring of the absolute values

```

## ● Touchstop Compliance

**TouchStop** ts1 = "Compliance" // Declare the touchstop control variable functioning as Compliance.

TPoint P0 = "JD",{0,0,0,0,0,0}

### (1) Single Axis

```
PTP("JPP",{0,0,90,0,90,0},50,200,0,false)
ts1.Single("Z", 100, 30, 30) // Set timeout to 10000ms
ts1.Ainput("ControlBox", 0, ">=", 3.3) // When ControlBox AI0 is greater than or equal to 3.3V
ts1.RecordPosPoint("P0", false) // Set the record touchstop coordinate value to P0.
int re = ts1.Start() // Start the touchstop compliance control motion
// Set direction Z with a moving distance of 100mm, a force of 30N, and a target speed of 30 mm/s, so it
// will move toward direction Z. When the stop criterion, ControlBox AI0 is greater than or equal to 3.3V,
// is comprehended, it stops the control motion, exits the function, and returns the result values.
```

### (2) Teach

```
TPoint P1 = {516.655,-147.754,445.381,179.444,-0.324,89.912},{0,0,90,0,90,0}
TPoint P2 = {517.419,-147.175,345.184,179.533,-0.319,89.918},{0.057,1.805,103.178,-
14.896,89.998,0.085}
TPoint P3 = {516.655,-147.740,445.388,-162.754,-0.281,89.819},{0.017,6.570,77.958,23.270,89.971,-
0.043}
PTP("JPP",{0,0,90,0,90,0},50,200,0,false)
ts1.Teach("Linear", "P1", "P2", 0, 40, 40) // Set the teach point P1 and P2 in the linear direction
ts1.Timeout(5000) // Set timeout to 5000ms
int re = ts1.Start() // Start the touchstop compliance control motion
// Teach the two points in the linear direction with a force of 40N and a target speed of 40 mm/s. Once
// the moving distance and the teach linear direction are comprehended, it stops the control motion,
// exits the function, and return the result values. Since the timeout is set, if it does not comprehend the
// moving distance in time, it stops the control motion, exits the function, and returns the result values.
PTP("JPP",{0,0,90,0,90,0},50,200,0,false)
ts1.Teach("Rotation", "P1", "P3", 0, 5000, 40)
// Set the teach point P1 and P3 in the rotating direction
re = ts1.Start()
// Teach the two points in the rotating direction with a force of 5000mNm and a target speed of 40
// mm/s. Once the moving distance and the teach rotating direction are comprehended, it stops the
// control motion, exits the function, and return the result values. Since the timeout is set (without
// resetting the timeout of ts1), if it does not comprehend the moving distance in time, it stops the control
// motion, exits the function, and returns the result values.
```

### (3) Advanced

```
PTP("JPP",{0,0,90,0,90,0},50,200,0,false)
ts1.AdvSet("X", true, 100, -100, 30, 30) // Set direction X with advanced parameters
ts1.AdvSet("Z", true, 100, -100, 30, 30) // Set direction Z with advanced parameters
ts1.AdvSet("X", true, 100, -100, 40, 40) // Set direction X with advanced parameters (will overwrite
// the previous AdvSet X parameter)
int re = ts1.Start() // Start the touchstop compliance control motion
// Set direction X and Z for compliance control. Once the moving distance is comprehended, it stops the
// control motion, exits the function, and returns the result values.
```

## ● Touchstop Line

**TouchStop** ts2 = "Line" // Declare the touchstop control variable functioning as Line

TPoint P0 = "JD",{0,0,0,0,0,0}

**(1) Single Axis**

```

PTP("JPP",{0,0,90,0,90,0},50,200,0,false)
ts2.Single("Z", 100, 30, 500) // Set direction Z with distance of 100mm
ts2.DInput("ControlBox", 0, "H") // When ControlBox DI0 is High
ts2.RecordPosPoint("P0", true) // Set the record touchstop coordinate value to P0.
int re = ts2.Start() // Start the touchstop control motion
// Set direction Z for control with a moving distance of 100mm, an accelerate time of 500ms, and a
// target speed of 30 mm/s, so it will move toward direction Z. Once the moving distance is
// comprehended, it stops the control motion, exits the function, and return the result values. When the
// stop criterion, ControlBox DI0 is High, is comprehended, it stops the control motion, exits the function,
// and returns the result values.

```

**(2) Teach**

```

TPoint P1 = {516.655,-147.754,445.381,179.444,-0.324,89.912},{0,0,90,0,90,0}
TPoint P2 = {517.419,-147.175,345.184,179.533,-0.319,89.918},{0.057,1.805,103.178,-
14.896,89.998,0.085}
TPoint P3 = {516.655,-147.740,445.388,-162.754,-0.281,89.819},{0.017,6.570,77.958,23.270,89.971,-
0.043}
PTP("JPP",{0,0,90,0,90,0},50,200,0,false)
ts2.Teach("Linear", "P1", "P2", 0, 40, 500) // Set the teach point P1 and P2 in the linear
 direction
ts2.Timeout(5000) // Set timeout to 5000ms
int re = ts2.Start() // Start the touchstop control motion
// Teach the two points in the linear direction with an accelerate time of 500ms and a target speed of 40
// mm/s. Once the moving distance and the teach linear direction are comprehended, it stops the
// control motion, exits the function, and return the result values. Since the timeout is set, if it does not
// comprehend the moving distance in time, it stops the control motion, exits the function, and returns
// the result values.

PTP("JPP",{0,0,90,0,90,0},50,200,0,false)
ts2.Teach("Rotation", "P1", "P3", 0, 40, 500) // Set the teach point P1 and P3 in the rotating
 direction
re = ts2.Start()
// Teach the two points in the rotating direction with an accelerate time of 500ms and a target speed of
// 40 mm/s. Once the moving distance and the teach rotating direction are comprehended, it stops the
// control motion, exits the function, and return the result values. Since the timeout is set (without
// resetting the timeout of ts1), if it does not comprehend the moving distance in time, it stops the control
// motion, exits the function, and returns the result values.

```

**● Touchstop Force**

```

FTSensor fts3 = "TMFT300" // Construct the device TMFT300.
TouchStop ts3 = "Force", "fts3" // Declare the touchstop control variable and tether fts3 as the
 // sensor name.

TPoint P0 = "JD",{0,0,0,0,0,0}
(1) Single Axis
PTP("JPP",{0,0,90,0,90,0},50,200,0,false)
ts3.Single("Z", 100, 30, 500) // Set direction Z with distance of 100mm

```

```

ts3.DInput("ControlBox", 0, "H") // When ControlBox DI0 is High
ts3.FTReached("Force", true, 2) // Enable monitoring of the resultant force of the control
 // direction to satisfy 2N.

ts3.RecordPosPoint("P0", true) // Set the record touchstop coordinate value to P0.
int re = ts3.Start() // Start the touchstop control motion

// Set direction Z for control with a moving distance of 100mm, an accelerate time of 500ms, and a
// target speed of 30 mm/s, so it will move toward direction Z. Once the moving distance is
// comprehended, it stops the control motion, exits the function, and returns the result values. When the
// stop criterion, ControlBox DI0 is High, is comprehended, it stop the control motion, exits the function,
// and returns the result values. Since the stop criterion on the monitoring of the resultant force is set, if
// it does not comprehend the moving distance in time, it stops the control motion, exits the function,
// and returns the result values.

```

## (2) Teach

```

TPoint P1 = {516.655,-147.754,445.381,179.444,-0.324,89.912},{0,0,90,0,90,0}
TPoint P2 = {517.419,-147.175,345.184,179.533,-0.319,89.918},{0.057,1.805,103.178,-
14.896,89.998,0.085}
TPoint P3 = {516.655,-147.740,445.388,-162.754,-0.281,89.819},{0.017,6.570,77.958,23.270,89.971,-
0.043}
PTP("JPP",{0,0,90,0,90,0},50,200,0,false)
ts3.Teach("Linear", "P1", "P2", 0, 40, 500) // Set the teach point P1 and P2 in the linear
 // direction

ts3.Timeout(5000) // Set timeout to 5000ms
ts3.FTReached("F3D", true, 2) // Enable monitoring of the force of the control direction to
 // satisfy 2N.

int re = ts3.Start() // Start the touchstop control motion

// Teach the two points in the linear direction with an accelerate time of 500ms and a target speed of 40
// mm/s. Once the moving distance and the teach linear direction are comprehended, it stops the
// control motion, exits the function, and return the result values. Since timeout is set, if it does not
// comprehend the moving distance in time, it stops the control motion, exits the function, and returns
// the result values. Since the stop criterion on the monitoring of the resultant force is set, if it does not
// comprehend the moving distance in time, it stops the control motion, exits the function, and returns
// the result values.

PTP("JPP",{0,0,90,0,90,0},50,200,0,false)
ts3.Timeout() // Cancel the timeout stop criterion.
ts3.Teach("Rotation", "P1", "P3", 0, 40, 500)
// Set the teach point P1 and P3 in the rotating direction
re = ts3.Start()

// Teach the two points in the rotating direction with an accelerate time of 500ms and a target speed of
// 40 mm/s. Once the moving distance and the teach rotating direction are comprehended, it stops the
// control motion, exits the function, and return the result values. Since the stop criterion on the
// monitoring of the resultant force is set (without resetting the stop criterion on the resultant force of
// ts3), if it does not comprehend the moving distance in time, it stops the control motion, exits the
// function, and returns the result values.

```



## 25. Force Control Functions

### 25.1 FTSensor Class

Use FTSensor class and declare variables to create a Force Torque Sensor device. The variable name will be the device name.

#### Construct 1

```
FTSensor VariableName = string, string, float[], float[], float
```

```
FTSensor VariableName = string, float[], float[], float
```

```
FTSensor VariableName = string, string
```

```
FTSensor VariableName = string
```

#### Parameters

|         |                                                                                                                                                                                                            |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| string  | Supported models from the sensor suppliers<br>"TMFT300"<br>"OnRobot_HEX-E"<br>"OnRobot_HEX-H"<br>"ROBOTIQ_FT300"<br>"SCHUNK_Axia80"<br>"SRI_M4313N3C"<br>"WACOH_WEF_WKF_115200"<br>"WEF-6A200-4-RG24_9600" |
| string  | connection description required when using serial port for communication                                                                                                                                   |
| float[] | Position setting: X(mm), Y(mm), Z(mm), RX(°), RY(°), RZ(°)                                                                                                                                                 |
| float[] | TCP value: X(mm), Y(mm), Z(mm)                                                                                                                                                                             |
| float   | Tool Mass: kg                                                                                                                                                                                              |

#### Note

```
FTSensor fts_1 = "TMFT300","COM2" // Construct the device TMFT300. The connection parameters
 are void for using EtherCAT communication.
```

```
FTSensor fts_2 = "TMFT300" // Construct the device TMFT300
```

```
FTSensor fts_3 = "ROBOTIQ_FT300","COM2"
```

```
// Construct the device ROBOTIQ_FT300. Require to assign the connection port for using serial port communication.
```

```
FTSensor fts_4 = "ROBOTIQ_FT300","COM2",{0,0,0,0,0,0},{0.12,0.24,0.36},1
```

```
// Construct the device ROBOTIQ_FT300. Require to assign the connection port for using serial port communication
and to configure the settings of the sensor position and tool mass center.
```

\* If not filling position setting, TCP value, or tool mass, it uses the parameters in the force sensor setting in the project.

\* After construction, either in flow projects or script projects, the device will not connect actively until proceeding to read or write.

#### Member Attributes

| Name | Type  | Mode | Description                      | Format |
|------|-------|------|----------------------------------|--------|
| X    | float | R    | The strength value of the X axis |        |
| Y    | float | R    | The strength value of the y axis |        |
| Z    | float | R    | The strength value of the z axis |        |
| TX   | float | R    | The torque value of the X axis   |        |
| TY   | float | R    | The torque value of the y axis   |        |



| Name                | Type    | Mode | Description                                                                        | Format                                         |
|---------------------|---------|------|------------------------------------------------------------------------------------|------------------------------------------------|
| TZ                  | float   | R    | The torque value of the z axis                                                     |                                                |
| F3D                 | float   | R    | The XYZ force strength value                                                       |                                                |
| T3D                 | float   | R    | The XYZ torque value                                                               |                                                |
| Value               | float[] | R    | The XYZ force strength value and torque value array.                               | {X, Y, Z, TX, TY, TZ}, Size = 6                |
| ForceValue          | float[] | R    | The XYZ force strength value array                                                 | {X, Y, Z}, Size = 3                            |
| TorqueValue         | float[] | R    | The XYZ torque value array                                                         | {TX, TY, TZ}, Size = 3                         |
| RefCoordX           | float   | R    | The X-axis strength value measured based on the reference base set in the node     |                                                |
| RefCoordY           | float   | R    | The Y-axis strength value measured based on the reference base set in the node     |                                                |
| RefCoordZ           | float   | R    | The Z-axis strength value measured based on the reference base set in the node     |                                                |
| RefCoordTX          | float   | R    | The X-axis torque value measured based on the reference base set in the node       |                                                |
| RefCoordTY          | float   | R    | The Y-axis torque value measured based on the reference base set in the node       |                                                |
| RefCoordTZ          | float   | R    | The Z-axis torque value measured based on the reference base set in the node       |                                                |
| RefCoordF3D         | float   | R    | The XYZ strength measured based on the reference base set in the node              |                                                |
| RefCoordT3D         | float   | R    | The XYZ torque measured based on the reference base set in the node                |                                                |
| RefCoordForceValue  | float[] | R    | The XYZ strength value matrix measured based on the reference base set in the node | {RefCoordX, RefCoordY, RefCoordZ}, Size = 3    |
| RefCoordTorqueValue | float[] | R    | The XYZ torque value matrix measured based on the reference base set in the node   | {RefCoordTX, RefCoordTY, RefCoordTZ}, Size = 3 |
| Model               | string  | R    | The Model name of the F/T sensor                                                   |                                                |
| Zero                | byte    | R/W  | Turn on or off F/T sensor offset                                                   | 0: Zero OFF, 1: Zero ON                        |

\*Attributes associated with RefCoord\* come with values when in Force Control motions.

## Member Methods

| Name    | Description                        |
|---------|------------------------------------|
| Open()  | Connect to the sensor device.      |
| Close() | Disconnect from the sensor device. |

### 25.1.1 Open()

Open the connection to the Force Torque Sensor.

#### Syntax 1

```
bool Open(
)
```

#### Parameters

void No parameter

#### Return

bool True open successfully

False open unsuccessfully (the project returns an error)

\* After opening the device, it proceeds to the connection to communicate. It may take a while to get the values due to different sensor models.

### 25.1.2 Close()

Close the connection from the Force Torque Sensor.

#### Syntax 1

```
bool Close(
)
```

#### Parameters

void No parameter

#### Return

bool True close successfully

False close unsuccessfully

#### Note

```
FTSensor fts_1 = "ROBOTIQ_FT300","COM2"
```

```
fts_1.Open() // Connect to the device.
```

```
fts_1.Close() // Disconnect from the device.
```

## 25.2 Force Class

Use Force class and declare variables for users to set the robot target force and torque parameters to reach a variety of associated force control motions.

#### Construct 1

```
Force VariableName = string
```

#### Parameters

string sensor name

#### Note

```
FTSensor fts1 = "TMFT300" // Construct the device TMFT300.
```

```
Force fc1 = "fts1" // Declare the force control variable and tether fts1 as the sensor name.
```

#### Member Methods

| Name                              | Defaults                                                                                                       | Description                                                                                                                                                             |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Reset()</a>           |                                                                                                                | Reset all the force control motion parameters to the default except the sensor name.                                                                                    |
| <a href="#">Frame()</a>           | 1                                                                                                              | Set the reference coordinate of the force control motion.                                                                                                               |
| <a href="#">Distance()</a>        | -                                                                                                              | The moving distance of the force control motion (Available only in the SetPoint F/T operation mode.)                                                                    |
| <a href="#">ProtectionSpeed()</a> |                                                                                                                | The speed protection of the force control motion                                                                                                                        |
| <a href="#">FTSet()</a>           | 0,false,5,2,-1<br>1,false,5,2,-1<br>2,false,5,2,-1<br>3,false,0.5,2,-1<br>4,false,0.5,2,-1<br>5,false,0.5,2,-1 | The force and the torque of the force control motion. The positive and negative values denote the force control direction. Users can adjust the PID control parameters. |
| <a href="#">Trajectory()</a>      | -                                                                                                              | The trajectory path of the force control motion. (Trajectory F/T operation mode.)                                                                                       |
| <a href="#">Timeout()</a>         | -                                                                                                              | Stop condition of timeout (Available only in the single F/T operation mode.)                                                                                            |
| <a href="#">AllowPosTol()</a>     | -                                                                                                              | Stop condition of tolerant position error                                                                                                                               |
| <a href="#">DInput()</a>          | -                                                                                                              | Stop condition of digital input                                                                                                                                         |
| <a href="#">AInput()</a>          | -                                                                                                              | Stop condition of analog input                                                                                                                                          |
| <a href="#">FTReached()</a>       | -                                                                                                              | The stop condition of monitoring the values of force, torque, or resultant force acquired.                                                                              |
| <a href="#">Condition()</a>       | -                                                                                                              | Stop condition of the conditional expression                                                                                                                            |
| <a href="#">Start()</a>           | true,false                                                                                                     | Start the force control motion.                                                                                                                                         |
| <a href="#">Stop()</a>            |                                                                                                                | Stop the force control motion.                                                                                                                                          |

### 25.2.1 Reset()

Reset all the force control motion parameters to the default except the sensor name.

#### Syntax 1

```
void Reset(
)
```

#### Parameters

void No parameter

#### Return

void No return

### 25.2.2 Frame()

Set the reference base of the force control motion.

#### Syntax 1

```
void Frame(
 int
)
```

#### Parameters

**int**            The base associated with the control motion

- 0**    The robot base
- 1**    Tool: the base is coupled with the orientation of the tool coordinate.  
(default)
- 2**    The current base
- 3**    Trajectory: the base changes with the path.

**Return**

**void**        No return

**Syntax 2**

```
void Frame(
 string
)
```

**Parameters**

**string**            Point name. Use the TCP coordinate of the point as the base associate with the control motion.

**Return**

**void**        No return

**Syntax 3**

```
void Frame(
 float[],
 string
)
```

**Parameters**

**float[]**            The TCP coordinate of the robot end point: X(mm), Y(mm), Z(mm), RX(°), RY(°), RZ(°)

**string**            Base name. Use the current base name if empty string.

**Return**

**void**        No return

**Syntax 4**

```
void Frame(
 float[]
)
```

**Note**

Same as syntax 3. Fill RobotBase in Base name.

**Syntax 5**

```
void Frame(
 float, float, float, float, float, float,
 string
)
```

**Note**

Same as syntax 3. Replace the type float[] with the parameters in float .

**Syntax 6**

```
void Frame(
 float, float, float, float, float, float
```

)

**Note**

Same as syntax 4. Replace the type float[] with the parameters in float .

**25.2.3 Distance()**

Set the moving distance of the force control motion. (Available only in the SetPoint F/T operation mode.)

**Syntax 1**

```
void Distance(
 int
```

)

**Parameters**

```
int Distance mm
```

```
< 0 No limit to the moving distance.
```

```
>= 0
```

Limited to the moving distance. (Distance calculated from the starting point where the force control motion begins.)

**Return**

```
void No return
```

**25.2.4 ProtectionSpeed()**

Set the speed protection of the force control motion.

**Syntax 1**

```
void ProtectionSpeed(
 int, int, int, int, int, int
```

)

**Parameters**

```
int, int, int, int, int, int
```

Set the speed protection against the axial force or torque: X(mm/s), Y(mm/s), Z(mm/s), RX(deg/s), RY(deg/s), RZ(deg/s)

```
<= 0 No limit to the speed.
```

```
> 0 Limited to the speed
```

\* FTSet() is required to enable speed protection against the respective axial force or torque.

\* Speed protection is available to configure in FTSet() as well.

**Return**

```
void No return
```

**25.2.5 FTSet()**

Set the force and the torque of the force control motion. The positive and negative values denote the force control direction. Users can adjust the PID control parameters.

**Syntax 1**

```
void FTSet(
```

```

int or string,
bool,
float,
int
int

```

```

)

```

### Parameters

```

int or string

```

Axis force or axis torque

```

0 or "FX" FX (N) 3 or "TX" TX (Nm)

```

```

1 or "FY" FY (N) 4 or "TY" TY (Nm)

```

```

2 or "FZ" FZ (N) 5 or "TZ" TZ (Nm)

```

```

bool

```

Whether to enable the control of the assigned axis force or axis torque

```

false Disable

```

```

true Enable

```

```

float

```

The control value of force or torque. The positive and negative values denote the force control direction.

```

int

```

PID control parameter

```

(weak) (strong)

```

```

0 1 2 3 4

```

```

int

```

Speed protection

```

<= 0 No limit to the speed.

```

```

> 0 Limited to the speed

```

### Return

```

void No return

```

### Syntax 2

```

void FTSet(
int or string,
bool,
float,
int
)

```

### Note

Same as syntax 1. Fill -1 to Speed protection parameter for no speed limit.

### Syntax 3

```

void FTSet(
int or string,
bool,
float
)

```

### Note

Same as syntax 1 parameter definition. Fill 2 to PID control parameter and -1 to Speed protection parameter for no speed limit.

### Syntax 4

```
void FTSet(
 int or string,
 bool
)
```

#### Note

Same as syntax 1 parameter definition. Use to set whether to enable the control of the assigned axis force or axis torque only. No change to other parameters (Axis force or axis torque control value, PID control parameter, and Speed protection parameter).

#### Syntax 5

```
void FTSet(
 int or string,
 float
)
```

#### Note

Same as syntax 1 parameter definition. Use to set the control value of force or torque. No change to other parameters (Whether to enable to the control, PID control parameter, and Speed protection parameter).

#### Syntax 6

```
void FTSet(
 int or string,
 float,
 int
)
```

#### Note

Same as syntax 1 parameter definition. Use to set Axis force or axis torque control value and speed control. only. No change to other parameters (Whether to enable to the control and PID control parameter).

#### Syntax 7

```
void FTSet(
 int or string,
 float[]
)
```

#### Parameters

int or string

Axis force or axis torque

|   |    |      |        |   |    |      |         |
|---|----|------|--------|---|----|------|---------|
| 0 | or | "FX" | FX (N) | 3 | or | "TX" | TX (Nm) |
| 1 | or | "FY" | FY (N) | 4 | or | "TY" | TY (Nm) |
| 2 | or | "FZ" | FZ (N) | 5 | or | "TZ" | TZ (Nm) |

float[]

PID control parameter {Kp, Ki, Kd}

#### Note

Same as syntax 1 parameter definition. Use to set PID control parameter (Kp, Ki, Kd) only. No change to other parameters (Whether to enable to the control, Axis force or axis torque control value, and speed control).

## 25.2.6 Trajectory()

Set the trajectory path of the force control motion. (Trajectory F/T operation mode.)

### Syntax 1

```
void Trajectory(
 string
)
```

#### Parameters

**string** Subflow name. Use in flow projects only. Return errors if use in script projects.

#### Return

**void** No return

### Syntax 2

```
void Trajectory(
 ?
)
```

#### Parameters

**?** Trajectory path. Be a statement or a customized function

#### Return

**void** No return

### Syntax 3

```
void Trajectory(
)
```

#### Parameters

**void**

No parameter for cancelling trajectory F/T operation mode. (Change to the SetPoint F/T operation mode.)

#### Return

**void** No return

## 25.2.7 Timeout()

Set the stop condition of timeout. (Available only in the SetPoint F/T operation mode.)

### Syntax 1

```
void Timeout(
 int
)
```

#### Parameters

**int** Timeout in milliseconds  
**< 0** Disable  
**>= 0** Timeout duration

#### Return

**void** No return

### Syntax 2



```
void Timeout(
)
```

**Parameters**

void No parameter for cancelling the stop condition.

**Return**

void No return

### 25.2.8 AllowPosTol()

Set the stop condition of the tolerant position error.

**Syntax 1**

```
void AllowPosTol(
 int
```

```
)
```

**Parameters**

int Error distance in mm  
 < 0 Disable  
 >= 0 Distance of the tolerant error.

**Return**

void No return

**Syntax 2**

```
void AllowPosTol(
)
```

**Parameters**

void No parameter for cancelling the stop condition.

**Return**

void No return

### 25.2.9 DInput()

Set the stop condition of the digital input.

**Syntax 1**

```
void DInput(
 string,
 int,
 int or string
```

```
)
```

**Parameters**

string Control module name  
 ControlBox The control box  
 EndModule The end module  
 ExtModuleN The external module (N = 0 .. n)

int Input channel 0 .. n

int or string

Set the stop condition to Low/High.

|   |    |     |      |
|---|----|-----|------|
| 0 | or | "L" | Low  |
| 1 | or | "H" | High |

**Return**

|      |           |
|------|-----------|
| void | No return |
|------|-----------|

**Syntax 2**

```
void DInput(
)
```

**Parameters**

|      |                                                 |
|------|-------------------------------------------------|
| void | No parameter for cancelling the stop condition. |
|------|-------------------------------------------------|

**Return**

|      |           |
|------|-----------|
| void | No return |
|------|-----------|

**25.2.10 AInput()**

Set the stop condition of the analog input.

**Syntax 1**

```
void AInput(
 string,
 int,
 int or string,
 float
)
```

**Parameters**

|               |                                  |
|---------------|----------------------------------|
| string        | Control module name              |
| ControlBox    | The control box                  |
| EndModule     | The end module                   |
| ExtModuleN    | The external module (N = 0 .. n) |
| int           | Input channel 0 .. n             |
| int or string | Set the condition to judge       |

|   |    |      |                                                                                                     |
|---|----|------|-----------------------------------------------------------------------------------------------------|
| 0 | or | ">"  | Greater than                                                                                        |
| 1 | or | ">=" | Greater than or equal to                                                                            |
| 2 | or | "==" | Equal to (Recommend not to use since it is not easy to hold the equal condition with analog input.) |

|   |    |      |                       |
|---|----|------|-----------------------|
| 3 | or | "<=" | Less than or equal to |
| 4 | or | "<"  | Less than             |

|       |                 |
|-------|-----------------|
| float | Condition value |
|-------|-----------------|

**Return**

|      |           |
|------|-----------|
| void | No return |
|------|-----------|

**Syntax 2**

```
void AInput(
)
```

**Parameters**

|      |                                                 |
|------|-------------------------------------------------|
| void | No parameter for cancelling the stop condition. |
|------|-------------------------------------------------|

**Return**

void      No return

**25.2.11 FTReached()**

Set the stop condition of monitoring the values of force, torque, or resultant force acquired.

**Syntax 1**

```
void FTReached(
 int or string,
 bool,
 float
)
```

**Parameters**

|              |                                                                                            |    |         |         |   |    |       |          |
|--------------|--------------------------------------------------------------------------------------------|----|---------|---------|---|----|-------|----------|
| <b>int</b>   | The values of force, torque, or resultant force.                                           |    |         |         |   |    |       |          |
|              | 0                                                                                          | or | "FX"    | FX (N)  | 3 | or | "TX"  | TX (Nm)  |
|              | 1                                                                                          | or | "FY"    | FY (N)  | 4 | or | "TY"  | TY (Nm)  |
|              | 2                                                                                          | or | "FZ"    | FZ (N)  | 5 | or | "TZ"  | TZ (Nm)  |
|              | 6                                                                                          | or | "F3D"   | F3D (N) | 7 | or | "T3D" | T3D (Nm) |
| <b>bool</b>  | Whether to enable monitoring the assigned the values of force, torque, or resultant force. |    |         |         |   |    |       |          |
|              | false                                                                                      |    | Disable |         |   |    |       |          |
|              | true                                                                                       |    | Enable  |         |   |    |       |          |
| <b>float</b> | The monitoring value                                                                       |    |         |         |   |    |       |          |

**Return**

void      No return

**Syntax 2**

```
void FTReached(
 int or string,
 bool
)
```

**Note**

Same as syntax 1. Use to whether to enable monitoring the assigned the values of force, torque, or resultant force.

**Syntax 3**

```
void FTReached(
 bool
)
```

**Parameters**

|             |                                                   |         |
|-------------|---------------------------------------------------|---------|
| <b>bool</b> | Whether to enable the absolute values monitoring. |         |
|             | false                                             | Disable |
|             | true                                              | Enable  |

**Syntax 4**

```
void FTReached(
)
```

**Parameters**

`void` No parameter for cancelling the stop condition.

**Return**

`void` No return

**25.2.12 Condition()**

Set the stop condition of the conditional expression.

**Syntax 1**

```
void Condition(
 bool or ?
)
```

**Parameters**

`bool or ?` The conditional can be true/false or a bool return of the statement.

**Return**

`void` No return

**Syntax 2**

```
void Condition(
)
```

**Parameters**

`void` No parameter for cancelling the stop condition.

**Return**

`void` No return

**25.2.13 Start()**

Start the force control motion.

**Syntax 1**

```
int Start(
 bool,
 bool
)
```

**Parameters**

`bool` Zero out force sensor before execution.

`true` Enable (default)

`false` Disable

`bool` Enable tool gravity compensation

`true` Enable

`false` Disable (default)

**Return**

`int` Return the result value after the force control motion stops.

`0` Not Working

`1` Working

|     |                               |
|-----|-------------------------------|
| 2   | Timeout                       |
| 3   | Distance reached              |
| 4   | IO triggered                  |
| 5   | Resisted                      |
| 6   | Error                         |
| 14  | Over Speed                    |
| 201 | Digital IO triggered          |
| 202 | Analog IO triggered           |
| 203 | Variable                      |
| 204 | Force is comprehended         |
| 205 | Allowable Position Tolerances |
| 206 | Motion Finish                 |

**Syntax 2**

```
int Start(
 bool
)
```

**Note**

Same as syntax 1. Enabling the tool gravity compensation parameter will retrieve the parameter values of the associated device name in the force sensor setting of the project. Fill false if not retrieving the associated device name.

**Syntax 3**

```
int Start(
)
```

**Note**

Same as syntax 1. Fill true to zero out force sensor before execution. Enabling the tool gravity compensation parameter will retrieve the parameter values of the associated device name in the force sensor setting of the project. Fill false if not retrieving the associated device name.

**25.2.14 Stop()**

Stop the force control motion.

**Syntax 1**

```
int Stop(
)
```

**Parameters**

void No parameter

**Return**

int Return the result value after the force control motion stops.

## ● Parameter Settings

```
FTSensor fts1 = " TMFT300" // Construct the device TMFT300.
```

```
Force fc1 = "fts1" // Declare the force control variable and tether fts1 as the sensor name.
```

```

(1)
fc1.Frame(3) // Set the force control coordinate as trajectory.
fc1.Frame({517.5,-147.8,442.45,180,0,90}) // Set the force control coordinate as point (will
// overwrite the previous setting)
fc1.Frame("P1") // Set the force control coordinate as point (will overwrite the
// previous setting)
fc1.Reset() // Reset all parameters (reserve tethering "fts1" as sensor name)

(2)
fc1.Frame(3)
fc1.Frame({517.5,-147.8,442.45,180,0,90})
fc1.Frame("P1")
fc1.FTSet(0, true, 5, 0) // Set axis FX with force control 5N with PID 0.
fc1.FTSet(1, true, 6, 1) // Set axis FY with force control 6N with PID 1.
fc1.FTSet(2, true, 7, 3) // Set axis FZ with force control 7N with PID 3.
fc1.FTSet("FZ", true, 8) // Set axis FZ with force control 8N with PID 2. (will overwrite
// the previous FZ setting)
fc1.Reset() // Reset all parameters (reserve tethering "fts1" as sensor name)

(3)
fc1.Frame(1) // Set the force control coordinate as tool.
fc1.StopDuration(300) // Set the duration of compliance stop. to 300ms
// Force and torque control
fc1.Distance(1000) // Set moving distance to 1000mm
fc1.FTSet("FZ", true, 5) // Set axis FZ with force control 5N
// Stop conditions
fc1.Timeout(10000) // Set timeout to 10000ms
fc1.AllowPosTol(100) // Set the tolerant error to 100mm
fc1.DInput("ControlBox", 0, "H") // When ControlBox DI0 is High
fc1.AInput("ControlBox", 0, ">=", 3.3) // When ControlBox AI0 is greater than or equal to 3.3V
fc1.FTReached(2, true, 1) // Axis FZ satisfies 1N
fc1.FTReached("FZ", true, 2) // Axis FZ satisfies 2N (will overwrite the previous FZ setting)
fc1.FTReached(true) // Enable the absolute value monitoring
int count = 0
fc1.Condition(count > 100) // conditional expression
fc1.Reset() // Reset all parameters (reserve tethering "fts1" as sensor name)

```

## ● SetPoint F/T Operation Mode

SetPoint mode applies mainly to the robot touching the target with force control.

**FTSensor** fts1 = "TMFT300"

**Force** fc1 = "fts1"

```

(1)
PTP("JPP",{0,0,90,0,90,0},50,200,0,false)
fc1.Distance(100) // Set moving distance to 100mm
fc1.FTSet("FZ", true, 5) // Set axis FZ with force control 5N
fc1.Start() // Start the force control motion. Zero out force sensor before
// execution.
// Set axis FZ with force control 5N and the moving distance of 100mm, and the robot will move along

```

with the Z-direction. The robot stops moving once it satisfies the moving distance. However, the motion is still under the force control, namely in the function.

(2)

```
PTP("JPP",{0,0,90,0,90,0},50,200,0,false)
fc1.FTSet("FZ", true, 5) // Set axis FZ with force control 5N
fc1.AllowPosTol(80) // The stop condition of the tolerant position error.
int re = fc1.Start() // Start the force control motion. Zero out force sensor before
 // execution.

// Set axis FZ with force control 5N, and the robot will move along with the Z-direction and monitor the
// stop conditions concurrently. By satisfying the stop condition, the robot stops moving once it reaches
// within the tolerant position error, stops the force control motion, exits the function, and returns the
// result value.
```

(3)

```
PTP("JPP",{0,0,90,0,90,0},50,200,0,false)
fc1.Frame(1) // Set the force control coordinate as工具
fc1.StopDuration(300) // Set the duration of compliance stop.to 300ms
// Force and torque control
fc1.Distance(200) // Set moving distance to 200mm
fc1.FTSet("FZ", true, 5) // Set axis FZ with force control 5N
// Stop conditions
fc1.Timeout(10000) // Set timeout to 10000ms
fc1.AllowPosTol(1000) // Set the tolerant error to 1000mm
fc1.DInput("ControlBox", 0, "H") // When ControlBox DI0 is High
fc1.AInput("ControlBox", 0, ">=", 3.3) // When ControlBox AI0 >= 3.3V
int count = 0
fc1.Condition(count > 100) // conditional expression
int re = fc1.Start() // Start the force control motion. Zero out force sensor before
 // execution.

// The robot stops moving once it satisfies the moving distance. However, the motion is still under the
// force control, and the system monitors the stop conditions continuously. Not until any stop conditions
// are comprehended will the force control motion be stopped, the function exit, and the result values
// after stopping be returned.
```

## ● Trajectory F/T Operation Mode

Trajectory mode applies mainly to the robot touching the target with robot base position control and force control.

**FTSensor** fts1 = " **TMFT300**"

**Force** fc1 = "fts1"

(1)

```
fc1.FTSet("FZ", true, 2) // Set axis FZ with force control 2N
fc1.Trajectory("FTSubflow0") // Set FTSubflow0 as the trajectory path. (Use in flow projects only.)
int re = fc1.Start() // Start the force control motion. Zero out force sensor before
 // execution.

// Suppose FTSubflow0 comes with a few point nodes to finish.
```

```
// Set axis FZ with force control 2N and the trajectory path as the subflow FTSubflow0. The robot
// follows the nodes in the subflow to control and with force control concurrently. When the subflow
// ends, it stops the force control motion, exits the function, and returns the result value.
```

(2)

```
fc1.FTSet("FZ", true, 2) // Set axis FZ with force control 2N
fc1.Frame(3) // Set the force control coordinate as trajectory
fc1.Trajectory("FTSubflow0") // Set FTSubflow0 as the trajectory path. (Use in flow projects only.)
int re = fc1.Start() // Start the force control motion. Zero out force sensor before
 // execution.

// Suppose FTSubflow0 comes with a few point nodes to finish.
// Set axis FZ with force control 2N and the trajectory path as the subflow FTSubflow0. The robot
// follows the nodes in the subflow to control and with force control concurrently. When the subflow
// ends, it stops the force control motion, exits the function, and returns the result value.
```

(3)

```
fc1.FTSet("FZ", true, 2) // Set axis FZ with force control 2N
fc1.Frame(1) // Set the force control coordinate as tool
fc1.Trajectory("FTSubflow1") // Set FTSubflow1 as the trajectory path. (Use in flow projects only.)
// Stop conditions
fc1.Timeout(10000) // Timeout is invalid in the trajectory mod.
fc1.AllowPosTol(1000) // Set the tolerant error to 1000mm
fc1.DInput("ControlBox", 0, "H") // When ControlBox DI0 is High
fc1.FTReached("FZ", true, 2) // Axis FZ satisfies 2N
fc1.FTReached(true) // Enable the absolute value monitoring
int re = fc1.Start() // Start the force control motion. Zero out force sensor before
 // execution.

// Suppose FTSubflow1 comes with a few point nodes to loop the loop.
// Set axis FZ with force control 2N and the trajectory path as the subflow FTSubflow1. The robot
// follows the nodes in the subflow to control and with force control concurrently. Since the subflow is
// looping the loop, the robot motion, the force control, and the stop condition monitoring keep on. Not
// until any stop conditions are comprehended will the force control motion be stopped, the the function
// exit, and the result values after stopping be returned.
```

(4)

```
define
{
 FTSensor fts1 = " TMFT300"
 Force fc1 = "fts1"
 int count = 0
}
main
{
 fc1.FTSet("FZ", true, 1, 0) // Set axis FZ with force control 8N with PID 0
 fc1.Frame(1) // Set the force control coordinate as tool
 fc1.Trajectory(FTMotion())//Set FTMotion() customized function as the trajectory path.
 // Stop conditions
 fc1.Timeout(10000) // Timeout is invalid in the trajectory mode.
 fc1.AllowPosTol(1000) // Set the tolerant error to 1000mm
```



```

fc1.DInput("ControlBox", 0, "H") // When ControlBox DI0 is High
fc1.FTReached("FZ", true, 2) // Axis FZ satisfies 2N
fc1.FTReached(true) // Enable the absolute value monitoring
fc1.Condition(count++ > 200000) // conditional expression
int re = fc1.Start()

 // Start the force control motion. Zero out force sensor before
 execution.

Display(re)
}
void FTMotion()
{
 while (true)
 {
 PTP("JPP",{15,0,90,0,90,0},50,200,100,false)
 PTP("JPP",{15,0,75,0,90,0},50,200,100,false)
 PTP("JPP",{ -15,0,75,0,90,0},50,200,100,false)
 PTP("JPP",{ -15,0,90,0,90,0},50,200,100,false)
 Sleep(10)
 }
}
// Set axis FZ with force control 1N, PID 0, and the trajectory path as FTMotion() customized function.
// The robot follows the contents of FTMotion() to control and with force control concurrently. Since this
// function is looping the loop, the robot motion, the force control, and the stop condition monitoring
// keep on. Not until any stop conditions are comprehended will the force control motion be stopped, the
// function exit, and the result values after stopping be returned.

```

## Revision History Table

| Revision | Date       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 00       | July, 2025 | <ul style="list-style-type: none"><li>● Modified TM Ethernet Slave data table and TMSVR descriptions</li><li>● Adjustment of chapter orders in External Script &amp; TM Ethernet Slave</li><li>● Added the chapter of CC-Link</li><li>● Added the FTSensor sensor model TMFT300</li><li>● Added Force.ProtectionSpeed() function</li><li>● Updated Force.FTSet() function</li><li>● Updated Velocity() function and Position() function</li><li>● Added the chapter of Real Time Remote Server (RTRS)</li></ul> |

